GE Medical Systems

Technical Publications

Direction 5133322

Revision 10

Signa® System Upgrades from Pre–EXCITE to EXCITE HD

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1	Mar 18, 2005	Initial Release.
2	Jun 30, 2005	Section 1 – Updated system configuration pictorial view with addition of HFD–s/PDU designation. Section 2 – Updated upgrade flowcharts to include IIQ Kit and RF Ground Strap. Also added alternative path for ACGD to HFD–S upgrade for applicable system upgrades. Section 4 – Updated Cable table notes for reusing Runs 762, 763, and 764 when upgrading ACGD to HFD–S. Updated applicable cable maps with same information. Added note in Section 4–7–1 for upgrading ACGD to HFD–S and updated cable location information on interface panel. Sections 6, 7, 8, & 9 – Added IIQ Kit (M3090NE) and RF Ground Strap (M3090NF) installation sub–sections for sites that have not already installed the two catalogs. Section 7 – For LCC and Cx magnets, added plastic replacement bracket in Rear Pedestal for connection of Runs 743 and 744 to RF Drive cables. Section 15 – Added information for upgrading ACGD to HFD–S. Added subsection 15–4 for connecting power cables to HFD–S. Section 18 – Revised catalog upgrade flowcharts and applicable pictorial views.
3	Jan 30, 2006	Section 4 – Revised information in cable removal table referring to Runs 762, 763, and 764 in equipment room. Added information to Note 6 on page 4–11 to address gradient cable installation for various cabinet upgrades. Added Pre–LCC LX cable maps that have Gradient/GRAM cabinets. Added to notes on all removal cable maps with information referring to upgrading grient cables in equipment room. Section 6 – Added instructions for replacing existing latch on RF Coil with new Draw Latch Assembly style, Section 6–4–3. New end bell has the mating Latch Keeper installed. Section 7 – Added instructions to add two new mounting holes to new RF Drive cable plastic brakcet, Section 7–5–1–D (MRIhc09538). Added alternative mounting for Ground Isolatoin Filter Boxes, Section 7–5–1–F (MRIhc09510).
4	Aug 18, 2006	Removed references to IIQ Key procedures in Sec 1–4, 2–3, and 3–1, since it is not part of product prior to upgrade (MRIhc15742). Sec 2 – Added “Power On” and “Host Monitor Setup and Cal” procedures. Sec 7 – Revised the fastener mounting hardware for the 16 Channel switch mounting plate installation.
5	Oct 26, 2006	Section 7 – Updated PAC replacement instructions with information on removing the PAC Control Housing (MRIhc17607, MRIhc18836).
6	Mar 7, 2007	Section 7: Removed PAC2B part number.
7	Aug 10, 2007	Sections 6, 7, 8 and 9: Updated the Hybrid Splitter upgrade by adding the Netcom Hybrid Splitter which has different Isolation Filter mounting brackets than the existing Teledyne Hybrid Splitter.
8	Sep 28, 2007	Section 1: Added sub–section 1–3 for non–English label instalaltion instructions. Section 2: Added LCC magnet with Horizon enclosure to flowchart in sub–section 2–3. Section 4: Sub–section 4–2: Revised information on Run 421 in cable table 4–1 for Cx and LCC magnets that need a longer Run 421. Also updated cable maps to reflect this information. Revised Run 421 cable connection to SRI–3 instructions in sub–section 7–1–3–C (PQR 13094054). Added information for connection of Mobile system Heat Exchanger power cable, Run 046, to PDU. Updated cable maps to reflect this information. Added heat Exchanger power cable terminal connection in PDU information to table in sub–section 4–7–1–C (PSR 13120695).
9	Jun 17, 2009	Sections 6, 7, 8, & 9:: Removed instructions on installation of rating plates (PQR 13246367). Added English label and information on attaching non–English labels (PQR 13246299).
10	Nov 20, 2010	Updated Ferrous Material Warnings for Dock, Blower Box, Rear Pedestal, Tools, and other components throught out manual. Section 1: Added 200 gauss zone illustration and notice about safety training. Sections 8 & 9: Inserted additional information for installation of blower box. Section 17–1: Added check for proper blower box anchoring. (CAPA 1859293).

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SECTION 1 – GETTING STARTED

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1–1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 1.5T pre–EXCITE system to Signa 1.5T EXCITE HD (Release 12.x). Refer to illustration 1–1 for pictorial overview of upgraded system.

NOTICE:

During this hardware upgrade, the magnet may be ramped to full field strength. Individuals working in the magnet room must be MR safety trained.

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE. THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.

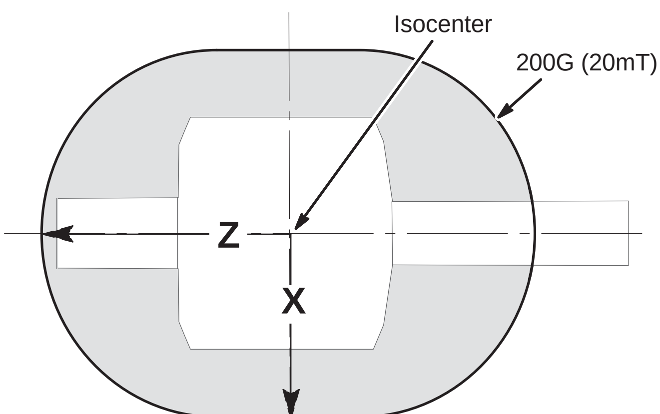

Illustration is not to scale

Approximate Distance from isocenter to 200G Line		
	X	**Z**
1.5T	150 cm (59 in)[1]	205 cm (81 in)[1]
Unshielded Magnet	Measure [2]	
Note 1: These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed. Note 2: For unshielded magnets, the magnetic field must be measured using a gauss meter.		

1–2 SITE READY CHECK

Pre–installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa 1.5T EXCITE system are listed in the *Signa 1.5T EXCITE HD Upgrade Pre–Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46–301450G1 Fiber Optic Cable Connector Repair kit

1–3 NON–ENGLISH LABEL INSTALLATION

Components and accessories delivered to this site may have English labels attached to them. For non–English sites, optional multi–language labels have also been supplied.

If required, non–English labels should be attached over the English labels.

- For Surface Coils and Coil Cables – Apply the appropriate label over the English label. The label is included with the coil component.
- For all other labels, follow the label update procedures in the manual or on the applicable multi–language label sheet.

Verify at the end of the installation that the original English labels are covered by new labels representing the native language of the customer.

1–4 PRE–DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 1.5T EXCITE HD Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1–5 UPGRADE SCHEDULE CONSIDERATIONS

FMIs, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man–hour and system down–time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1–6 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The “Shipping Document” lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1–7 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **“damage in shipment”** written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1–8 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1–8–1 MATERIAL POLICY OVERVIEW

1–8–1–1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1–8–1–2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1–8–2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the EXCITE HD upgrade to GoldSeal which includes:

- All removed cabinets
- Penetration Panel
- SRI module
- PAC module
- All removed Magnet Enclosure parts
- All removed Rear Pedestal parts
- LPCC, Carriage, and/or removed Bridge and related parts
- Octane or Indigo or Operator Console, and Computer cabinet with all associated cables and modules

1–8–3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1–8–4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1–8–5 REGION SPECIFIC RETURN INFORMATION

1–8–5–1 Sites in Americas

A GoldSeal return pre–printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1–877–251–4468 or 262–513–4147

1–8–5–1 Sites in Americas (Continued)

For non–GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)–747–6997 or Dial Comm 8*579–6997
Fax: (414)–747–6855
Field ASPEN: 1–800–525–1516, Box #26041

1–8–5–2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS–Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1–8–5–3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS–E – GOLD SEAL OPERATION & RECYCLING CENTER
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL
Quai 45 à 49 – 91090 LISSES – FRANC
Tel: (+33) 1–60–91–57–35 or (+33) 1 64 97 55 41
Fax: (+33) 1–60–86–12–31

For non–Gold Seal returns in Europe:

European Renewable Resources Center
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2
Quai 45 à 49 – 91090 LISSES – FRANCE
TEL = (+33)–1–64–97–55–41 FAX = (+33)–1–60–86–12–31

1–8–6 PRODUCT LOCATOR INORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1–9.

1–9 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans–shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
   http://gein2.med.ge.com/gib

   The FE Site Verification consists of three components that are available on the web from the main menu. They are:

   - Install/deinstall product locator model and serial numbers
   - Add/modify ship to address information
   - Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page – http://supportcentral.ge.com/15563

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans–shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1–10 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on–site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un–cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1–11 UPGRADE SYSTEM CONFIGURATION

Illustrations 1–1 and 1–2 show the major equipment installed for a Signa 1.5T EXCITE HD Upgrade.

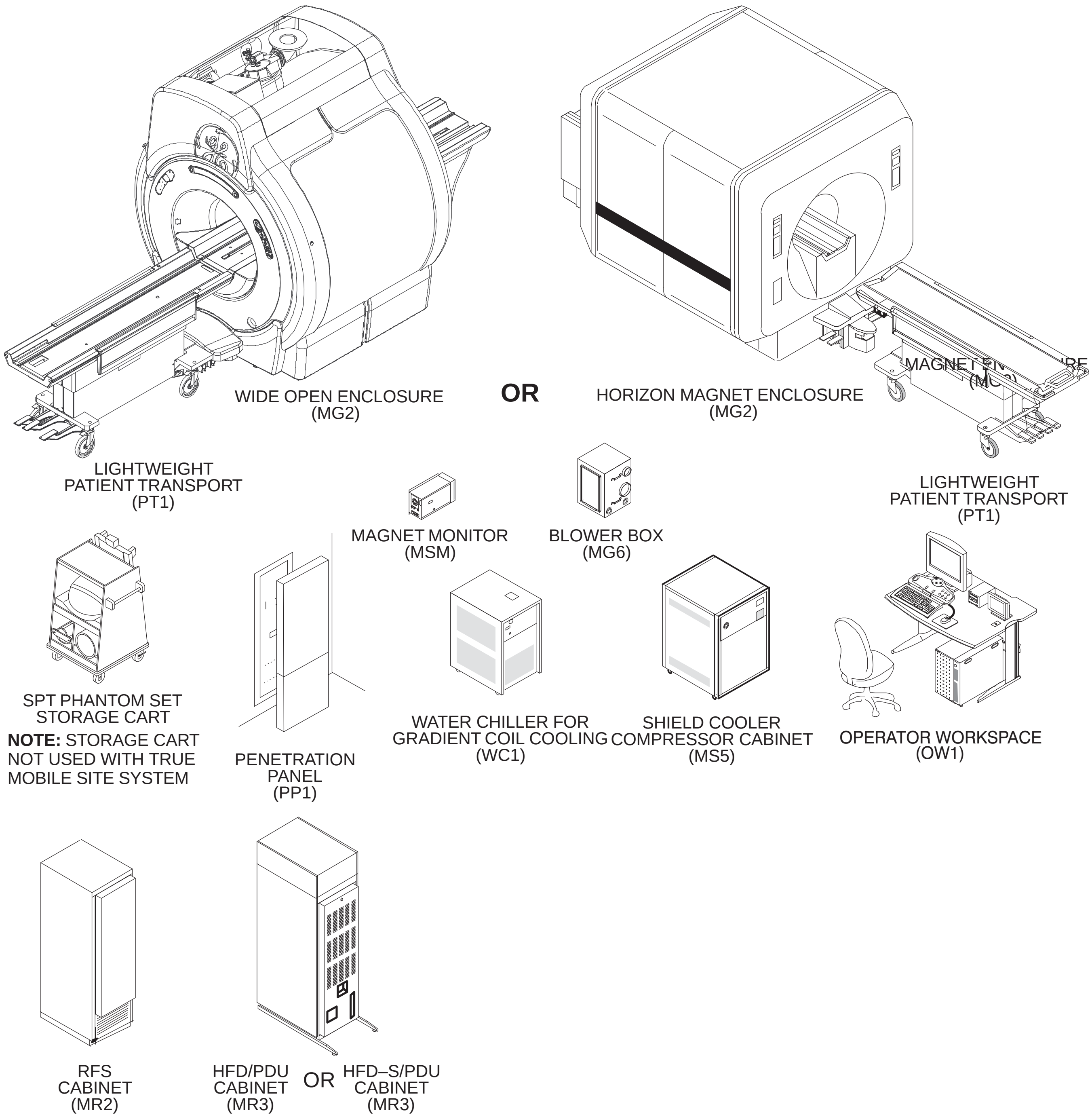

ILLUSTRATION 1–2
PRE–EXCITE UPGRADED TO EXCITE HD NON–TWINSPEED SYSTEM MAJOR EQUIPMENT

1–12 UPGRADE SYSTEM CONFIGURATION (Continued)

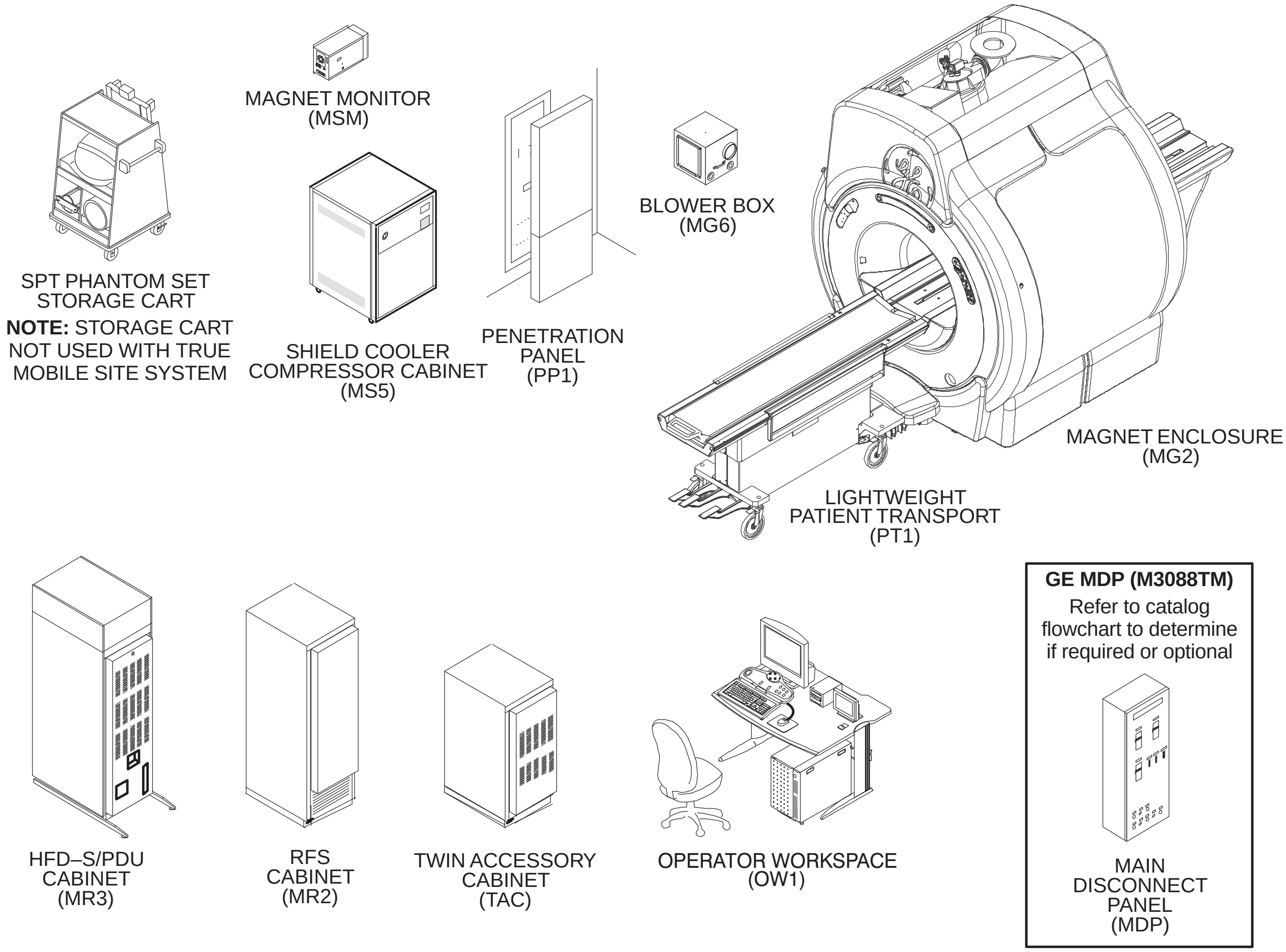

ILLUSTRATION 1–3
PRE–EXCITE UPGRADED TO Signa 1.5T EXCITE HD TwinSpeed System

SECTION 2 – UPGRADE INSTALLATION PROCESS

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2–1 SITE READINESS

All on–site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

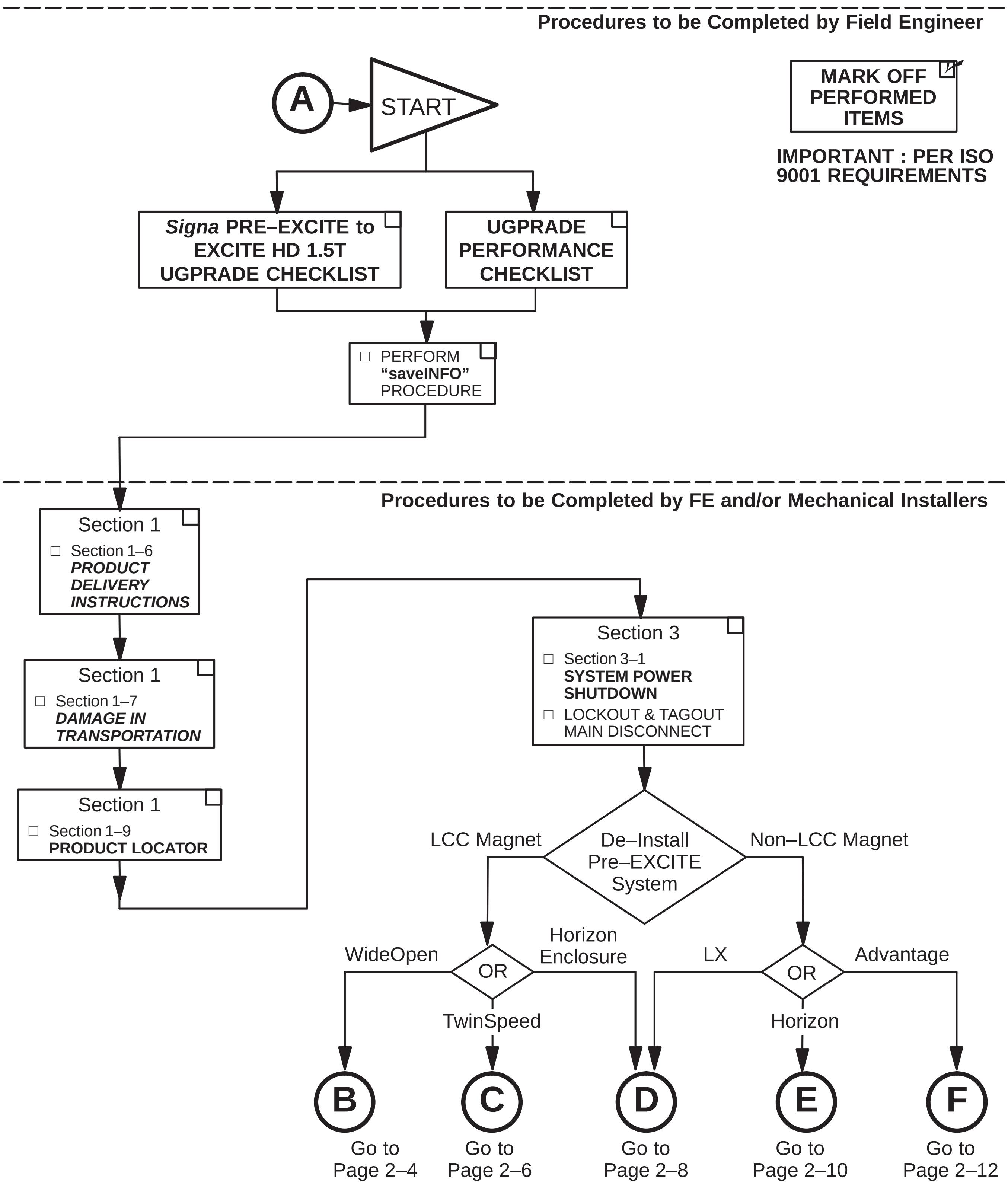
2–2 UPGRADE PROCEDURE

The following upgrade flow charts contain information for upgrading to Signa EXCITE HD 1.5T Release 12.x.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

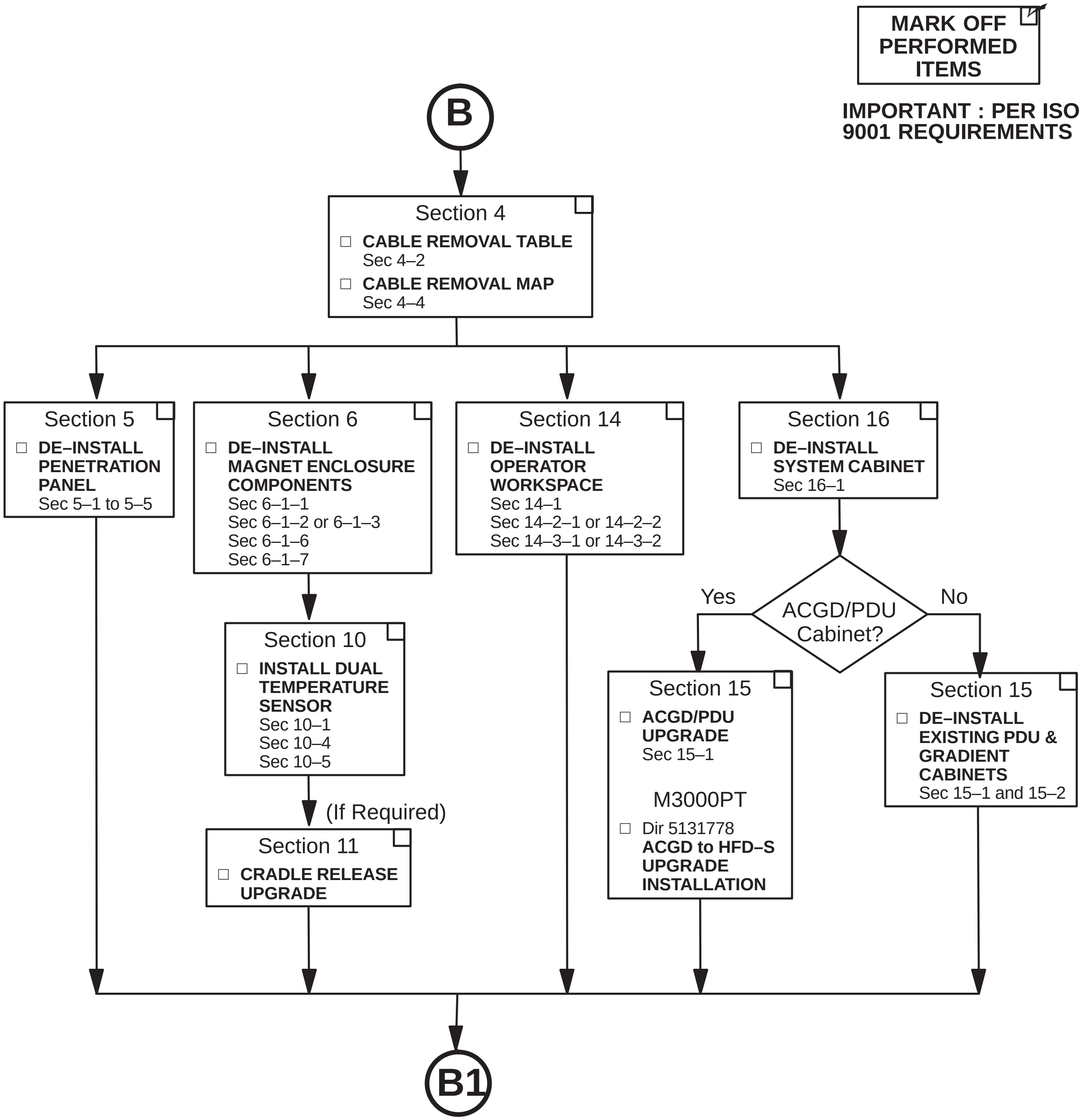
This upgrade flowchart has been developed assuming that all Signa EXCITE HD equipment has been delivered together. **Make sure that every part required is available before starting.**

2–3 SYSTEM MECHANICAL UPGRADE START FLOWCHART



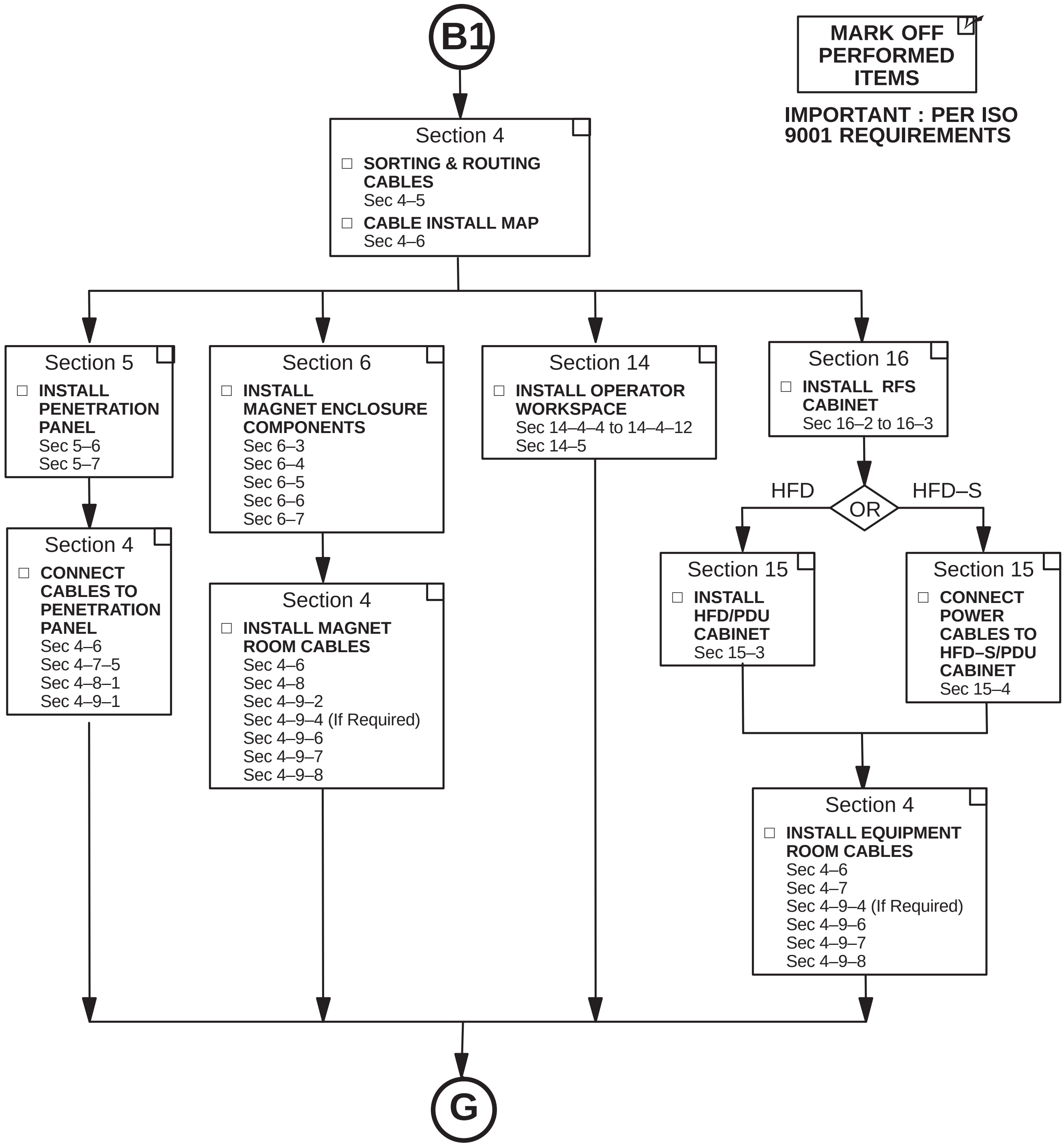
2–3–1 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR WideOpen

De–Installation Procedures to be Completed by FE and/or Mechanical Installers



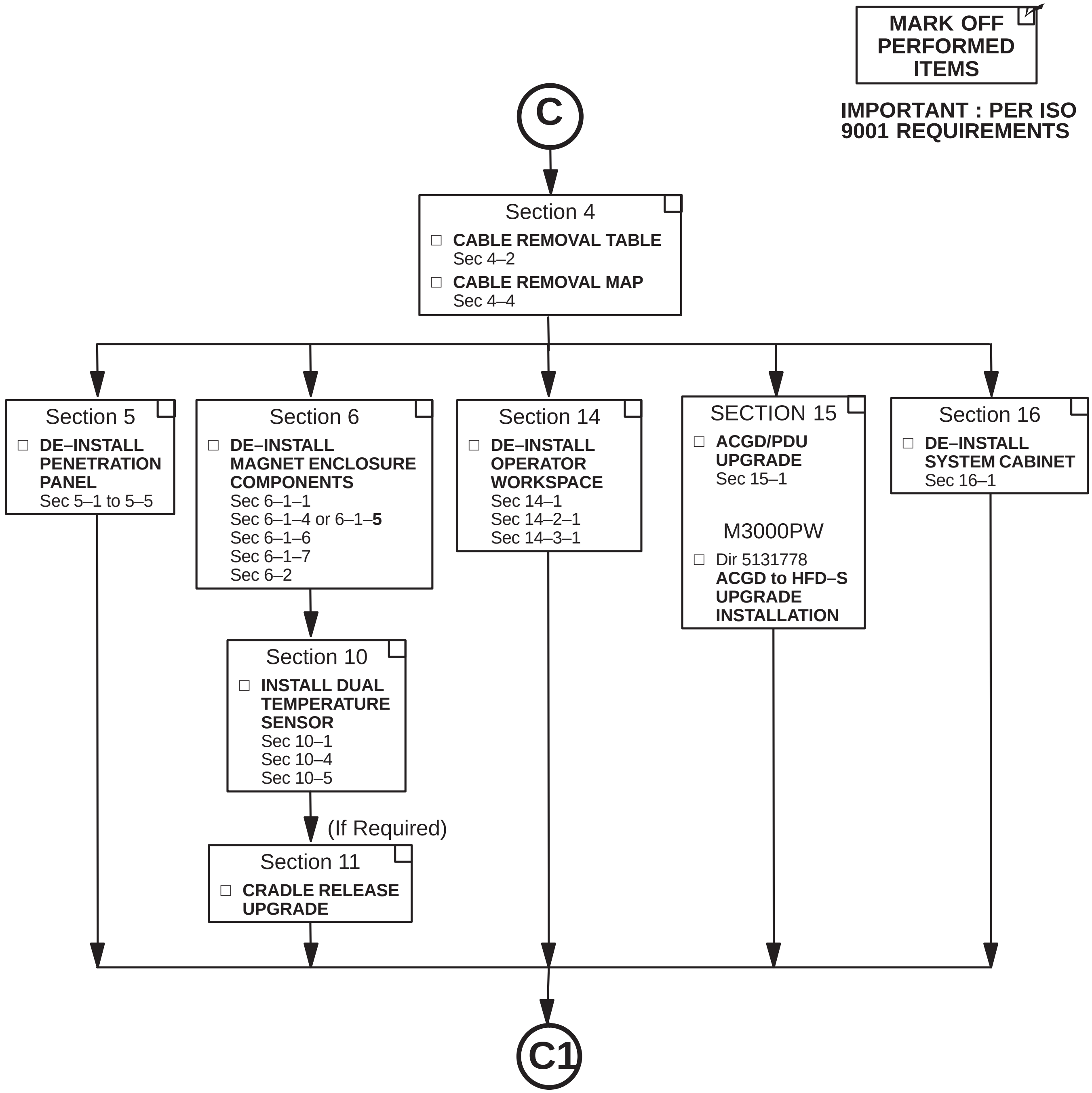
2–3–1 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR WideOpen (Continued)

De–Installation Procedures to be Completed by FE and/or Mechanical Installers



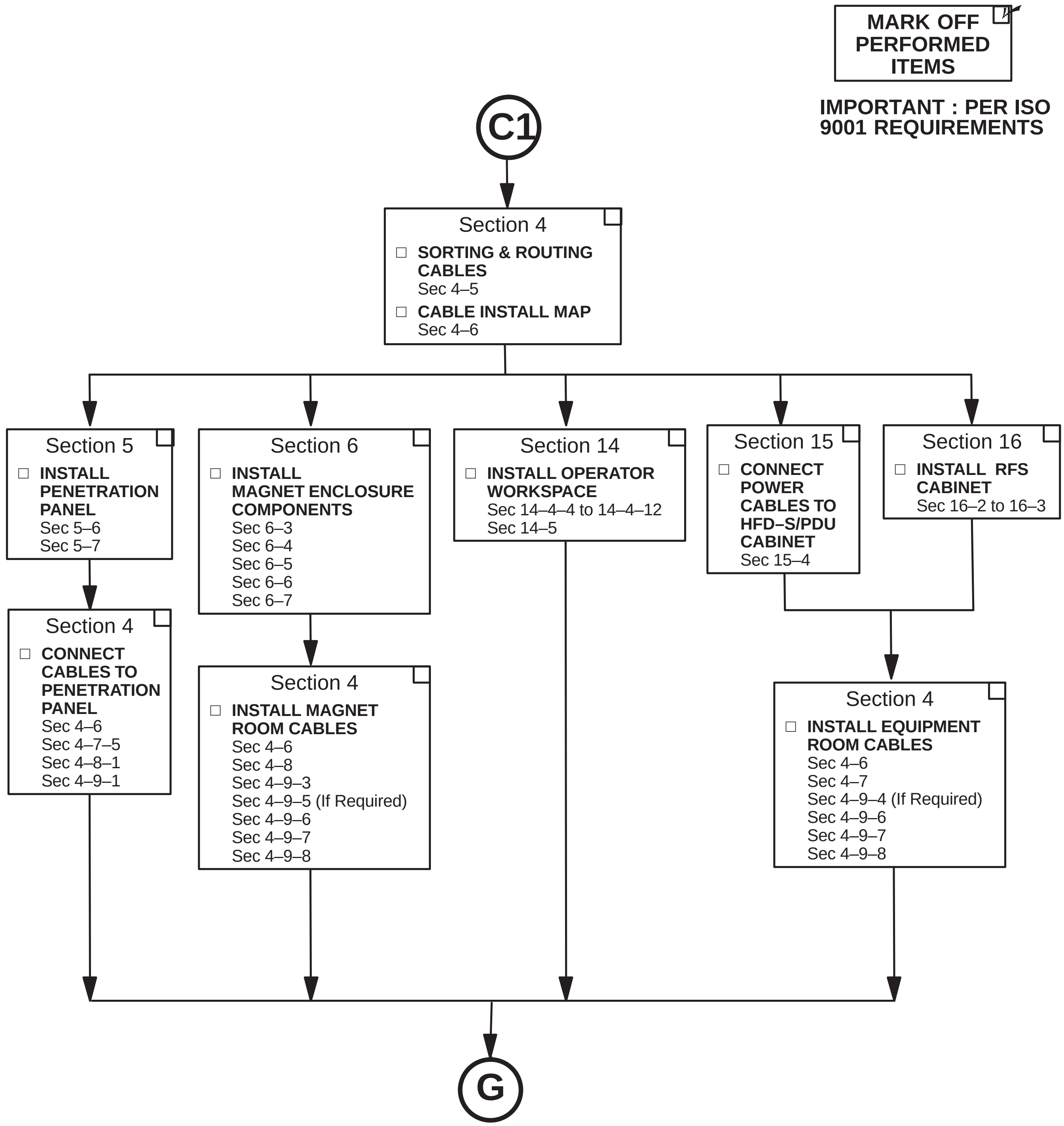
2–3–2 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR TwinSpeed

De–Installation Procedures to be Completed by FE and/or Mechanical Installers



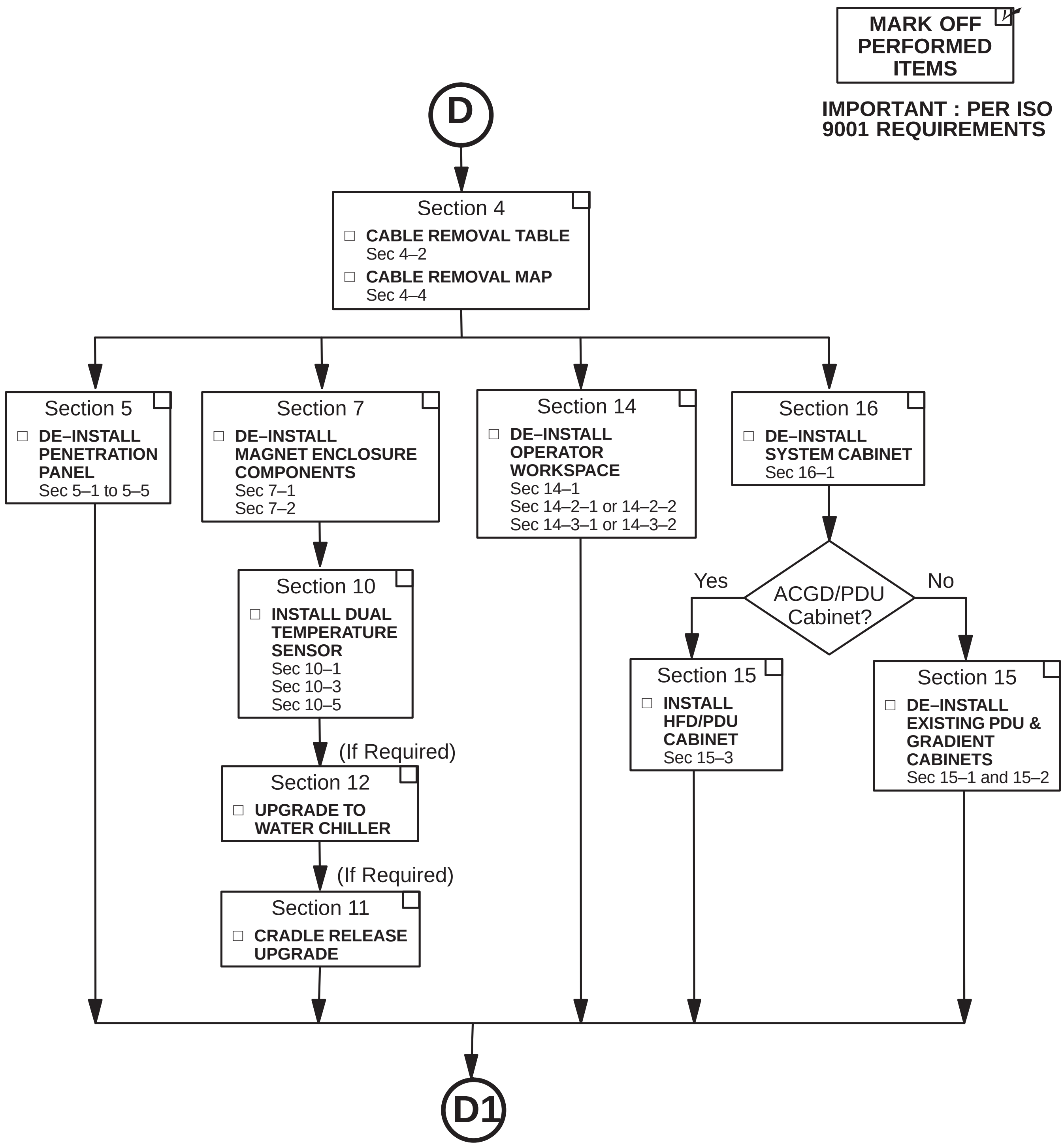
2–3–2 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR TwinSpeed (Continued)

De–Installation Procedures to be Completed by FE and/or Mechanical Installers



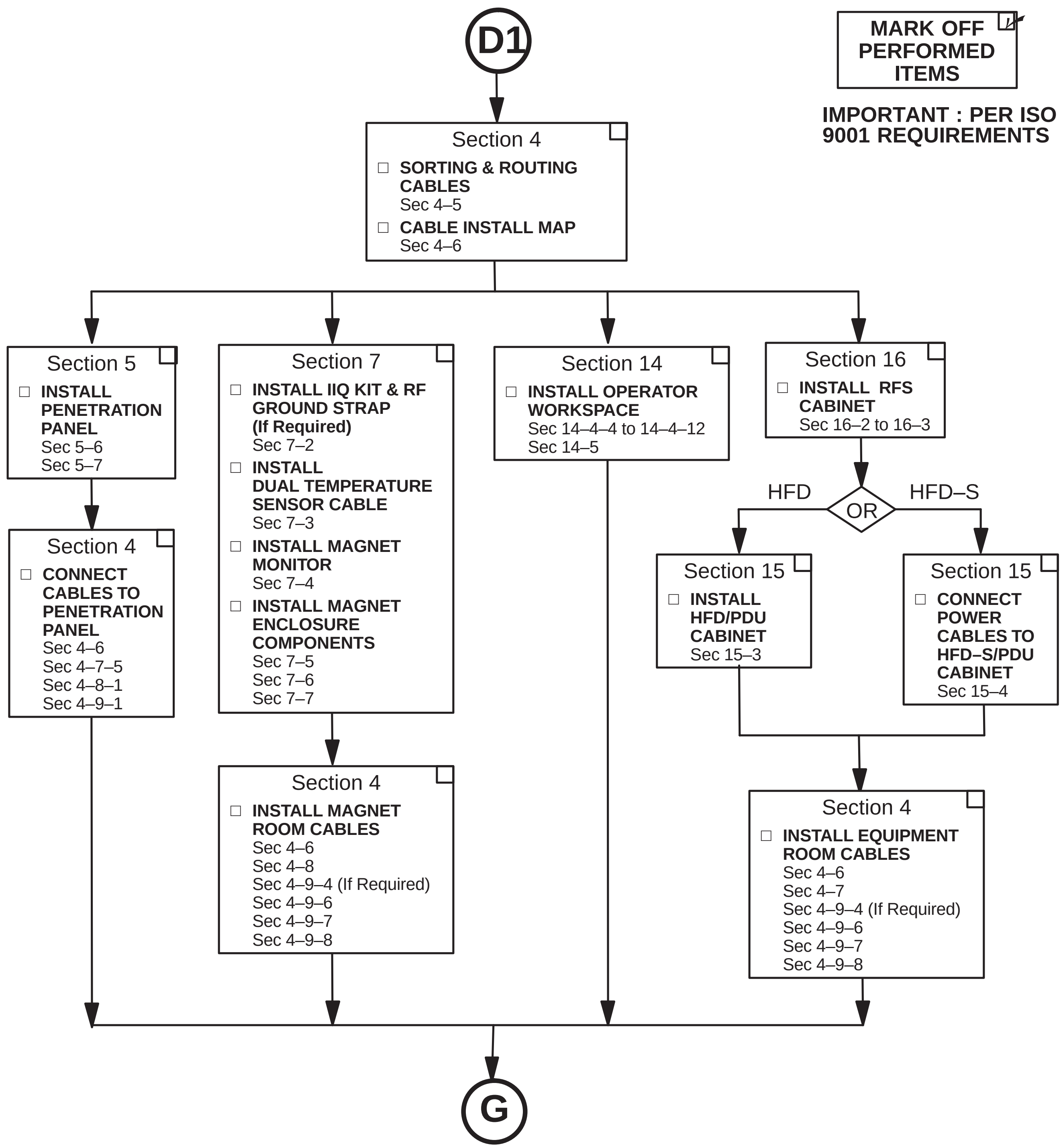
2–3–3 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON–LCC OR LCC WITH HORIZON ENCLOSURE LX

De–Installation Procedures to be Completed by FE and/or Mechanical Installers



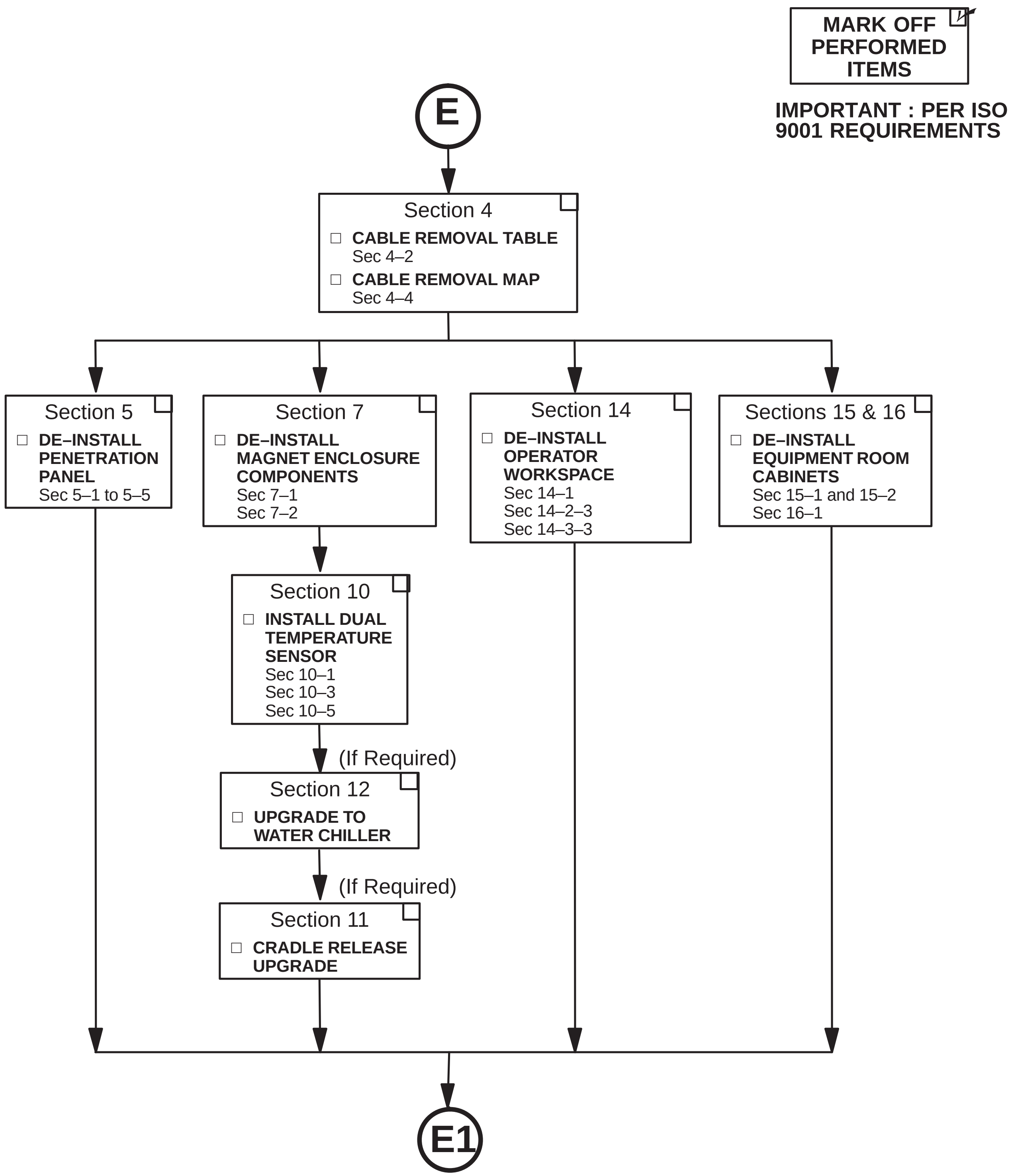
2–3–3 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON–LCC OR LCC WITH HORIZON ENCLOSURE LX (Continued)

De–Installation Procedures to be Completed by FE and/or Mechanical Installers



2–3–4 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON–LCC HORIZON

De–Installation Procedures to be Completed by FE and/or Mechanical Installers

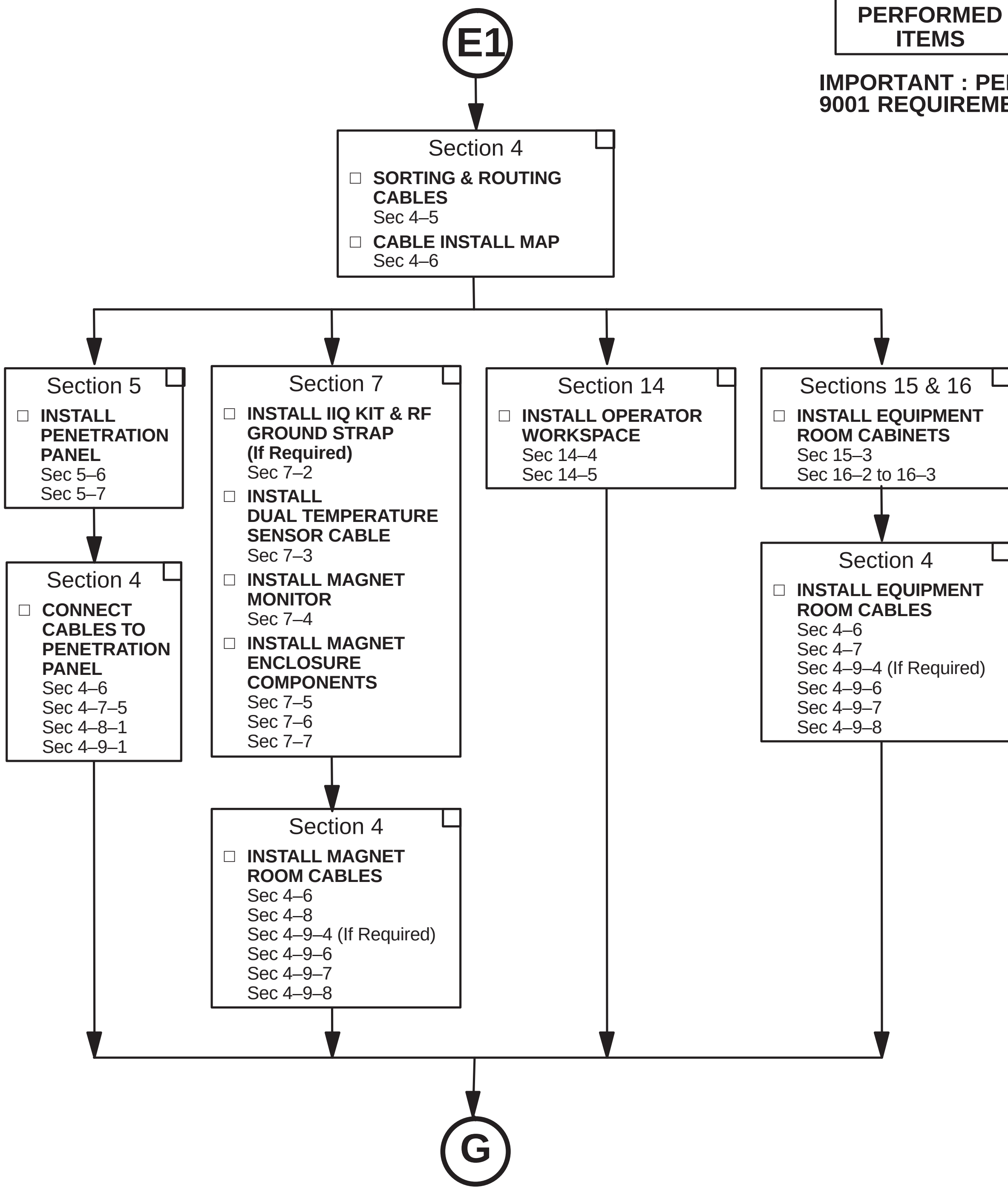


2–3–4 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON–LCC HORIZON (Continued)

De–Installation Procedures to be Completed by FE and/or Mechanical Installers

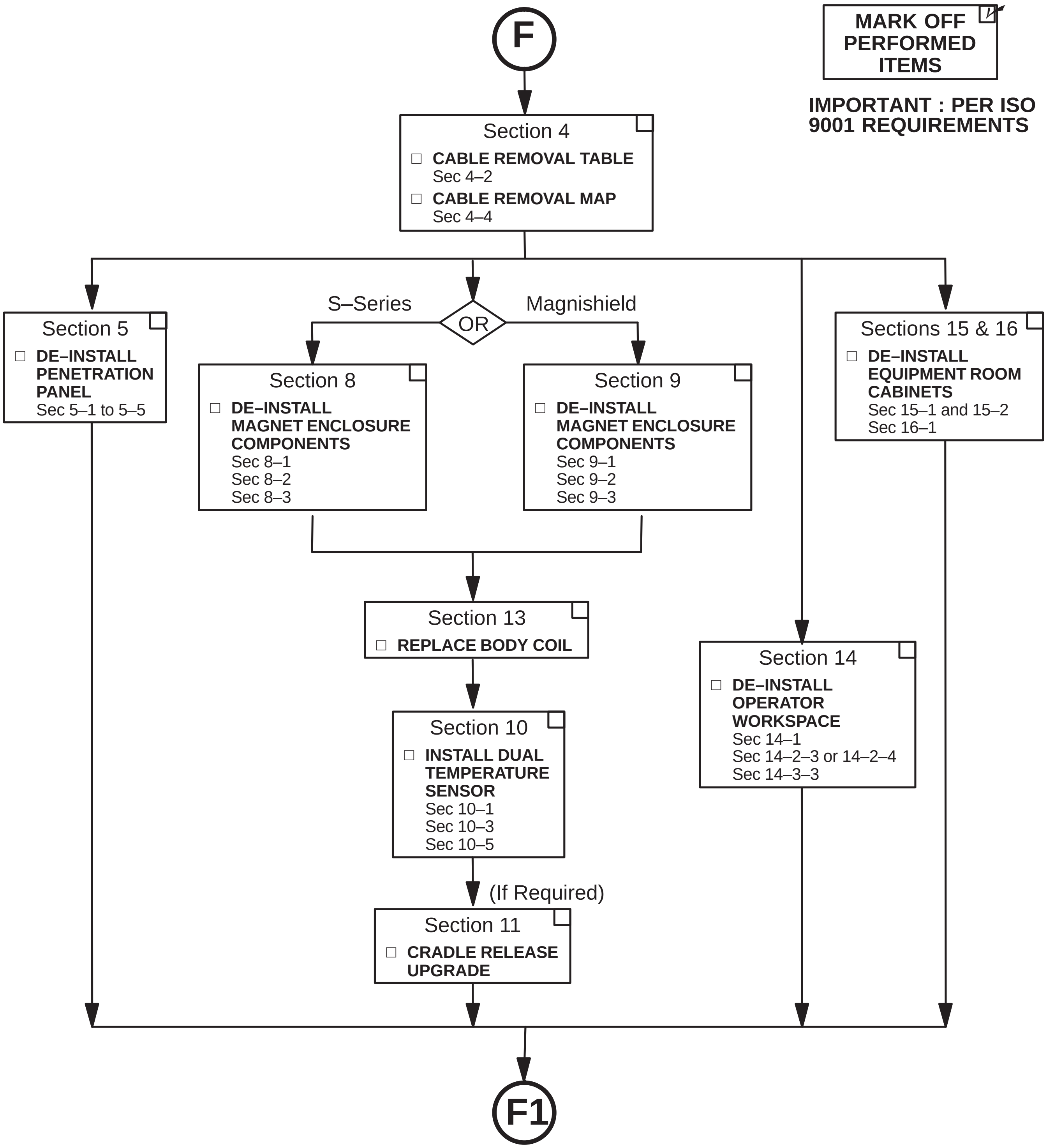
MARK OFF PERFORMED ITEMS

IMPORTANT : PER ISO 9001 REQUIREMENTS



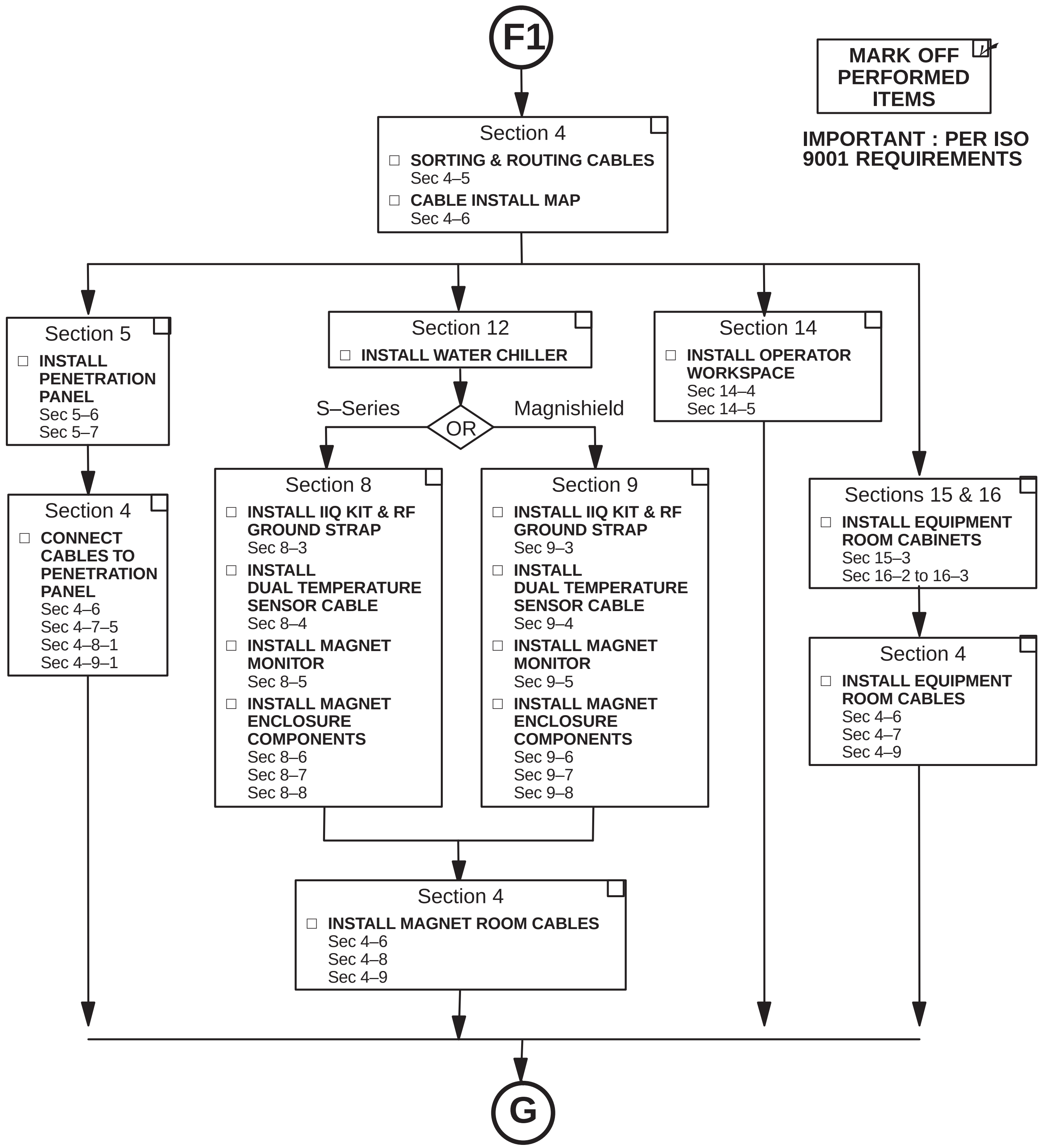
2–3–5 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON–LCC ADVANTAGE

De–Installation Procedures to be Completed by FE and/or Mechanical Installers



2–3–5 SYSTEM MECHANICAL UPGRADE FLOWCHART FOR NON–LCC ADVANTAGE (Continued)

De–Installation Procedures to be Completed by FE and/or Mechanical Installers



2–4 UPGRADE COMPLETION CHECKS AND POWER ON FLOWCHART

Procedures to be Completed by FE and/or Mechanical Installers

MARK OFF PERFORMED ITEMS ☐

IMPORTANT : PER ISO 9001 REQUIREMENTS

G

Section 17 ☐

- ☐ Section 17–1 **MECHANICAL INSTALLATION COMPLETION CHECKLIST**
- ☐ Section 17–2 **RETURNING SHIPPING EQUIPMENT**
- ☐ Section 17–3 **MATERIAL DISPOSITION GUIDANCE**
- ☐ Section 17–4 **FINAL COORDINATION**

MECHANICAL INSTALLATION COMPLETE

Procedures to be Completed by FE and Applications

Section 14 ☐

- ☐ Direction 5133322>Section 14–6 **POWER ON INFORMATION**

Section 11 ☐

- ☐ Direction 5133322>Section 17–5>**POWER ON**

or

MR Service Methods CD–ROM – Direction 5133315

- ☐ Adjustments/Calibrations>System Level Procedures>***System Power On***

MR Service Methods CD–ROM – Direction 5133315 ☐

- ☐ Adjustments/Calibrations>***Host LCD Monitor Setup and Calibration***

MR Service Methods CD–ROM – Direction 5133315 ☐

- ☐ Installation>EXCITE 1.5 HD Installation Calibration Flows> **EXCITE 1.5T HD System Upgrade Callibration Flow**

Applications/Training ☐

- ☐ **Turn site over to Applications who will instruct users.**

TURN OVER TO CUSTOMER

SECTION 3 – POWER OFF

3–1 SYSTEM POWER SHUTDOWN

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute “Save Info”.
2. Verify that System performance baseline measurements have been obtained, especially:
   - SPT Full Test Mode
   - White Pixel Test
   - LV Shim
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – SYSTEM CABLES

TABLE OF CONTENTS

(Continued on next page)

TABLE OF CONTENTS (Continued)

4–1 OVERVIEW

This section provides information on removing and re–labeling Pre–EXCITE system cables, and installing new cables for upgrading to Signa 1.5T EXCITE HD.

4–2 CABLE REMOVAL FOR UPGRADING TO SIGNA EXCITE HD

Table 4–1 lists all cables to be removed from various early system configurations for upgrading to a Signa 1.5T EXCITE HD system.

Signa Advantage 4.x systems that are being upgraded to Signa Horizon LX or EXCITE require many more cables to be removed or re–labeled than indicated on the cable removal maps on pages 4–9 thru 4–12. Table 4–1 lists all cables removed from various early system configurations, including Advantage 5.4 and Horizon 5.x, for upgrading to Signa EXCITE HD.

On the following table 4–1, the letters at the top of each column on the right side of the table are defined by the following cable removal explanations. An “X” in the box indicates that the cable should be removed.

For upgrading to EXCITE HD, remove these cables if existing sytem is:

A – Signa LCC WideOpen LX with Octane Computer, ACGD and Magnet Monitor

B– Signa LCC TwinSpeed LX with Octane Computer, ACGD and Magnet Monitor

C – Signa Pre–LCC LX with Octane Computer and ACGD, and without Magnet Monitor

D – Signa LCC LX with Octane Computer, RF/PDU, SGD, and Magnet Monitor

E – Signa Pre–LCC LX with Octane Computer, RF/PEN, PDU, SGD (or Grad/GRAM), and w/o Magnet Monitor

F – Signa Pre–LCC LX with Indigo Computer, RF/PEN, PDU, SGD (or Grad/GRAM), and w/o Magnet Monitor

G – Signa Horizon 5.x with Operator Console/Computer, RF/PEN, PDU, Grad/GRAM and w/o Magnet Monitor

H– Signa Advantage 5.x with Operator Console/Computer, RF, PEN, PDU, Gradient and w/o Magnet Monitor

I – Signa Advantage 4.5 or later but pre–5.4 with Operator Console, Computer, RF, PEN, PDU, and Gradient

J – Signa Advantage 4.x but pre–4.5 with Operator Console, Computer, RF, PEN, PDU, and Gradient

Note
Discard means to remove and return to GE for recycling

4–2 CABLE REMOVAL FOR UPGRADING TO SIGNA EXCITE HD (continued)

TABLE 4–1
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
002	PD1–A2–TS1–32/NEUT/GND	MR2–A12–SERVICE									X	X	
003	PD1–A2–TS1–1,2,3/NEUT/GND	MR3–A10–AC IN								X	X	X	
007	PD1–FEEDER 15	PP2–A6–AC IN										X	
008	PD1–FEEDER 16	PP2–A6–SERVICE										X	
009	PD1–A2–TS1–10,11,12/NEUT/GND	MR1–A7–J1					X	X	X	X	X	X	
011	PD1–A2–TS1–7,8,9	RESISTIVE SHIM CABINET											SEE NOTE 1
012	PD1–A2–TS1–21/NEUT/GND	OC1–A8–J3 (AC IN)									X	X	
013	PD1–A2–TS1–33/NEUT/GND	OC1–A8–J6 (SERVICE)									X	X	
014	PD1–A2–TS1–34/NEUT/GND	CC1–A20–SERVICE									X	X	
015	PD1–A2–TS1–13,14,15/NEUT/GND	CC1–A20–AC IN									X	X	
016	PD1–A2–TS1–16,17,18/NEUT/GND	CC2–A20–AC IN										X	
017	PD1–A2–TS1–34/NEUT/GND	CC2–A20–SERVICE										X	
020	PD1–A2–TS1–28,29,30	RESISTIVE SHIM CABINET											SEE NOTE 1
021	PD1–A2–GND	CC1–A20–GND									X	X	
022	PD1–A2–GND	CC2–A20–GND										X	
023	PD1–GND BUS	OC1–GND							X	X	X	X	
027	PD1–A2–TS1–32/NEUT/GND	PP2–A15–J1–SERVICE								X	X	X	
028	PD1–A2–TS1–31/NEUT/GND	PP2–A15–J2–AC IN								X	X	X	
030	PDI–A2–TS1–16,17,18,NEUT,GND	MR2–A12–J1 AC IN				X	X	X	X	X	X	X	
035	PD1–A2–TS1–22,23,24/NEUT/GND	CC2–A20–AC IN										X	
036	PD1–FEEDER 5–TS1–4,5,6	MR6										X	
037	PD1–GND BUS	MR2–A11 GND	X	X	X	X	X	X	X	X	X	X	
038	PD1–GND BUS	MR1–A7 GND	X	X	X		X	X	X	X	X	X	
041	PD1–TS1–1,2,3,GND	MR3–CABI/FJ6				X	X	X	X				
042	PD1–TS1–4,5,6,GND	MR3–CABI/FJ7				X	X	X	X				
043*	MR7–A4–J7	PD1–A2					X	X	X				
044	PP1–GND	MR1–A7–GND	X	X	X	X	X	X	X				
045	PP1–GND	MG6–GND	X	X	X	X	X	X	X				
046	PD1–TS1–20,NEUT,GND	HE1					X	X	X				SEE NOTE 13
047	PD1–TS1–X,X,NEUT,GND	OW1–A2–A5–J1						X					
048	PD1–GND BUS	OW1–A3–A5–GND	X	X	X	X	X	X					
049	OW1 A1 A1	OW1 A2 A5 J2	X	X	X	X	X	X					
050	OC1–AC–IN	PD1–TS1–13,14,15,N,GND							X	X			
064	MR2–A12–J1–AC OUT	CC1							X	X			
065	PD1–TS1–26,27,GND	WC1											SEE NOTE 13
200	MR2–A11–J19	MR3–A9–J1									X	X	
201	MR2–A11–J20	MR3–A9–J2									X	X	
202	MR2–A11–J21	MR3–A9–J3									X	X	
203	MR2–A11–J22	MR3–A9–J4									X	X	
207	MR2–A11–J6	RF DOOR SWITCH											SEE NOTE 2
223	OC1–A8–J13	MR2–A1–J36									X	X	
224	CC1–A21–J19	MR1–A7–J8									X	X	

4–2 CABLE REMOVAL FOR UPGRADING TO SIGNA EXCITE HD (continued)

TABLE 4–1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
226	MR1–A7–J5	MR2–A11–J14									X	X	
227	PD1–A1–A2–J2	MR2–A11–J16									X	X	
228	CC1–A21–J10	MR2–A11–J17									X	X	
229	MR2–A11–J1	MR1–A7–J3	X	X	X	X	X	X	X	X	X	X	
231	MR2–A11–J2	PP1–J8	X	X	X	X	X	X	X	X	X	X	
233	MR2–A1–J5	PP1–J7									X	X	
235	MR1–A7–J13	PP1–J44	X	X	X	X	X	X	X	X	X	X	
236	MR1–A7–J12	PP1–J4	X	X	X	X	X	X	X	X	X	X	
238	CC1–A21–J14	OC1–A8–J8									X	X	
239	CC1–A21–J15	OC1–A8–J10									X	X	
244	CC1–A21–J13	BTI–HOST										X	
248	CC2–A21–J1	OC1–A9–J1										X	
249	CC2–A21–J22	OC1–A8–J12										X	
256	OC1–A8–J11	PP1–J12									X	X	
257	MG3–A3–J1	PP1–J44	X	X	X	X	X	X	X	X	X	X	
258	MG3–A17–J1	PP1–J4	X	X	X	X	X	X	X	X	X	X	
260	PP1–J7	MG3–A17–J2									X	X	
261	PP1–J6	MG2–A11–A1–J1										X	
262	PP1–J8	MG3–A3–A5–OUTPUT	X	X	X	X	X	X	X	X	X	X	
274	CC1–A21–J20	MR1–A7–J7									X	X	
280	PP2–A1–J2	PP1–J15										X	
281	PP2–A1–J3	PP1–J11										X	
282	MG3–A2–J8	PP1–J12	X	X	X	X	X	X	X	X	X	X	
287	CC2–A21–J15	CC1–A21–J2										X	
296	PP1–J18	EO2											SEE NOTE 8
297	PP1–J18	EO1											SEE NOTE 8
300	PP1–J15	MG2–A15											
301	CC2–A21–J20	CC1–A21–J28										X	
302	CC2–A21–J21	CC1–A21–J29										X	
318	PP1–J11	MG3–A7–LONG DRIVE	X	X	X	X	X	X	X	X	X	X	
322	MR2–A11–J11	PP2–A6–J18										X	
325	CC1–A21–J27	CC2–A21–J19										X	
327	CC1–A21–J25	CC2–A21–J17										X	
346	MR2–A11–J41	CM1–FRONT									X	X	
347	MR2–A11–J42	CM1–BACK									X	X	
354	CM1	ST1									X	X	
370	MR1–A7–J19	PP1–A9–J46										X	
384	PP2–A15–J17	PP1–J20								X	X	X	
385	PP1–J20	MG2–A30–TS1,2,3											
386	PP2–A15–J21	PP1–J47								X	X	X	
387	MG2–A29–P1	PP1–J47	X	X	X	X	X	X	X	X	X	X	
388	PP2–A15–J22	PP1–J11								X	X	X	

4–2 CABLE REMOVAL FOR UPGRADING TO SIGNA EXCITE HD(continued)

TABLE 4–1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
389	PP2–A15–J20	PP1–J15								X	X	X	
404	MG3 A17 J1	MG2 A16 A5 J6	X	X	X	X	X	X	X	X	X	X	
405	MR3–A9–TS1–1/2	PP1–A7–1,2								X	X	X	
405	MG3–TS1–1,2	PP1–A7–1,2								X	X	X	
406	MR3–A9–TS1–5/6	PP1–A7–3,4								X	X	X	
406	MG3–TS1–3,4	PP1–A7–3,4								X	X	X	
407	MR3–A9–TS1–5/6	PP1–A7–5,6								X	X	X	
407	MG3–TS1–5,6	PP1–A7–5,6								X	X	X	
408	PP1–J16	PD1–A14–A1–J1								X	X	X	
409	PP1–J16	MG2–A12–A1–J7								X	X	X	
410	PP1–J14	PP2–A15–J23										X	
411	PP1–J14	MG3–A13–J1										X	
412	PP1–J51	MG3–A32–J1								X	X	X	
413	PP1–J51	PP2–A15–J28(A16–J9)								X	X	X	
414	MG2–A35–MIC	MG3–A2–J9											
415	PP1–A8–J45	MG3–A33–J7										X	
416	MR1–A7–J21	PP1–A8–J45										X	
417	PP2–A15–J19	MR2–A11–J32									X	X	
418	MR2–A11–J39	PP2–A15–J										X	
419	MR2–A11–J26	PP1–A10–J1									X	X	
420	MG3–A33–J1	PP1–A10–J2									X	X	
421	MG2–A33–J3	MG2–A31–J1											SEE NOTE 14
422	MG2–A33–J2	MG3–A1–J4											
424	PP1–A10–J5/J6	MG2–A33–J11/J12									X	X	
425	MR2–A11–J12	PP1–J6										X	
426	MR1–A7–J9	MR3–A9–J5									X	X	
427	MR2–A11–J17	CC1–A21–J13									X	X	
428	MR2–A11–J44	CC1–A21–J35									X	X	
429	BT1	CC1–A21–J25									X	X	
430	MR2–A11–J24	CC1–A21–J26									X	X	
431	MR2–A11–J25	CC1–A21–J27									X	X	
433	OC1–A9–J1	CC1–A21–J10									X	X	
434	CC1–A8–J12	CC1–A21–J29									X	X	
435	MR2–A11–J13	PP1–A10–J3									X	X	
436	MG2–A12–A1–J1	MG3–A33–J3									X	X	
437	MG2–A12–A1–J2	MG3–A33–J4									X	X	
439	MG3–A3–J4	MG3–A33–J1									X	X	
440	MG3–A3–J3	MG3–A34–J2									X	X	
441	MG3–A33–J7	MG3–A34–J8									X	X	
446	PP1–A9–J46	MG2–A12–A13–J5										X	
447	PP1–A9–J46	MG2–A12–A13–J3										X	
448	MR2–A11–J44	CC1–A21–J1										X	

4–2 CABLE REMOVAL FOR UPGRADING TO SIGNA EXCITE HD(continued)

TABLE 4–1 (CONT'D)

CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
449	MR2–A11–J24	CC1–A21–J34										X	
450	MR2–A11–J25	CC1–A21–J35										X	
453	PP1–J50	MG2–A11–A1–J4									X	X	
454	MR2–A11–J34	PP1–J50									X	X	
457	OM1	PP1–J17											SEE NOTE 3
458	OM3	PP1–J17											SEE NOTE 3
470	MR1–A7–J17	PP1–A11–J71								X	X	X	
474	PP1–A11–J72	MG3–A12–A1–J3								X	X	X	
475	PP1–A11–J73	MG3–A12–A1–J4								X	X	X	
476	PP1–A11–J74	MG3–A33–J7								X	X	X	
477	PP1–A11–J75	MG3–A12–A1–J5								X	X	X	
478	PP1–A11–J76	MG3–A12–A1–J6								X	X	X	
485	MG3 A17 J2	MG2 A16 A3 OUT	X	X	X	X	X	X	X	X	X	X	
486	PP1–A14–J85	MR1–A7–J22	X	X	X	X	X	X	X	X	X	X	
487	PP1–A15–J78	MR1–A7–J20	X	X	X	X	X	X	X	X	X	X	
488	PP1–J80	MR2–A11–J8	X	X	X	X	X	X	X	X	X	X	
489	PP1–J81	MR2–A11–J9	X	X	X	X	X	X	X	X	X	X	
490	PP1–J82	MR2–A11–J10	X	X	X	X	X	X	X	X	X	X	
491	PP1–J80	MG3–A36–J7	X	X	X	X	X	X	X	X	X	X	
492	PP1–J81	MG3–A36–J8	X	X	X	X	X	X	X	X	X	X	
493	PP1–J82	MG3–A36–J9	X	X	X	X	X	X	X	X	X	X	
494	PP1–A14–J87	MG3–A36–J6	X	X	X	X	X	X	X	X	X	X	
495	PP1–A15–J78	MG3–A36–J10	X	X	X	X	X	X	X	X	X	X	
499	PP1–A14–J84	MR2–A11–J5	X	X	X	X	X	X	X	X			
605	MR2–A11–J11	PP1–J10			X		X	X	X	X	X	X	
605	MS1–A3–A1–P403	PP1–J10			X		X	X	X	X	X	X	
611	PP1–SHIM	TAC–J22											
612	PP1–SHIM	MG2–A12–A1											SEE NOTE 9
700	PP1 A14 J86	MG2/MG3 A17 J2	X	X	X	X	X	X	X	X	X	X	
701	MR2–J11–J15	RF DOOR SWITCH											SEE NOTE 2
702	MR2–J11–J18	MR1–A7–J5	X	X	X	X	X	X	X	X	X	X	
703	MR2 A11 J23	PD1 A14 A1 J2	X		X	X	X	X	X	X			
704	PP1–J12	OC1–A10–J9							X	X			
706	MR2 A11 J14	PD1 A14 A1 J3 / MR1 PD1 J3	X		X	X	X	X	X	X			
707	MR2–A11–J26–ESTOP	OC1–J8							X	X			
708	MR2–A11–J16	MR1–A7–J27	X	X	X	X	X	X	X	X			
709	MR1–A7–J26	MR3–A9–J1	X	X	X	X	X	X	X	X			
710	MR2–A11–J17	MR3–A9–J2	X	X	X	X	X	X	X	X			
711	PP1–A17–J91	MG2–A33								X	X	X	
712	PP1–A17–J91	MG2–A11–A1								X	X	X	
711/712	PP1–A17–J91	MR2–A11–J13	X	X	X	X	X	X	X				
711/712	PP1–A17–J91	MG2–A33–J11,J12,J13 MG2–A11–J9,J10											

4–2 CABLE REMOVAL FOR UPGRADING TO SIGNA EXCITE HD (continued)

TABLE 4–1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
715	PP1–J14	MG2–A33–J1											
716	PP1–J89	MG2–A11–J2											
718	MR2–A11–J29	OC1–A10–J4							X	X			
719	PP1–J6	MR2–A11–J4								X			
726	PP1–A16–J77	MR1–A7–J18	X	X	X	X	X	X	X	X	X	X	
727	PP1–A16–J77	MG3–A36–J5	X	X	X	X	X	X	X	X	X	X	
728	MG3 A36 J11	MG2 A16 OUT J1	X	X	X	X	X	X	X	X	X	X	
729	MG3 A36 J12	MG2 A16 OUT J2	X	X	X	X	X	X	X	X	X	X	
730	MG3 A36 J13	MG2 A16 OUT J3	X	X	X	X	X	X	X	X	X	X	
731	MG3 A36 J14	MG2 A16 OUT J4	X	X	X	X	X	X	X	X	X	X	
732	MG3 A36 J15	MG2 A16 OUT J5	X	X	X	X	X	X	X	X	X	X	
733	MG3 A36 J16	MG2 A16 OUT J6	X	X	X	X	X	X	X	X	X	X	
734	MG3 A36 J17	MG2 A16 OUT J7	X	X	X	X	X	X	X	X	X	X	
735	MG3 A36 J18	MG2 A16 OUT J8	X	X	X	X	X	X	X	X	X	X	
736	MG3 A36 J19	MG2 A16 A8–J9	X	X	X	X	X	X	X	X	X	X	
743	MG2–A12–A2–J2	MG3–A3–J3								X	X	X	
744	MG2–A12–A2–J1	MG3–A3–J4								X	X	X	
749	MG2–A33–J10	MG2–A36	X	X	X	X	X	X	X	X	X	X	
753*	MR7–A4–J1	MR3–A11–J5					X	X	X				SEE NOTE 4
754*	MR7–A4–J2	MR3–A11–J4					X	X	X				SEE NOTE 4
755*	MR7–A4–J3	MR3–A11–J3					X	X	X				SEE NOTE 4
756*	MR7–A4–J4	MR3–A11–J8					X	X	X				SEE NOTE 4
757*	MR7–A4–J5	MR3–A11–J7					X	X	X				SEE NOTE 4
758*	MR7–A4–J6	MR3–A11–J6					X	X	X				SEE NOTE 4
759*	MR3–11–J12	MR7–A4–J8					X	X	X				SEE NOTE 4
760*	MR3–11–J11	MR7–A4–J9					X	X	X				SEE NOTE 4
761*	MR3–11–J10	MR7–A4–J10					X	X	X				SEE NOTE 4
762	MR3CABI/FJ3	PP1–A7–1/2				X	X	X	X				SEE NOTE 6
762	MG2–A12–A1–1/2	PP1–A7–1/2											SEE NOTE 5
763	MR3CABI/FJ4	PP1–A7–3/4				X	X	X	X				SEE NOTE 6
763	MG2–A12–A1–3/4	PP1–A7–3/4											SEE NOTE 5
764	MR3CABI/FJ5	PP1–A7–5/6				X	X	X	X				SEE NOTE 6
764	MG2–A12–A1–5/6	PP1–A7–5/6											SEE NOTE 5
767	MR1–A7–J37	PP1–J51								X			
768	MR1–A7–J31	PP1–J20	X	X	X	X	X	X	X				
769	MR1–A7–J34	PP1–J15	X	X	X	X	X	X	X				
770	MR1–A7–J41	PP1–J11	X	X	X	X	X	X	X				
771	MR1–A7–J36	PP1–J47	X	X	X	X	X	X	X				
772	MR1–A7–J40	PP1–J14	X	X	X	X	X	X	X				
773	MR1–A7–J35	PP1–A17–J1	X	X	X	X	X	X	X				
774	MR1–A7–J32	MR2–A11–J25	X	X	X	X	X	X	X				
775	MR1–A7–J42	PP1–A11–J92	X	X	X	X	X	X	X				

4–2 CABLE REMOVAL FOR UPGRADING TO SIGNA EXCITE HD (continued)

TABLE 4–1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
776	MR1–A7–J44	PP1–A11–J93	X	X	X	X	X	X	X				
777	MR1–A7–J43	PP1–A11–J94	X	X	X	X	X	X	X				
778	MG2–A12–A2–J3	PP1–A11–J72	X	X	X	X	X	X	X				
779	MG2–A12–A2–J4	PP1–A11–J73	X	X	X	X	X	X	X				
780	MG3–A3–J6	PP1–A11–J74	X	X	X	X	X	X	X				
781	MG2–A12–A2–J5	PP1–A11–J75	X	X	X	X	X	X	X				
782	MG2–A12–A2–J6	PP1–A11–J76	X	X	X	X	X	X	X				
783	MR1–A7–J37	PP1–F5,F6	X	X	X	X	X	X	X				
784	MG6–J1	PP1–F5,F6	X	X	X	X	X	X	X				
785	MR1–A7–J47	PP1–F7,F8	X		X	X	X	X	X				
786	MG6–J2	PP1–F7,F8	X		X	X	X	X	X				
787	MR2–A11–J30	CC1–A1–A1–SERIAL A									X	X	
788	PP1–J12	OW1–A1–A4–J7	X	X	X	X	X	X					
789	MR2–A11–J26	OW1–A1–A2–J18	X	X	X	X	X	X					
790	OW1–A2–A6–XO,RO,X1,R1	MR2–A11–J21						X					
791	MR2–A11–J30	OW1–A2–A4–EXP1	X	X	X	X	X	X					
792	OW1–A2–A1–MIC/AUDIO	OW1–A1–A4–J11,J13				X	X	X					
793	OW1–A2–A1–COM 1	OW1–A1–A4–J9						X					
794	OW1–A2–A3–MIC/AUDIO	OW1–A1–A4–J12,J14						X					
795	OW1–A2–A3–PWR	OW1–A1–A4–J17						X					
796	OW1–A2–A3–KBD/MS	OW1–A1–A4–J8						X					
797	OW1 A2 A1 PC	OW1 A3 LCD	X	X	X	X	X	X					
799	OW1–A15–MONITOR (OR 820)	OW1–A4 (B/W) (OR 820)				X	X	X					
800	OC1–J1	CC1–A2–J1 (MUX)							X	X			
801	OC1–J10	CC1–A2–J2 (SECUR. KEY)							X	X			
802	OC1–J12	CC1–A1–J9 (PLASMA)							X	X			
803	OC1–J13	CC1–A1–A10 (VIDEO)							X	X			
804	OC1–J3	CC1–A1–A1 (BODY)							X	X			
805	OW1–A1–A4–J6	OW1–A7 (NON–SCIM KYBRD)	X	X	X	X	X	X					
806	OW1–A2–A3–SER 1	OW1–A13 (MODEM)						X					
807	OW1–A16	OW1–A5 (DASM)	X	X	X	X	X	X					
809	OW1–A15–MIC	OW1–A1–A4–J12,J14	X		X	X	X						
810	OW1–A15–MD,KYBD	OW1–A1–A4–J8	X	X	X	X	X						
811	OW1–A2–A4–PORT2	OW1–A15–ETHERNET	X	X	X	X	X						
812	OW1–A15–COM1	OW1–A13 (MODEM)	X	X	X	X	X						
813	OW1–A2–A1–COM2	OW1–A15–COM2	X	X	X	X	X						
814	OW1–A1–A4–J9	OW1–A2–A1–KBD,MOUSE	X	X	X	X	X						
815	OW1–A16–SCSI	OW1–A15–SCSI (PCI)	X	X	X	X	X						
816	OW1–A2–A6–J1	OW1–A15–BIT3	X	X	X	X	X						
817	OW1–A1–A2–J17	OW1–A2–A5–J3	X	X	X	X	X						
818	MR2–A11–J31	OW1–A16–PORT6,7,8	X	X	X	X	X						

4–2 CABLE REMOVAL FOR UPGRADING TO SIGNA EXCITE HD (continued)

TABLE 4–1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	COMMENTS
819	OW1–A2–A7–SCSI	OW1–A15–SCSI	X	X	X	X	X						
820	OW1–A15–MONITOR	OW1–A4 (B/W)				X	X	X					
836	MR2–A11–J21	OW1–A15–BIT3	X	X	X	X	X						
848	MG2–A4–J5	MG2–A11–J4											
861	MG2–A4–J4	MG2–A33–J5,6,7,8,9	X	X	X	X	X	X	X				
862	MG2–A4–J1	MG2–A5–J1											
863	MG2–A4–J2	MG2–A5–J2											
897	MG2–A4–J3	MG2–A33–J4											
939	MDS–UPS	MSM4											
941	MSM1–J11	SCC–J1											
942	MSM1–J5	OW1–A2–A4–PORT 7											
966	MR1–A7–J1	MR3–PD1–7,8,9,GND	X	X	X								
967	MR1–A7–CB1	MR3–PD1–10,11,12,GND	X	X	X								
968	MR2–A11–J1	MR3–PD1–4,5,6,GND	X	X	X								
969	OW1–A2–A5–GND	MR3–PD1–18,19,GND,N	X	X	X	X	X						
970	WC1	MR3–PD1–16,17,GND	X		X	X							SEE NOTE 13
971	MR2–A11–J23	MR3–PD1–J2	X	X	X	X	X						
972	MR2–A11–J14	MR3–PD1–J3	X	X	X	X	X						
974	MR3–PD1–J5	TAC–J12											
976	MR3–CAB I/F–J3	TAC +/– X											SEE NOTE 10
977	MR3–CAB I/F–J4	TAC +/– Y											SEE NOTE 10
978	MR3–CAB I/F–J5	TAC +/– Y											SEE NOTE 10
979	TAC–X+,WB–,ZM–	PP1–XWB–.ZM–											SEE NOTE 10
980	TAC–Y+,WB–,ZM–	PP1–YWB–.ZM–											SEE NOTE 10
981	TAC–Z+,WB–,ZM–	PP1–ZWB–.ZM–											SEE NOTE 10
982	PP1–ZZM+.ZM–	MG2–A12–A1–7/8											SEE NOTE 10
983	PP1–YZM+.ZM–	MG2–A12–A1–9/10											SEE NOTE 10
984	PP1–XZM+.ZM–	MG2–A12–A1–11/12											SEE NOTE 10
985	PP1–ZWB+,WB–	MG2–A12–A1–1/2											SEE NOTE 10
986	PP1–YWB+,WB–	MG2–A12–A1–3/4											SEE NOTE 10
987	PP1–XWB+,WB–	MG2–A12–A1–5/6											SEE NOTE 10
988	PP1–A7–GND	TAC–GND											SEE NOTE 10
989	MR3–PD1–16,17,N,GND	TAC–J8											
990	PP1–J52	VACUUM SENSOR											SEE NOTE 11
991	PP1–J52	TAC–J9											SEE NOTE 11
992	TSCC–J3	TAC–J10											
1004	OW1–A16–PORT 2	TAC–J27											SEE NOTE 12
1152	PP1–SHIM	MG2–A12–A1											SEE NOTE 9
	MG3–A2–J7	MG2–A11–A1–MIC	X	X	X	X	X	X	X				SEE NOTE 7

4–2 CABLE REMOVAL FOR UPGRADING TO SIGNA EXCITE HD (continued)

Table 4–1 Notes:

1. If part of existing system, the shim and magnet power supply cables, Runs 011 and 020, need to be reused for the new HFD/PDU Cabinet. Refer to Section 15–2–2 for instructions on removing and reterminating the cables.
2. A new RF Door Switch cable, Run 701, is provided for upgrade. If existing cable is routed within wall, it may be reused. If existing cable in labeled as Run 207, replace with new Run 701 or relabel per instructions in Section 4.
3. If optional Oxygen Monitor is installed on site, retain Runs 457 and 458 for connection to new Penetration Panel.
4. These cable Runs are removed only if GRAM Cabinet is part of existing site being deinstalled.
5. Magnet Room side of Gradient cables, Runs 762, 763, and 764 may be reused. Butt Splices and Shrink Wrap Tubing are not provided with upgrade for reterminating Gradient cables at rear of magnet. Disconnect only at Penetration Panel and reconnect to new EXCITE HD Penetration Panel.
6. Equipment Room side of Gradient cables, Runs 762, 763, and 764:

   For sites with ACGD/PDU cabinet:

   - Cables that are connected to ACGD/PDU cabinet will be reused. Disconnect only at Penetration Panel and reconnect to new EXCITE HD Penetration Panel.

   For sites with SGD cabinet:

   - Early production SGD cabinets were installed using the larger #1/0 gradient cables with braided mesh clamped in the interface panel for grounding. These cables must be removed and replaced with the new #2 gradient cables provided with upgrades.
   - If SGD cabinet was installed using the #2 cables, then these may be reused providing that the length is usable for connection to the new HFD cabinet. If needed, install the new #2 gradient cables provided with this upgrade.

   For sites with Gradient/GRAM cabinets:

   - De–install existing #1/0 cables and replace with new #2 cables to be routed and connected to HFD cabinet.
7. Existing microphone cable will be replaced with upgrade.
8. Existing Emergency Off Switch cables, Runs 296 and 297, may be retained and reused.
9. Early system configurations installed Run 612, later configurations installed Run 1152 for Shim option.
10. Existing cable Runs 982 to 987 in Magnet Room will not be replaced. They will be reused and connected to new Penetration Cabinet. Runs 976 to 981 in Equipment Room can also be reused with upgraded HFD–S/PDU Cabinet and new EXCITE HD Penetration Panel. The TAC Cabinet will not be replaced, so the cables will not be disconnected from the cabinet.
11. Runs 991 and 992 may not be present on all systems. Later versions of TwinSpeed moved the vacuum sensor into the TAC cabinet.
12. Run 1004 will reconnect to cable 2382771 inside the GOC cabinet.
13. For Mobile systems, the Heat Exchanger power cord will be retained and connected to TS–20, NEU, and GND on the PDU.
14. For Cx or LCC magnet with Horizon enclosure. Run 421 will be replaced by a longer Run 421, 46–287908G3. This cable has to be ordered separately.

4–3 RE–LABELING OF EXISTING CABLES

The Signa Advantage 4.x RF Door Switch cable, Run 207, may be reused but must be re–labeled. Refer to Table 4–2.

TABLE 4–2
CABLES TO BE RE–LABELED

OLD RUN	SYSTEM TYPE	ONE END OLD LABEL	ONE END NEW LABEL	OTHER END OLD LABEL	OTHER END NEW LABEL
207	4.x	RUN #207 MR2–A11–J6 46–243775G411	RUN #701 MR2–A11–J15 46–243775G807	RUN #207 DOOR SWITCH 46–243775G411	RUN #701 RF DOOR SWITCH 46–243775G807

4–4 CABLE REMOVAL INTERCONNECT MAPS

The following “Cable Removal” interconnect maps refer only to cables removed, relabeled, and/or relocated when upgrading:

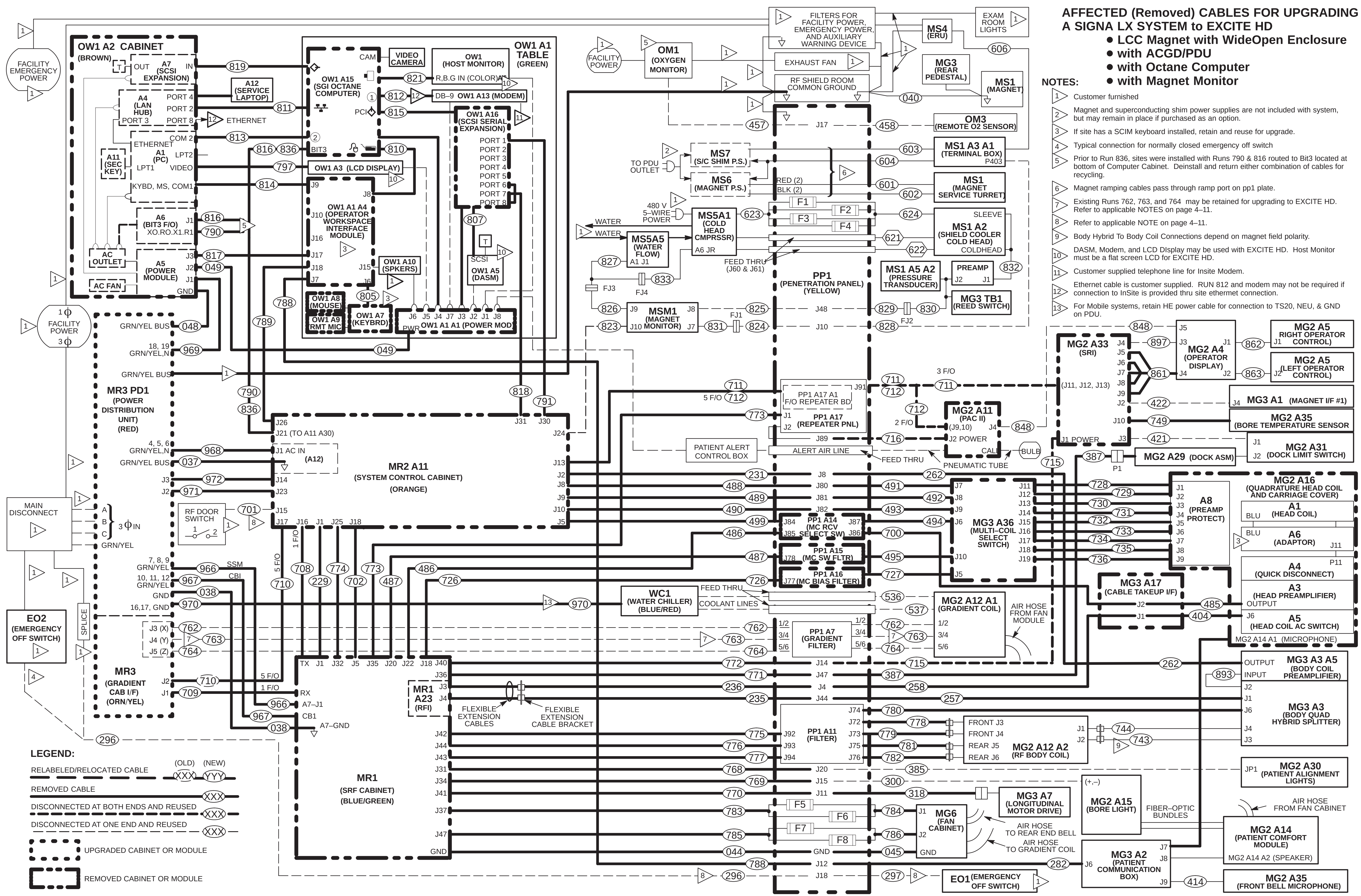
- Signa LCC WideOpen with Octane, ACGD, and Magnet Monitor .... Page 4–13
- Signa LCC TwinSpeed with Octane, ACGD, and Magnet Monitor .... Page 4–14
- Signa pre–LCC LX with Octane, ACGD and without Magnet Monitor .... Page 4–15
- Signa LCC WideOpen with Octane, RF/PDU, SGD, and Magnet Monitor .... Page 4–16
- Signa LCC Pre–WideOpen with Octane, RF/PDU, SGD, and Magnet Monitor .... Page 4–17
- Signa pre–LCC LX with Octane, SGD, RF/PEN, PDU and w/o Magnet Monitor .... Page 4–18
- Signa pre–LCC LX with Indigo, SGD, RF/PEN, PDU and w/o Magnet Monitor .... Page 4–19
- Signa pre–LCC LX with Octane, Grad/GRAM, RF/PEN, PDU and w/o Magnet Monitor .... Page 4–20
- Signa pre–LCC LX with Indigo, Grad/GRAM, RF/PEN, PDU and w/o Magnet Monitor .... Page 4–21
- Signa Horizon 5.x with Operator Console/Computer, Grad/GRAM, RF/PEN, PDU, w/o Mag Mon . Page 4–22
- Signa Advantage 5.x with Oper. Console/Computer, Grad/GRAM, RF, PEN, PDU, w/o Mag Mon . Page 4–23
- Signa Advantage 4.x with Operator Console. computer, RF, REN, PDU, and Gradient –
  Cable removal map is not provided. Refer to Table 4–1, Columns H and I, for cable removal information on Advantage 4.x systems

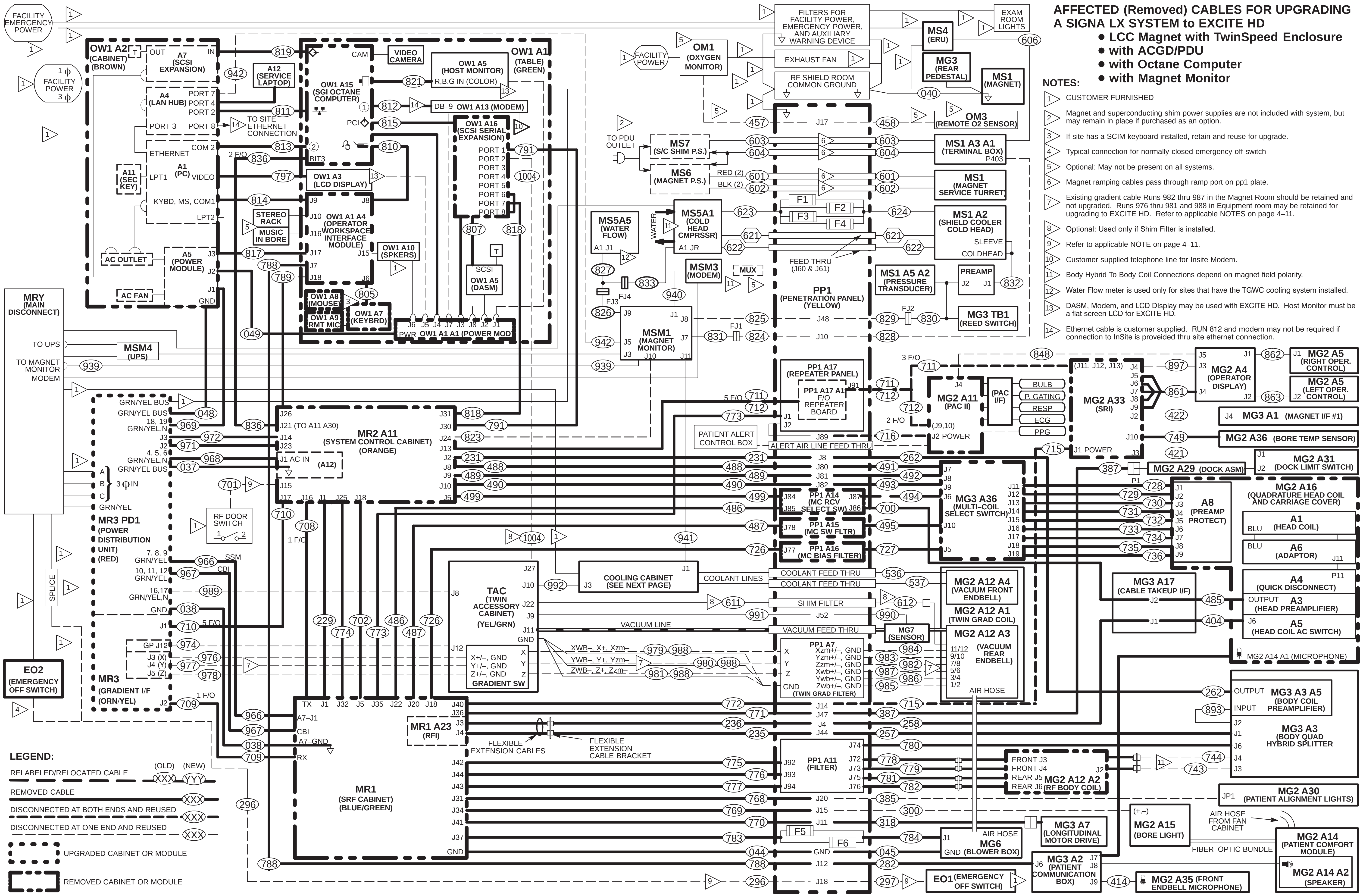
AFFECTED (Removed) CABLES FOR UPGRADING A SIGNA LX SYSTEM to EXCITE HD

- LCC Magnet with WideOpen Enclosure
- with ACGD/PDU
- with Octane Computer
- with Magnet Monitor

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. If site has a SCIM keyboard installed, retain and reuse for upgrade.
4. Typical connection for normally closed emergency off switch
5. Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. Deinstall and return either combination of cables for recycling.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing Runs 762, 763, and 764 may be retained for upgrading to EXCITE HD. Refer to applicable NOTES on page 4–11.
8. Refer to applicable NOTE on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. DASM, Modem, and LCD DIsplay may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.
13. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.





AFFECTED (Removed) CABLES FOR UPGRADING A SIGNA LX SYSTEM to EXCITE HD

- LCC Magnet with TwinSpeed Enclosure
- with ACGD/PDU
- with Octane Computer
- with Magnet Monitor

NOTES:

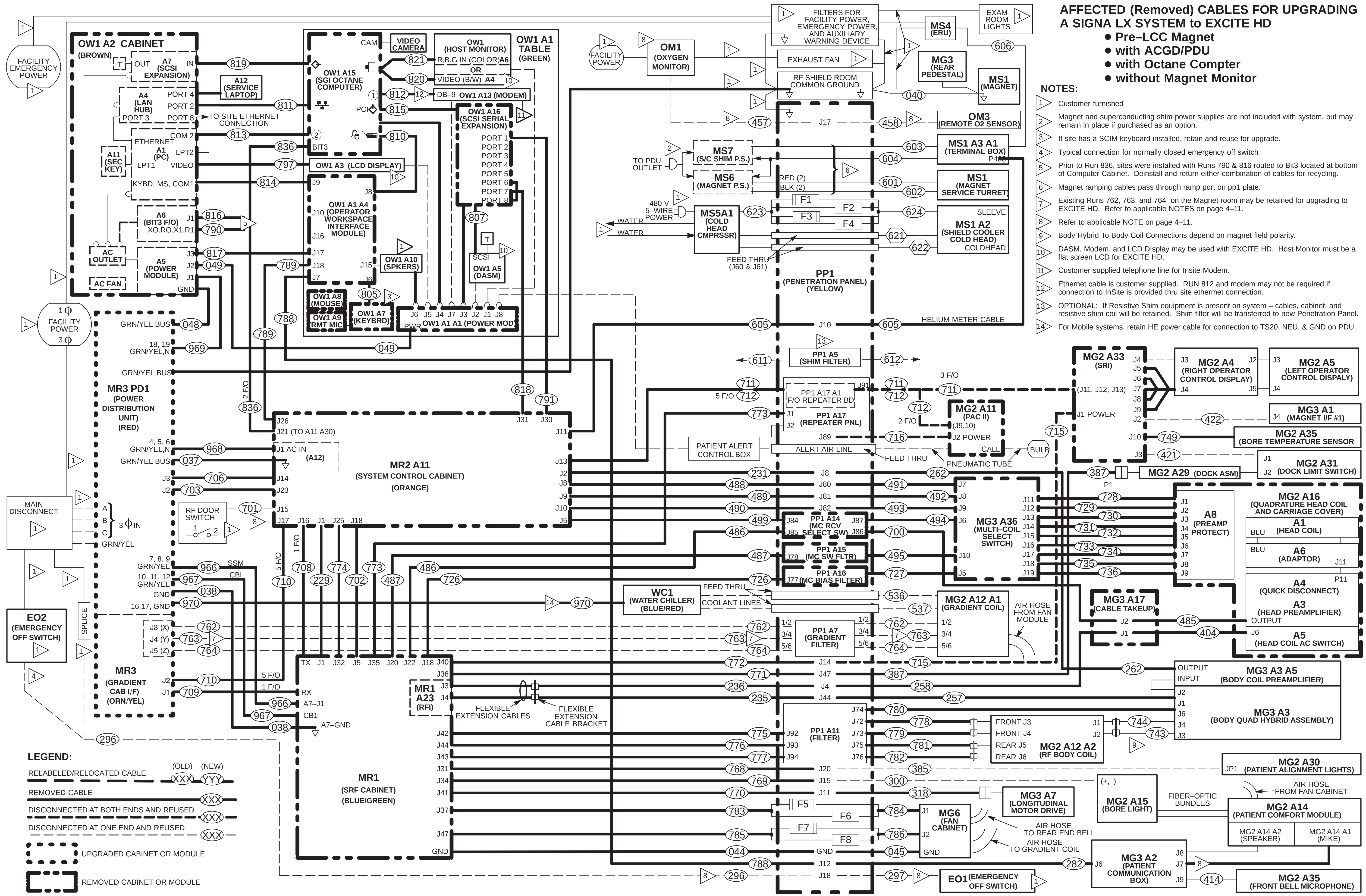
1. CUSTOMER FURNISHED
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. If site has a SCIM keyboard installed, retain and reuse for upgrade.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing gradient cable Runs 982 thru 987 in the Magnet Room should be retained and not upgraded. Runs 976 thru 981 and 988 in Equipment room may be retained for upgrading to EXCITE HD. Refer to applicable NOTES on page 4–11.
8. Optional: Used only if Shim Filter is installed.
9. Refer to applicable NOTE on page 4–11.
10. Customer supplied telephone line for Insite Modem.
11. Body Hybrid To Body Coil Connections depend on magnet field polarity.
12. Water Flow meter is used only for sites that have the TGWC cooling system installed.
13. DASM, Modem, and LCD Display may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
14. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is proveided thru site ethernet connection.

AFFECTED (Removed) CABLES FOR UPGRADING A SIGNA LX SYSTEM to EXCITE HD

- Pre–LCC Magnet
- with ACGD/PDU
- with Octane Compter
- without Magnet Monitor

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. If site has a SCIM keyboard installed, retain and reuse for upgrade.
4. Typical connection for normally closed emergency off switch
5. Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. Deinstall and return either combination of cables for recycling.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Refer to applicable NOTES on page 4–11.
8. Refer to applicable NOTE on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. DASM, Modem, and LCD DIsplay may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.
13. OPTIONAL: If Resistive Shim equipment is present on system – cables, cabinet, and resistive shim coil will be retained. Shim filter will be transferred to new Penetration Panel.
14. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.

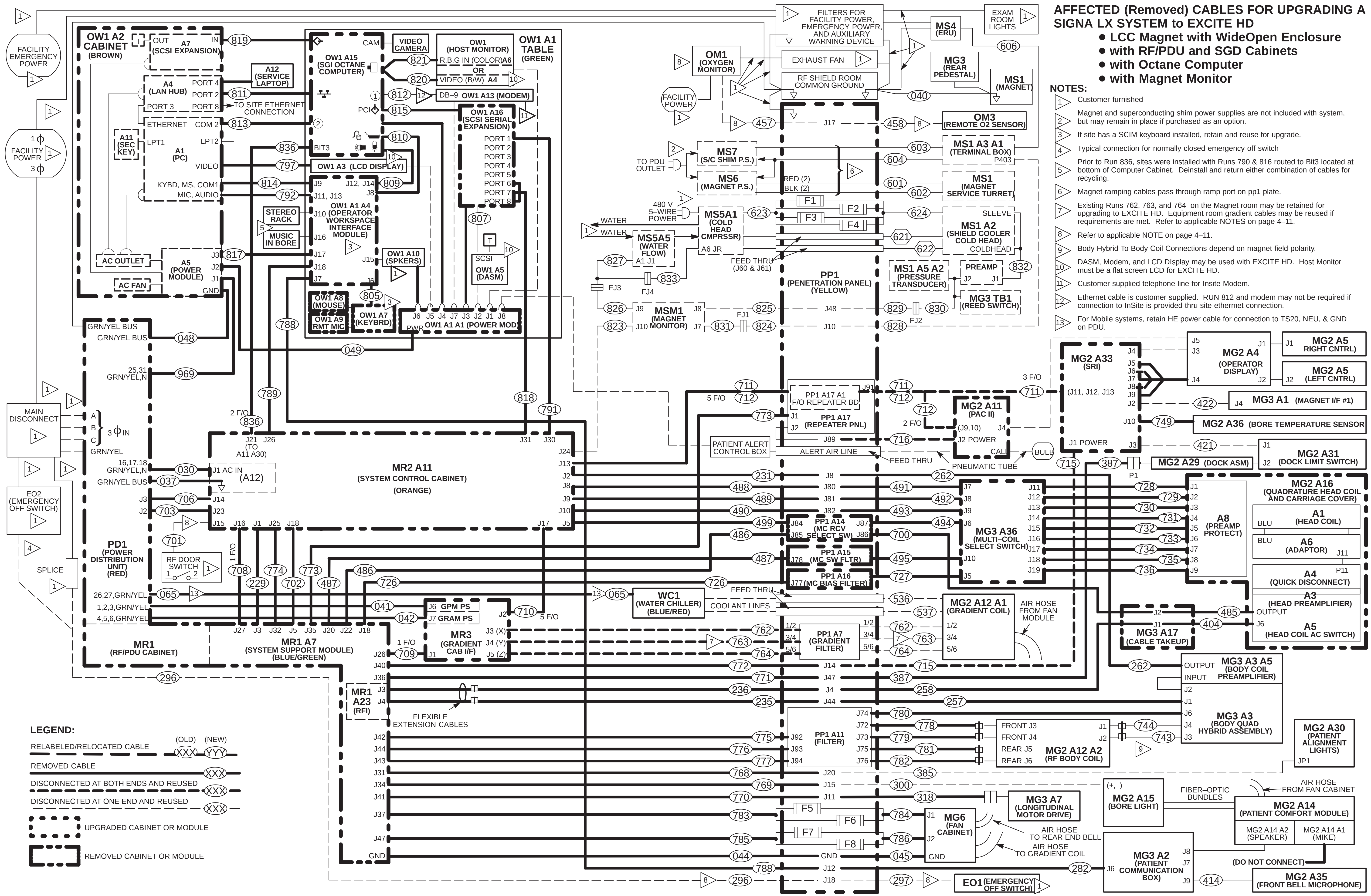


LEGEND:

RELABELED/RELOCATED CABLE	(OLD) XXX (NEW) YYY
REMOVED CABLE	XXX
DISCONNECTED AT BOTH ENDS AND REUSED	XXX
DISCONNECTED AT ONE END AND REUSED	XXX
UPGRADED CABINET OR MODULE	
REMOVED CABINET OR MODULE	

AFFECTED (Removed) CABLES FOR UPGRADING A SIGNA LX SYSTEM to EXCITE HD

- LCC Magnet with WideOpen Enclosure
- with RF/PDU and SGD Cabinets
- with Octane Computer
- with Magnet Monitor



NOTES:

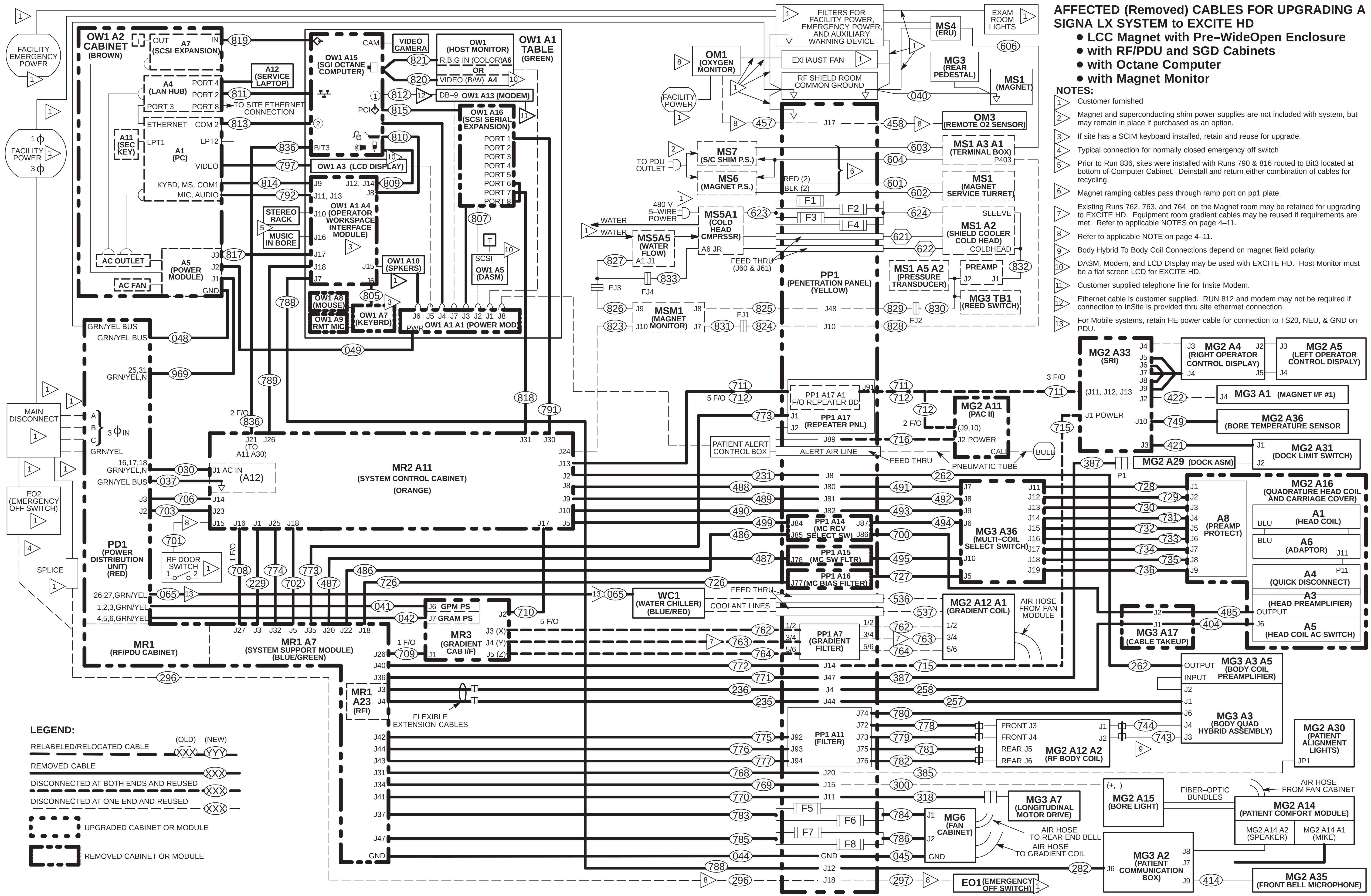
1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. If site has a SCIM keyboard installed, retain and reuse for upgrade.
4. Typical connection for normally closed emergency off switch
5. Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. Deinstall and return either combination of cables for recycling.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Equipment room gradient cables may be reused if requirements are met. Refer to applicable NOTES on page 4–11.
8. Refer to applicable NOTE on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. DASM, Modem, and LCD DIsplay may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.
13. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.

AFFECTED (Removed) CABLES FOR UPGRADING A SIGNA LX SYSTEM to EXCITE HD

- LCC Magnet with Pre–WideOpen Enclosure
- with RF/PDU and SGD Cabinets
- with Octane Computer
- with Magnet Monitor

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. If site has a SCIM keyboard installed, retain and reuse for upgrade.
4. Typical connection for normally closed emergency off switch
5. Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. Deinstall and return either combination of cables for recycling.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Equipment room gradient cables may be reused if requirements are met. Refer to applicable NOTES on page 4–11.
8. Refer to applicable NOTE on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. DASM, Modem, and LCD DIsplay may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.
13. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.

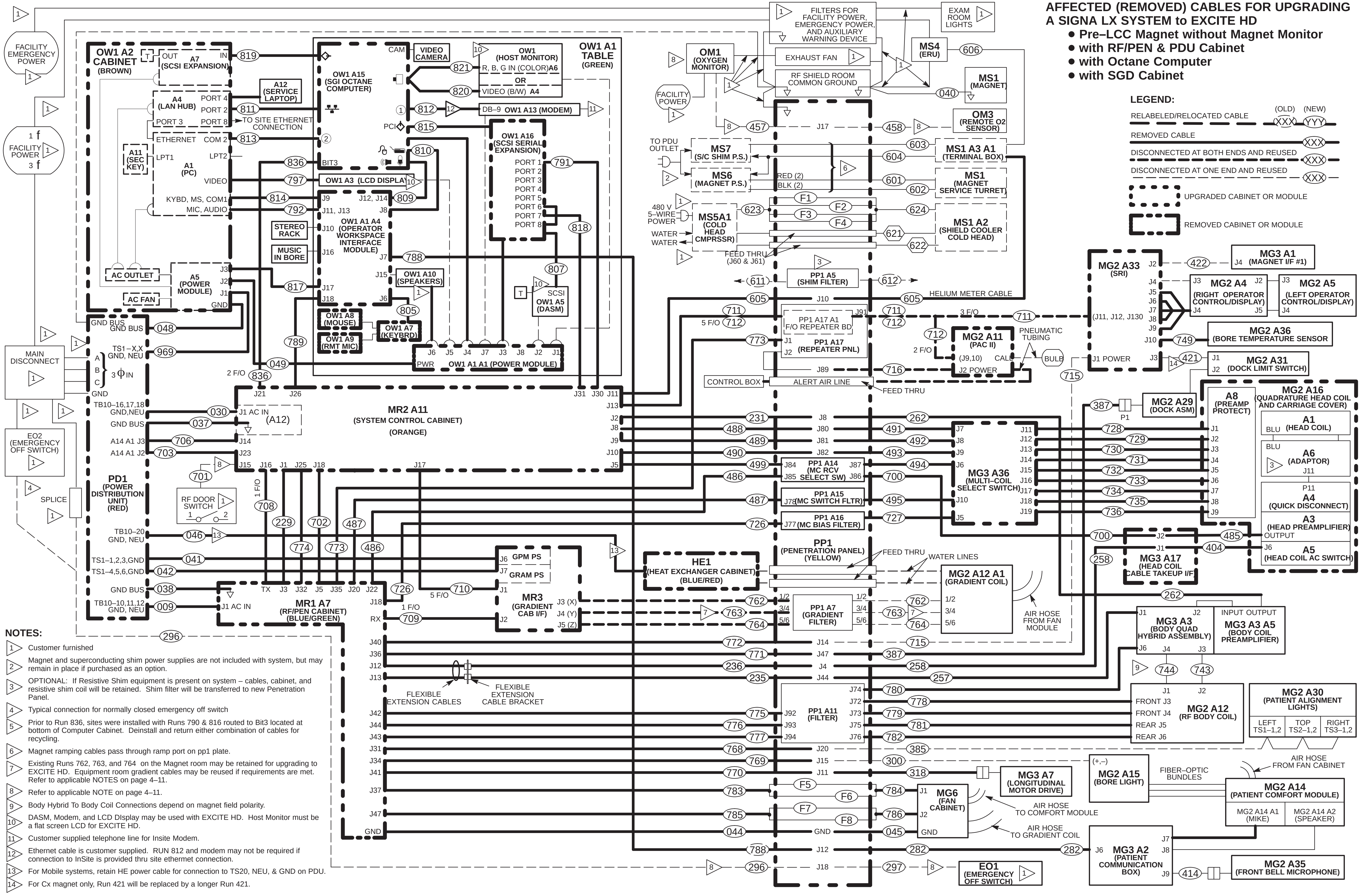


AFFECTED (REMOVED) CABLES FOR UPGRADING A SIGNA LX SYSTEM to EXCITE HD

- Pre–LCC Magnet without Magnet Monitor
- with RF/PEN & PDU Cabinet
- with Octane Computer
- with SGD Cabinet

LEGEND:

- RELABELED/RELOCATED CABLE (OLD) XXX (NEW) YYY
- REMOVED CABLE XXX
- DISCONNECTED AT BOTH ENDS AND REUSED XXX
- DISCONNECTED AT ONE END AND REUSED XXX
- UPGRADED CABINET OR MODULE
- REMOVED CABINET OR MODULE

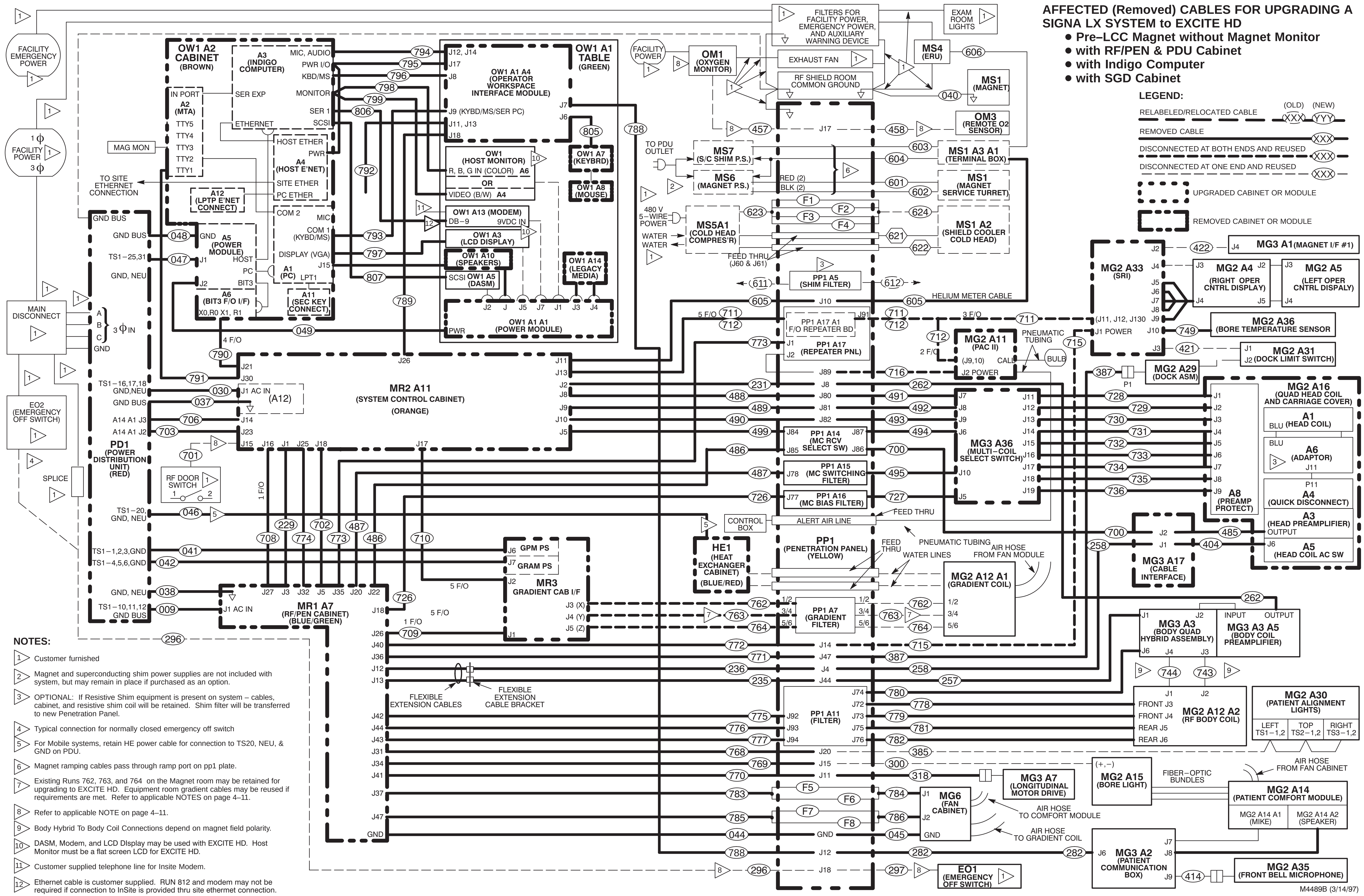


NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. OPTIONAL: If Resistive Shim equipment is present on system – cables, cabinet, and resistive shim coil will be retained. Shim filter will be transferred to new Penetration Panel.
4. Typical connection for normally closed emergency off switch
5. Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. Deinstall and return either combination of cables for recycling.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Equipment room gradient cables may be reused if requirements are met. Refer to applicable NOTES on page 4–11.
8. Refer to applicable NOTE on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. DASM, Modem, and LCD DIsplay may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.
13. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.
14. For Cx magnet only, Run 421 will be replaced by a longer Run 421.

AFFECTED (Removed) CABLES FOR UPGRADING A SIGNA LX SYSTEM to EXCITE HD

- Pre–LCC Magnet without Magnet Monitor
- with RF/PEN & PDU Cabinet
- with Indigo Computer
- with SGD Cabinet

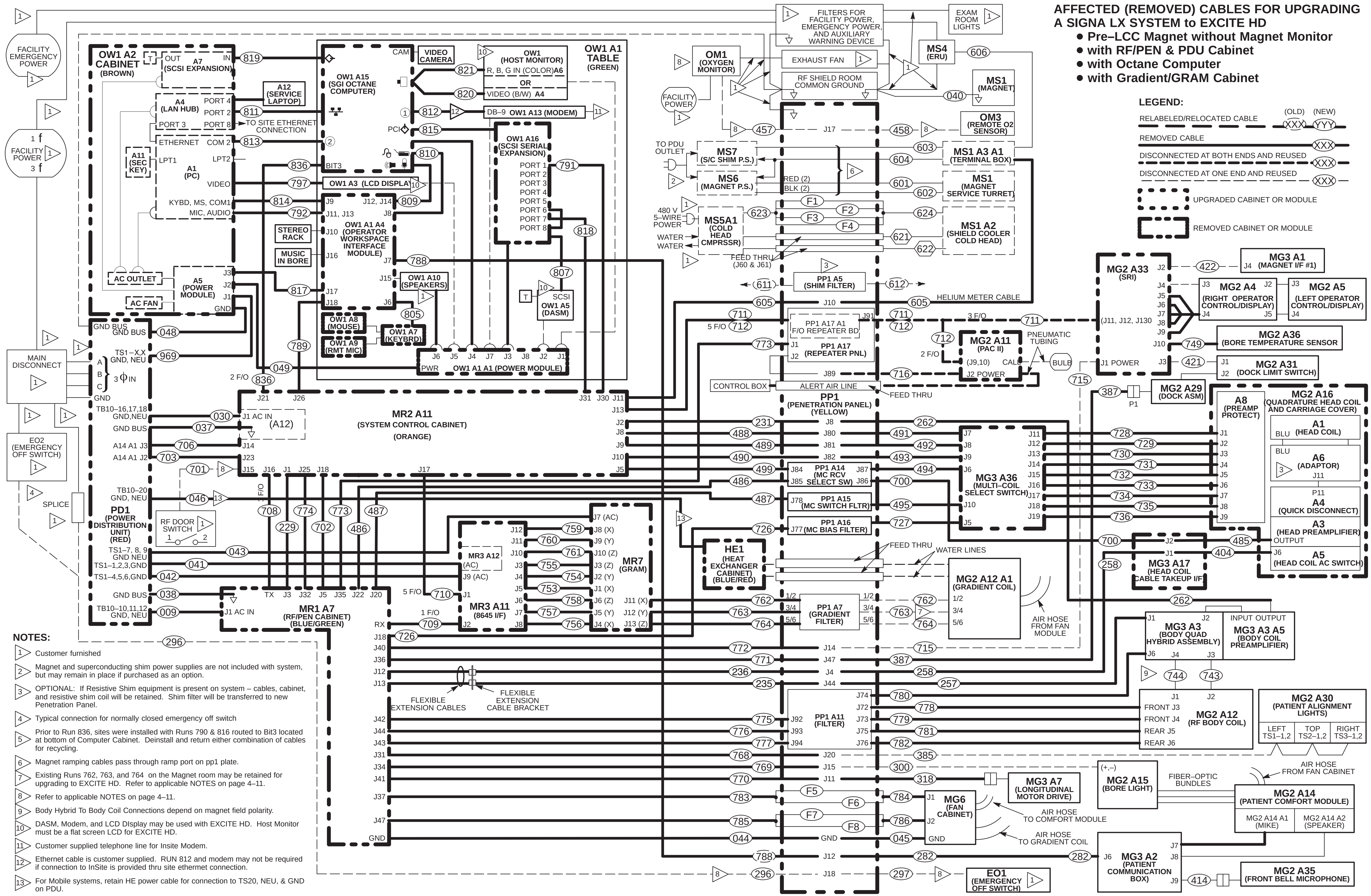


NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. OPTIONAL: If Resistive Shim equipment is present on system – cables, cabinet, and resistive shim coil will be retained. Shim filter will be transferred to new Penetration Panel.
4. Typical connection for normally closed emergency off switch
5. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Equipment room gradient cables may be reused if requirements are met. Refer to applicable NOTES on page 4–11.
8. Refer to applicable NOTE on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. DASM, Modem, and LCD DIsplay may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethermet connection.

AFFECTED (REMOVED) CABLES FOR UPGRADING A SIGNA LX SYSTEM to EXCITE HD

- Pre–LCC Magnet without Magnet Monitor
- with RF/PEN & PDU Cabinet
- with Octane Computer
- with Gradient/GRAM Cabinet



LEGEND:

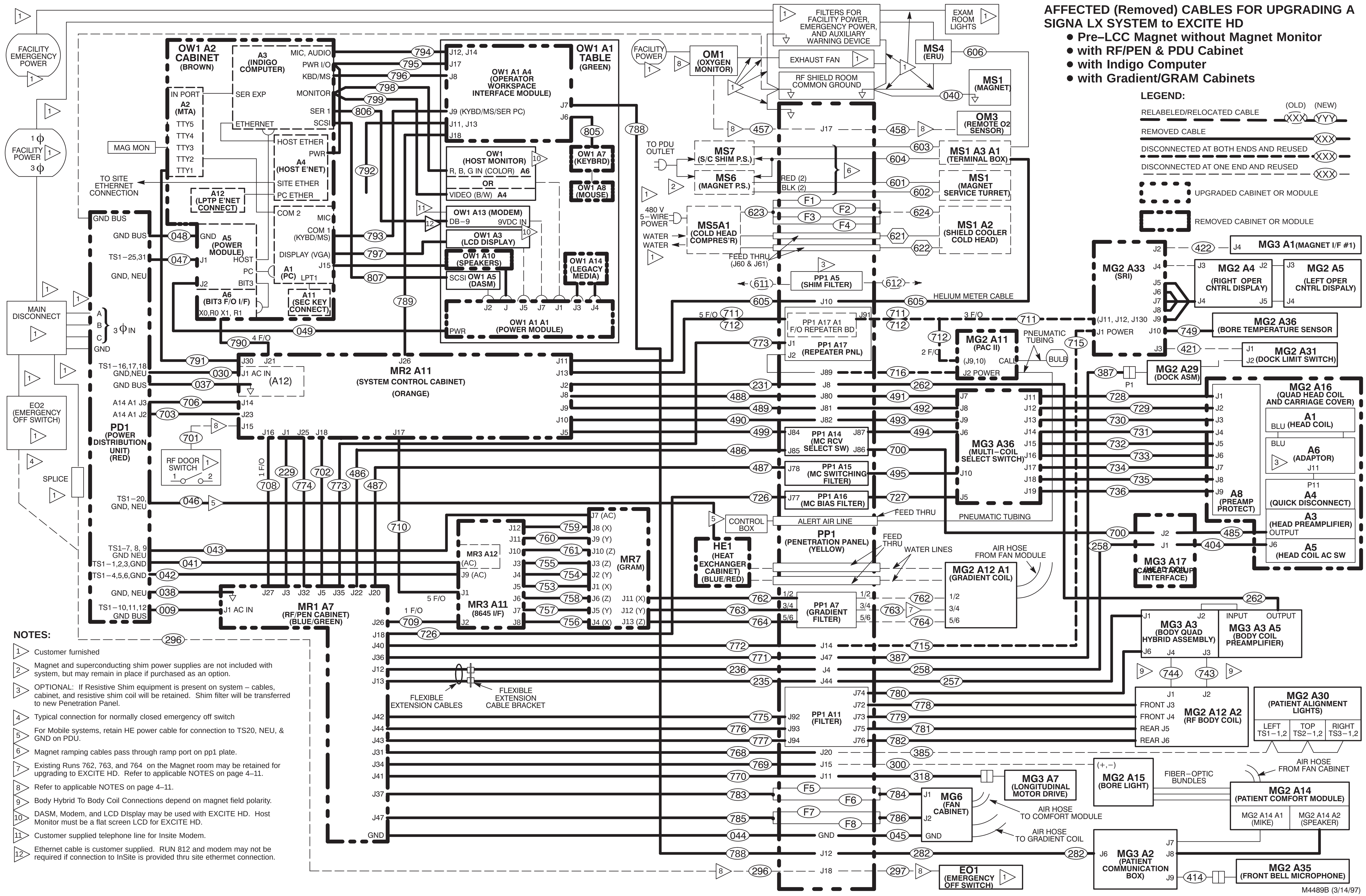
RELABELED/RELOCATED CABLE	(OLD) XXX (NEW) YYY
REMOVED CABLE	XXX
DISCONNECTED AT BOTH ENDS AND REUSED	XXX
DISCONNECTED AT ONE END AND REUSED	XXX
UPGRADED CABINET OR MODULE	
REMOVED CABINET OR MODULE	

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. OPTIONAL: If Resistive Shim equipment is present on system – cables, cabinet, and resistive shim coil will be retained. Shim filter will be transferred to new Penetration Panel.
4. Typical connection for normally closed emergency off switch
5. Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. Deinstall and return either combination of cables for recycling.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Refer to applicable NOTES on page 4–11.
8. Refer to applicable NOTES on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. DASM, Modem, and LCD DIsplay may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.
13. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.

AFFECTED (Removed) CABLES FOR UPGRADING A SIGNA LX SYSTEM to EXCITE HD

- Pre–LCC Magnet without Magnet Monitor
- with RF/PEN & PDU Cabinet
- with Indigo Computer
- with Gradient/GRAM Cabinets



NOTES:

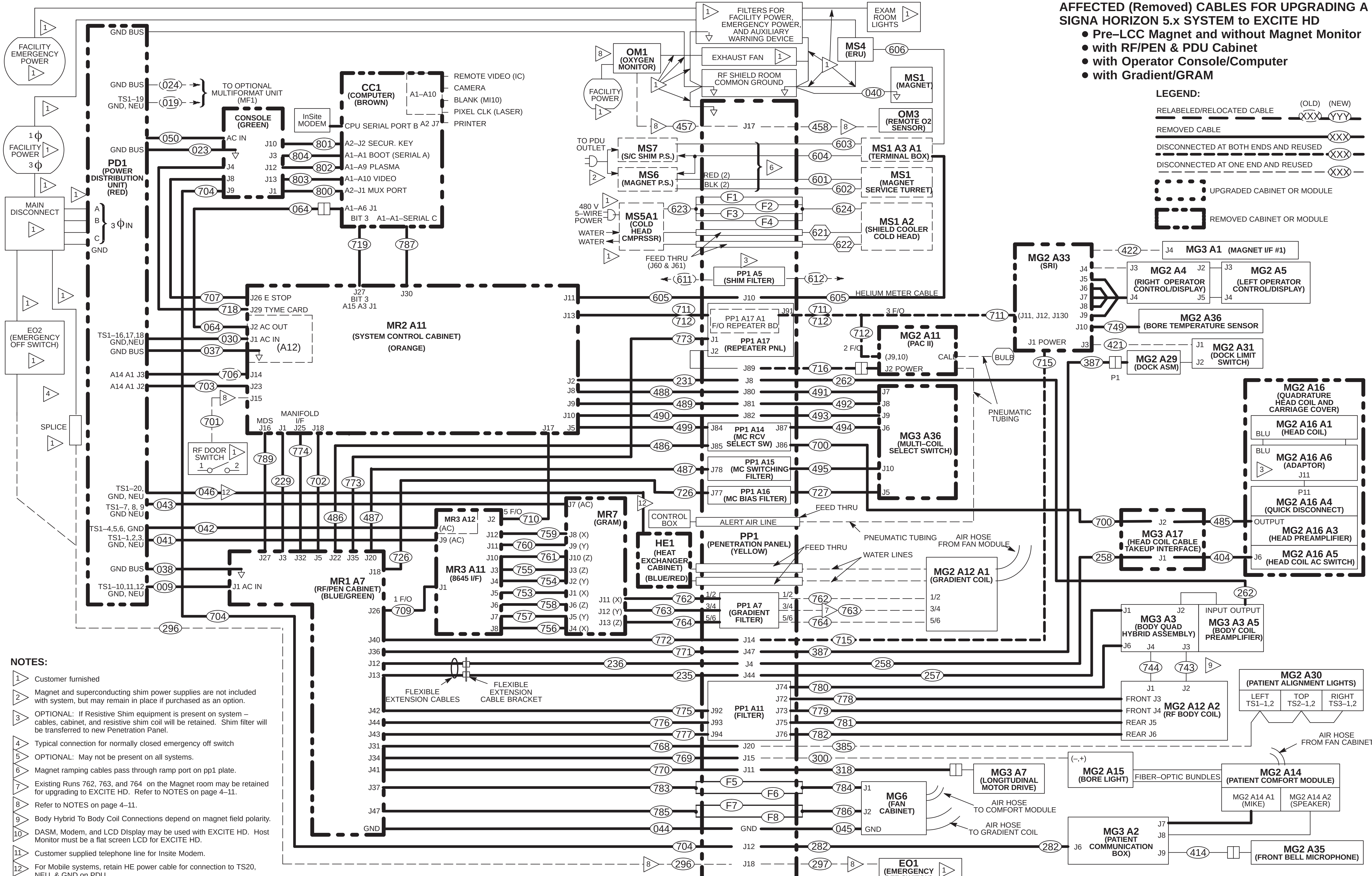
1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. OPTIONAL: If Resistive Shim equipment is present on system – cables, cabinet, and resistive shim coil will be retained. Shim filter will be transferred to new Penetration Panel.
4. Typical connection for normally closed emergency off switch
5. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Refer to applicable NOTES on page 4–11.
8. Refer to applicable NOTES on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. DASM, Modem, and LCD DIsplay may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

AFFECTED (Removed) CABLES FOR UPGRADING A SIGNA HORIZON 5.x SYSTEM to EXCITE HD

- Pre–LCC Magnet and without Magnet Monitor
- with RF/PEN & PDU Cabinet
- with Operator Console/Computer
- with Gradient/GRAM

LEGEND:

- RELABELED/RELOCATED CABLE (OLD) XXX (NEW) YYY
- REMOVED CABLE XXX
- DISCONNECTED AT BOTH ENDS AND REUSED XXX
- DISCONNECTED AT ONE END AND REUSED XXX
- UPGRADED CABINET OR MODULE
- REMOVED CABINET OR MODULE



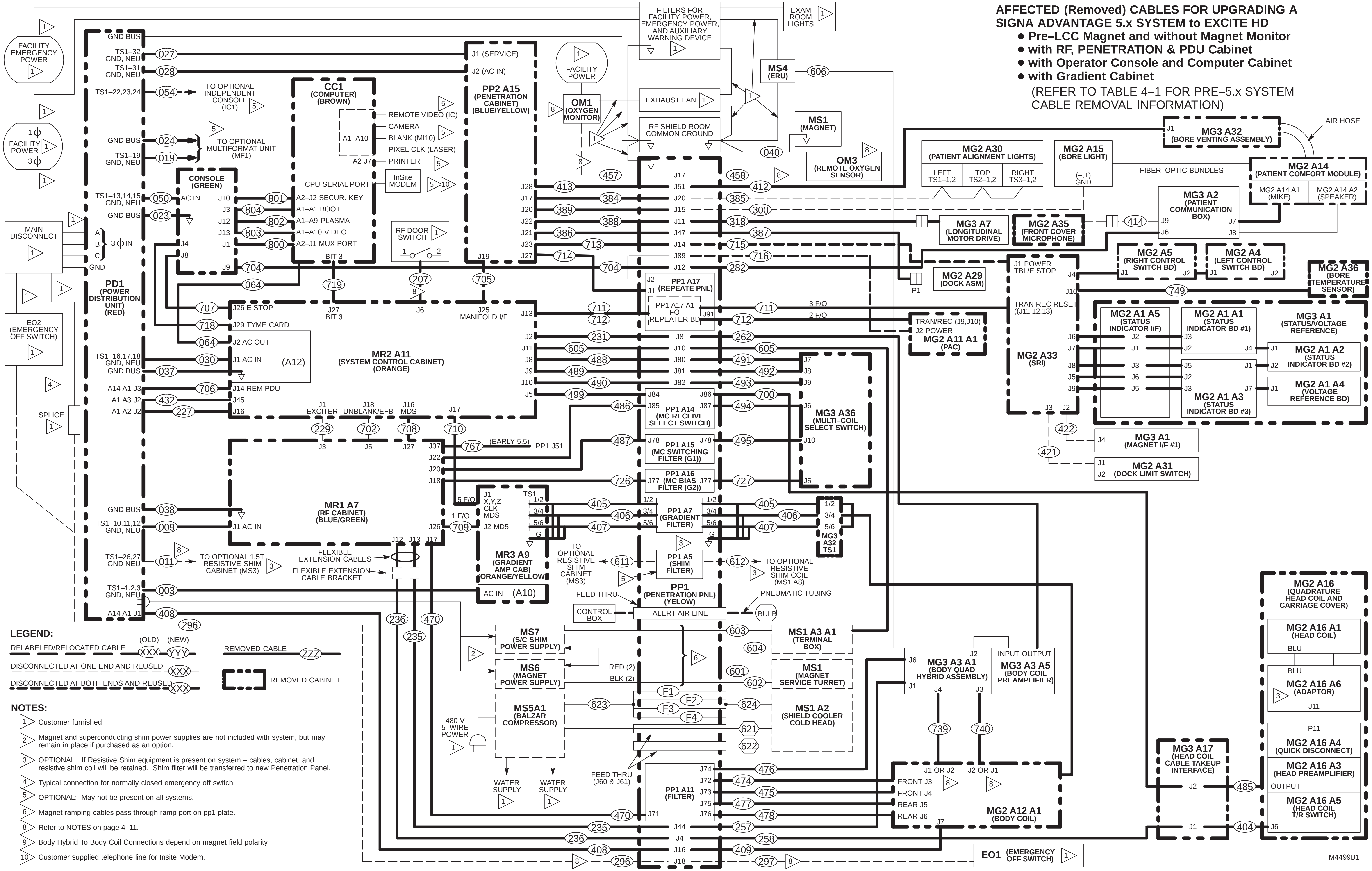
NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. OPTIONAL: If Resistive Shim equipment is present on system – cables, cabinet, and resistive shim coil will be retained. Shim filter will be transferred to new Penetration Panel.
4. Typical connection for normally closed emergency off switch
5. OPTIONAL: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. Existing Runs 762, 763, and 764 on the Magnet room may be retained for upgrading to EXCITE HD. Refer to NOTES on page 4–11.
8. Refer to NOTES on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. DASM, Modem, and LCD DIsplay may be used with EXCITE HD. Host Monitor must be a flat screen LCD for EXCITE HD.
11. Customer supplied telephone line for Insite Modem.
12. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.

AFFECTED (Removed) CABLES FOR UPGRADING A SIGNA ADVANTAGE 5.x SYSTEM to EXCITE HD

- **Pre–LCC Magnet and without Magnet Monitor**
- **with RF, PENETRATION & PDU Cabinet**
- **with Operator Console and Computer Cabinet**
- **with Gradient Cabinet**

(REFER TO TABLE 4–1 FOR PRE–5.x SYSTEM CABLE REMOVAL INFORMATION)



LEGEND:

RELABELED/RELOCATED CABLE (OLD) XXX (NEW) YYY	REMOVED CABLE ZZZ
DISCONNECTED AT ONE END AND REUSED XXX	REMOVED CABINET
DISCONNECTED AT BOTH ENDS AND REUSED XXX	

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. OPTIONAL: If Resistive Shim equipment is present on system – cables, cabinet, and resistive shim coil will be retained. Shim filter will be transferred to new Penetration Panel.
4. Typical connection for normally closed emergency off switch
5. OPTIONAL: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
8. Refer to NOTES on page 4–11.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.

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4–5 SORTING AND ROUTING CABLES

CAUTION

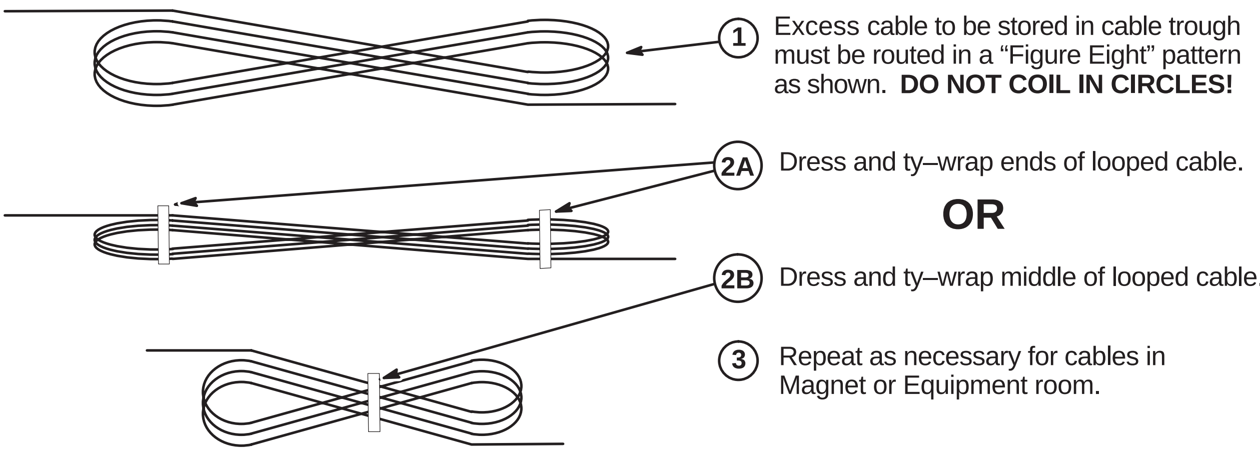
Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 2–1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber–optic cables.



PROPER STORAGE OF EXCESS CABLES
ILLUSTRATION 2 – 1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

4–6 EXCITE HD CABLE INTERCONNECT MAPS

The following cable maps illustrate the Signa EXCITE HD 1.5T system interconnect for all supplied cables and interface with customer supplied wiring.

- Interconnects for additional cables supplied with other options are illustrated in the installation manual shipped with the option.
  - Instructions for connections for optional laser cameras are supplied with option.
  - Connection procedures for other options are covered in the installation manual shipped with the option.
  - Spectroscopy option cabling can be found on page 6–7.
- Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation section within this manual.
- Interconnect details for magnet subsystem are covered in the applicable magnet subsystem manual.

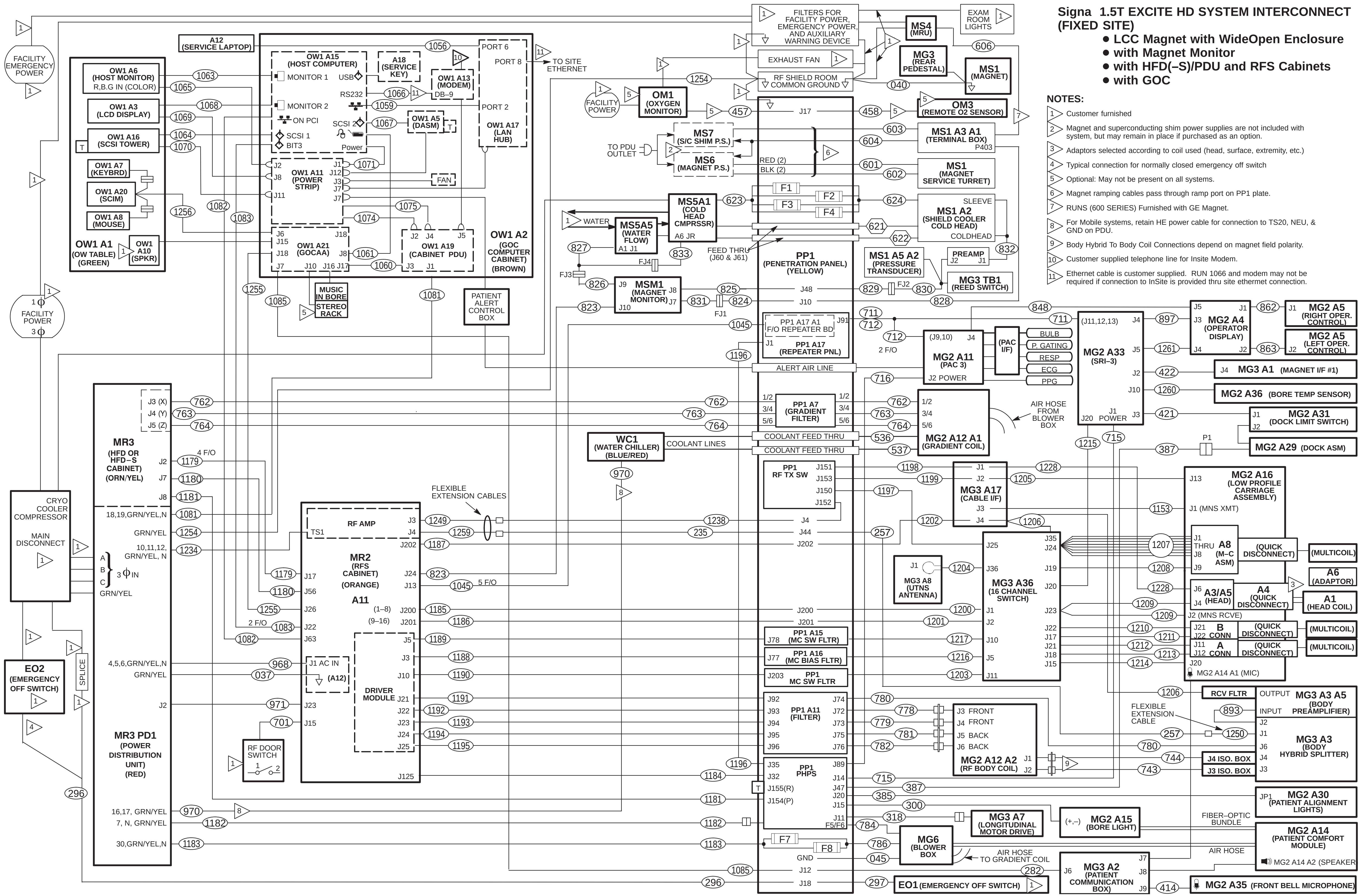
For the convenience of the mechanical installers, a duplicate System Interconnect is provided. This will allow the install team to post one interconnect map in the Magnet Room and the other map in the Equipment/Operator Room. This will improve the efficiency of the installation when marking off cables when their connections are complete.

Fold–out EXCITE HD 1.5T System Interconnect Cable Maps are provided on the following pages:

Tape maps in a convenient location in the Magnet Room and the Equipment/Operator Room to check off routed cables. Additional information on each cable is located in the Appedndix. Upon completion of connecting cables, return fold–out maps to the binder for future reference.

Signa 1.5T EXCITE HD SYSTEM INTERCONNECT (FIXED SITE)

- LCC Magnet with WideOpen Enclosure
- with Magnet Monitor
- with HFD(–S)/PDU and RFS Cabinets
- with GOC



NOTES:

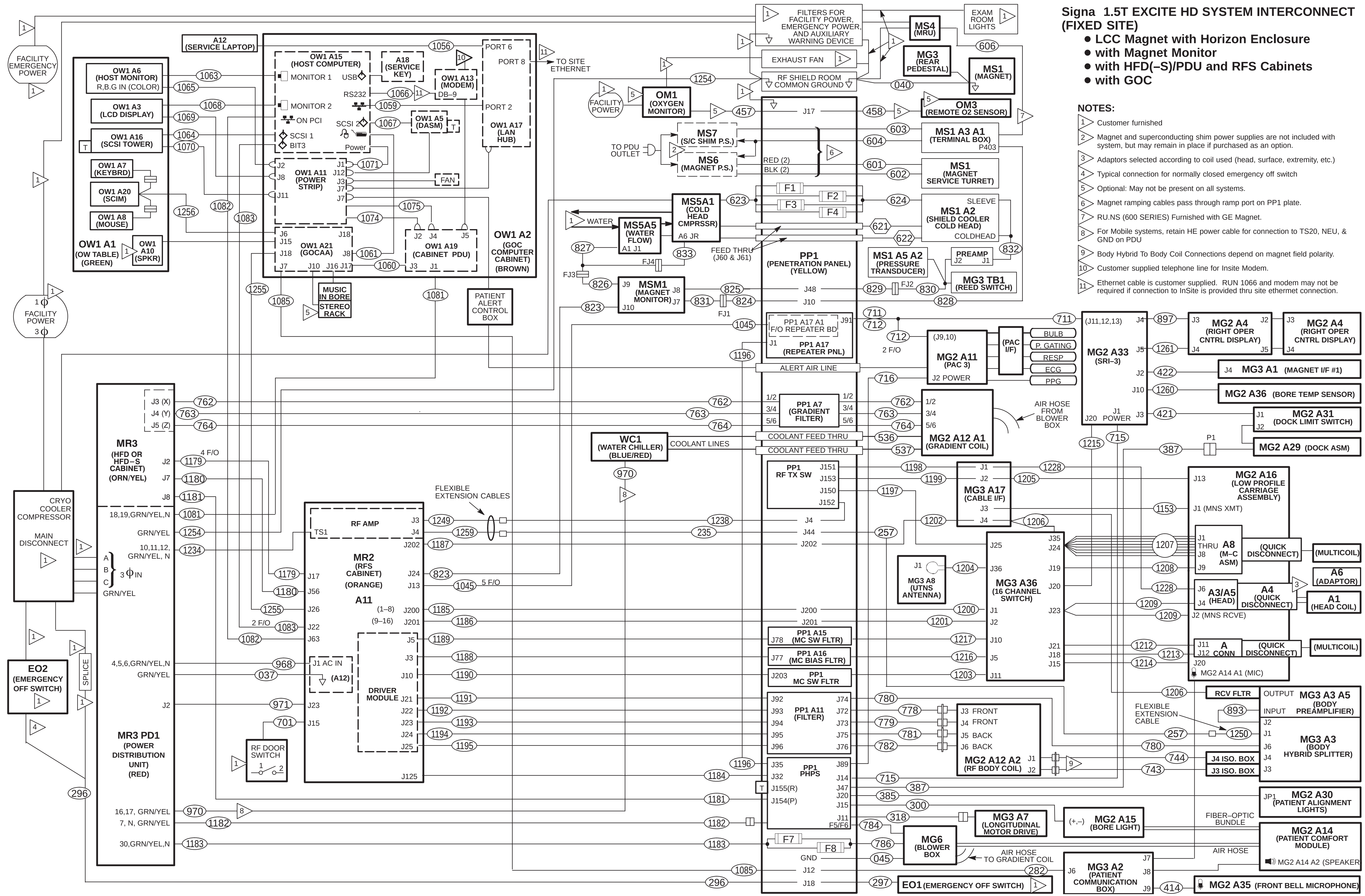
1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

Signa 1.5T EXCITE HD SYSTEM INTERCONNECT (FIXED SITE)

- LCC Magnet with Horizon Enclosure
- with Magnet Monitor
- with HFD(–S)/PDU and RFS Cabinets
- with GOC

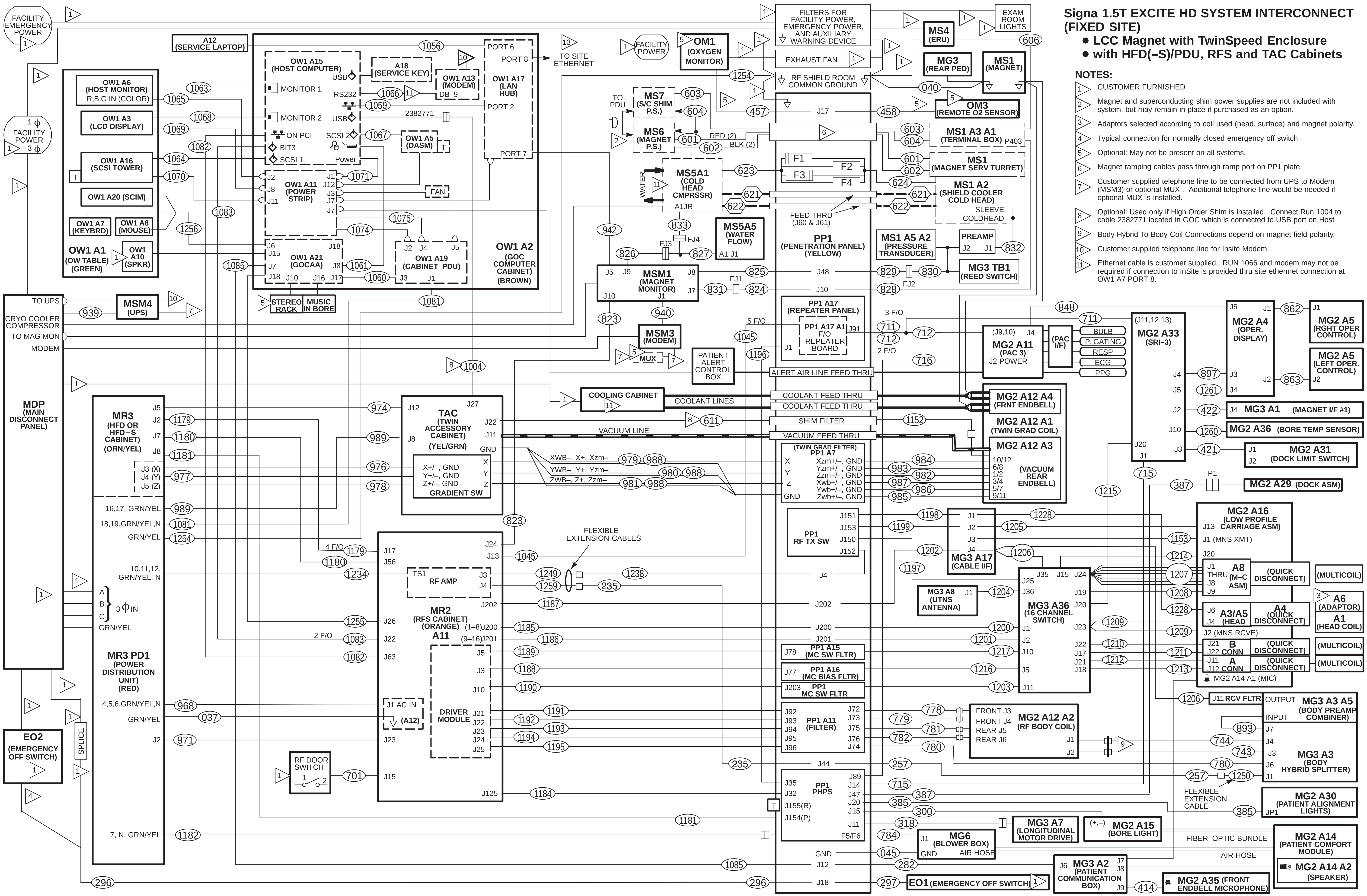
NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. RU.NS (600 SERIES) Furnished with GE Magnet.
8. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.



Signa 1.5T EXCITE HD SYSTEM INTERCONNECT (FIXED SITE)

- LCC Magnet with TwinSpeed Enclosure
- with HFD(–S)/PDU, RFS and TAC Cabinets

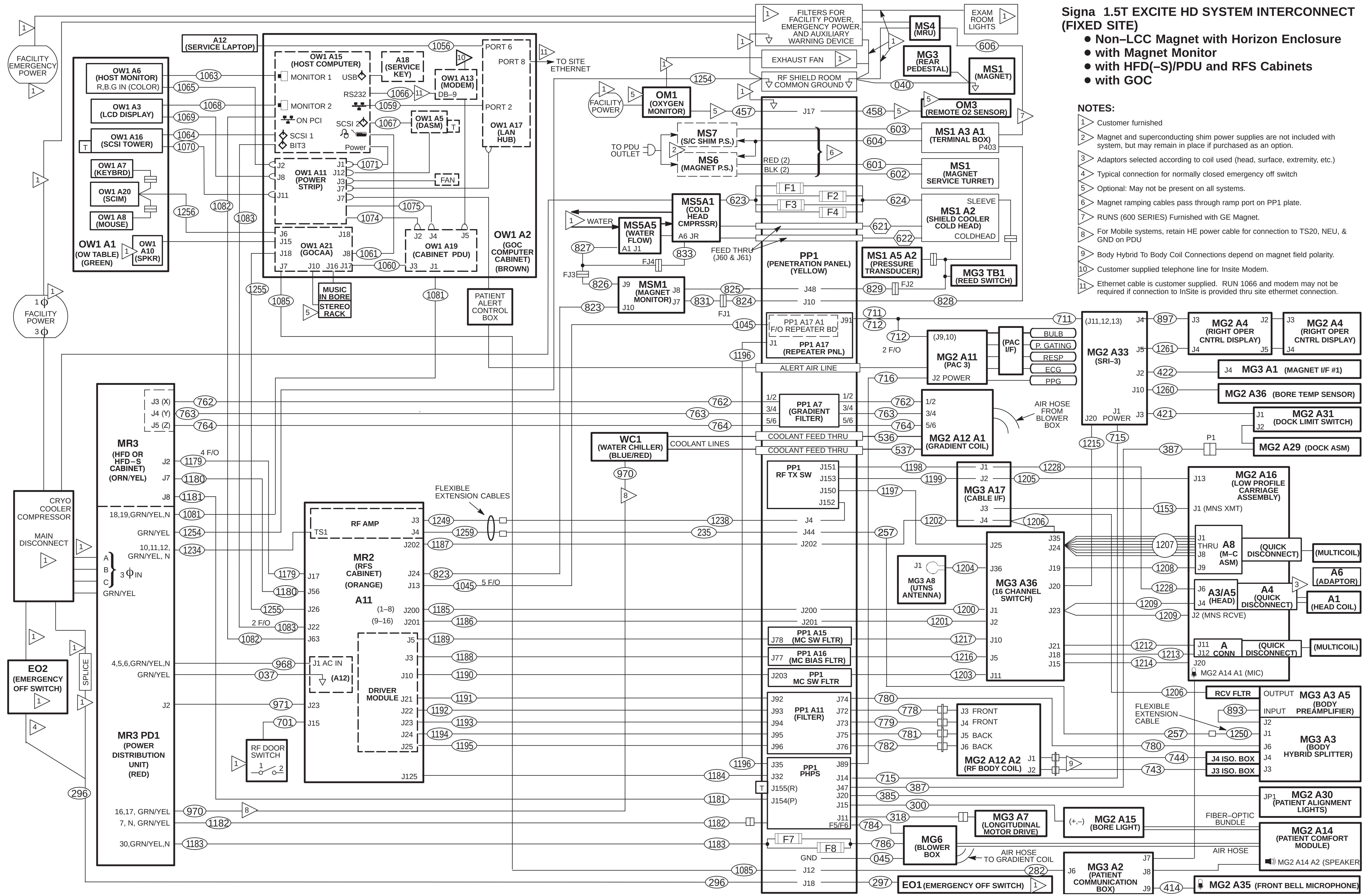


NOTES:

1. CUSTOMER FURNISHED
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX. Additional telephone line would be needed if optional MUX is installed.
8. Optional: Used only if High Order Shim is installed. Connect Run 1004 to cable 2382771 which is connected to USB port on Host
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1 A7 PORT 8.

Signa 1.5T EXCITE HD SYSTEM INTERCONNECT (FIXED SITE)

- Non–LCC Magnet with Horizon Enclosure
- with Magnet Monitor
- with HFD(–S)/PDU and RFS Cabinets
- with GOC



NOTES:

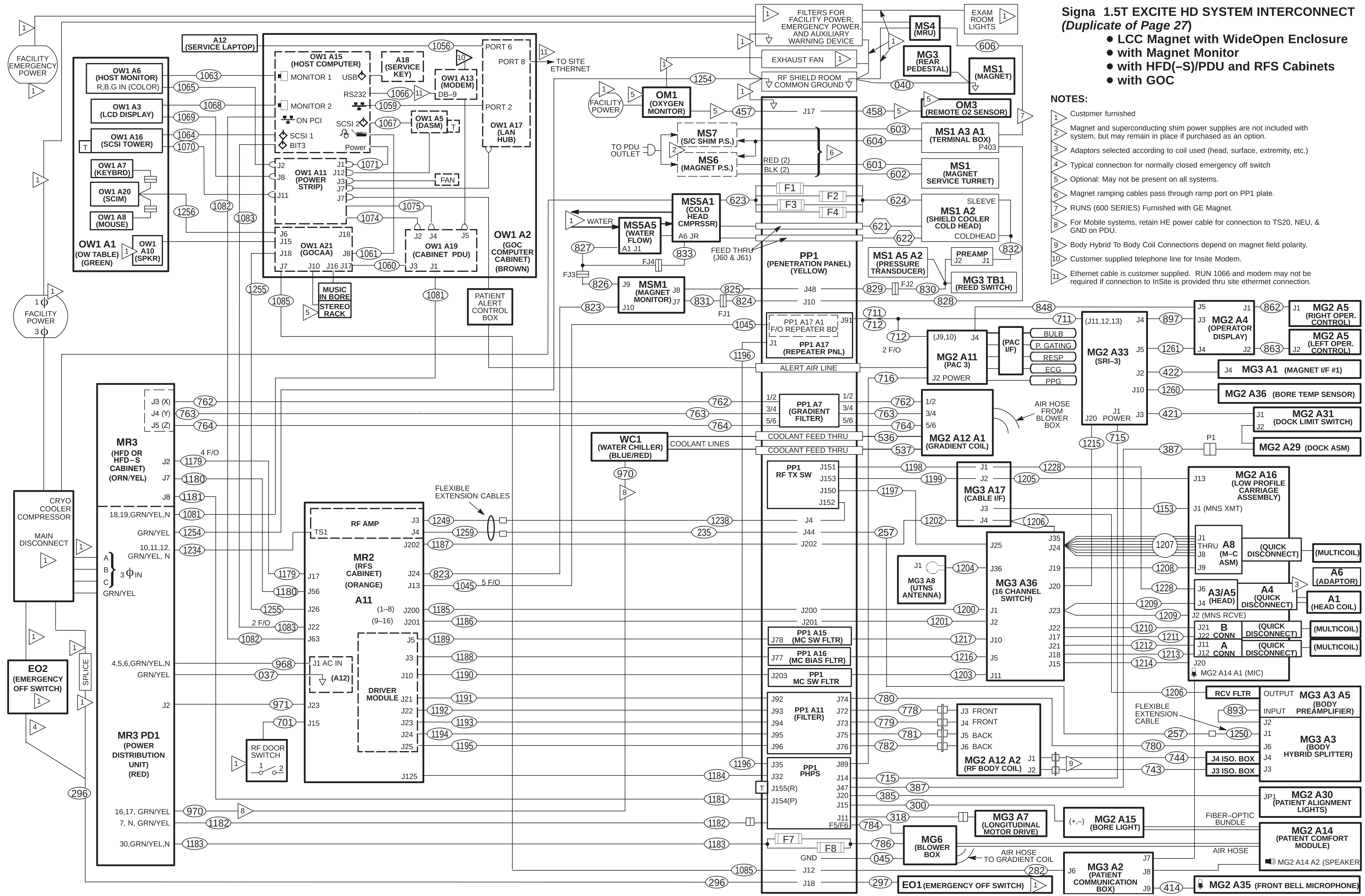
1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

Signa 1.5T EXCITE HD SYSTEM INTERCONNECT
(Duplicate of Page 27)

- LCC Magnet with WideOpen Enclosure
- with Magnet Monitor
- with HFD(–S)/PDU and RFS Cabinets
- with GOC

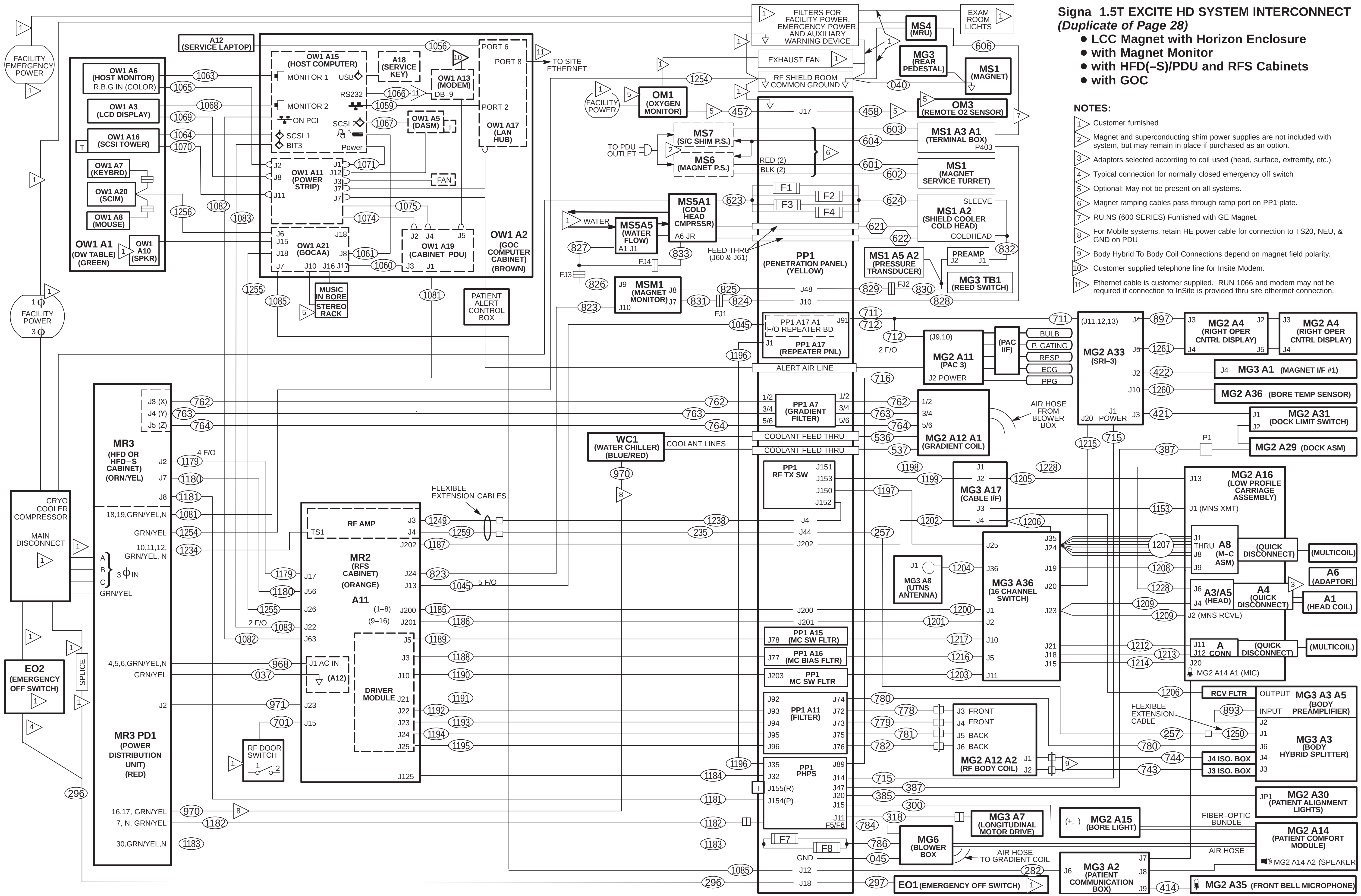
NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.



Signa 1.5T EXCITE HD SYSTEM INTERCONNECT
(Duplicate of Page 28)

- **LCC Magnet with Horizon Enclosure**
- **with Magnet Monitor**
- **with HFD(–S)/PDU and RFS Cabinets**
- **with GOC**



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. RU.NS (600 SERIES) Furnished with GE Magnet.
8. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

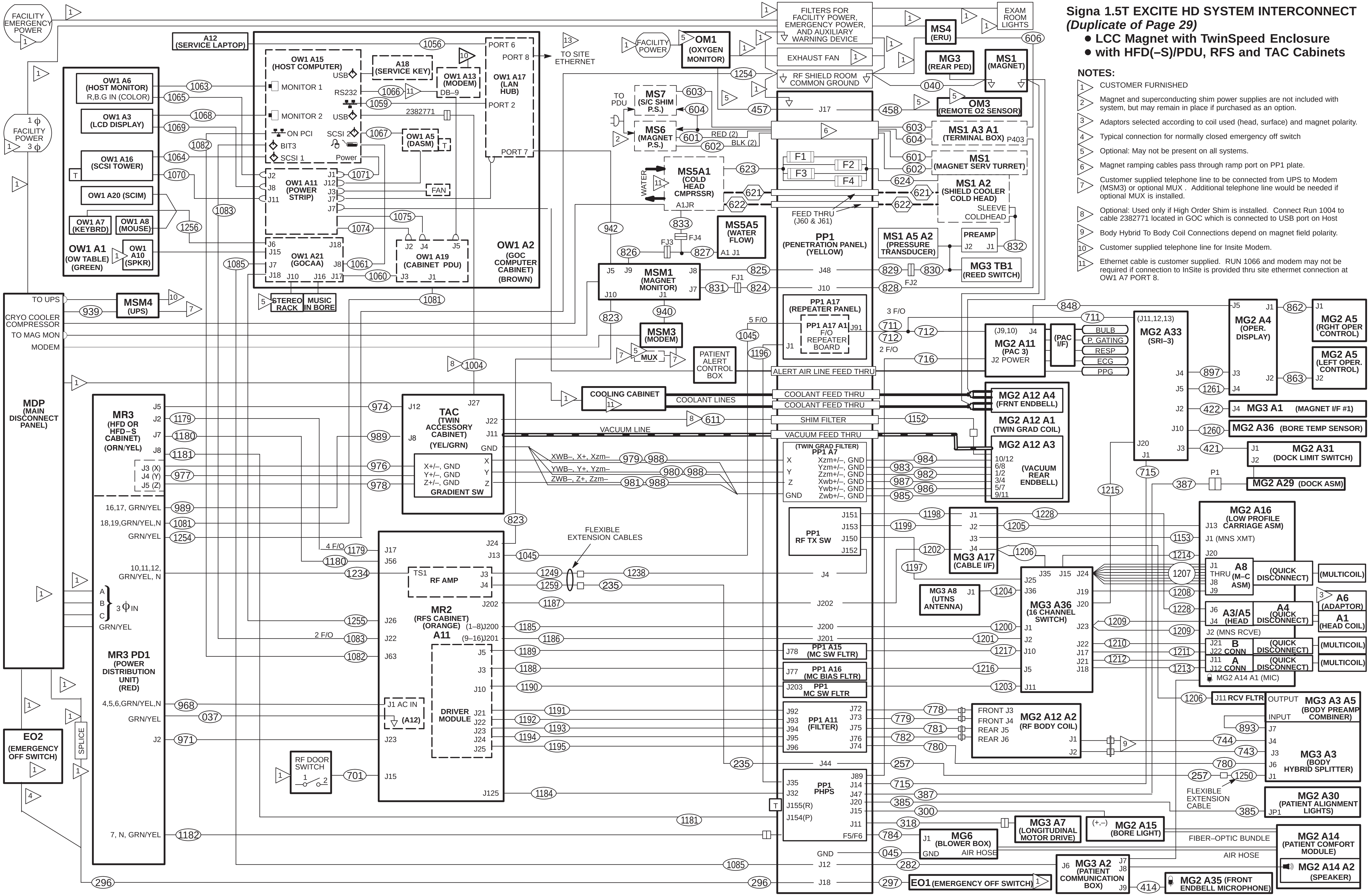
Signa 1.5T EXCITE HD SYSTEM INTERCONNECT

(Duplicate of Page 29)

- **LCC Magnet with TwinSpeed Enclosure**
- **with HFD(–S)/PDU, RFS and TAC Cabinets**

NOTES:

1. CUSTOMER FURNISHED
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
8. Optional: Used only if High Order Shim is installed. Connect Run 1004 to cable 2382771 located in GOC which is connected to USB port on Host
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1 A7 PORT 8.



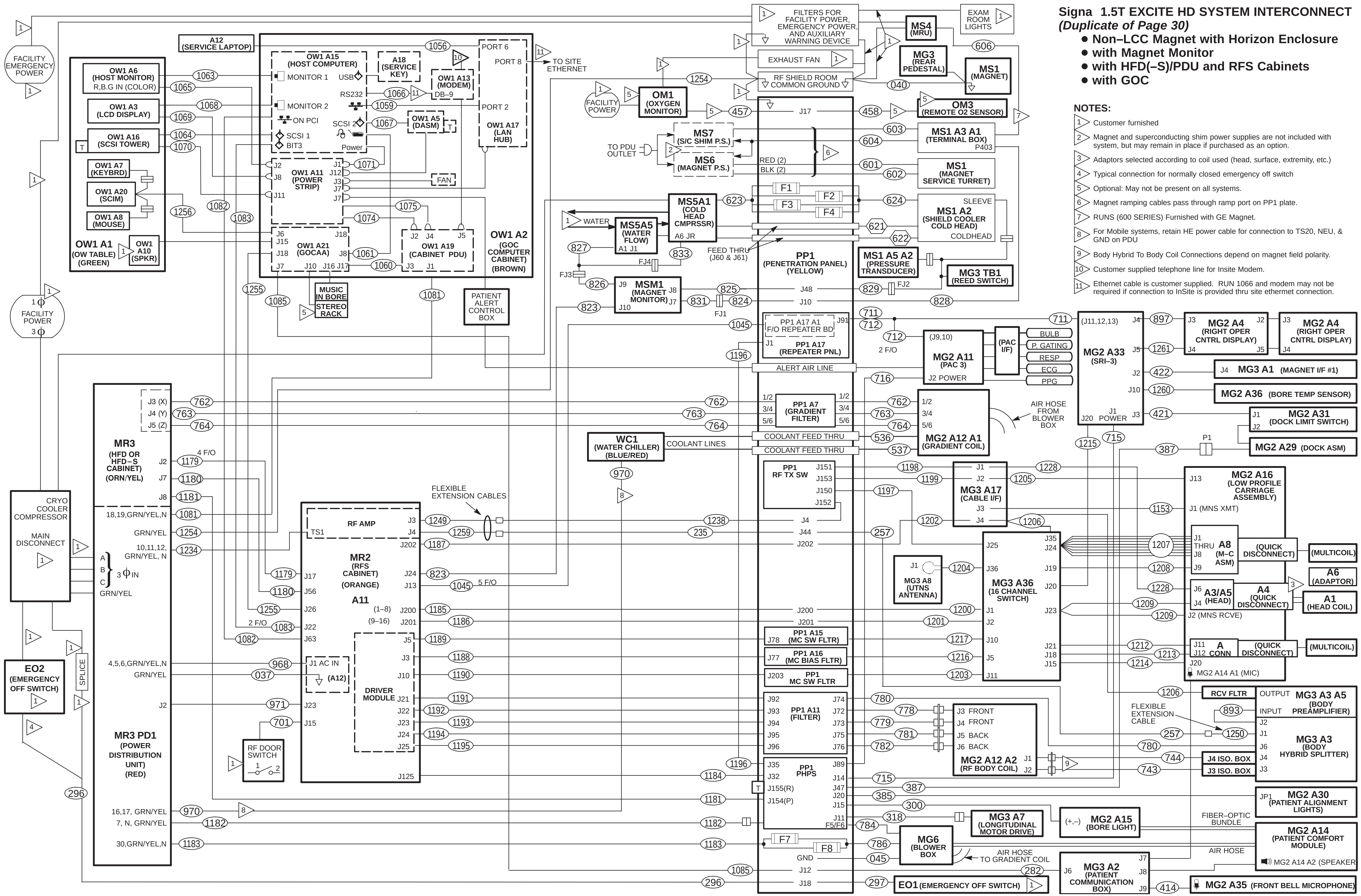
Signa 1.5T EXCITE HD SYSTEM INTERCONNECT

(Duplicate of Page 30)

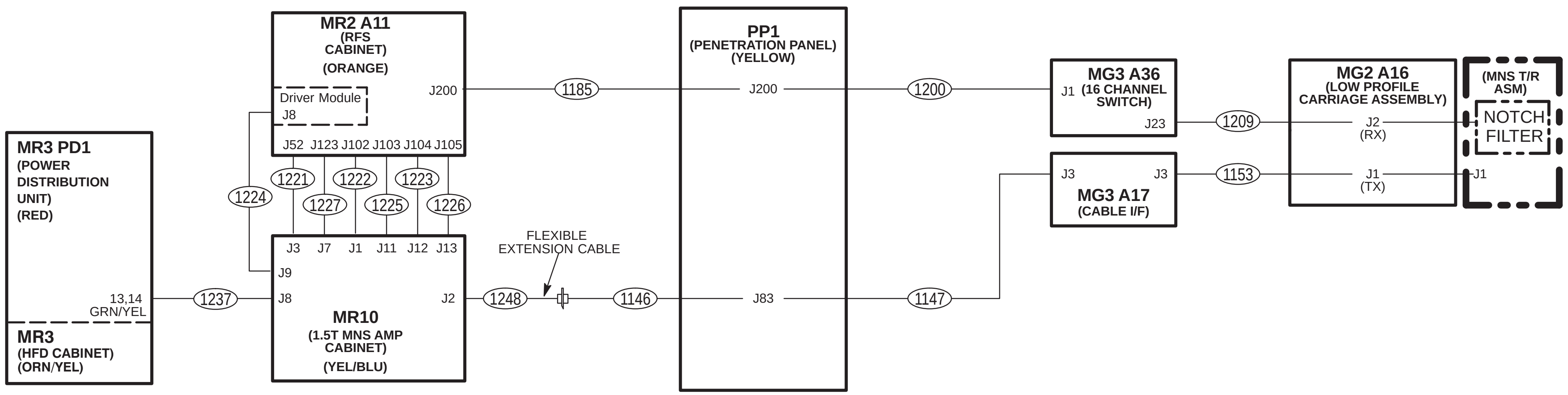
- Non–LCC Magnet with Horizon Enclosure
- with Magnet Monitor
- with HFD(–S)/PDU and RFS Cabinets
- with GOC

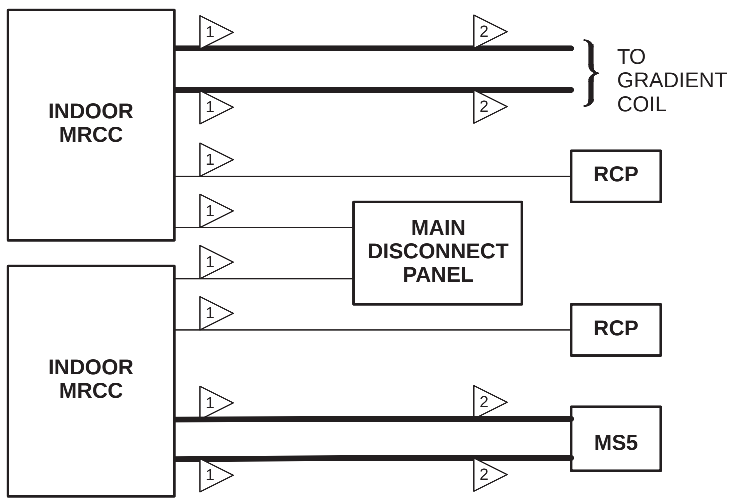
NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. For Mobile systems, retain HE power cable for connection to TS20, NEU, & GND on PDU
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.



Signa 1.5T EXCITE HD, Release 12.0: 2KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS

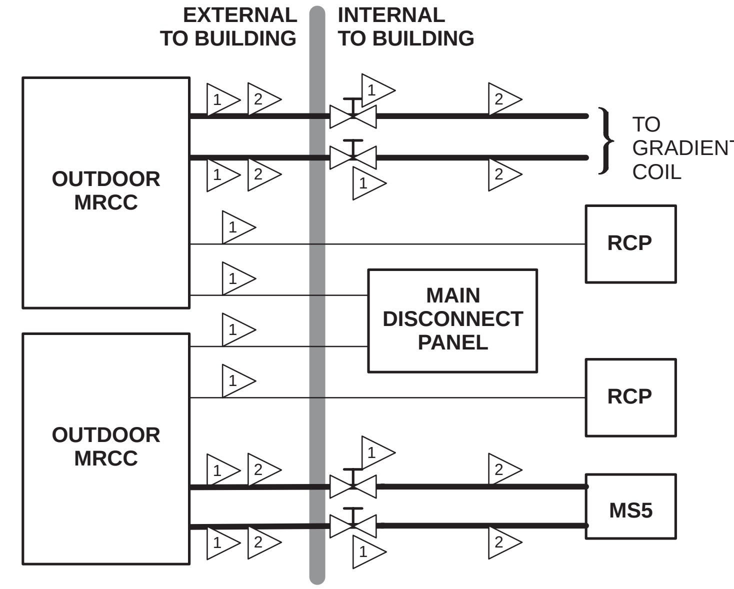




NOTES:
1. CUSTOMER FURNISHED.
2. THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

NOTE:
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

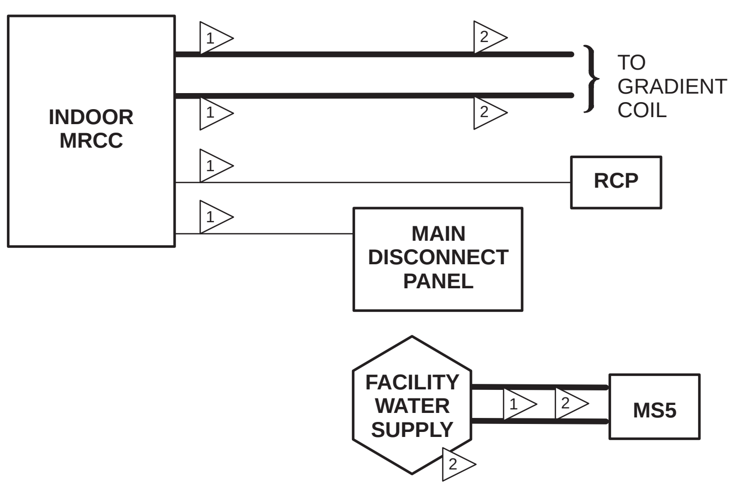
DUAL INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



NOTES:
1. CUSTOMER FURNISHED.
2. THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

NOTE:
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

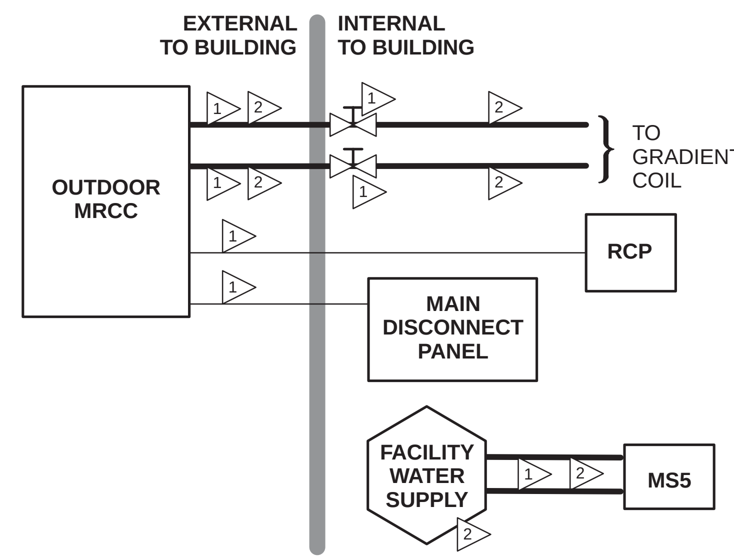
DUAL OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



NOTES:
1. CUSTOMER FURNISHED.
2. THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

NOTE:
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

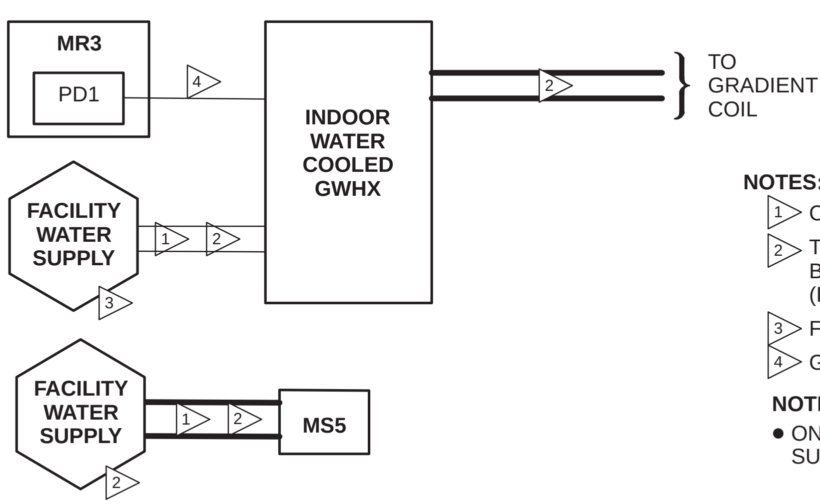
INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



NOTES:
1. CUSTOMER FURNISHED.
2. THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

NOTE:
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



NOTES:
1. CUSTOMER FURNISHED.
2. THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
3. FOR INDOOR WATER COOLED GWHX ONLY.
4. GWHX PROVIDED CABLE.

NOTE:
- ONLY INTERCONNECTS SPECIFIC TO GWHX SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR GWHX SUBSYSTEM GROUP INTERCONNECT DIAGRAM

4–7 EQUIPMENT ROOM CABLE INSTALLATION

The equipment room cables and cabinets can be installed simultaneously. Cable locations depend on where cabinets are installed.

4–7–1 HFD CABINET CABLE CONNECTIONS

Note: If site has an ACGD/PDU cabinet and is being upgraded to HFD–S/PDU, refer to Section **15–4** for connection of cables to PDU module.

DANGER!!

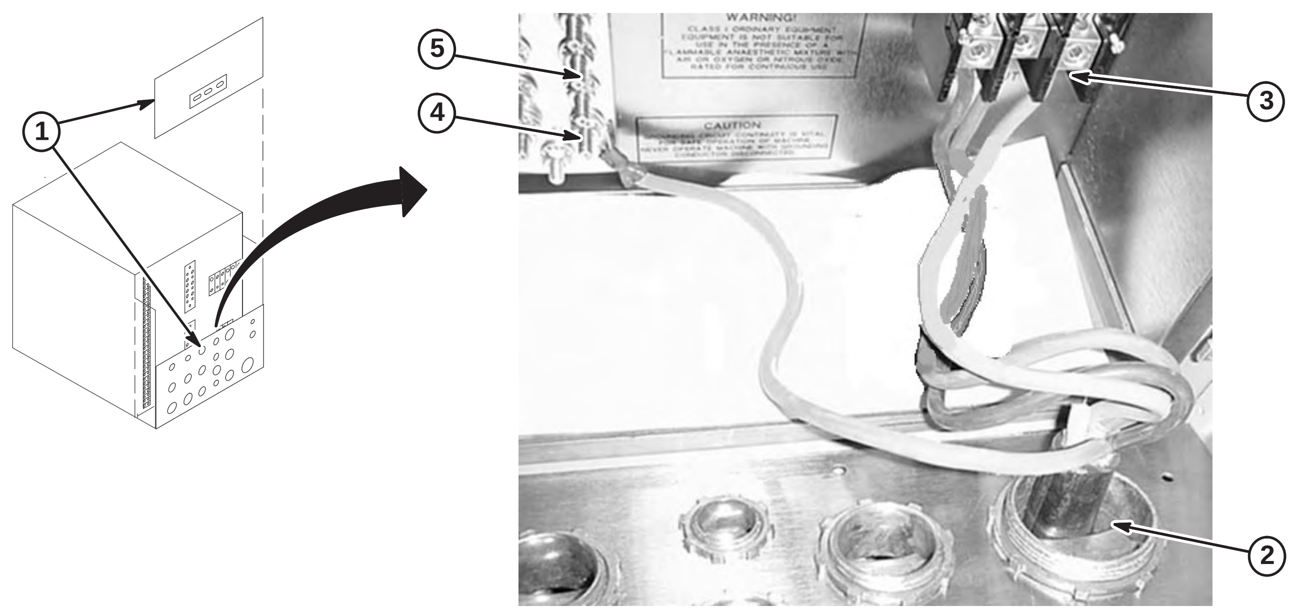
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE CONNECTING WIRES.

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE HFD CABINET.

4–7–1–A HFD Cabinet Power and Ground Cable Installation

1. Remove rear PDU module covers.
2. Route facility input power cable and ground wire thru interface panel.
3. Connect L1–L3 conductors to terminal block.
4. Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.
5. Connect RF screen room ground conductor to Ground Bus Bar.

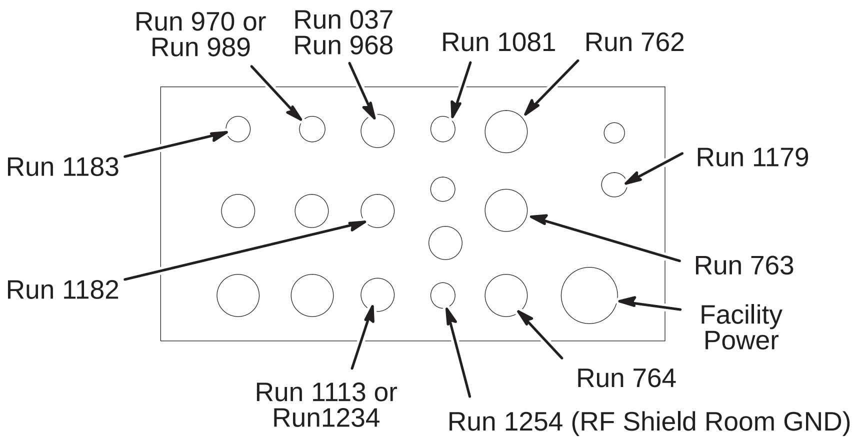


4–7–1–B Route Power Cables Thru Interface Panel

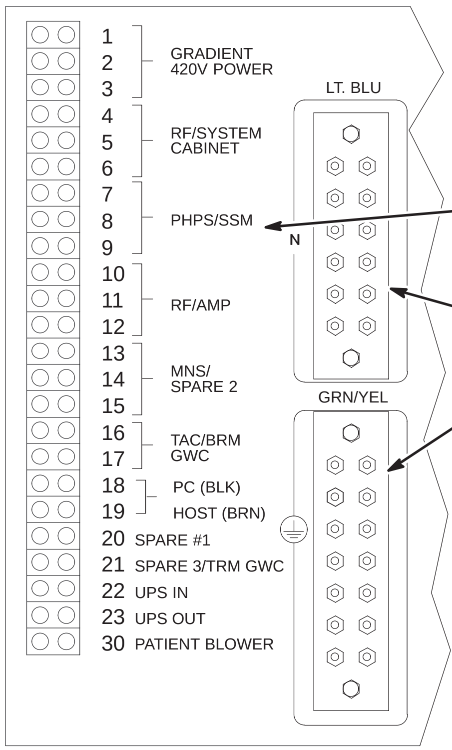
1. Route Runs thru interface panel as marked.

Note: The GRN/YEL ground wire, Run 037, **MUST** be routed with Run 968 power cable thru the same port.

2. Route any option Runs as marked.



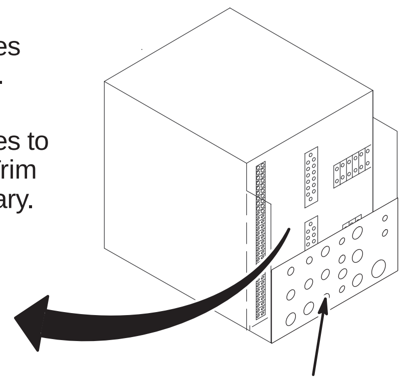
4–7–1–C Connect Power Cables to PDU Module



WARNING!

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD CABINET.

1. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in applicable Table below
2. Connect neutral wire cables to Neutral bus connectors.
3. Connect ground wire cables to Ground bus connectors. Trim and reterminate if necessary.



4. Tighten clamps of interface panel to secure cables in place.

	CONNECT TO	RUN	PHASE	DESTINATION	
NON–TwinSpeed	4,5,6	968	ϕ A, B, C	RFS Cabinet	3ϕ, Neutral, GND
	7	1182	ϕ A	PHPS Module	Single ϕ Neutral, GND
	10,11,12	1234	ϕ A, B, C	RF Amp in RFS Cabinet	3ϕ, Neutral, GND
	16,17	970	ϕ A,B	Water Chiller	Single ϕ GND
	18,19	1081	ϕ C,A	Operator Workspace	Single ϕ, Neutral, GND
	20	046/973	ϕ A	Heat Exchanger (Mobile)	Single ϕ, Neutral, GND
	30	1183	ϕ A	Blower Box	Single ϕ, Neutral, GND
	GRN/YEL	037		System Cabinet Ground Wire	
	GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)	
TwinSpeed	4,5,6	968	ϕ A, B, C	RFS Cabinet	3ϕ, Neutral, GND
	7	1182	ϕ A	PHPS Module	Single ϕ Neutral, GND
	10,11,12	1113	ϕ A, B, C	NB RF Amp Cabinet	3ϕ, Neutral, GND
	16,17	989	ϕ A,C	TAC Cabinet	Single ϕ, GND
	18,19	1081	ϕ C,A	Operator Workspace	Single ϕ, Neutral, GND
	(MDP)* or		ϕ A or	AirSys Water Cooled Cabinet or	Single ϕ, Neutral, GND
	30 (PDU)		ϕ A	AirSys Heat Exchanger	Single ϕ, Neutral, GND
	GRN/YEL	037		System Cabinet Ground Wire	
	GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)	

COLOR CODE: for remainder of cables:
3ϕ = Black, Red, Orange Neutral = Light blue Single ϕ = Brown, Black Gnd = Grn/Yel

* Connection to Main Disconnect Panel to be completed by customer electrician.

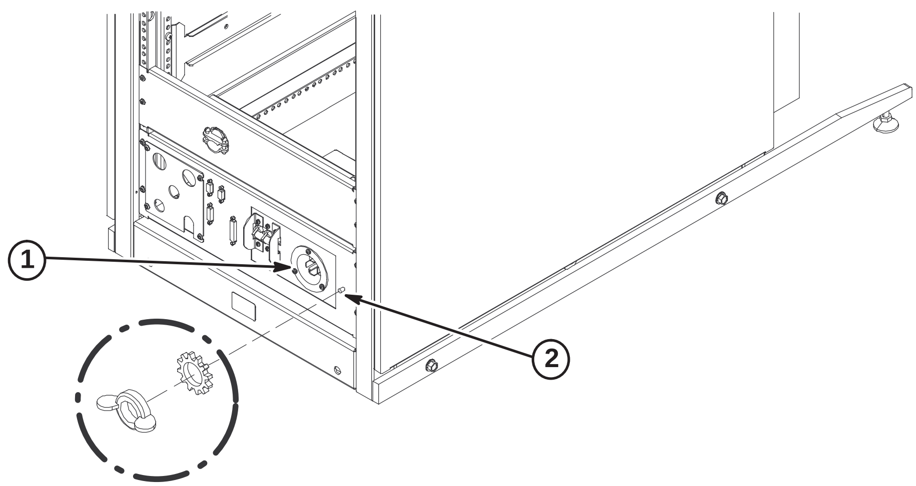
4–7–2 EQUIPMENT ROOM CABINET POWER CABLE CONNECTIONS

WARNING!

CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD CABINET.

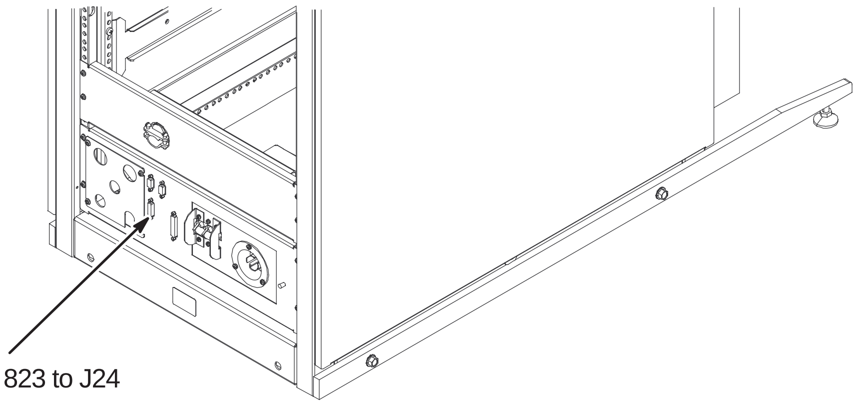
4–7–2–A RF/System Cabinet Power and Ground Cable Connections

1. Connect Run 968 cable plug to “J1”. Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.



4–7–3 INSTALLATION OF MAGNET MONITORING CABLES

Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual,* for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.

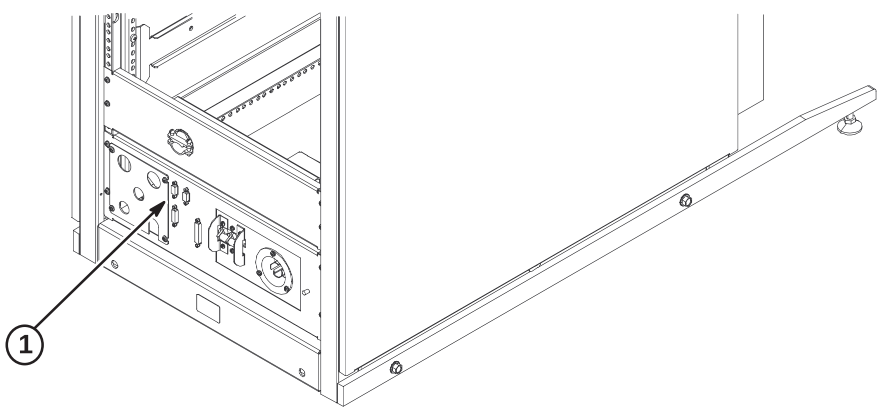


1 Connect Run 823 to J24

4–7–4 INSTALL RUN 701 – RF DOOR SWITCH (If Required)

Run 701 is a 100 ft cable with a 9–pin subminature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminature D plug.

1. Connect Run #701 to J15 on System Cabinet and route to RF Door Switch. Refer to site engineering drawings for cable routing path.
2. Connect black lead on Run #701 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run #701 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run #701 to RF Door Switch N.C. (normally closed) contact.

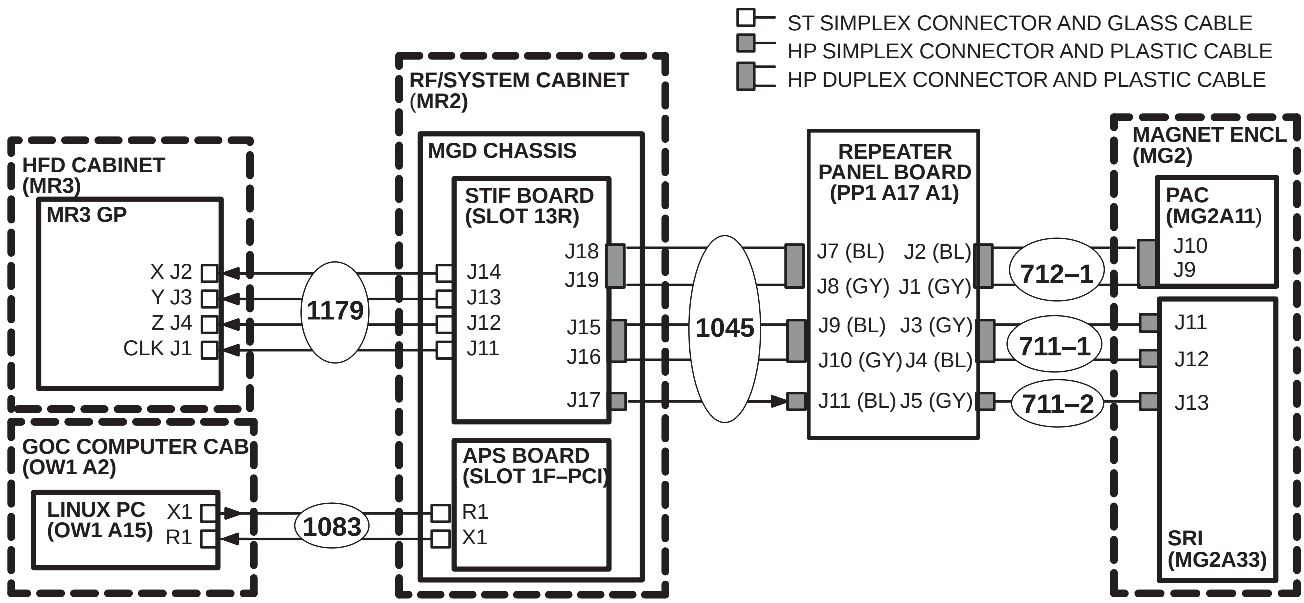


4–7–5 INSTALLATION OF FIBER OPTIC CABLES

Plan for storage of excess Run 1179 conduit, as it cannot be reterminated. For most Horizon amd LX configurations, Runs 711/712 in the magnet room will be retained for the upgrade to EXCITE HD.

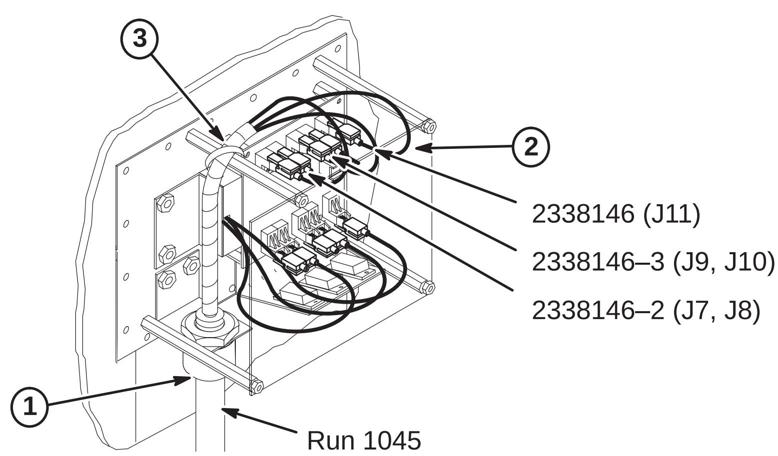
4–7–5–A Routing of Fiber Optic Cables

1. Unroll coiled fiber optic conduits for Runs 711/712, 1045, 1083, and 1179 along intended routes.
2. Connection of Runs is shown in diagram below.



4–7–5–B Connect Run 1045 to Penetration Panel

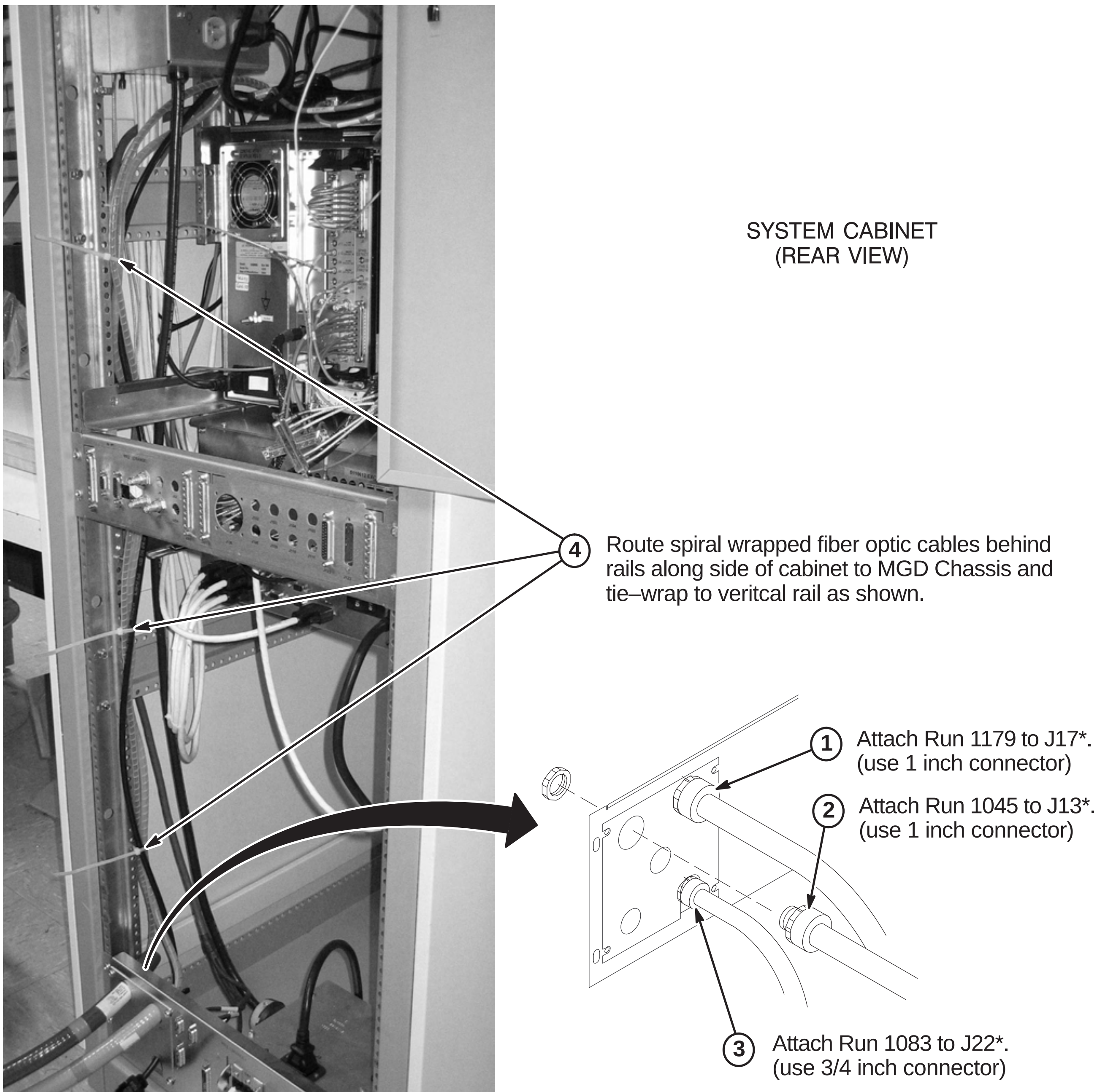
1. Move Run 1045 to the Equipment Room side and attach “PP1–A17” end to access bracket on Penetration Panel. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
2. Adjust route of individual fiber optic runs to maintain large radius and connect Run 1045 as shown.
3. Adjust position of protective spiral wrap and secure with tie–wrap.
4. Route other end of Run 1045 to System Cabinet.



4–7–5–C Route Fiber Optic Runs 1045, 1083, and 1179 in RF/System Cabinet

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



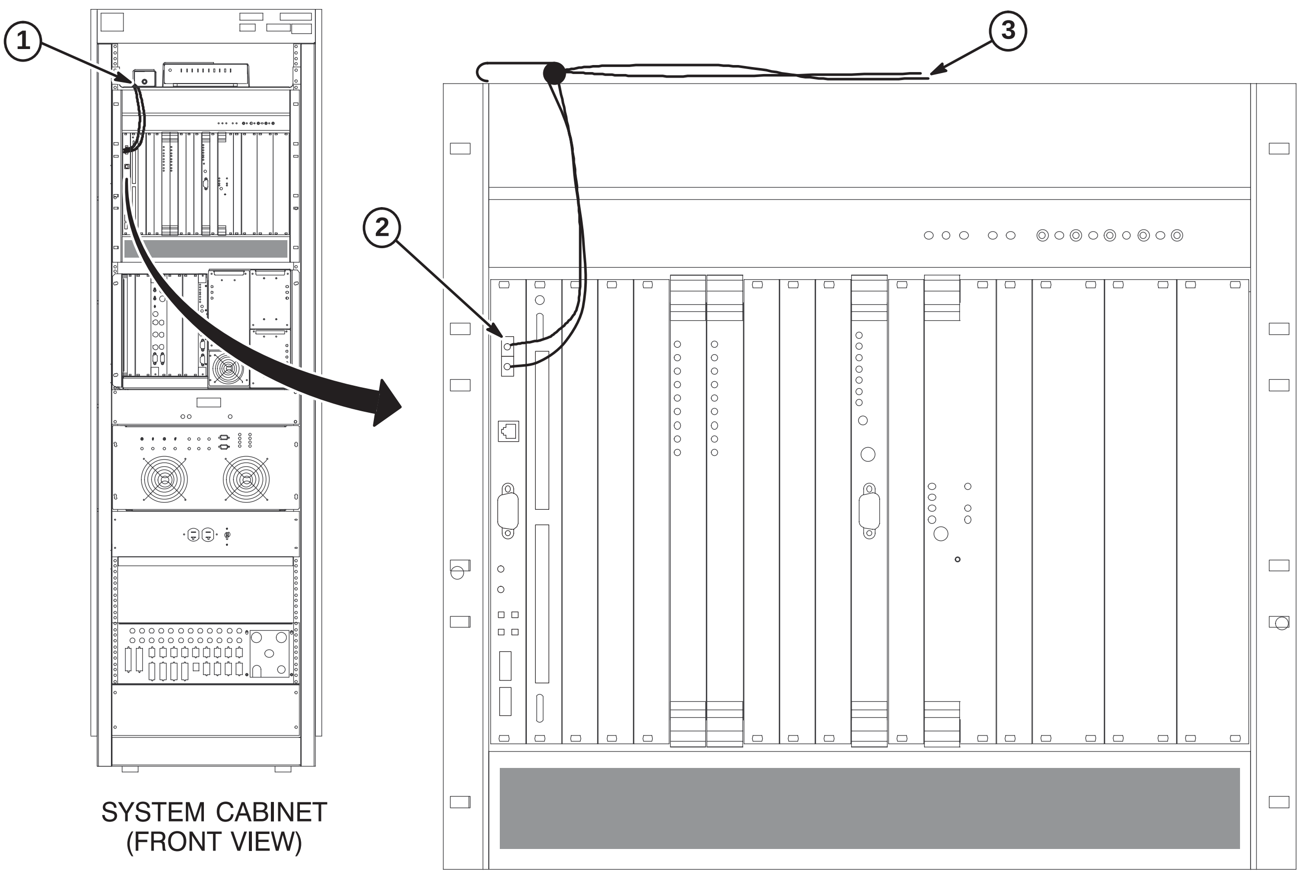
* Instructions provided with connectors.

4–7–5–D Route/Connect Fiber Optic Run 1083 to APS Board

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route Run 1083 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet. **Do not bend fiber optic cables to radius smaller than two inches (50mm).**
2. Connect the Orange and Blue fiber optic cables to the TI–MGD Daughter Board as indicated.
3. Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.

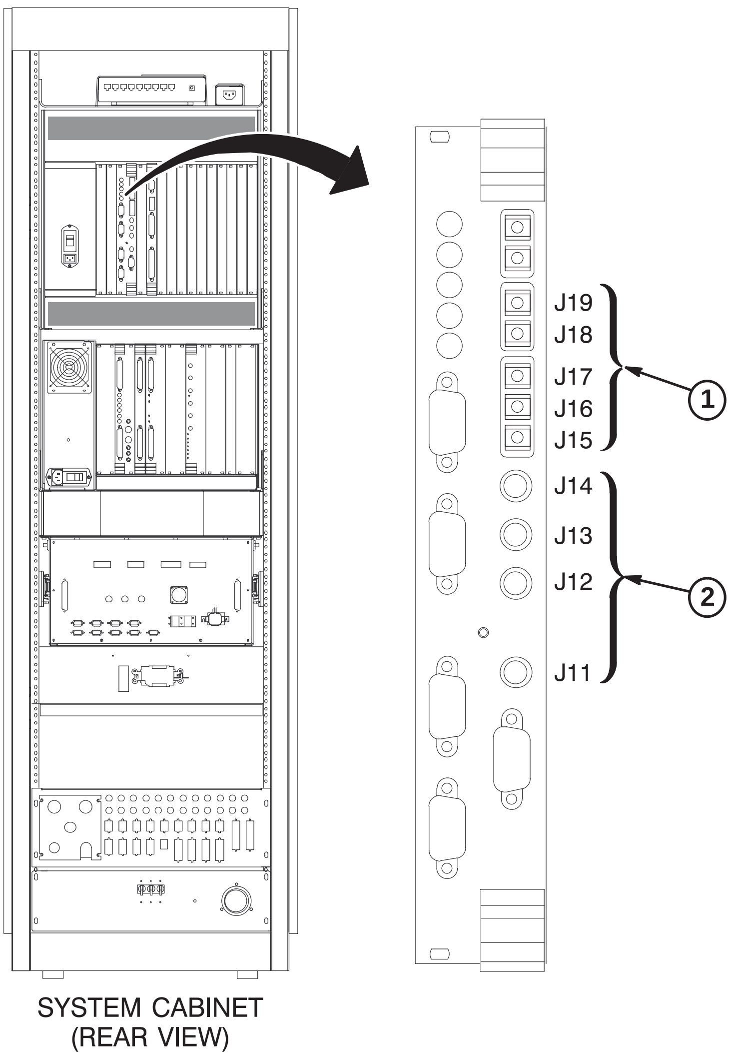


4–7–5–E Route/Connect Fiber Optic Runs1079 and 1045 to STIF Board

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
2. Route/Connect Run 1179 fiber optic cables to J11, J12, J13, and J14 on the STIF Board.



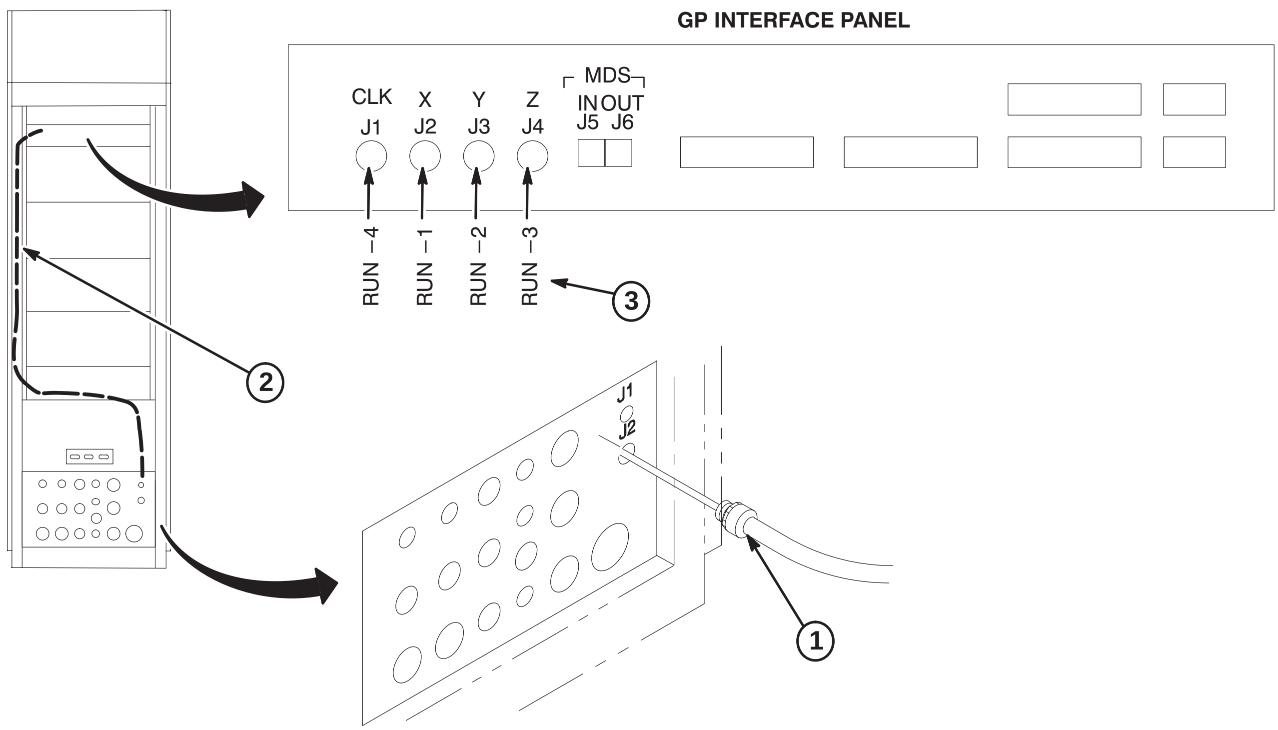
Note:
Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

4–7–5–F Route/Connect Fiber Optic Run 1179 to GP3 Module in HFD

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.

1. Attach Run 1179 to I/F J2.
2. Route fiber optic cables to GP3 Module. Leave enough slack for extending GP module forward.
3. Connect fiber optics to GP3 module according to table.

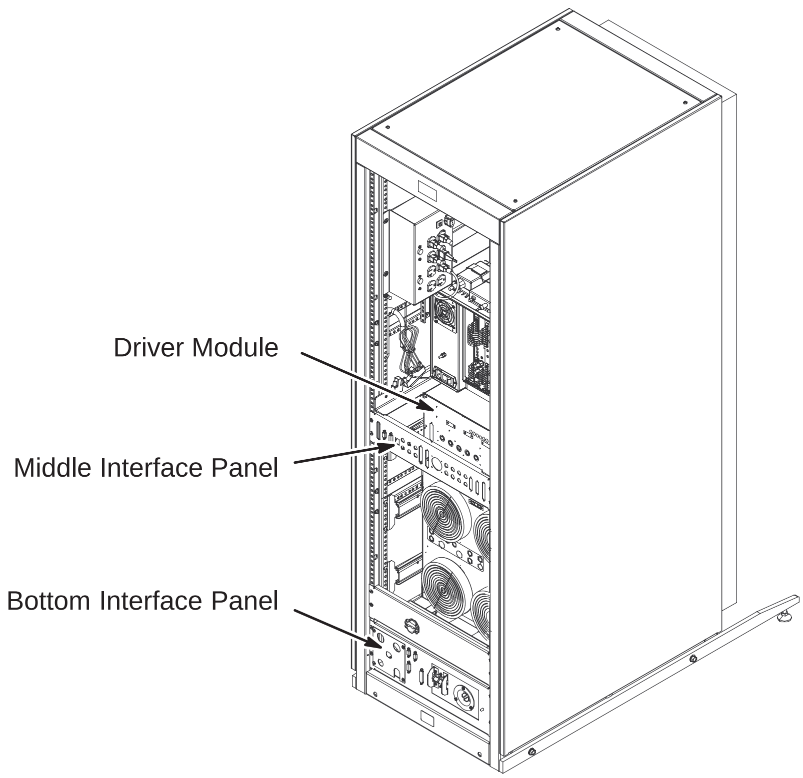


4–7–6 INSTALLATION OF REMAINING EQUIPMENT ROOM CABLES

Connect remaining Runs within equipment room. Dress and secure all cables after connections are made. Reference the applicable interconnect map in Section 4 and Cable Table in Appendix for to/from designator information and Run number reference.

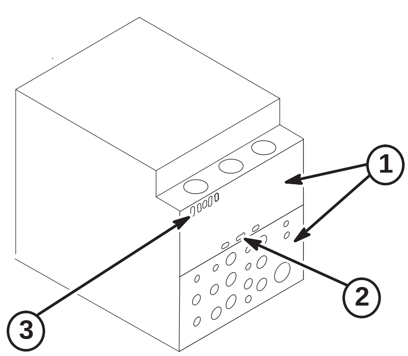
4–7–6–A RF/System Cabinet Connections

1. Connect routed cables to Interface Panels at bottom and middle of cabinet and to the Driver Module as marked. Refer to cable map.
2. Dress and secure cables. Adjust as necessary.



4–7–6–B HFD Cabinet PDU Module Connections

1. Attach rear PDU module covers removed earlier.
2. Connect Run 1179 to J2.
3. Connect Runs 974, 1180, and 1181 to J5, J7, and J8.



4–8 MAGNET ROOM CABLE INSTALLATION

The magnet enclosure and magnet room cables must be installed simultaneously.

Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.

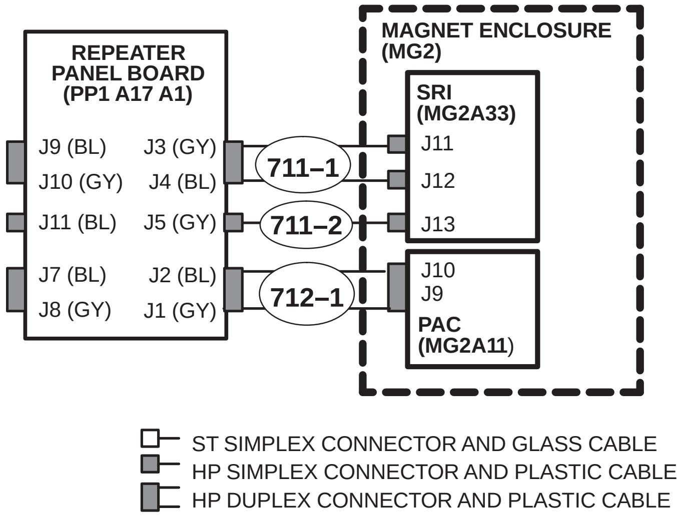
WARNING!

FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AS FAR FROM THE MAGNET AS POSSIBLE.

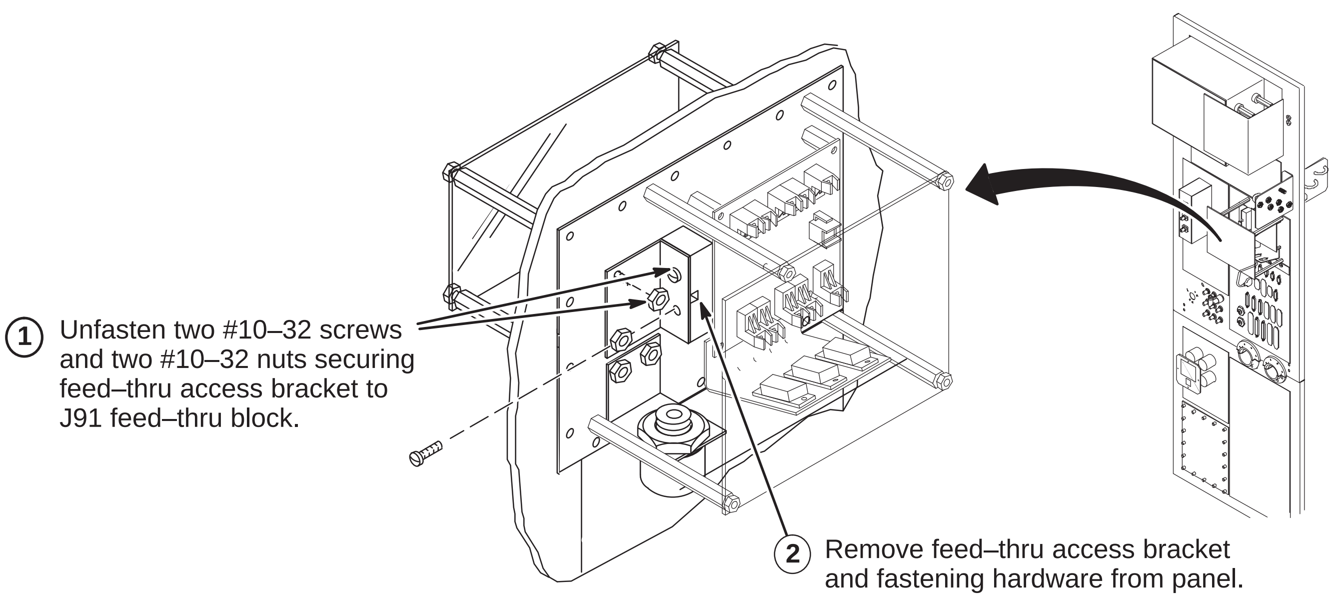
4–8–1 MAGNET ROOM FIBER OPTIC PAC/SRI INSTALLATION

For Horizon and LX configuratoins, existing Run 711/712 will be retained and connected to the new EXCITE HD Penetration Panel.

Shown below are the connections for the fiber optics cables routed in the magnet room from the Penetration Panel, thru the cable trough to the Rear Pedestal, and under the magnet to the PAC/SRI platform.



4–8–1–A Repeater Panel Feed–thru Bracket Removal



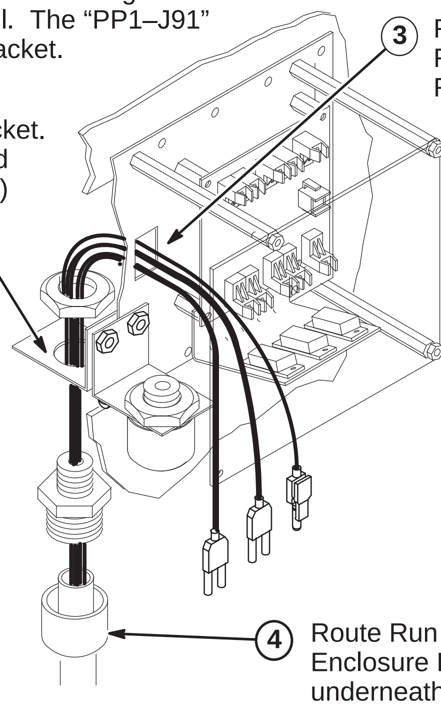
4–8–1–B Routing Run 711/712 in Magnet Room

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cut and ends terminated with the Fiber Optic Connector Repair Kit. Refer to Appendix for instructions. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

1 Move Run 711/712 (46–328079G9) to the Magnet Room side of the Penetration Panel. The “PP1–J91” end will be attached to interface bracket.

2 Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.

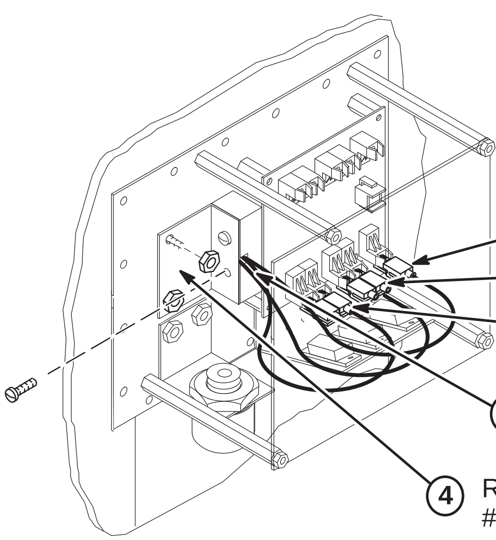
3 Route ends of individual fiber–optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)



4 Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

4–8–1–C Connecting Fiber Optic Run 711/712 to Repeater Panel

Note: Take care to maintain the minimum 2 inch bend radius.



1 Adjust route of individual fiber optic runs to maintain large radius.

2 Connect Runs 712–1, 711–1, and 711–2 as shown.

Run 711–2 (J5)

Run 711–1 (J3, J4)

Run 712–1 (J1, J2)

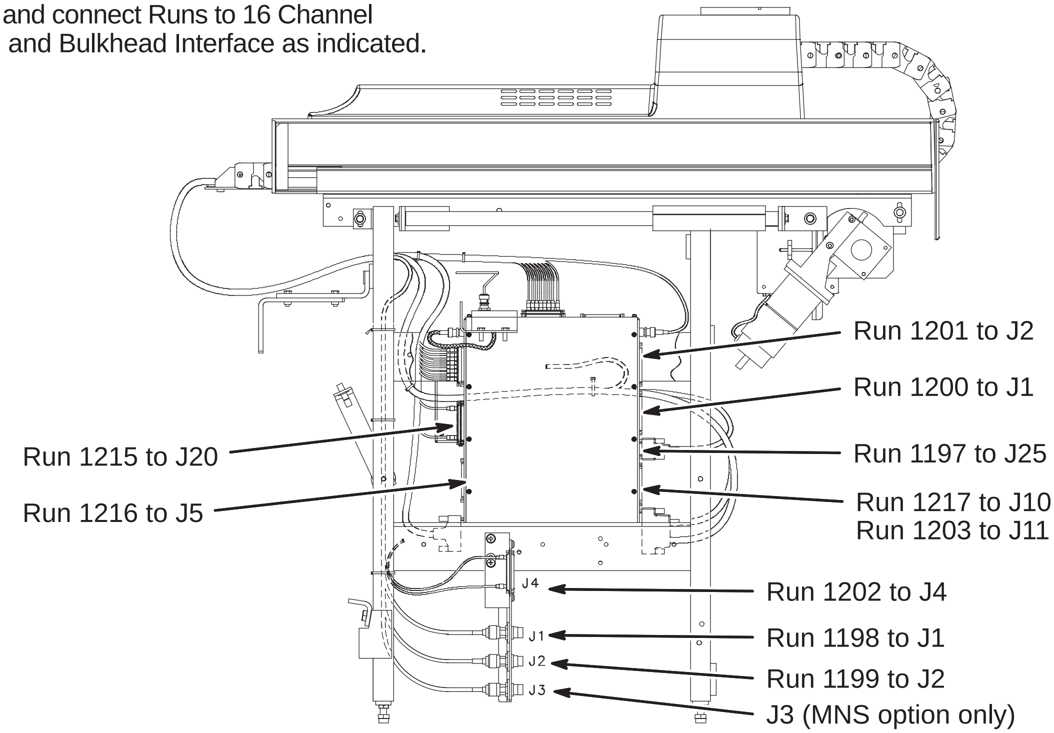
3 Gather fiber optic runs and route through wave guide notch in feed–thru block.

4 Replace feed–thru access bracket and fasten with two #10–32 screws and two #10–32 nuts removed in step A1.

4–8–2 CONNECT REMAINING CABLES TO REAR PEDESTAL

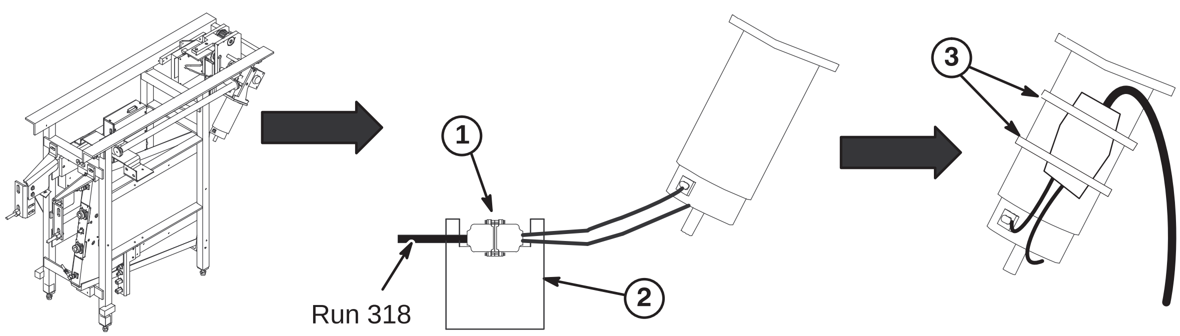
4–8–2–A Connect Cables to 16 Channel Switch and Bulhead Interface

1 Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



4–8–2–B Route/Connect Run 318 to Longitudinal Drive Motor

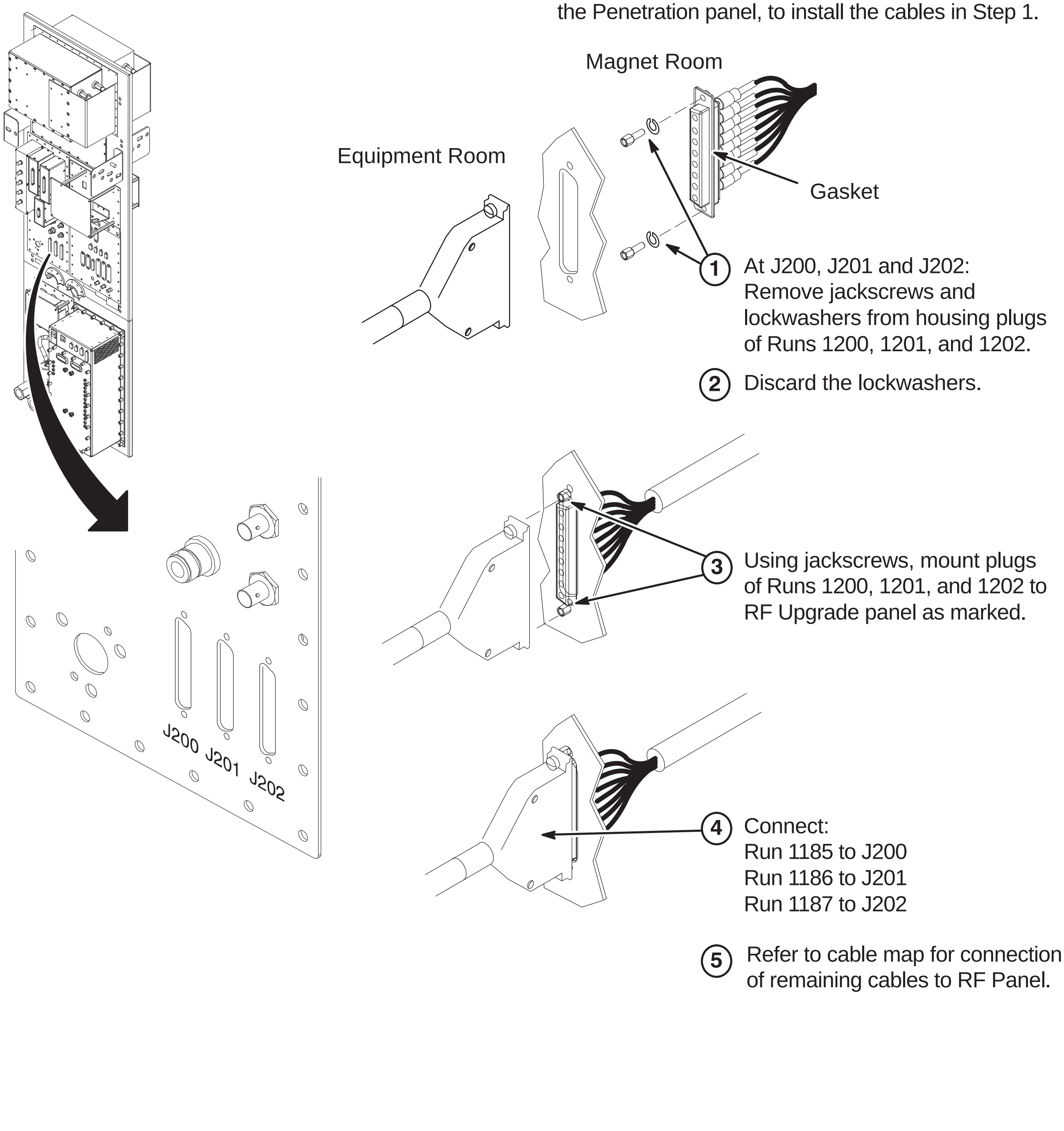
1. Route Run 318 from J11 on Penetration Panel tand connect to Longitudinal Drive Motor cable.
2. Wrap both connectors with 177 x 200 Connector Wrap (2236130–2).
3. Ty–wrap cable to motor to prevent movement.



4–9 INSTALLATION OF CABLES THAT RUN BETWEEN EQUIPMENT ROOM AND MAGNET ROOM

4–9–1 CONNECTION OF CABLES AT PENETRATION PANEL J200, J201, and J202

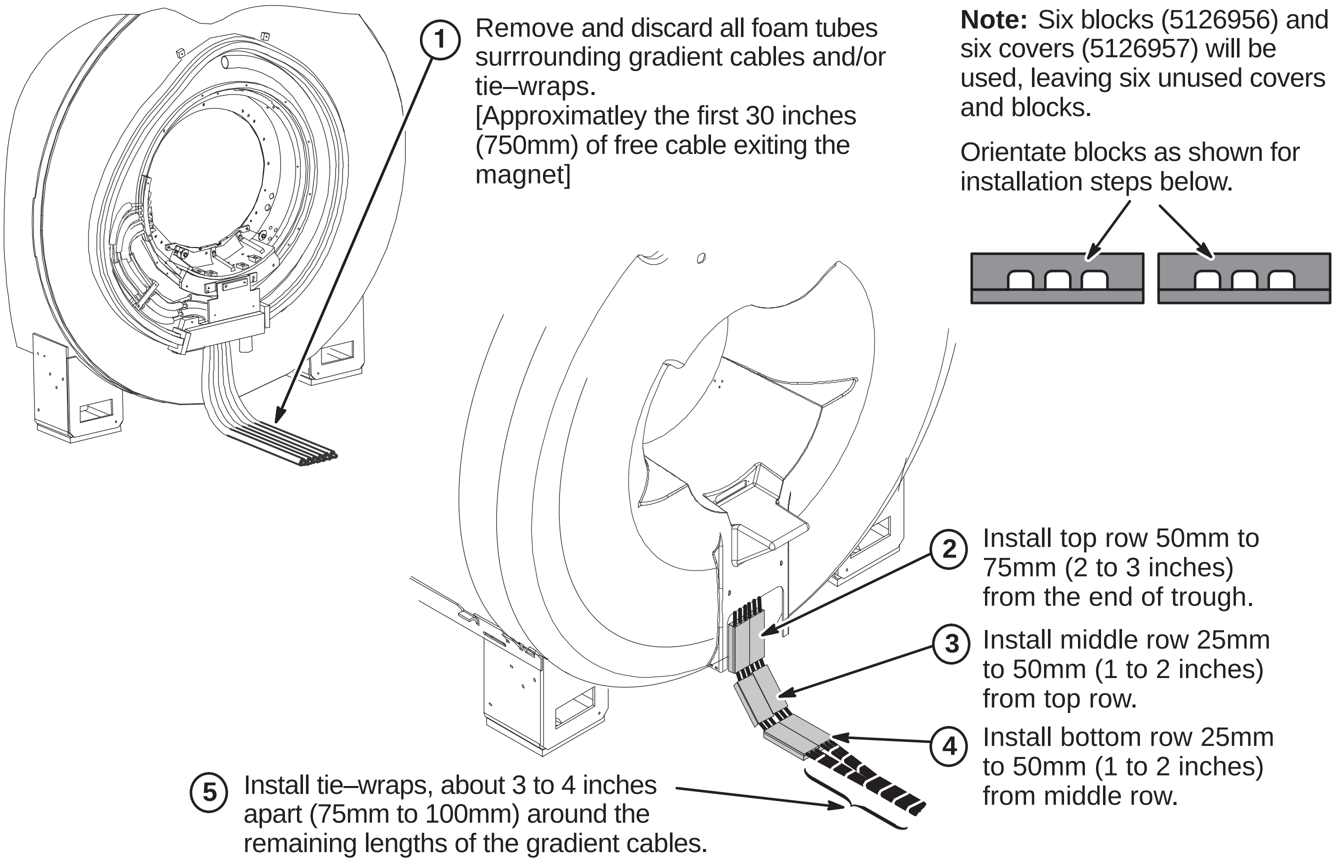
Note: Two people are required, one on each side of the Penetration panel, to install the cables in Step 1.



1. At J200, J201 and J202: Remove jackscrews and lockwashers from housing plugs of Runs 1200, 1201, and 1202.
2. Discard the lockwashers.
3. Using jackscrews, mount plugs of Runs 1200, 1201, and 1202 to RF Upgrade panel as marked.
4. Connect:
   Run 1185 to J200
   Run 1186 to J201
   Run 1187 to J202
5. Refer to cable map for connection of remaining cables to RF Panel.

4–9–2 INSTALL GRADIENT CABLE RESTRAINT BLOCKS FOR WideOpen

The existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.



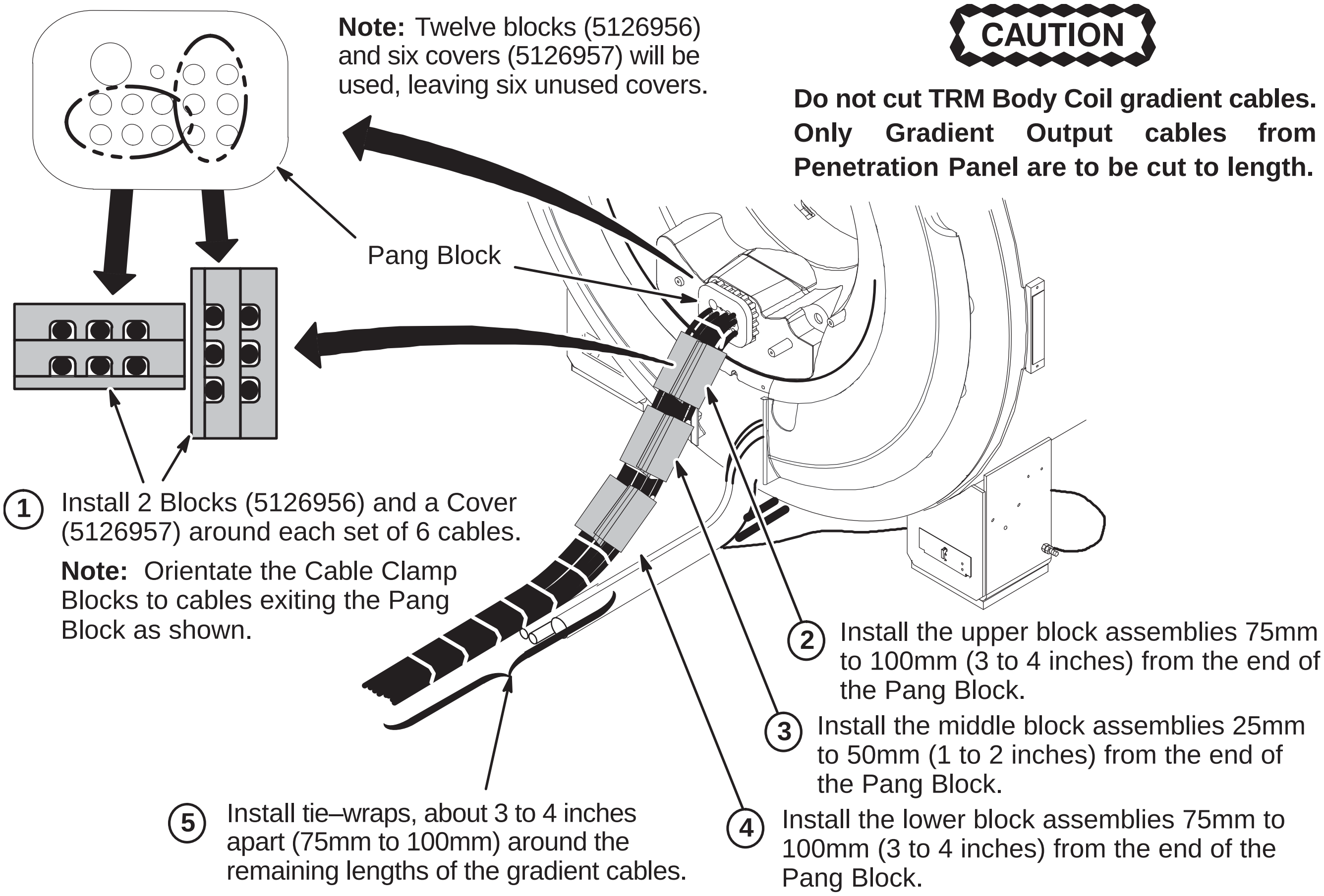
4–9–3 INSTALL GRADIENT CABLE RESTRAINT BLOCKS FOR TwinSpeed

Any existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.

4–9–3–A Attach Gradient Cable Blocks

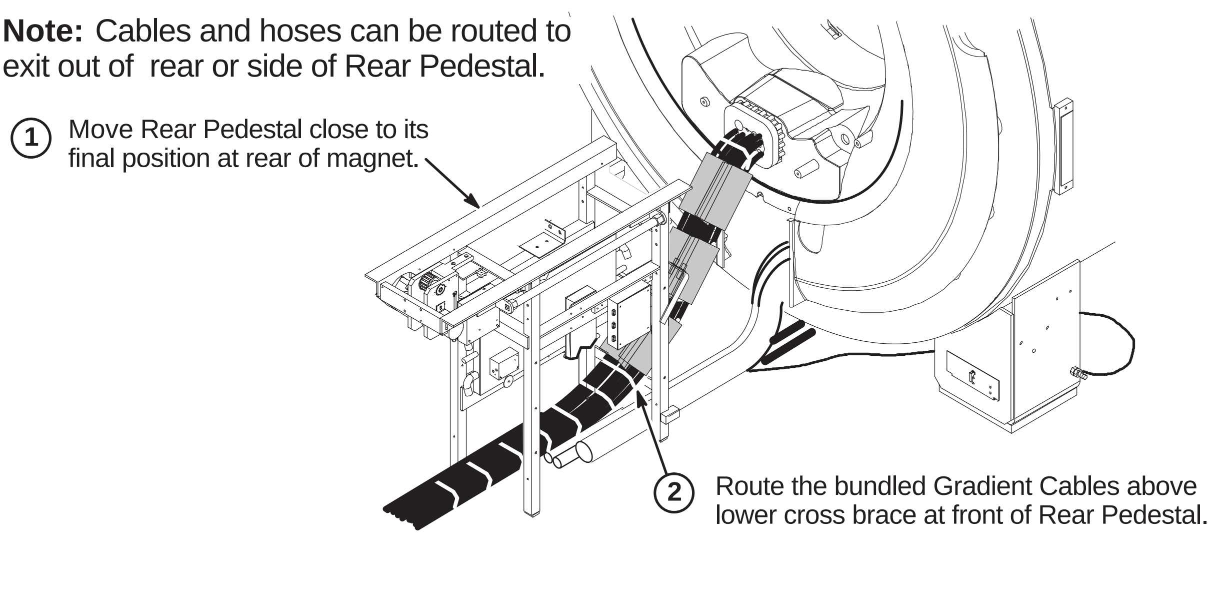
CAUTION

Do not cut TRM Body Coil gradient cables. Only Gradient Output cables from Penetration Panel are to be cut to length.



4–9–3–B Route Body Coil Gradient Cables Thru Rear Pedestal

Note: Cables and hoses can be routed to exit out of rear or side of Rear Pedestal.



4–9–4 GRADIENT CABLE INSTALLATION FOR NON–TwinSpeed

NOTE: Sites that already have a BRM–D Body Coil installed with leads crimped to gradient cables routed from Penetration Panel WILL NOT replace the existing cables with new gradient cables in Magnet Room. Sections **4–9–4–E** thru **4–9–4–H** will not be completed for these sites.

Three cables have been delivered to site. All have three lead wires with terminals at each end. The leads on this cable end are connected to the Equipment room and Magnet room sides of the Gradient Filter on the Penetration Panel. This procedure cuts the cables to length and connects them to the HFD cabinet and the cable leads from the Gradent Coil.

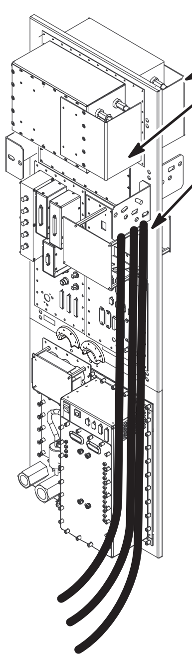
Verify that the length of supplied Run 762, 763, and 764 cable is sufficient to route from HFD Cabinet to Penetration Panel and, if required, Penetration Panel to Body Coil Gradient Cables at rear of magnet. **Each cable has identical terminations at each end for connection to both sides of the Penetration Panel Gradient Filter**.

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

4–9–4–A Connect Gradient Cables to Equipment Room Side of Penetration Panel



1 Remove gradient filter terminal covers.

2 Route one end of Runs 762, 763, and 764 to gradient filter on Penetration Panel.

4 Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft–lbs.

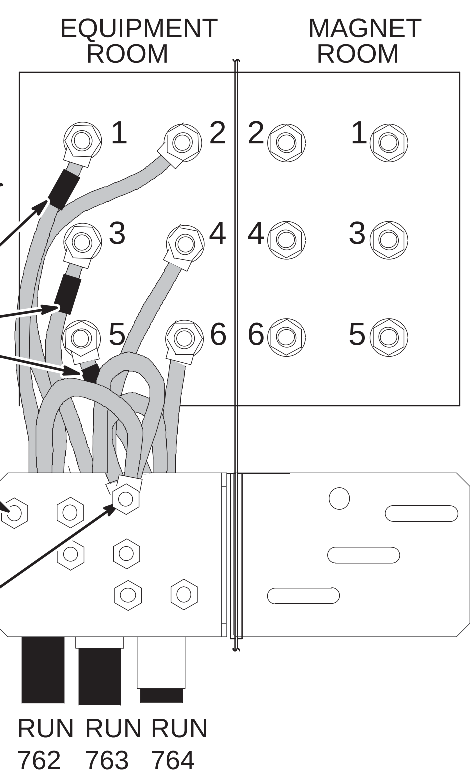
Note: Red tape indicates positive (+) wires #1, #3, and #5.

3 Secure each cable Run to strain–relief bracket with U–bolts.

Note: Tighten U–bolts to strain–relief cables only. Do not over tighten; cable may become damaged.

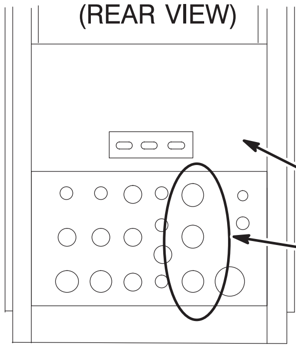
5 Secure GND lead wire on Ground Stud for each cable Run.

6 Replace gradient filter terminal covers.



4–9–4–B Route Runs 762, 763, and 764 to HFD Cabinet

HFD CABINET
(REAR VIEW)

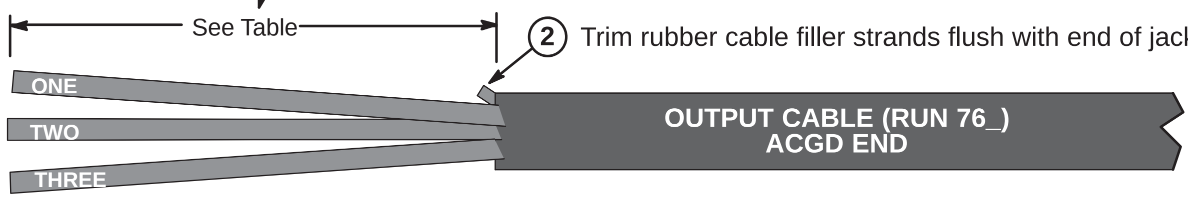


1. Unroll and route each cable from equipment room side of Penetration Panel to HFD Cabinet. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at HFD Cabinet.
2. Mark location for cutting.
3. Cut cables to length.
4. Remove Rear Cover Plate by removing seven screws that hold it in place.
5. Loosen screws on clamps for gradient output cables. After termination of cables in Procedure **4–9–4–C**, insert ends of cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

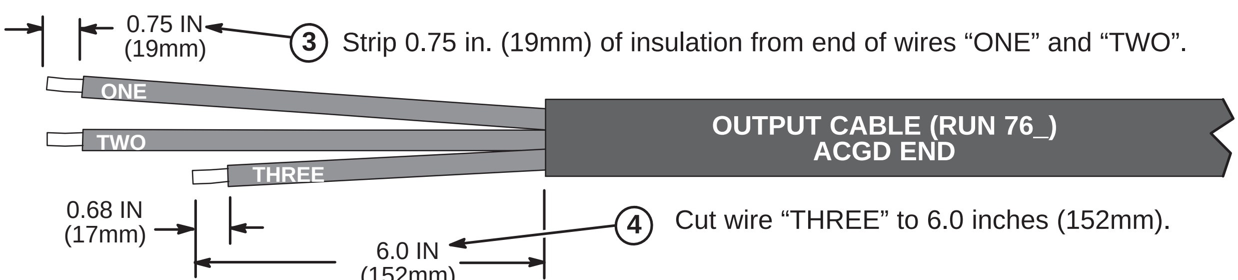
4–9–4–C Terminate Runs 762, 763, and 764 for HFD Cabinet

1. Refer to above table to determine length of jacket and foil to be removed from end of cable Runs 762 (X), 763 (Y), and 764 (Z).

	Run 762	**Run 763**	**Run 764**
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)



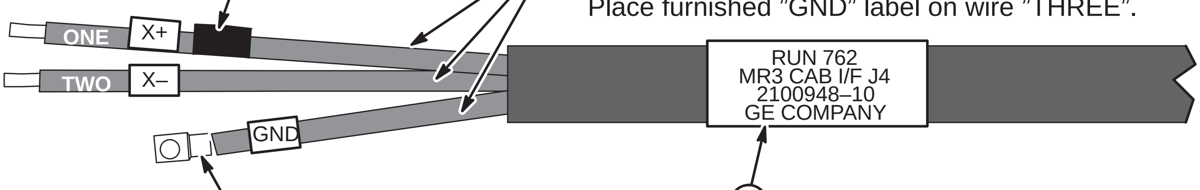
2. Trim rubber cable filler strands flush with end of jacket.



3. Strip 0.75 in. (19mm) of insulation from end of wires “ONE” and “TWO”.
4. Cut wire “THREE” to 6.0 inches (152mm).

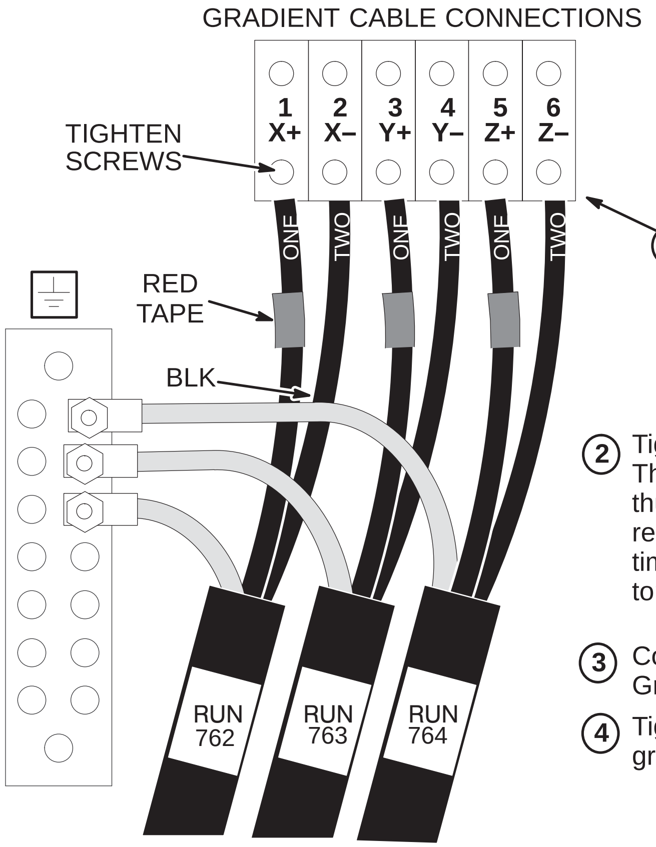
Note:
Steps below show completion of Run 762. Repeat same process for Runs 763 (Y+, Y–) and 764 (Z+, Z–).
Red tape, Y+ (RUN 763) **and Z+** (RUN 764) **labels are attached to wire ”ONE”.**

6. Wrap red tape around wire ”ONE”.
5. Place furnished ”X+” label on wire ”ONE”. Place furnished ”X–” label on wire ”TWO”. Place furnished ”GND” label on wire ”THREE”.



7. Crimp a furnished T&B #G2–14 terminal on GND wire end using a Ratcheting Crimper with #2 die.
8. Place furnished ”RUN 762” label on cable.

4–9–4–D Connect Runs 762, 763, and 764 at HFD Cabinet



1. Connect gradient output cables with red–taped (+) #1 wires to + output terminals and black (–) #2 wires to – output terminals.

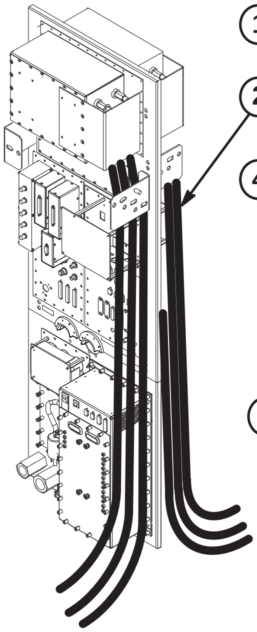
   NOTE: OVER TIGHTENING OF SCREWS IN STEP 2 CAN DAMAGE TERMINAL BLOCK.

2. Tighten screws to 52–65 inch–lbs (5,88–7,34 N–M). This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.

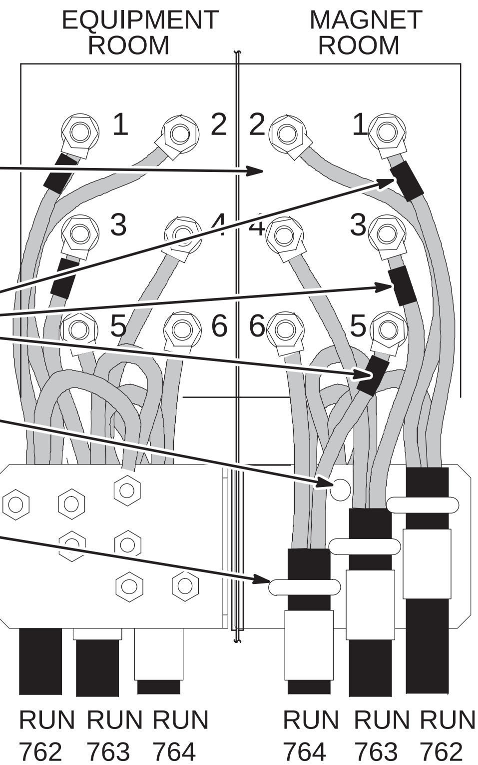
3. Connect the GND leads of each run to the Ground Bus Bar studs.

4. Tighten clamps on Interface Panel to secure gradient output cables.

4–9–4–E Connect Gradient Cables to Magnet Room Side of Penetration Panel

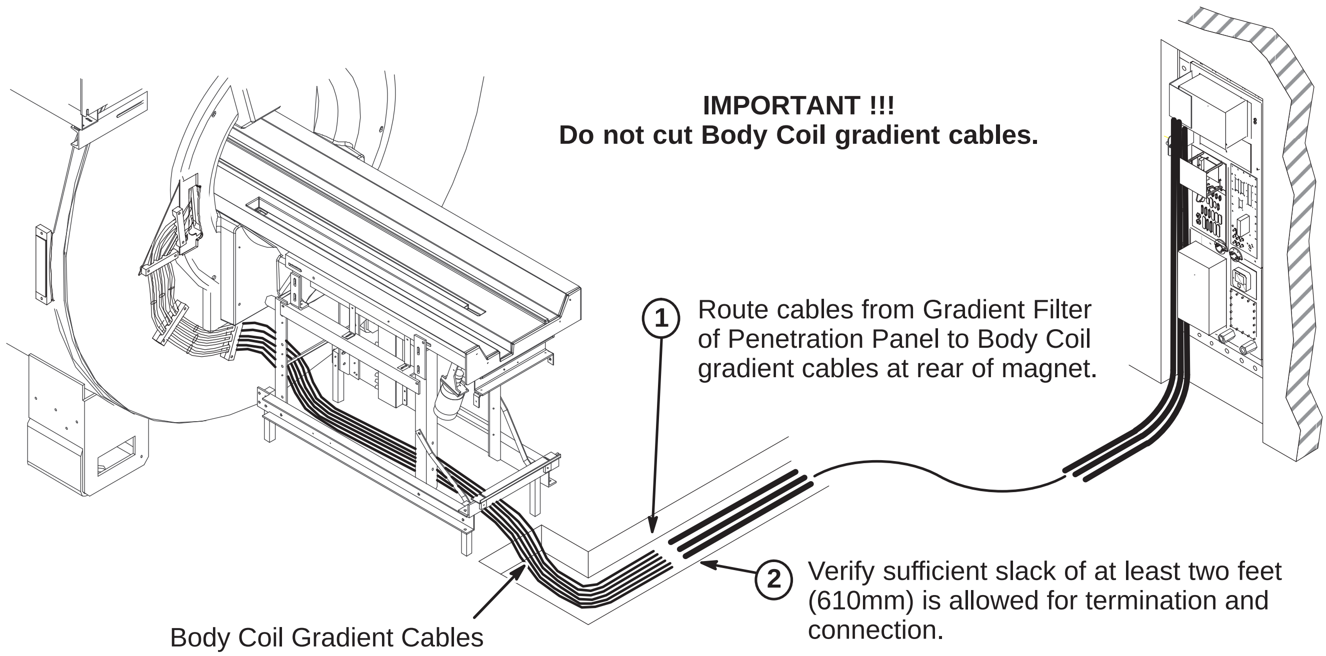


1. Move remainder of gradient cables to Magnet Room.
2. Route Run 762, 763, and 764 to gradient filter on Penetration Panel.
3. Secure each cable Run to strain–relief bracket with U–bolts.
   Note: Tighten U–bolts to strain–relief cables only. Do not over tighten; cable may become damaged.
4. Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft–lbs.
   Note: Red tape indicates positive (+) wires #1, #3, and #5.
5. Secure GND lead wire on Ground Stud for each cable Run.
6. Replace gradient filter terminal covers.



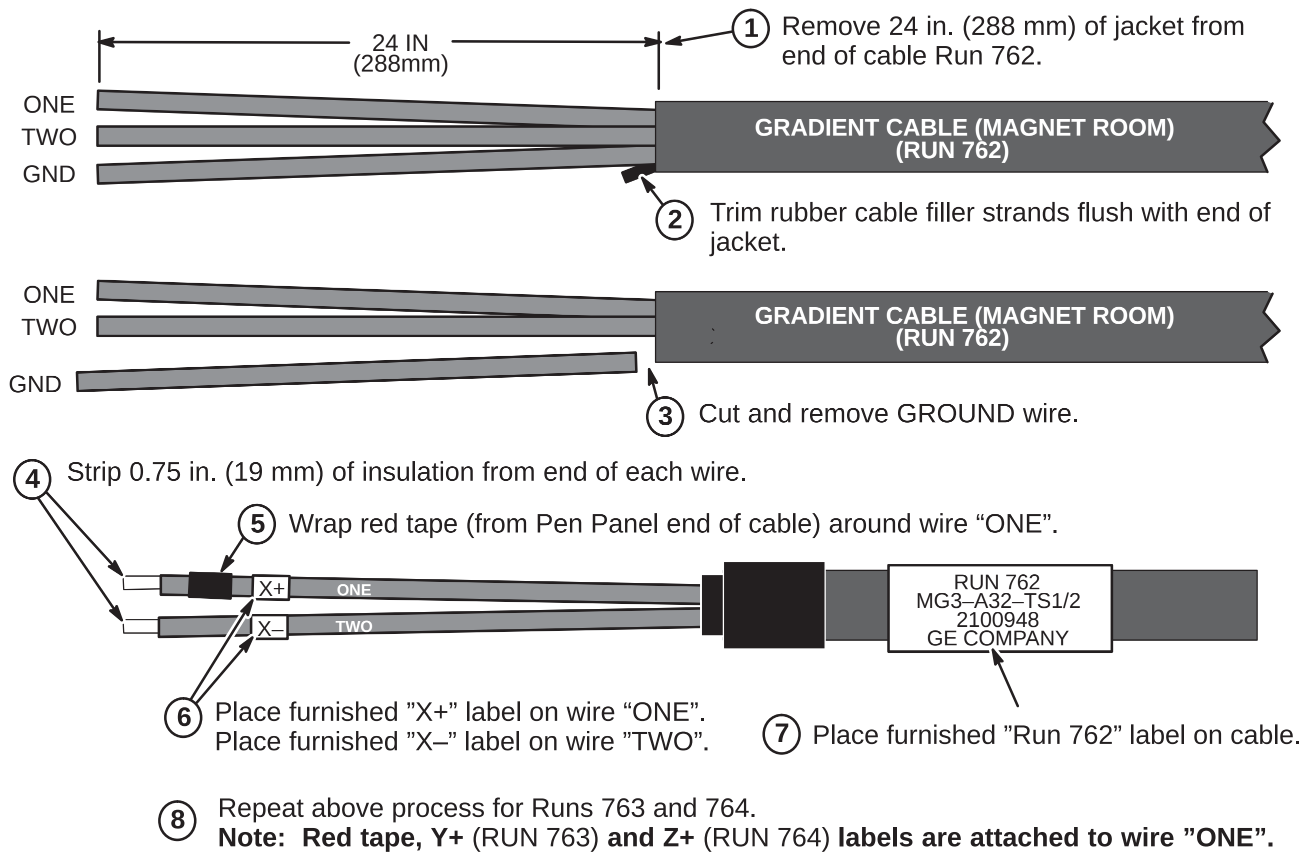
4–9–4–F Route Runs 762, 763, and 764 to Rear Pedestal Area

IMPORTANT !!!
Do not cut Body Coil gradient cables.

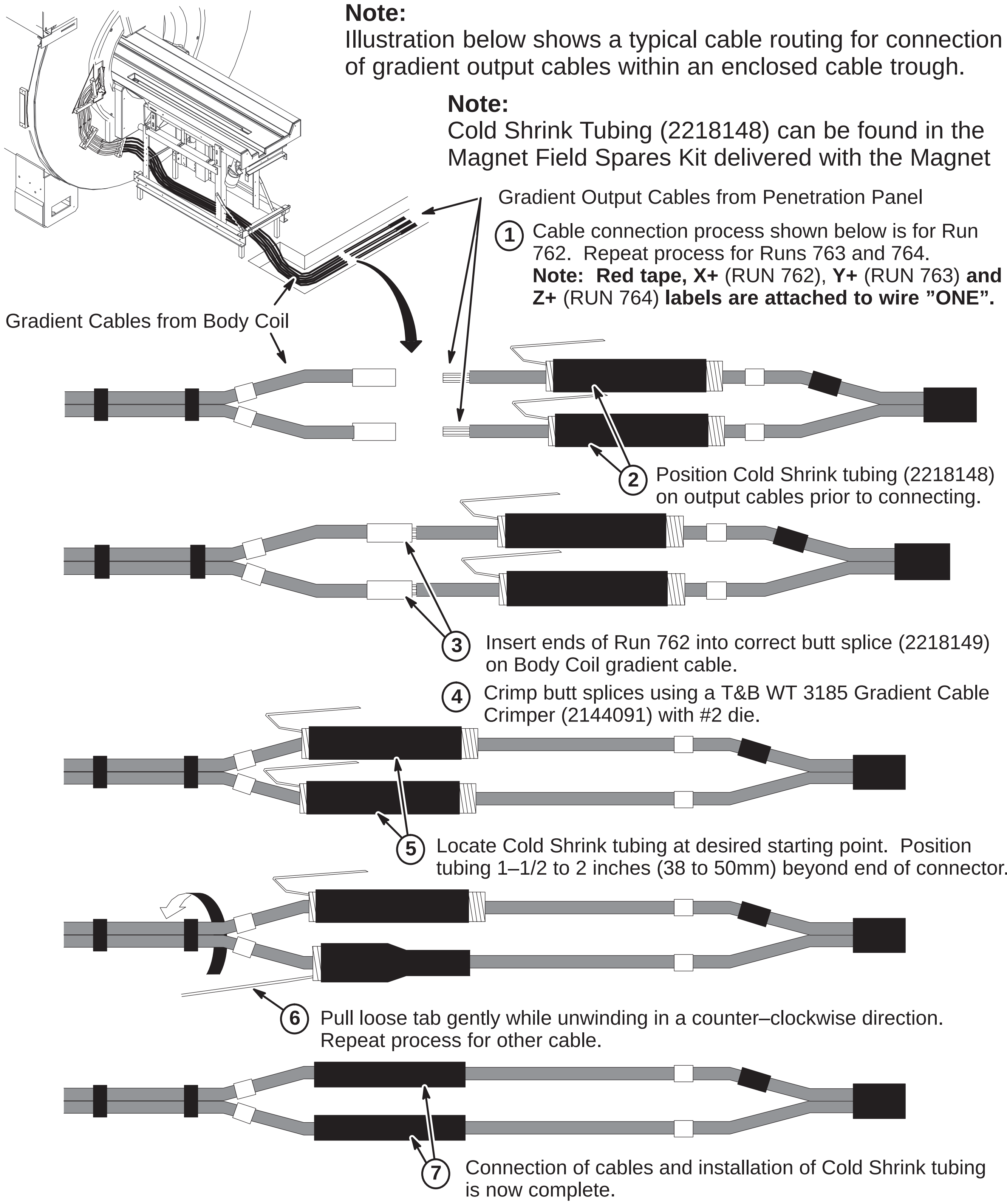


1. Route cables from Gradient Filter of Penetration Panel to Body Coil gradient cables at rear of magnet.
2. Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection.
3. Mark location for cutting and cut cables to length.
4. Label cut ends with applicable run number.

4–9–4–G Terminate Runs 762, 763, and 764 for Magnet Room



4–9–4–H Connection of Body Coil Gradient Cables to runs 762, 763, and 764



4–9–5 GRADIENT CABLE INSTALLATION FOR TwinSpeed

NOTE: Sites that already have a TRM Body Coil installed with leads crimped to gradient cables routed from Penetration Panel WILL NOT replace the existing cables with new gradient cables in Magnet Room. Sections **4–9–5–L** thru **4–9–5–Q** will not be completed for these sites.

Also, the gradient cables in the Equipment Room do not have to be replaced. Existing gradient cables can be reused.

The following TwinSpeed gradient cables installation procedures are provided for those sites that have to install gradient cables.

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

Nine cables have been delivered to site. Nine of the labeled pre–terminated leads will be connected to the Twin Gradient Filter on the Penetration Panel. The other three labeled pre–terminated ends will be connected to the TAC Cabinet. The other ends of all the cables will be cut to length, labeled, and terminated.

- Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. **Do not cut Body Coil gradient cables.**
- Locate furnished bag of parts that contain the terminals and labels required to be attached to the cables.

The nine cables with supplied labels are as follows:

RUN	LABEL	SUPPLIED LABELS
976/979	PP1–A7–X	(1) RUN 976, (1) RUN 979, (2) X+, (1) X–, (1) GND, (1) X_{ZM-}, (1) X_{WB-}
977/980	PP1–A7–Y	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y–, (1) GND, (1) Y_{ZM-} (1) Y_{WB-}
978/981	PP1–A7–Z	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z–, (1) GND, (1) Z_{ZM-} (1) Z_{WB-}
982	PP1–A7–Z_{ZM}	(1) RUN 982, (1) Z_{ZM-} (1) Z_{ZM+}
983	PP1–A7–Y_{ZM}	(1) RUN 983, (1) Y_{ZM-} (1) Y_{ZM+}
984	PP1–A7–X_{ZM}	(1) RUN 984, (1) X_{ZM-} (1) X_{ZM+}
985	PP1–A7–Z_{WB}	(1) RUN 985, (1) Z_{WB-} (1) Z_{WB+}
986	PP1–A7–Y_{WB}	(1) RUN 986, (1) Y_{WB-} (1) Y_{WB+}
987	PP1–A7–X_{WB}	(1) RUN 987, (1) X_{WB-} (1) X_{WB+}

4–9–5–A General Steps for Cutting Cables to Length and Routing in Equipment Room

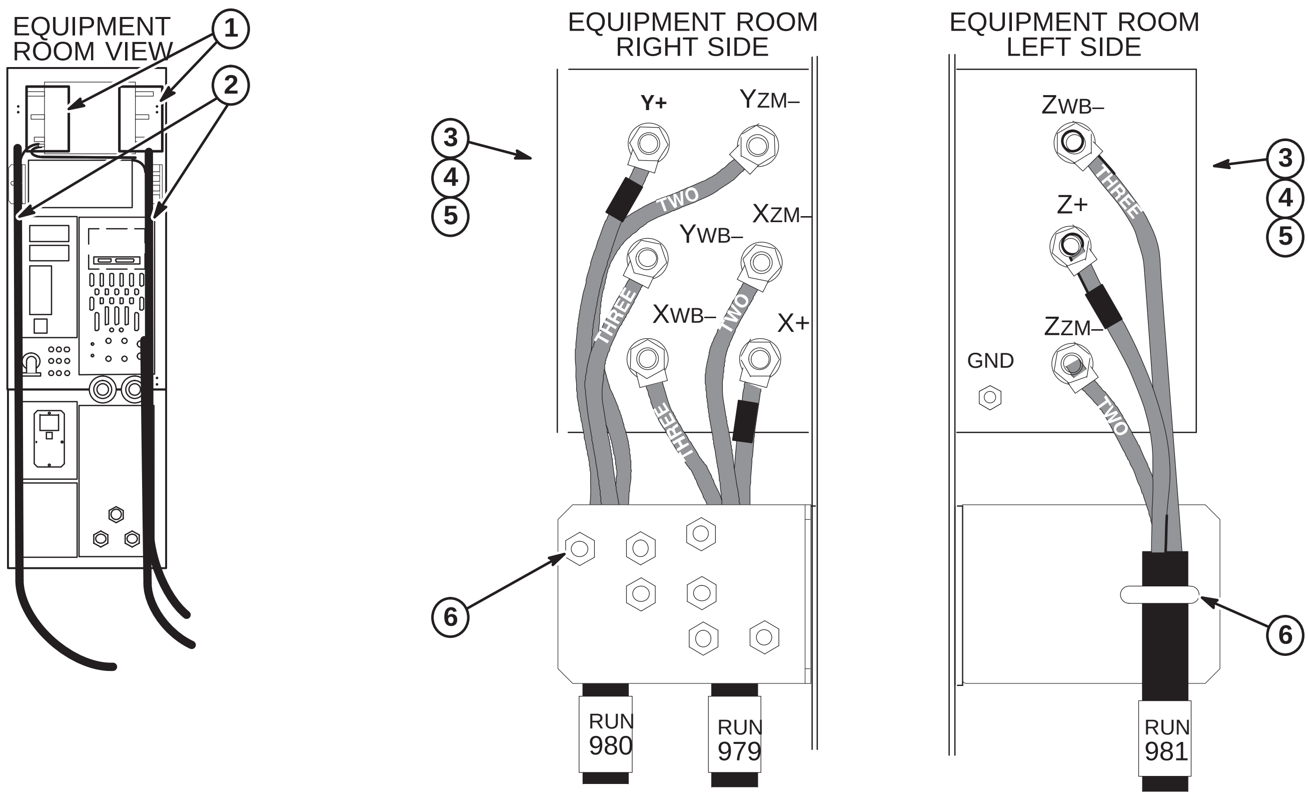
1. Move Run 979, 980, and 981 cables to equipment room. The end of cable with terminals will be connected to Penetration Panel.
2. Route cables from Gradient Filter of Penetration Panel to right side of interface panel at rear of Twin Accessory Cabinet (TAC).
3. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at TAC Cabinet. Mark location for cutting. Cut cables to length and label cut ends with applicable run number.
4. Refer to Procedure **4–9–5–E** to terminate and label TAC end cable Runs 979, 980, and 981
5. Refer to Procedure **4–9–5–F** for connection of Runs 979, 980, and 981 to TAC cabinet.
6. Route remainder of cables from rear of TAC to rear of HFD cabinet. Make sure enough slack is provided for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
7. Refer to Procedure **4–9–5–H** for connection of Runs 976, 977, and 978 to TAC cabinet.
8. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at HFD Cabinet. Mark location for cutting. Cut cables to length and label cut ends with applicable run number.
9. Refer to Procedure **4–9–4–H** for connection of Runs 976, 977, and 978 to HFD cabinet.

4–9–5–B Connect Runs 979, 980, and 981 to Penetration Panel

1. Remove gradient filter terminal covers.
2. Route Run 981 (Z) to left side and Runs 979 (X) and 980 (Y) to right side of gradient filter on Penetration Panel.

Note: The three X and three Y terminals are routed to the right side and three Z terminals to the left side.

3. Connect +, ZM– (cable marked as TWO), & WB– (cable marked as THREE) wires to gradient filter terminals as marked on each side of gadient filter.
   Note: Red tape indicates positive (+) wires.
4. Align cables to obtain maximum spacing from other cables and conductive metal objects.
   Note: When tightening nuts, it is important to hold screw body.
5. Tighten nuts to a torque of 35 ft–lbs (67–74Nm) or after tightening, verify that nothing is loose.
   Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.
6. Route and secure each cable Run to strain–relief bracket with U–bolts.

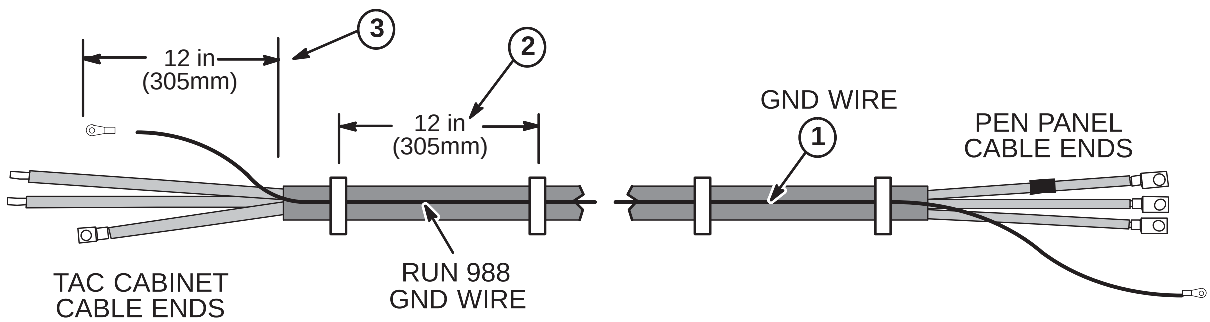


4–9–5–C Attach Ground Cables to Runs 979, 980, and 981

For each gradient cable routed between the Penetration Panel and TAC Cabinet, a ground wire, Run 988 (2283297), has to be ty–wrapped to each cable.

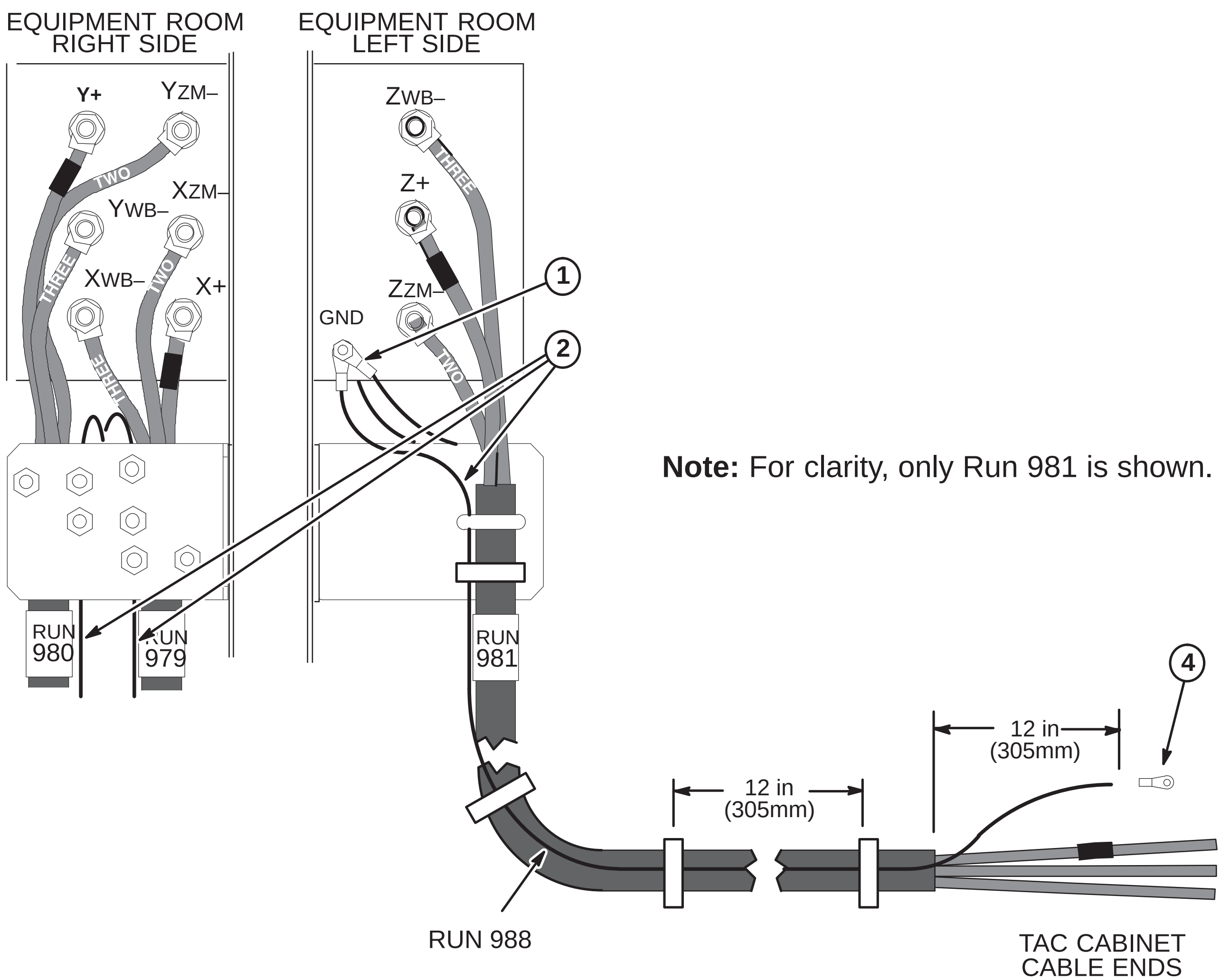
1. One Ground Wire will be routed along each of the Gradient Cables, Runs 979, 980, and 981.
2. Each Run 988 must be ty–wrapped about every 12 inches along entire length of each gradient cable.
   Note: Run 988 must be routed parallel to the gradient cable. It **MUST NOT** be routed spirally around the gradient cable.
3. End of Run 988 at TAC cabinet should extend about 12 inches (305mm) past end of outer jacket after trimming.
   Note: Refer to Procedure **4–9–5–E** for length of gradient cable jacket material to be removed.

Excess ground cable can not be coiled. It must be cut to length and reterminated.



4–9–5–D Connect Run 988 Cables Ground Wires to Gradient Filter

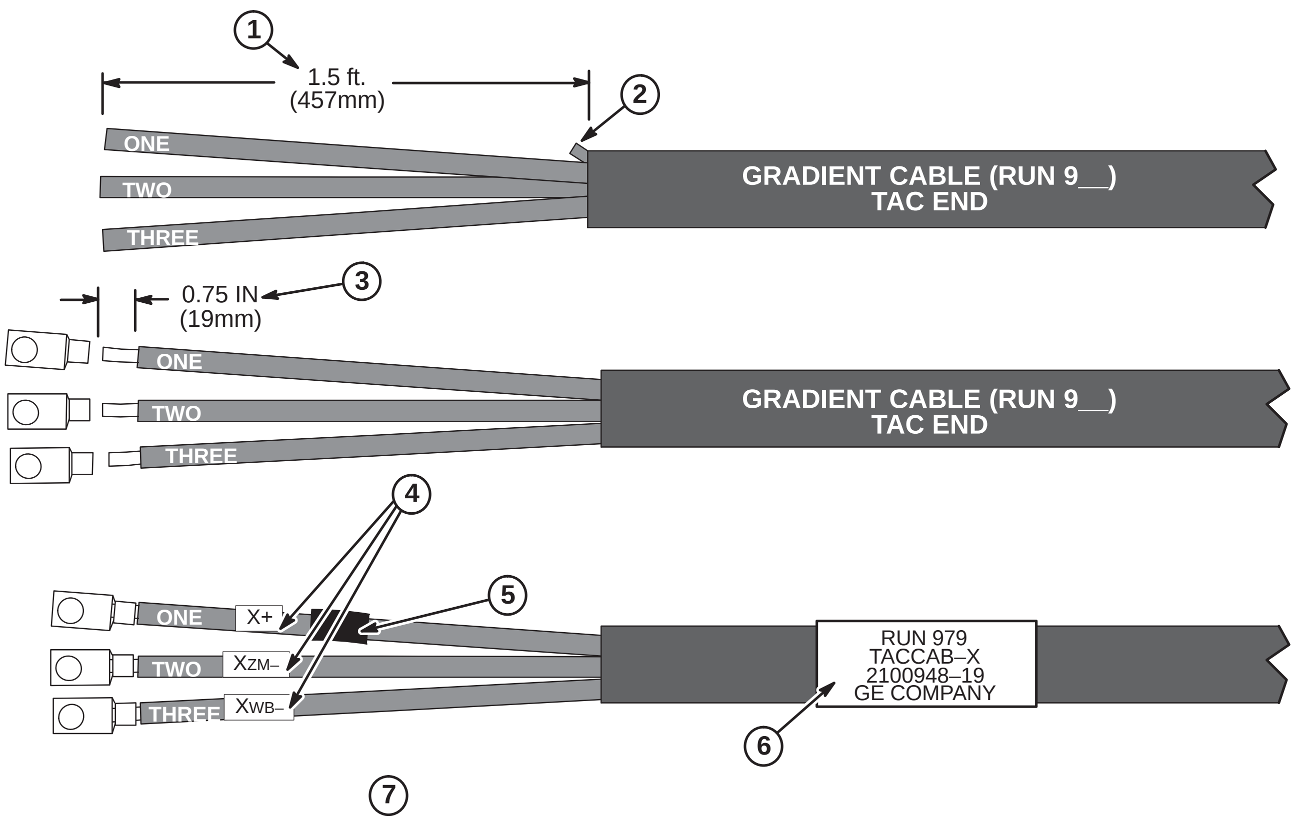
1. Connect the three Run 988 Gound Wires to GND on left side of Gradient Filter.
2. Each Run 988 must be ty–wrapped about every 12 inches along entire length of each gradient cable.
   Note: Run 988 must be routed parallel to the gradient cable. It **MUST NOT** be routed spirally around the gradient cable.
3. Re–install gradient filter terminal covers.
4. Cut each Run 988 to length, reterminate with 3/8 ring terminal for connection to GND stud on TAC Cabinet.



4–9–5–E Terminate Runs 979, 980, and 981 for TAC Cabinet Connection

The following steps are required to terminate the end of the cable that was cut in Procedure **4–9–4–A**. The other end of the cable is connected to the Penetration Panel.

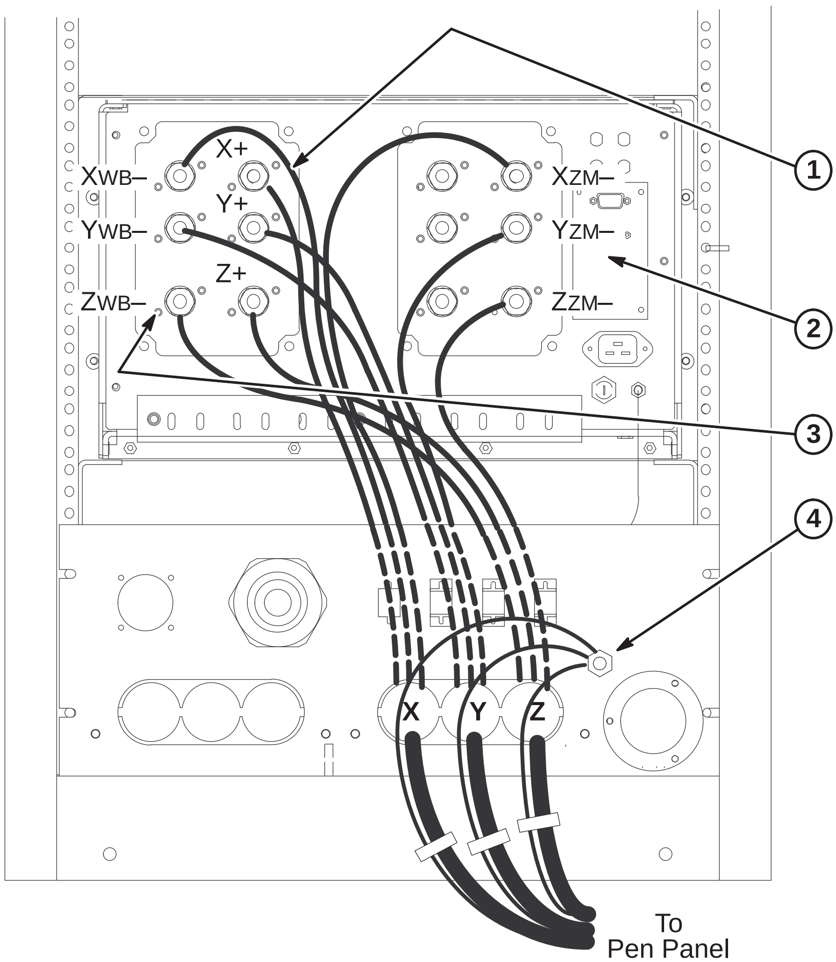
1. Remove 18 inches of jacket from end of cable Runs 979 (X), 980 (Y), and 981 (Z).
2. Trim rubber cable filler strands flush with end of jacket.
3. Strip 0.75 in. (19mm) of insulation from end of wires “ONE”, “TWO” and “THREE” and crimp a furnished T & B 1/2” (G2–3–12) terminal to each using Ratcheting Crimper with #2 die.
4. Place furnished ”X+” label on wire ”ONE”.
   Place furnished ”XZM–” label on wire ”TWO”.
   Place furnished ”XWB–” label on wire ”THREE”.
5. Wrap red tape around wire ”ONE”.
6. Place furnished ”RUN 979” label on cable.
7. Steps above show completion of Run 979. Repeat same process for Runs 980 (Y+, YZM–, YWB–) and 981 (Z+, ZZM–, ZWB–).



4–9–5–F Connect Runs 979, 980, and 981 to TAC Cabinet

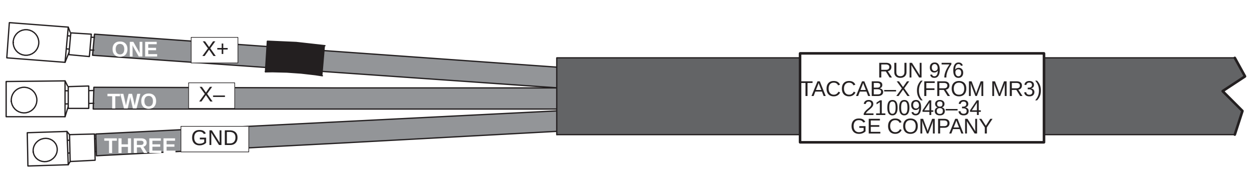
To make installation easier, route and connect Run 981 (Z terminals) first, then Run 980 (Y), and last Run 979 (X).

1. Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
2. Connect XZM–, YZM–, and ZZM– cables as marked.
3. Connect XWB–, YWB–, and ZWB– cables as marked.
4. Route and connect the three Run 988 Ground wires to GND stud on interface panel.



4–9–5–G Pre–terminated cable ends for Runs 976, 977, and 978

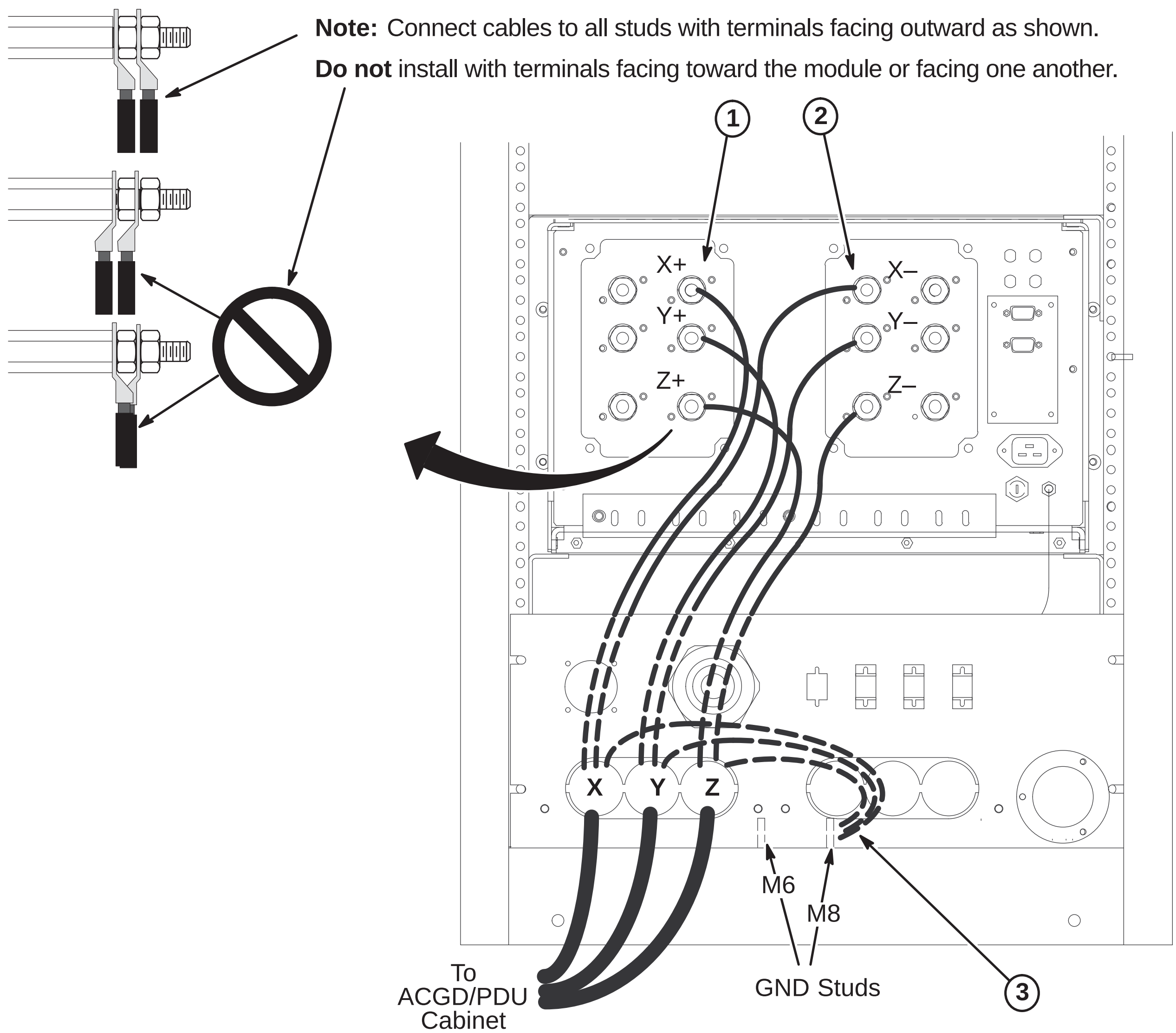
Cables delivered to site will be pre–terminated for the TAC Cabinet connection of Runs 976, 977, and 978. Refer to Procedure **4–9–4–A** for cutting cables to length for connection to HFD Cabinet. A typical pre–terminated end for each cable is shown below.



4–9–5–H Connect Runs 976, 977, and 978 to TAC Cabinet

To make installation easier, route and connect Run 978 (Z terminals) first, then Run 977 (Y), and last Run 976 (X).

1. Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
2. Connect X–, Y–, and Z– cables as marked.
3. Connect Ground leads from each cable to M8 GND stud located at bottom of cabinet.



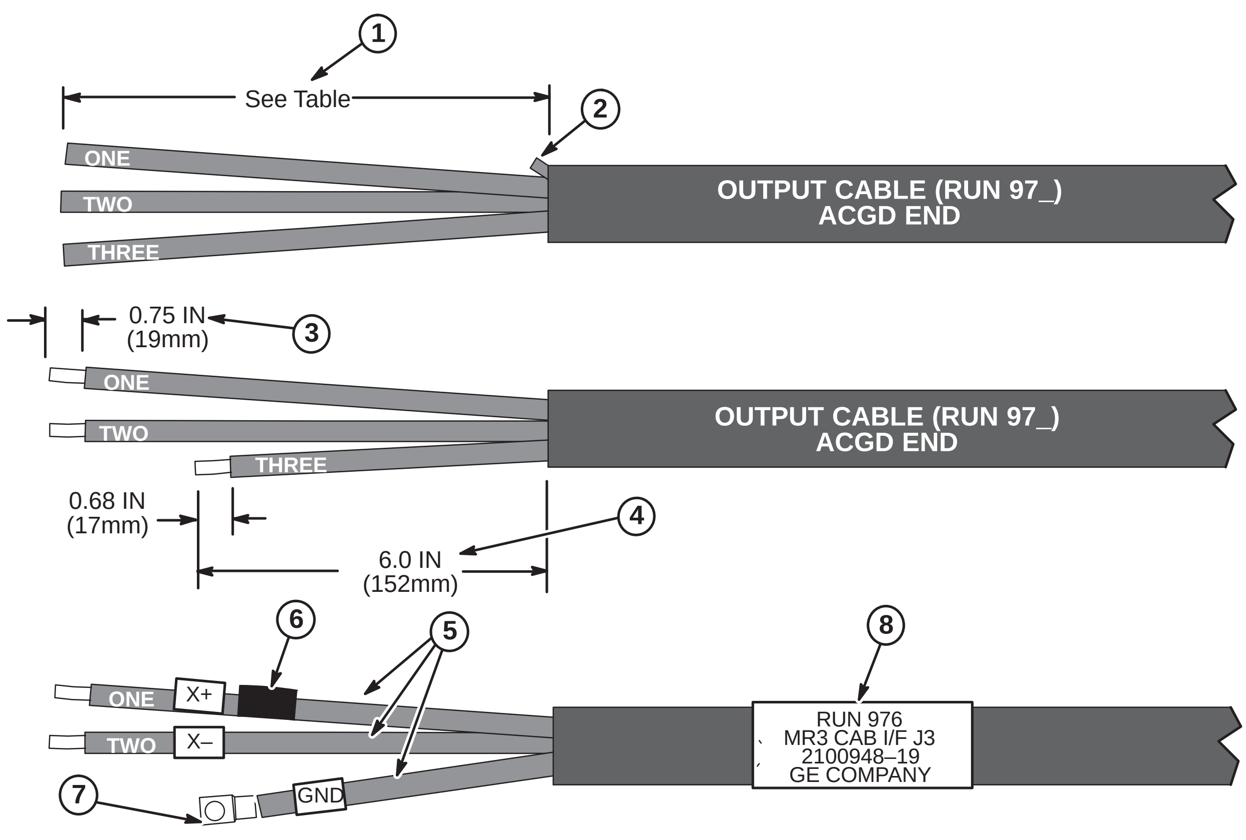
4–9–5–I Terminate Runs 976, 977, and 978 for Connection to HFD Cabinet

The following steps are required to terminate the end of the cable that is to be connected to the HFD Cabinet. The other end of the cable is connected to the TAC Cabinet.

	Run 976	Run 977	Run 978
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)

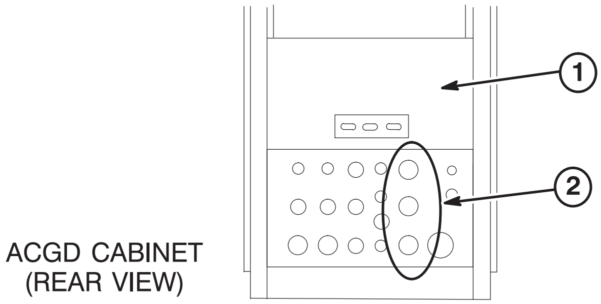
1. Refer to above table to determine length of jacket and foil to be removed from end of cable Runs 976 (X), 977 (Y), and 978 (Z).
2. Trim rubber cable filler strands flush with end of jacket.
3. Strip 0.75 in. (19mm) of insulation from end of wires “ONE” and “TWO”.
4. Cut wire “THREE” to 6.0 inches (152mm).
5. Place furnished ”X+” label on wire ”ONE”.
   Place furnished ”X–” label on wire ”TWO”.
   Place furnished ”GND” label on wire ”THREE”.
6. Wrap red tape around wire ”ONE”.
7. Crimp a furnished T&B #G2–38 terminal on GND wire end using a Ratcheting Crimper with #2 die.
8. Place furnished ”RUN 976” label on cable.

Steps in illustration below show completion of Run 976. Repeat same process for Runs 977 (Y+, Y–) and 978 (Z+, Z–). **Red tape, Y+** (RUN 977) **and Z+** (RUN 978) **labels are attached to wire ”ONE”.**



4–9–5–J Gradient Output Cable Interface Panel Access at HFD Cabinet

1. Remove Rear Cover Plate by removing seven screws that hold it in place.
2. Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 976 into J3, Run 977 into J4, and Run 978 into J5)

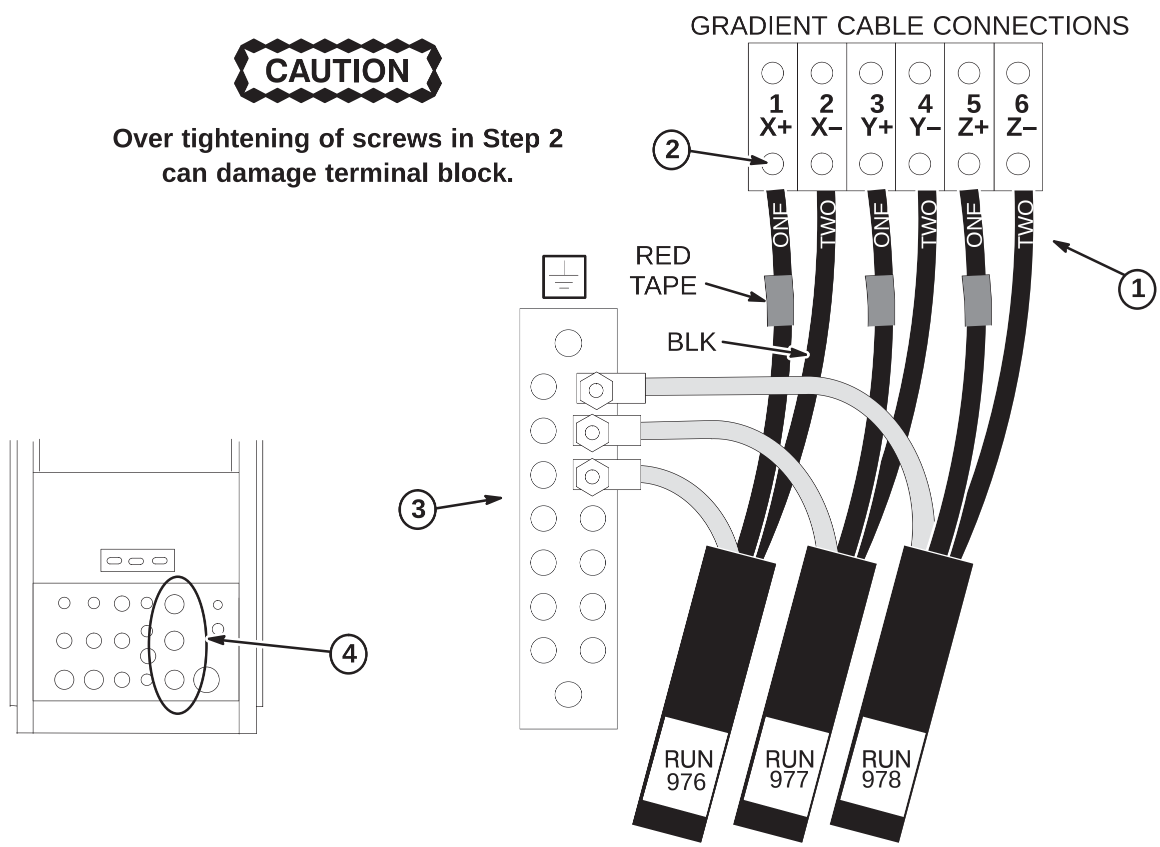


4–9–5–K Gradient Output Cable Routing and Connection in HFD Cabinet

1. Connect gradient output cables with red–taped (+) #1 wires to + output terminals and black (–) #2 wires to – output terminals.
2. Tighten six screws to <u>52–65 inch–lbs (5,88–7,34 N–M).</u> This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
3. Connect the GND leads of each run to the Ground Bus Bar studs.
4. Tighten clamps on Interface Panel to secure gradient output cables.

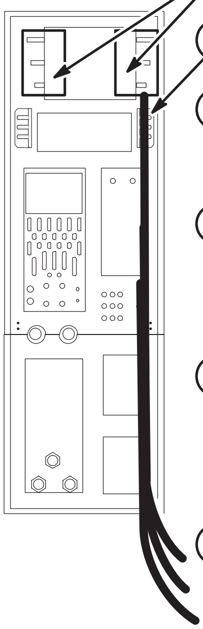
CAUTION

Over tightening of screws in Step 2 can damage terminal block.



4–9–5–L Connect ZM (Zoom) Cables to Penetration Panel

MAGNET ROOM VIEW



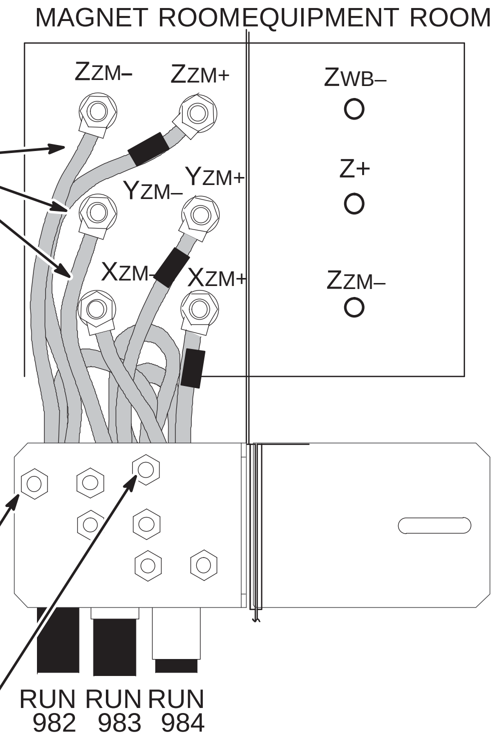
1. Remove gradient filter terminal covers.
2. Route Runs 982, 983, and 984 to right side of gradient filter on Penetration Panel.
3. Connect six ZM– and ZM+ wires to gradient filter terminals as marked.

   Note: Red tape indicates positive (+) wires.
4. Align cables to obtain maximum spacing from other cables and conductive metal objects.

   Note: When tightening nuts, it is important to hold screw body.
5. Tighten nuts to a torque of 35 ft–lbs (67–74Nm) or after tightening, verify that nothing is loose.

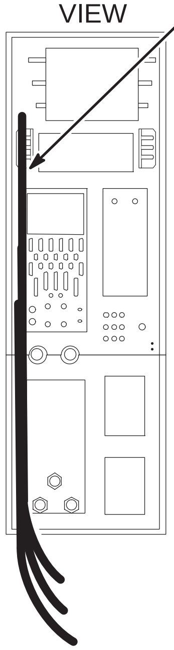
   Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.
6. Route and secure each cable Run to strain–relief bracket with U–bolts.

   Note: Tighten U–bolts to strain–relief cables only. Do not over tighten; cables may become damaged.
7. Secure GND lead wire on Ground Stud for each cable Run.



4–9–5–M Connect WB (Whole Body) Cables to Penetration Panel

MAGNET ROOM VIEW



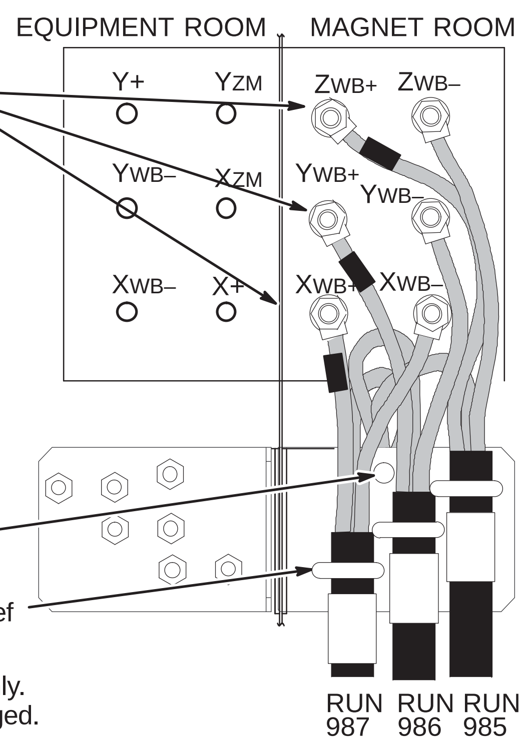
1. Route Runs 985, 986, and 987 to left side of gradient filter on Penetration Panel.
2. Connect six WB– and WB+ wires to gradient filter terminals as marked.

   Note: Red tape indicates positive (+) wires.
3. Align cables to obtain maximum spacing from other cables and conductive metal objects.

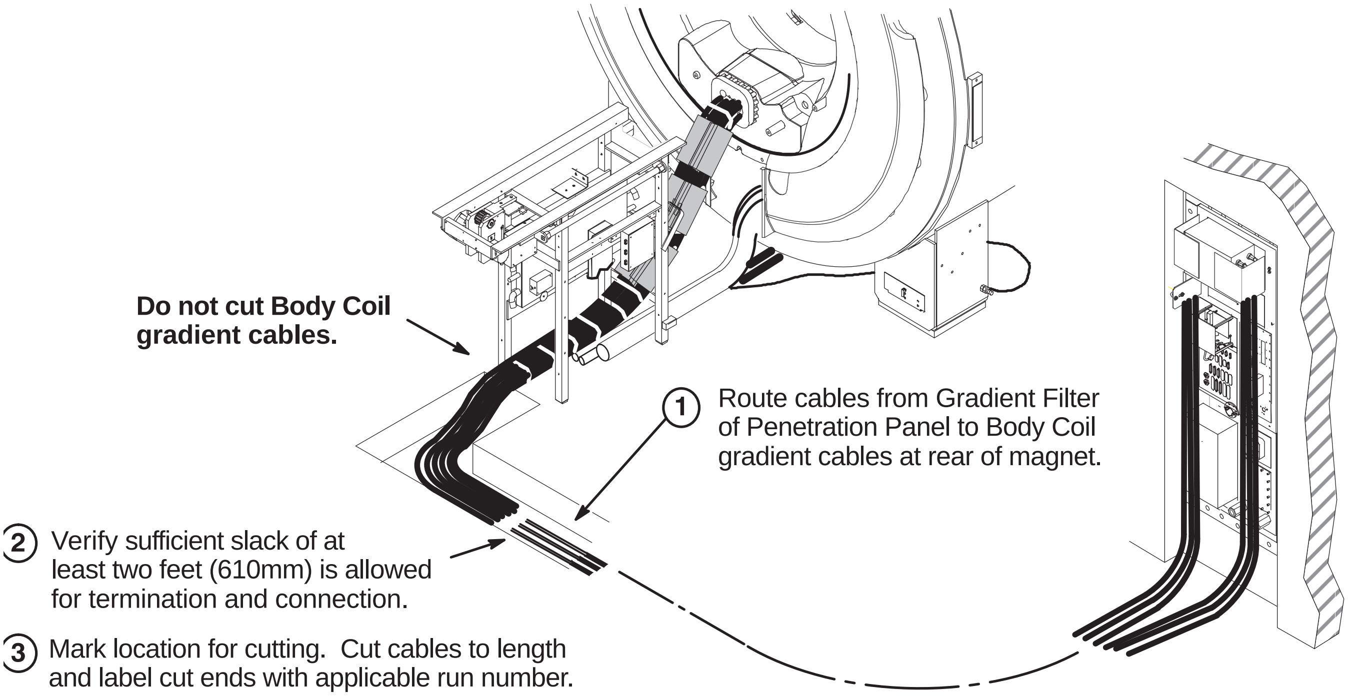
   Note: When tightening nuts, it is important to hold screw body.
4. Tighten nuts to a torque of 35 ft–lbs (67–74Nm) or after tightening, verify that nothing is loose.

   Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.
5. Secure GND lead wire on Ground Stud for each cable Run.
6. Route and secure each cable Run to strain–relief bracket with U–bolts.

   Note: Tighten U–bolts to strain–relief cables only. Do not over tighten; cables may become damaged.
7. Replace gradient filter terminal covers.

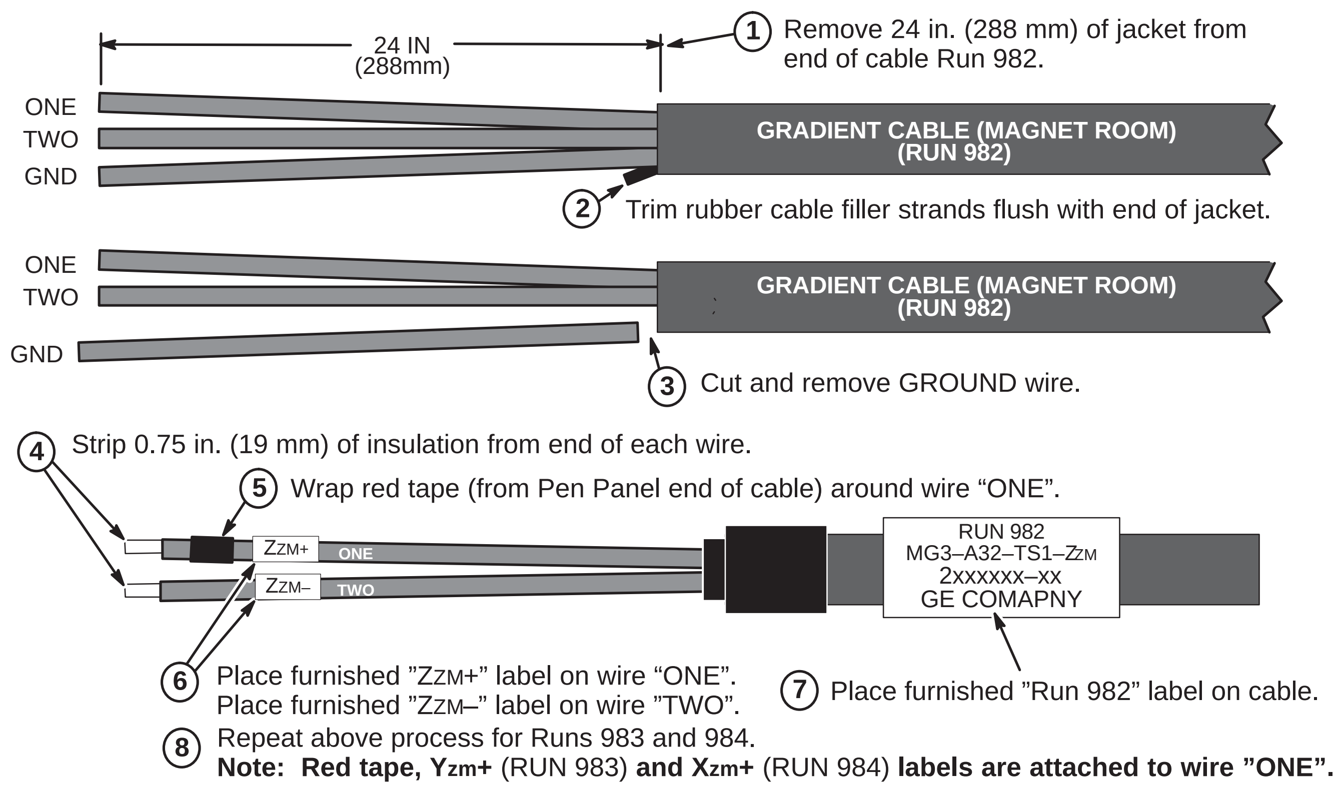


4–9–5–N Route Gradient Cables to Rear of Magnet



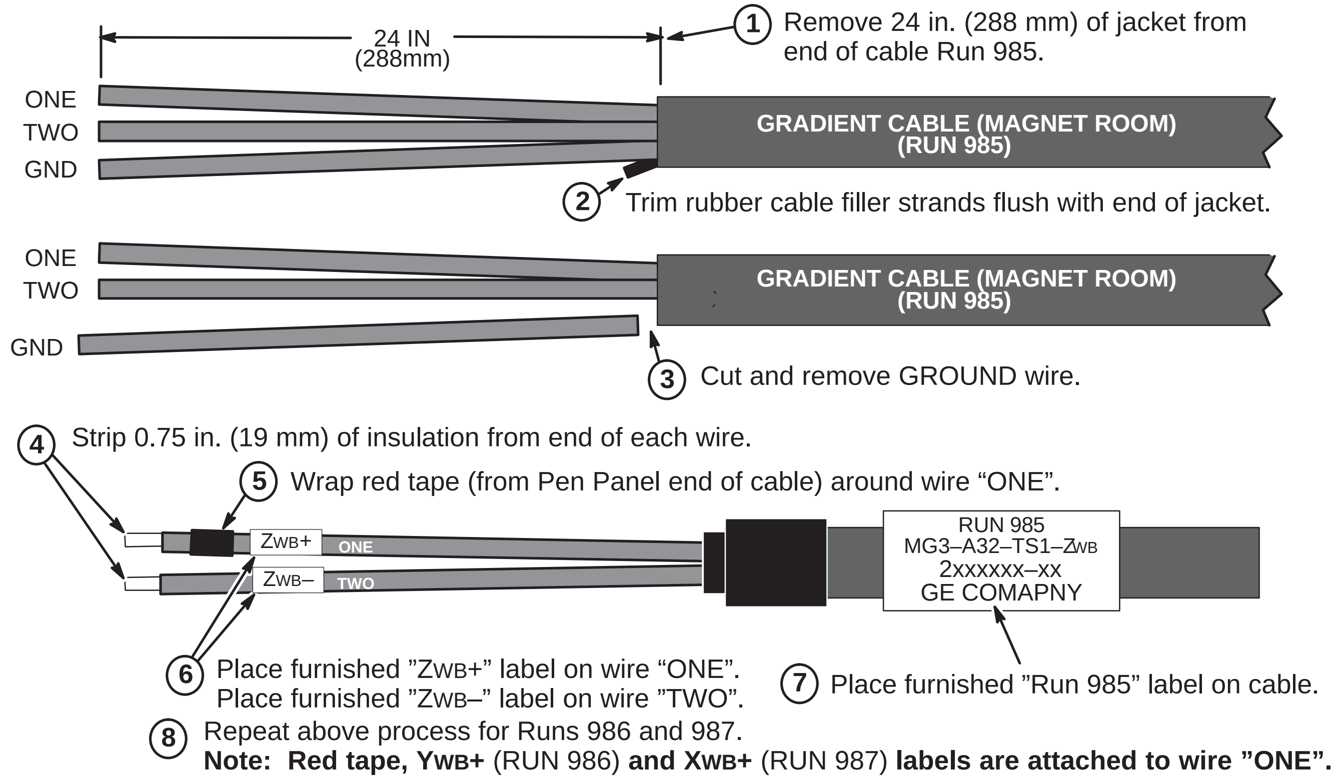
4–9–5–O Terminate Runs 982, 983, And 984 for Magnet Room

Note: The following steps are required to terminate the end of the cable that was cut in Section **4–9–5–N**. The other end of the cable is connected to the Penetration Panel.



4–9–5–P Terminate Runs 985, 986, And 987 for Magnet Room

Note: The following steps are required to terminate the end of the cable that was cut in Section **4–9–5–N**. The other end of the cable is connected to the Penetration Panel.



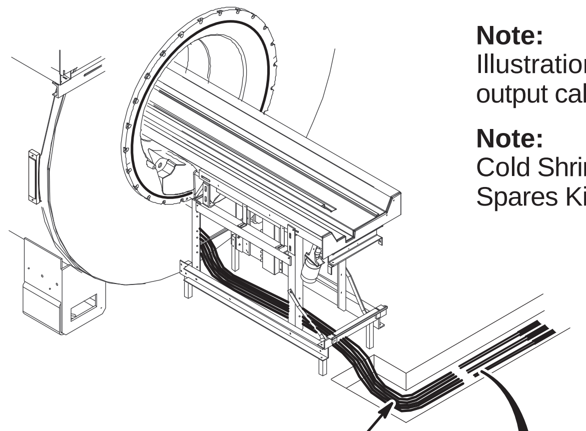
WARNING!

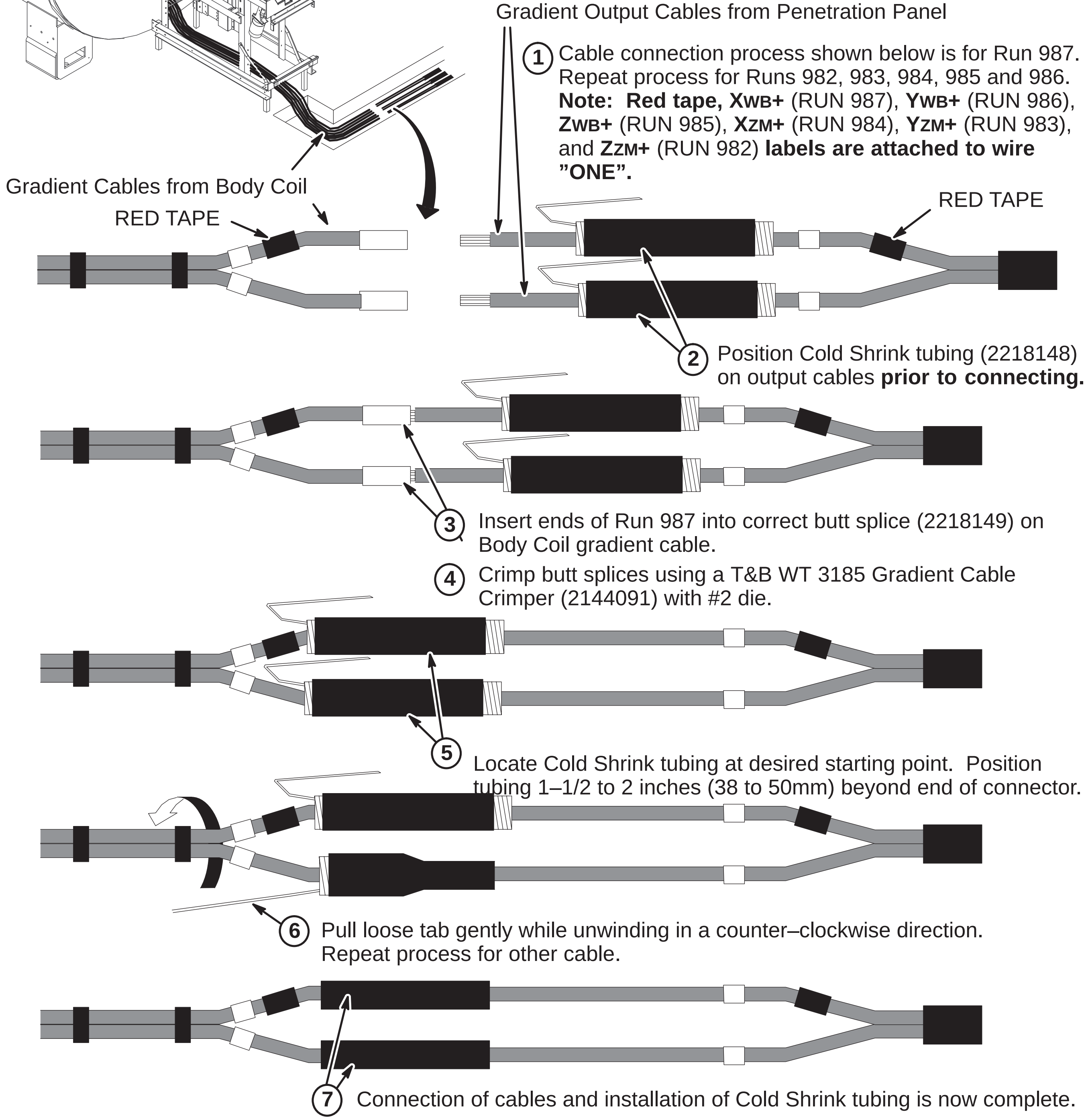
THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET WHICH COULD CAUSE INJURY OR DAMAGE TO THE EQUIPMENT.

4–9–5–Q Connect Magnet Room Gradient Cables

Note:
Illustration below shows a typical cable routing for connection of gradient output cables within an enclosed cable trough.

Note:
Cold Shrink Tubing (2218148) can be found in the Magnet Field Spares Kit delivered with the Magnet





4–9–6 TIMES MICROWAVE RF CABLES INSTALLATION

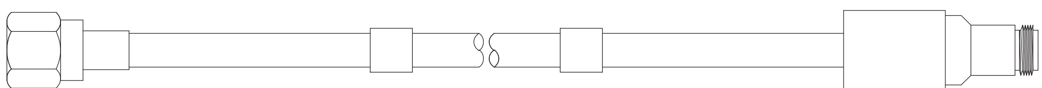
4–9–6–A Cable Information

Five cables, three extension cables, and other associated parts to be installed are provided for this procedure. Also provided are two Times Microwave stripping tools for LMR 1200 (2384858) and the LMR 600 (2352193), and the Times Microwave cutting tool (5111565).

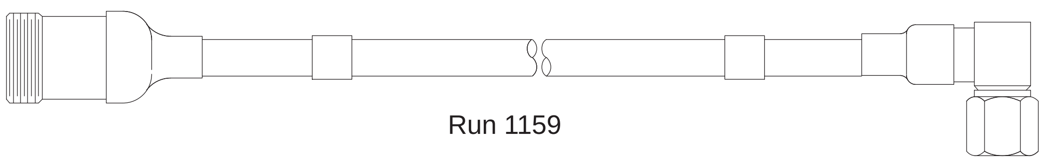
CAUTION

Do not subject LMR–600FR cable (Head) to a bend radius less than 3 inches (75mm). Do not subject LMR–1200FR cable (Body) to a bend radius less than 6.5 inches (165mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

Two extension cables, 5116213–2, Run 1249 (Head) and 5121269, Run 1159 (Body) are connected to the RF Amplifier module in the RF/System Cabinet. The other extension cable, Run 1250, 5121245, is connected to the Body Hybrid Splitter in the Rear Pedestal.

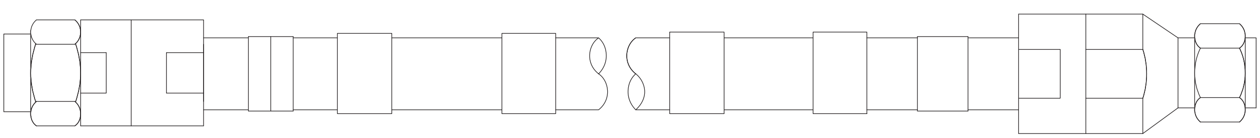


Run 1249 and Run1250



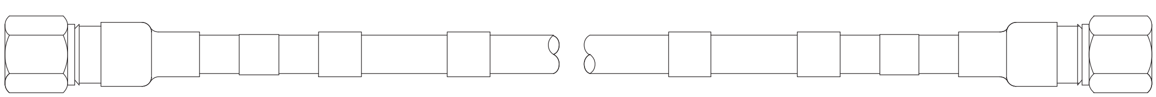
Run 1159

Cable 2316584–2 (Body, Runs 235 & 257), will be cut to length to make two cables and terminated for routing in the equipment and scan rooms.



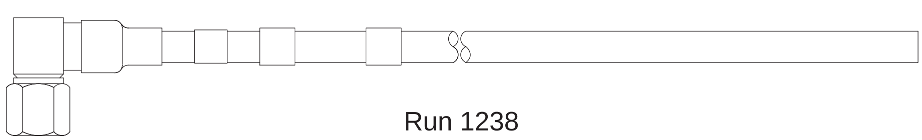
Run 235 and Run 257

Cable 5115915 (Head, Runs 1198 & 1199), will be cut to length to make two cables and terminated for routing in the scan room.



Run 1198 and Run 1199

Cable 5115916 (Head, Run 1238), will be cut to length and terminated for routing in the equipment room.



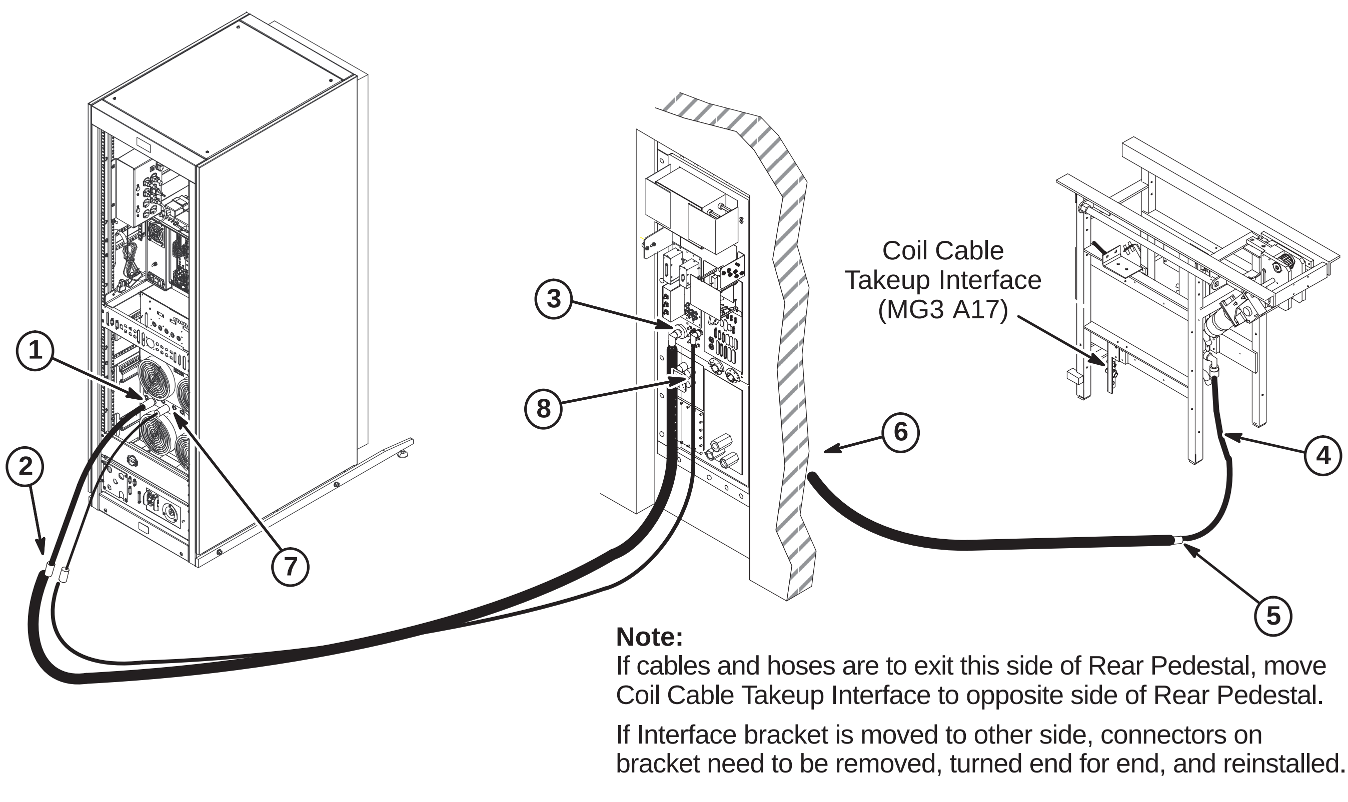
Run 1238

The remaining connectors, fasteners, and labels will be installed as directed in the following steps.

4–9–6–B Install Runs 235, 257, 1238, 1249, 1250, and 1259

RF cables will be routed and connected as indicated in the following steps.

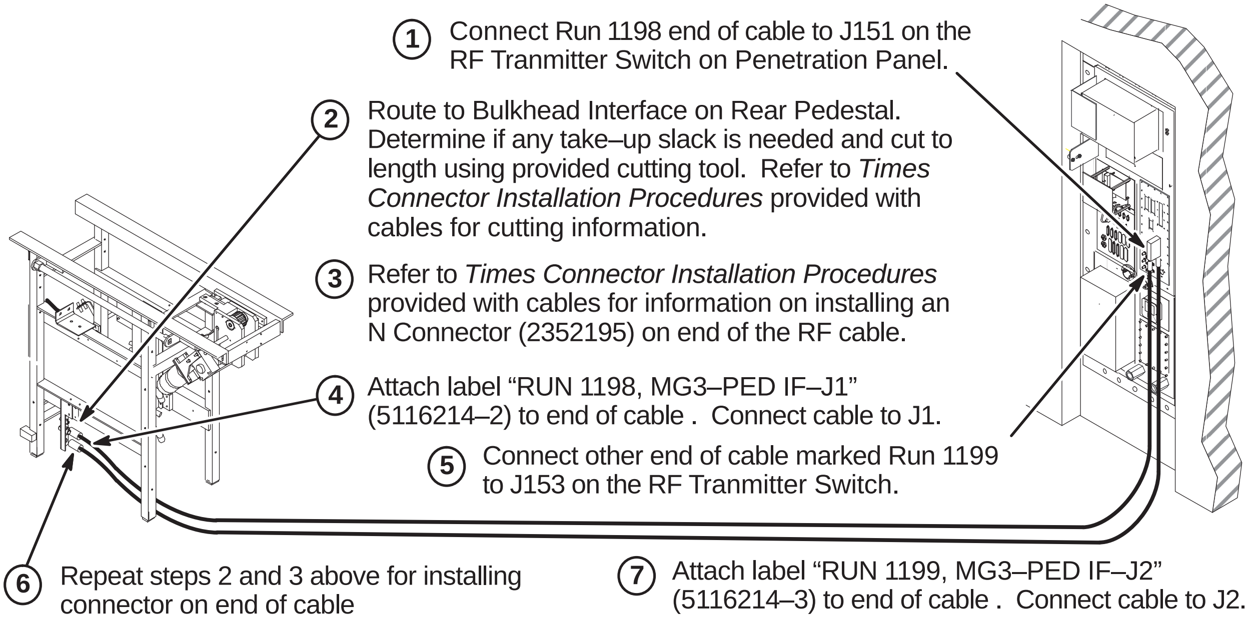
1. Connect extension cable Run 1259 to J4 Body Output at rear of RF Amp in RFS cabinet.
2. Connect end of Times RF Body cable marked "(RUN 235) FJ5" to Run 1259, Body extension cable. Attach label "RUN 235, Connect to Run 1259" (5121244) over existing label.
3. Route cable to J44 at Penetration Panel. Determine if any take–up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4–9–6–E** for termination procedure of cut end.
4. Connect Run 1250 to J1 on Hybrid Splitter.
5. Move remainder of RF Body cable to Scan Room and connect other end of cable marked "(RUN 257) MG3A3–J1" to Run 1250 at Rear Pedestal as shown below. Attach label "RUN 257, Connect to Run 1250" (5121244–2) over existing label.
6. Route remainder of cable to J44 at Penetration Panel. Determine if any take–up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4–9–6–E** for termination instructions of cut end.
7. Connect extension cable Run 1249 to J3 Head Output at rear of RF Amp IN RFS cabinet.
8. Connect end of Times RF Head cable marked "(RUN 1238) PP1–J4" to J4 at Penetration Panel. Route to RF Head extension cable at rear of SRF cabinet. Determine if any take–up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4–9–6–D** for termination procedure of cut end.



Note:
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.

If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

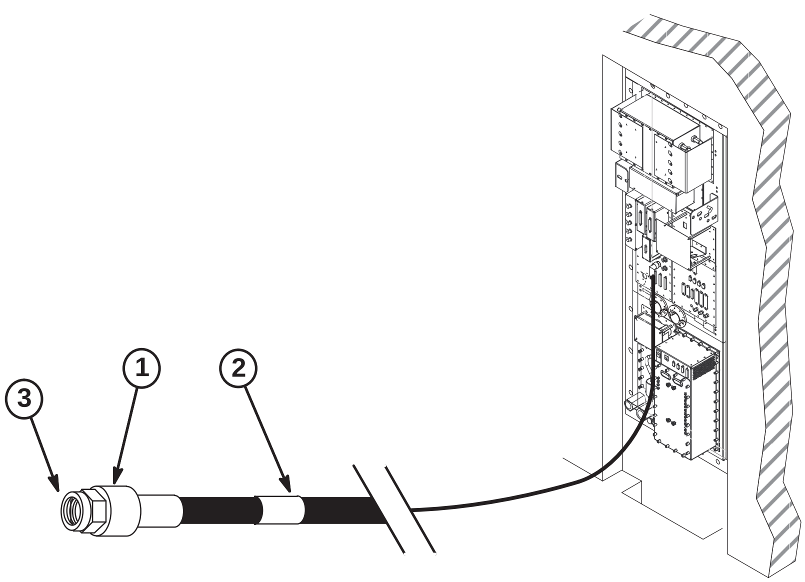
4–9–6–C Install Runs 1198 and 1199



4–9–6–D Terminate and Connect RF Head Coil Cable

Perform the following steps for terminating and connecting the Head RF cables in the Equipment room.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an Straight Male N Connector (2352195) on each end of the RF cable. Use provided cutting tool (5111565) and the provided LMR600 Stripping Tool (2352193).
2. Attach label "RUN 1238 CONNECT TO RUN 1249" (5116214) to cable as shown in Equipment room.
3. Connect cable to Run 1249.



4–9–6–E Terminate and Connect RF Body Cables at Penetration Panel

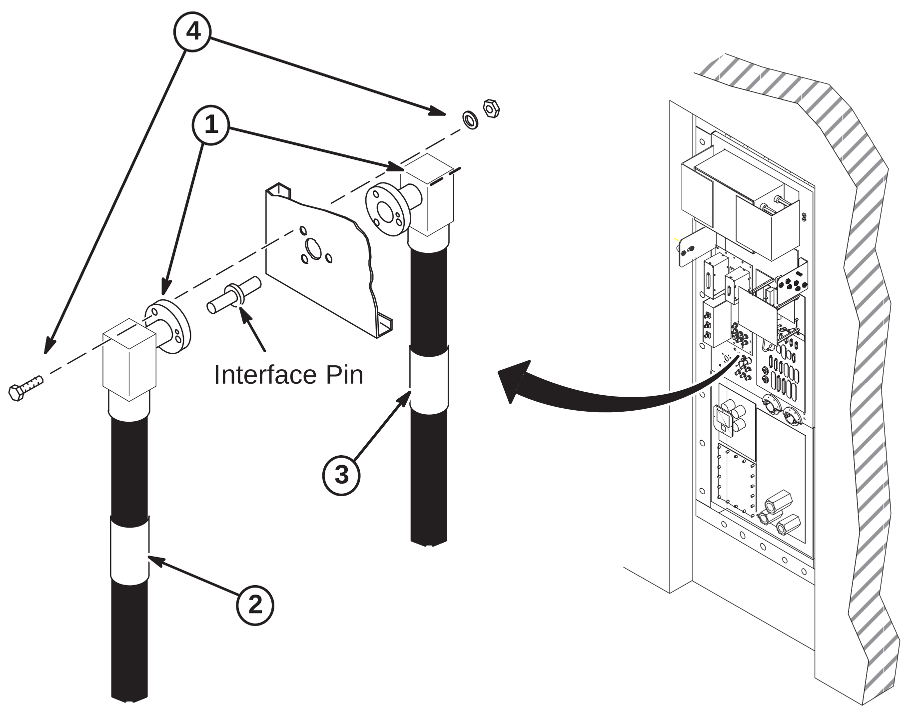
Perform the following steps for terminating and connecting the RF Body cables at the Penetration Panel J44.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an EIA Right Angle Connector (2384859) on each end of the RF cables. Use provided cutting tool (5111565) and the LMR900 Stripping Tool (2384858).

Note

Make sure the fastener holes in the connector flanges align with the holes in the Penetration Panel so the cables will route toward floor when installed.

2. Attach label “RUN 235, PP1–J44” (2357230) to cable as shown on Equipment room side of Penetration Panel.
3. Attach label “RUN 257, PP1–J44” (2357230–2) to cable as shown on Scan room side of Penetration Panel.
4. Insert Interface Pin into one of the connectors. At J44, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
   - 1/4–20 x 1.25 Hex Head Bolt (brass)
   - 1/4 Screw Size Washer (phos bronze)
   - 1/4–20, 7/16” Hex x 3/16” thk Nut (brass)



4–9–7 PNEUMATIC PATIENT ALERT INSTALLATION

For most existing Horizon and LX configurations, the Patient Alert system has been installed prior to the eXCITE HD upgrade. For these upgrades, route and connect existing tubes and associated parts as needed to complete system.

For new installation of Patient Alert sytem, the customer and users should be involved in the following decisions:

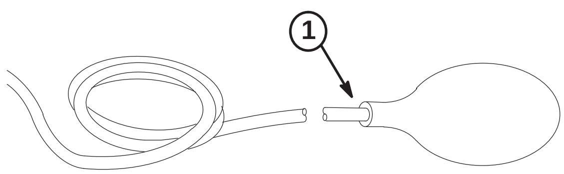
- Routing of Pneumatic Tubing
- Location of Control Box

CAUTION

Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

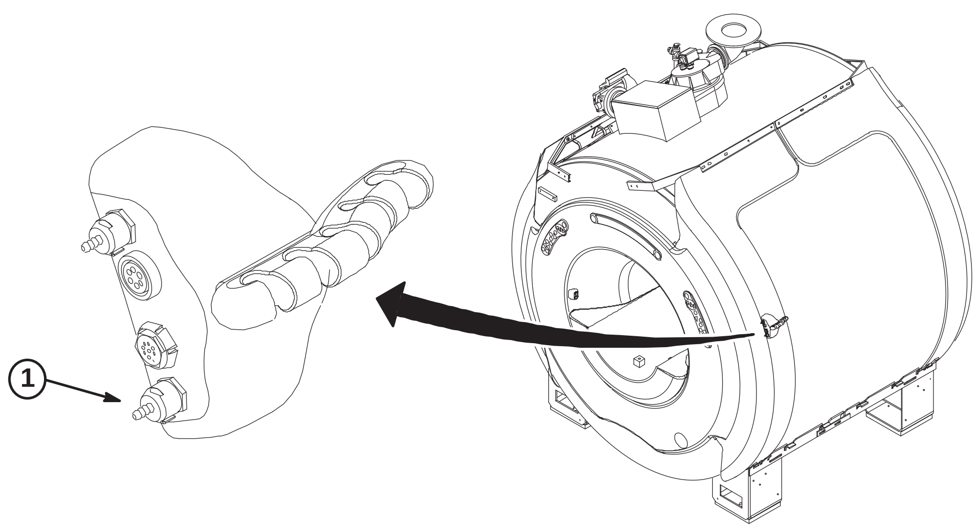
4–9–7–A SQUEEZE BULB INSTALLATION

1. Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.
2. Before cutting pneumatic tubing from roll to length required, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the “CALL” connector on the Remote PAC Interface Assembly without getting in the way of operator or patient during use.



4–9–7–B CONNECT TUBE TO REMOTE PAC INTERFACE ASSEMBLY

1. Connect end of tubing to ”CALL” connection on Remote PAC Interface Assembly.



4–9–7–C MAGNET ROOM ROUTING OF TUBING & WAVE GUIDE INSTALLATION

CAUTION

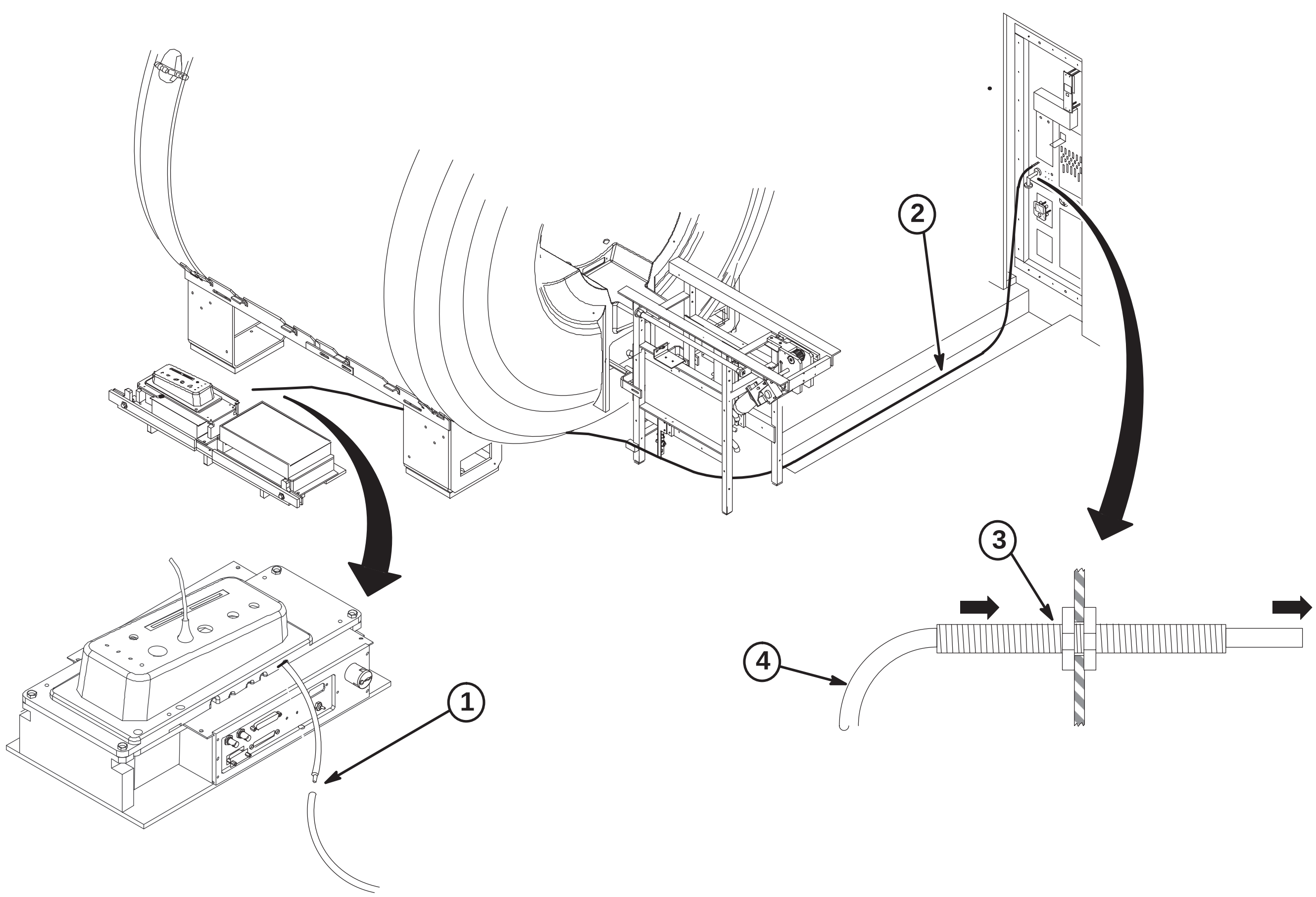
Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

Also, make sure that the pneumatic tubing can be routed from the PAC connector through the Rear Pedestal, and to the Penetration panel without being pinched along the route by other cables.

1. Push end of pneumatic tubing onto PAC–II tubing connector.

Note: If the pneumatic tubing is to be pulled thru fiber–optic conduit, coat with baby powder, not grease, to allow easier pulling.

2. Route Pneumatic Tubing from Rear Pedestal to Penetration Panel with other system cables.
3. Insert Wave Guide thru “ALERT AIR LINE” hole and fasten to Penetration Panel with nut on each side.
4. Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide into the Equipment Room where the Control Box will be installed.



4–9–7–D DETERMINING NEED OF EXTENDER BOX

If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator's Console), Extender Kit, 46–317758P2, must be ordered.

- **If the pneumatic tubing is long enough to reach control box:**
  The extender is not needed. Proceed to Procedure **4–9–7–F**, CONTROL BOX INSTALLATION.
- **If the pneumatic tubing is not long enough to reach control box**:
  The extender is needed. Proceed to Procedure **4–9–7–E**, INSTALLATION OF EXTENDER KIT.

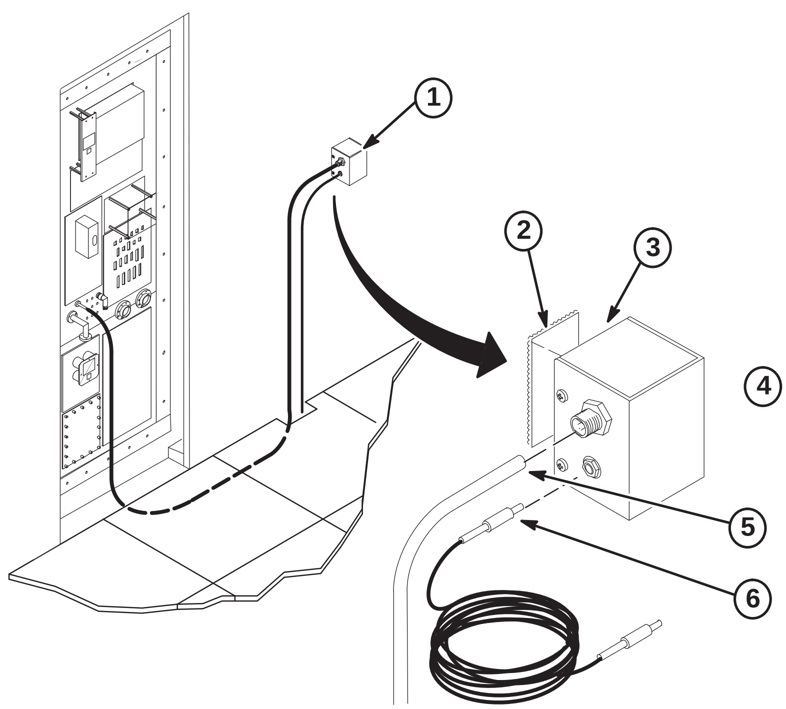
4–9–7–E INSTALLATION OF EXTENDER KIT (46–317758P2)

CAUTION

Do not mount Extender Box in Magnet Room because the box contains ferrous parts.

Install Extender Box

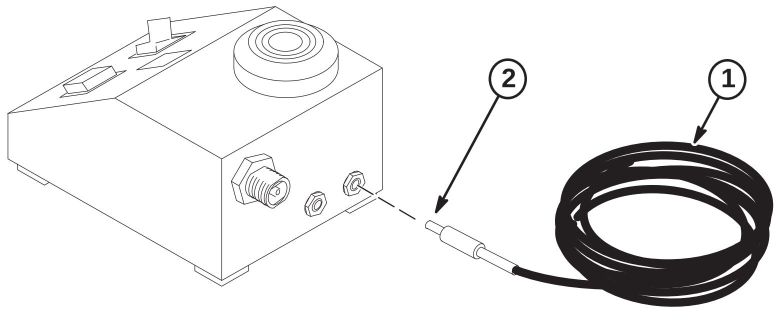
1. Choose the mounting location for extender box on equipment room wall next to Penetration Panel. Extender box will be mounted with pneumatic and electrical jacks on side of extender box.
2. Cut a two inch long piece of Velcro loops and a two inch long piece of Velcro hooks. Both are obtained from the Pneumatic Patient Alert Kit.
3. Clean the back of extender box, remove protective paper from back of Velcro loops, and attach Velcro to center of extender box back.
4. Clean wall in the area where extender box will be mounted, remove protective paper from back of Velcro hooks, and attach velcro to wall in same orientation as Velcro loops on extender box.
5. Push end of pneumatic tubing fully onto Feed–thru connector on Extender Box.
6. Push the jack on one end of the 65 foot Extender Wire into the plug on side of Extender Box.



4–9–7–E Installation of Extender Kit (46–317758P2) (Continued)

Connection of Extender Wire to Control Box

1. Route the extender wire from extender box to control box. Control box will be installed near Operator Console, within five feet of an electrical outlet.
2. Push the jack at other end of extender wire into the plug on back of control box.



4–9–7–F CONTROL BOX INSTALLATION OPTIONS

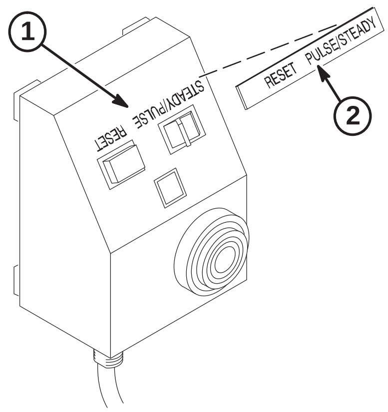
Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

- Mount control box on a wall or other vertical surface per Procedure **4–9–7–G**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Mount control box under shelf per Procedure **4–9–7–G**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed <u>within sight of operator</u>. Proceed to Procedure **4–9–7–K**, FINAL CONNECTIONS TO CONTROL BOX.

Note: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

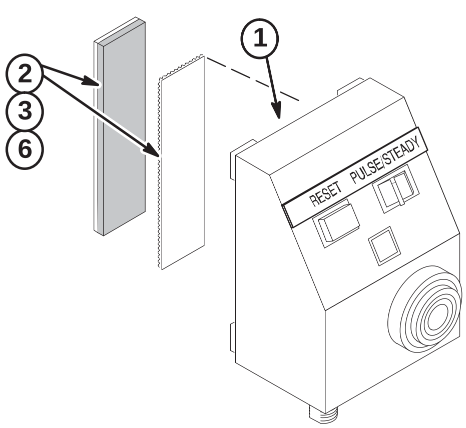
4–9–7–G CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF

1. Clean Control Box surface.
2. Remove adhesive backing from metalized nameplate and apply over silk screened lettering as shown.
3. Select one of the following for installation of Control Box:
   - If control box will be mounted to WALL or SHELF with Velcro, go to Procedure **4–9–7–H**
   - If control box will be mounted to WALL with screws, go to Procedure **4–9–7–I**
   - If control box will be mounted to SHELF with screws, go to Procedure **4–9–7–J**



4–9–7–H Control Box Mounted With Velcro

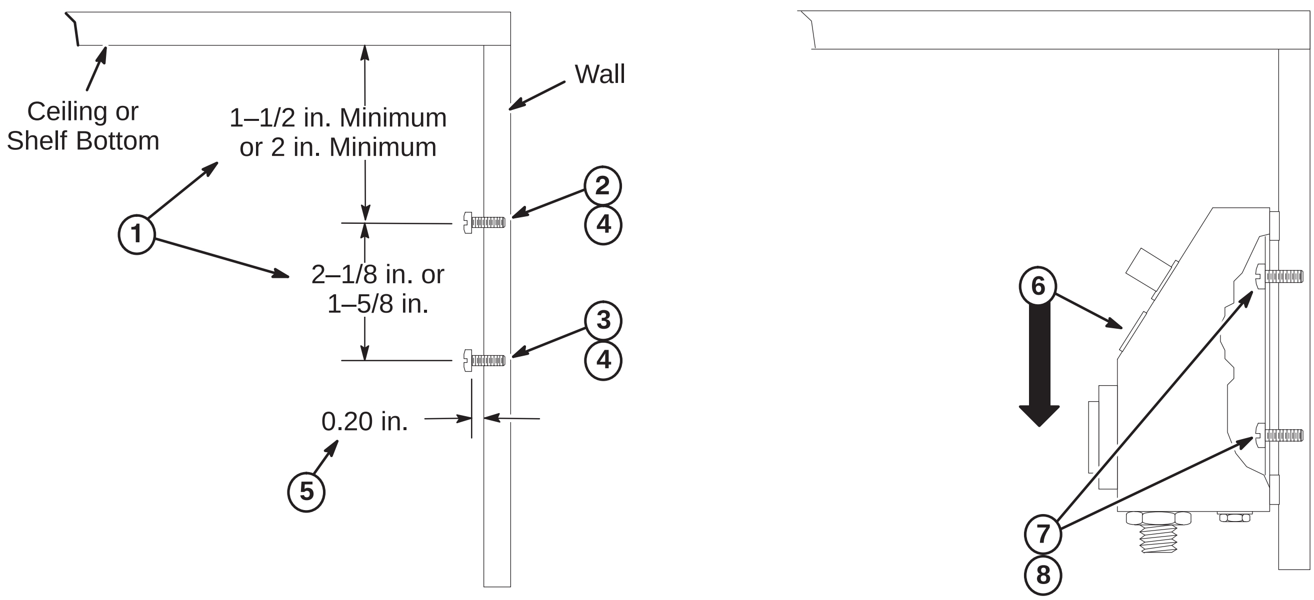
1. Clean the bottom surface of the Control Box.
2. From the Pneumatic Patient Alert Kit, cut a three inch long piece of Velcro loops and a three inch piece of Velcro hooks.
3. Remove protective paper from back of Velcro loops, and attach Velcro to center bottom of control box.
4. Choose the mounting location for control box on wall (or under shelf) next to operator console. Control box should be mounted within sight of operator.
5. Clean the area where control box will be mounted.
6. Remove protective paper from back of Velcro hooks and attach to mounting location in same orientation as Velcro on box.
7. Mount the control box by pressing Velcro on control box against Velcro on mounting location and twisting slightly.
8. Proceed to Procedure **4–9–7–K**, FINAL CONNECTIONS TO CONTROL BOX.



4–9–7–I Installing Control Box with Mounting Screws on Vertical Surface

Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

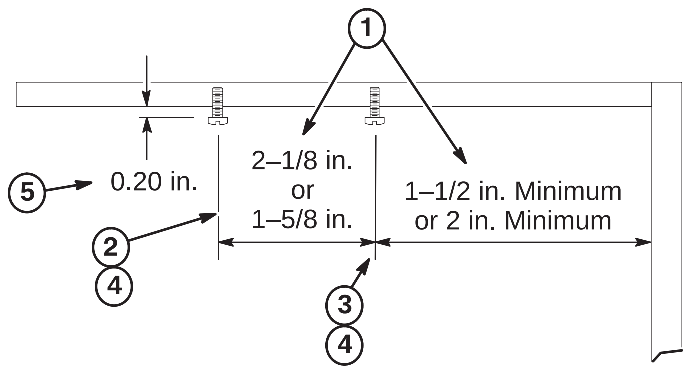
1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark top mounting hole position on wall.
3. Mark bottom mounting hole position on wall.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **4–9–7–K**, FINAL CONNECTIONS TO CONTROL BOX.

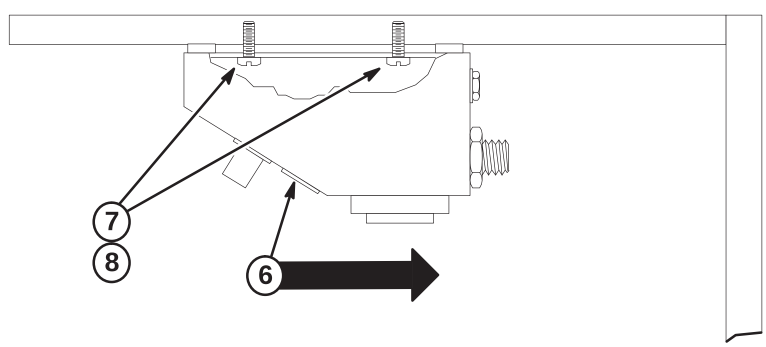


4–9–7–J Installing Control Box with Mounting Screws Under Shelf

Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

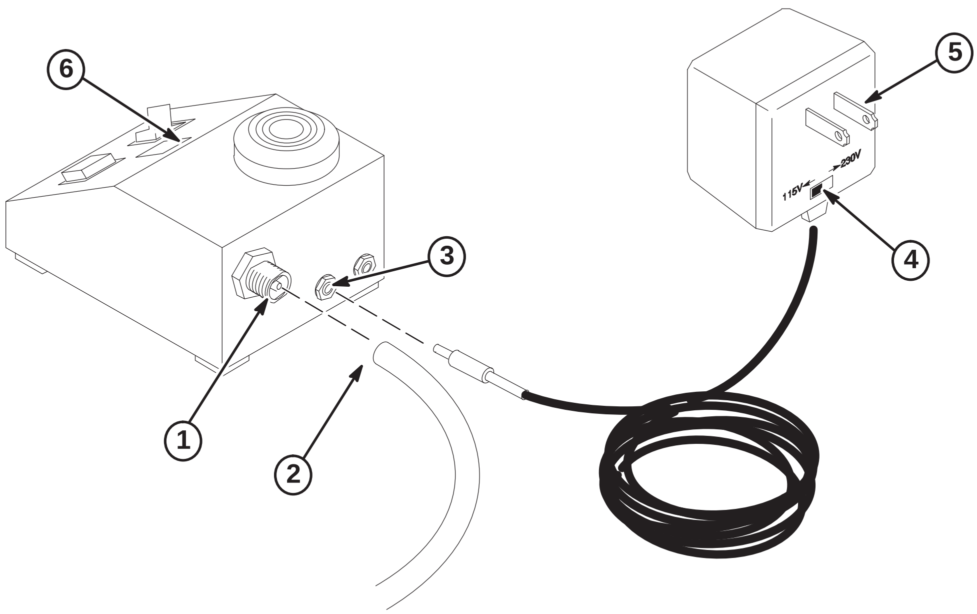
1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark front mounting hole position on underside of shelf.
3. Mark rear mounting hole position on underside of shelf.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push inward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **4–9–7–K**, FINAL CONNECTIONS TO CONTROL BOX.





4–9–7–K FINAL CONNECTIONS TO CONTROL BOX

1. Bring the pneumatic tubing to back of control box and mark length. Leave two inches of extra pneumatic tubing for connection.
2. Cut tubing and push fully onto feed–thru connector on back of control box.
3. Plug Power Supply Jack into Control Box.
4. Set Power Supply to outlet voltage (115VAC or 230VAC).
5. Plug Power Supply into AC outlet.
6. Check that green LED is on.



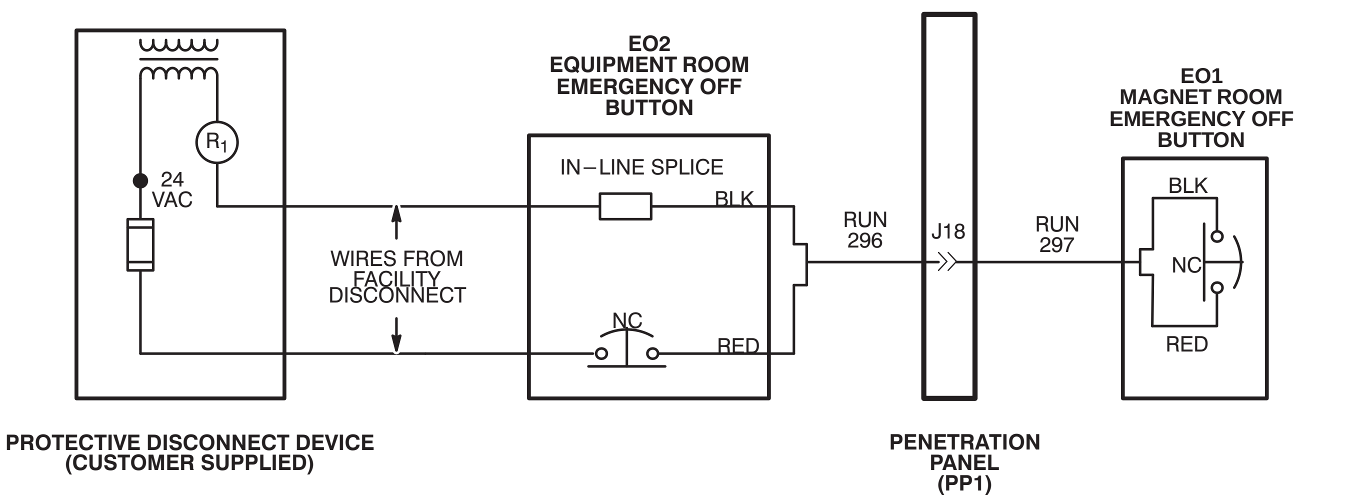
4–9–8 EMERGENCY OFF CONNECTIONS

Run 296 is routed to EO2 (Equipment Room Emergency Off) and connected to EO2 and wire from Facility Disconnect. Run 297 is routed from Penetration Panel to EO1 (Magnet Room Emergency Off Button). Refer to *Direction 2355598, Signa EXCITE 3.0T Pre–Installation*, Section 5, Protective Disconnect Device, for detailed wiring diagram.

1. Complete routing of Run 296 to EO2 (Equipment Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.

Note: In Illustration below, black and red wires are used for connections in Runs 295 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. Runs 296 and 297 are actually nine wire cables.

2. At equipment room room "Emergency Off" (EO2) location, terminate red wire at end of Run 296 with local supplied terminal and connect to customer supplied Equipment Room Emergency Off Button (EO2).
3. Terminate customer supplied wire (from fuse in Protective Disconnect Device) with local supplied applicable terminal and connect to customer supplied Emergency Off Button (EO2).
4. Terminate customer supplied wire from R1 in Protective Disconnect Device with local supplied push–on terminal.
5. Terminate black wire at end of Run 296 with local supplied "push–on" terminal.
6. Connect black wire from end of Run 296 to customer supplied wire from protective disconnect device with local supplied in–line splice.
7. Complete routing of Run 297 to EO1 (Magnet Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.
8. Terminate red and black wires at end of Run 297 with local supplied terminals and connect to customer supplied Magnet Room Emergency Off Button (EO1).



SECTION 5 – PENETRATION PANEL

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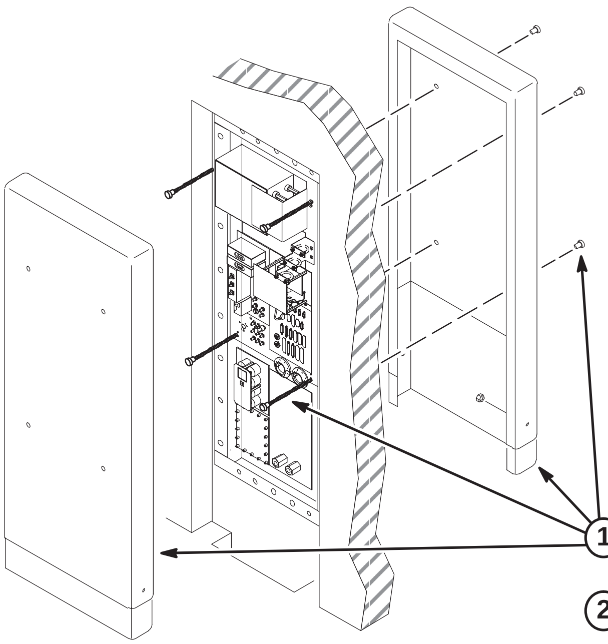
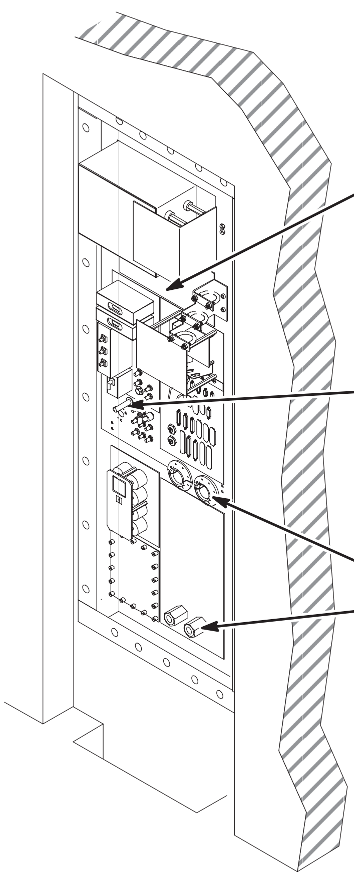
5–1 REMOVE PENETRATION PANEL COVERS



1 If installed, unfasten and save Penetration Panel covers, stand–offs, and mounting hardware.

2 Move above items to a safe storage area. They will be reinstalled at end of upgrade.

5–2 REMOVE PARTS FOR TRANSFER TO NEW PENETRATION PANEL



1 If installed on Penetration Panel, transfer the Shim Filter, only if it's a 46–271282G1, (if not this model, order one) to new Penetration Panel.

2 Disconnect and remove Patinet Alert tubing. Transfer Patient Alert Wave Guide to new Penetration Panel re–using mounting hardware.

3 If Patient Alert Extender Box is mounted on Penetration Panel, remove and re–install on new Penetration Panel.

CAUTION! – OLDER SYSTEMS USED BERYLIUM PASTE ON WAVE GUIDES AND GAS LINES. WASH HANDS IF YOU COME IN CONTACT WITH THIS MATERIAL.

5–3 DISCONNECT COOLANT HOSES

CAUTION

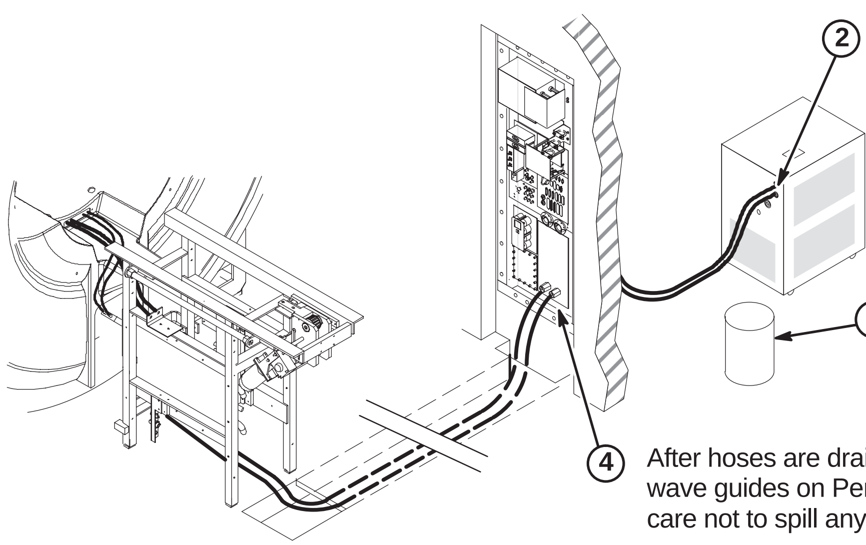
Coolant spillage possibility. The gradient coil cooling system is pressurized. Turn off the heat exchanger or water chiller before disconnecting the hoses. This takes pressure off the system and helps avoid coolant from being sprayed all ove the immediate area.

WARNING!

POSSIBLE EYE INJURY! DRAINING OF COOLANT (ANTI–FREEZE) MAKES SPLASHING POSSIBLE. THIS COULD CAUSE EYE INJURY. BE SURE TO WEAR LATEX GLOVES AND SAFETY GLASSES.

Note: Four hands are better than two – While it is possible for one person to disconnect the coolant hoses, it is easier if two people do it. There would be less chance of spilling the coolant.

1. Prior to disconnection of coolant hoses, calculate the amount of coolant that will be drained from system to determine if the 5 gallon bucket provided with upgrade will be sufficient for collection of coolant to be drained. Approximately two gallons of coolant will remain in gradient coil after disconnection. If more containers are needed for collection of drained coolant, they must be obtained locally.



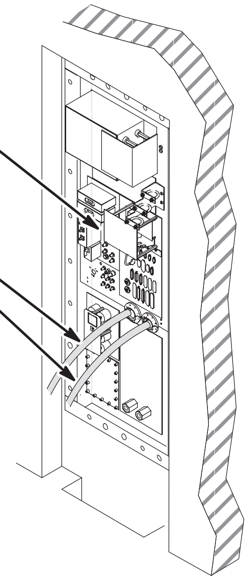
2. Remove Hose Clamps (2128542–2) and disconnect two coolant hoses.

Note: If existing system has a Heat Exchanger, it will be deinstalled and a new Water Chiller will replace it. Refer to Section **12**.

3. Place ends of coolant hoses in provided bucket (2324086–2) for collection of coolant.

4. After hoses are drained, pull hoses out of wave guides on Penetration Panel. Take care not to spill any excess coolant.

5–4 DEINSTALL PENETRATION PANEL



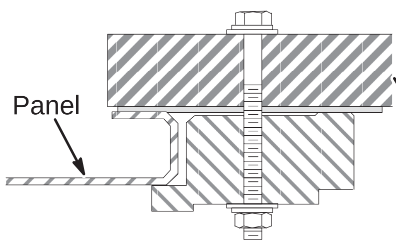
1. Disconnect all cables attached to both sides of Penetration Panel.

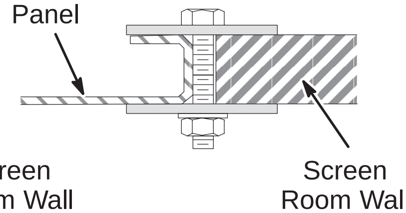
   Note: Refer to Cable Section in this manual to determine which cables are retained and which are returned to GE for recycling.

2. The two helium gas lines will be reused and MUST NOT be disconnected. Remove and save Bronze Wool (46–318068P1) from inside of J60 and J61.

3. Deinstall Penetration Panel by removing the mounting hardware located on outside edge of panel. Two different methods are shown below.

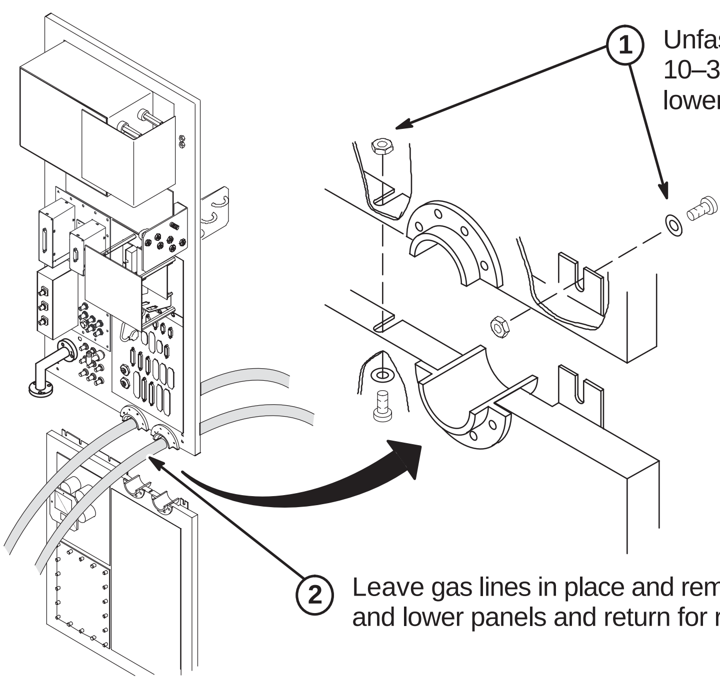
Hold Down Method **Brass Sealing Plate Method**





4. Save the grade 8 bolts and related mounting hardware for installing the new Penetration Panel.

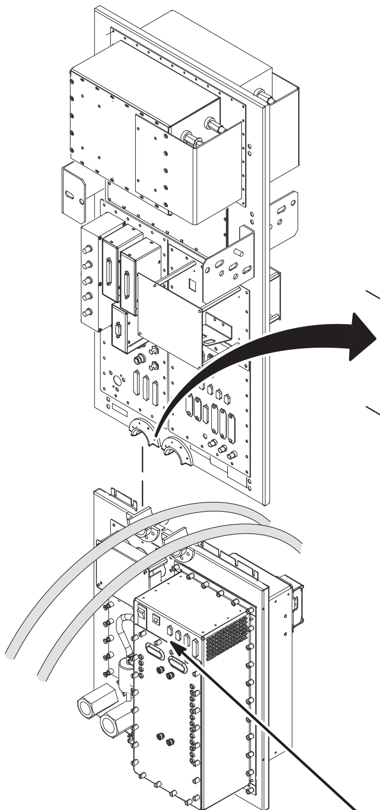
5–5 UNFASTEN PENETRATION PANEL AROUND HELIUM GAS LINES



1. Unfasten Nylock Lock Nuts, Flat Washers, and 10–32 x 3/8 Screws that attach the upper and lower panels around the gas lines.

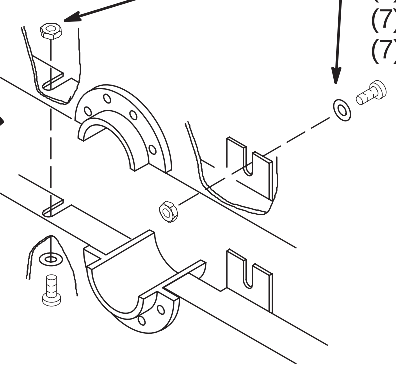
2. Leave gas lines in place and remove upper and lower panels and return for recycling.

5–6 INSTALL EXCITE HD PENETRATION PANEL



1 To make sure of good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**

2 Assemble Penetration Panel around helium lines (Runs 621 and 622) using:
(7) #10–32 Nylock Lock Nuts (46–208937P6)
(7) 0.2 ID, 0.5 OD Flat Washers (1000904P425)
(7) #10–32 x 3/8 Screws (46–208560P82)



3 Install new Penetration Panel using same method and hardware as removed penetration panel.

4 Seal all joints using copper foil tape with a conductive adhesive (3M #1181 or 46–258218P4, 5, or 6) for RF–tight integrity.

5 Fill inside of J60 and J61 guides with Bronze Wool (46–318068P1).

6 Install Patient Alert Wave Guide into “ALERT AIR LINE” hole or another available hole. If required, install Extender Box.

7 Install CAN Terminator (2303317) at J155.

8 If required, install previously removed Shim Filter and cables.

5–7 CONNECT CABLES TO PENETRATION PANEL

For information and instructions on cable and hose connections at the Penetration Panel, refer to applicable:

- Installation Flowchart in Section 1
- Coolant Hose installation in Section 12
- Cable Interconnect Map in Section 4
- Cable and System Installation in Section 7 .

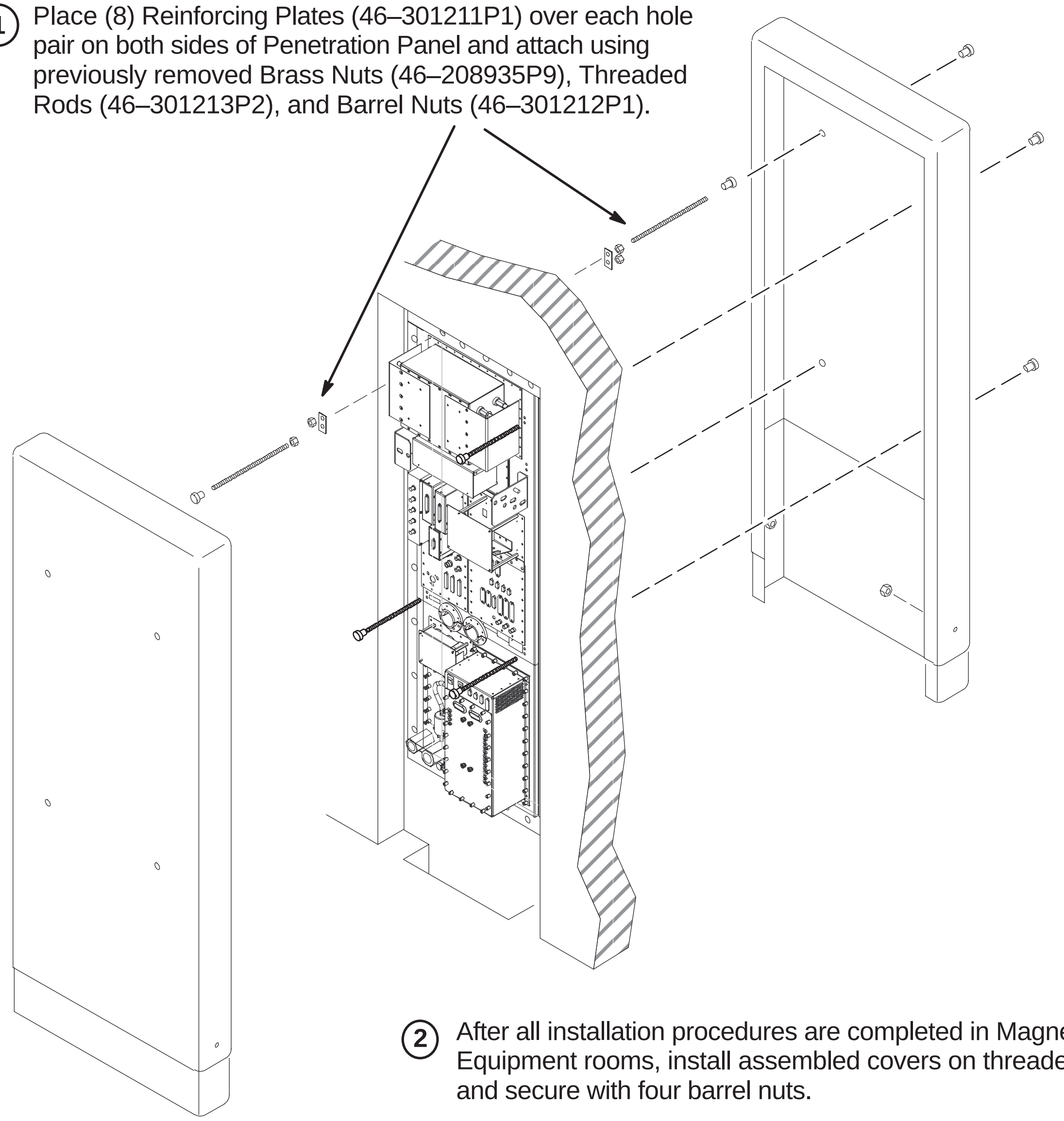
Make sure to reinstall all customer added option connections.

5–8 INSTALL PENETRATION PANEL COVERS (If Previously Removed)

Note

The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

1 Place (8) Reinforcing Plates (46–301211P1) over each hole pair on both sides of Penetration Panel and attach using previously removed Brass Nuts (46–208935P9), Threaded Rods (46–301213P2), and Barrel Nuts (46–301212P1).



2 After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.

SECTION 6 – UPGRADE FROM WideOpen & TwinSpeed LX

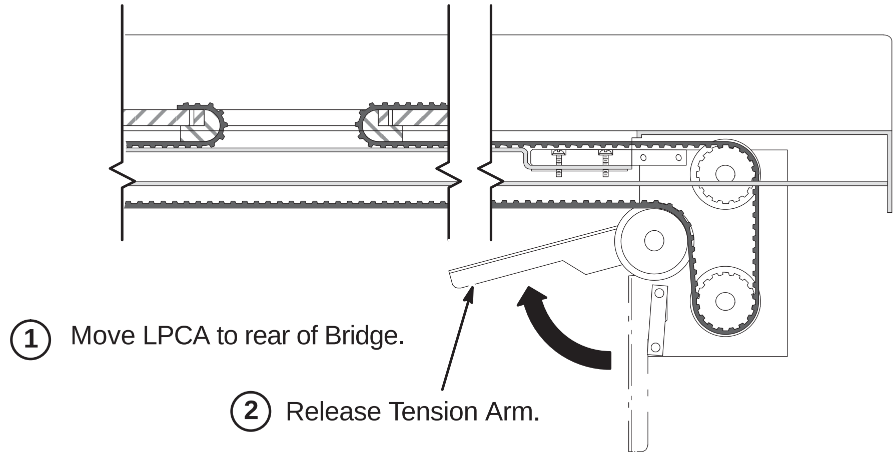
TABLE OF CONTENTS

SECTION PAGE

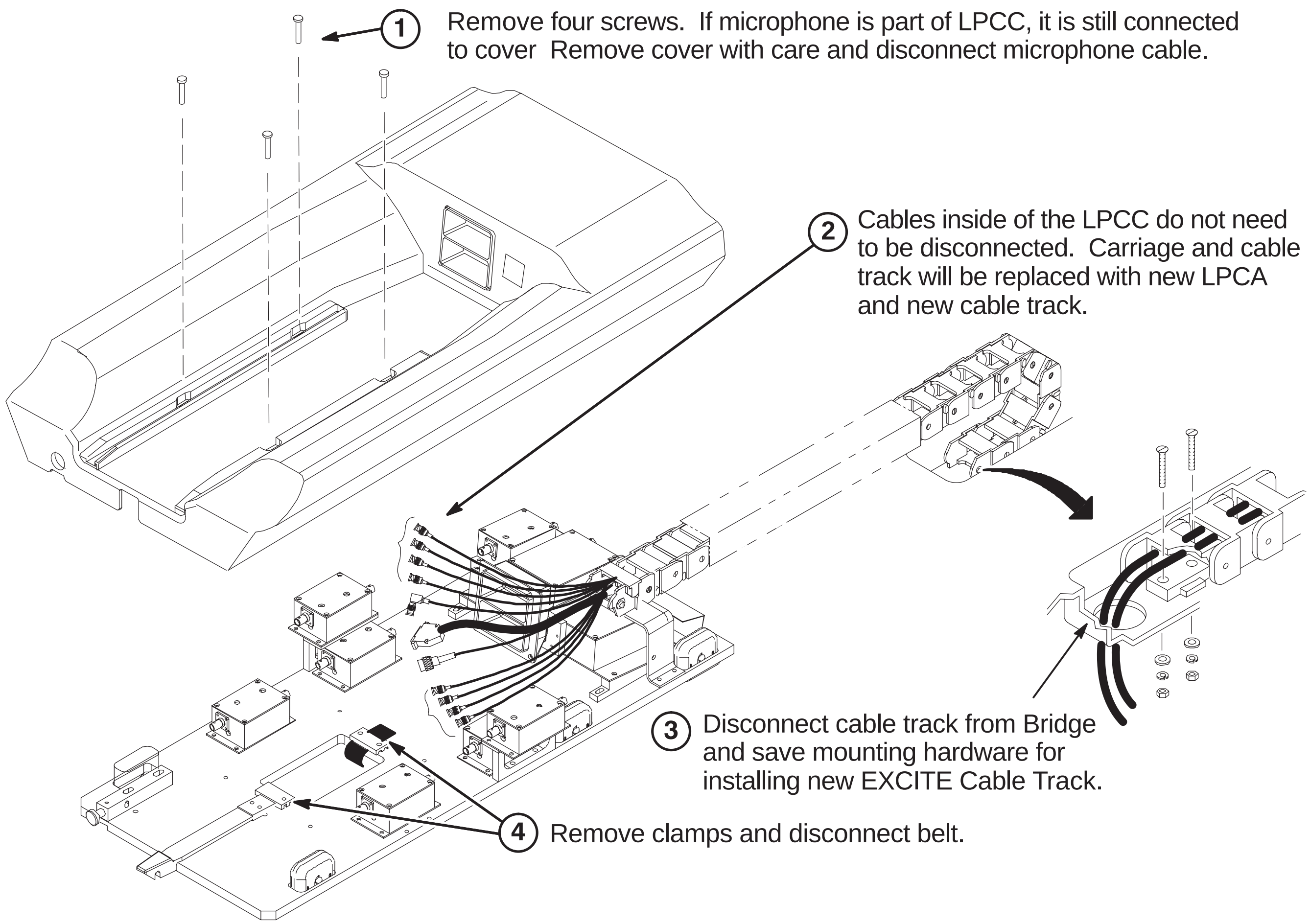
6–1 DE–INSTALL COMPONENTS ON WideOpen OR WideOpen ENCLOSURE

6–1–1 DE–INSTALL EXISTING LPCA & CABLE TRACK

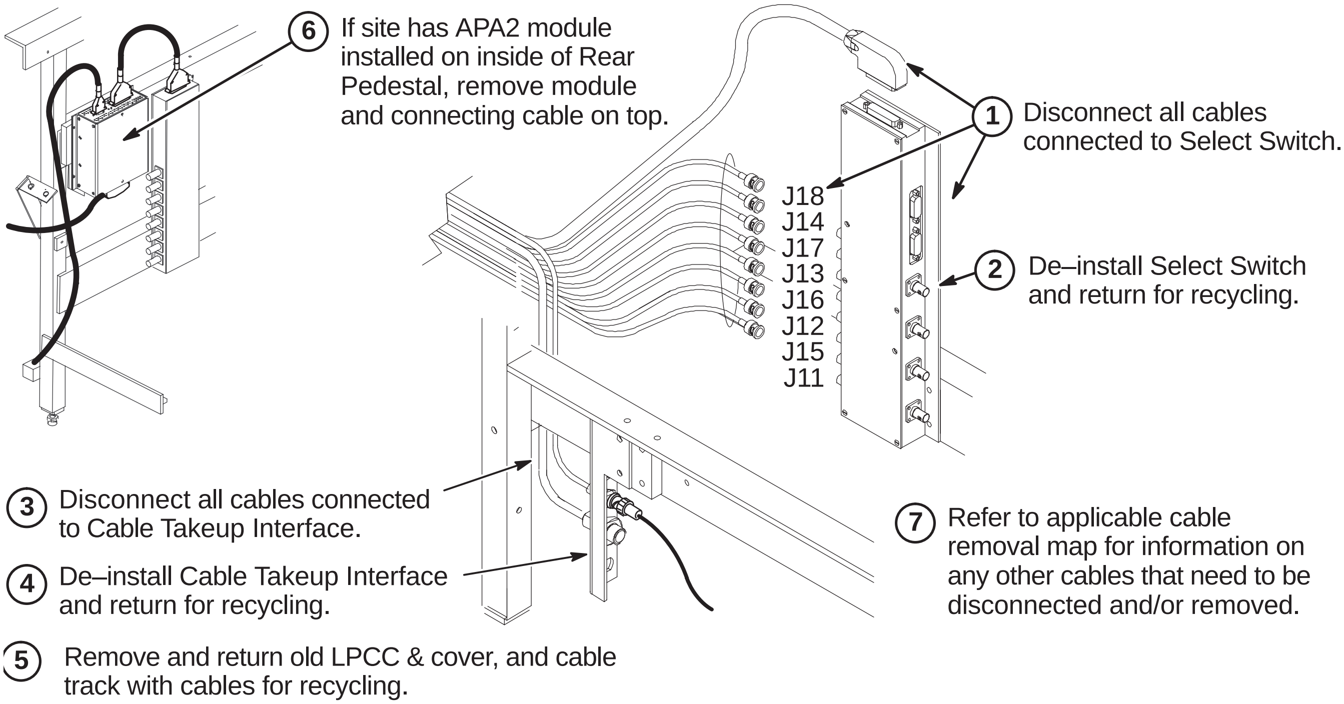
6–1–1–A Relese Tension Arm



6–1–1–B Disconnect Low Profile Carriage Cover & Cable Track



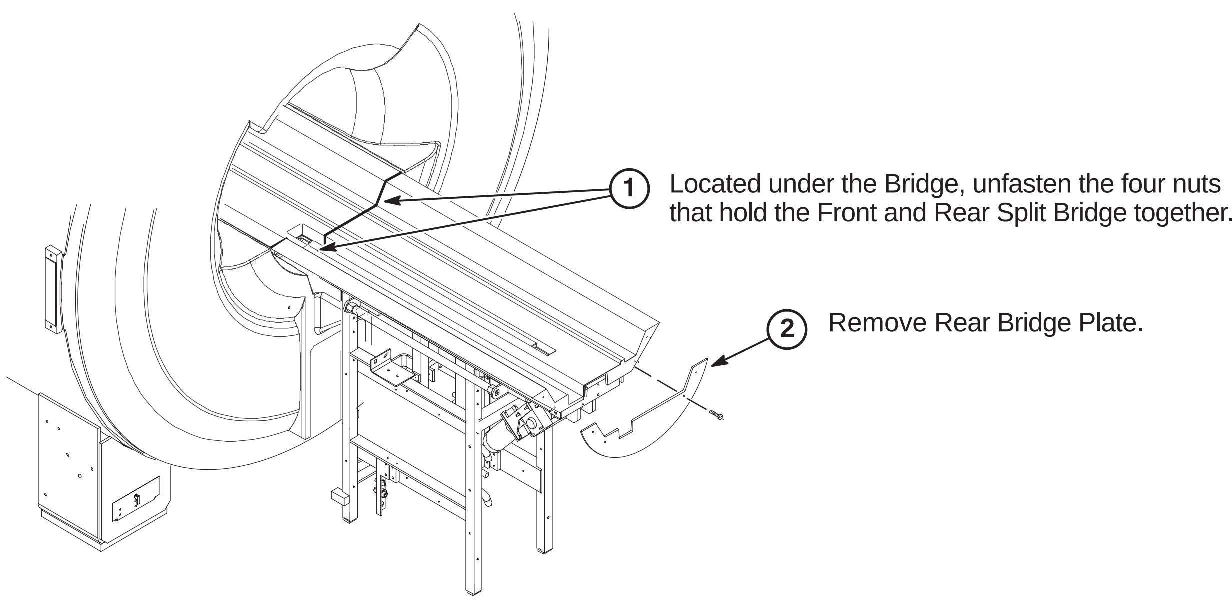
6–1–1–C Disconnect Cables In Rear Pedestal



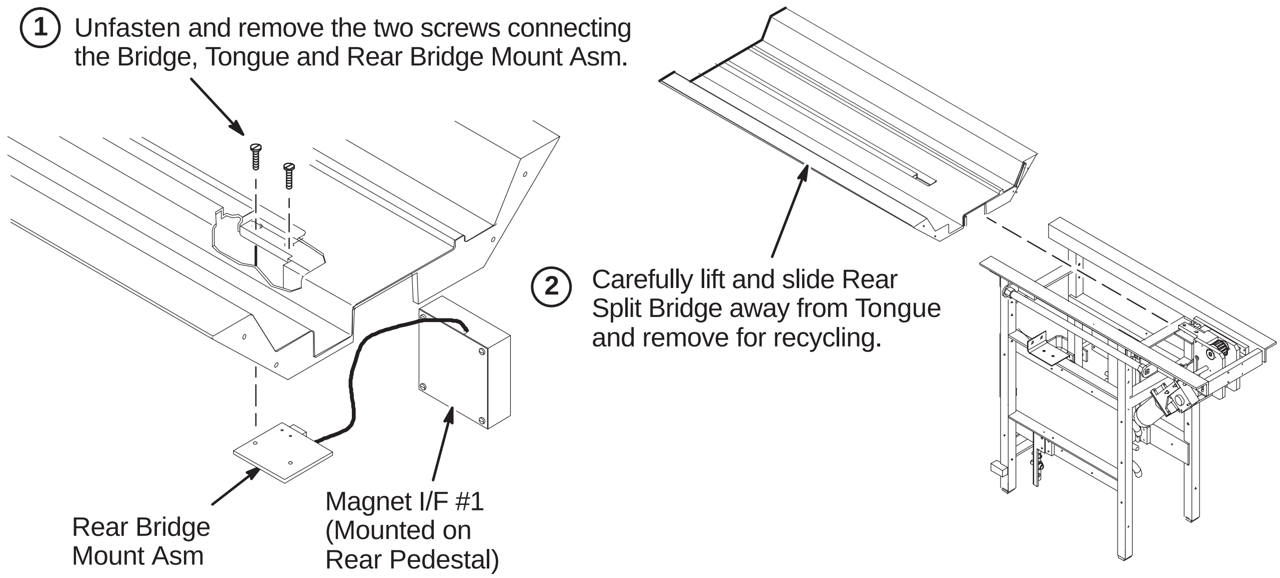
6–1–2 DEINSTALL SPLIT BRIDGE FOR WideOpen

Upgrading to EXCITE HD requires a new Front and Rear Split Bridge if the site has an LCC magnet with a WideOpen or TwinSpeed enclosure. The following procedures are for removing either the original split bridge designed for a single cable track or the older style single piece long bridge. The WideOpen enclosure with a long bridge will also have the front end bell removed and replaced.

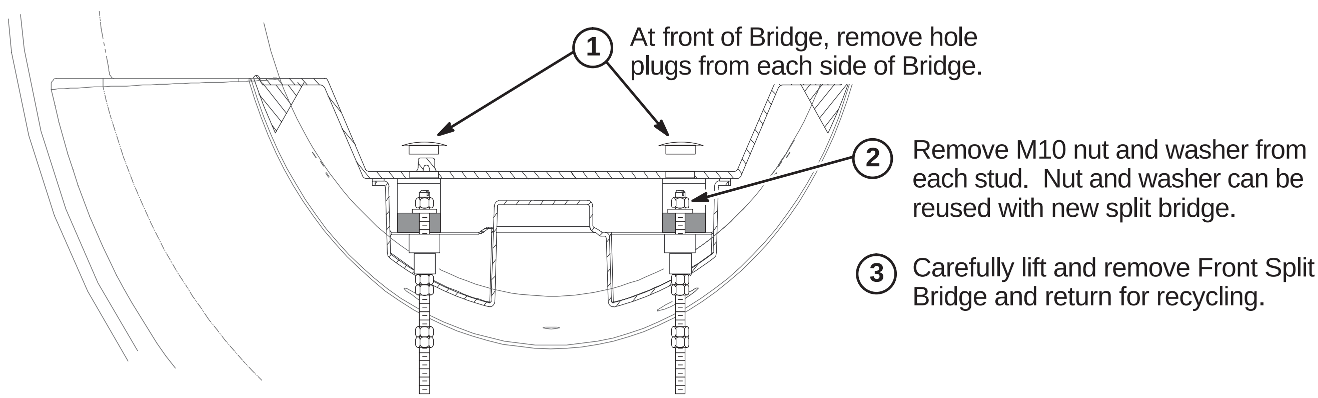
6–1–2–A Unfasten Rear Split Bridge



6–1–2–B Remove Rear Split Bridge



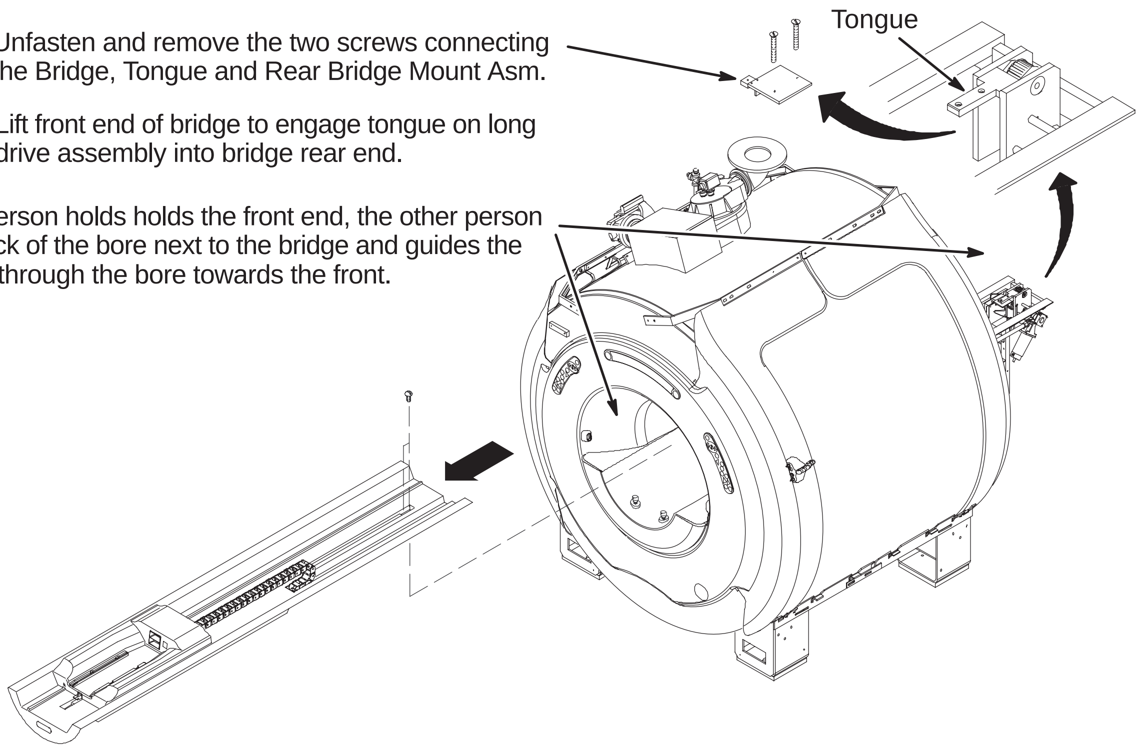
6–1–2–C Remove Front Split Bridge



6–1–3 DE–INSTALL LONG BRIDGE FOR WideOpen

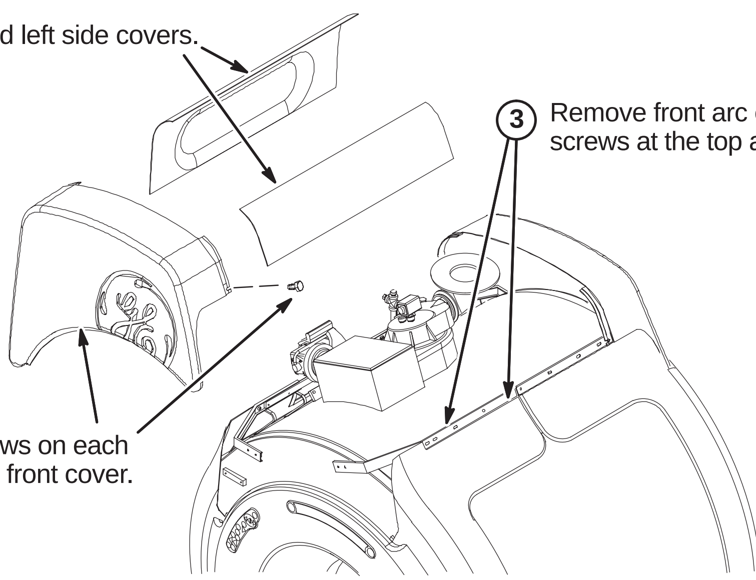
6–1–3–A Remove Long Bridge

1. Unfasten and remove the two screws connecting the Bridge, Tongue and Rear Bridge Mount Asm.
2. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.
3. While one person holds holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore towards the front.

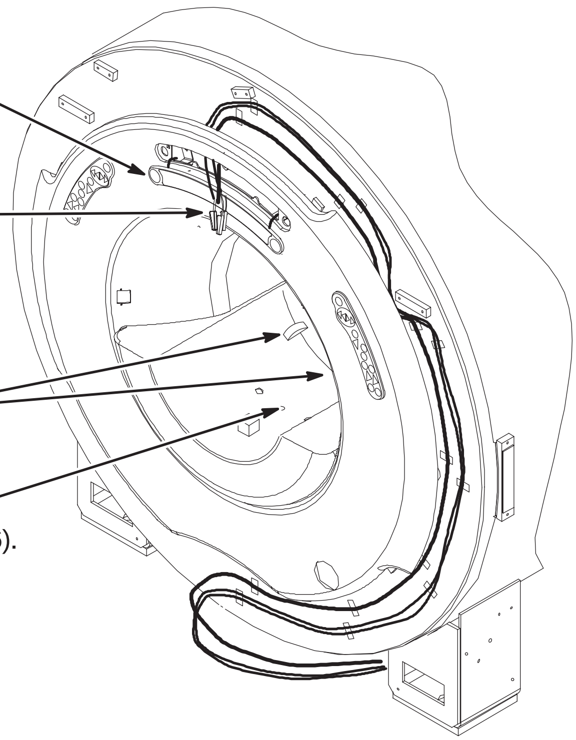


6–1–3–B Remove WideOpen Enclosure Covers

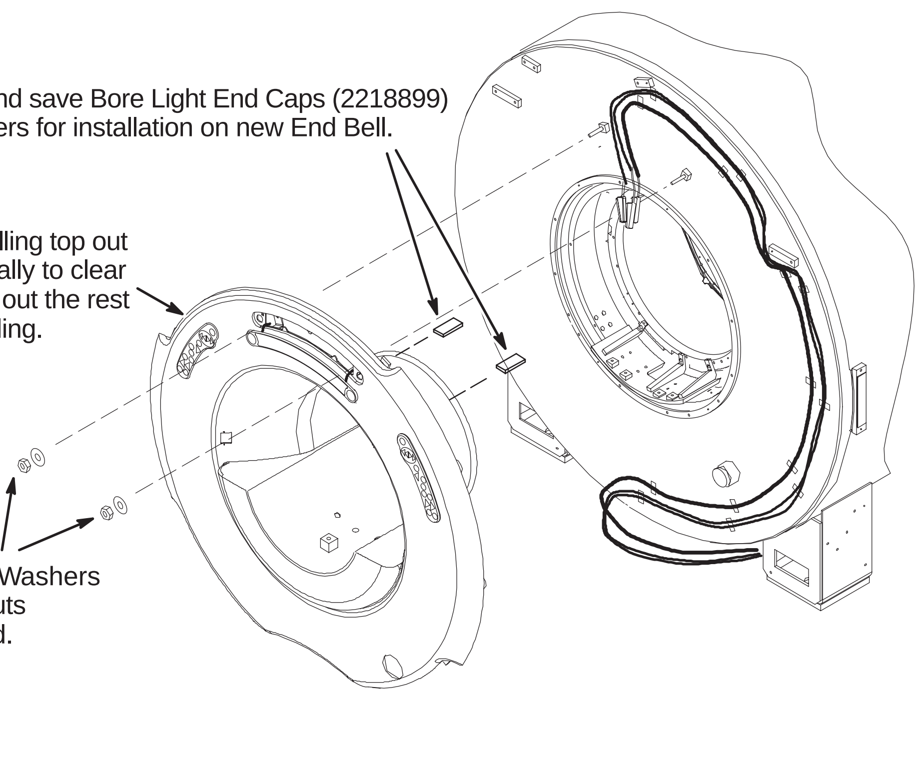
1. Remove tight and left side covers.
2. Remove screws on each side, then top front cover.
3. Remove front arc covers by removing two screws at the top and sliding cover to the side.



6–1–3–C Disconnect Operator Control Panel and Unfasten Front End Bell



6–1–3–D Remove Front End Bell

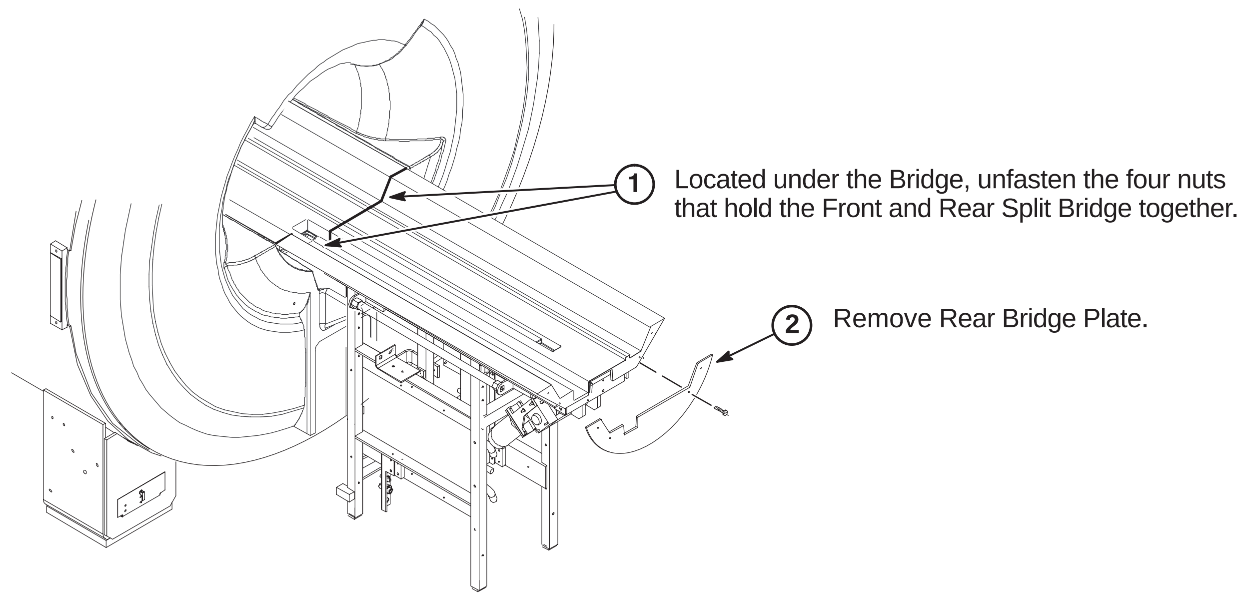


6–1–4 DE–INSTALL SPLIT BRIDGE FOR TwinSpeed

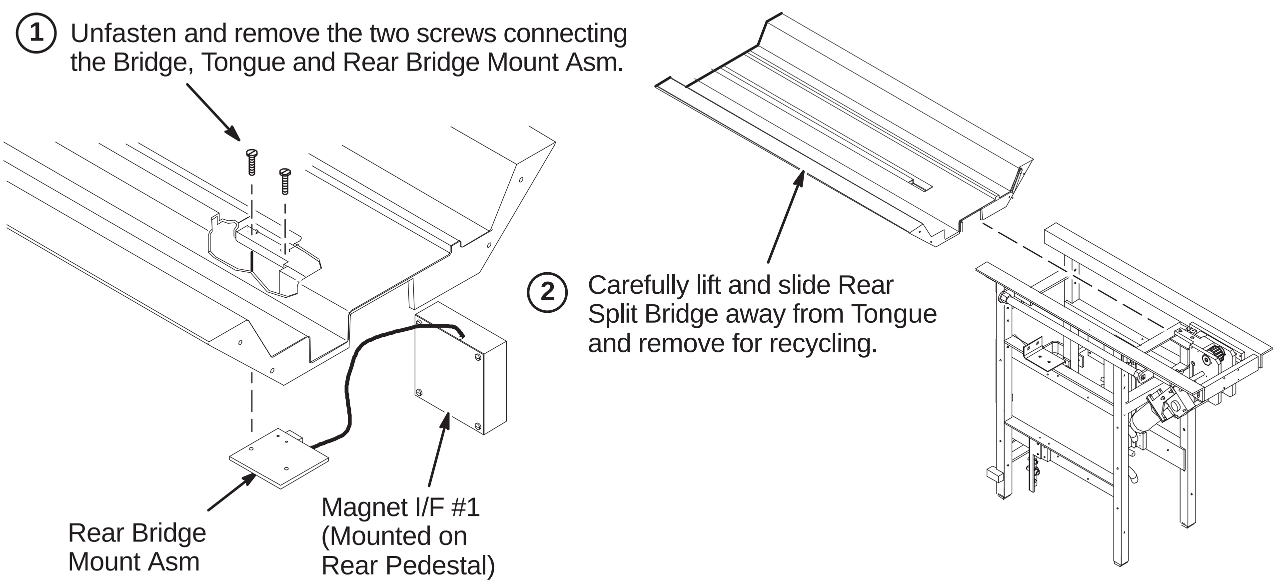
Upgrading to EXCITE HDTwinSpeed requires a new Front and Rear Split Bridge. The method of attachment at the front of the Bridge is different between a bridge that was installed for a forward production (new) installation and a Long bridge that was upgraded from a non–TwinSpeed WideOpen enclosure to TwinSpeed.

- Section 6–1–4 shows the deinstallation for a forward production Split Bridge.
- Section 6–1–5 shows the deinstallation of the upgraded Long Bridge.

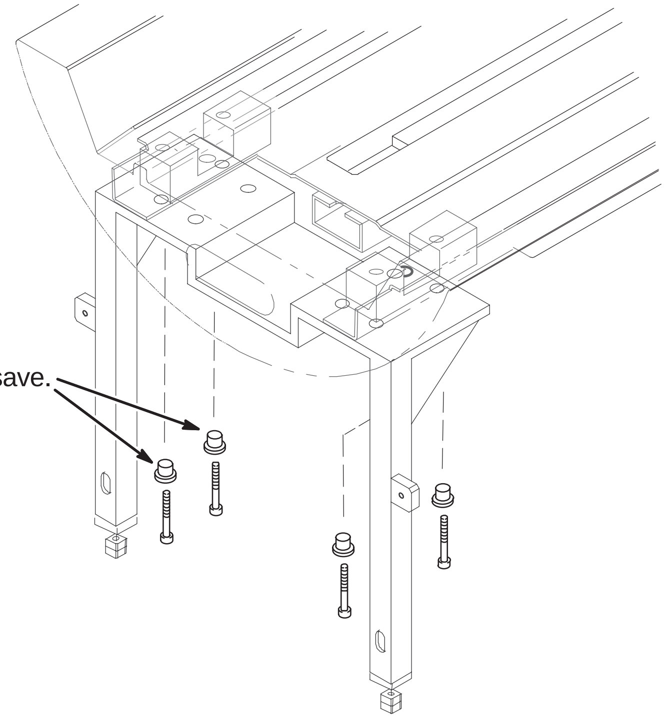
6–1–4–A Unfasten Rear Split Bridge



6–1–4–B Remove Rear Split Bridge



6–1–4–C Remove Forward Production Installed Front Split Bridge

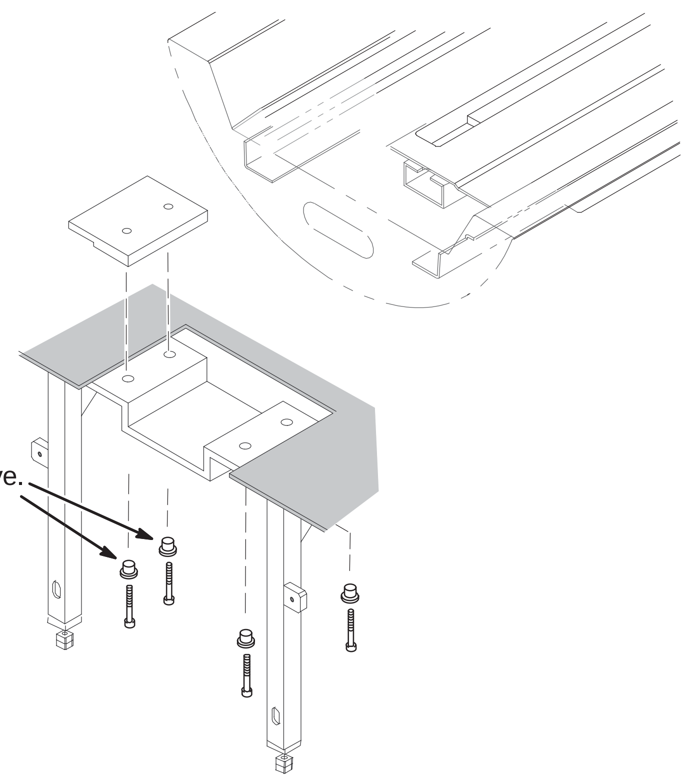


1. Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867–17) and Support Bushings and save. These will be reused for installing the new Bridge.

2. Carefully lift and remove Front Split Bridge and return for recycling.

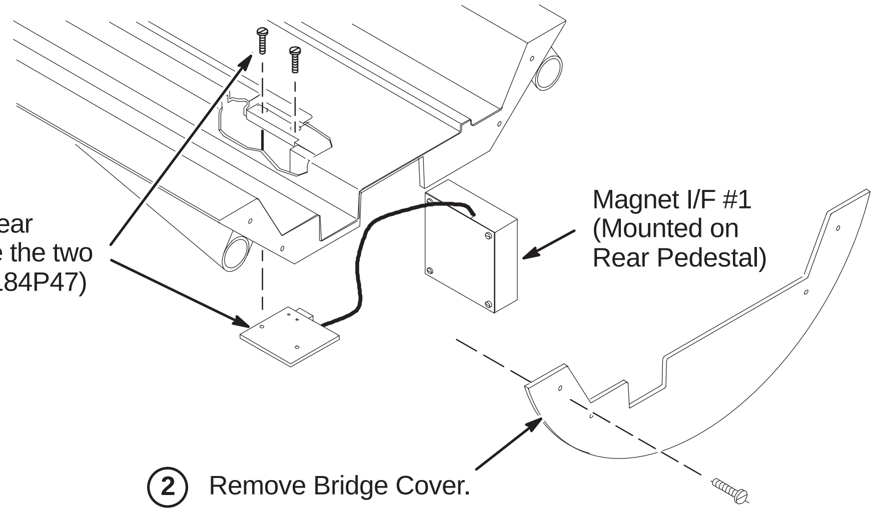
6–1–5 DE–INSTALL LONG BRIDGE FOR TwinSpeed

6–1–5–A Unfasten Front of Long Bridge



1. Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867–17) and Support Bushings and save. These will be reused for installing the new Bridge.

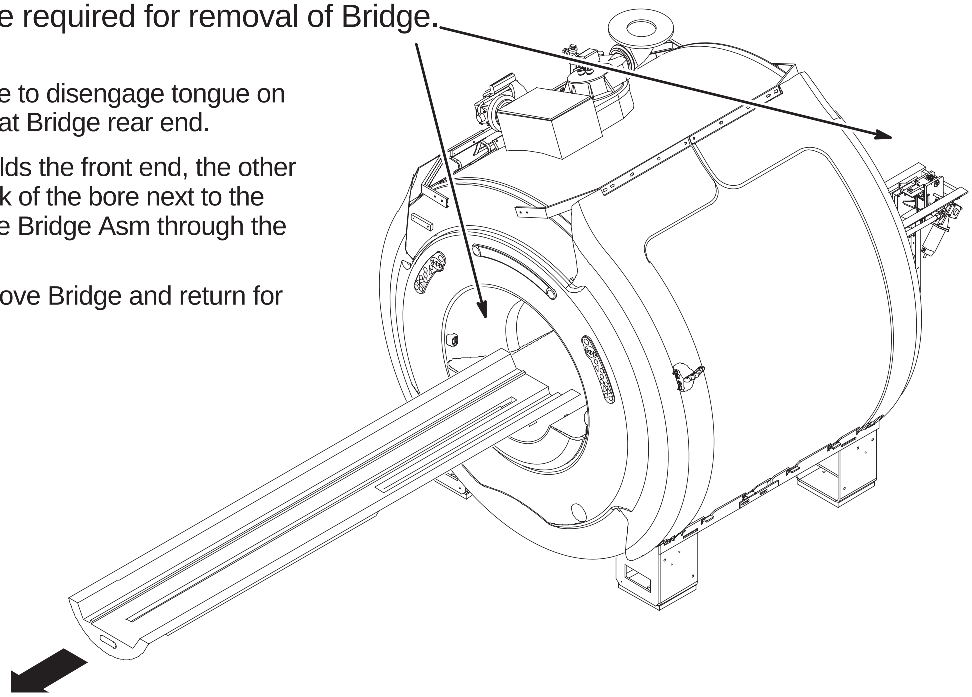
6–1–5–B Disconnect Bridge Mount Asm



1. Unfasten the Bridge, Tongue and Rear Bridge Mount Asm (2132034). Save the two (2) 10–32 x 1–3/8 Screws (46–220184P47) for refastening later.
2. Remove Bridge Cover.

6–1–5–C Remove Upgrade Long Bridge

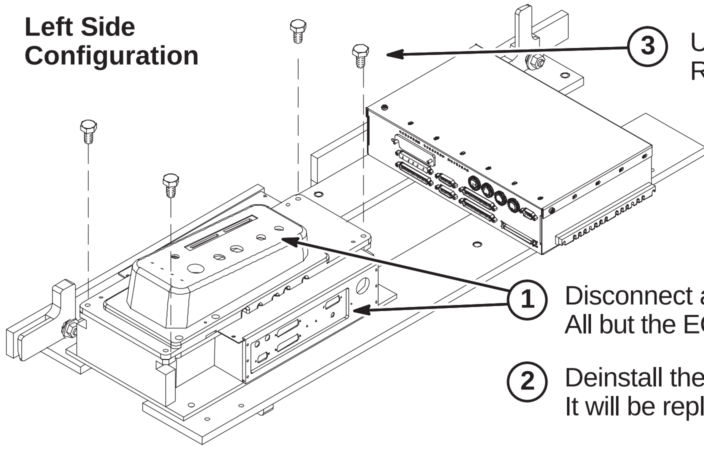
NOTE: Two people are required for removal of Bridge.



1. Lift front end of bridge to disengage tongue on long drive assembly at Bridge rear end.
2. While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore to the front end.
3. Carefully lift and remove Bridge and return for recycling.

6–1–6 DE–INSTALL PAC MODULE

Left Side Configuration

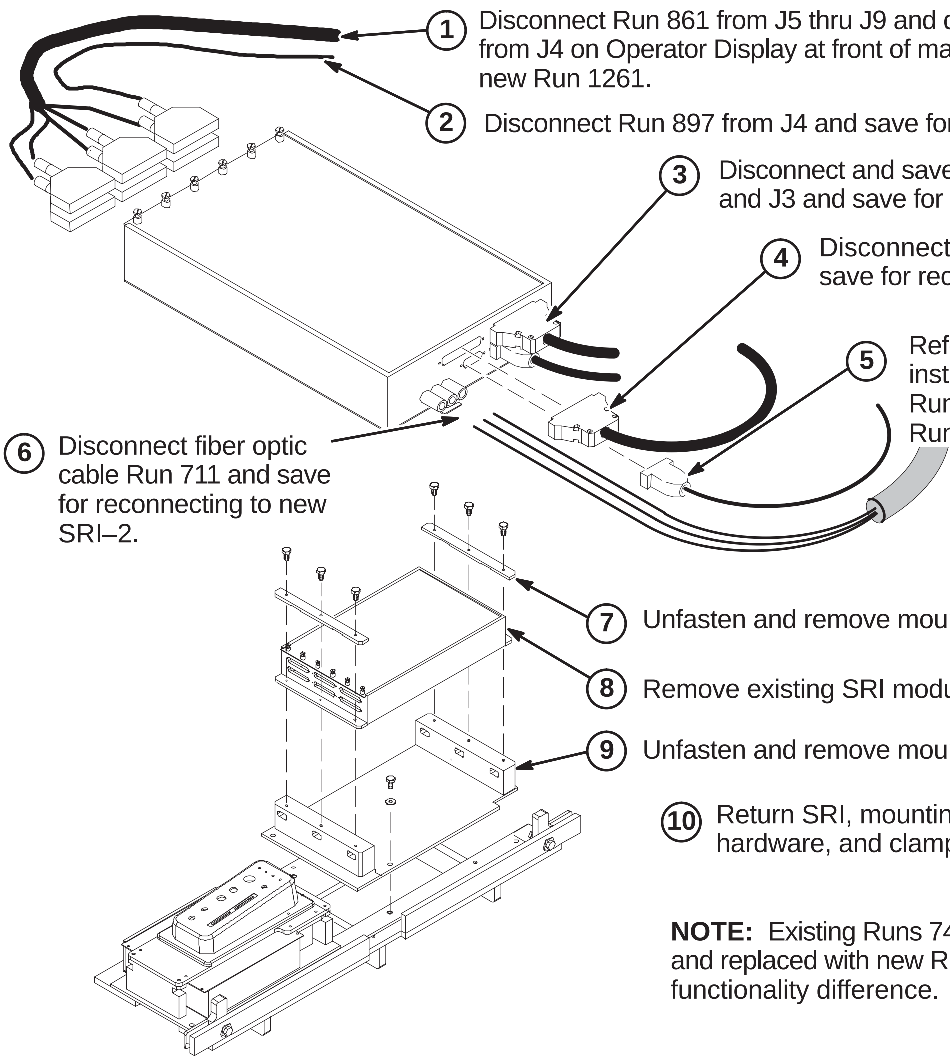


3 Unfasten 4 screws and remove PAC. Return to GoldSeal for recycling.

1 Disconnect all cables connected to existing SRI Module. All but the ECG Cable will be reconnected to new PAC.

2 Deinstall the ECG Cable and return to GoldSeal for recycling. It will be replaced by the new ECG Cable (5122500).

6–1–7 DE–INSTALL SRI MODULE



1 Disconnect Run 861 from J5 thru J9 and discard cable. Also disconnect from J4 on Operator Display at front of magnet. Cable will be replaced by new Run 1261.

2 Disconnect Run 897 from J4 and save for reconnecting to new SRI–3.

3 Disconnect and save Runs 421 and 422 from J4 and J3 and save for reconnecting to new SRI–2.

4 Disconnect Run 715 from J14 and save for reconnecting to new SRI–3.

5 Refer to Section **10–4** for instructions on removing Run 749 and replacing with Run 1260.

6 Disconnect fiber optic cable Run 711 and save for reconnecting to new SRI–2.

7 Unfasten and remove mounting hardware and clamps.

8 Remove existing SRI module.

9 Unfasten and remove mounting hardware and SRI platform.

10 Return SRI, mounting platform, hardware, and clamps for recycling.

NOTE: Existing Runs 749 and 1011 must be removed and replaced with new Runs1260 and 1261. There is a functionality difference.

6–2 INSTALLATION OF TwinSpeed SPECIFIC PARTS (TRM ONLY)

This upgrade section is performed only on a TwinSpeed system which uses a TRM coil. The Twin RF Body Coil will be replaced and the Manifold Protector Plate will be installed.

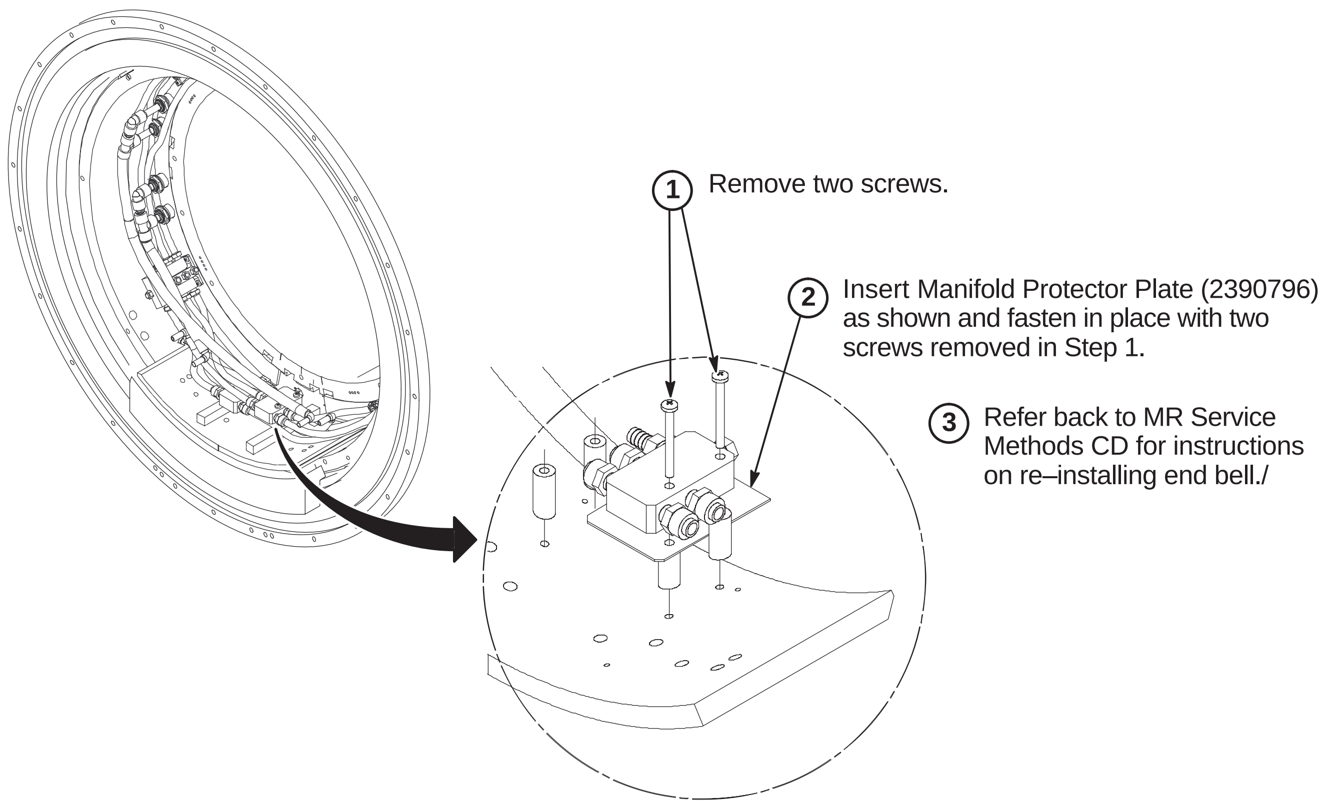
6–2–1 1.5T TwinSpeed RF BODY COIL REPLACEMENT

For replacement procedure, refer to applicable MR Service Methods CD–ROM and go to:

Replacements/1.5T Twin RF Body Coil Replacement Procedure.

When front End Bell is removed, perform Procedure 6–2–2 for installing the Manifold Protector plate.

6–2–2 INSTALL MANIFOLD PROTECTOR PLATE



6–3 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) has been installed. This kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It would be required for earlier versions of the WideOpen style enclosure. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and it must be installed at this point of the EXCITE HD ugprade.

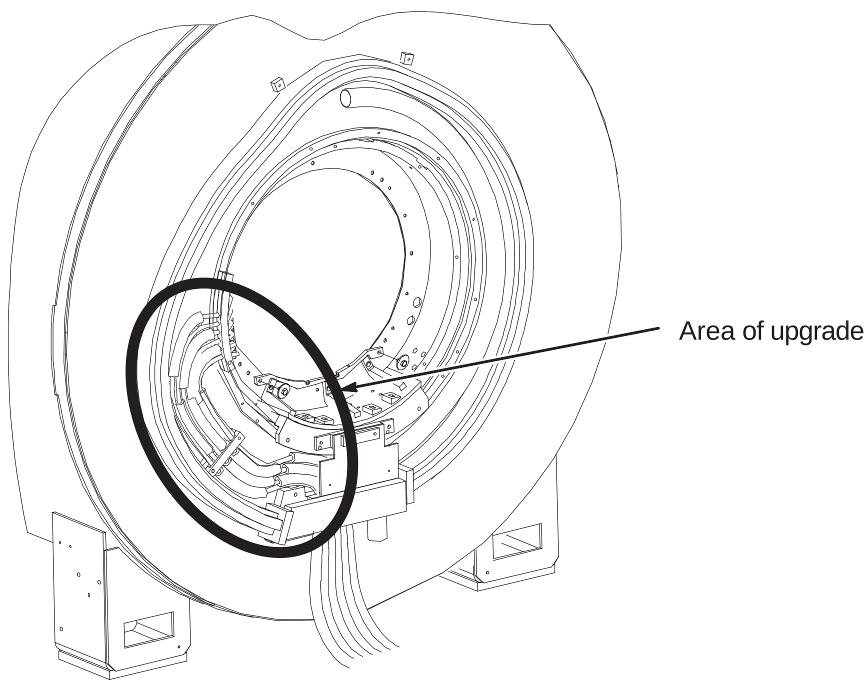
Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2318581, *Signa Cx/K4 Magnet WideOpen Enclosure Bridge Support* and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.

Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade,* for instructions on installing this kit.

6–4 INSTALL NEW COMPONENTS AND CABLES

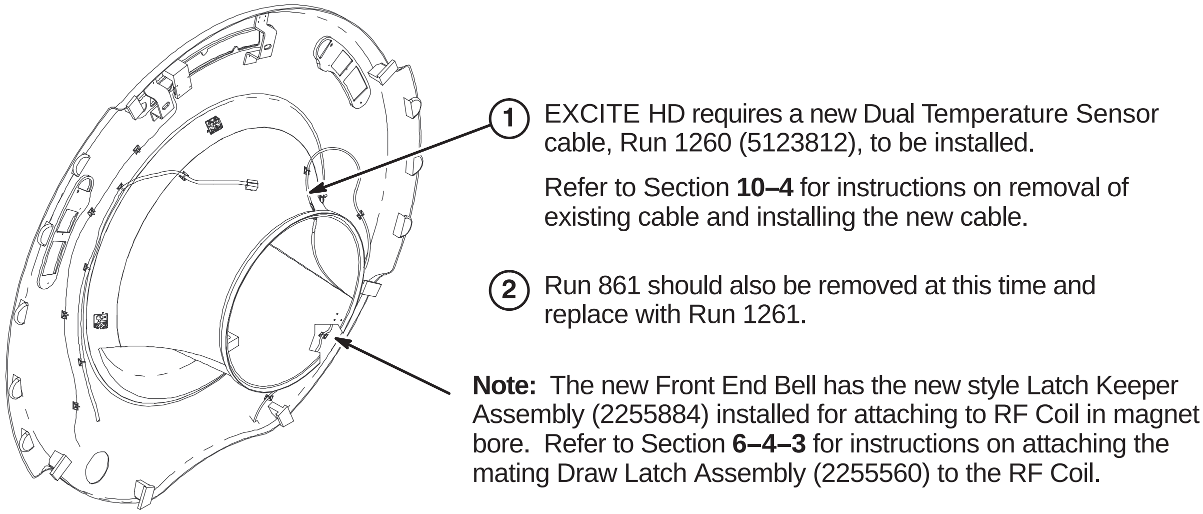
6–4–1 INSTALL BRM GRADIENT CABLE LEAD BOARD (MR/i or CV/i Systems Only)

The mounting of the gradient cables at the rear of the magnet must be upgraded. Refer to Direction 2404642, provided with Gradient Cable Lead Board collector (5110336), for instructions on how to install the new design.

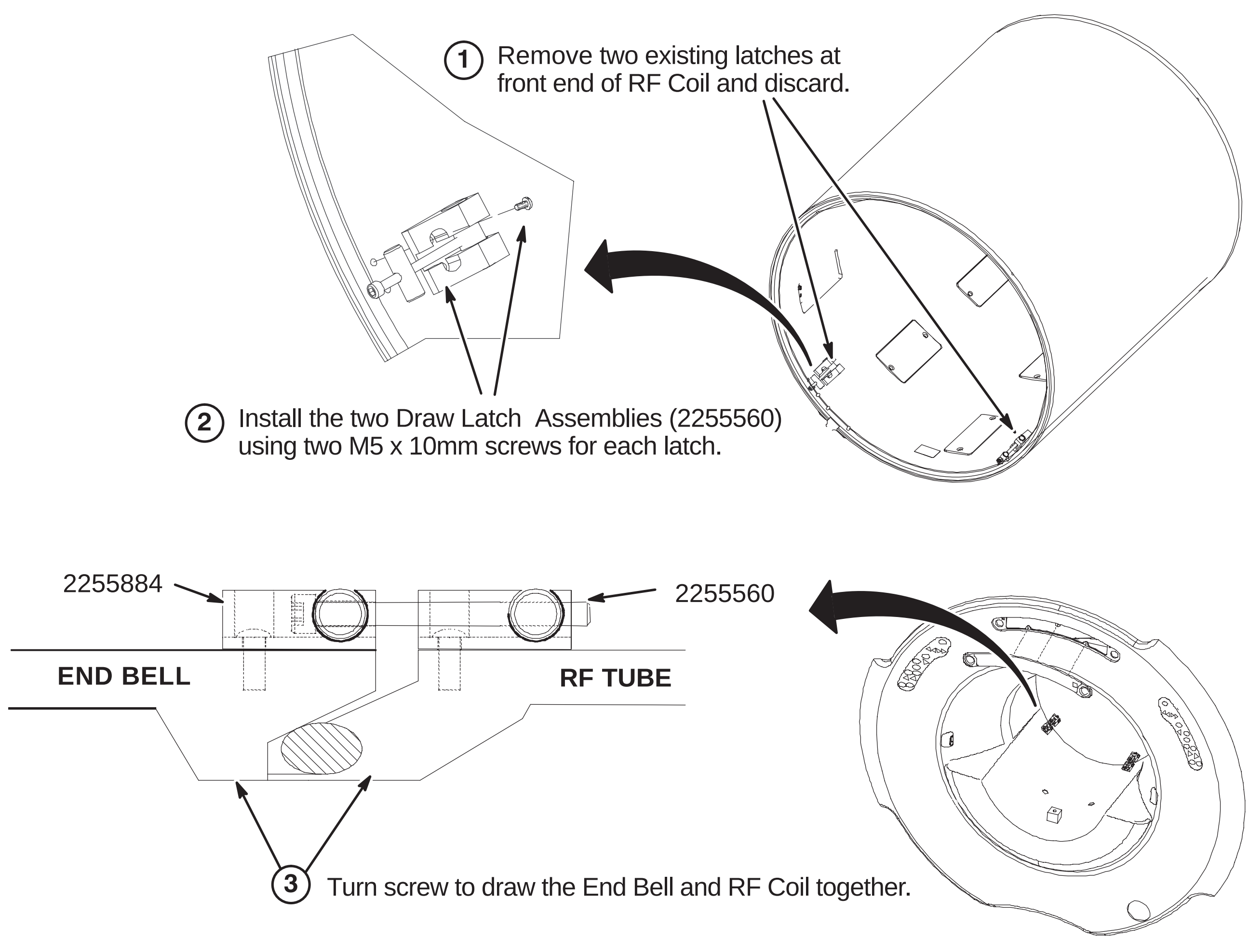


6–4–2 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL

This procedure must be completed if existing or new Front End Bell does not have Run 1260 installed on it. To verify cable installed, check cable attached to J10 of SRI Module. It must be Run 1260.

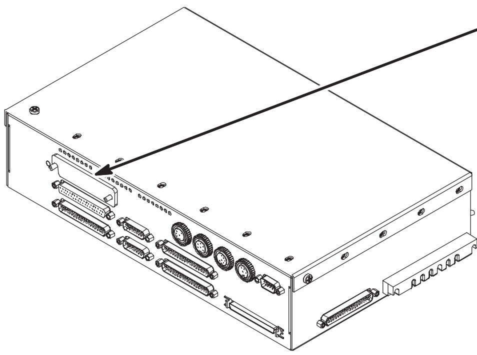


6–4–3 INSTALL LATCH KEEPER ASSEMBLY ON RF COIL

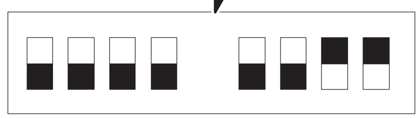


6–4–4 INSTALL PAC AND SRI MODULES

6–4–4–A Verify Jumper Positions for SRI



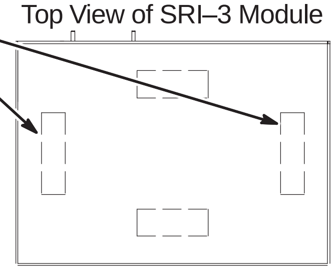
1. Slide cover plate away on new SRI–3 (5105270) and verify jumper settings for 1.5T with Coil ID.
2. For the eight jumpers, the two jumpers on the right should be in the “ON” or located in the up position. All others should be “OFF” or down.



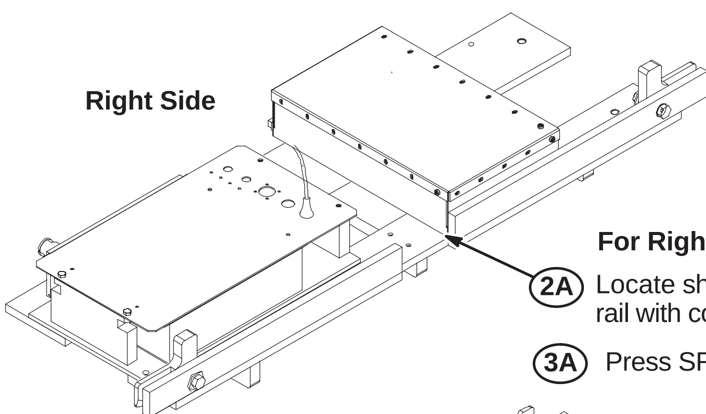
6–4–4–B Install New PAC and SRI On Right or Left Side of Enclosure

1. On bottom of SRI Module, remove protective cover from Velcro at these two locations.

Top View of SRI–3 Module



Right Side



For Right Side Configuration

2A. Locate short side of SRI against the vertical rail with corner aligned to end of rail.

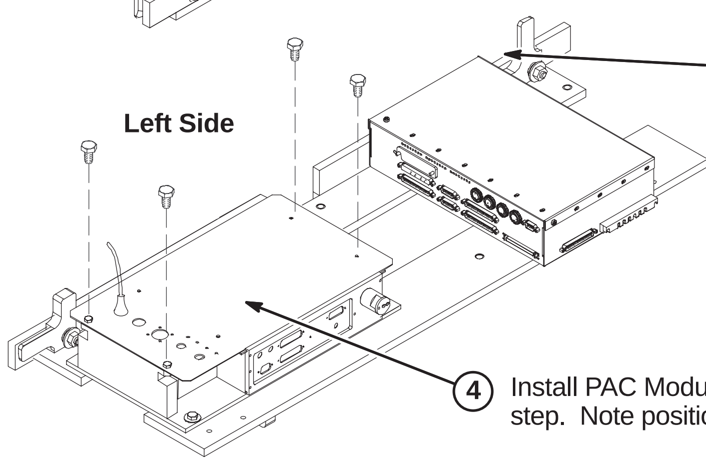
3A. Press SRI module into place.

For Left Side Configuration

2B. Place short side of SRI against the vertical rail with corner located 25mm (1 inch) from latch.

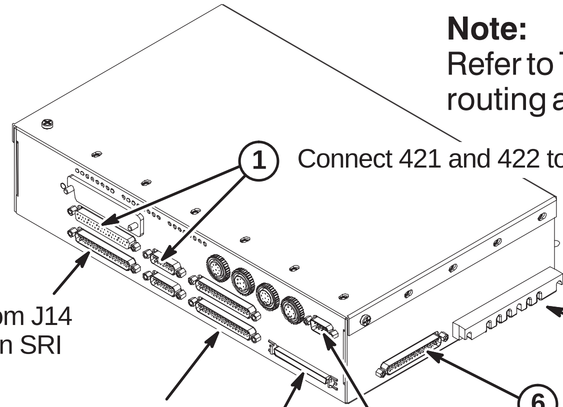
3B. Press SRI module into place.

Left Side

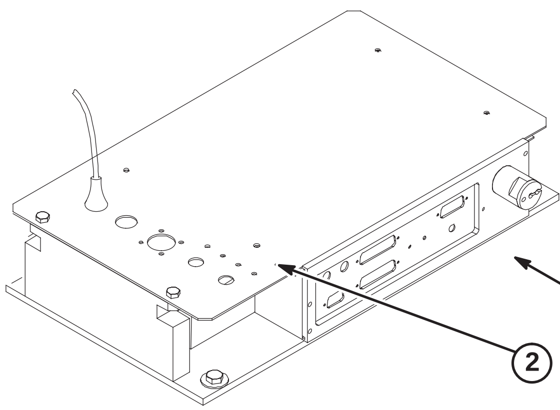


4. Install PAC Module (5112668–2) using fasteners removed in previous step. Note position of module for left or right side configuration.

6–4–4–C Connect Cables to SRI–3 Module

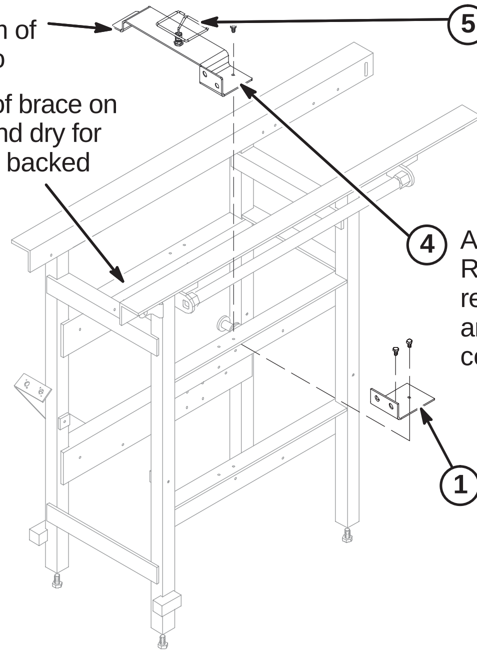


6–4–4–D Connect Cables to PAC Module



6–4–5 INSTALL NEW COMPONENTS ON REAR PEDESTAL

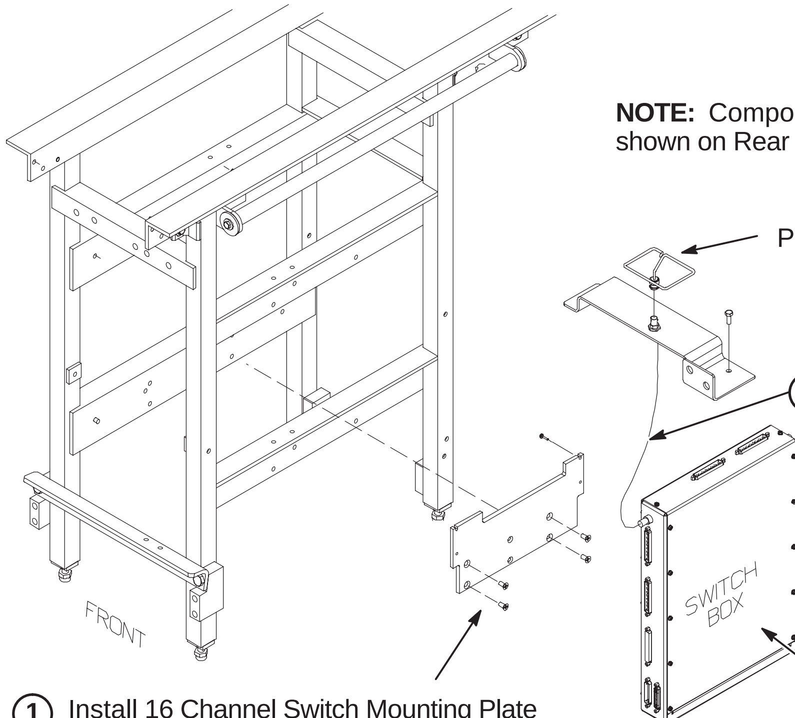
6–4–5–A UTNS Antenna Installation



3. Peel of liner from bottom of adhesive backed Velcro
2. Make sure top surface of brace on opposite side is clean and dry for attachment of adhesive backed Velcro.
5. Attach UTNS Antenna (2355200) to bracket.
4. Attach UTNS Antenna Bracket (2384445) to Rear Pedestal using mounting hardware removed in step 1 on this side of Rear Pedestal and Velcro on other side. Install BNC cable connectors.
1. At existing Mounting Bracket, disconnect Runs 743 and 744 and cables from RF Coil. Remove and save BNC cable connectors. Remove and discard bracket. Save mounting hardware for installation of new UTNS Antenna bracket.

6–4–5–B Install 16 Channel Switch

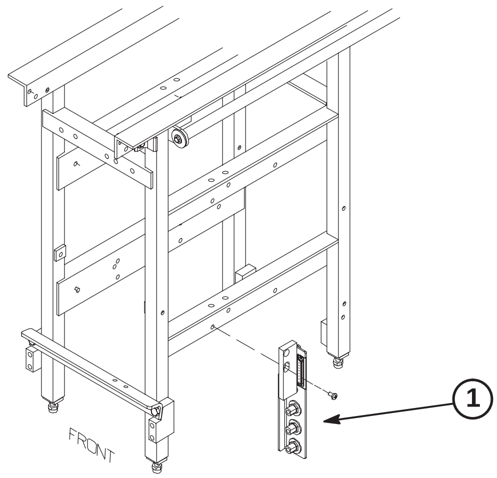
NOTE: Components and cables not shown on Rear Pedestal for clarity.



Part of Rear Pedestal

3. Connect Run 1204 (5116105) as shown.
1. Install 16 Channel Switch Mounting Plate (5110428) using Loctite 242 (46–170686P2) and four M6 x 12mm Slotted Flat Head Screws (25120369–86) in the four larger outside holes.
2. Attach 16 Channel Switch (5109645) to Mounting Plate using Loctite 242 (46–170686P2) and four M3 x 12mm Pan Head screws (46–312342P24).

6–4–5–C Install New Bulkhead Interface



6–4–6 HYBRID SPLITTER UPGRADE

Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

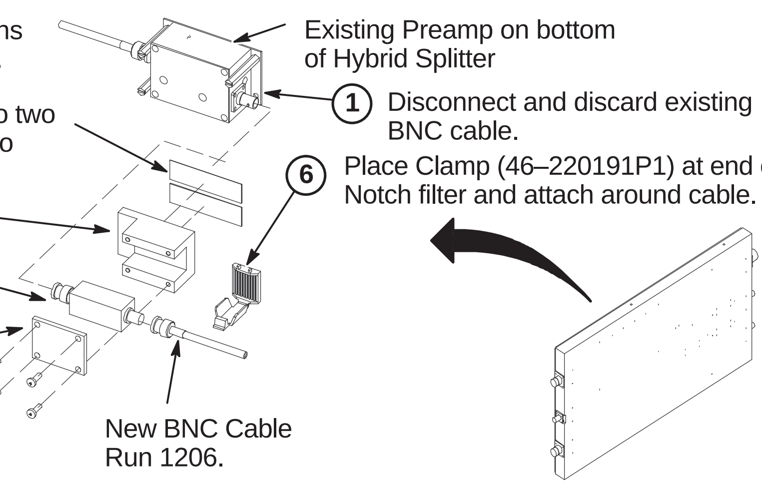
- Refer to Section 6–4–6–A for instructions on installing the Notch Filter and mounting block
- Refer to Section 6–4–6–B for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 6–4–6–C for instructions on installing Isolation Filters on systems with Teledyne Splitter

The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils. Do NOT install on systems with TRM gradient coils (TwinSpeed).

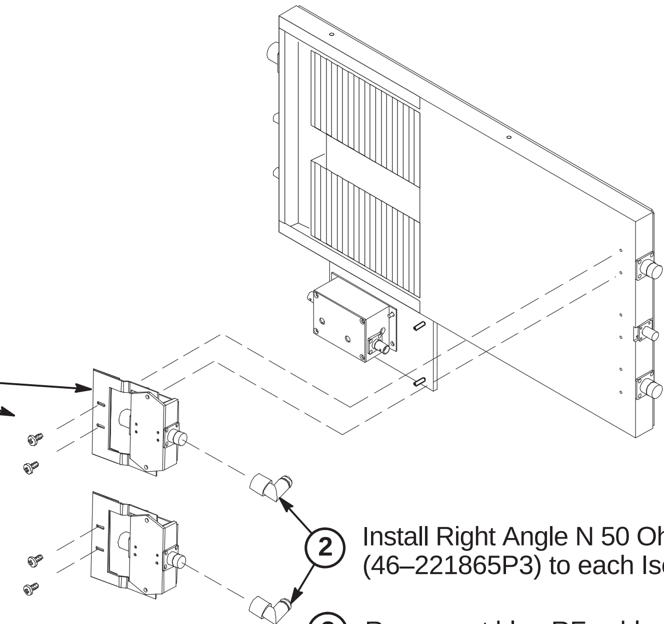
6–4–6–A Install Notch Filter and Filter Mount Block Asm

This Filter Mount assembly may not be required on newer Netcom Hybrid Spitter as it may already have a Notch Filter mounting clamp of a different style installed.

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.



6–4–6–B Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)



FOR BRM's ONLY

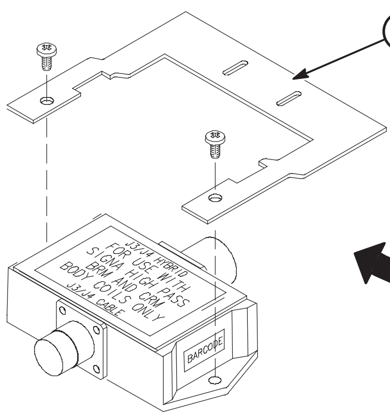
1. Attach and connect each Ground Isolation Filter Module (5118952–2) to the Hybrid Splitter using two screws (46–220184P5).

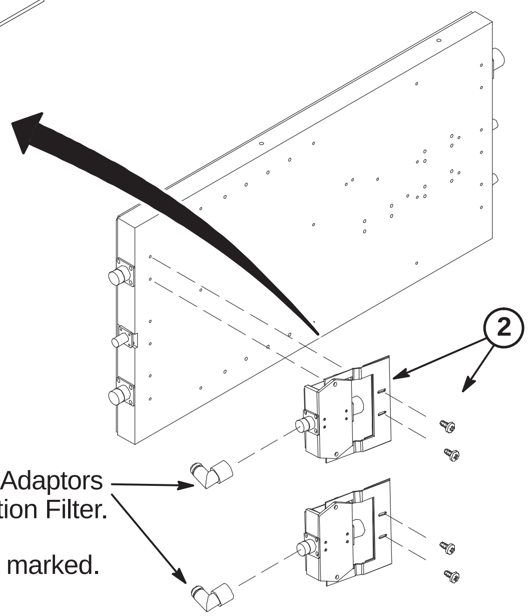
   Do NOT install on TRM gradient coil systems.

2. Install Right Angle N 50 Ohm Adaptors (46–221865P3) to each Isolation Filter.

3. Reconnect blue RF cables as marked.

6–4–6–C Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)





1. For Teledyne Hybrid Splitter only, remove existing bracket that is attached to Isolation Filter and replace with Mounting Bracket Assembly (5237198–2) provided with upgrade kit. (Bracket by itself is stamped with 2254629)

FOR BRM's ONLY

2. Attach and connect each Ground Isolation Filter Module (5118952) to the Hybrid Splitter using two screws (46–220184P5) for each module.

   Do NOT install on TRM gradient coil systems.

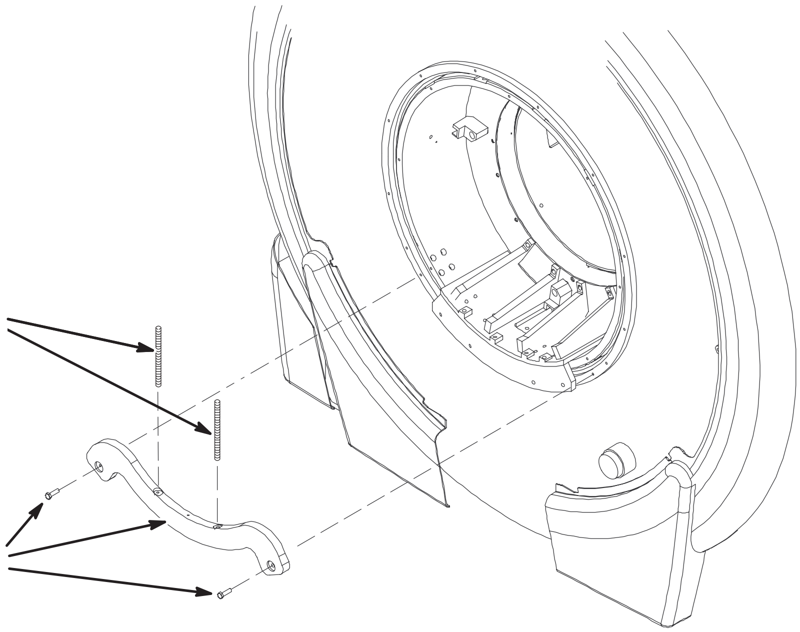
3. Install Right Angle N 50 Ohm Adaptors (46–221865P3) to each Isolation Filter.

4. Reconnect blue RF cables as marked.

6–4–7 INSTALL FRONT END BELL

The following procedures must be completed if new Front End Bell is installed on WideOpen enclosure that had an existing Long Bridge removed for this upgrade.

6–4–7–A Install Split Bridge Support

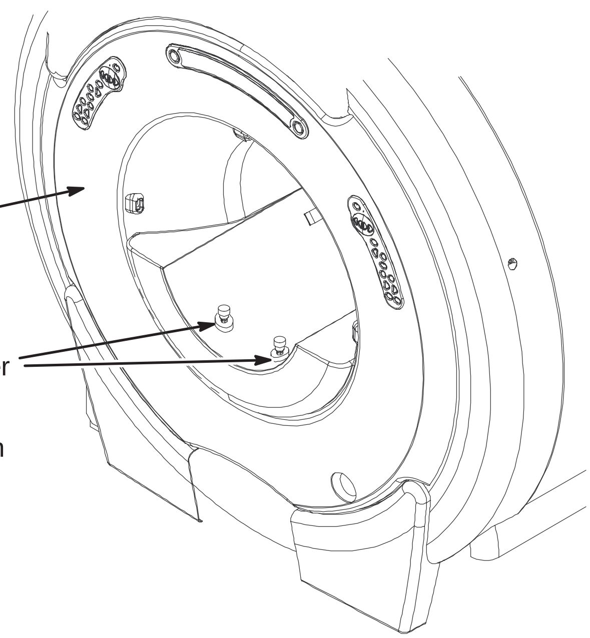


2 Install two M10 x 165mm Studs into top of bracket as shown.

1 Install new Split Bridge Support (2293379)on front ring with two M10 x 30mm hex Head Screws.

6–4–7–B Install Front End Bell

NOTE:
Make sure new Front End Bell has Dual Tempreature Sensor cable, Run 1260, attached before installation**.**

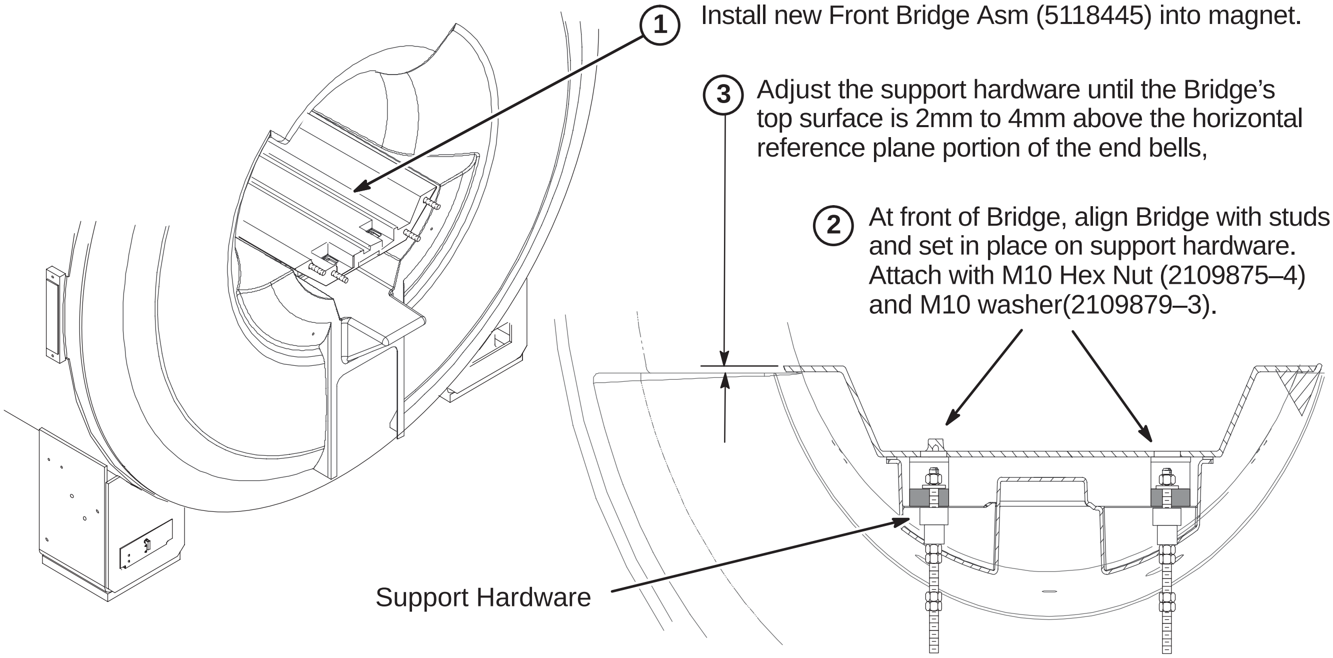


1 Install new Front End Bell (2306247) by reversing the de–installation procedures in Sections **6–1–3–C** and **6–1–3–D**.

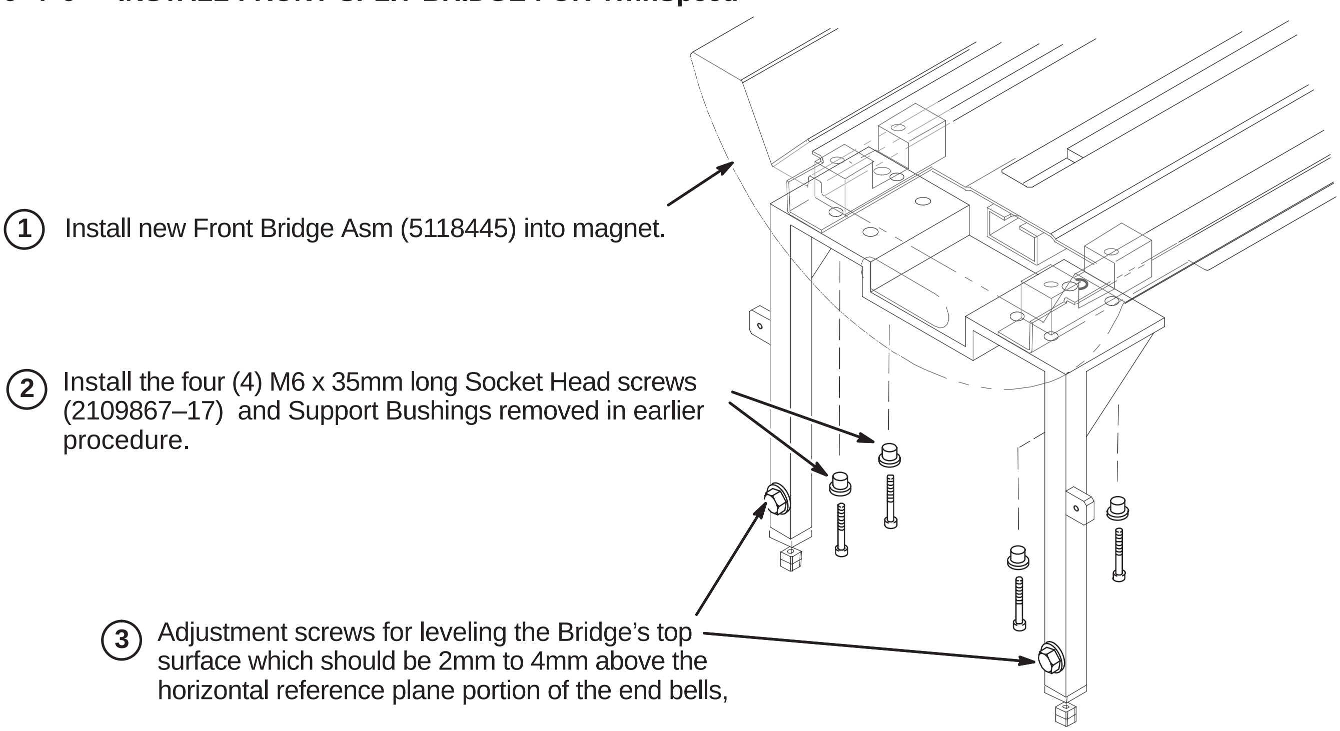
2 Remove PSA backing of each Foam Gasket (2226752–2) and stretch over stud. Center over end bell access hole and secure to end bell.

3 Install one Bridge Adjustment Nut (2293381) on each stud.

6–4–8 INSTALL FRONT SPLIT BRIDGE FOR WideOpen

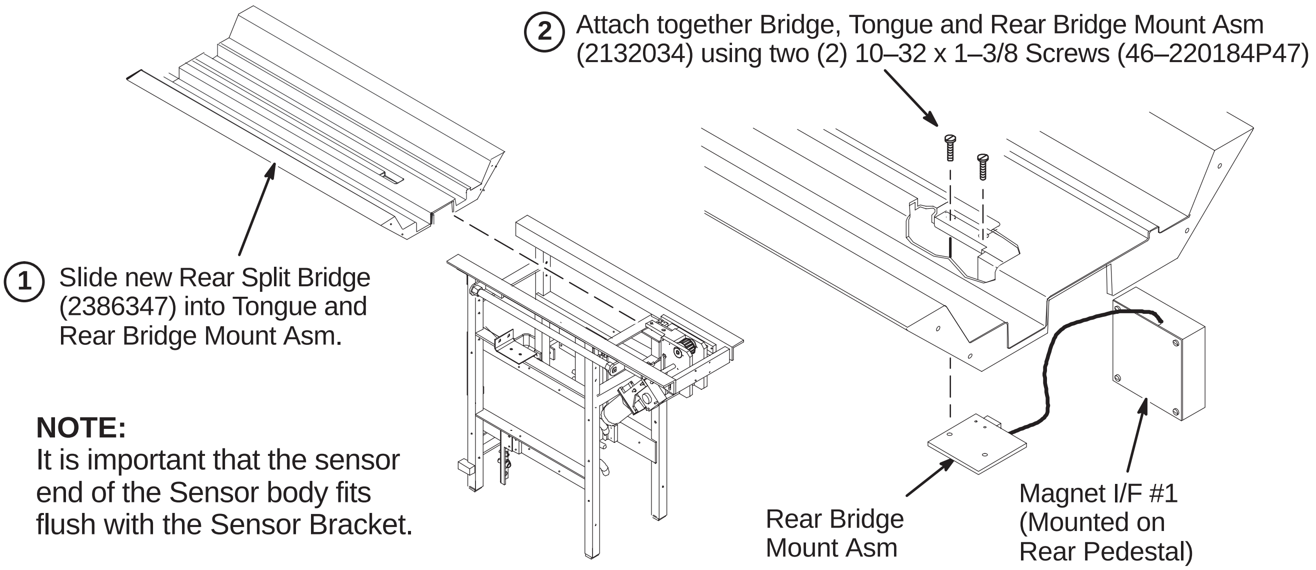


6–4–9 INSTALL FRONT SPLIT BRIDGE FOR TwinSpeed

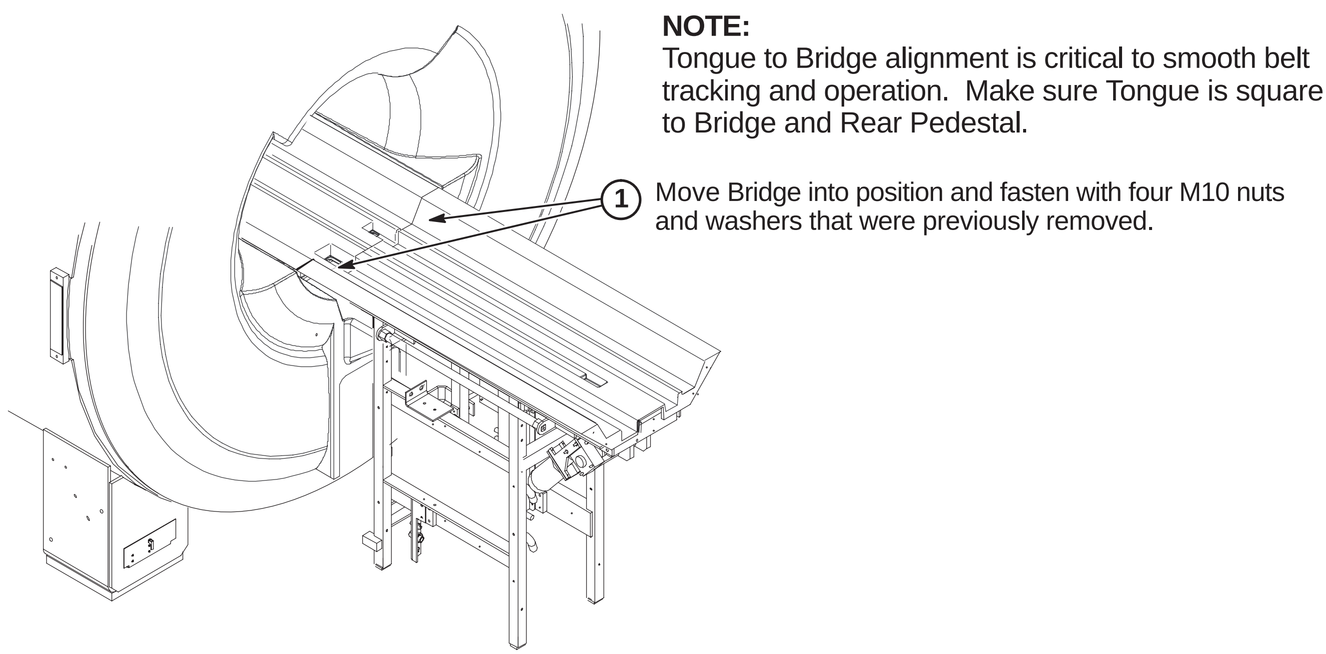


6–4–10 INSTALL REAR SPLIT BRIDGE

6–4–10–A Install Rear Split Bridge on Rear Pedestal



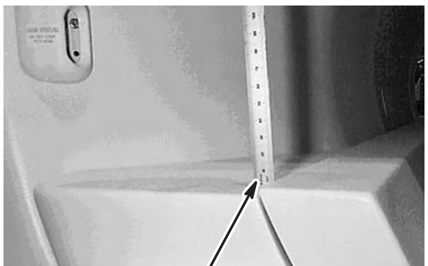
6–4–10–B Connect Rear Split Bridge to Front Split Bridge



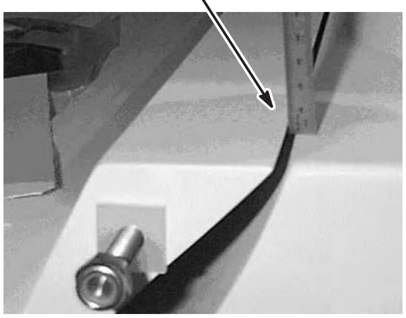
6–4–11 LEVEL BRIDGE

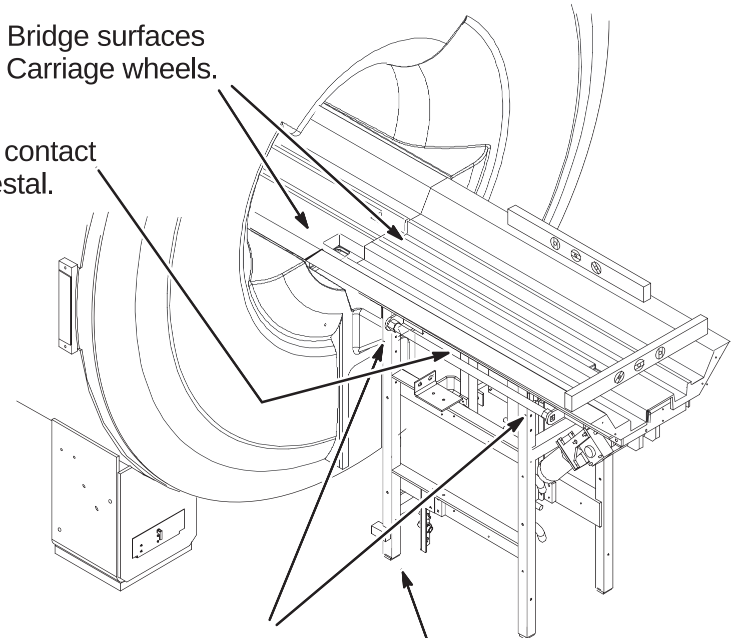
NOTE:
Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:
Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.



2mm to 4mm above End Bell surface.





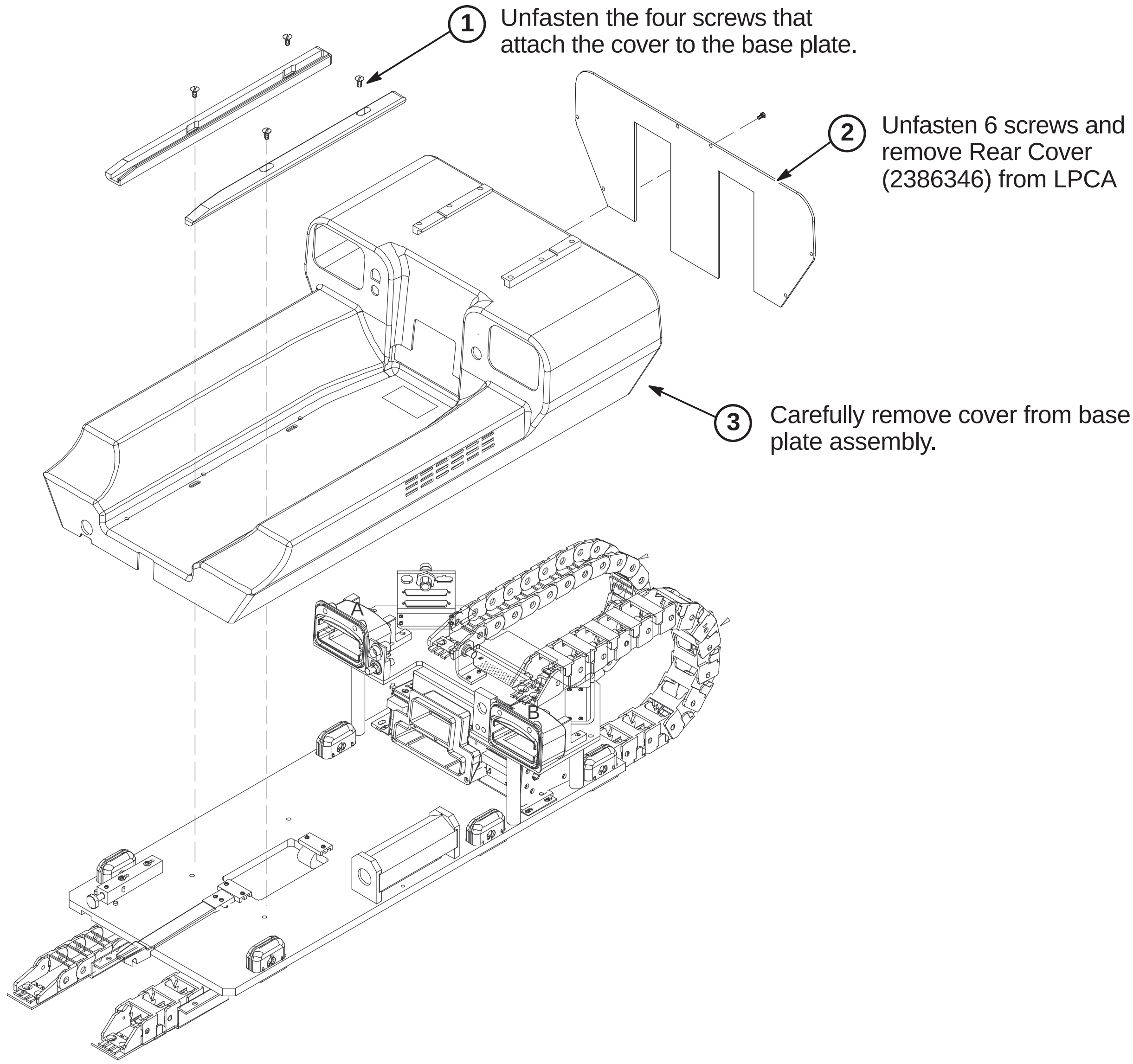
1. Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of front and rear End Bells, as indicated in photos at left. Tighten Vertical Adjustment Fasteners.

2. Adjust leg screws on both sides as needed to level Rear Pedetal.

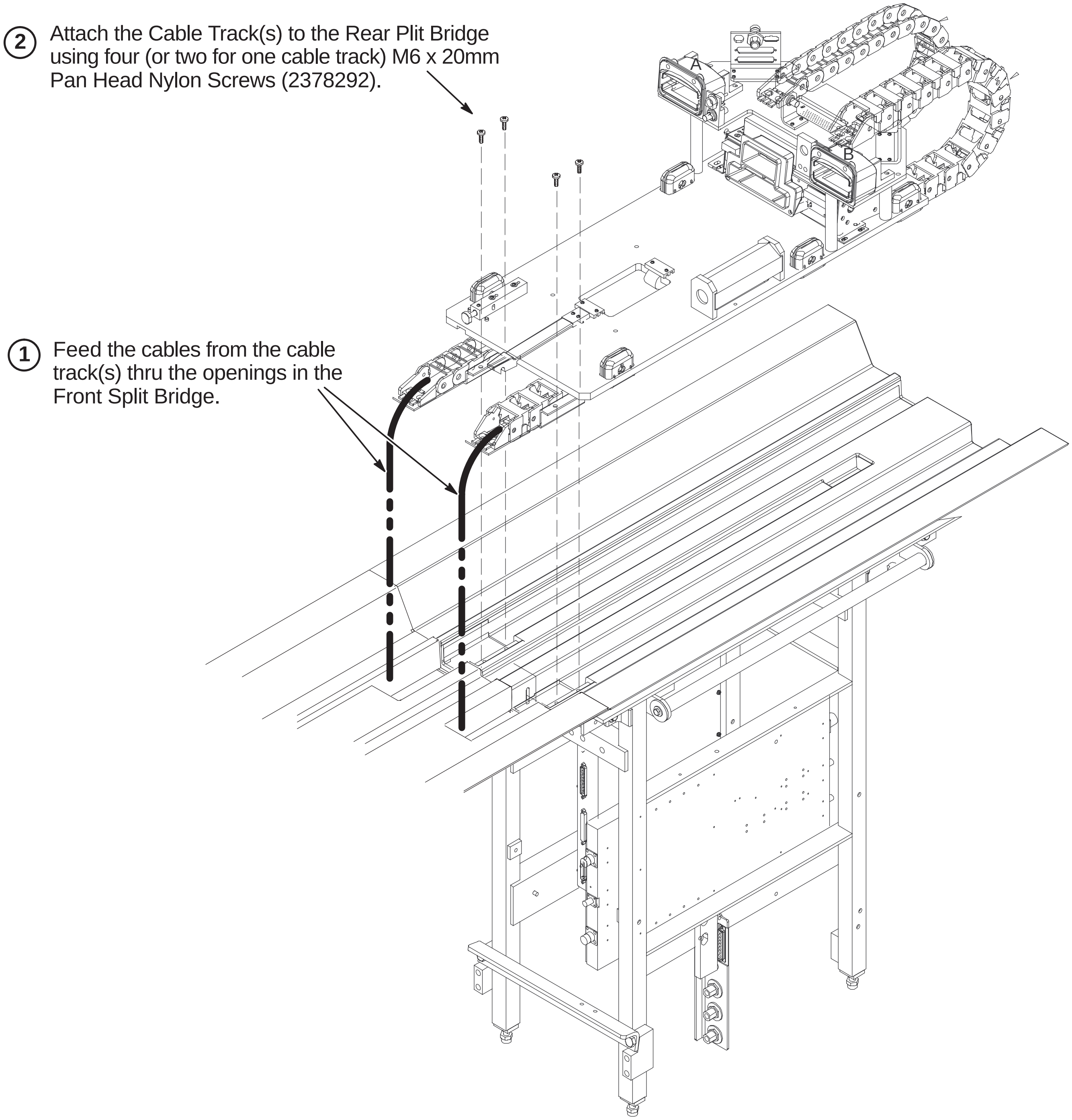
6–4–12 INSTALL LPCA AND CABLE TRACKS

The WideOpen and TwinSpeed style enclosures will be upgraded to an LPCA with a dual cable track. The S–Series and Cx magnet enclosures will install a new LPCA with only one cable track.

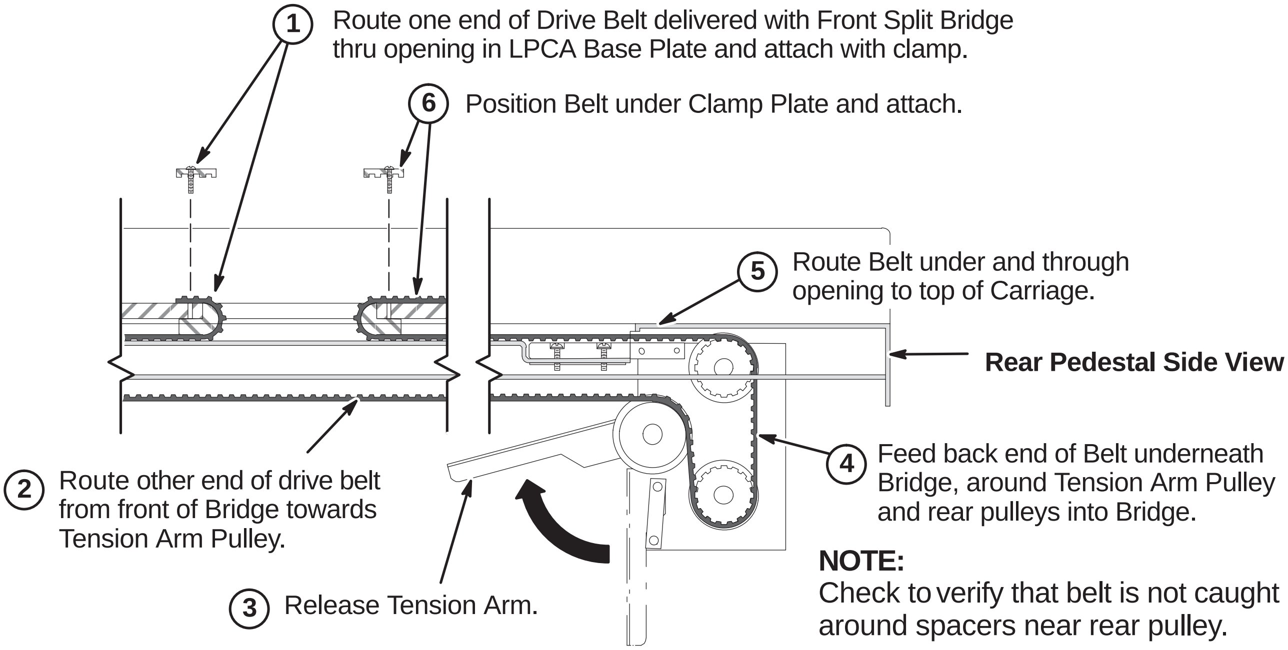
6–4–12–A Remove LPCA Cover



6–4–12–B Attach Cable Tracks to Bridge

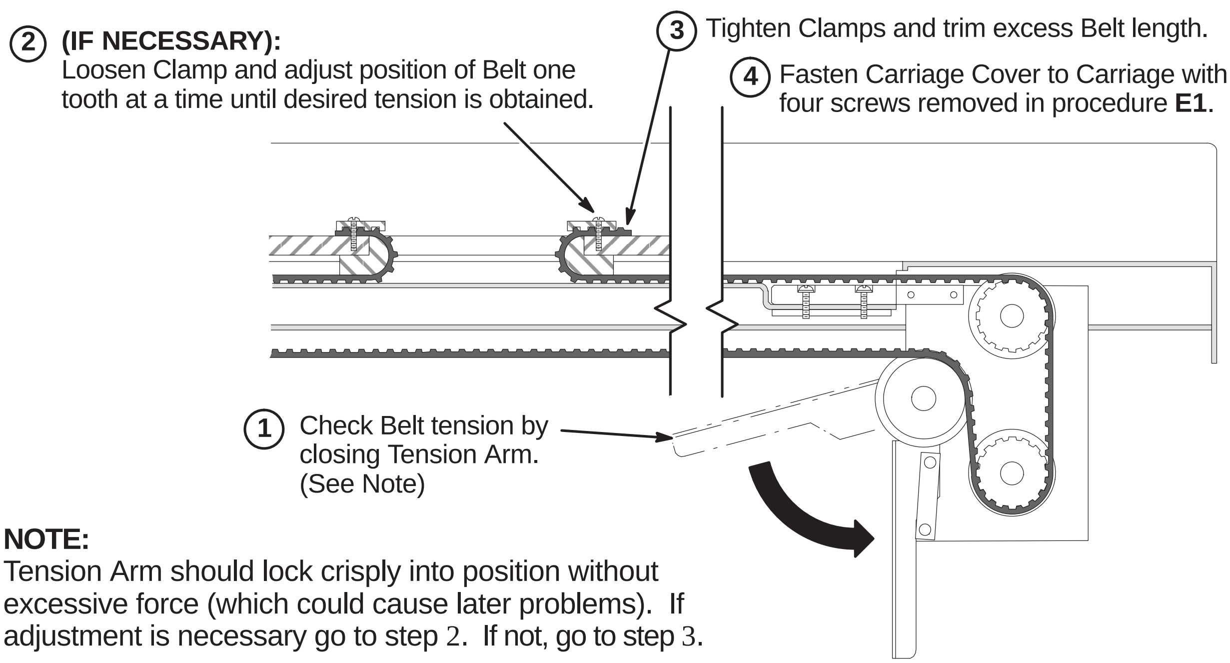


6–4–12–C Route and Attach Drive Belt



NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

6–4–12–D Final Adjustment of Drive Belt



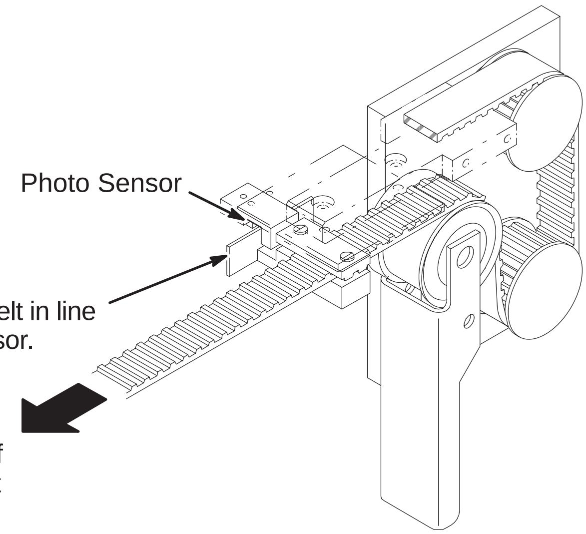
NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

6–4–12–E Installation of Sensor Flag

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

1. Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

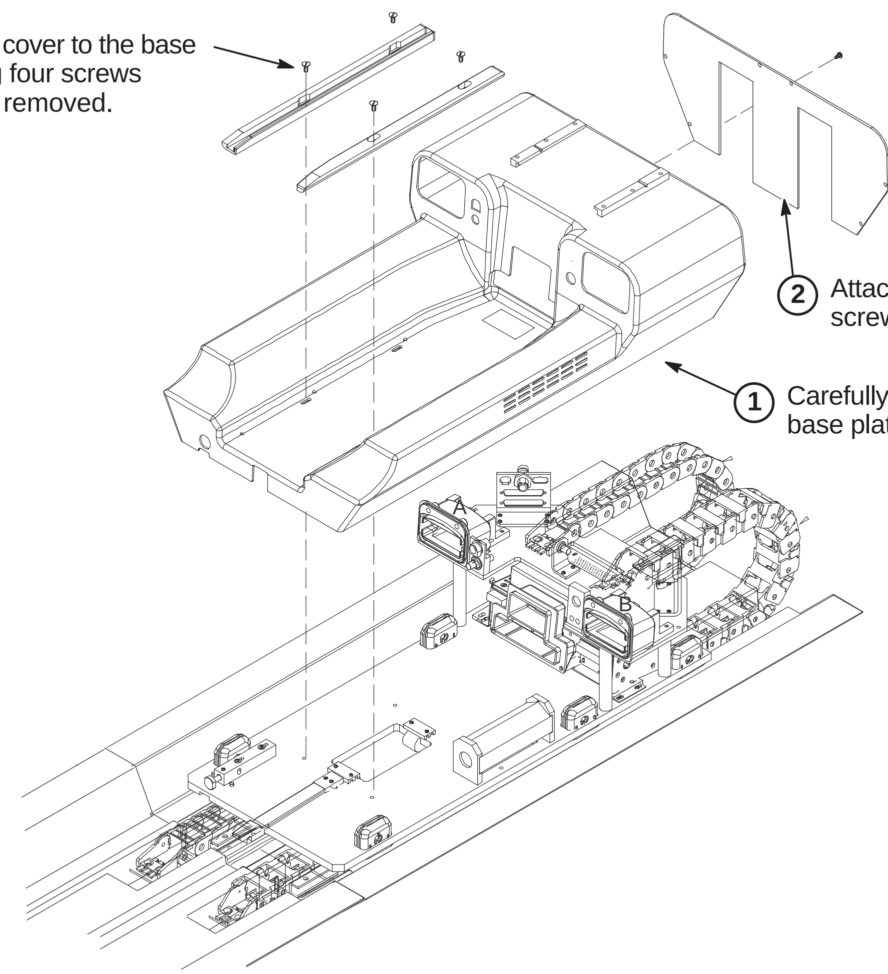


6–4–12–F Installation of LPCA Cover

3. Attach the cover to the base plate using four screws previously removed.

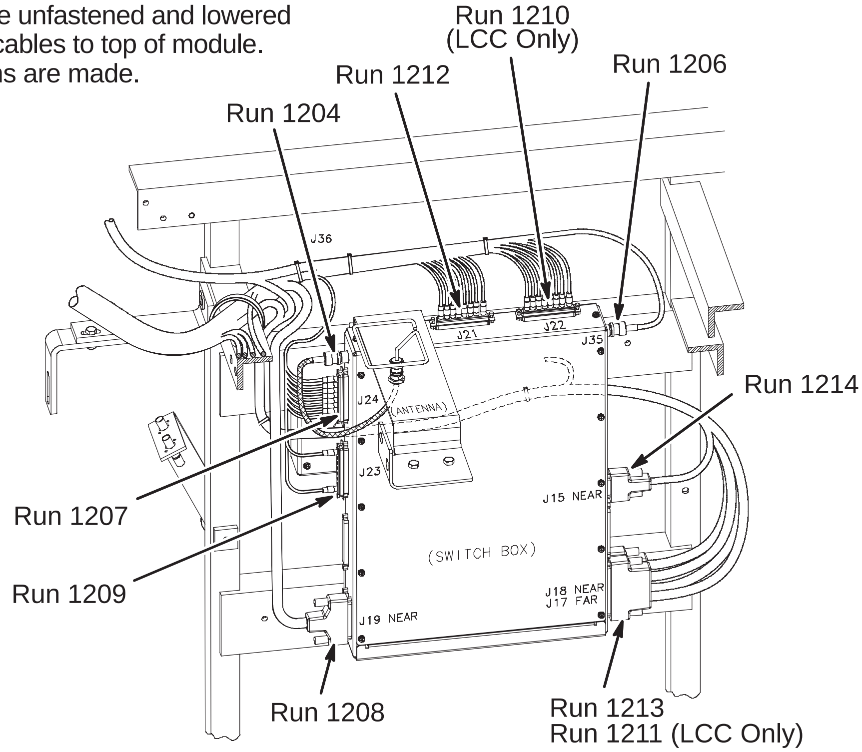
2. Attach Rear Cover using 6 screws previously removed.

1. Carefully place cover on base plate assembly.



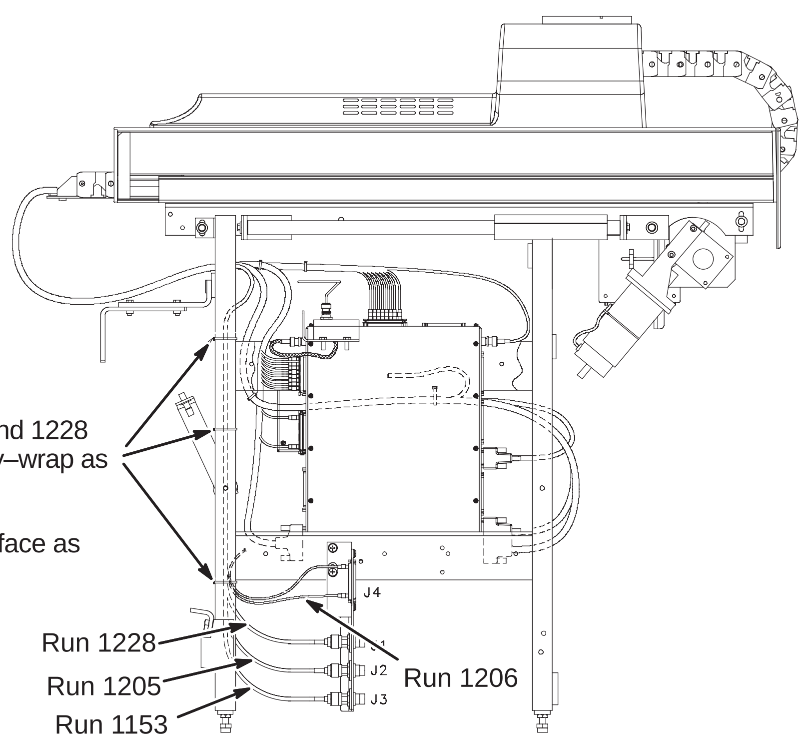
6–4–12–G Connect Cables from Cable Track to 16 Channel Switch Module

1. 16 Channnel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re–mount module after connections are made.
2. Connect cable Runs as indicated.



6–4–12–H Connect Cables from Cable Track to Bulkhead Interface

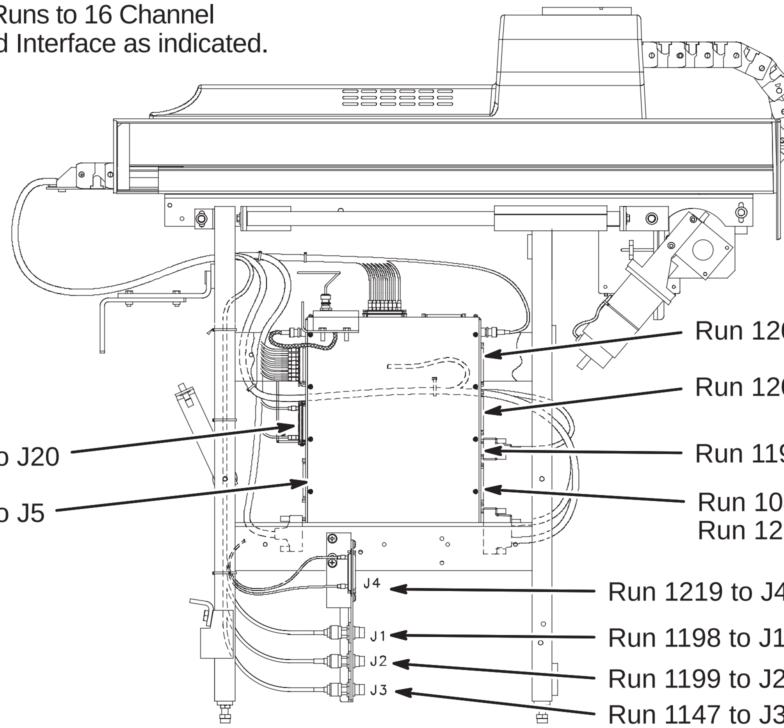
1. Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty–wrap as indicated to leg.
2. Connect Runs to Bulkhead Interface as indicated.



6–4–13 CONNECT REMAINING CABLES TO REAR PEDESTAL

6–4–13–A Cable Connections to 16 Channel Switch and Bulkhead Interface

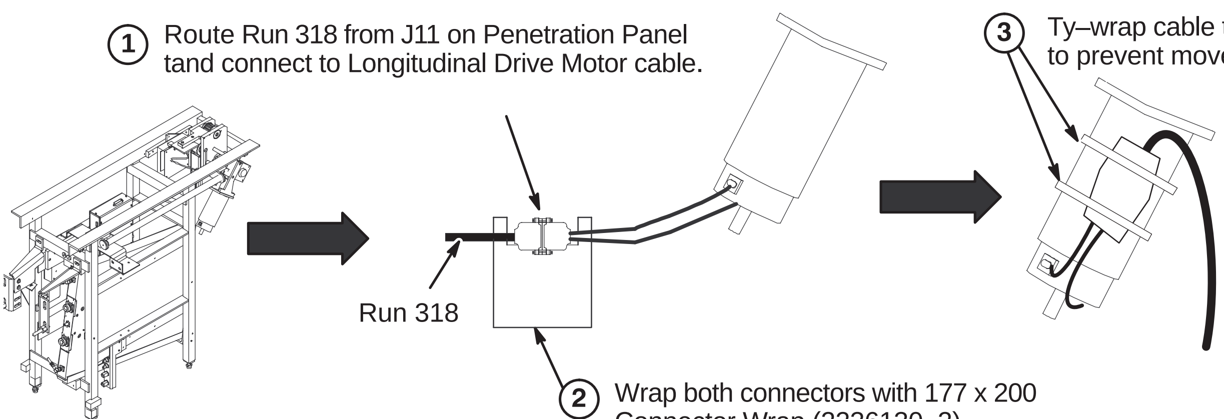
1 Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



6–4–13–B Route/Connect Run 318 to Longitudinal Drive Motor

1 Route Run 318 from J11 on Penetration Panel tand connect to Longitudinal Drive Motor cable.

3 Ty–wrap cable to motor to prevent movement.



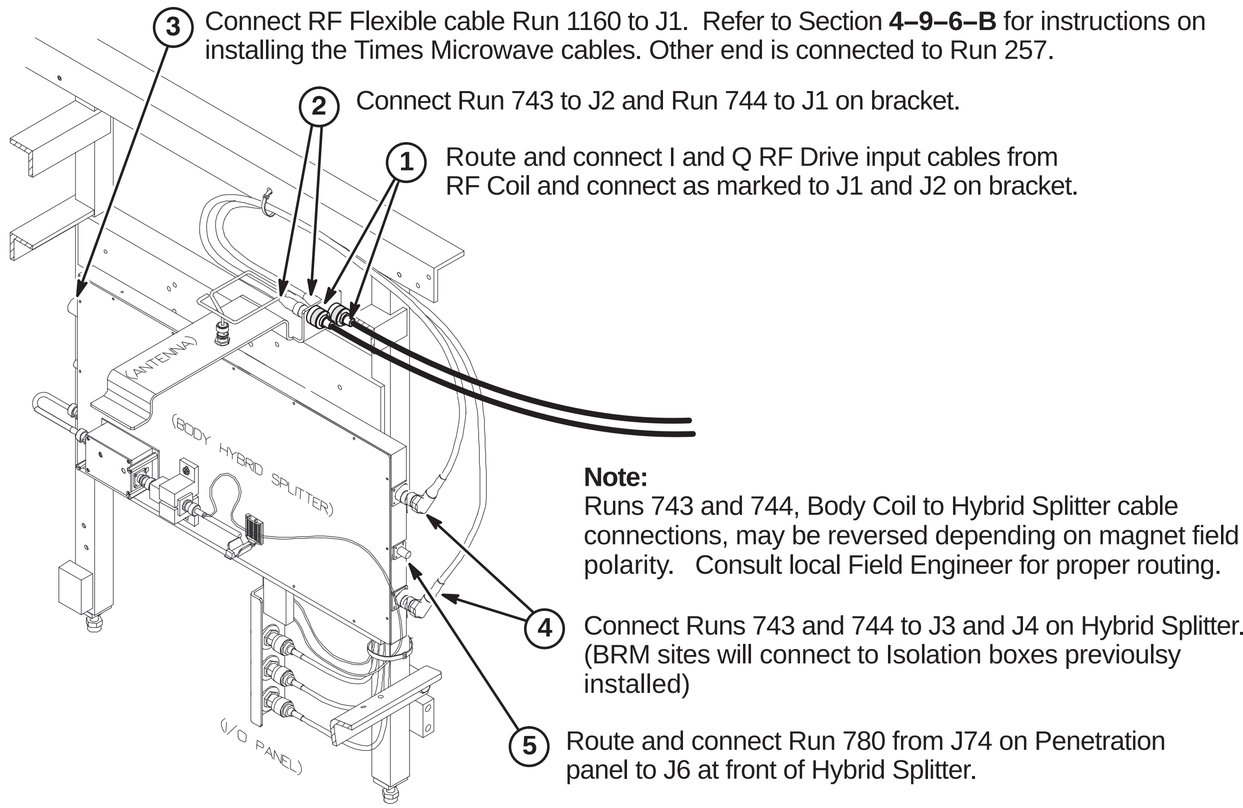
2 Wrap both connectors with 177 x 200 Connector Wrap (2236130–2).

6–4–13–C Connect Cables to Hybrid Splitter

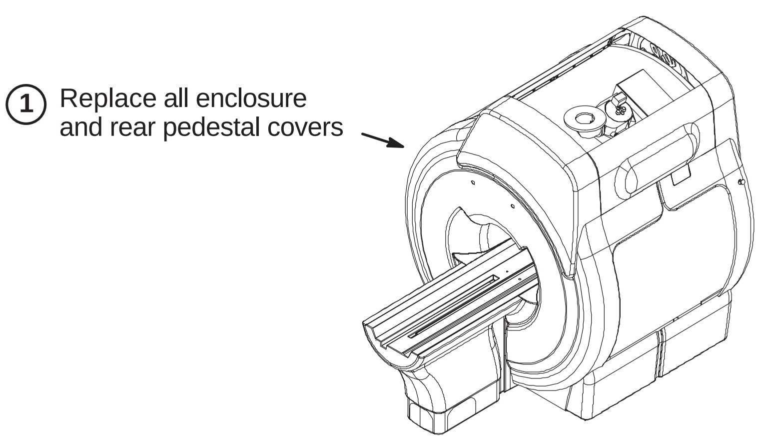
CAUTION

The RF Drive Input cables utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cables to the RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



6–5 INSTALL COVERS

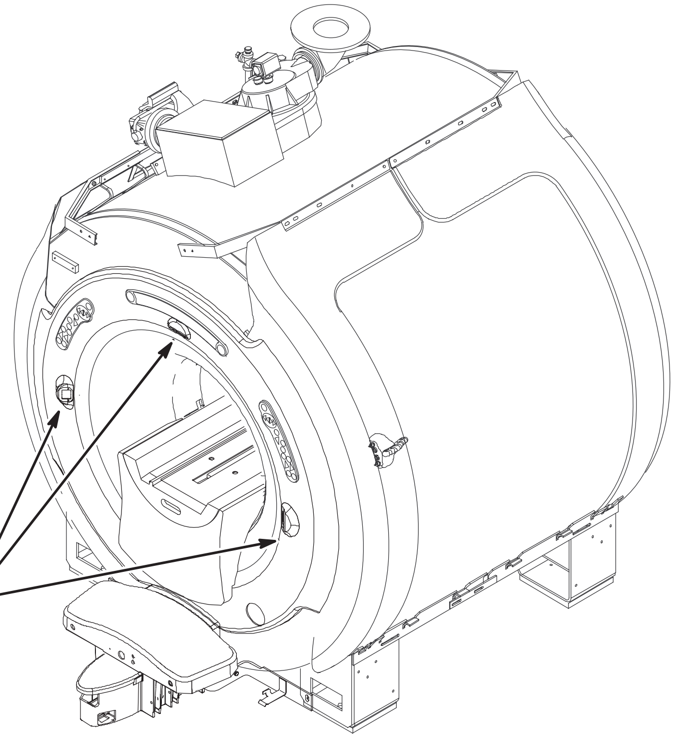


6–6 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

English



1. For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.

French

German

Portuguese

Spanish

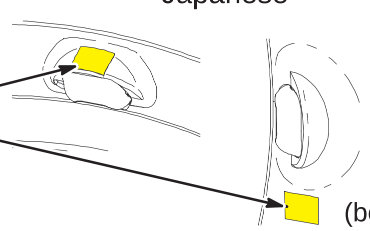
Italian

Swedish

Japanese

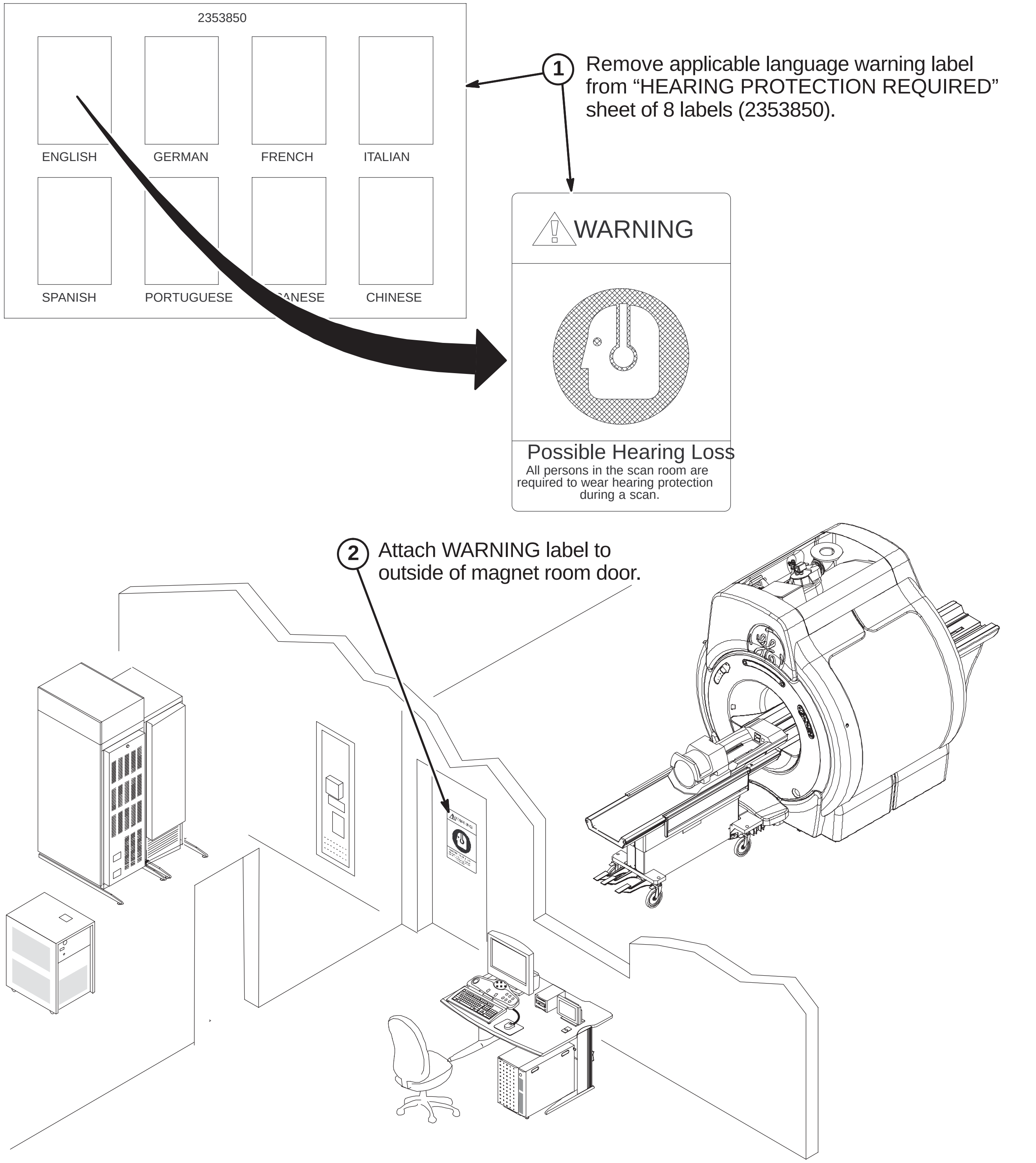
Chinese

2. If enclosure does not have labels attached, attach new labels as indicated.



(both sides)

6–7 INSTALL HEARING LOSS WARNING SIGN



SECTION 7 – S–SERIES/Cx/LCC HORIZON OR LX UPGRADE

TABLE OF CONTENTS

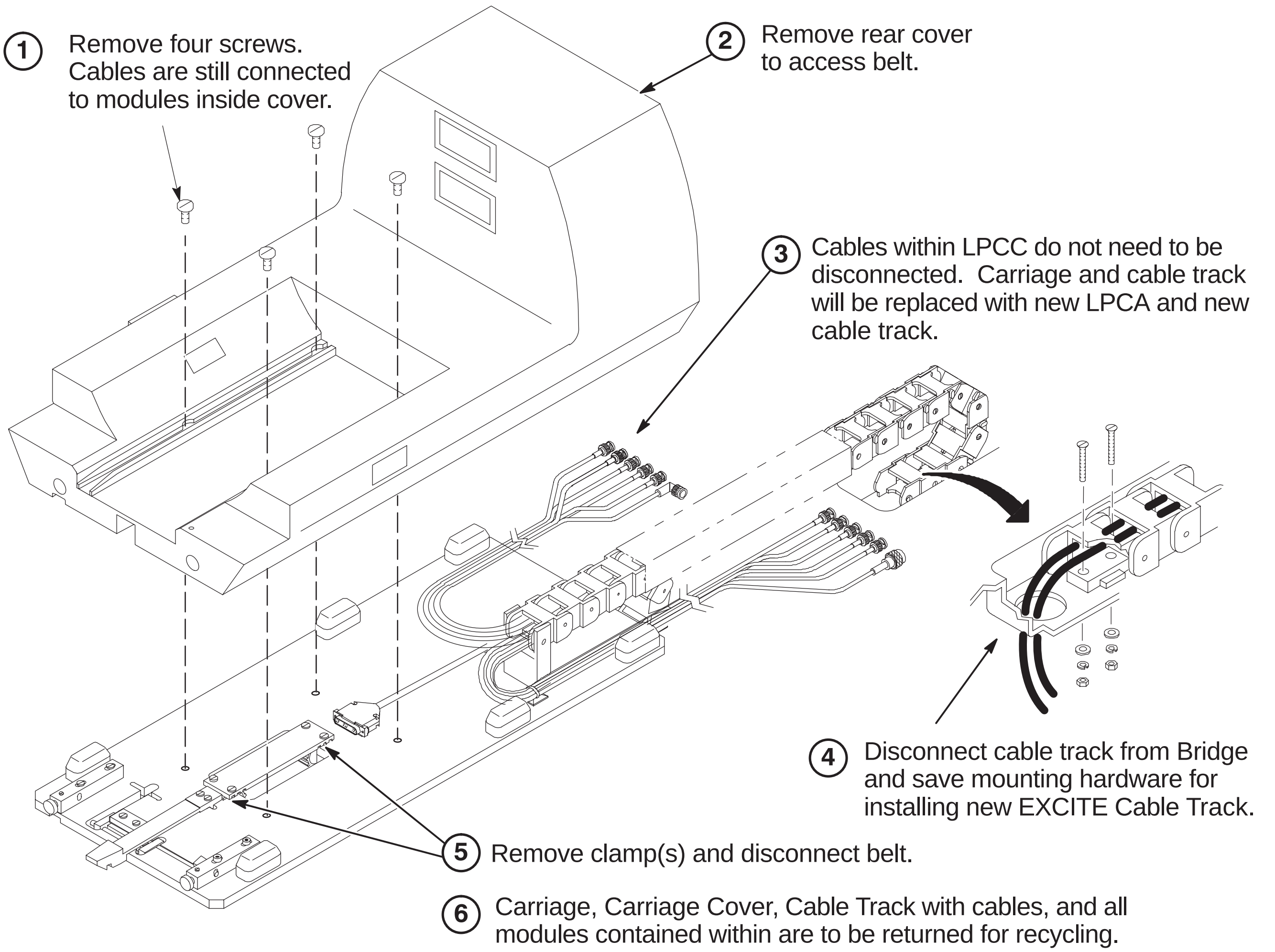
7–1 DE–INSTALL COMPONENTS ON HORIZON OR LX MAGNET ENCLOSURE

7–1–1 REMOVE EXISTING CARRIAGE COVER & CABLE TRACK

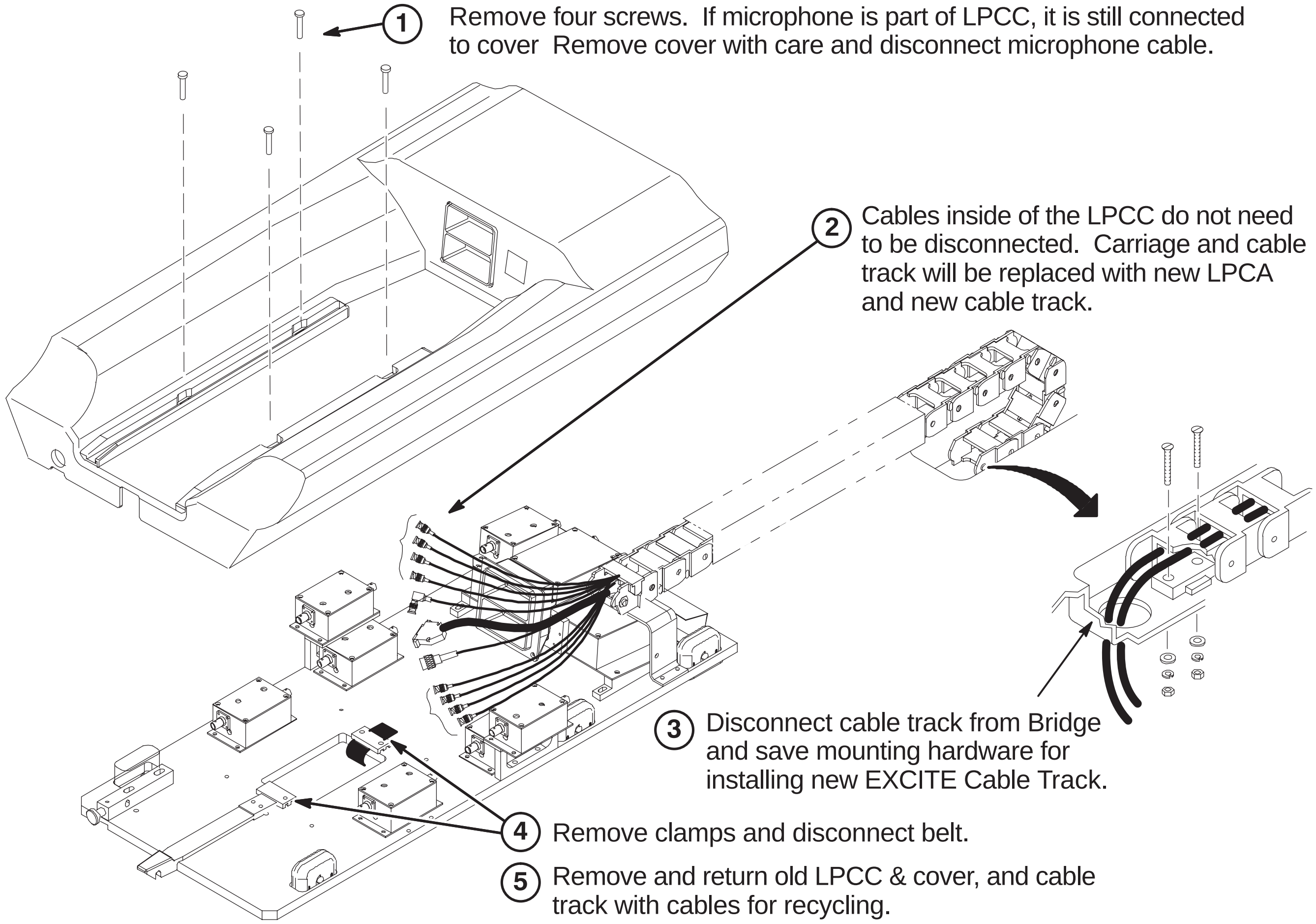
The site being upgraded may have either a High Profile Carriage Cover or a Low Profile Carriage Cover that will need to be replaced with the new LPCA and cable track.

Refer to sections **7–1–1–A** or **7–1–1–B** for the applicable removal procedure.

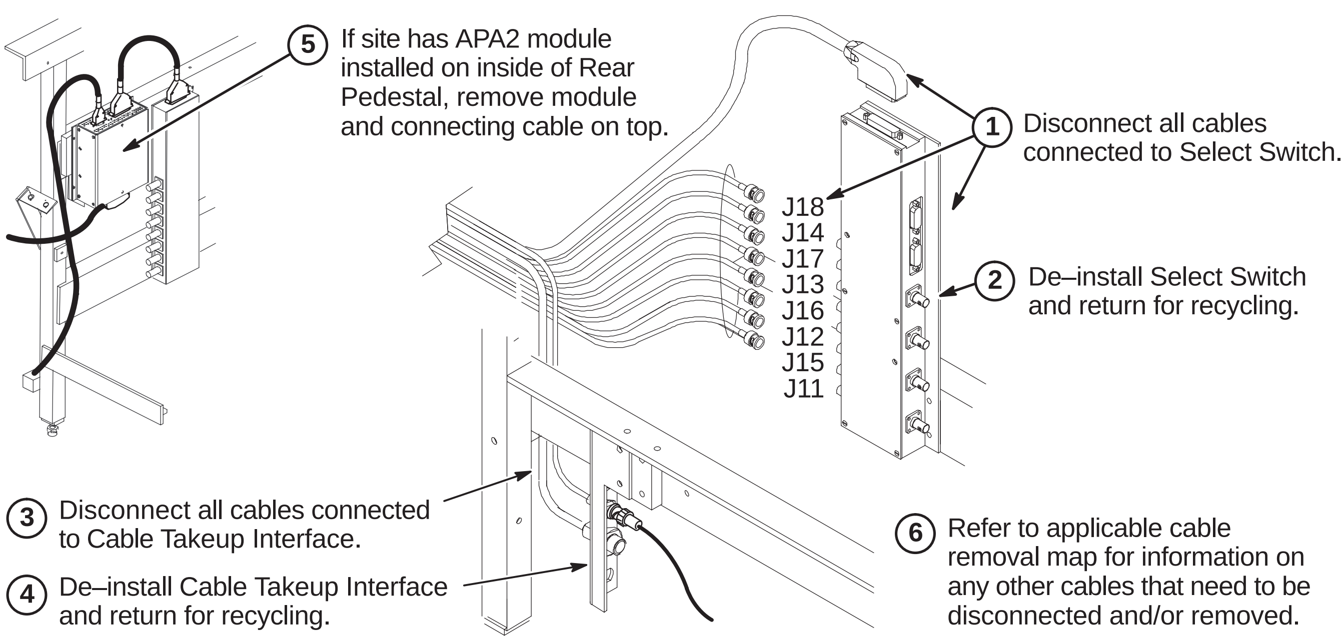
7–1–1–A Remove Existing High Profile Carriage Cover & Cable Track



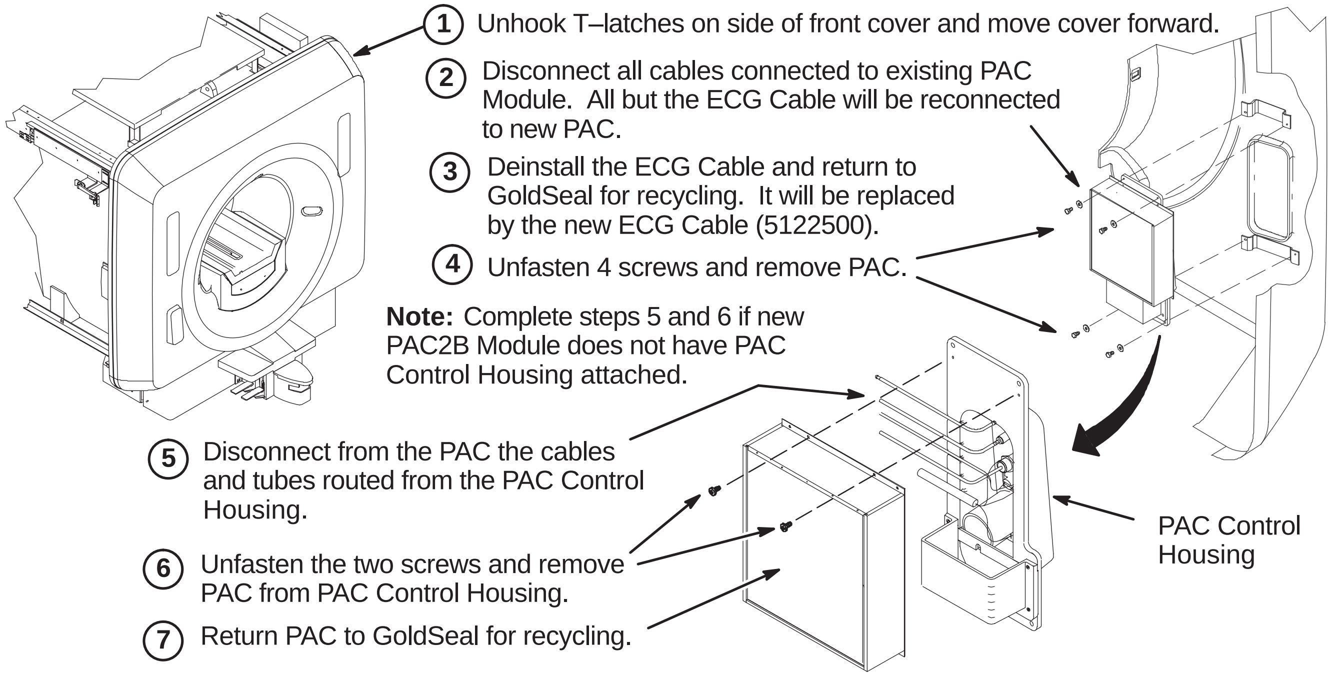
7–1–1–B Remove Existing Low Profile Carriage Cover & Cable Track



7–1–1–C Disconnect Cables In Rear Pedestal



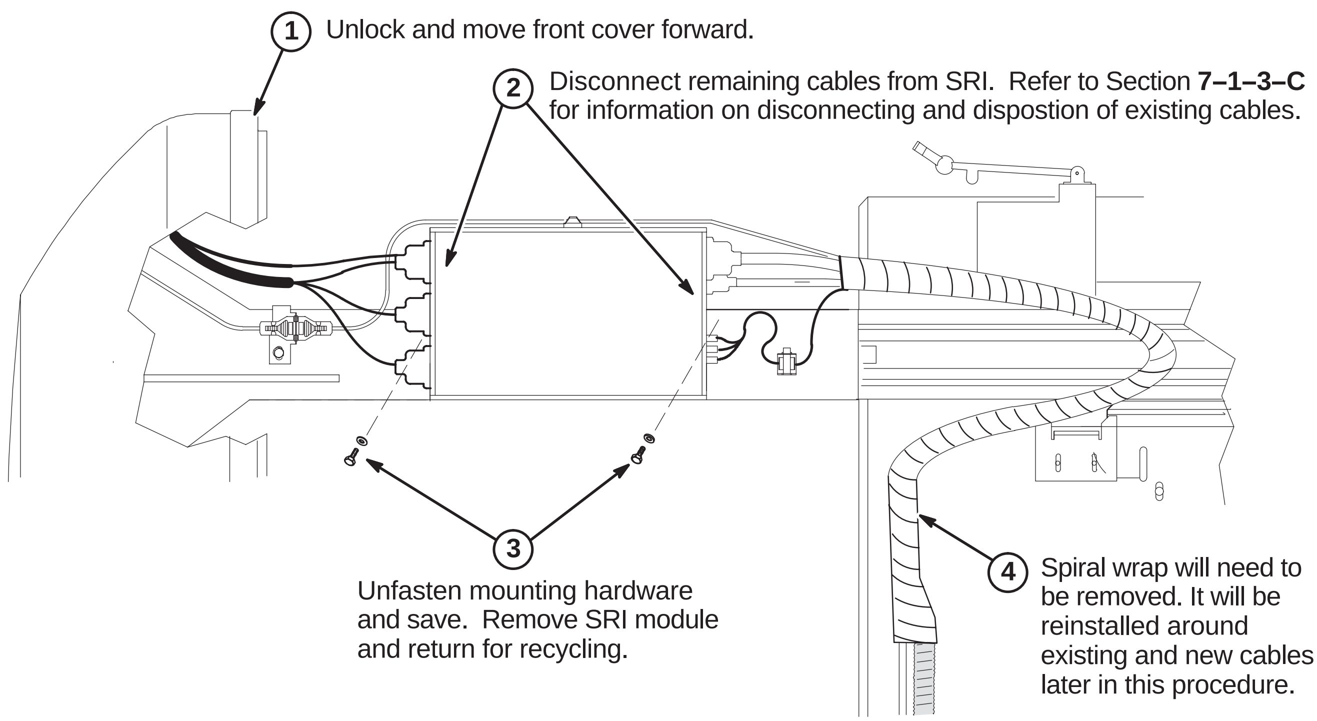
7–1–2 REMOVE PAC MODULE



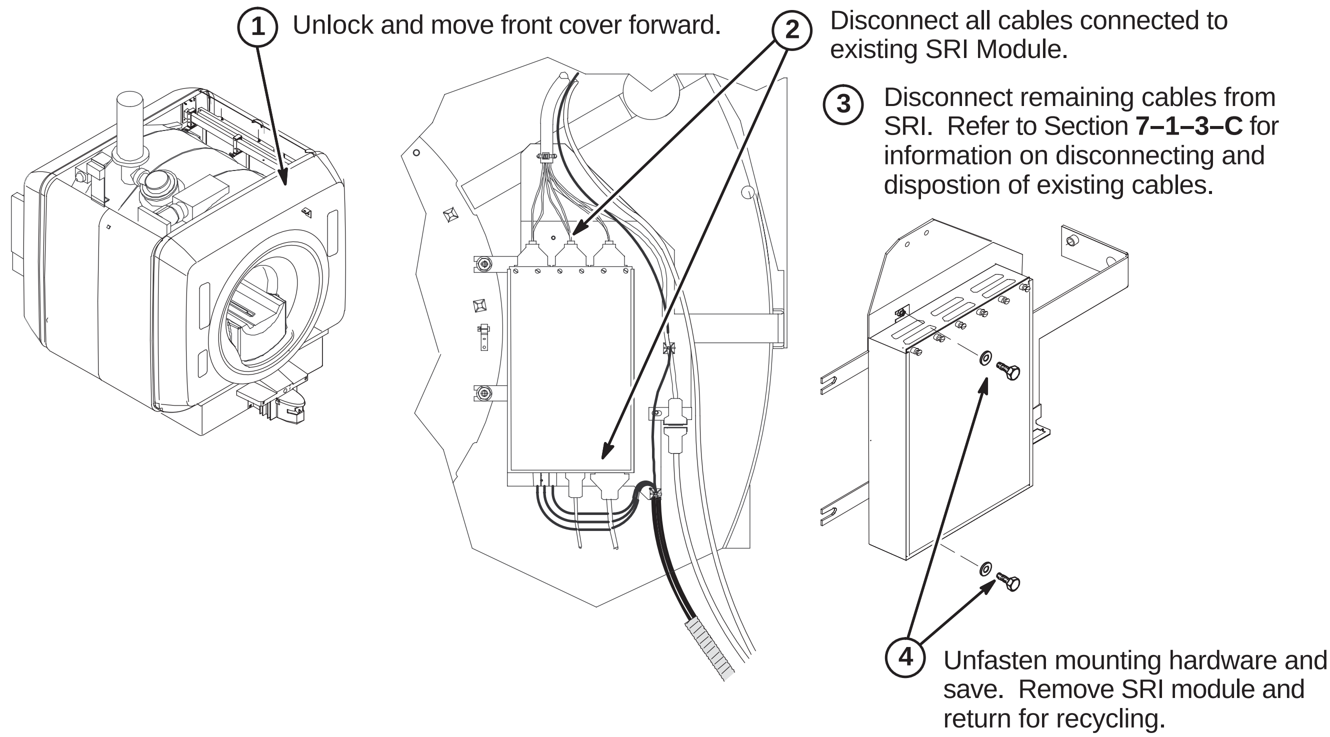
7–1–3 REMOVE SRI MODULE

Section **7–1–3–A** contains instructions for removing the SRI module on an S–Series magnet. Section **7–1–3–B** contains instructions for removing the SRI module on a Cx magnet.

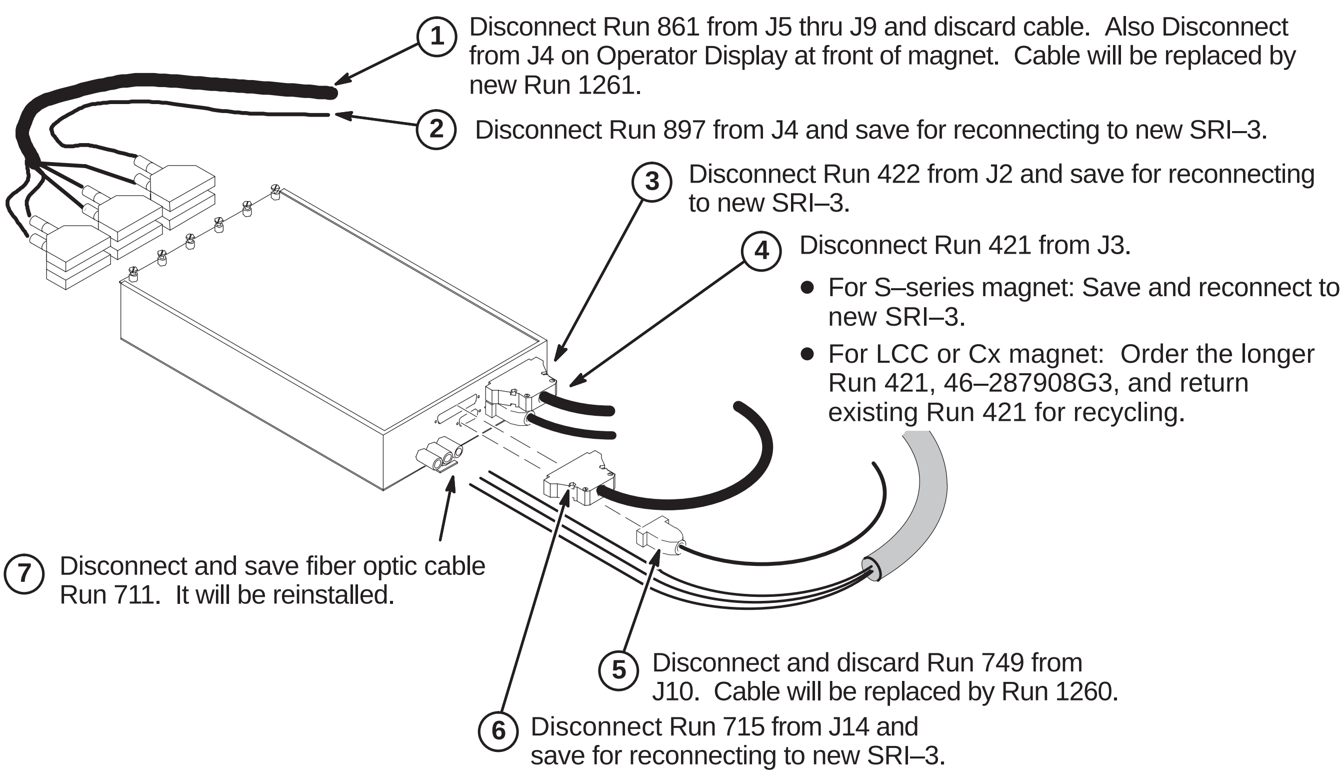
7–1–3–A De–Install SRI Module on S–Series Enclosure



7–1–3–B Remove SRI Module for Cx or LCC Enclosure



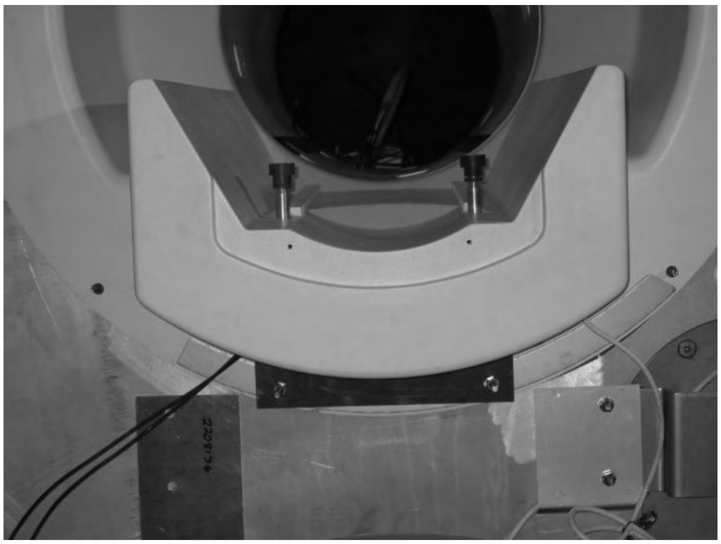
7–1–3–C SRI Cable Disposition



7–2 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

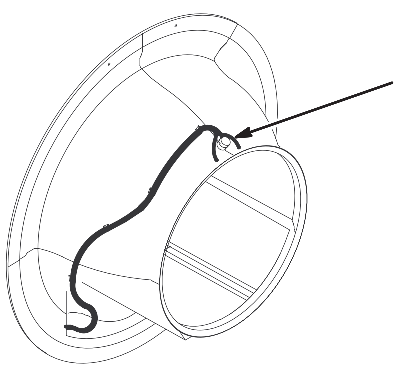
Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) has been installed. This kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It is required for allmagnets with Horizon style enclosures. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and it must be installed at this point of the EXCITE HD ugprade.

Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2401984, *S–Series Magnet Bridge Support* (shown below) and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.



Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade,* for instructions on installing this kit.

7–3 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL



EXCITE HD requires a new Dual Temperature Sensor cable, Run 1260 (5123812), to be installed.

Refer to Sections **10–3–7–B, 10–3–8**, and **10–3–9** for instructions on removal of existing cable and installing the new cable.

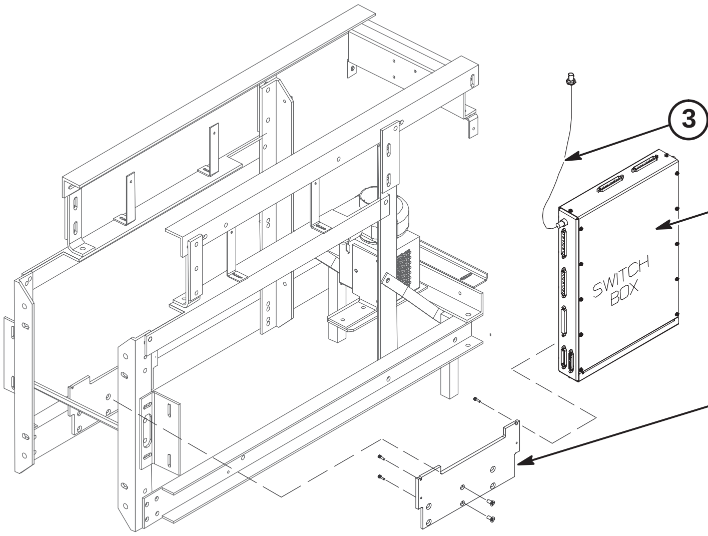
7–4 INSTALL MAGNET MONITOR

EXCITE HD requries Magnet Monitor on all systems. For this upgrade, all the parts and instructions are included in parts collector 5127838, Magnet Monitor Kit for Non–LCC Systems. Refer to Direction 2230681, included with collector, for instructions on installing magnet monitor on this magnet.

7–5 INSTALL NEW COMPONENTS AND CABLES

7–5–1 INSTALL NEW COMPONENTS ON REAR PEDESTAL

7–5–1–A Install 16 Channel Switch



NOTE: Components and cables not shown on Rear Pedestal for clarity.

3 Connect Run 1204 (5116105) as shown.

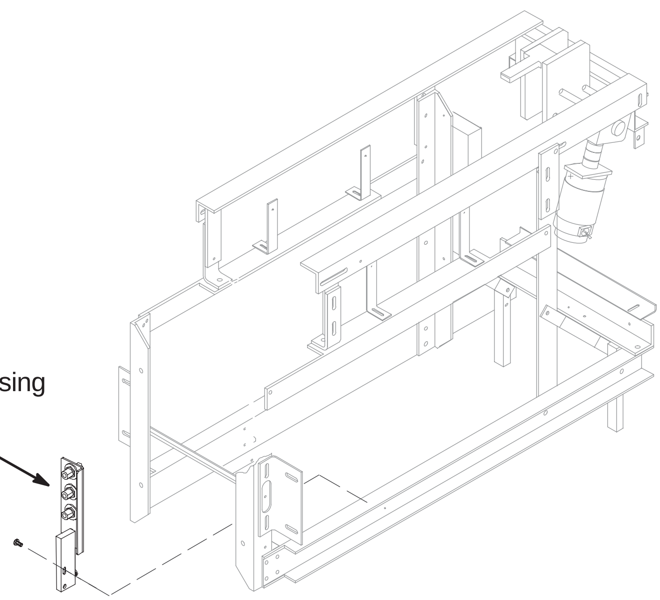
2 Attach 16 Channel Switch (5109645) to Mounting Plate using Loctite 242 (46–170686P2) and four M3 x 12mm Pan Head screws (46–312342P24).

1 Using the two center holes, install the 16 Channel Switch Mounting Plate (5110428) using Loctite 242 (46–170686P2) and:

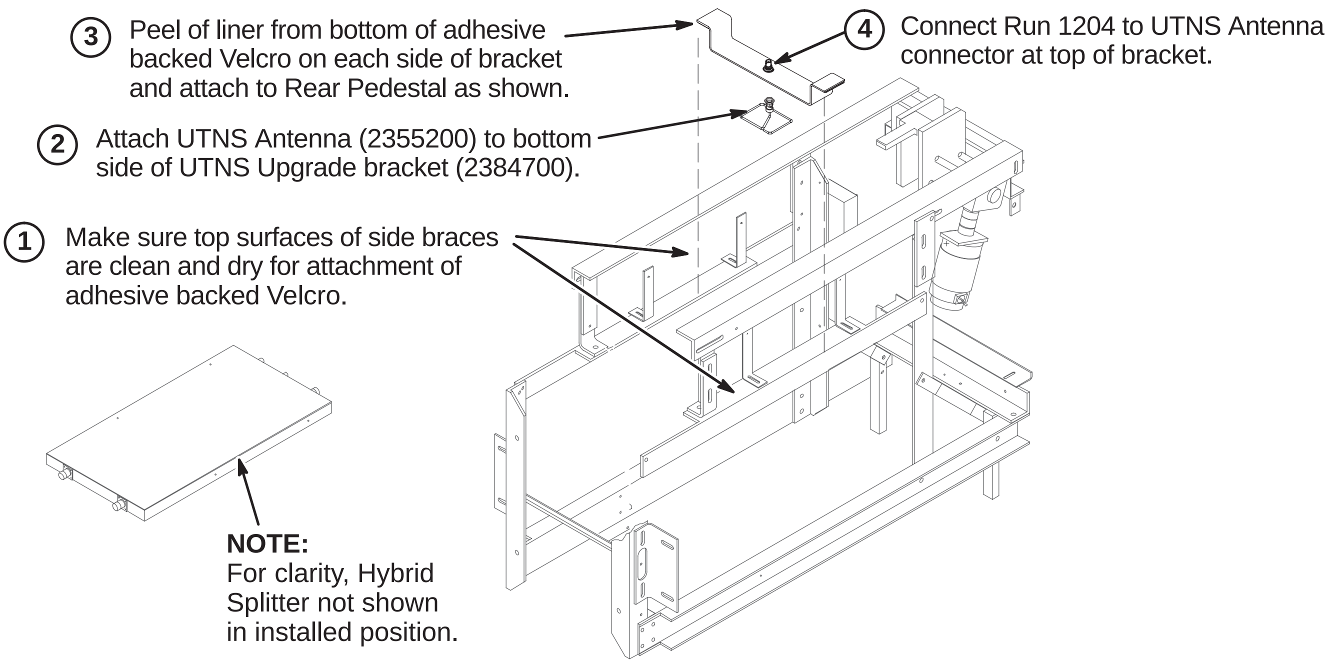
- *If Welded Pedestal* – two M6 x 12mm Slotted Flat Head Screws (25120369–86)
- *If Bolted Pedestal* – two 10–32 x 1/2 Slotted Flat Screws (46–208913P12)

7–5–1–B Install Bulkhead Interface

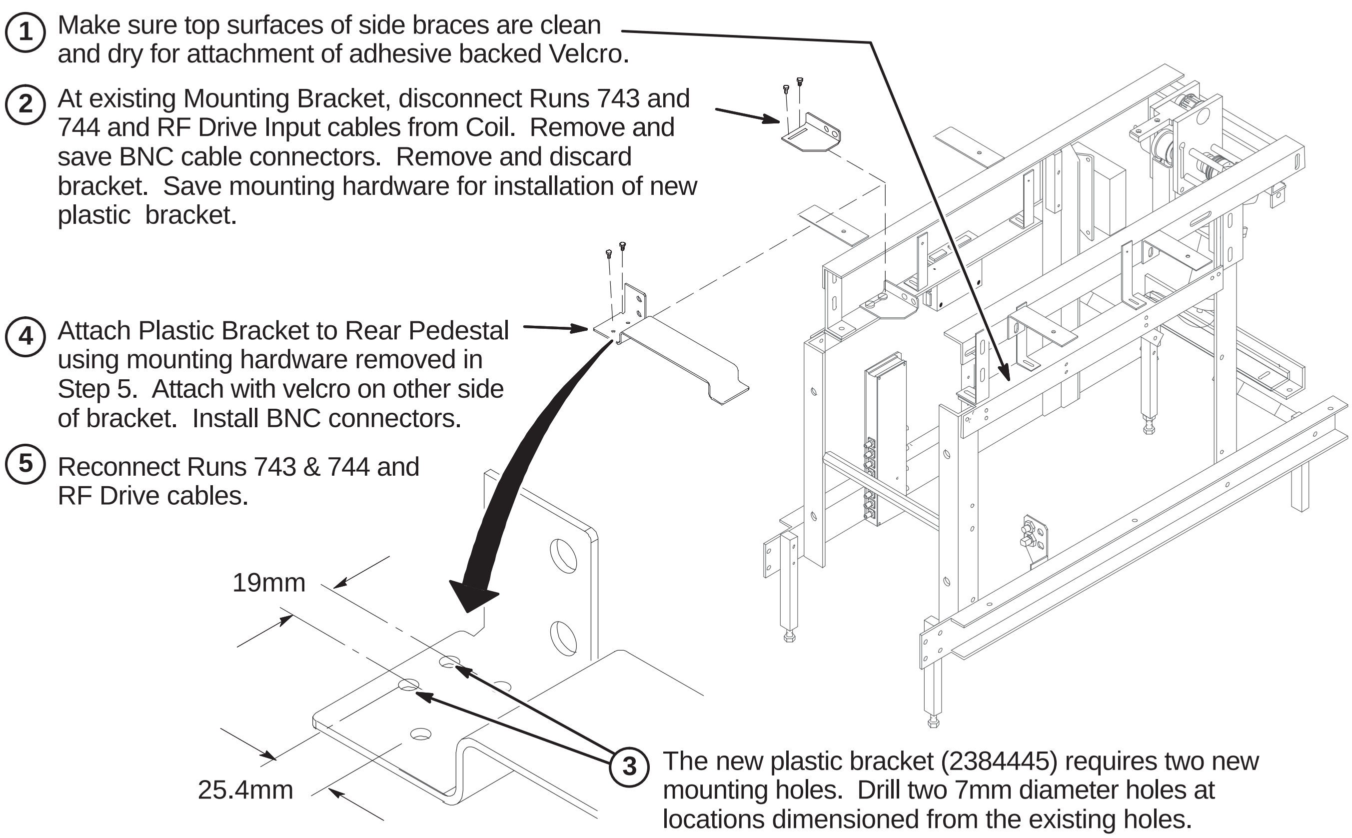
1 Install new Bulkhead Interface (5113250) using Loctite 242 (46–170686P2) and two M6 x 12mm Pan Head Screws (2109873–24).



7–5–1–C Install UTNS Antenna



7–5–1–D Install RF Drive Cable Mounting Bracket for LCC and Cx Magnets Only



7–5–2 HYBRID SPLITTER UPGRADE

Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

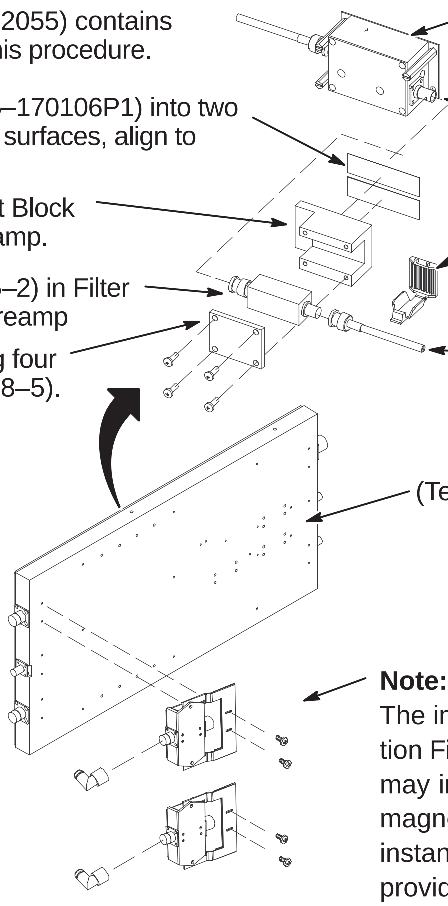
- Refer to Section 7–5–2–A for systems that do not have mount assembly for the Notch Filter
- Refer to Section 7–5–2–B for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 7–5–2–C for instructions on installing Isolation Filters on systems with Teledyne Splitter

The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils.

7–5–2–A Install Filter Mount Asm and Isolation Filters to Body Hybrid Splitter

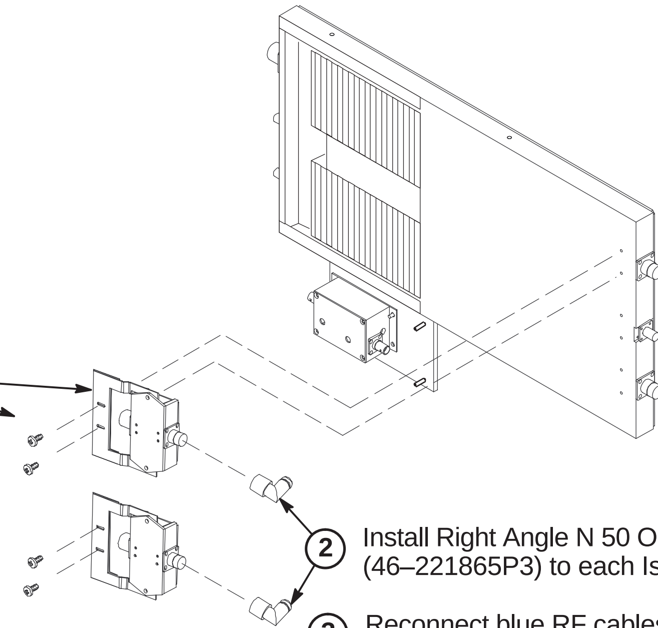
This Filter Mount assembly may not be required on newer Netcom Hybrid Spitter as it may already have a Notch Filter mounting clamp of a different style installed.

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.



Note:
The installation procedure for the Ground Isolation Filters as shown in the following procedures may interfere with the rear end bell on an LCC magnet with a Horizon enclosure. For these instances, an alternative installation method is provided in **Section 7–5–2–D**.

7–5–2–B Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)



FOR BRM's ONLY

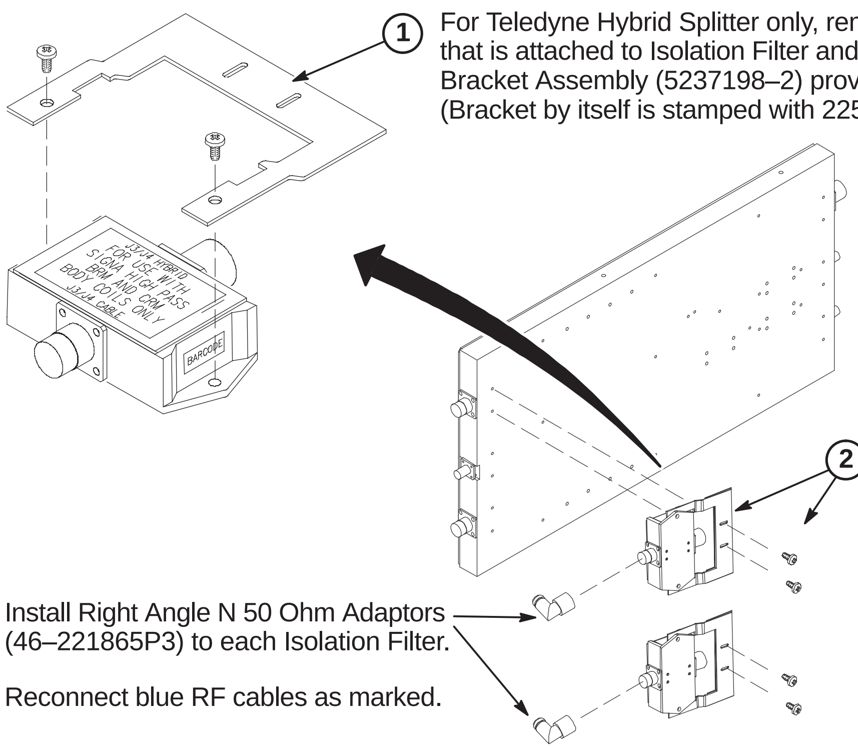
1. Attach and connect each Ground Isolation Filter Module (5118952–2) to the Hybrid Splitter using two screws (46–220184P5).

   Do NOT install on TRM gradient coil systems.

2. Install Right Angle N 50 Ohm Adaptors (46–221865P3) to each Isolation Filter.

3. Reconnect blue RF cables as marked.

7–5–2–C Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)



1. For Teledyne Hybrid Splitter only, remove existing bracket that is attached to Isolation Filter and replace with Mounting Bracket Assembly (5237198–2) provided with upgrade kit. (Bracket by itself is stamped with 2254629)

FOR BRM's ONLY

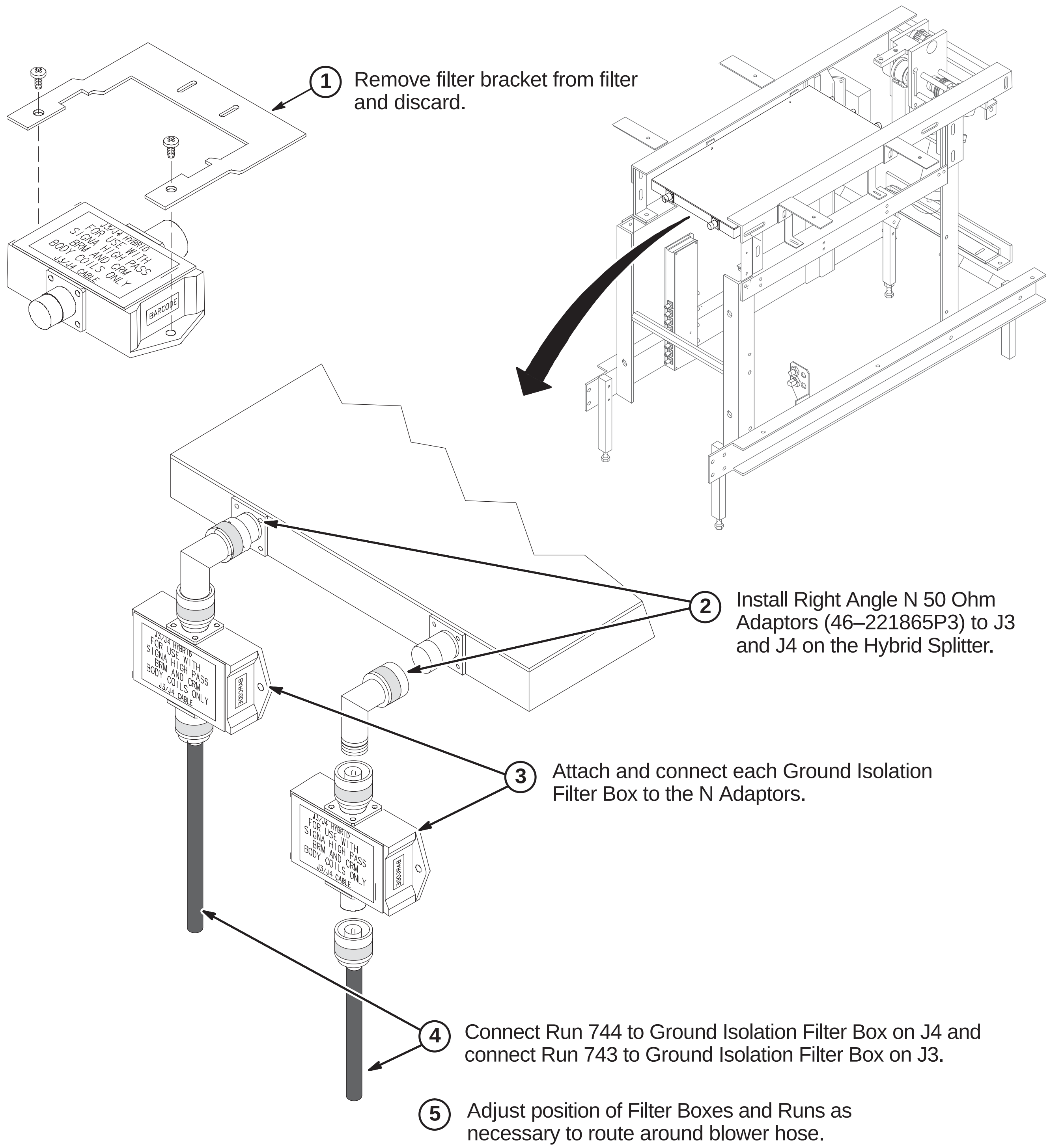
2. Attach and connect each Ground Isolation Filter Module (5118952) to the Hybrid Splitter using two screws (46–220184P5) for each module.

   Do NOT install on TRM gradient coil systems.

3. Install Right Angle N 50 Ohm Adaptors (46–221865P3) to each Isolation Filter.

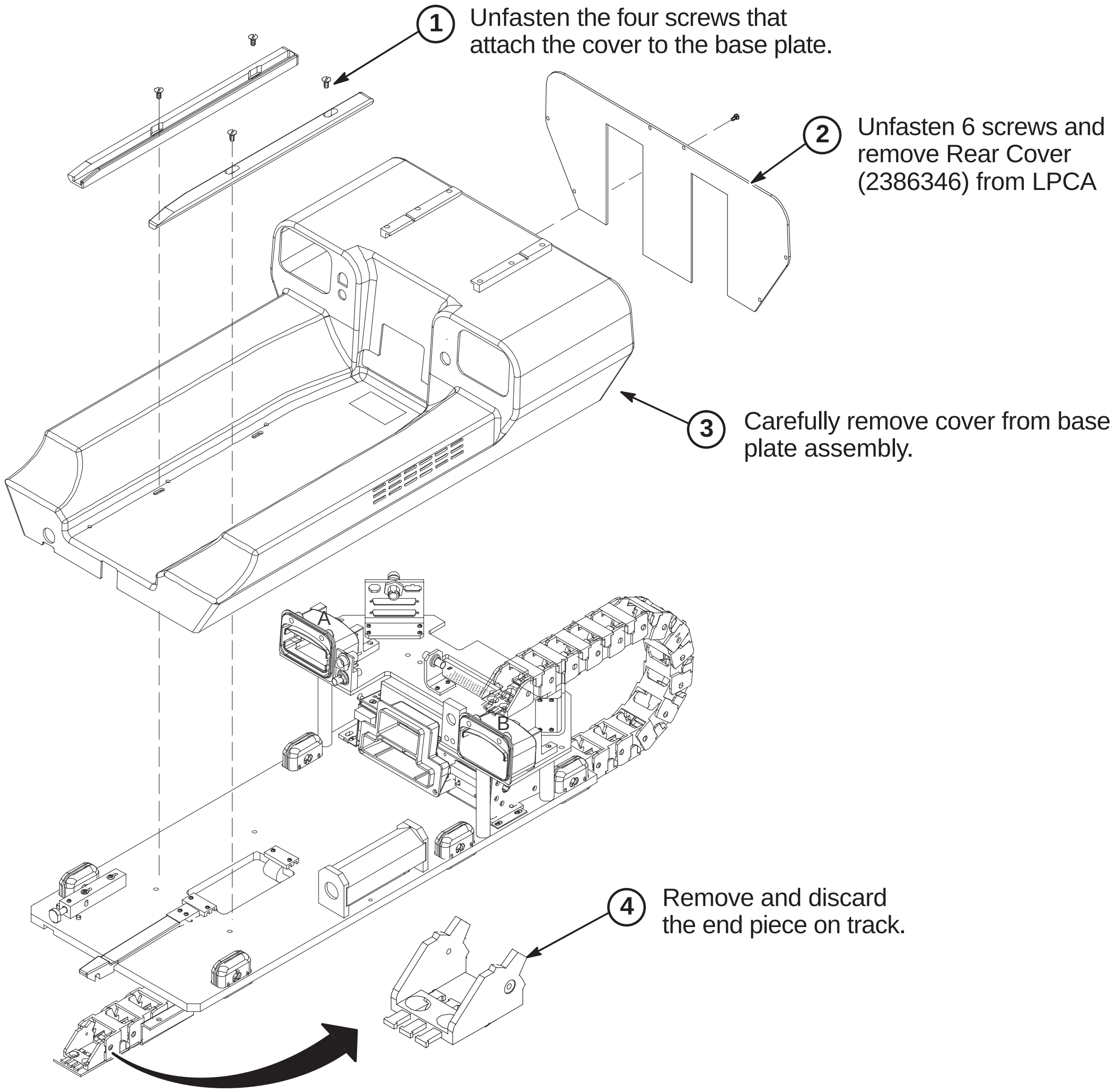
4. Reconnect blue RF cables as marked.

7–5–2–D Alternative Ground Isolation Filter Installation for LCC Magnet with Horizon Enclosure

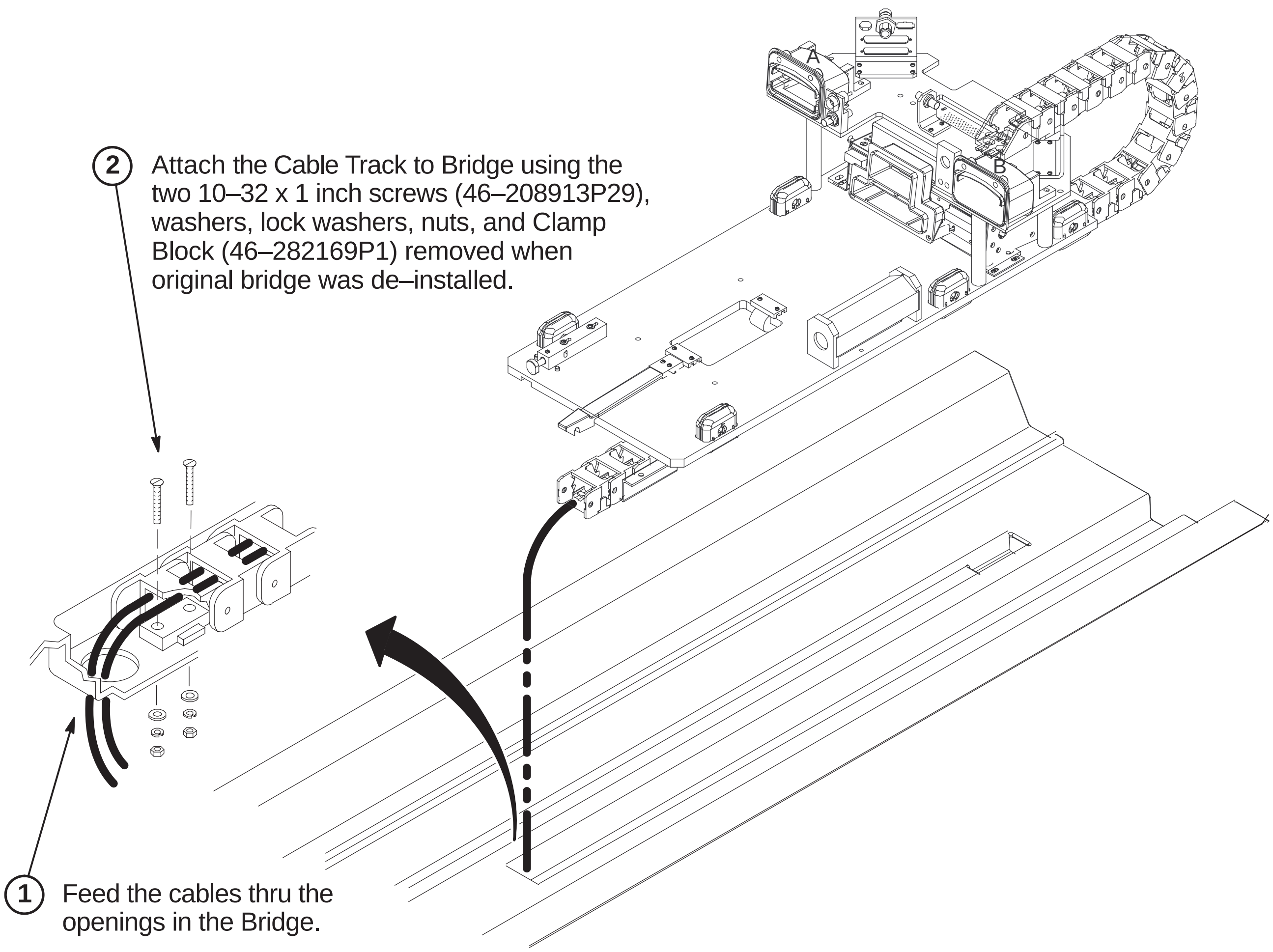


7–5–3 INSTALL LPCA AND CABLE TRACK

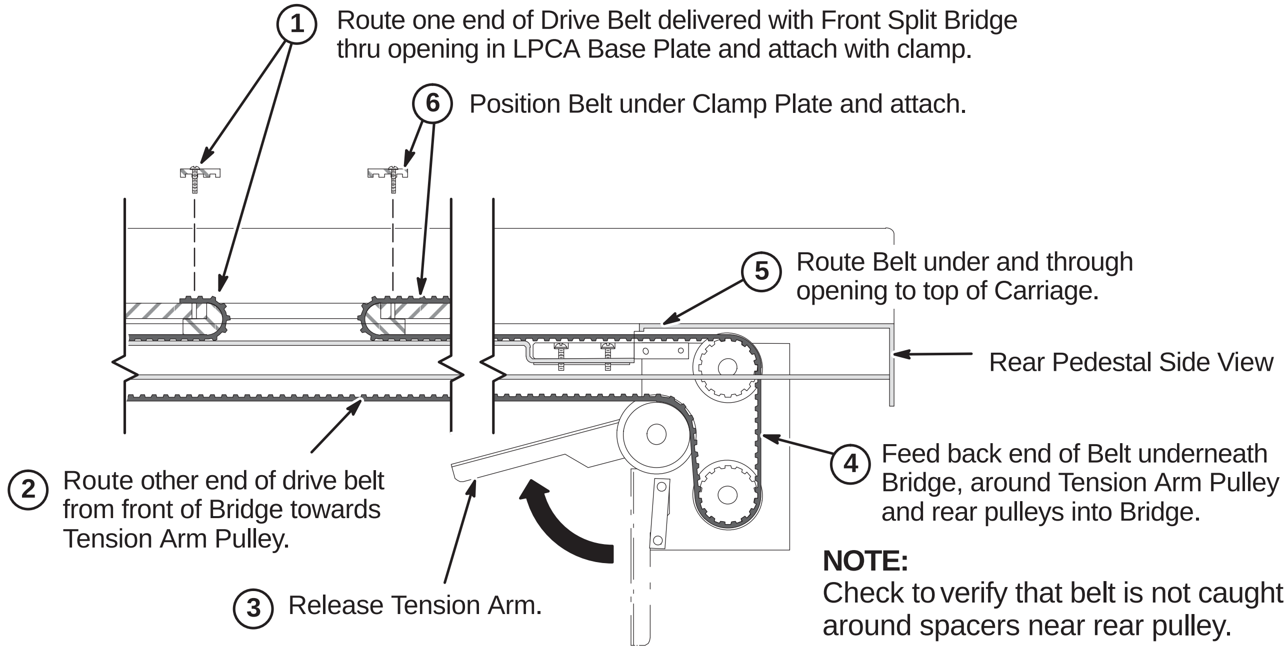
7–5–3–A Remove LPCA Cover



7–5–3–B Attach Cable Track to Bridge

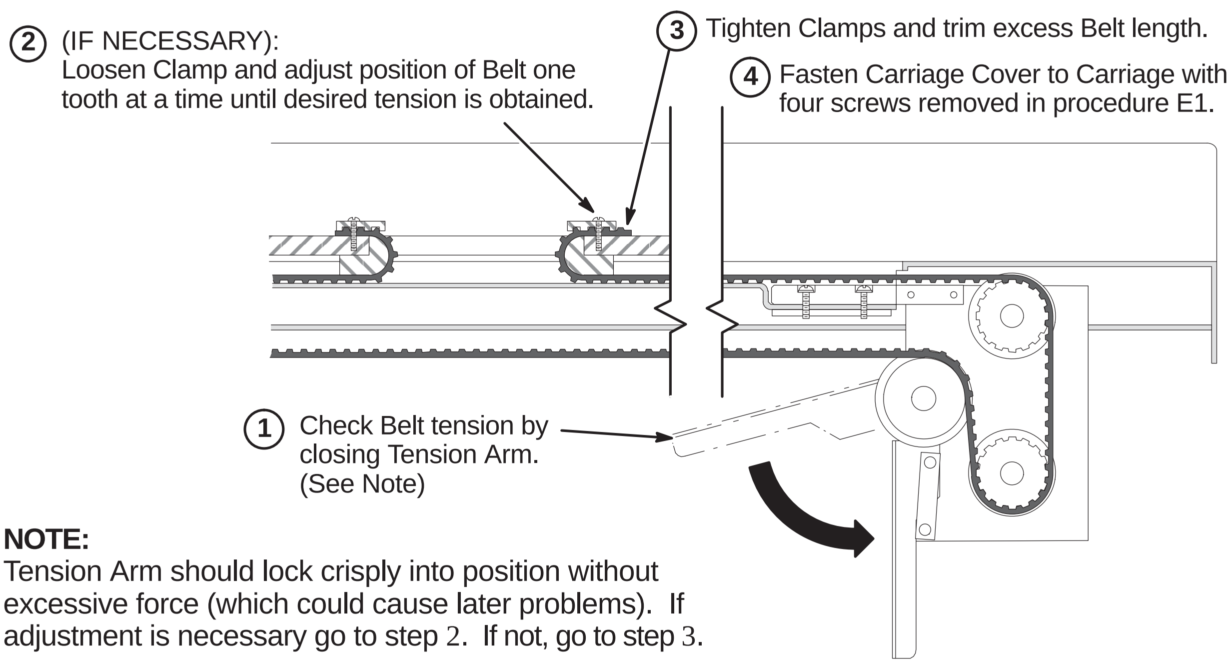


7–5–3–C Route and Attach Drive Belt



NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

7–5–3–D Final Adjustment of Drive Belt



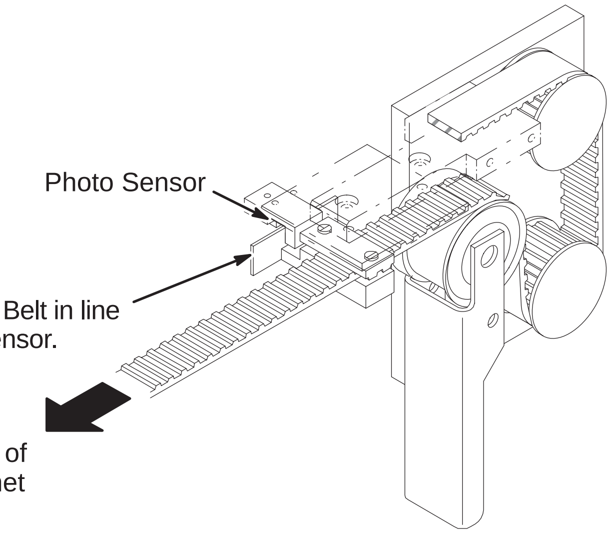
NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

7–5–3–E Installation of Sensor Flag

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

1 Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

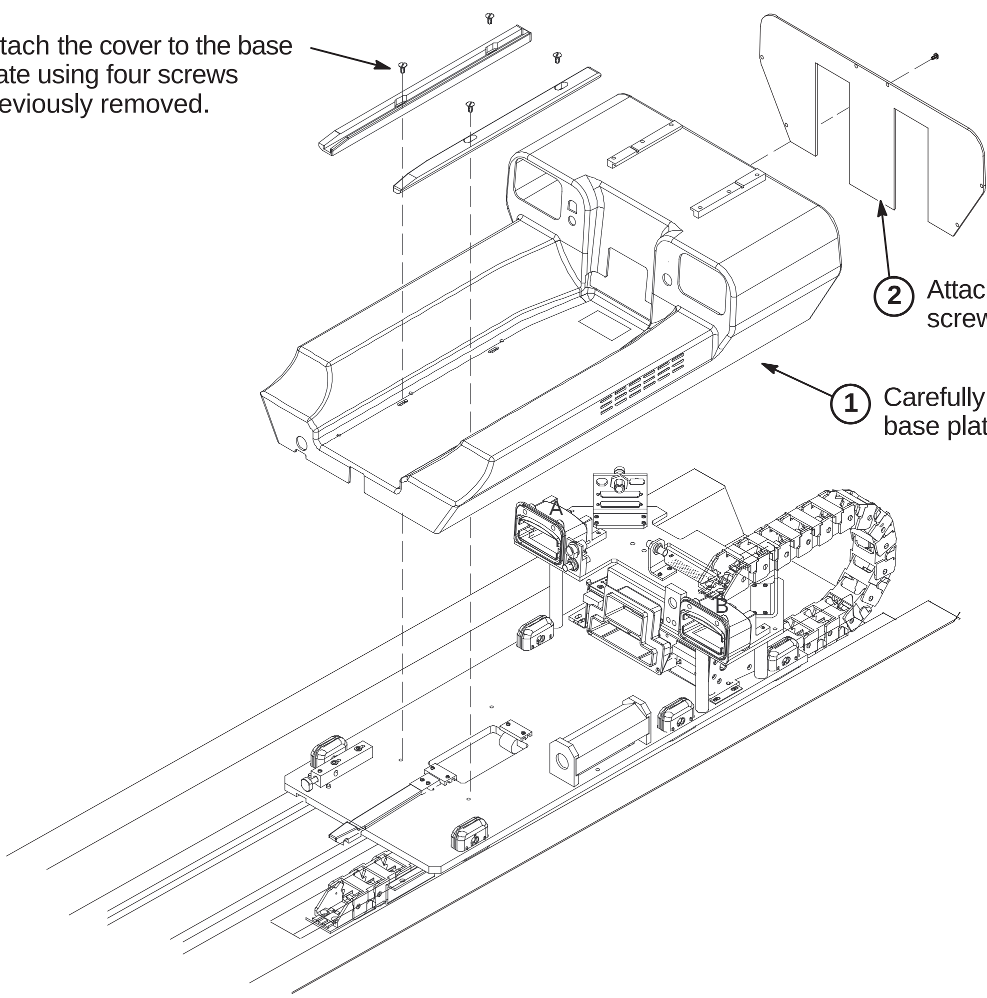


7–5–3–F Installation of LPCA Cover

3 Attach the cover to the base plate using four screws previously removed.

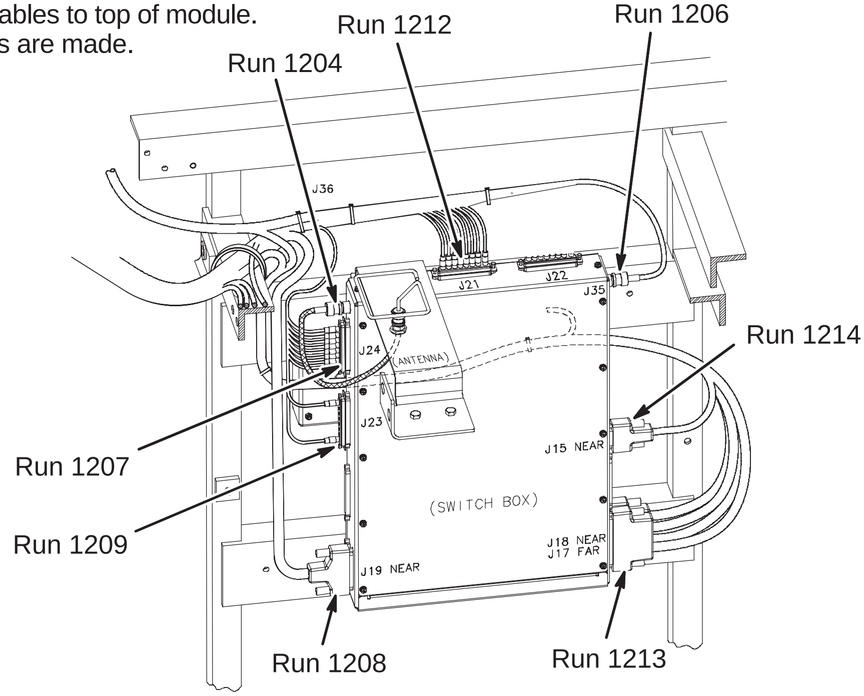
2 Attach Rear Cover using 6 screws previously removed.

1 Carefully place cover on base plate assembly.



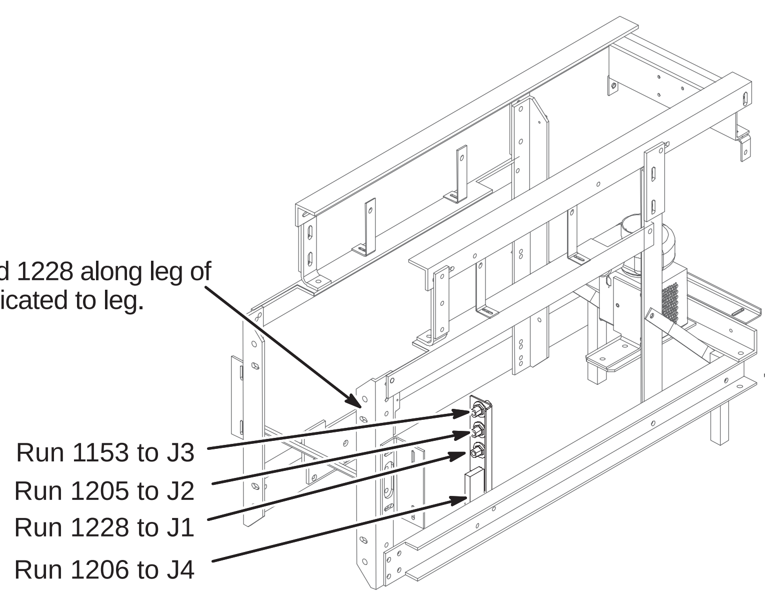
7–5–3–G Connect Cables from Cable Track to 16 Channel Switch Module

1. 16 Channnel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re–mount module after connections are made.
2. Connect cable Runs as indicated.



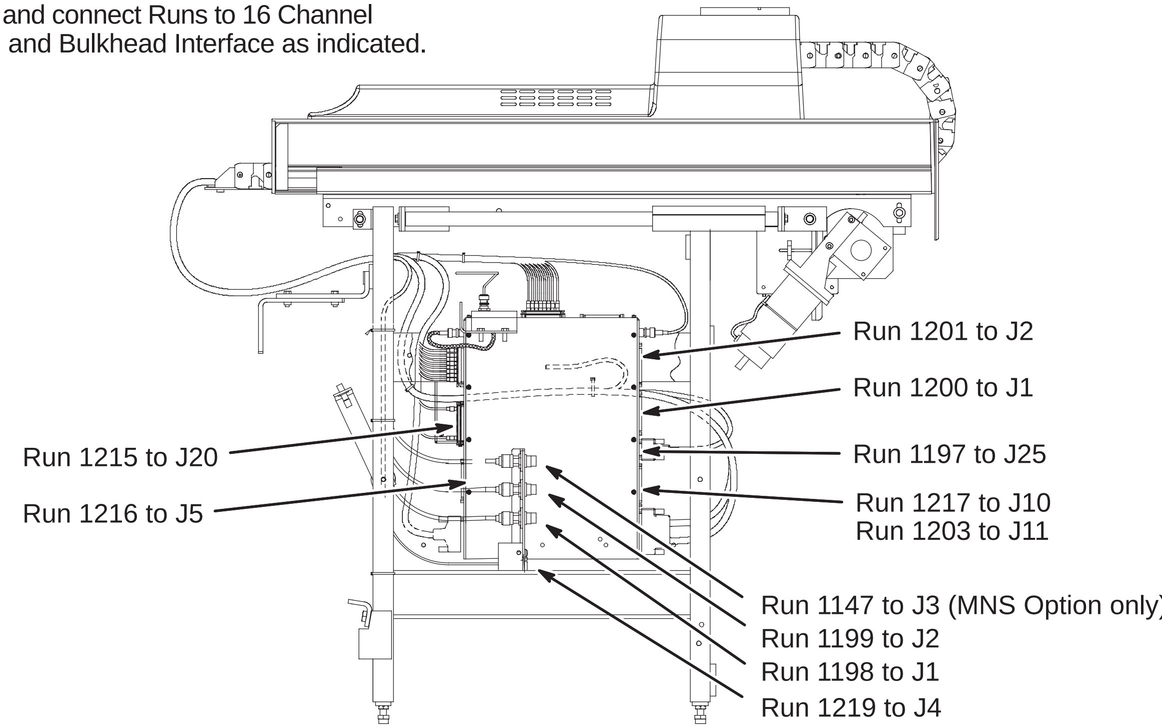
7–5–3–H Connect Cables from Cable Track to Bulkhead Interface

1. Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty–wrap as indicated to leg.
2. Connect Runs to Bulkhead Interface as indicated.

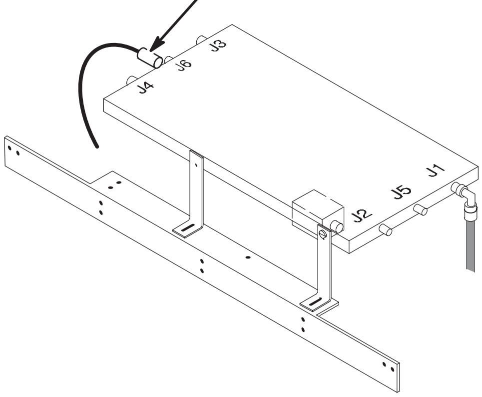


7–5–4 CONNECT REMAINING CABLES TO REAR PEDESTAL

1. Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



2. Route/connect Run 780 to J6 on RF Splitter.

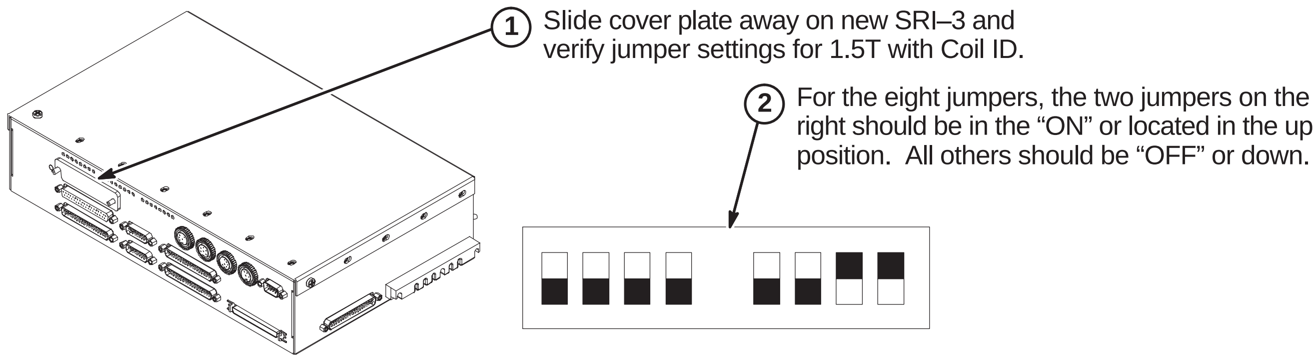


3. Connect Run 1250 to J1 on RF Splitter:

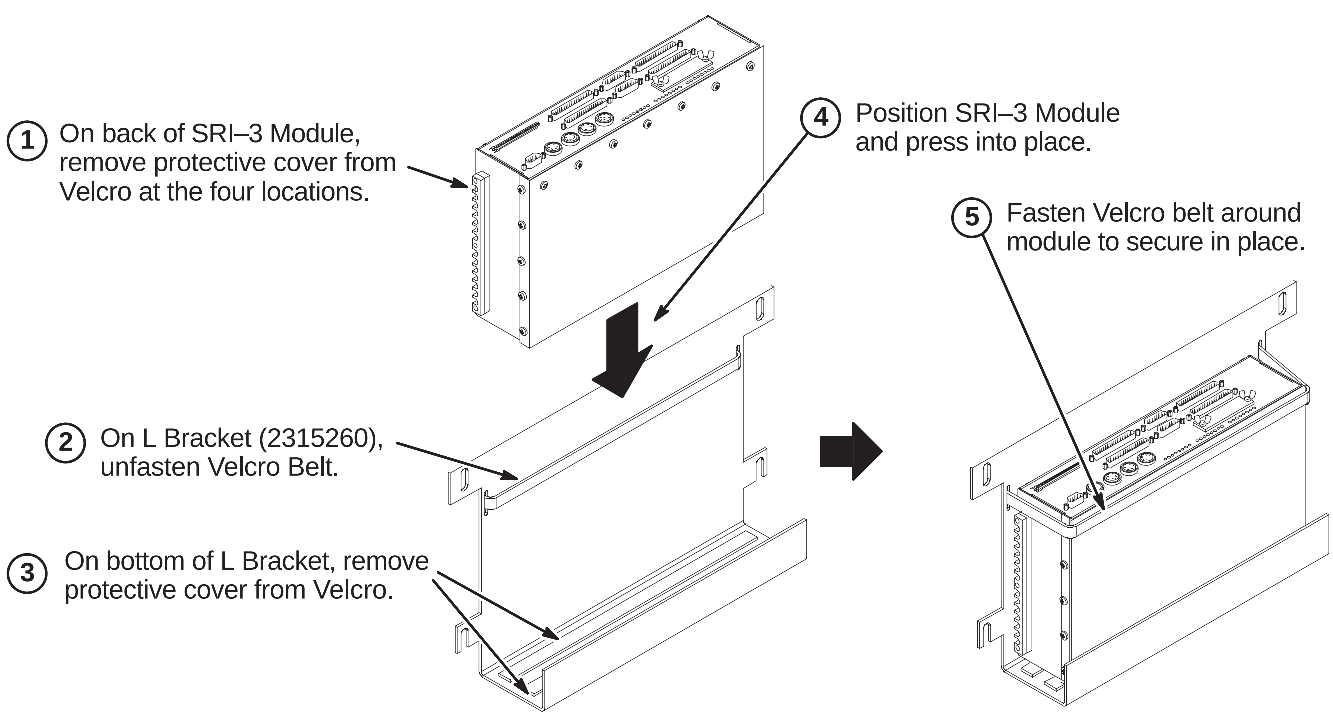
7–5–5 INSTALL SRI–3 MODULE

Sections **7–5–5–C** contains instructions for installing the SRI module on an S–Series magnet. Section **7–5–5–D** contains instructions for installing the SRI module on a Cx or LCC magnet. The remaining procedures in this section are applicable to either magnet type.

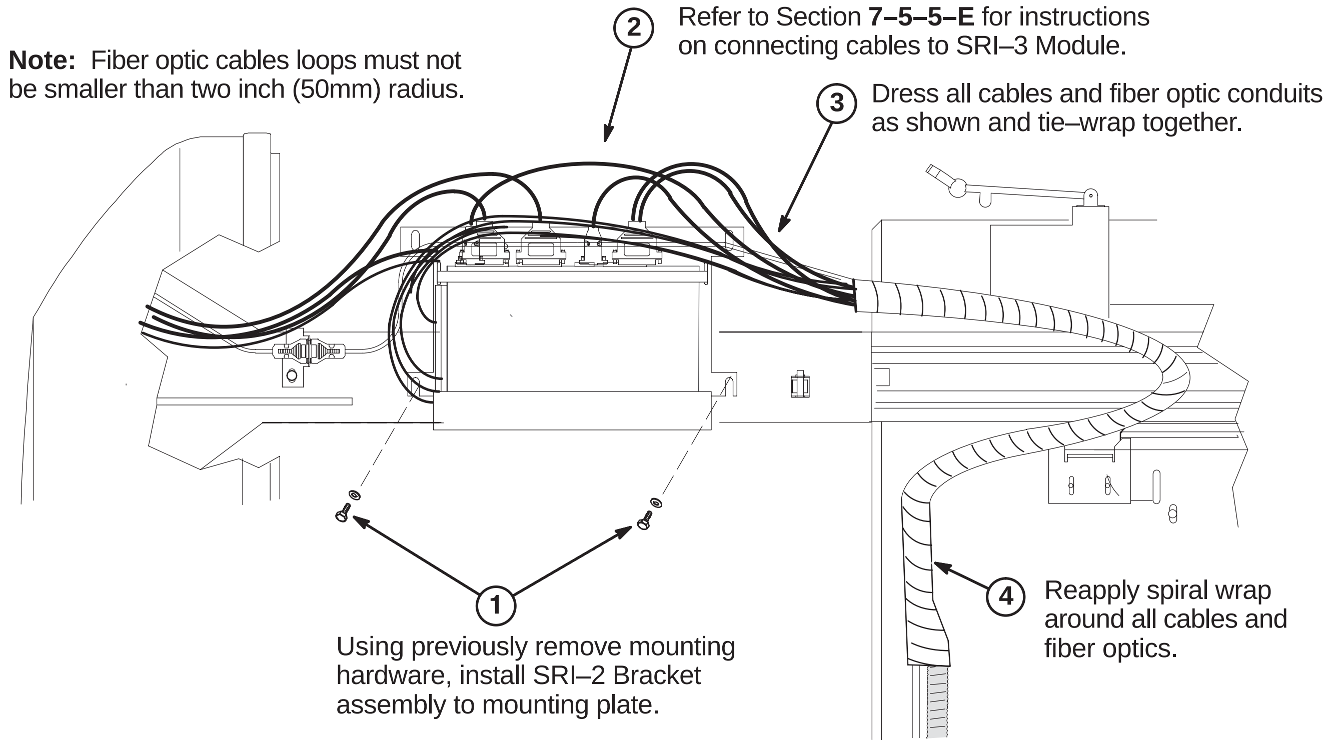
7–5–5–A Verify SRI–3 Jumper Settings



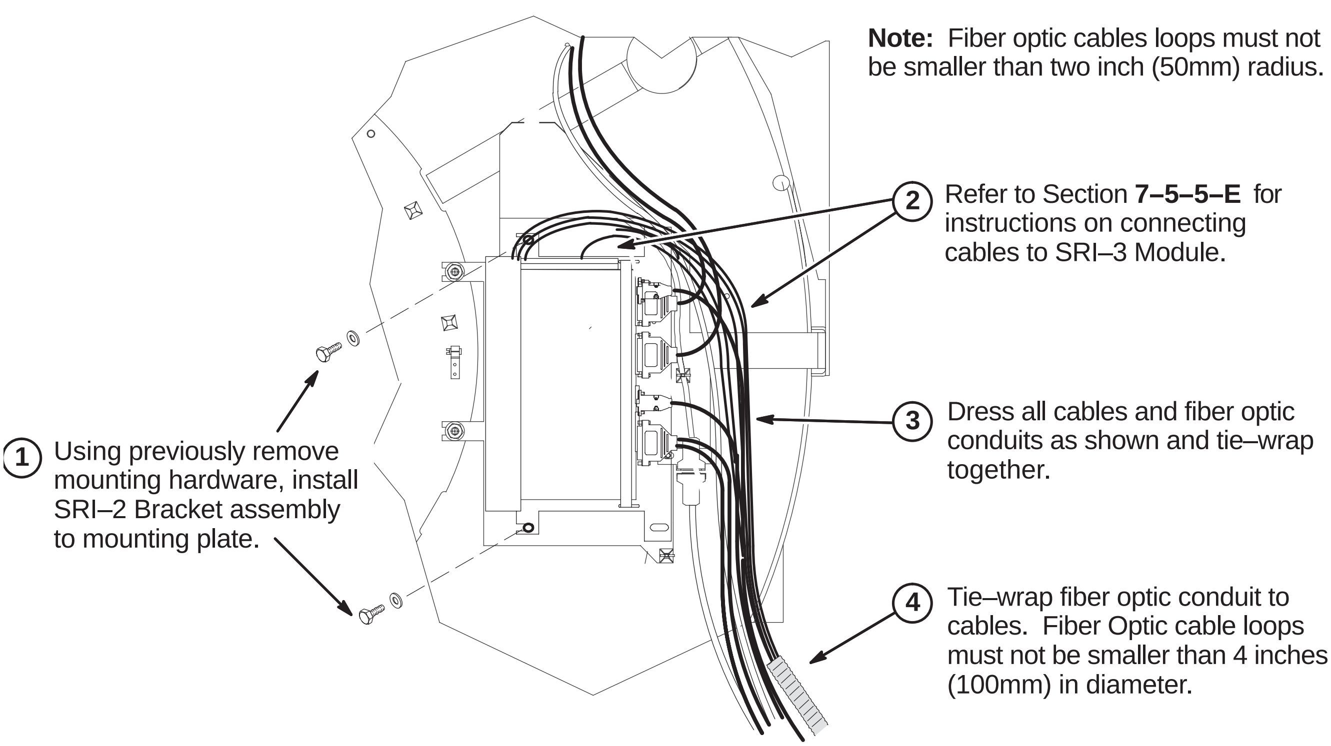
7–5–5–B Position SRI–3 in L Bracket



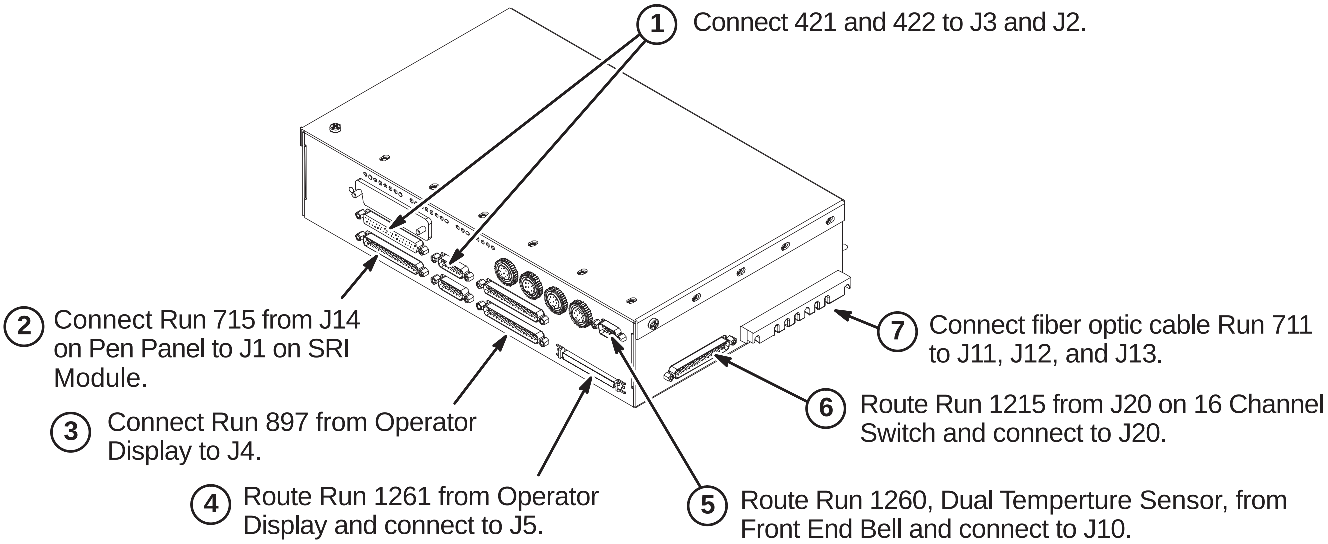
7–5–5–C Install SRI–3 on S–Series Magnet



7–5–5–D Install SRI–3 on Cx or LCC Magnet

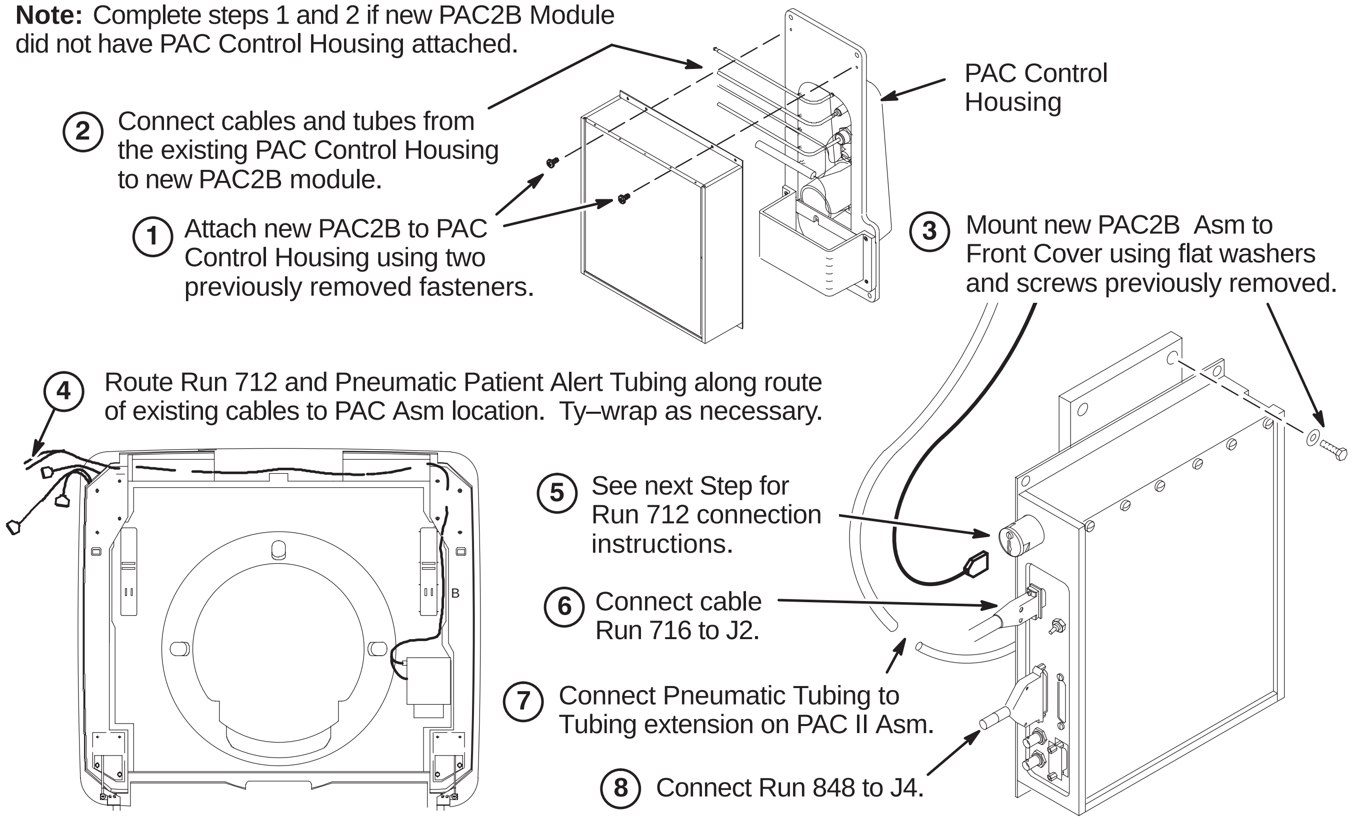


7–5–5–E Connect Cables at SRI–3 Module Location

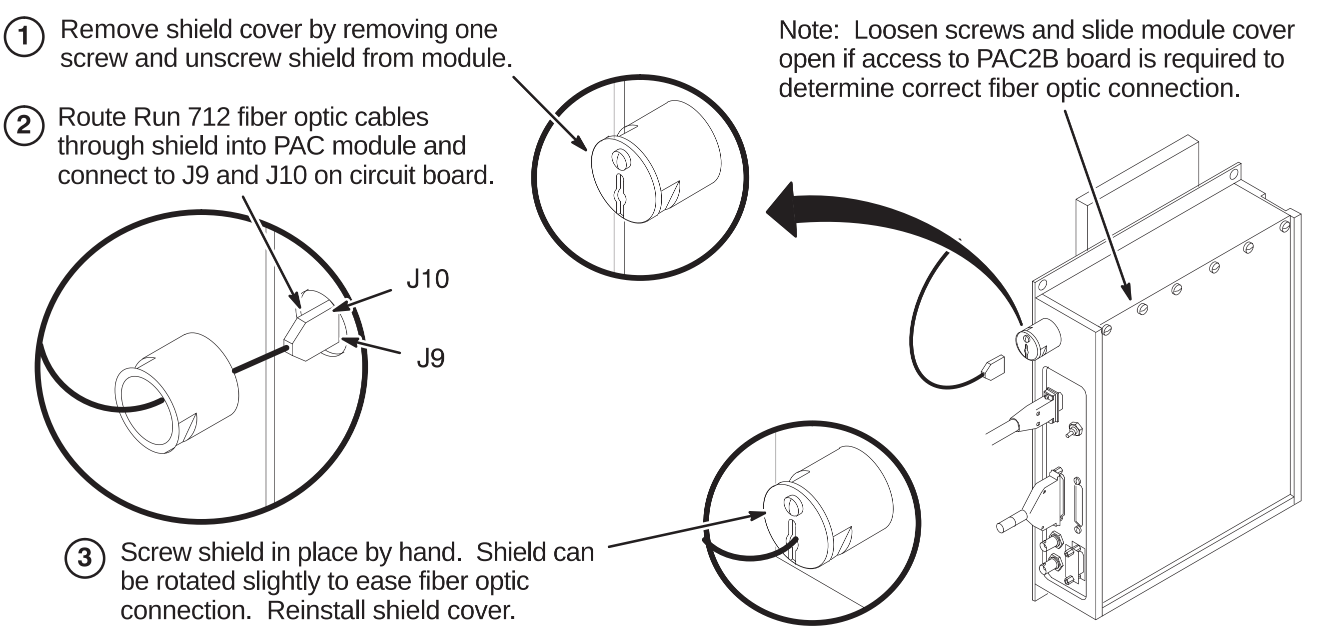


7–5–6 INSTALL PAC MODULE

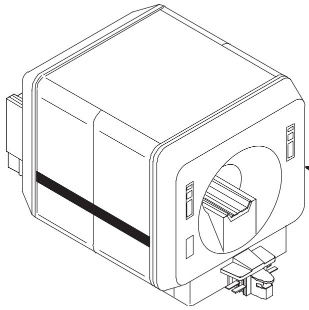
7–5–6–A Connect Run 712, 716, & Pneumatic Patient Alert Tubing



7–5–6–B Connect Run 712

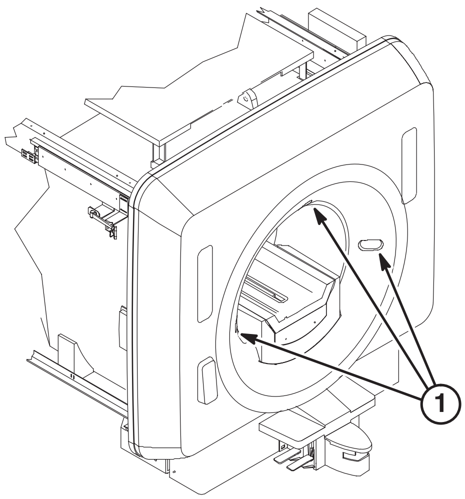


7–6 INSTALL COVERS



1 Replace all covers on enclosure and rear pedestal.

7–7 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

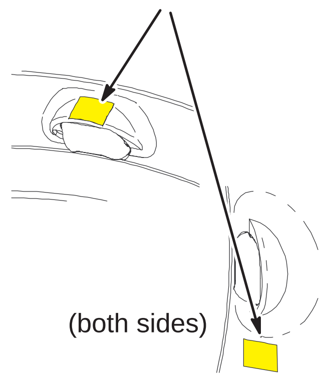


A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

English

2 If enclosure does not have labels, attach new labels as indicated.



(both sides)

For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.

French

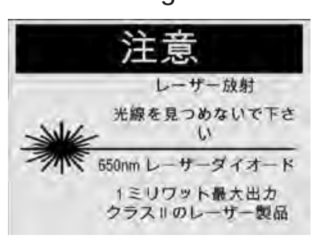
German

Portuguese

Spanish

Italian

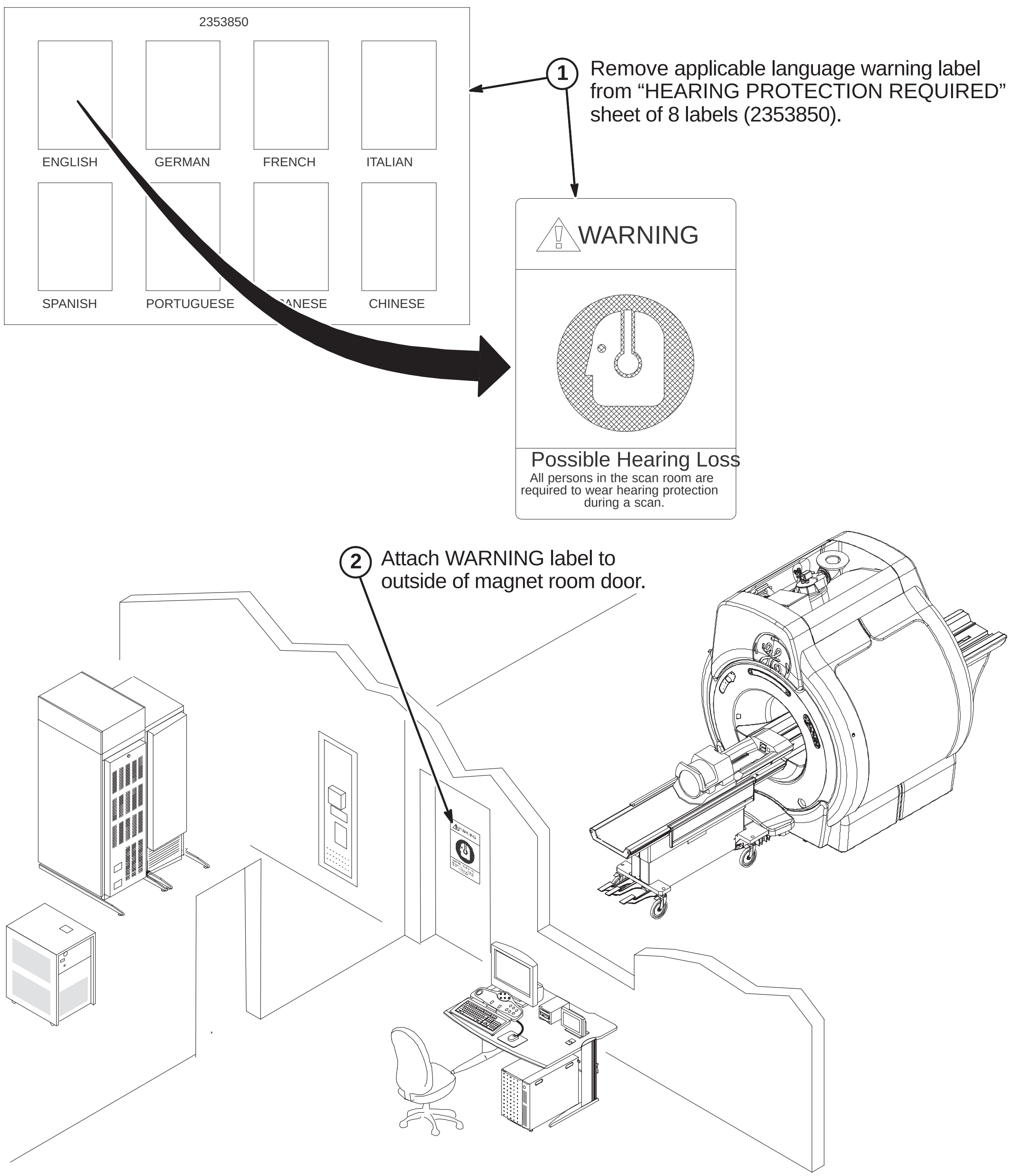
Swedish



Japanese

Chinese

7–8 INSTALL HEARING LOSS WARNING SIGN



SECTION 8 – S–SERIES ADVANTAGE UPGRADE

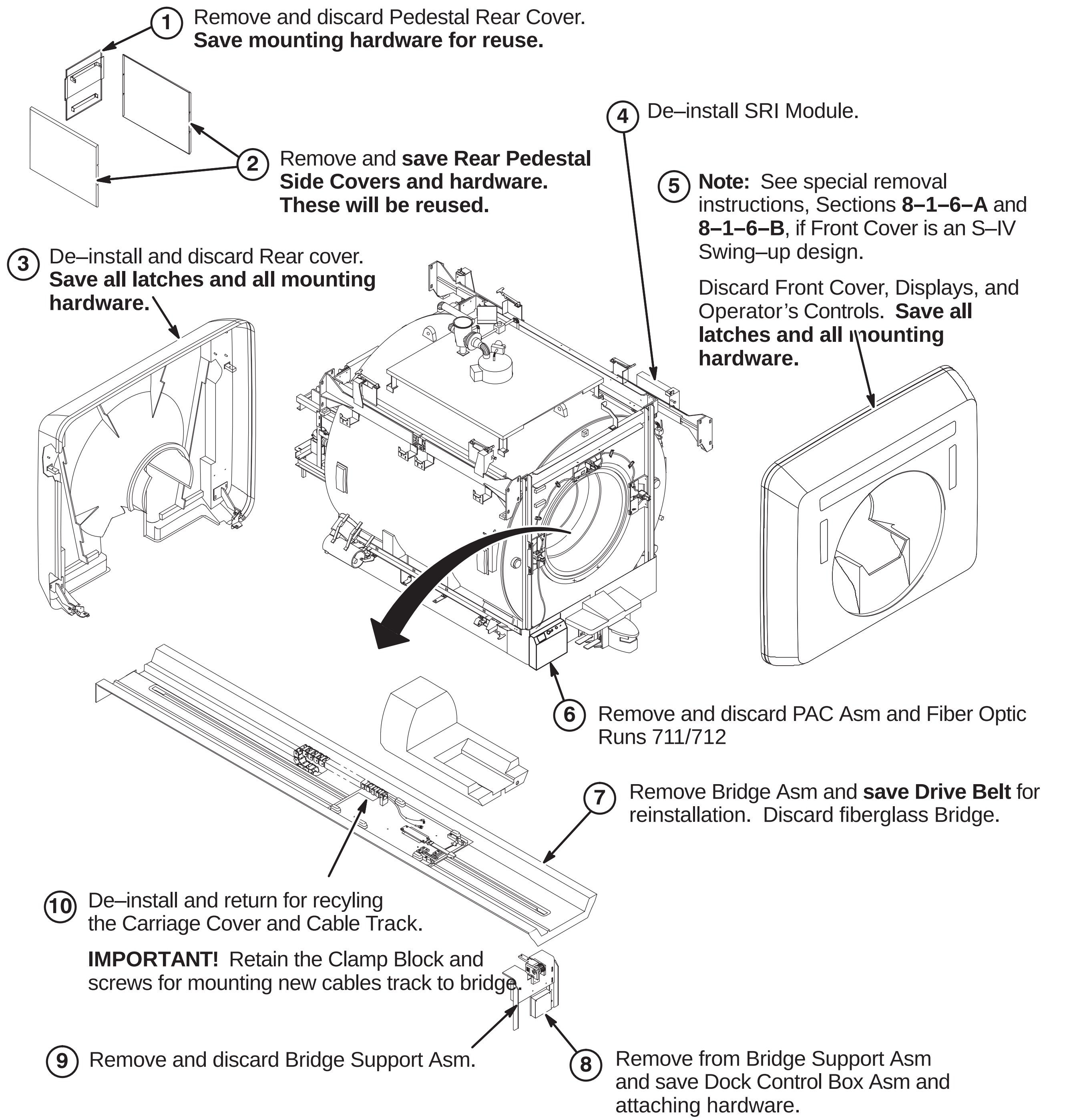
TABLE OF CONTENTS

8–1 DE–INSTALL ADVANTAGE S–SERIES MAGNET ENCLOSURE

Refer to Section 4, Table 4–1, for information on cable removal. Also refer to applicable cable map for additional information on cable removal.

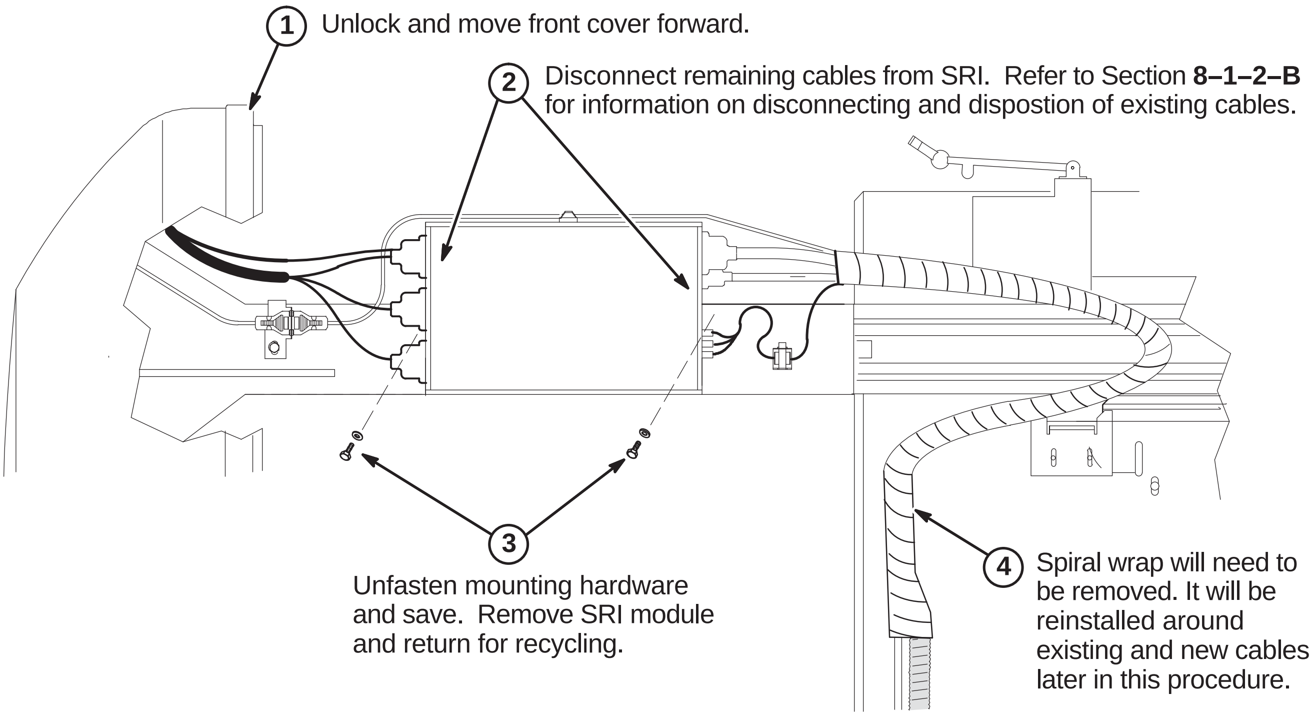
Discarded Material Return Policy – Refer to Section for procedures on how to handle all de–installed material that is designated as “discard” in the following procedures.

8–1–1 DE–INSTALL COVERS, BRIDGE, BRIDGE SUPPORT, DSIM/SRI AND PAC ASM

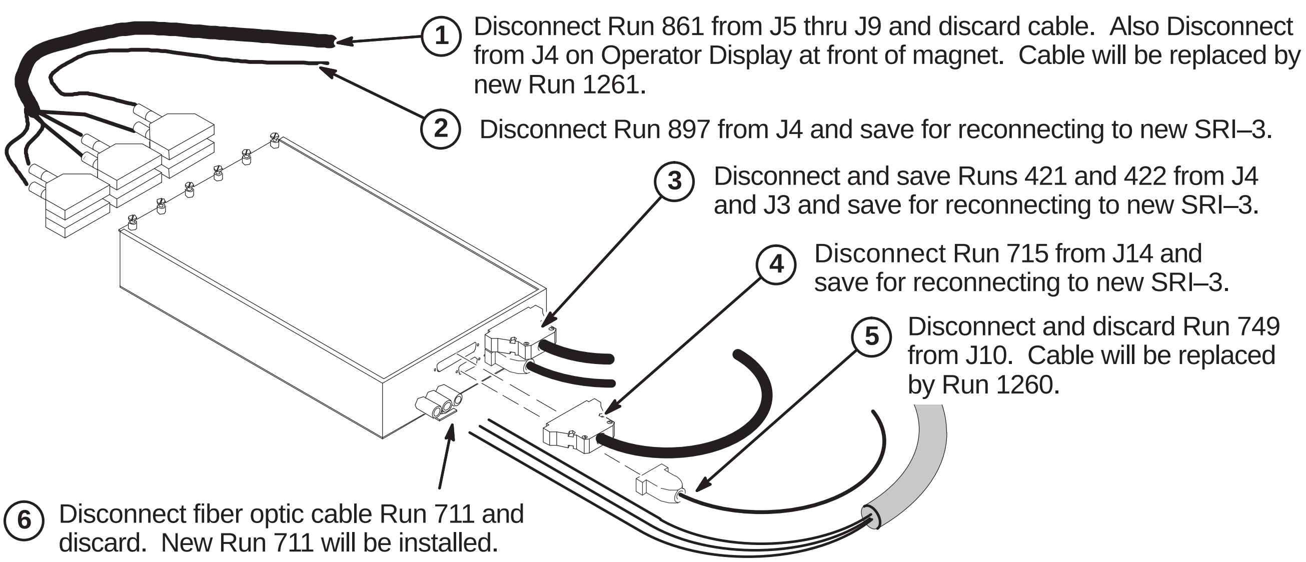


8–1–2 REMOVE EXISTING SRI MODULE

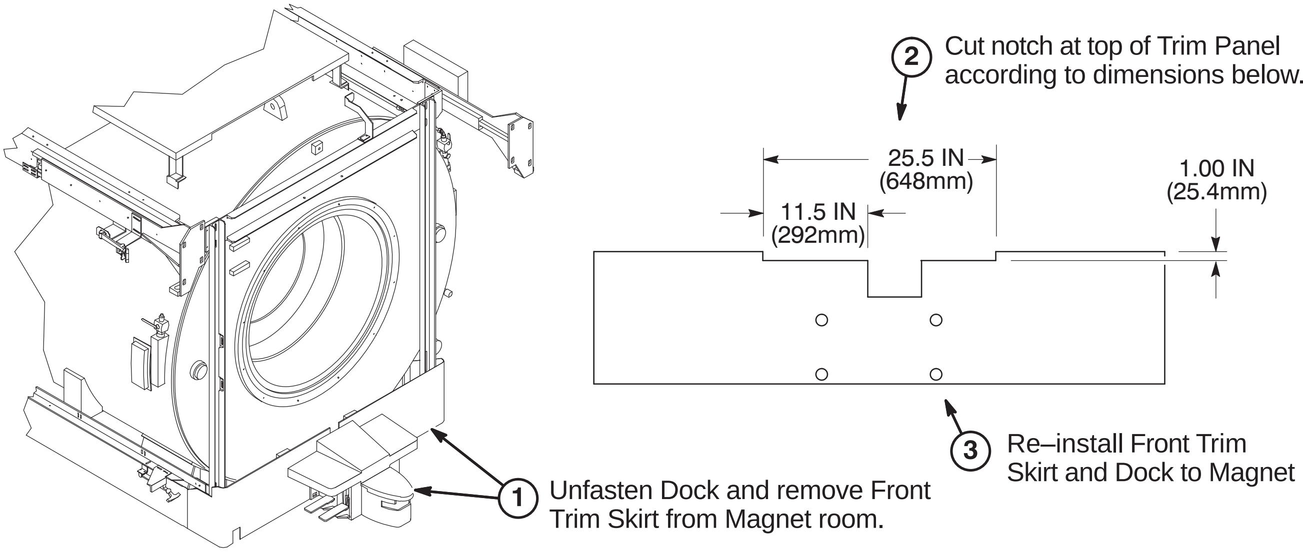
8–1–2–A De–Install SRI Module



8–1–2–B SRI Cable Disposition



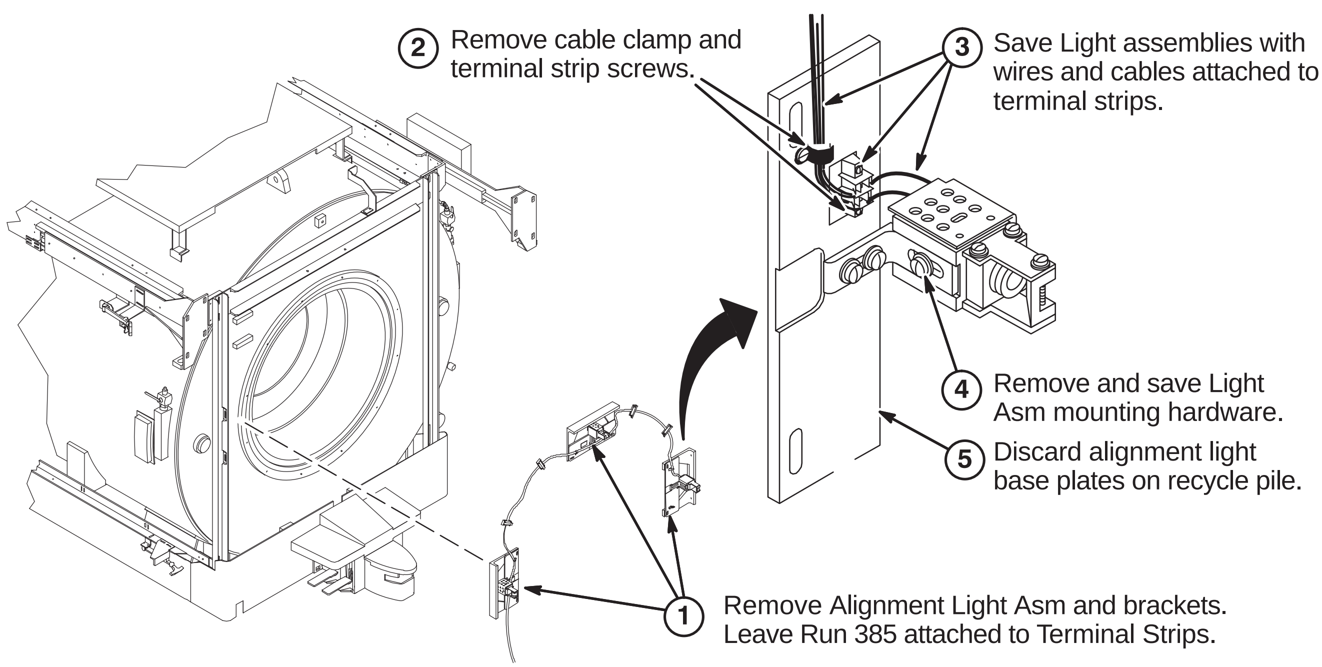
8–1–3 REMOVE AND NOTCH FRONT TRIM COVER



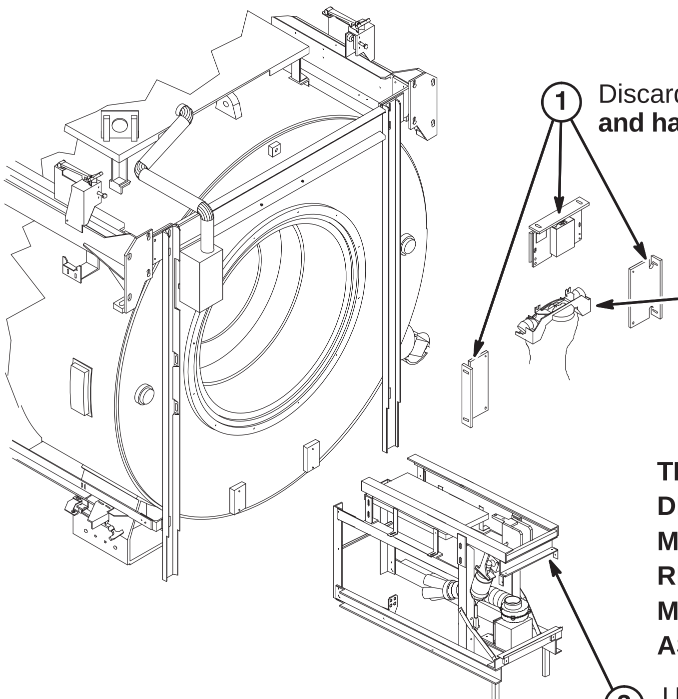
WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION. IF MAGNET IS RAMPED, WHENEVER MOVING DOCK, DOWNWARD PRESSURE MUST BE CONSTANTLY APPLIED TO THE TOP OF THE DOCK ASSEMBLY BY TWO PEOPLE. DOCK MUST BE MOVED SLOWLY ALONG THE GROUND AND NOT LIFTED VERTICALLY.

8–1–4 REMOVE ALIGNMENT LIGHTS



8–1–5 REMOVE REAR PEDESTAL, BORE LIGHT, & COMFORT MODULE



1. Discard Bore Light brackets.**Save Bore Light and hardware for transfer to new Rear Ring.**
2. Remove asnd discard Comfort Module.

WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

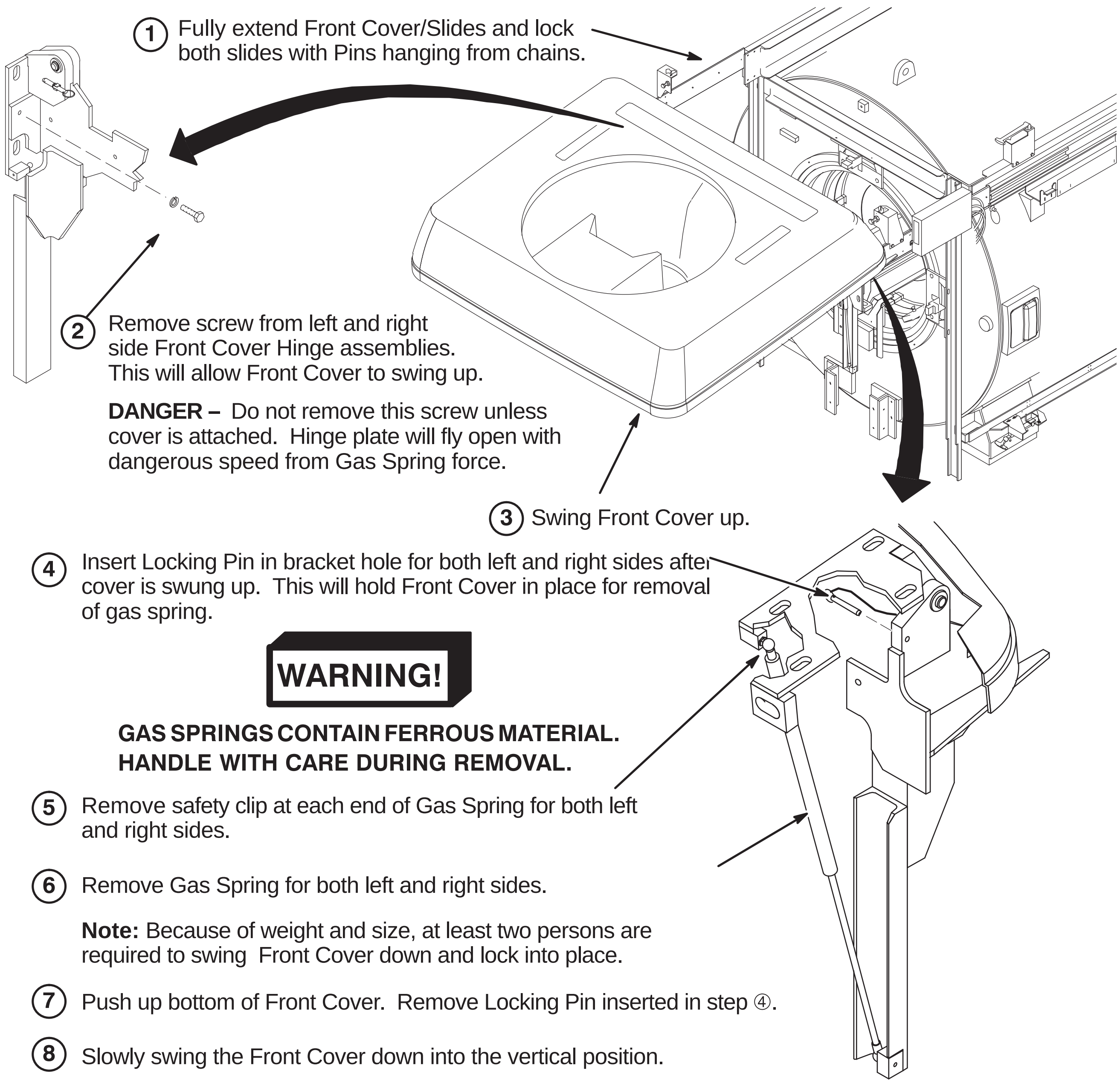
3. Unfasten Rear Pedestal and remove from Magnet Room. **Many of the components of the removed Rear Pedestal HAVE to be transferred to the new Rear Pedestal.**
Refer to Section NO TAG for details.
4. Discard old Rear Pedestal frame after transfer of components.

8–1–6 DE–INSTALL FRONT COVER GAS SPRING (S–IV & S–X Magnets)

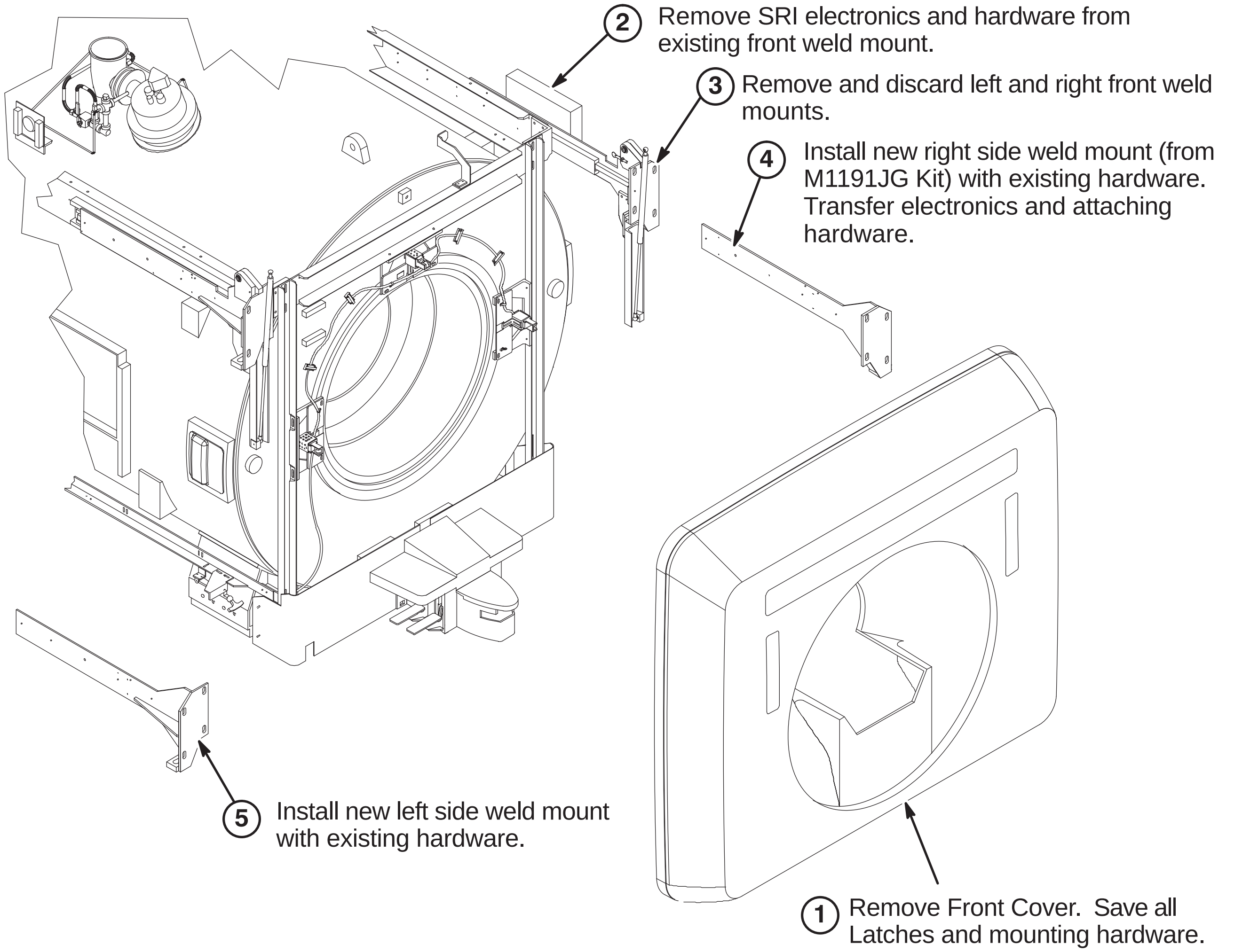
8–1–6–A Preliminary Front Cover Removal Instructions

CAUTION

MAGNET MATERIAL HAZARD! The cover dolly was not designed for cover installation with magnet at field. The caster assemblies and fastening hardware are magnetic. If magnet is at field, three installers are required to remove front cover from dolly and carry cover into position at front of magnet.

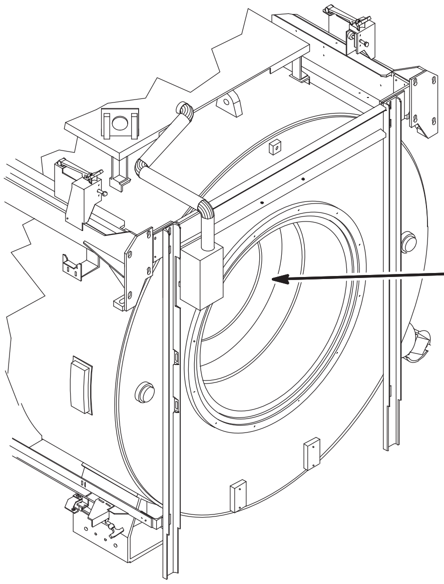


8–1–6–B Remove Front Cover, Install New Weld Mounts (S–IV & S–X Magnets)



8–1–7 BODY COIL REPLACEMENT

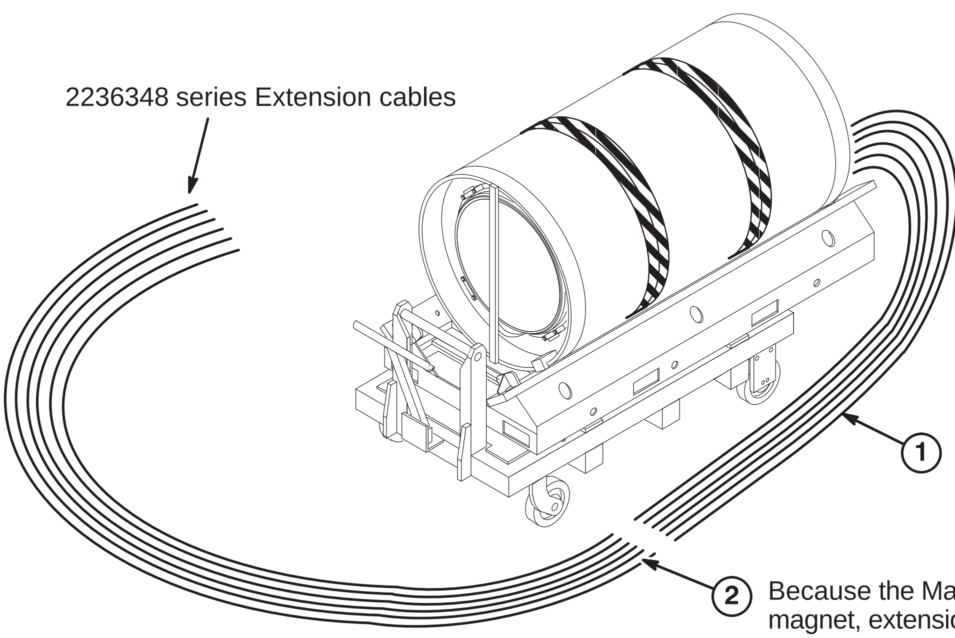
8–1–7–A Coil Removal Instructions



1 Remove Gradient Body Coil.

Note: Refer to procedures in Section **13** and *Signa EXCITE HD 1.5T Service Methods CD–ROM,* Direction 5133315 Replacement>Coil Replacements>Gradient Coil Replacement for Non–Cx Magnets, for instructions on de–installing the existing Body Coil.

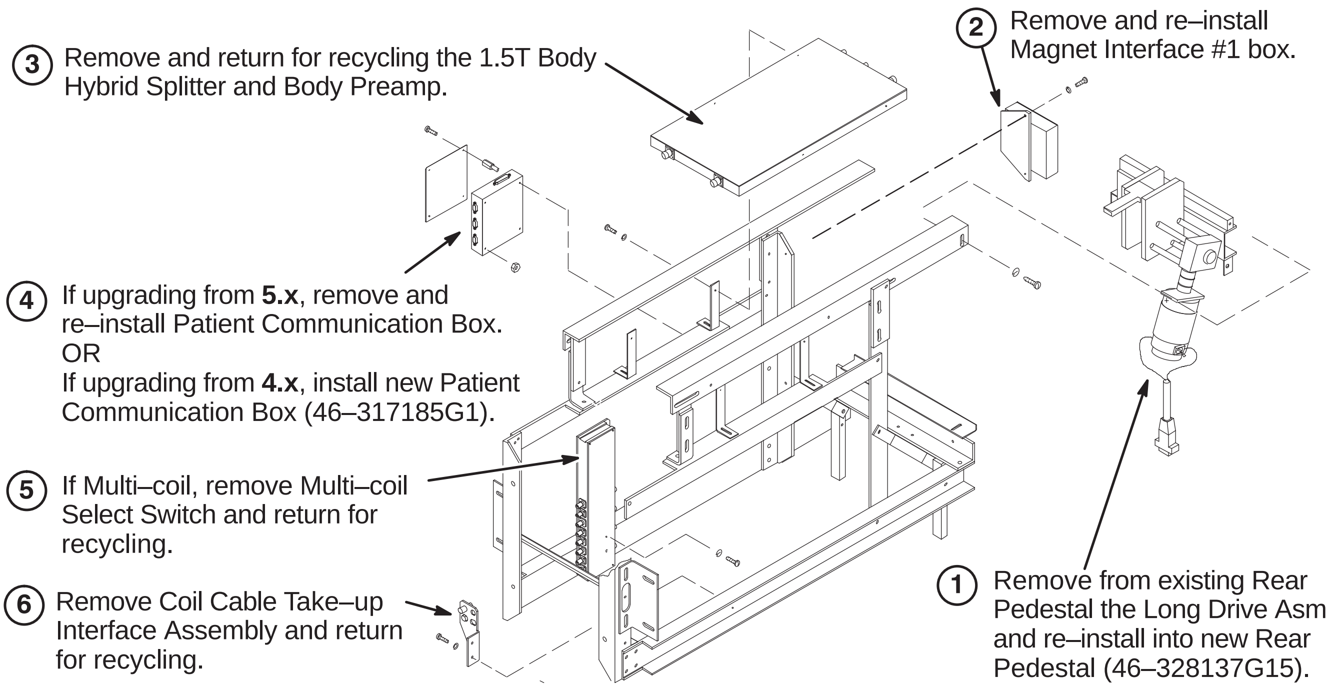
8–1–7–B BRM–D Gradient Coil Pre–Installation Instructions



1 The new BRM–D coil will have long gradient cable leads.

2 Because the Magnishield is not an Active–Shield magnet, extension gradient leads have to be added. The procedure for attaching the extension leads is located in Section **13–2** must be completed outside of the magnet room.

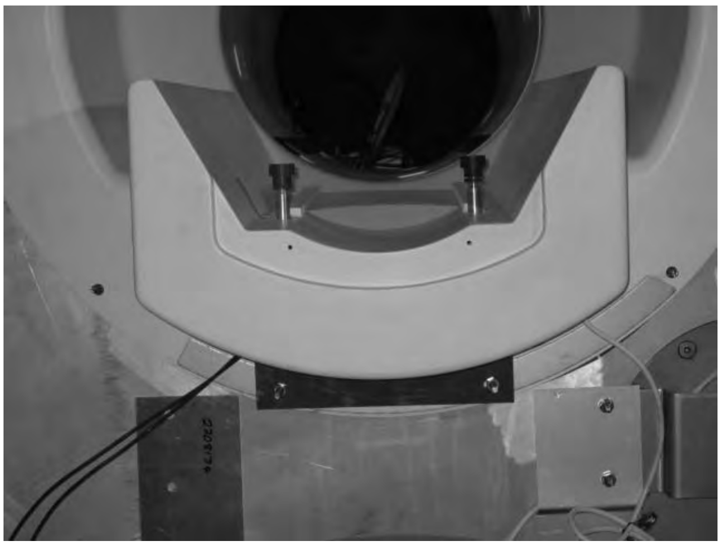
8–2 TRANSFER EXISTING COMPONENTS TO NEW REAR PEDESTAL FRAME



8–3 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

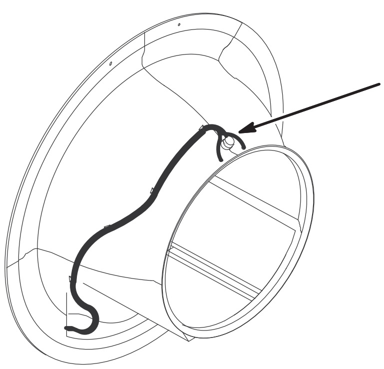
Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) has been installed. This kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It is required for allmagnets with Horizon style enclosures. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and it must be installed at this point of the EXCITE HD ugprade.

Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2401984, *S–Series Magnet Bridge Support* (shown below) and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.



Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade,* for instructions on installing this kit.

8–4 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL



The new Front End Bell is equipped with a Dual Temperature Sensor cable, Run 749, that has to be replaced with a new cable, Run 1260 (5123812).

EXCITE HD requires Run 1260 to be installed.

Refer to Sections **10–3–7–B, 10–3–8,** and **10–3–9** for instructions on removal of existing cable and installing the new cable.

8–5 INSTALL MAGNET MONITOR

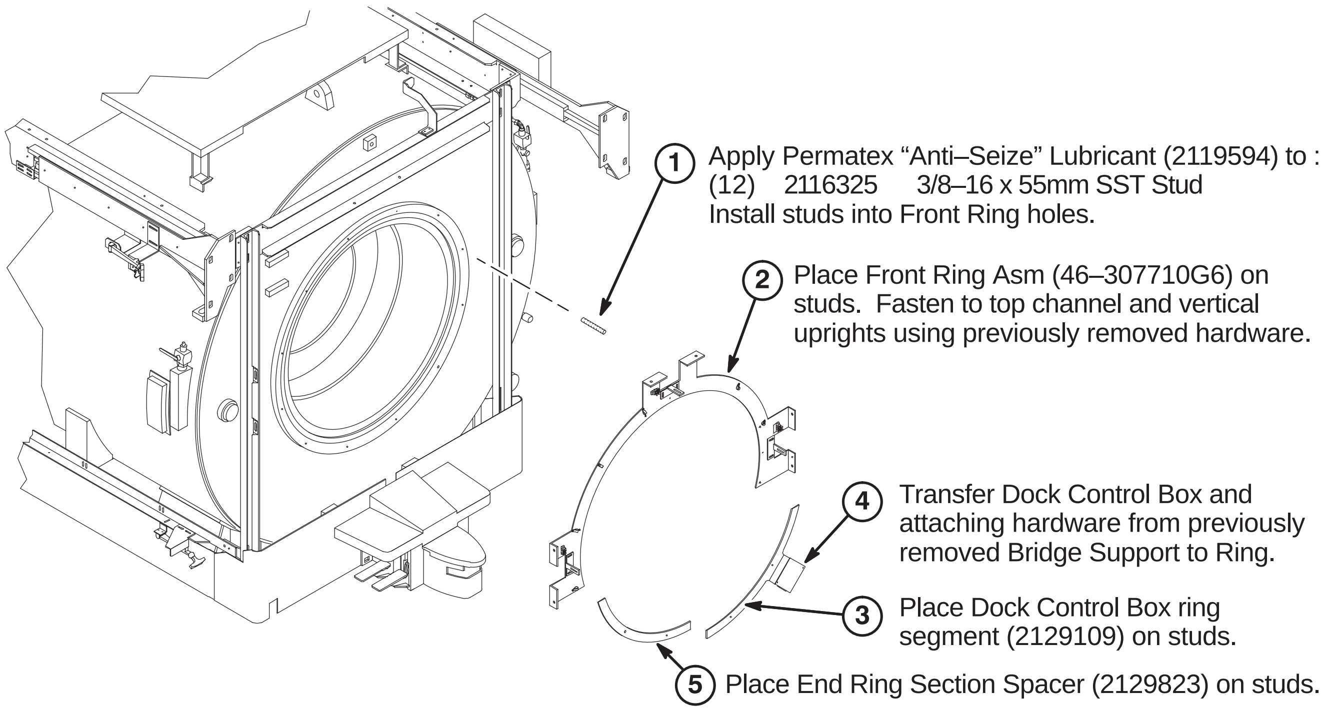
EXCITE HD requries Magnet Monitor on all systems. For this upgrade, all the parts and instructions are included in parts collector 5127838, Magnet Monitor Kit for Non–LCC Systems. Refer to Direction 2230681, included with collector, for instructions on installing magnet monitor on this magnet.

8–6 INSTALL NEW COVERS, COMPONENTS, AND CABLES

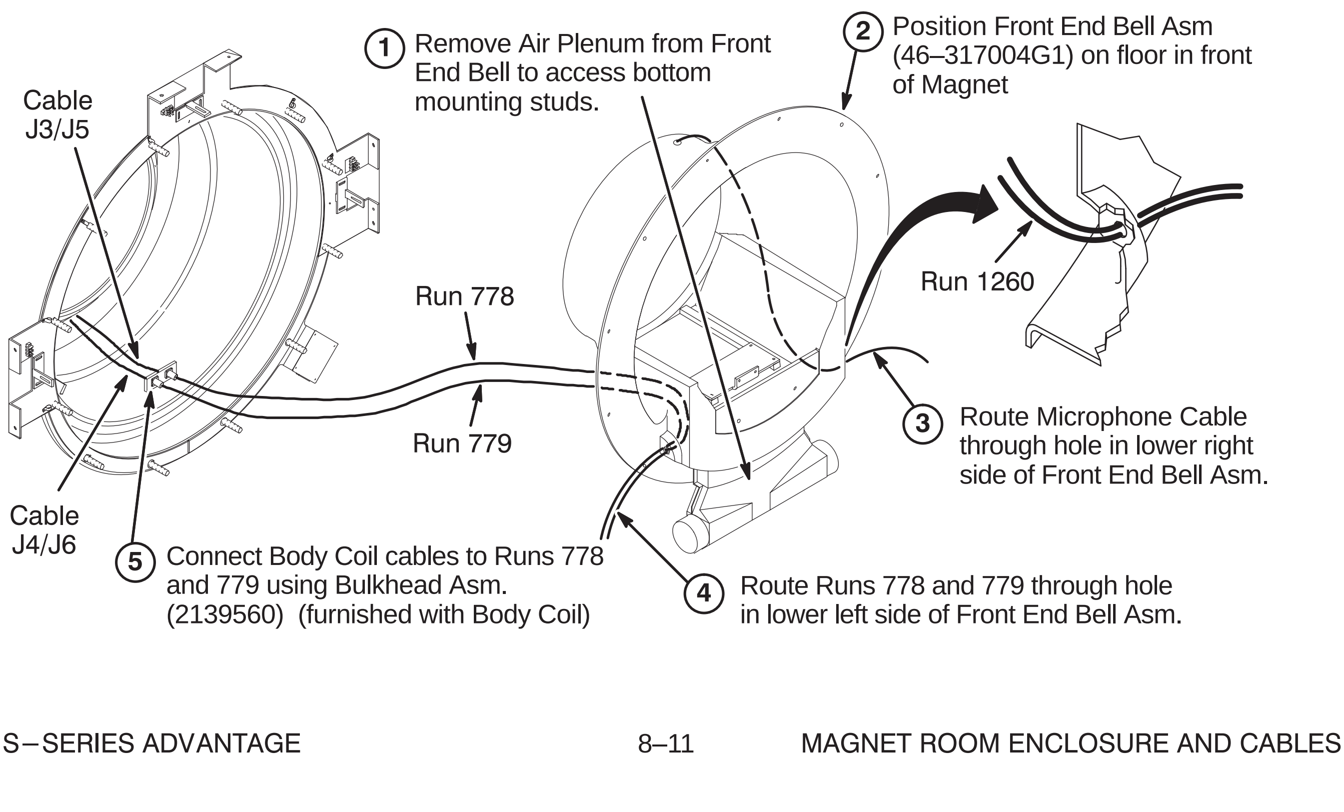
The new Gradient Coil must be installed before starting the installation of new parts referred to in the following steps. Refer to Section for Gradient Coil installation instructions.

8–6–1 INSTALL COMPONENTS AT FRONT OF MAGNET

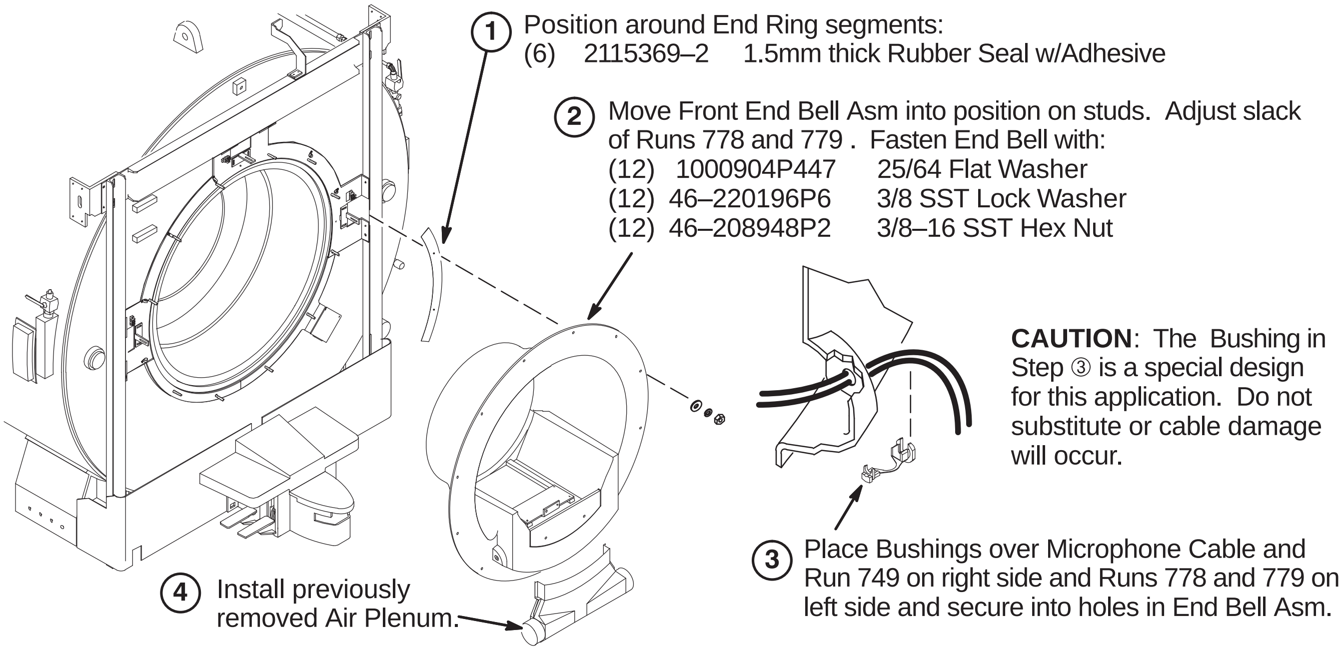
8–6–1–A Install Front Ring Assembly



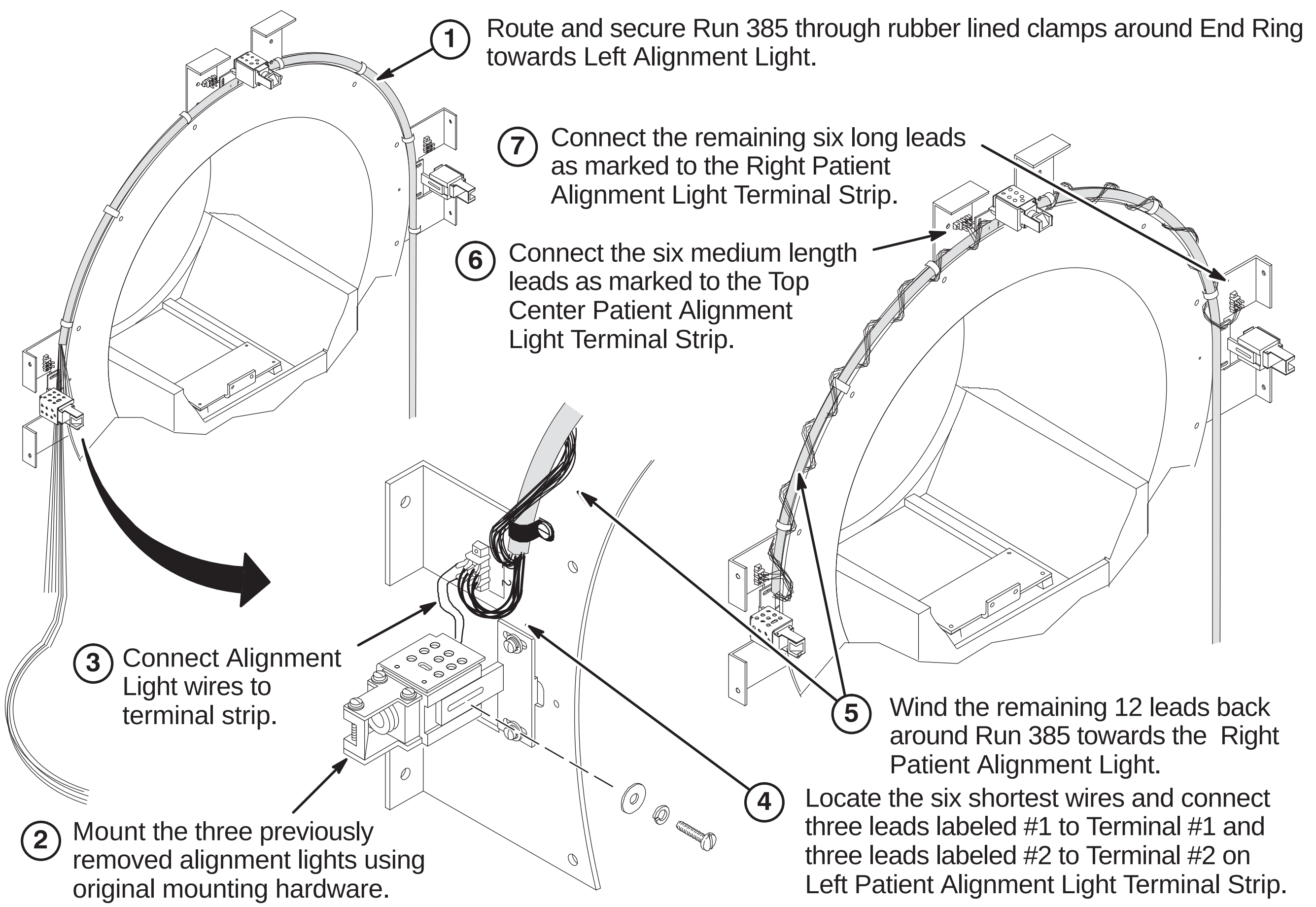
8–6–1–B Route and Connect Runs 778 and 779



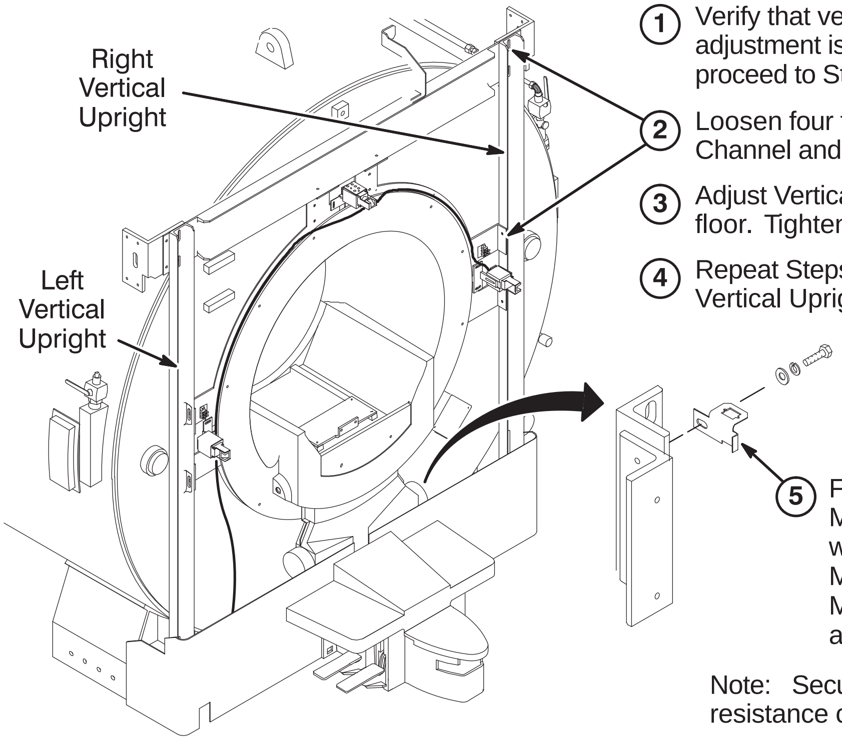
8–6–1–C Install Front Gasket and Mount End Bell



8–6–1–D Install and Connect Alignment Lights



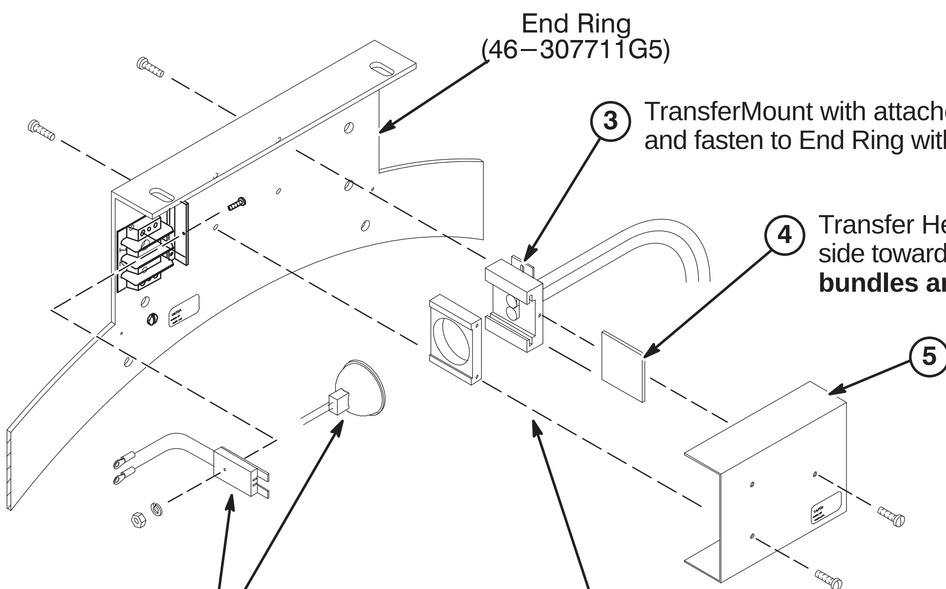
8–6–1–E Adjust Front Vertical Uprights & Installing Microphone Bracket



1. Verify that vertical uprights contact the floor. If adjustment is needed, proceed to Step ②. If not, proceed to Step ⑤.
2. Loosen four fasteners attaching Upright to Top Channel and Front Ring Asm.
3. Adjust Vertical Upright to make firm contact with the floor. Tighten fasteners.
4. Repeat Steps ② and ③ for adjustment of Left Vertical Upright.
5. From the Hardware Box, install Front Microphone Connector Bracket (2109027) with existing hardware. Insert (2) Cable Mounts (46–265067P1) and connect Microphone cable from End Bell to Run 414 at this bracket.

Note: Secure electrical ground required. Insure low resistance connection and tighten fasteners securely.

8–6–1–F Transfer & Assemble Bore Light to New Rear End Ring

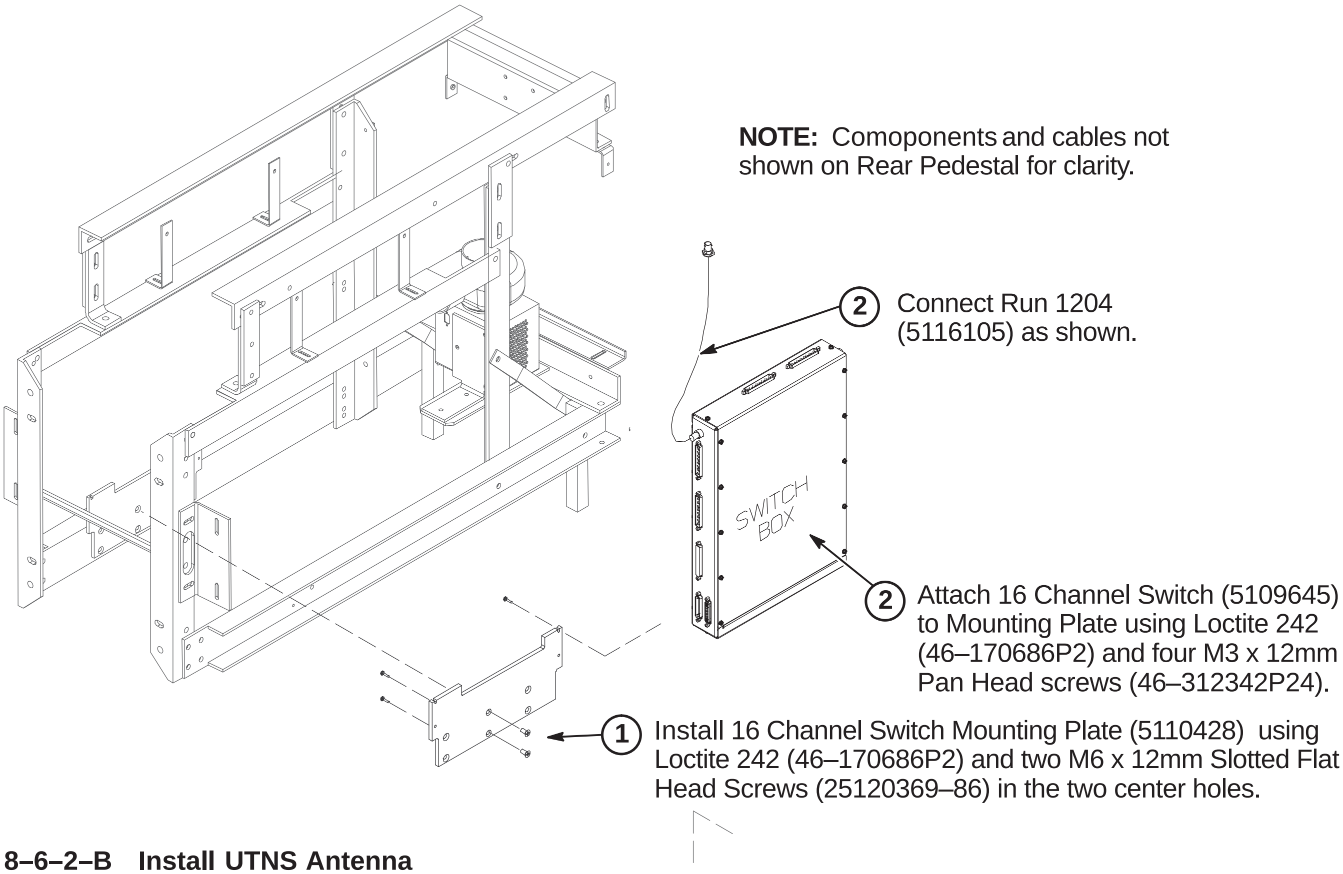


3. TransferMount with attached Fiber Optic Bundle and fasten to End Ring with 6–32 screws.
4. Transfer Heat Reflecting Mirror with coated side toward bulb. **Note that Fiber Optic bundles are NOT to touch glass.**
5.

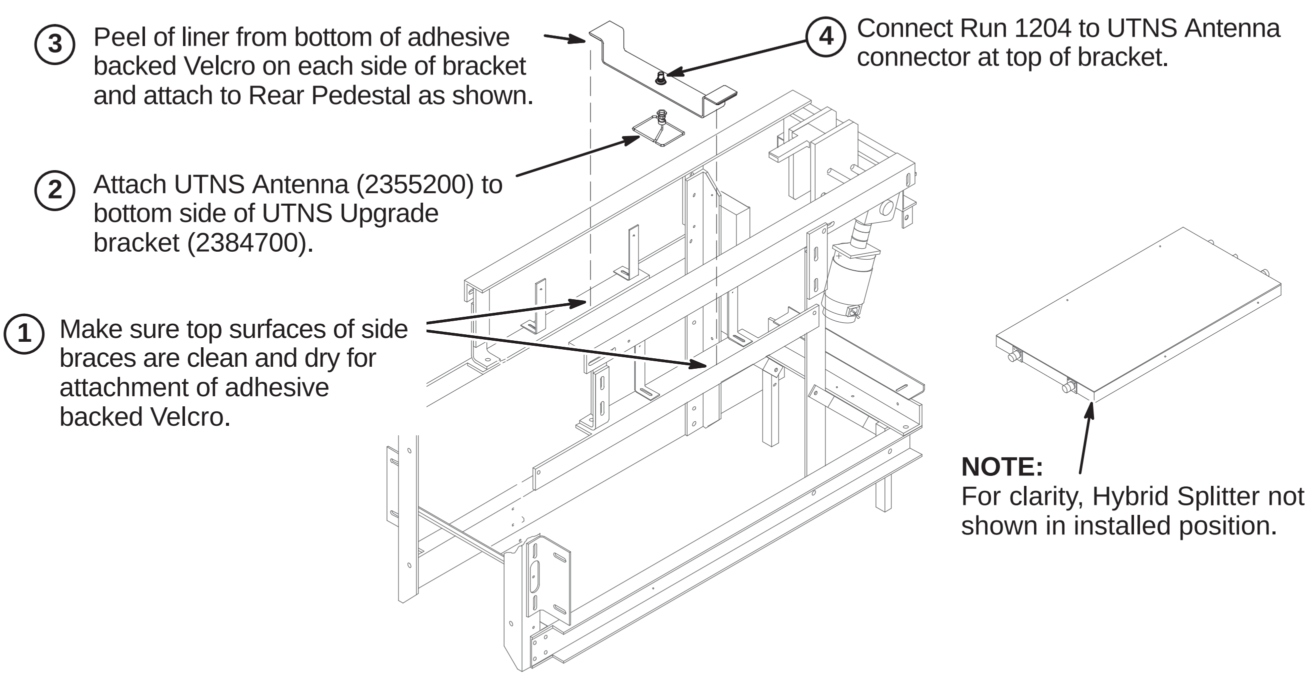
1. Transfer socket to new mounting bracket and attach with 4–40 hardware. Connect wires to terminal strip; plug in Bore Lamp.
2. Transfer Bulb Retainer and attach to End Ring with 6–32 screws.

8–6–2 INSTALL NEW COMPONENTS ON REAR PEDESTAL

8–6–2–A Install 16 Channel Switch

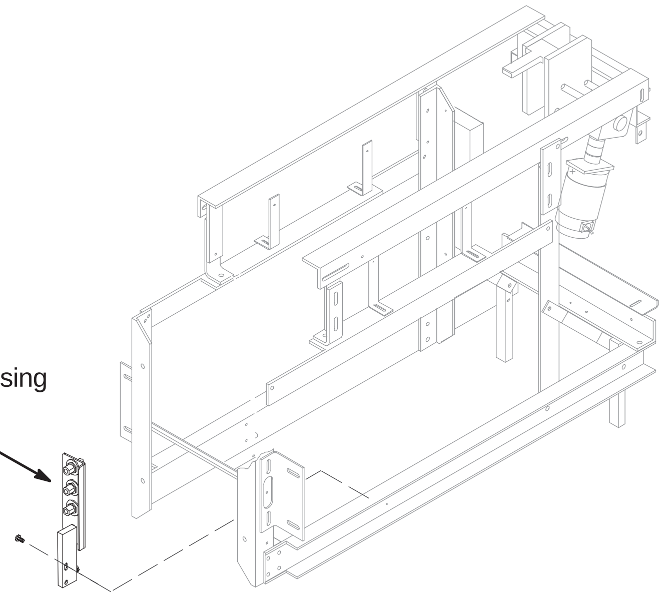


8–6–2–B Install UTNS Antenna



8–6–2–C Install Bulkhead Interface

1 Install new Bulkhead Interface (5113250) using Loctite 242 (46–170686P2) and two M6 x 12mm pan Head Screws (2109873–24).



8–6–3 HYBRID SPLITTER UPGRADE

Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

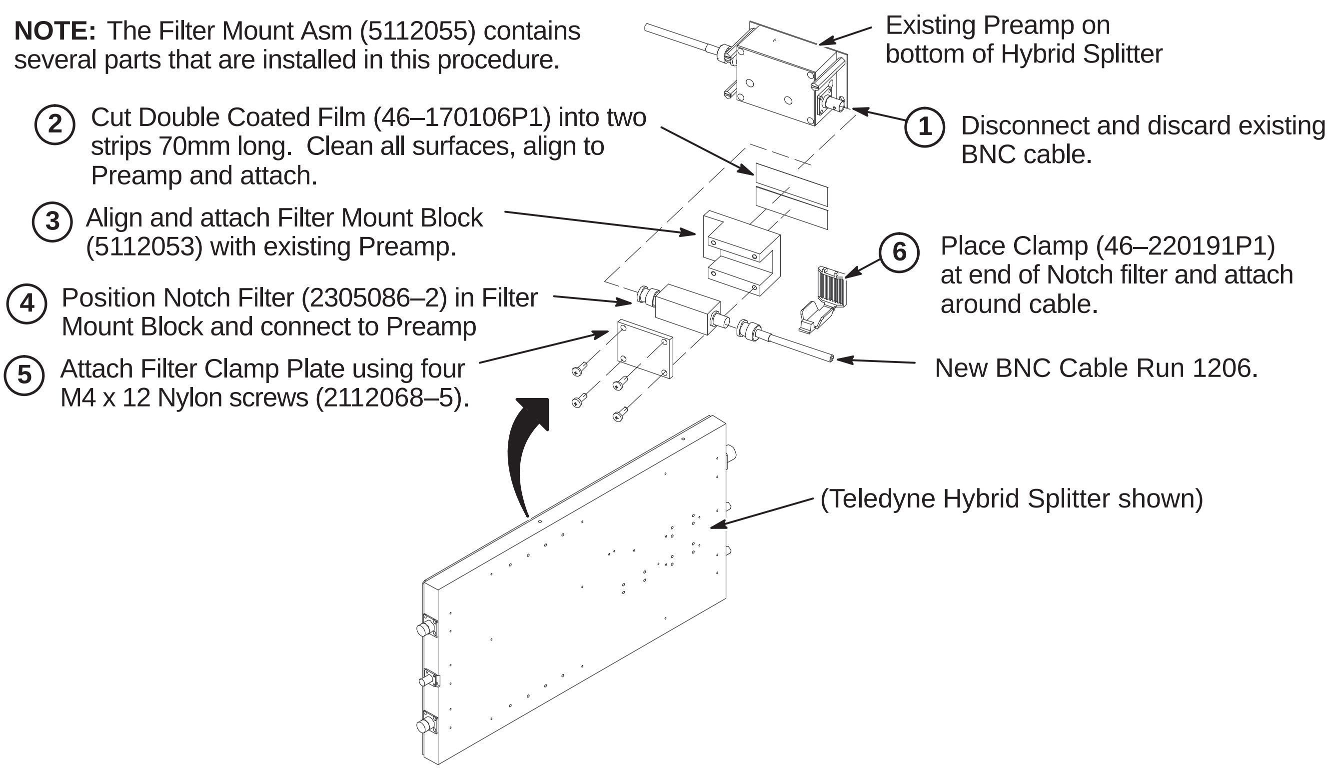
- Refer to Section 8–6–3–A for instructions on installing the Notch Filter and mounting block
- Refer to Section 8–6–3–B for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 8–6–3–C for instructions on installing Isolation Filters on systems with Teledyne Splitter

The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils.

8–6–3–A Install Filter Mount Asm and Isolation Filters to Body Hybrid Splitter

This Filter Mount assembly may not be required on newer Netcom Hybrid Spitter as it may already have a Notch Filter mounting clamp of a different style installed.

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.

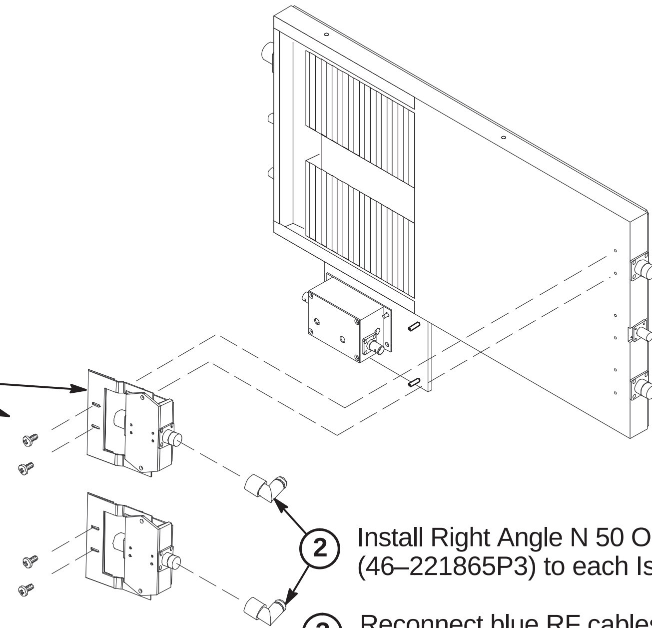


8–6–3–B Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)

FOR BRM's ONLY

1 Attach and connect each Ground Isolation Filter Module (5118952–2) to the Hybrid Splitter using two screws (46–220184P5).

Do NOT install on TRM gradient coil systems.

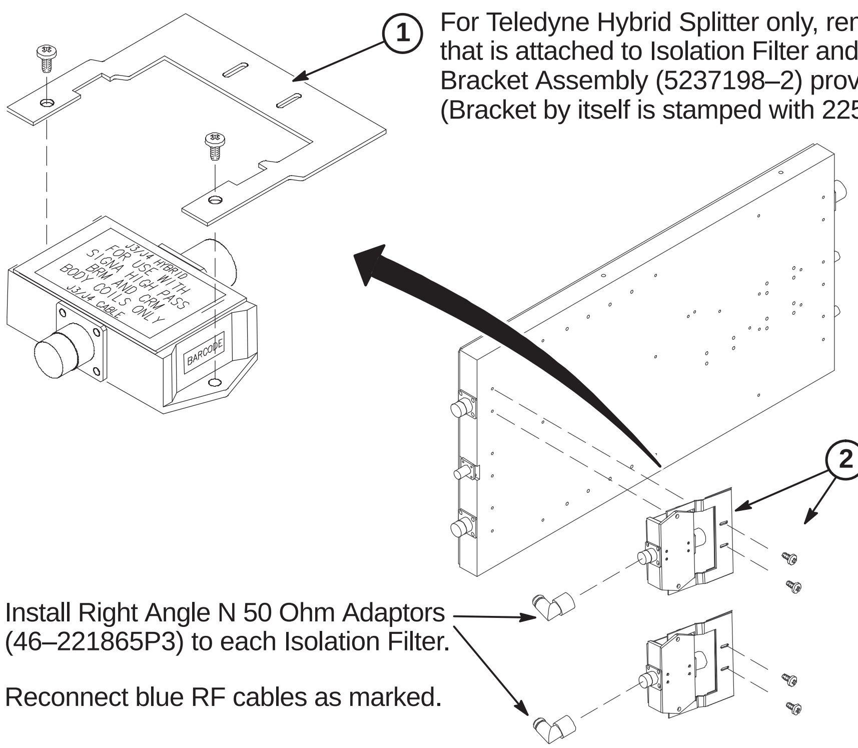


2 Install Right Angle N 50 Ohm Adaptors (46–221865P3) to each Isolation Filter.

3 Reconnect blue RF cables as marked.

8–6–3–C Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)

1 For Teledyne Hybrid Splitter only, remove existing bracket that is attached to Isolation Filter and replace with Mounting Bracket Assembly (5237198–2) provided with upgrade kit. (Bracket by itself is stamped with 2254629)



FOR BRM's ONLY

2 Attach and connect each Ground Isolation Filter Module (5118952) to the Hybrid Splitter using two screws (46–220184P5) for each module.

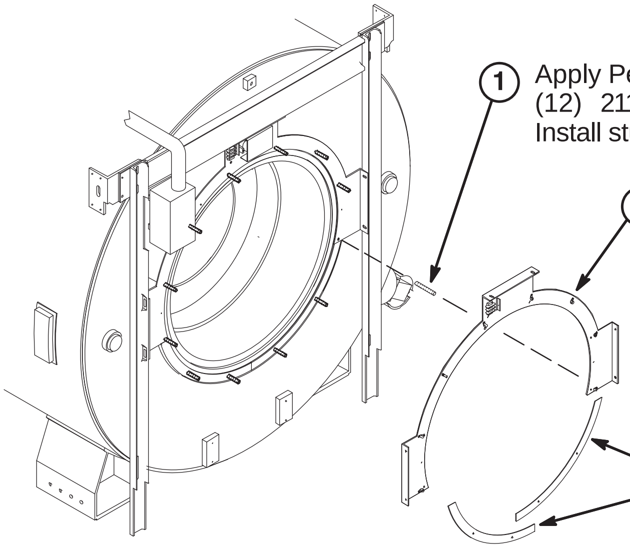
Do NOT install on TRM gradient coil systems.

3 Install Right Angle N 50 Ohm Adaptors (46–221865P3) to each Isolation Filter.

4 Reconnect blue RF cables as marked.

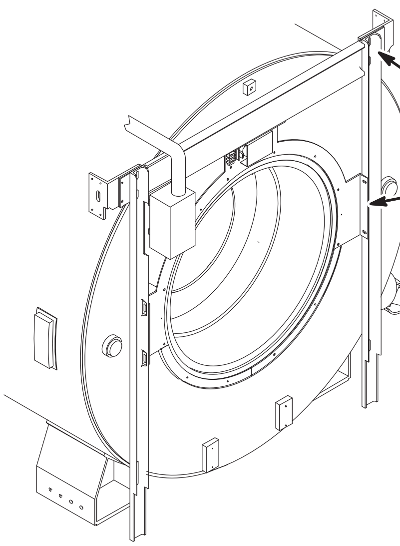
8–6–4 INSTALL COMPONENTS AT REAR OF MAGNET

8–6–4–A Install Rear End Rings



1. Apply Permatex “Anti–Seize” Lubricant (2119594) to :
(12) 2116325–4 3/8–16 x 105mm SST Stud
Install studs into Rear Ring holes.
2. Install End Ring Asm (46–307711G5) on studs.
3. Install End Ring Section Spacers (2129823) to studs on lower part of Magnet End Ring.

8–6–4–B Adjusting Rear Vertical Uprights



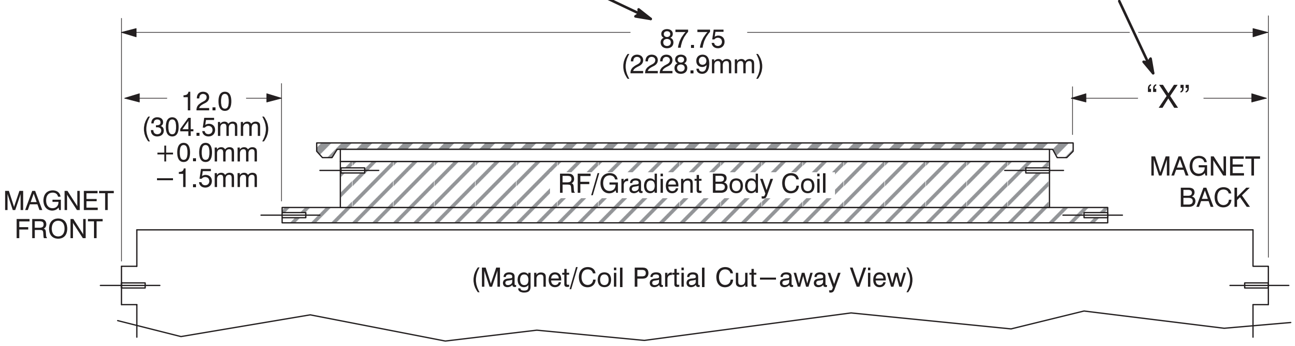
1. Verify that vertical uprights contact the floor.
If adjustment is needed, proceed to Step ②.
If not, proceed to Step ④.
2. Loosen four fasteners attaching Upright to Top Channel and Front Ring Asm.
3. Adjust Vertical Upright to make firm contact with the floor. Tighten fasteners.
4. Repeat Steps ② and ③ if adjustment is needed for Right Rear Vertical Upright.

8–6–4–C Formula For Determining Rear Spacers

1. With RF/Gradient Body Coil in place, measure in millimeters the distance from RF Tube to mounting flange on Magnet. This is distance “X”. **Make sure the distance is measured to the Magnet mounting flange, not the Rear Frame Asm End Ring installed in Step D3.**

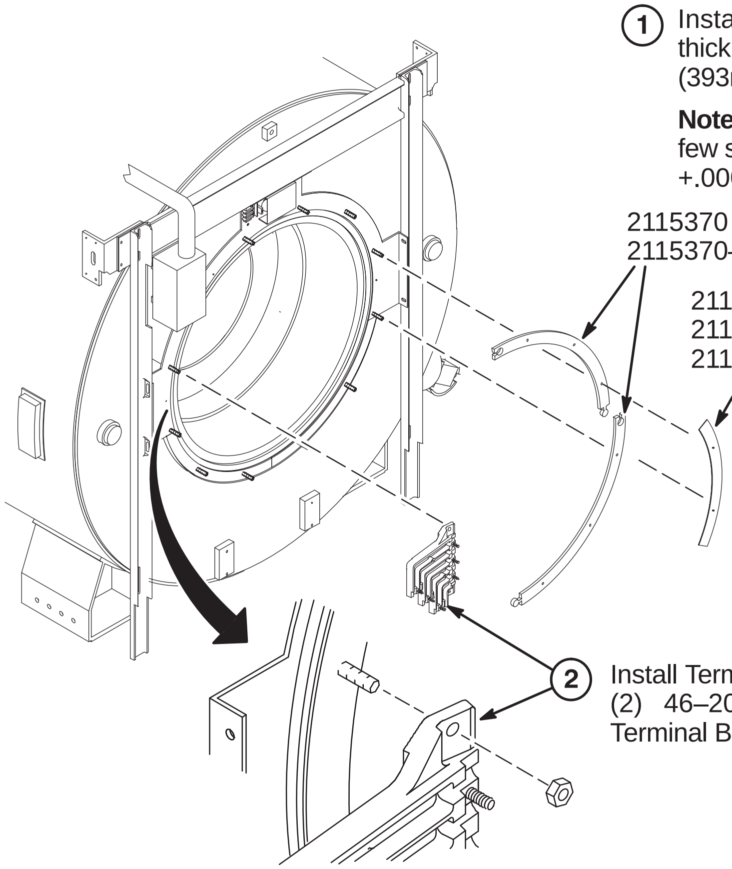
Nominal Magnet length from mounting flange to mounting flange.

2. Subtract distance “X” from 393mm to determine the correct shim thickness.



Note: Values shown here are for GE S–II, S–III, S–IV Magnets only.

8–6–4–D Install Rear Spacers And Gradient Terminal Block



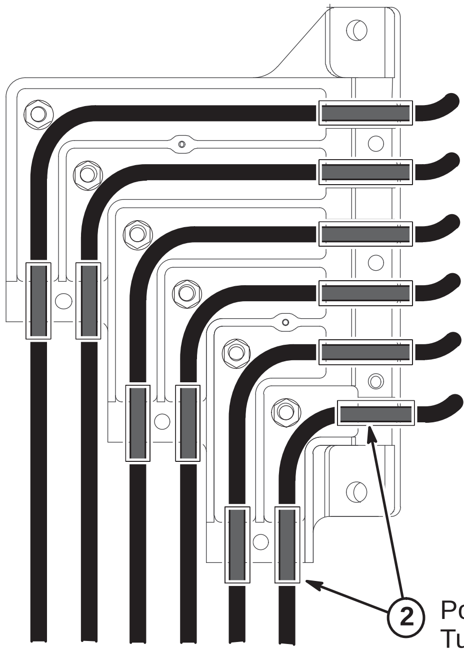
1. Install the following Rubber Seals as required to obtain a thickness derived from the formula in the previous step: (393mm – “X” = Shim Thickness)

   Note: Better air sealing results will occur with having too few shims than too many. The tolerance on distance “X” is +.000mm, –.75mm.

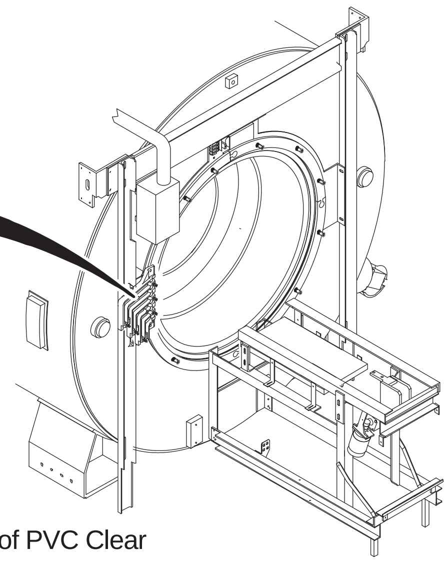
2115370	12.7mm thick Interlocking Rubber Seal
2115370–2	7.9mm thick Interlocking Rubber Seal
2115369	3/4mm thick Rubber Seal w/adhesive
2115369–2	1.5mm thick Rubber Seal w/adhesive
2115369–3	2.25mm thick Rubber Seal w/adhesive

2. Install Terminal Block to magnet End Ring with:
(2) 46–208948P2 3/8–16 SST Hex Nut
Terminal Block Access Cover fits over top of this nut.

8–6–4–E Route and Install Gradient Cables in Terminal Block

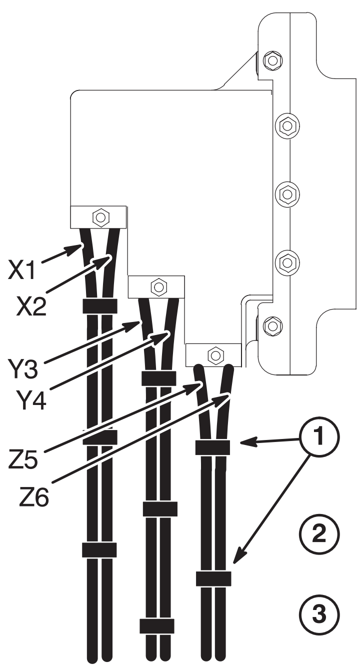


1. Route the six (6) Body Coil Gradient cables thru the Terminal Block as shown.

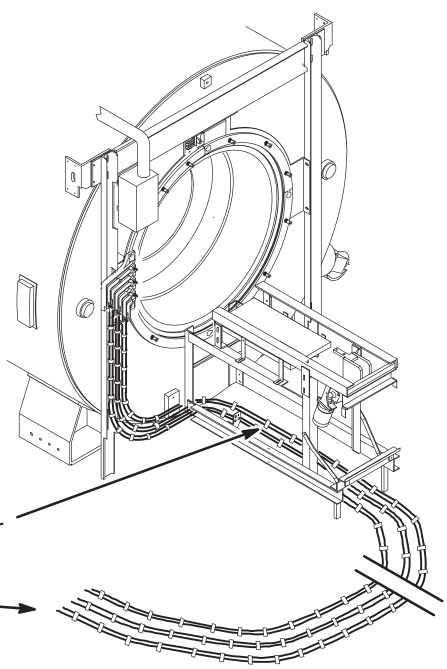


2. Position the twelve (12) pieces of PVC Clear Tubing (46–221986P3) as shown.
3. Install Terminal Block strain reliefs over PVC tubing. Install cover.

8–6–4–F Secure X, Y, and Z Gradient Cables

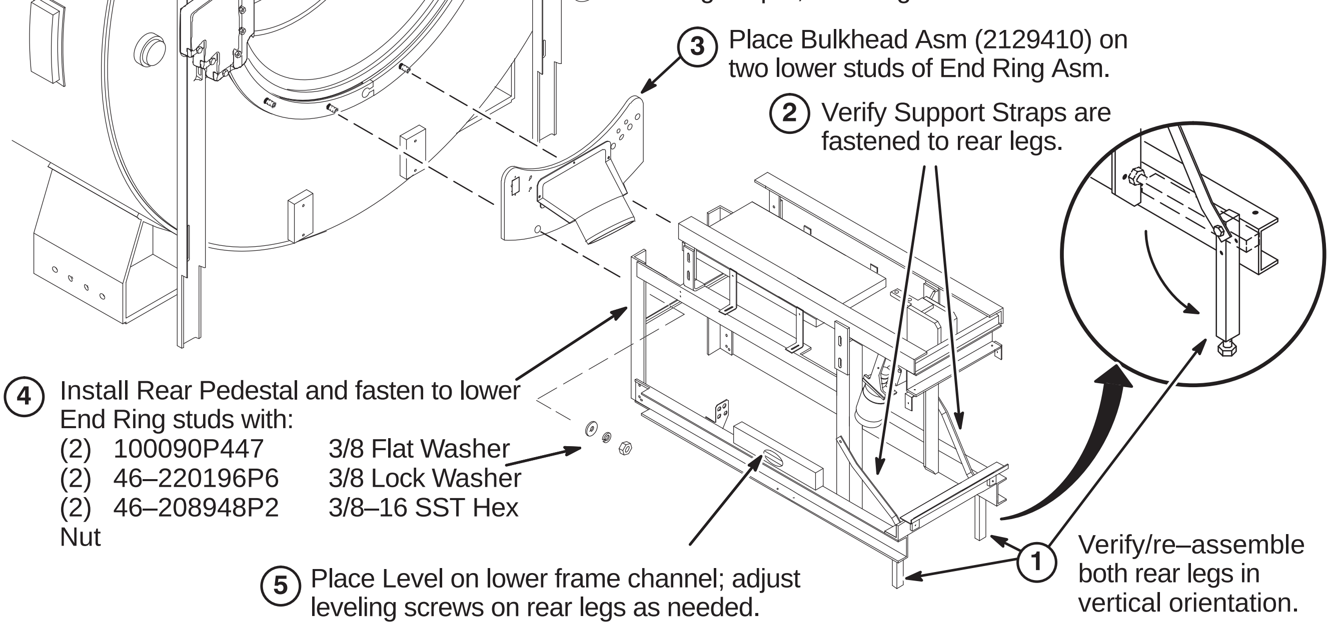


1. Tape cables together as shown with electrical tape, approximately 100mm (4 in.) apart, all the way back to cable jacket.
2. Route Body Coil cables under area where Rear Pedestal will be installed.
3. **NON–ACTIVE SHIELD MAGNETS**
For connection to Gradient cables from Penetration Panel, continue to route Body Coil cables to farthest point away from Magnet. (If possible, outside of Magnet Room)



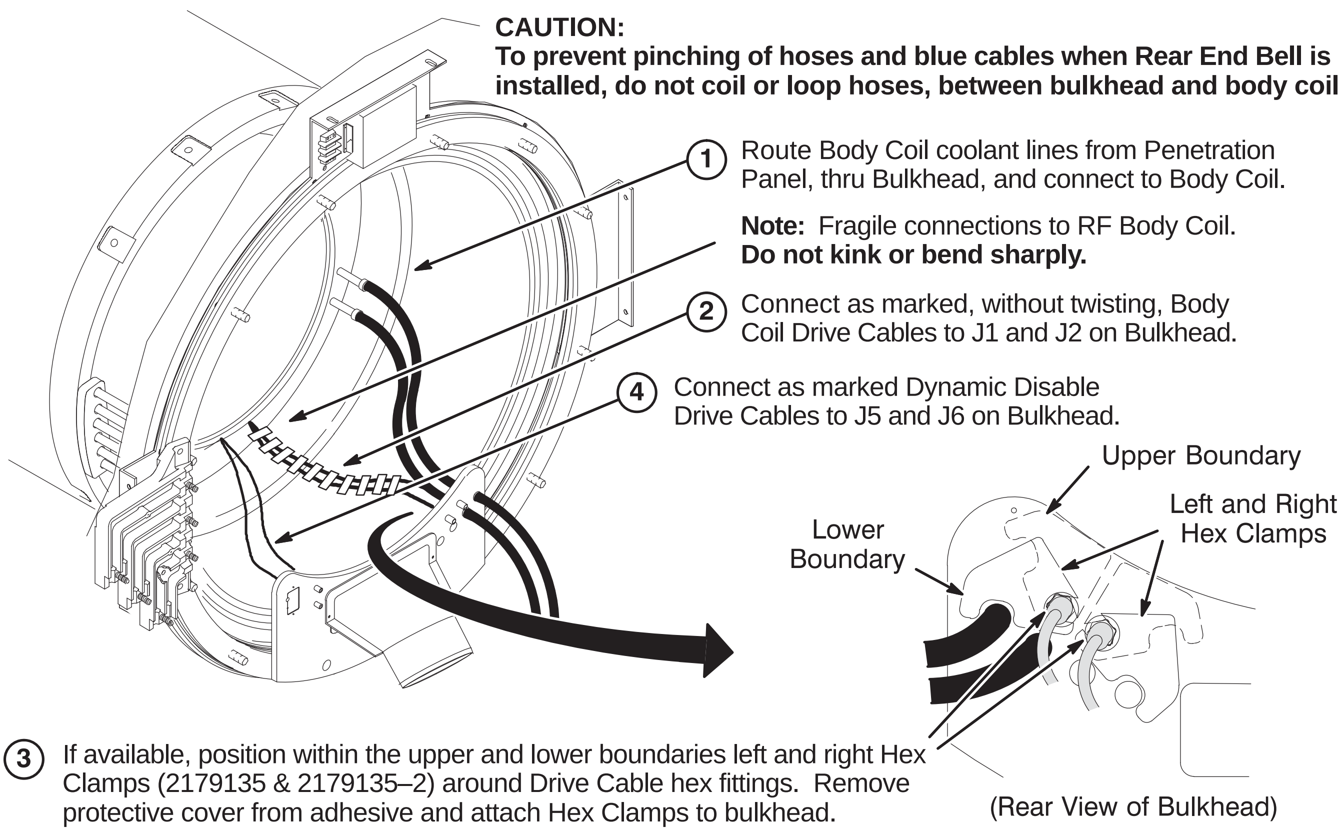
8–6–4–G Install Rear Bulkhead And Rear Pedestal

Note: If desired, installation procedures in Section **8–6–4–H** can be performed after completion of Step 3 below and prior to starting Step 4, installing Rear Pedestal.



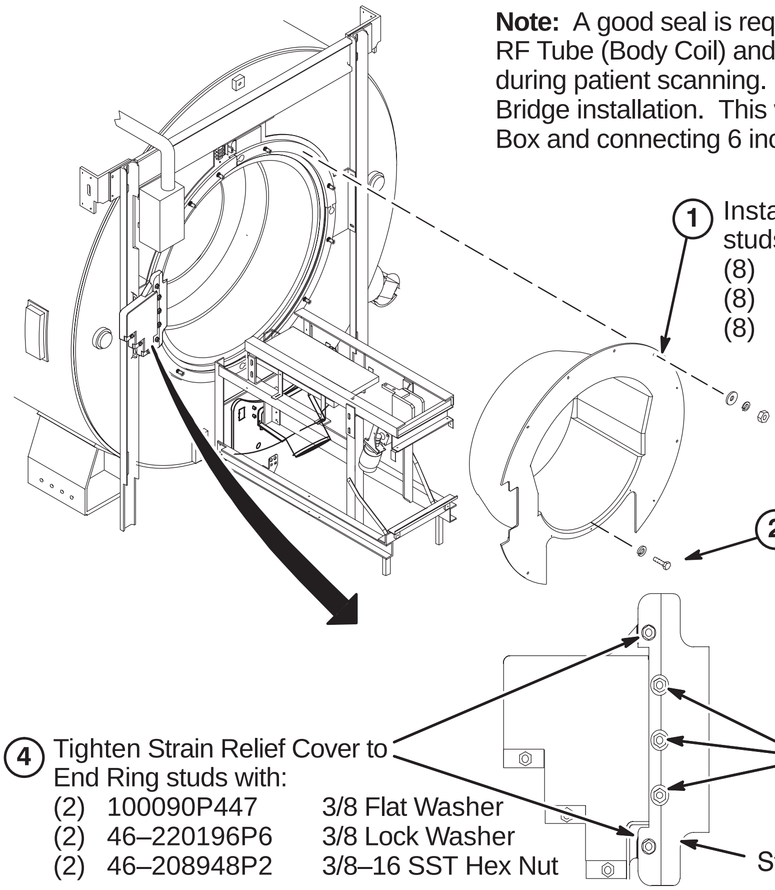
8–6–4–H Route And Connect Bulkhead Cables And Hoses

CAUTION:
To prevent pinching of hoses and blue cables when Rear End Bell is installed, do not coil or loop hoses, between bulkhead and body coil.



8–6–4–I Install Rear End Bell

Note: A good seal is required to prevent air leaks. Any air leaks between RF Tube (Body Coil) and End Bell joint can cause a loud humming noise during patient scanning. If possible, conduct an air leakage test prior to Bridge installation. This would require connecting power to the Blower Box and connecting 6 inch blower hose from Blower Box to Bulkhead.

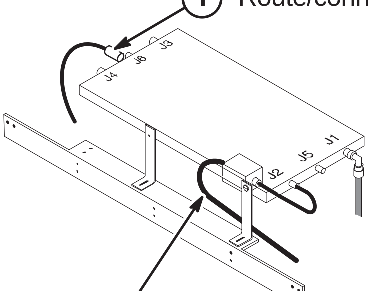


1. Install Rear End Bell (2113076) and fasten to studs with:
   (8) 1000904P447 25/64 Flat Washer
   (8) 46–220196P6 3/8 SST Lock Washer
   (8) 46–208948P2 3/8–16 SST Hex Nut
2. Install Rear End Bell to Bulkhead Asm with:
   (3) 2109878–3 M6 SST Flat Washer
   (3) 2109866–13 M6 SST Hex Screw
3. Install Strain Relief Cover to Terminal Block using hardware provided with assembly.
4. Tighten Strain Relief Cover to End Ring studs with:
   (2) 100090P447 3/8 Flat Washer
   (2) 46–220196P6 3/8 Lock Washer
   (2) 46–208948P2 3/8–16 SST Hex Nut

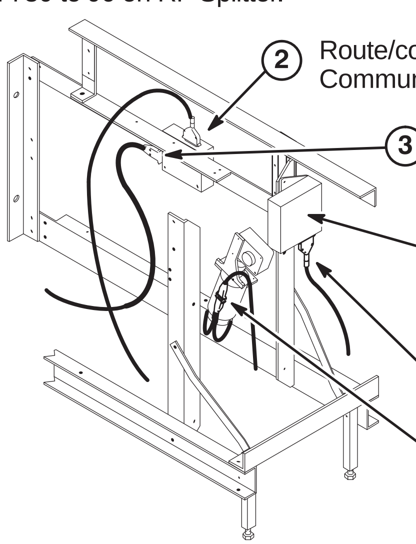
Strain Relief Cover

8–6–4–J Connect Runs 282, 318, 414, 422, 780, And 1206

1. Route/connect Run 780 to J6 on RF Splitter.



6. Route/connect Run 1206 to Notch Filter.



2. Route/connect Run 282 to J6 on Patient Communications Box (MG3 A2).
3. Route/connect Run 414 to J9.

Magnet Interface #1 (MG3 A1)

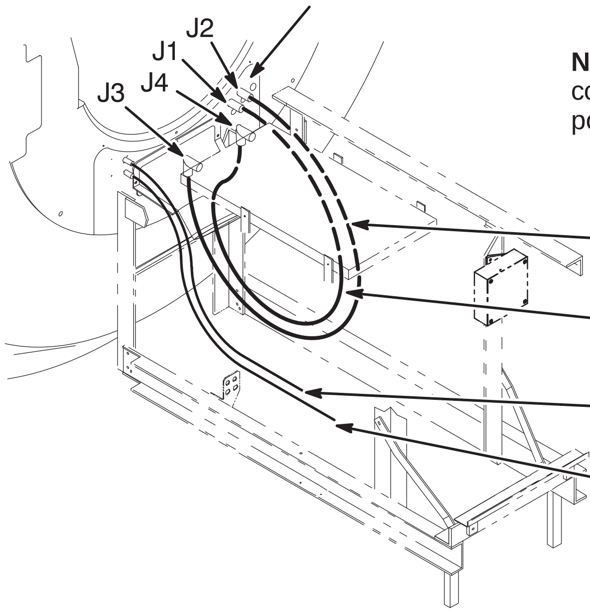
4. Route/connect Run 422 to J4 on Magnet Interface #1.
5. Route/connect Run 318 to short cable on Longitudinal Motor Drive (MG3 A7).

8–6–4–K Route/Connect Runs 743, 744, 781, And 782

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

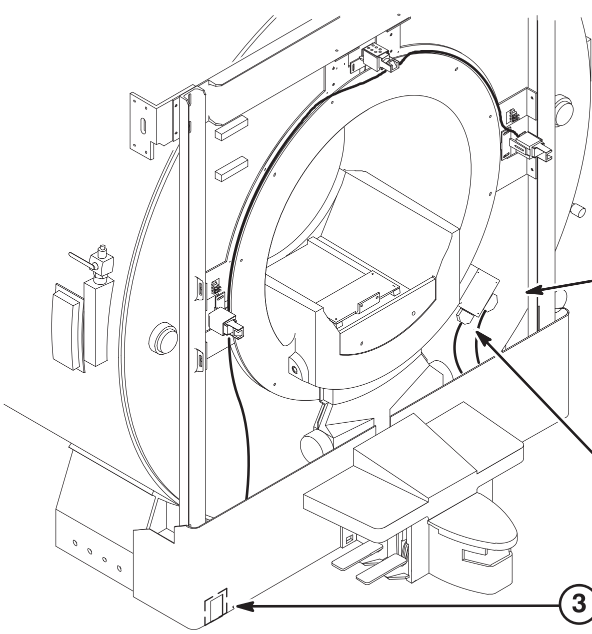
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

1. Route and connect Run 744 (2105691–5) from J1 on the Bulkhead to J4 on the RF Splitter.
2. Route and connect Run 743 (2105691–4) from J2 on the Bulkhead to J3 on the RF Splitter.
3. Route and connect Run 781 (46–328000G919) from J5 on the Bulkhead to J75 on the Penetration Panel.
4. Route and connect Run 782 (46–328000G920) from J6 on the Bulkhead to J76 on the Penetration Panel.

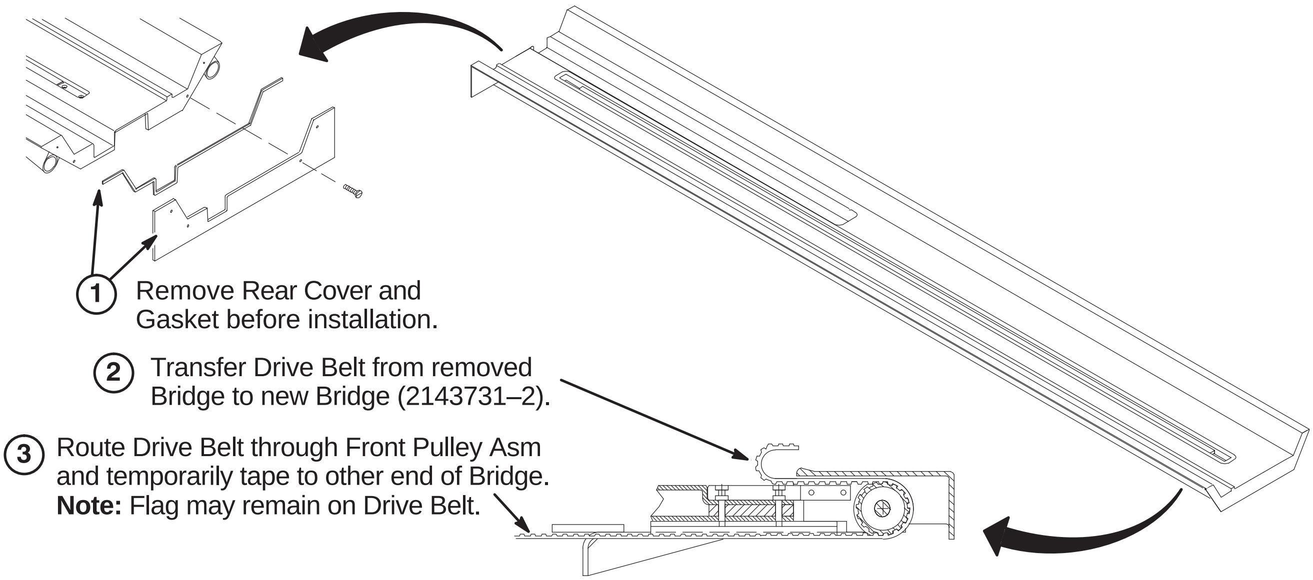
8–6–4–L Route/Connect Runs 421 And Dock Cable



1. Attach Cable Extension (2143072) to Dock Cable and connect to J2 on Dock Limit Module (MG3 A31).
2. Route/connect Run 421 from SRI Module to J1 on Dock Limit Module (MG3 A31).
3. Remove protective coating and place Front Skirt Trim Plate (2139555–3) behind slot.

8–6–5 INSTALL BRIDGE

8–6–5–A Bridge Assembly Preparation Before Installing in Magnet



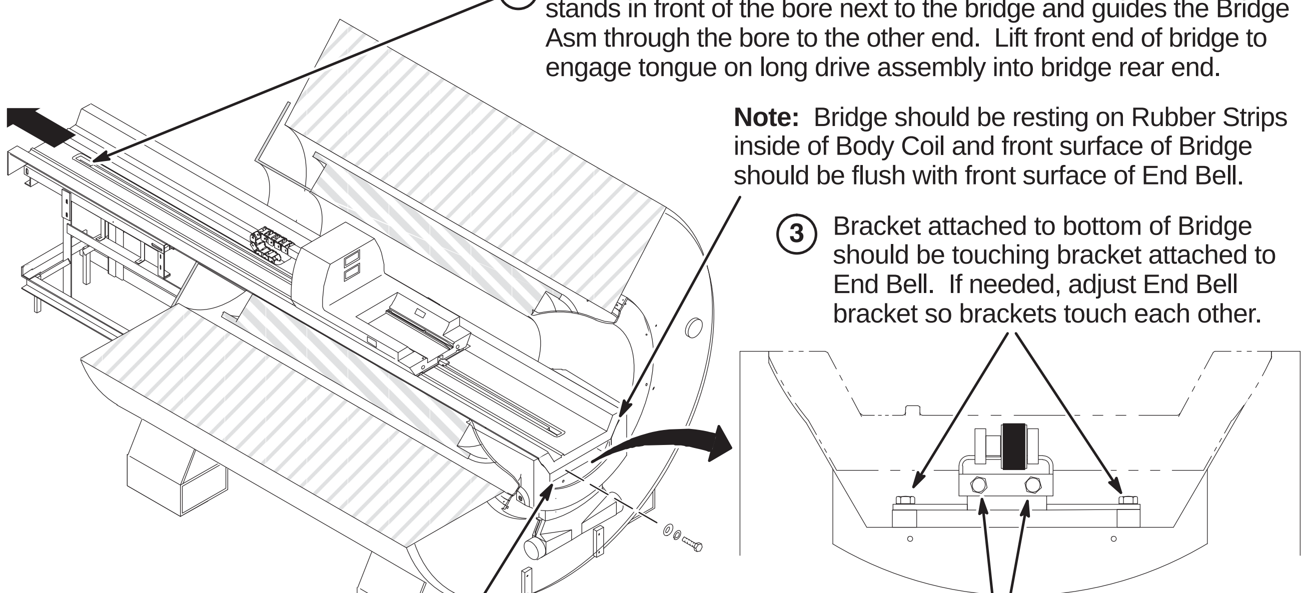
1. Remove Rear Cover and Gasket before installation.
2. Transfer Drive Belt from removed Bridge to new Bridge (2143731–2).
3. Route Drive Belt through Front Pulley Asm and temporarily tape to other end of Bridge. **Note:** Flag may remain on Drive Belt.

8–6–5–B Securing Front of Bridge

1. Position rear end of Bridge Asm at front of Body Coil.
2. While one person holds holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.

Note: Bridge should be resting on Rubber Strips inside of Body Coil and front surface of Bridge should be flush with front surface of End Bell.

3. Bracket attached to bottom of Bridge should be touching bracket attached to End Bell. If needed, adjust End Bell bracket so brackets touch each other.



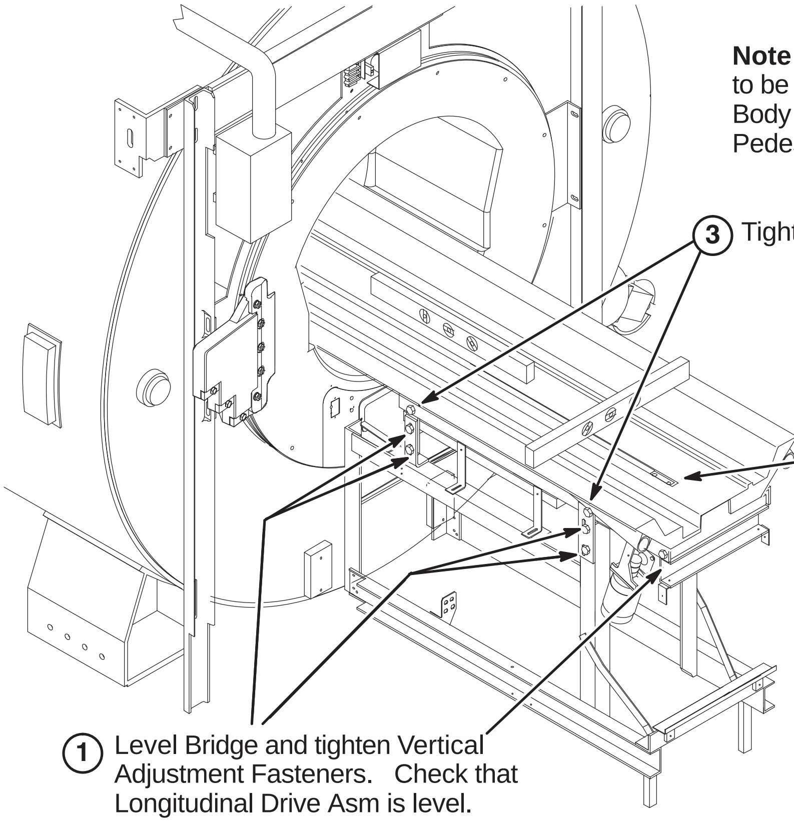
4. Fasten front end of Bridge Asm to Front End Bell using:

(2)	2109878–4	M10 SST Flat Washer
(2)	2109880–4	M10 SST Lock Washer
(2)	2109866–20	M10 SST Hex Screw

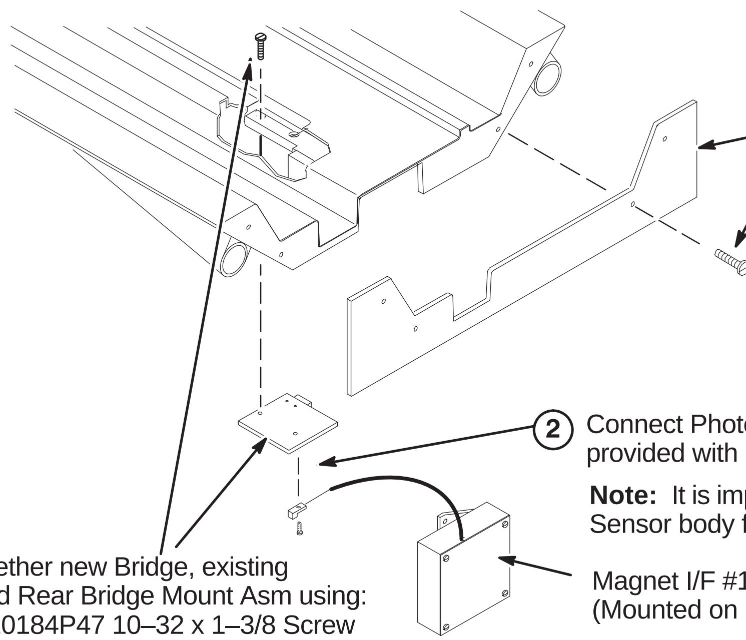
Note: Bridge and End Bell front surfaces should still be flush.

5. Install previously removed End Bell Panel with:
(2) 2109873–27 M6 x 25 Pan Screw

8–6–5–C Leveling Bridge

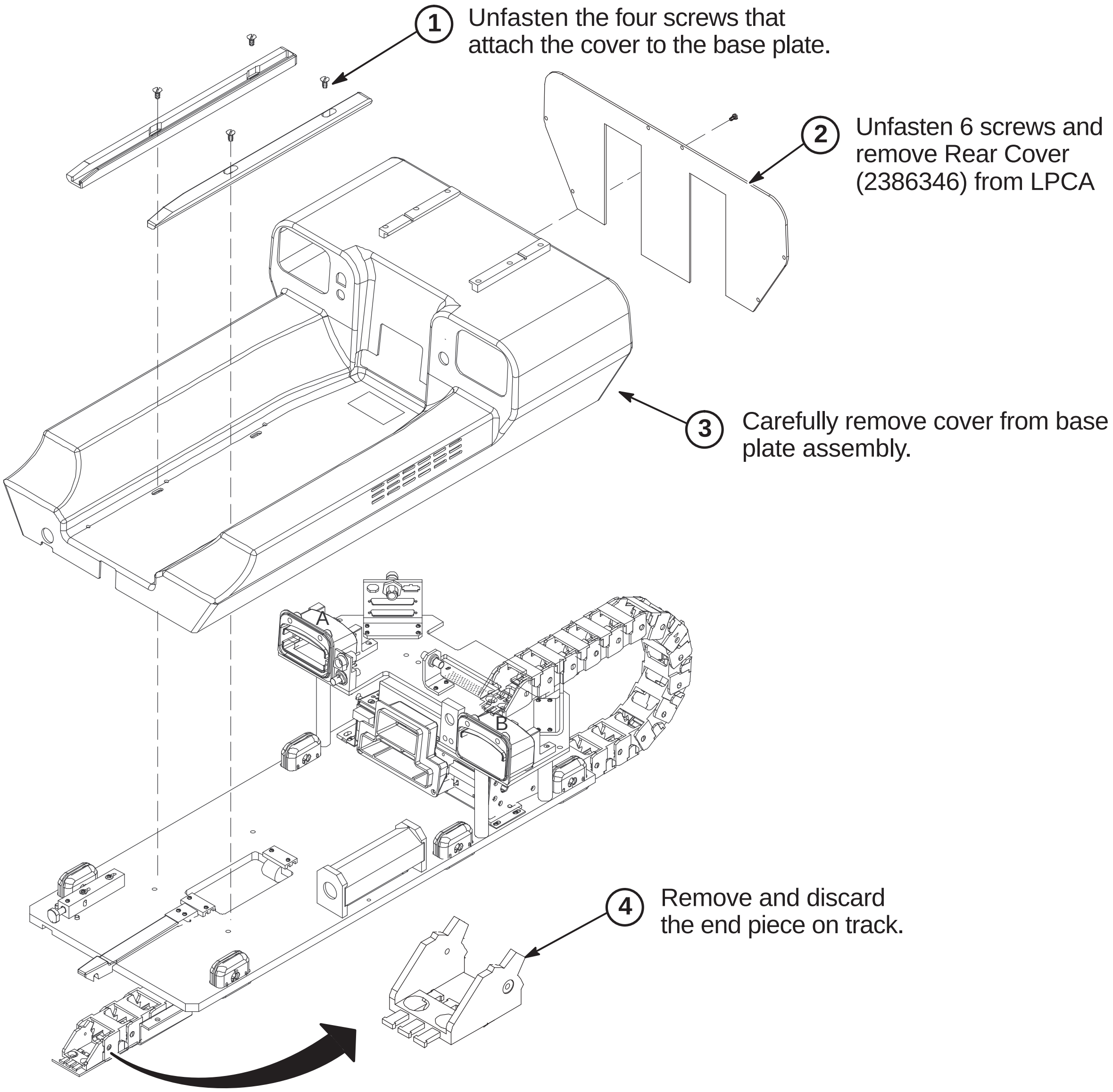


8–6–5–D Connect Photo Sensor & Bridge End Cover

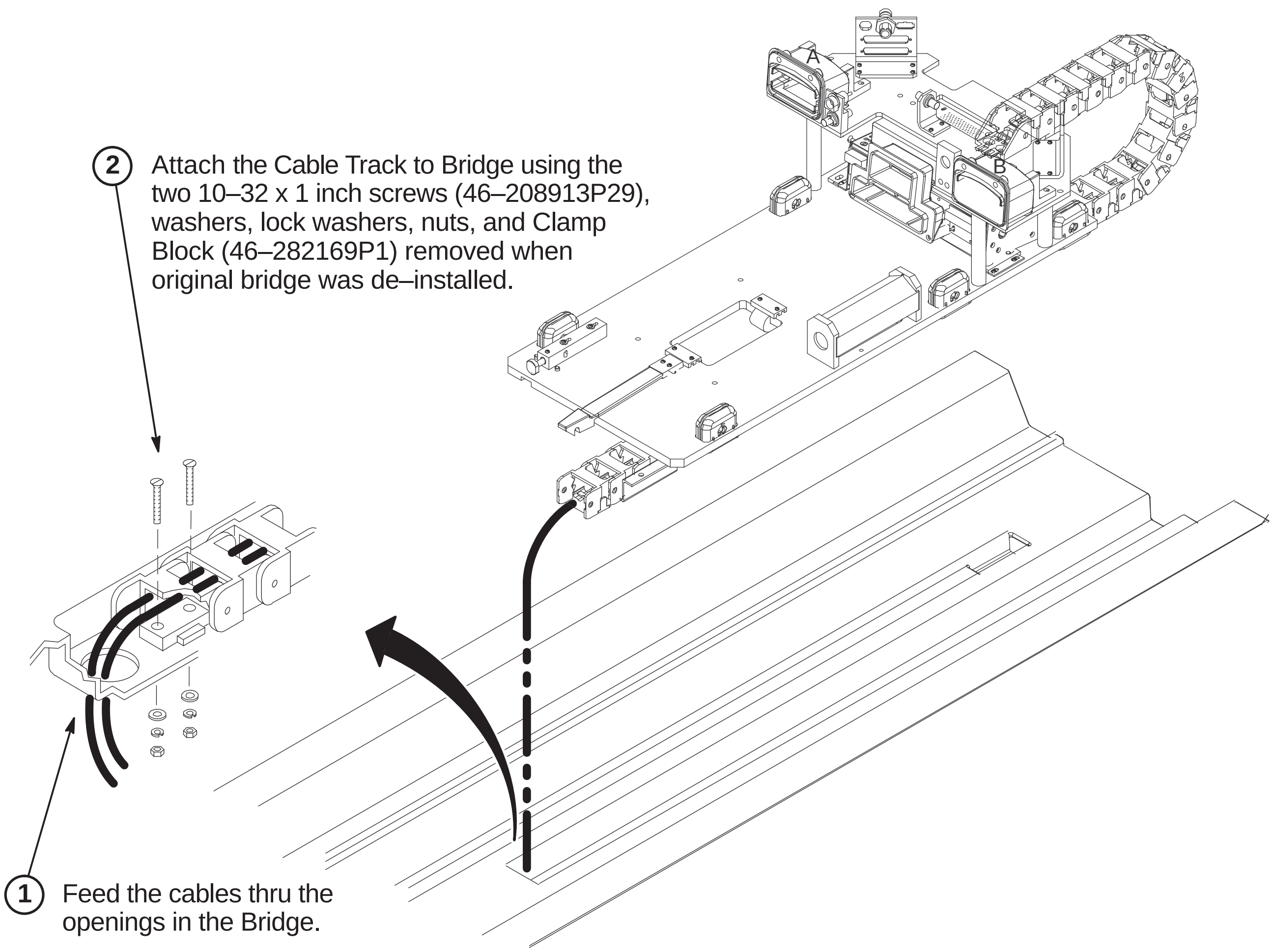


8–6–6 INSTALL LPCA AND CABLE TRACK

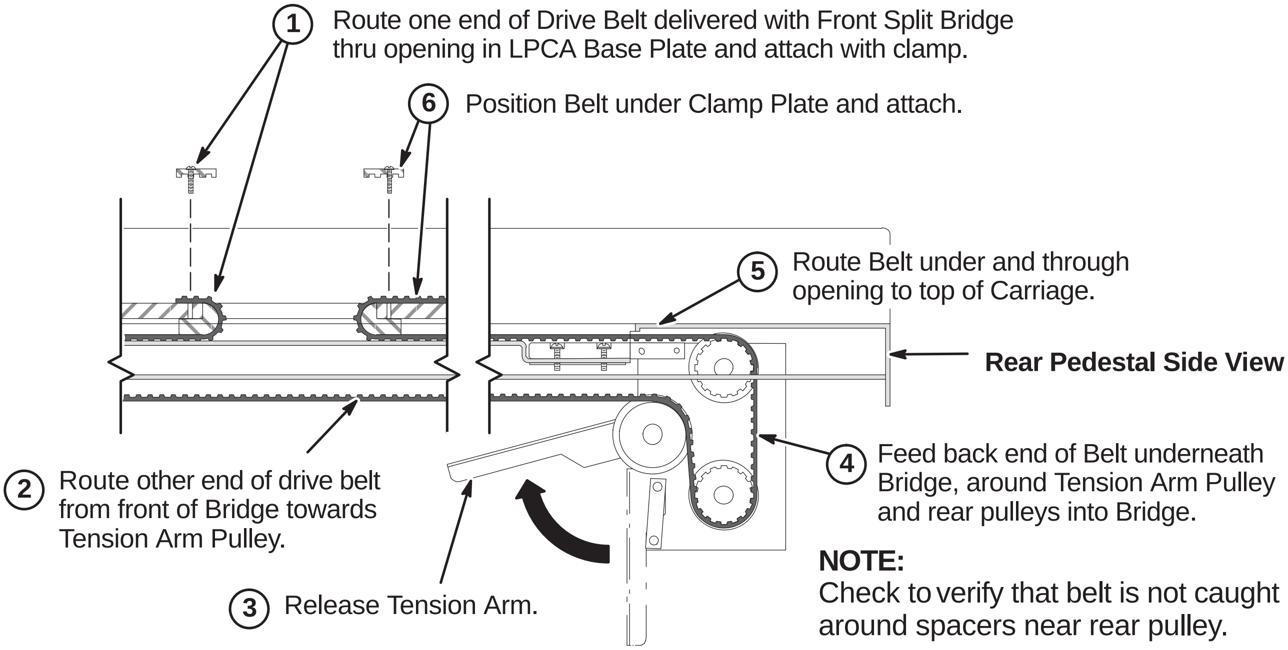
8–6–6–A Remove LPCA Cover



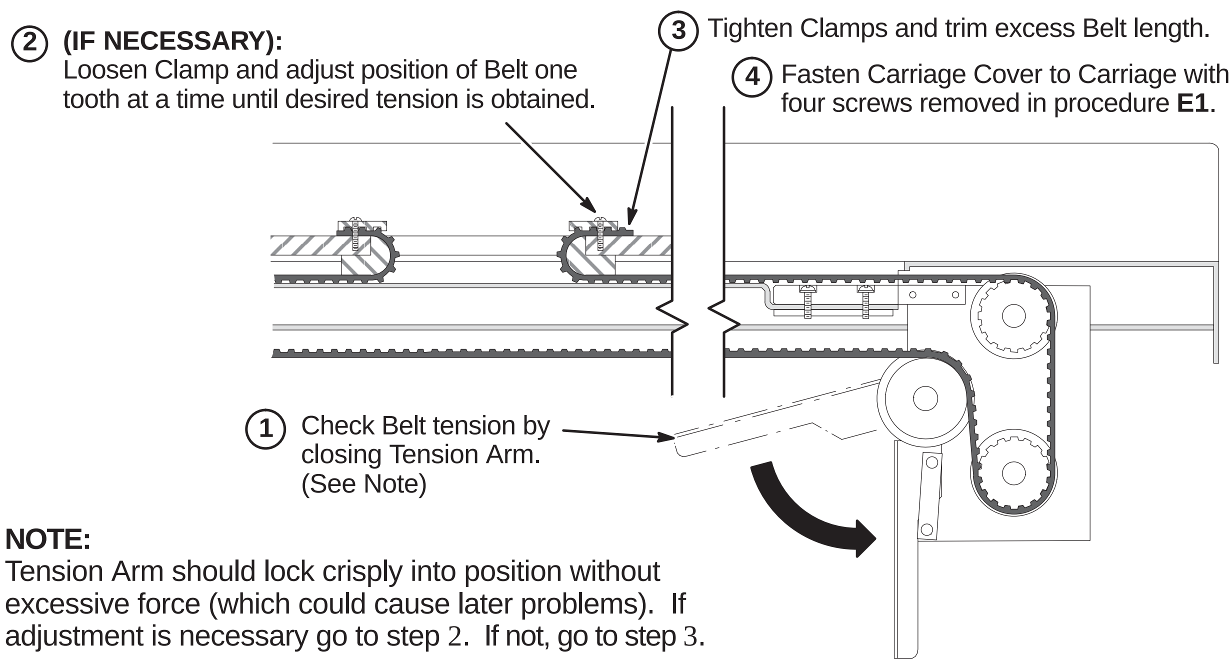
8–6–6–B Attach Cable Track to Bridge



8–6–6–C Route and Attach Drive Belt



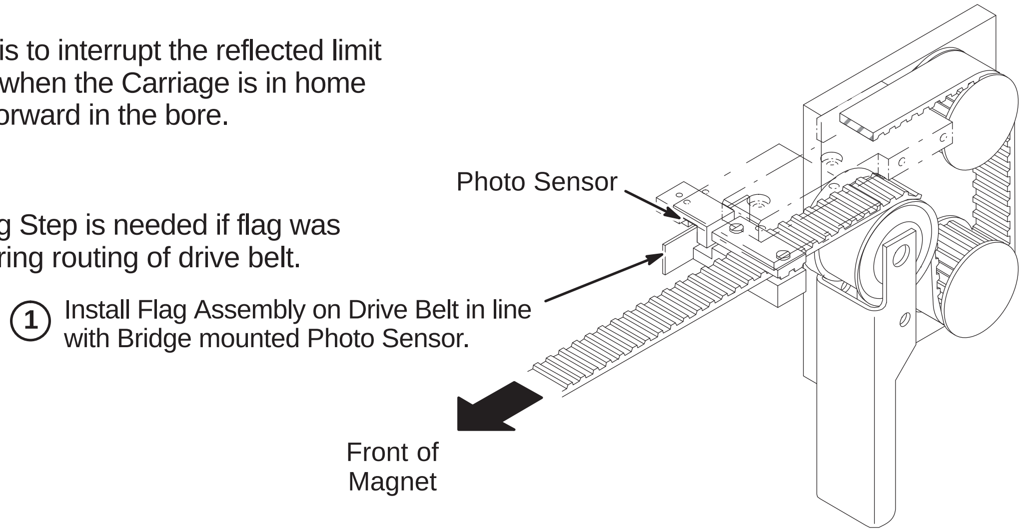
8–6–6–D Final Adjustment of Drive Belt



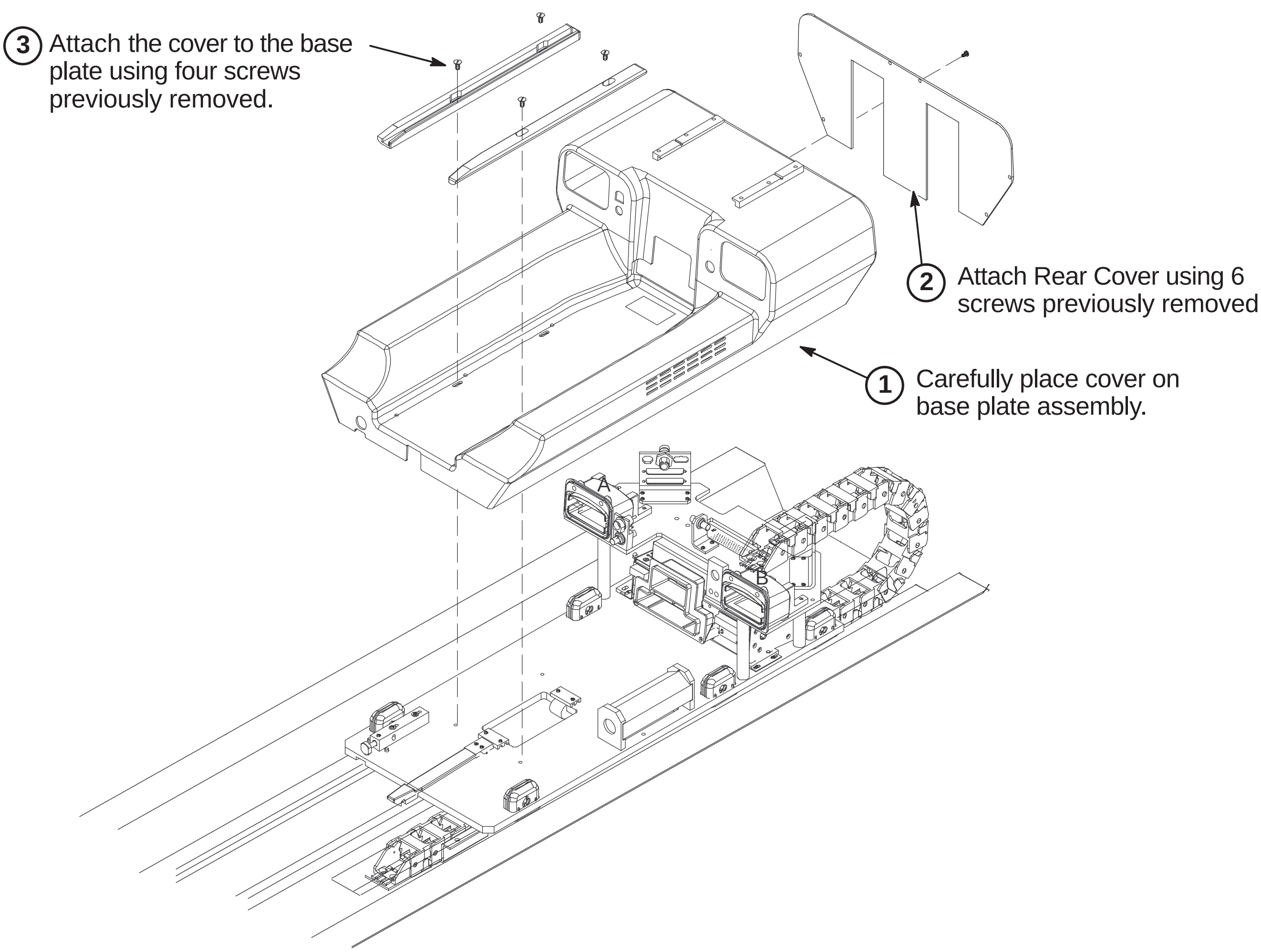
8–6–6–E Installation of Sensor Flag

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

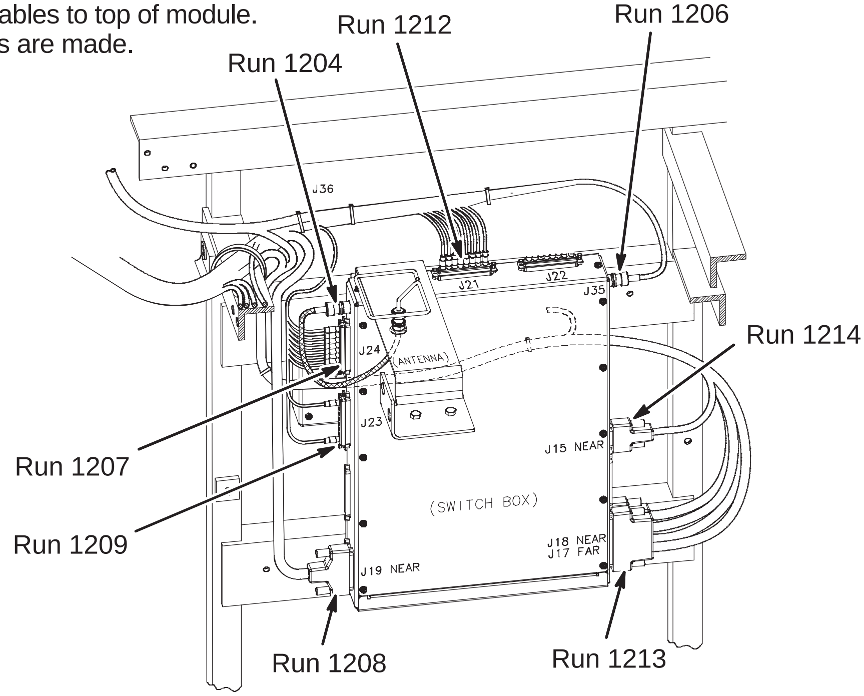


8–6–6–F Installation of LPCA Cover



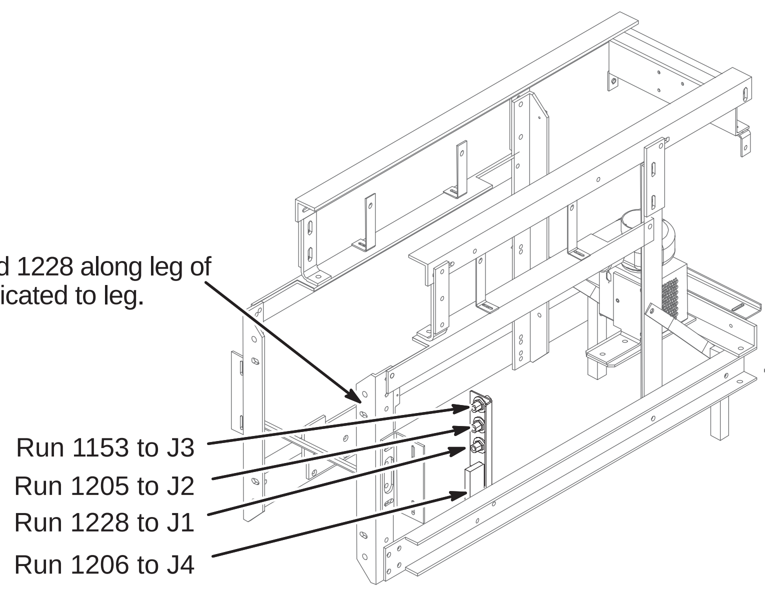
8–6–6–G Connect Cables from Cable Track to 16 Channel Switch Module

1. 16 Channnel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re–mount module after connections are made.
2. Connect cable Runs as indicated.



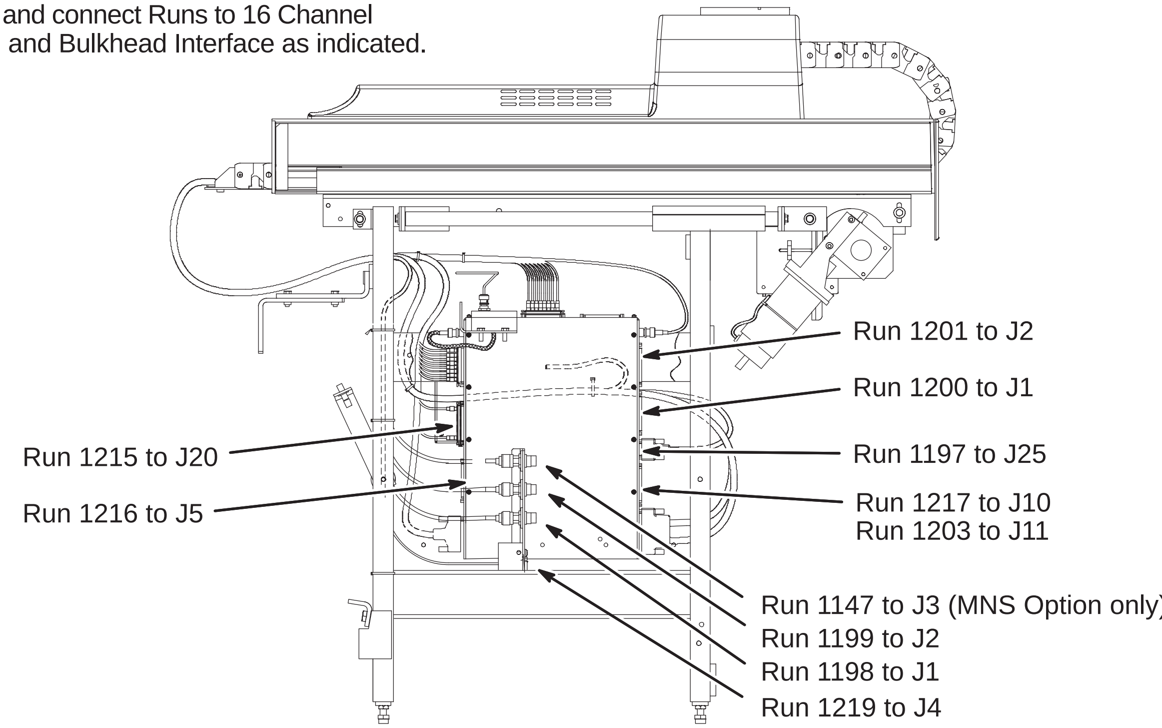
8–6–6–H Connect Cables from Cable Track to Bulkhead Interface

1. Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty–wrap as indicated to leg.
2. Connect Runs to Bulkhead Interface as indicated.



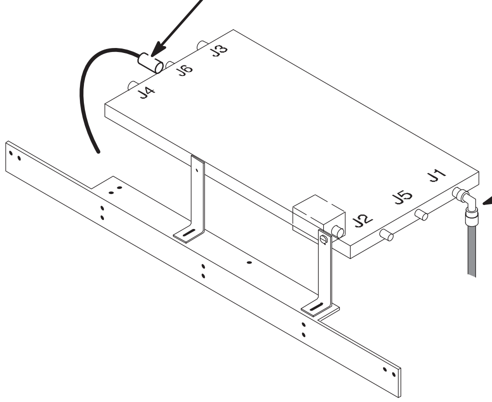
8–6–7 CONNECT REMAINING CABLES TO REAR PEDESTAL

1. Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



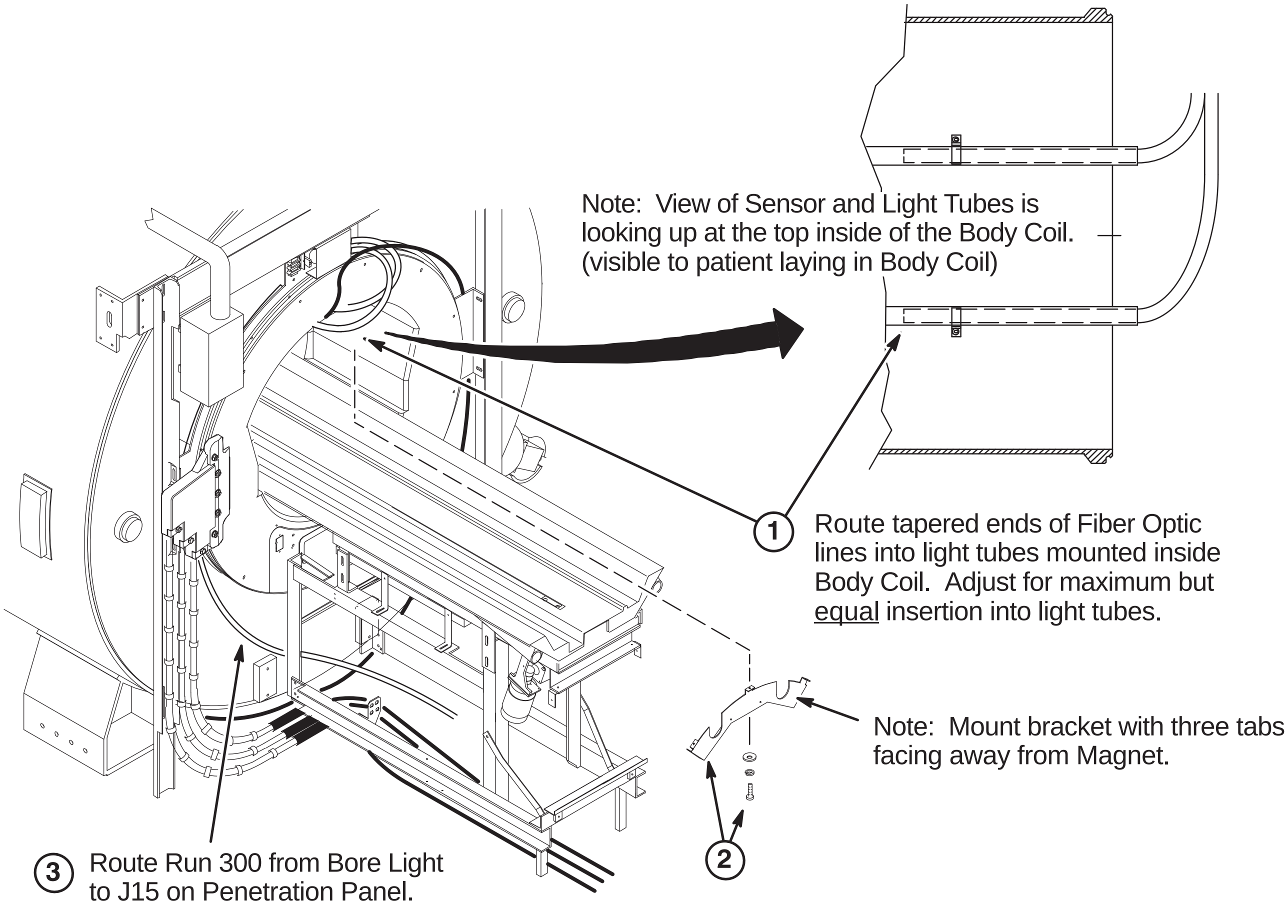
2. Route/connect Run 780 to J6 on RF Splitter.

3. Connect Run 1250 to J1 on RF Splitter.

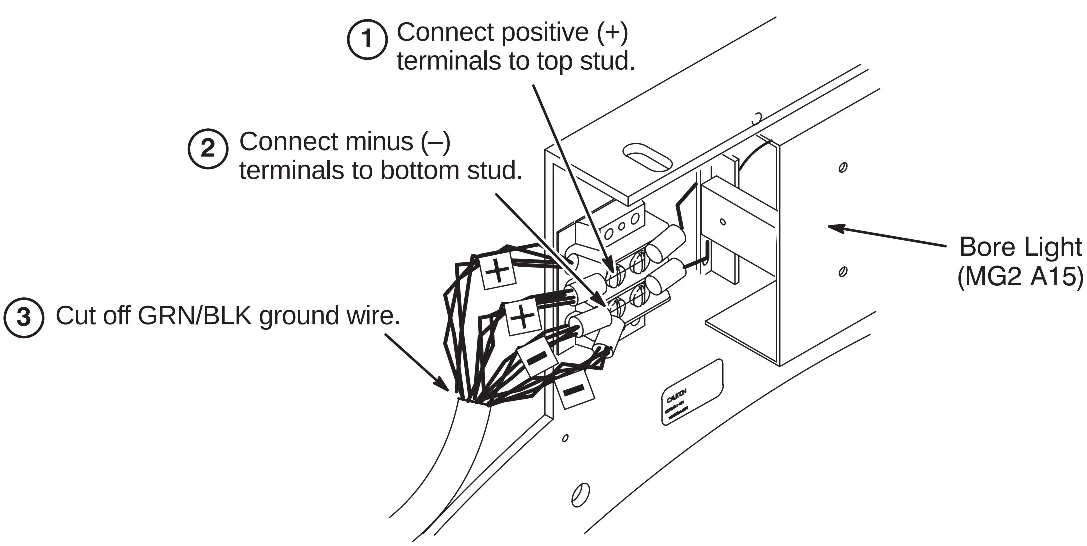


8–6–8 INSTALL BORE LIGHT

8–6–8–A Route Run 300 and Install Light Tube Fiber Optics

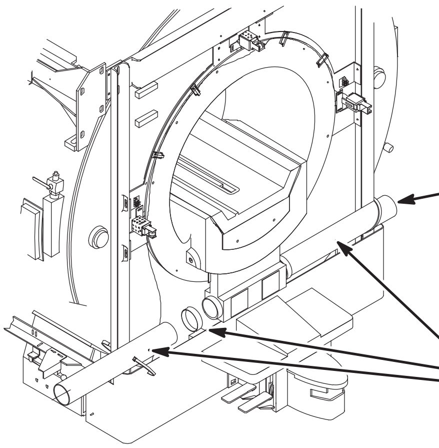


8–6–8–B Connect Run 300 To Bore Light



8–6–9 INSTALL PATIENT COMFORT AIR SYSTEM

8–6–9–A Connecting Front Air Plenum Hoses



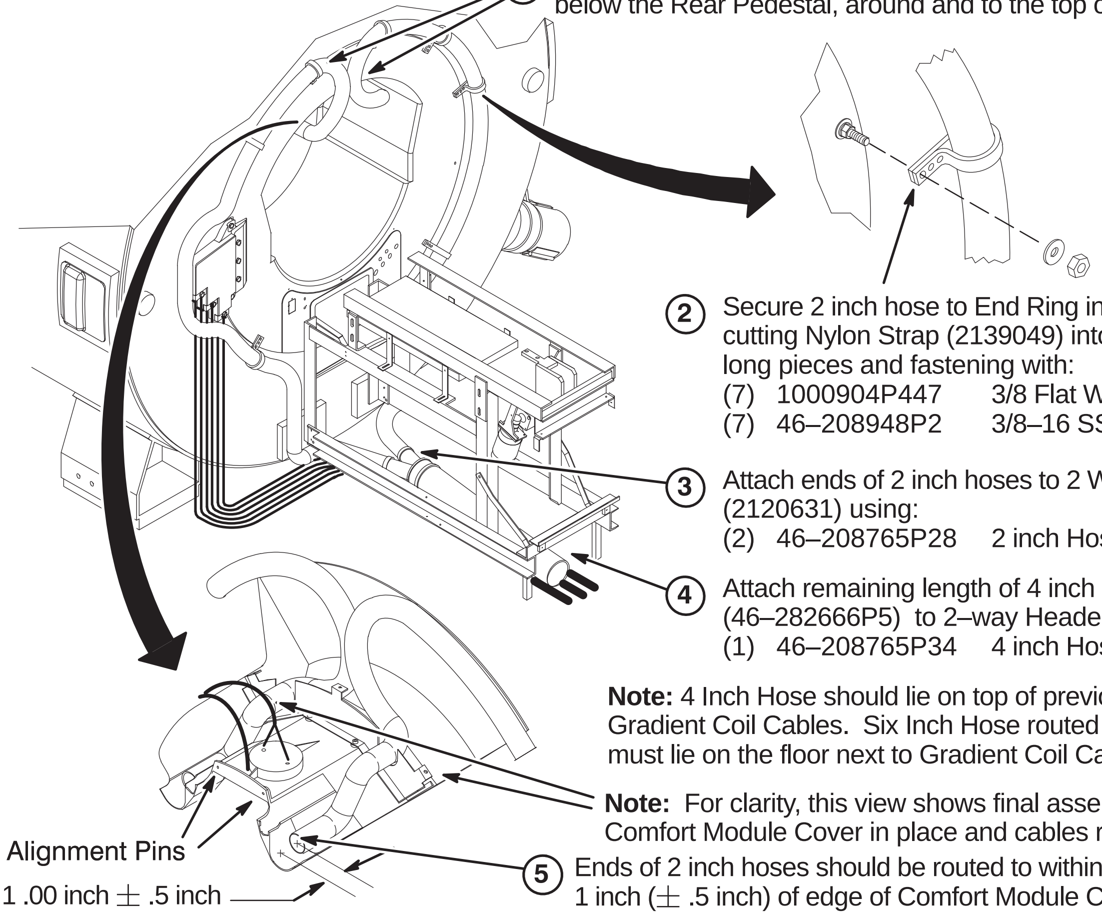
2 Ty–wrap each hose to Vertical Uprights on each side.

Note: All the air hoses and clamps in the following steps are located in the Blower Hose Kit (2129412).

1 Cut two (2) 26 inch lengths of 4 Inch Hose (46–282666P5) and attach one length to each side of Front Air Plenum using:
(2) 46–208765P34 4 inch Hose Clamp

8–6–9–B Routing/Connection Of Patient Comfort Air Hoses

1 Route the (2) 2 inch diameter hoses (46–282668P18) as shown from below the Rear Pedestal, around and to the top of the End Ring.



2 Secure 2 inch hose to End Ring in seven places by cutting Nylon Strap (2139049) into seven 12 inch long pieces and fastening with:
(7) 1000904P447 3/8 Flat Washer
(7) 46–208948P2 3/8–16 SST Hex Nut

3 Attach ends of 2 inch hoses to 2 Way Header (2120631) using:
(2) 46–208765P28 2 inch Hose Clamps

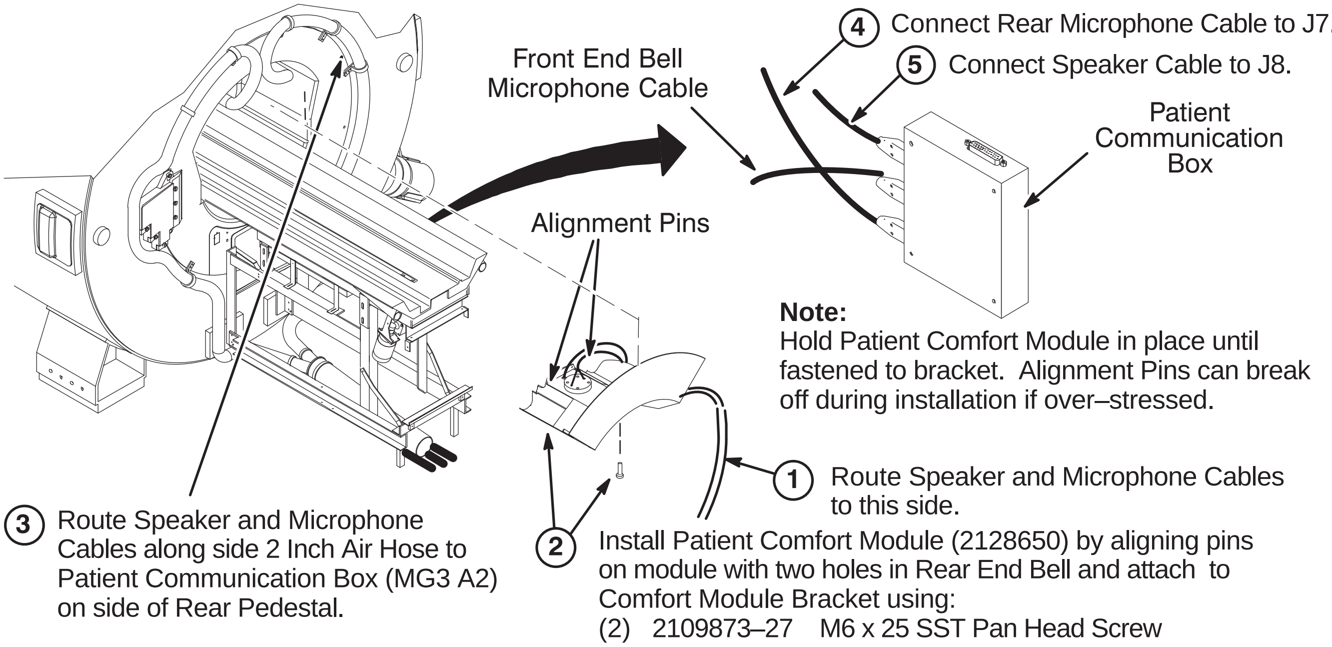
4 Attach remaining length of 4 inch Hose (46–282666P5) to 2–way Header using:
(1) 46–208765P34 4 inch Hose Clamp

Note: 4 Inch Hose should lie on top of previously routed Gradient Coil Cables. Six Inch Hose routed in Step L4 must lie on the floor next to Gradient Coil Cables.

Note: For clarity, this view shows final assembly with Comfort Module Cover in place and cables routed to side.

5 Ends of 2 inch hoses should be routed to within 1 inch (± .5 inch) of edge of Comfort Module Cover.

8–6–9–C Install Patient Comfort Module and Cables



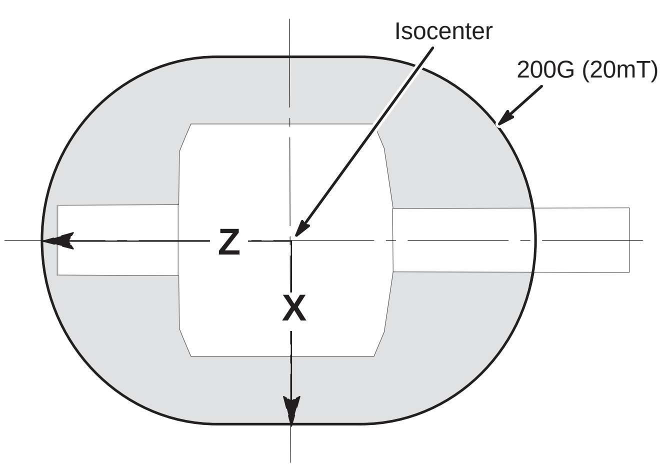
8–6–9–D 200 Gauss Zone Limit for Movement of Blower Box

Refer to the illustration and table below for applicable 200 Gauss zone for magnet at site.

DANGER!!

THE BLOWER BOX IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.

Illustration is not to scale



Approximate Distance from isocenter to 200G Line		
	X	Z
1.5T	150 cm (59 in)[1]	205 cm (81 in)[1]
1.0T & Lower	145 cm (57 in)[1]	195 cm (77 in)[1]
Unshielded Magnet	Measure [2]	

Note 1:
These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed.

Note 2:
For unshielded magnets, the magnetic field must be measured using a gauss meter.

8–6–9–E Installation of Blower Box

WARNING!

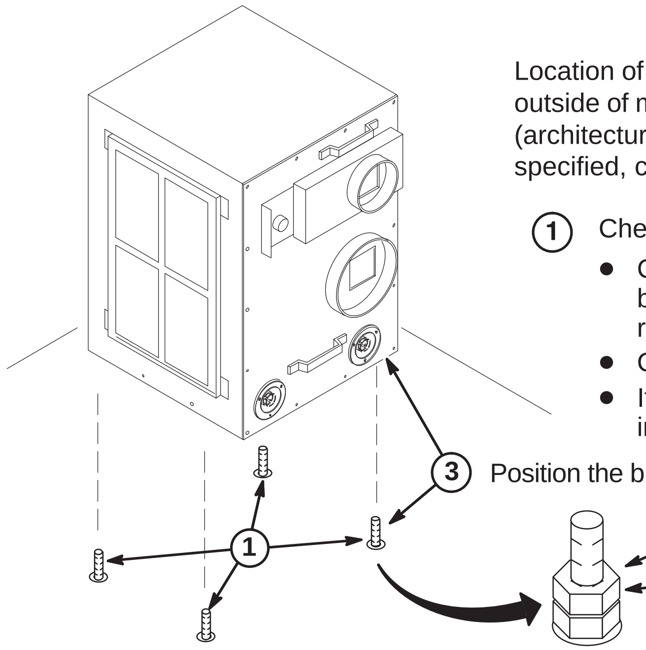
RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).



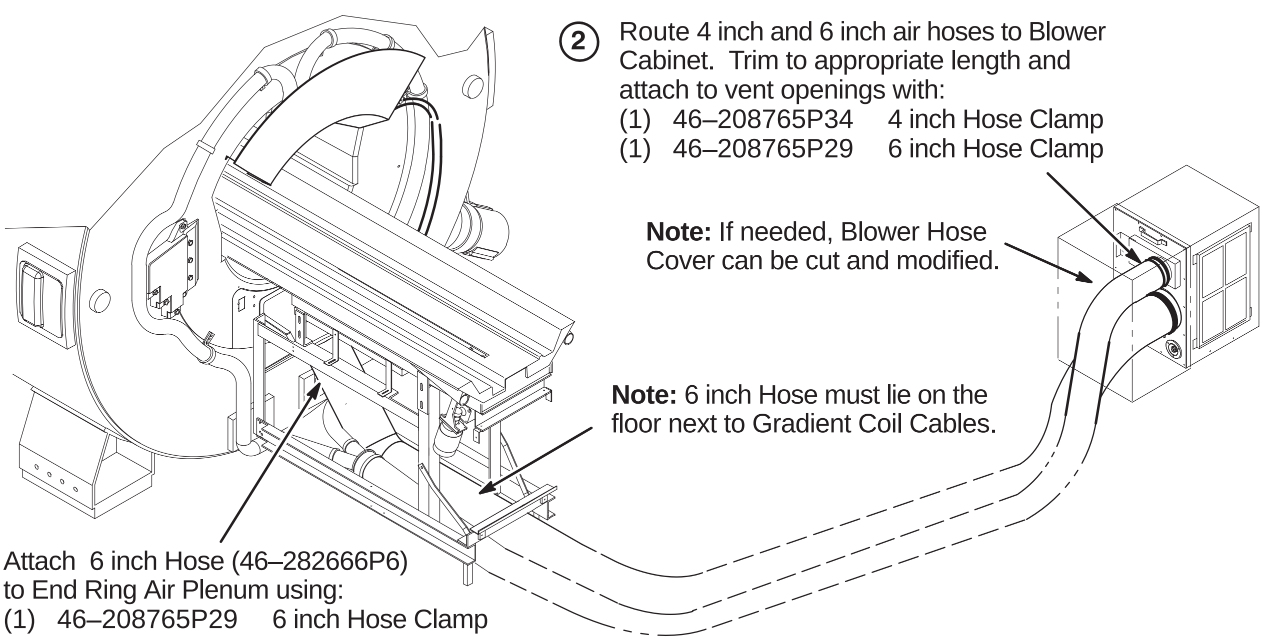
1. Check for correct installation of the anchors:
   - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
   - Confirm there are at least four (4) anchors.
   - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.

3. Position the blower box on top of anchors.

2. Remove any mounting hardware from the anchors.

4. Secure the blower box using the hardware provided by the RF shield room vendor to attach.

8–6–9–F Install Blower Box Air Hoses



2. Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:
   (1) 46–208765P34 4 inch Hose Clamp
   (1) 46–208765P29 6 inch Hose Clamp

Note: If needed, Blower Hose Cover can be cut and modified.

Note: 6 inch Hose must lie on the floor next to Gradient Coil Cables.

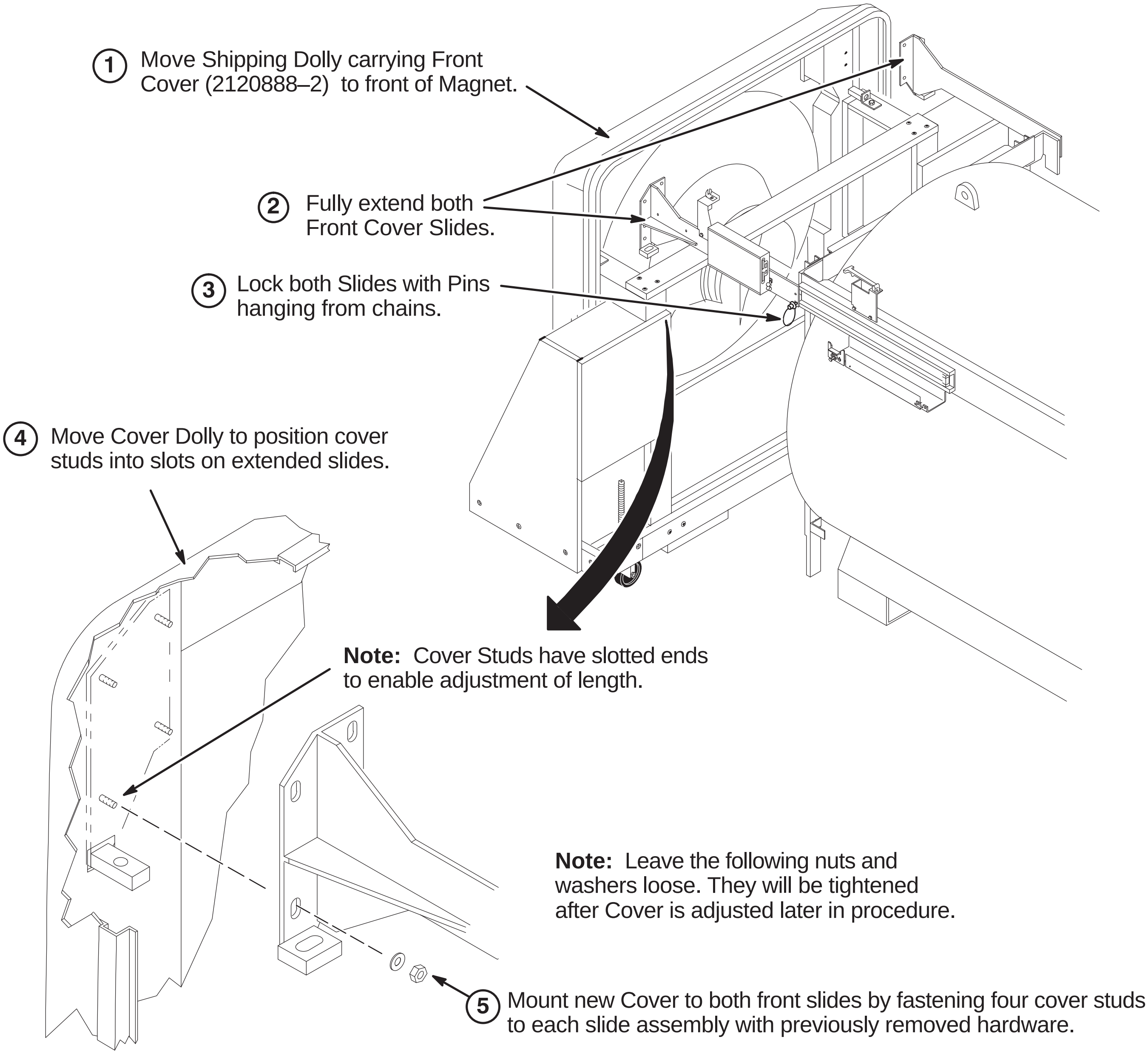
1. Attach 6 inch Hose (46–282666P6) to End Ring Air Plenum using:
   (1) 46–208765P29 6 inch Hose Clamp

8–6–10 INSTALL FRONT COVER

8–6–10–A Mount Front Cover

DANGER!!

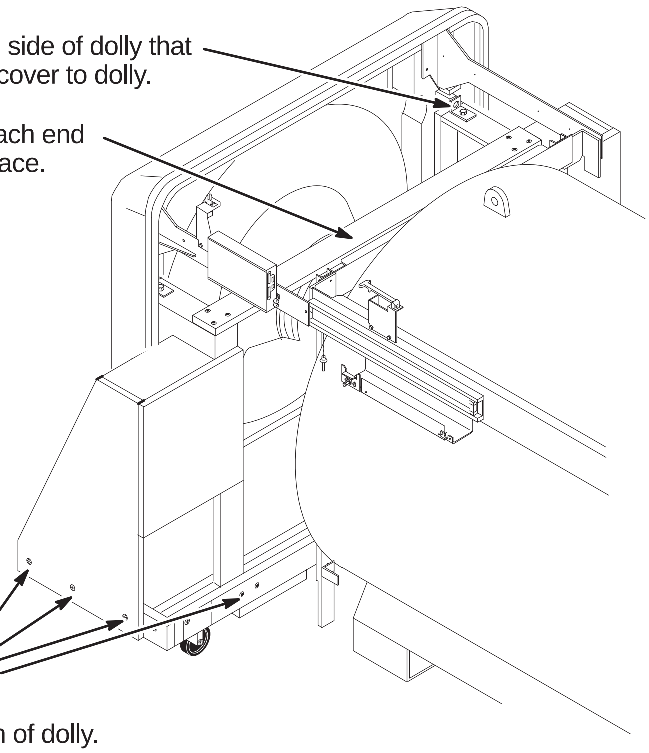
FERROUS MATERIAL HAZARD! THE COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE FRONT COVER FROM DOLLY AND CARRY COVER INTO POSITION AT FRONT OF MAGNET.



8–6–10–B Disassemble Cover Dolly

1 Remove bolts from each side of dolly that were used for attaching cover to dolly.

2 Remove three screws from each end of cross brace and remove brace.



3 Remove screws holding dolly side section. Side section will slide down when last bolt is removed.

Repeat above for removal of opposite side section of dolly.

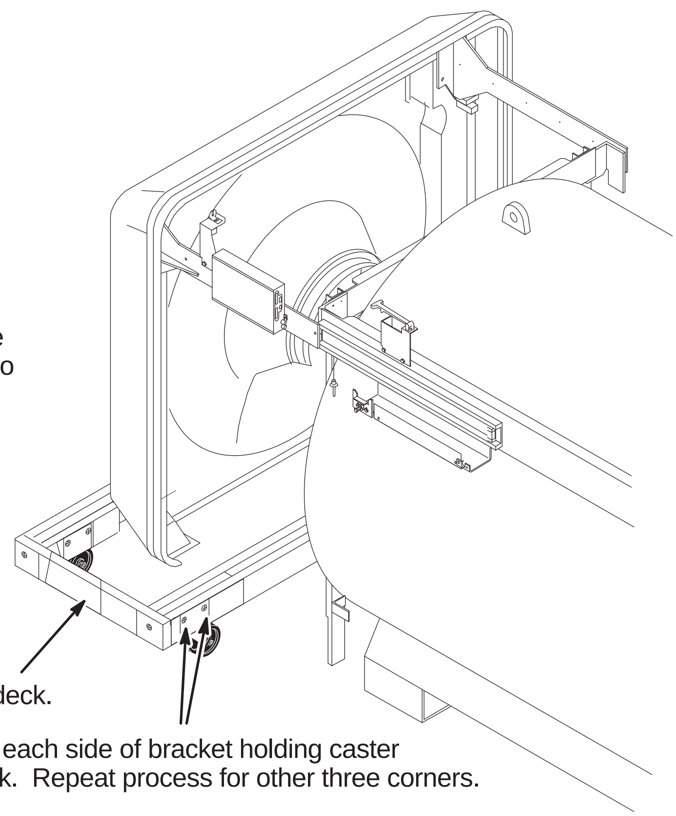
8–6–10–C Remove Cover Dolly

6 Place remaining dolly parts on recycle pile.

5 Place the four bracket and caster assemblies in pre–addressed box provided with dolly for shipment.

4 With casters removed, slide frame away from magnet. Use care not to damage floor when sliding and removing dolly.

3 Remove caster assemblies.

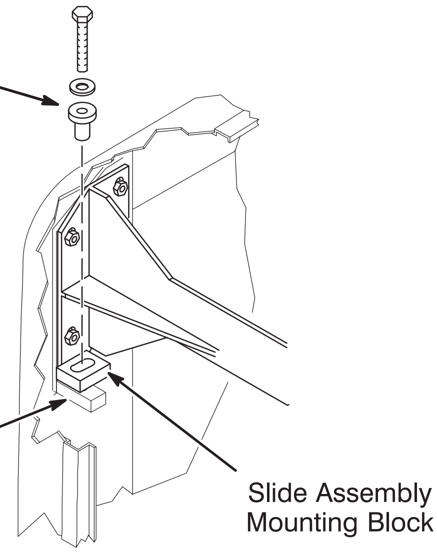


2 Lift this end of dolly deck.

1 Remove screws from each side of bracket holding caster assembly to dolly deck. Repeat process for other three corners.

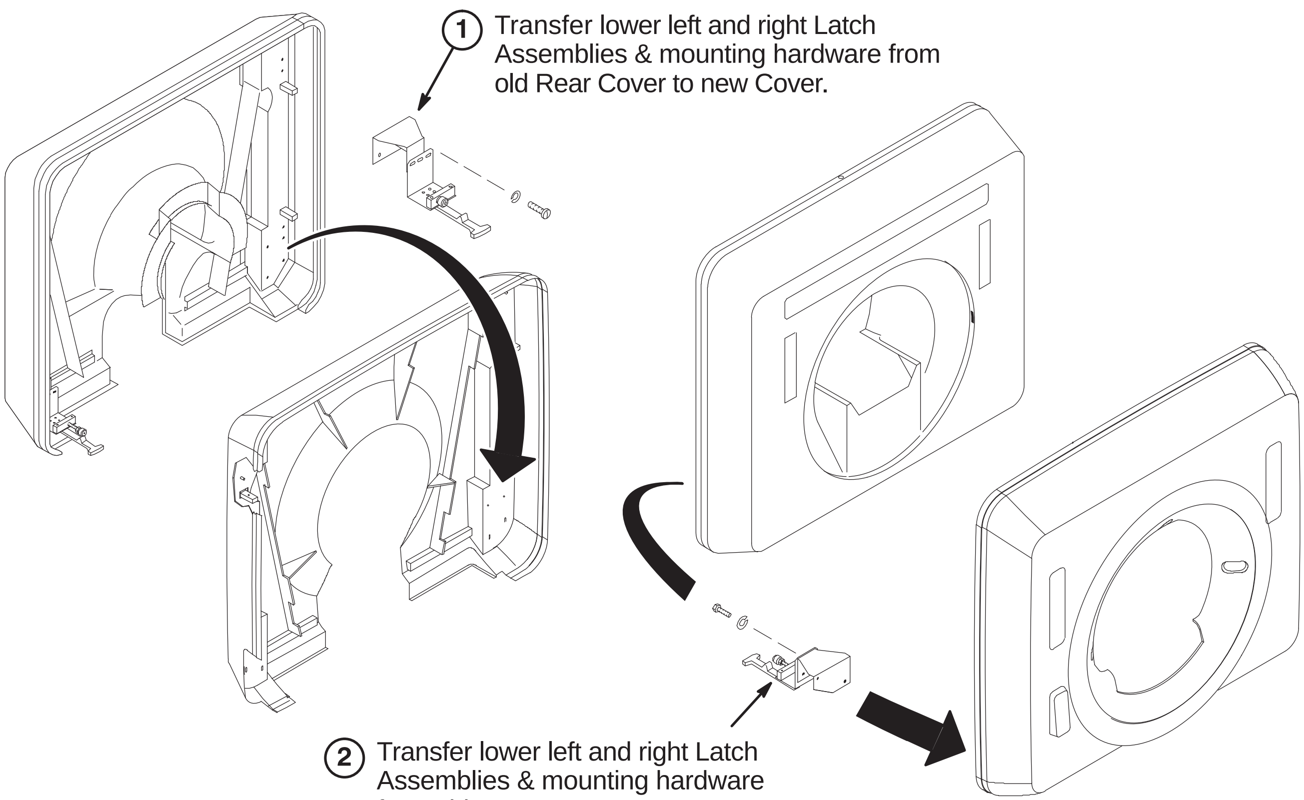
8–6–10–D Install Front Cover Alignment Hardware

1. Transfer and insert the Alignment Hardware from the old Front Cover into the oval hole on the Slide Assembly Mounting Block

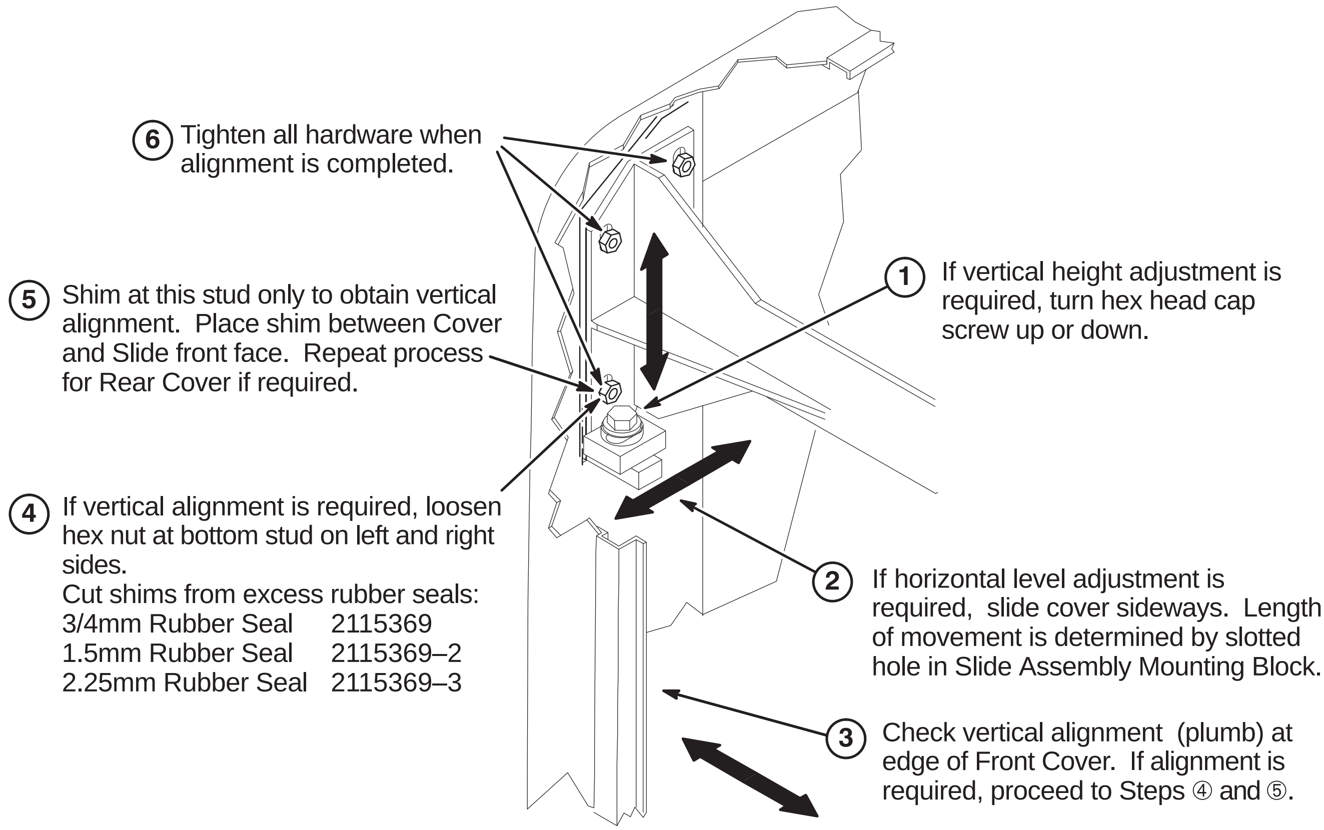


2. Engage threads in Cover Mounting Block.
3. Repeat process for other side of Front Cover.

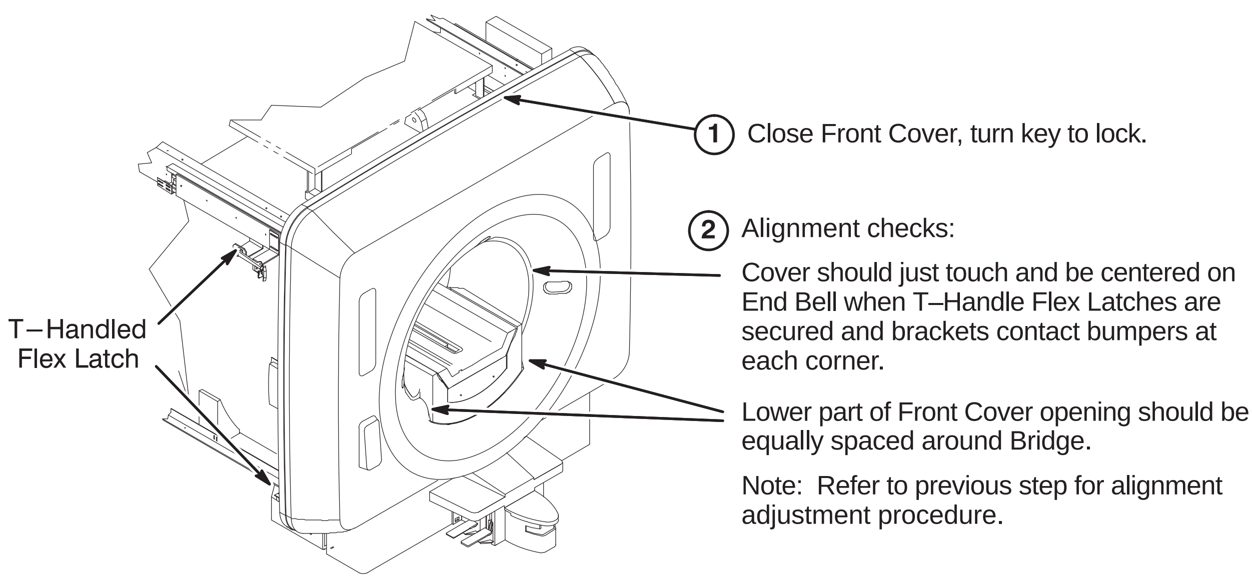
8–6–10–E Transfer Lower Latch Assemblies to New Front & Rear Covers



8–6–10–F Adjust Front Cover Alignment



8–6–10–G Align Front Cover



8–6–11 INSTALL REAR COVER

DANGER!!

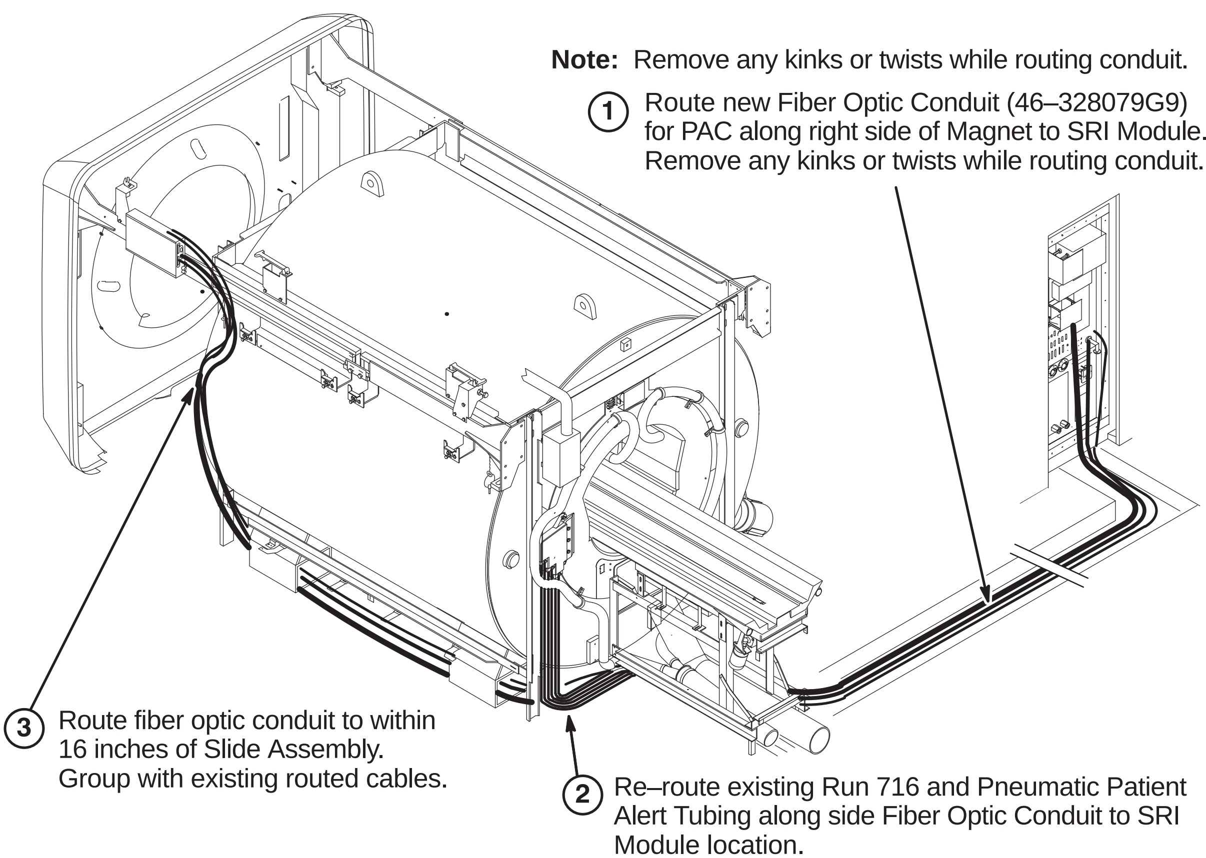
FERROUS MATERIAL HAZARD! THE REAR COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE COVER FROM DOLLY AND CARRY COVER INTO POSITION AT REAR OF MAGNET.

THE STEPS ARE THE SAME AS THE JUST COMPLETED FRONT COVER INSTALLATION PROCEDURE
Note: Experience has shown that the Rear Cover does not require shimming for vertical levelness.

8–6–12 ROUTE/CONNECT RUNS 711/712, 716, AND PATIENT ALERT TUBING

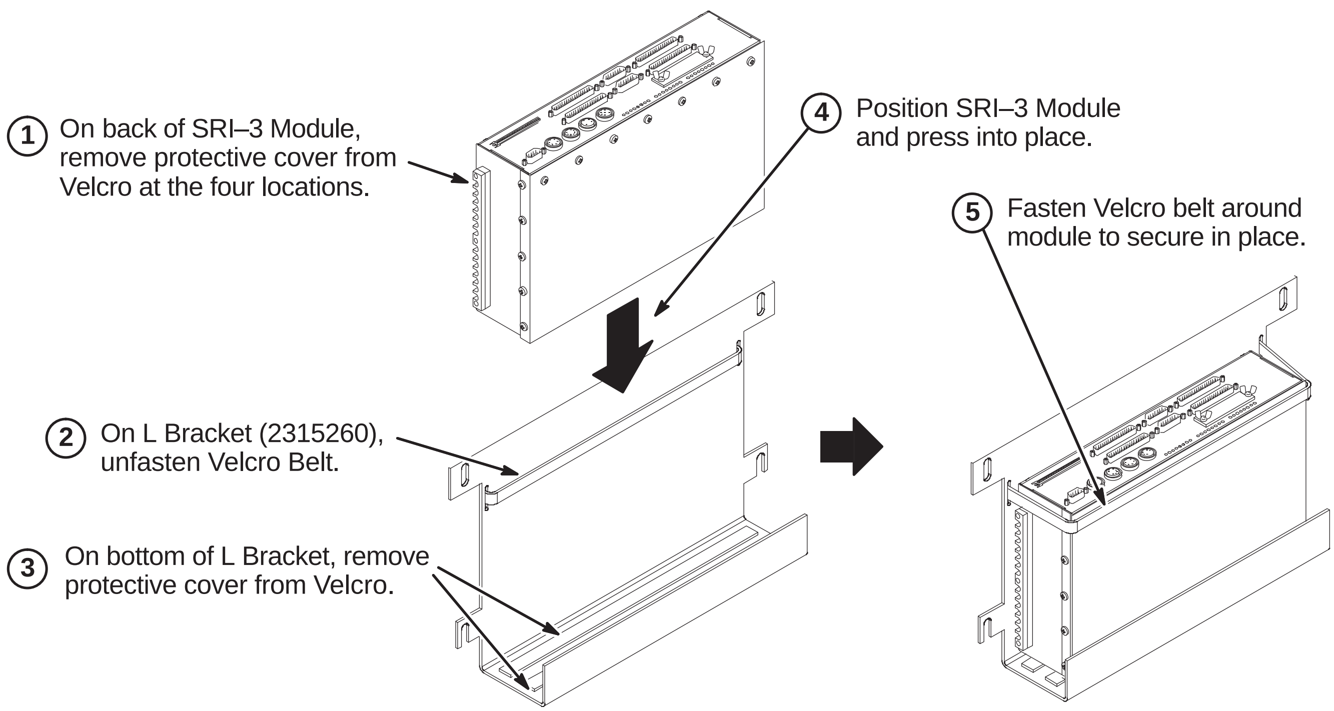
CAUTION

EQUIPMENT DAMAGE POSSIBILITY. Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend to radius smaller than two inches. Do not pull connector from cable. Avoid scratching connector ends. Keep ends protected until ready to connect.

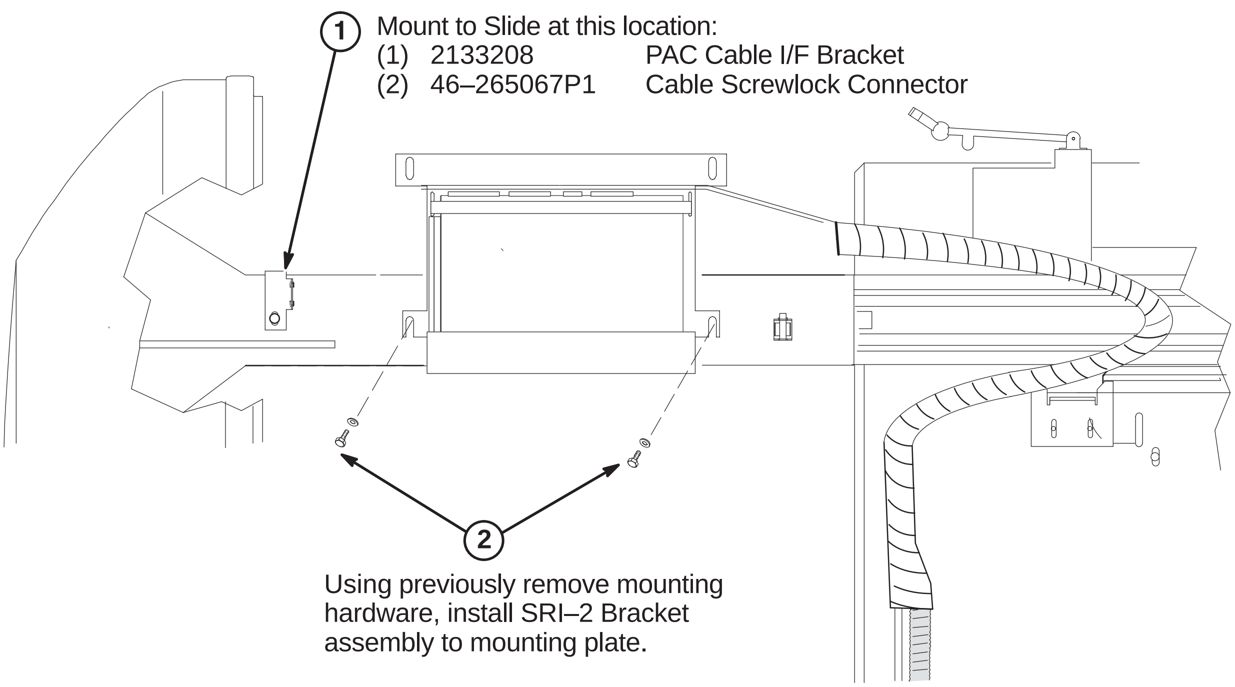


8–6–13 INSTALL SRI–3 MODULE

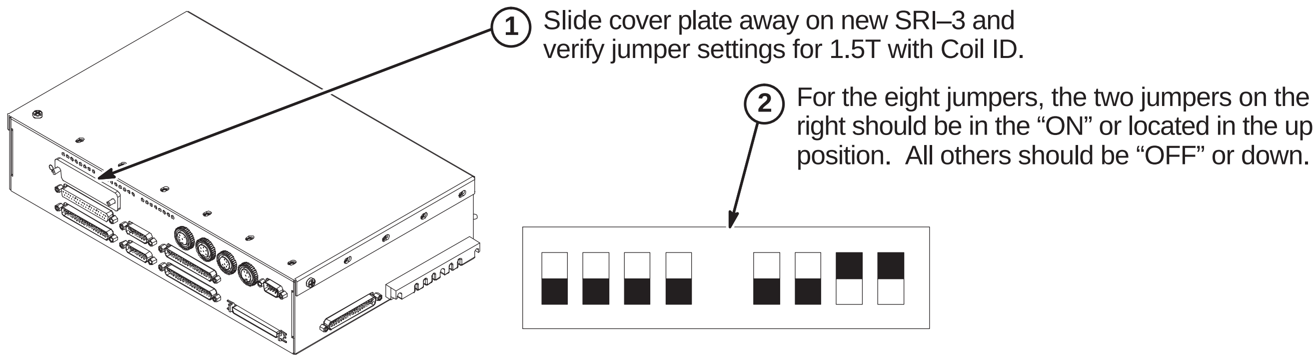
8–6–13–A Position SRI–3 in L Bracket



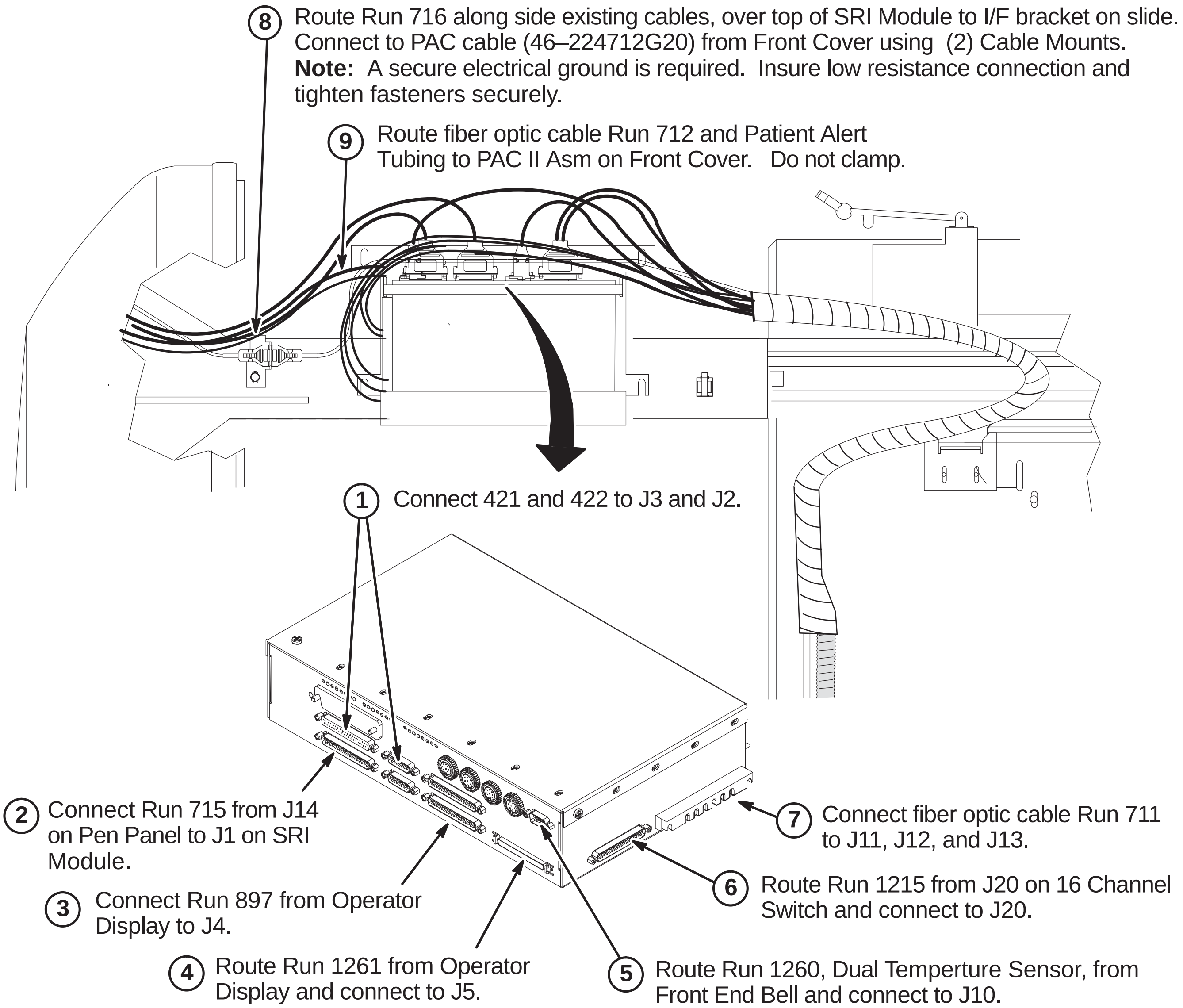
8–6–13–B Install SRI–3 on Magnet



8–6–13–C Verify SRI–3 Jumper Settings

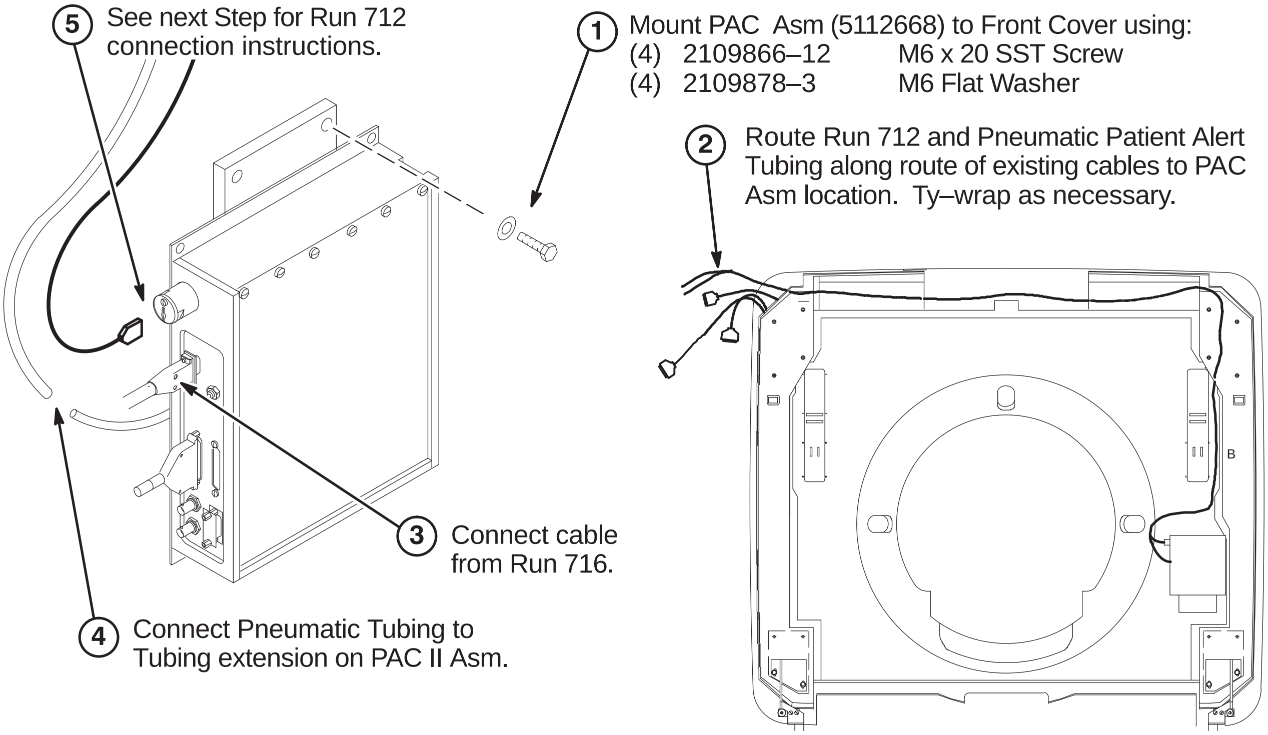


8–6–13–D Connect Cables at SRI–3 Module Location



8–6–14 INSTALL NEW PAC MODULE

8–6–14–A Connect Run 712, 716, & Pneumatic Patient Alert Tubing



1 Mount PAC Asm (5112668) to Front Cover using:
(4) 2109866–12 M6 x 20 SST Screw
(4) 2109878–3 M6 Flat Washer

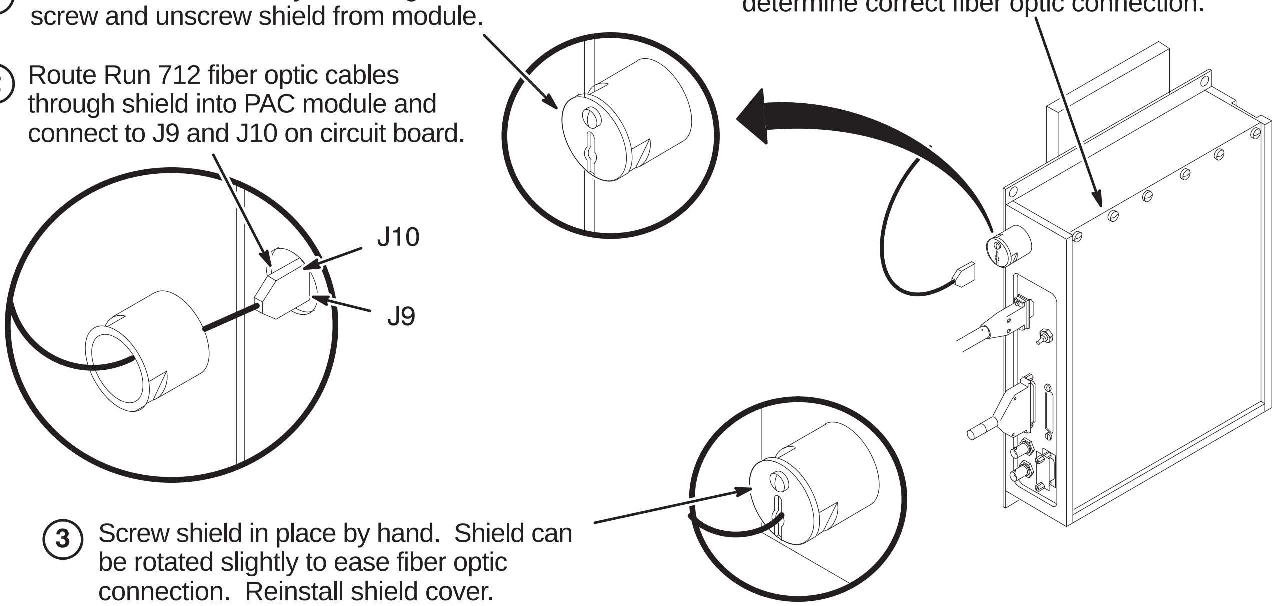
2 Route Run 712 and Pneumatic Patient Alert Tubing along route of existing cables to PAC Asm location. Ty–wrap as necessary.

3 Connect cable from Run 716.

4 Connect Pneumatic Tubing to Tubing extension on PAC II Asm.

5 See next Step for Run 712 connection instructions.

8–6–14–B Connect Run 712



1 Remove shield cover by removing one screw and unscrew shield from module.

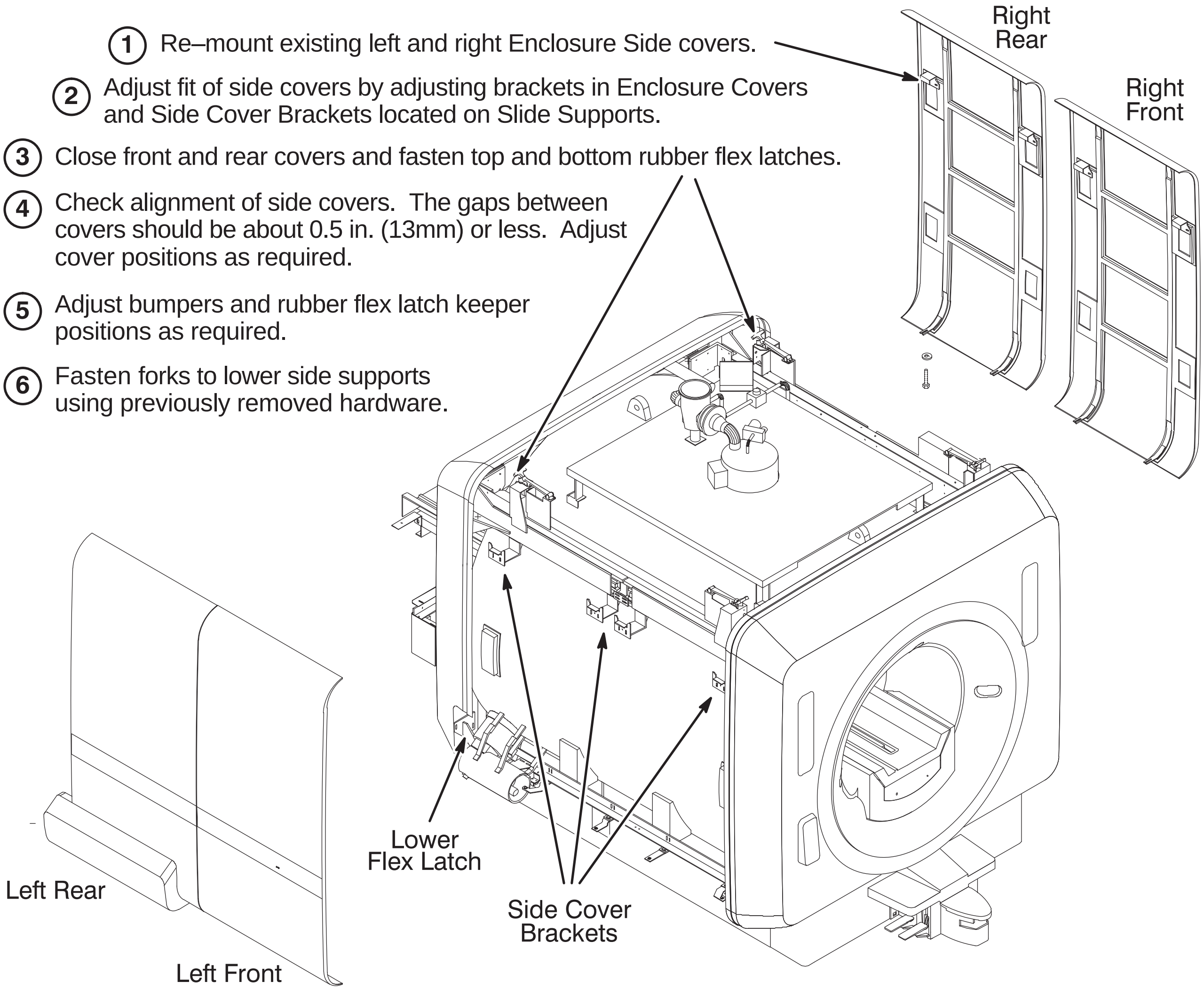
2 Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.

3 Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.

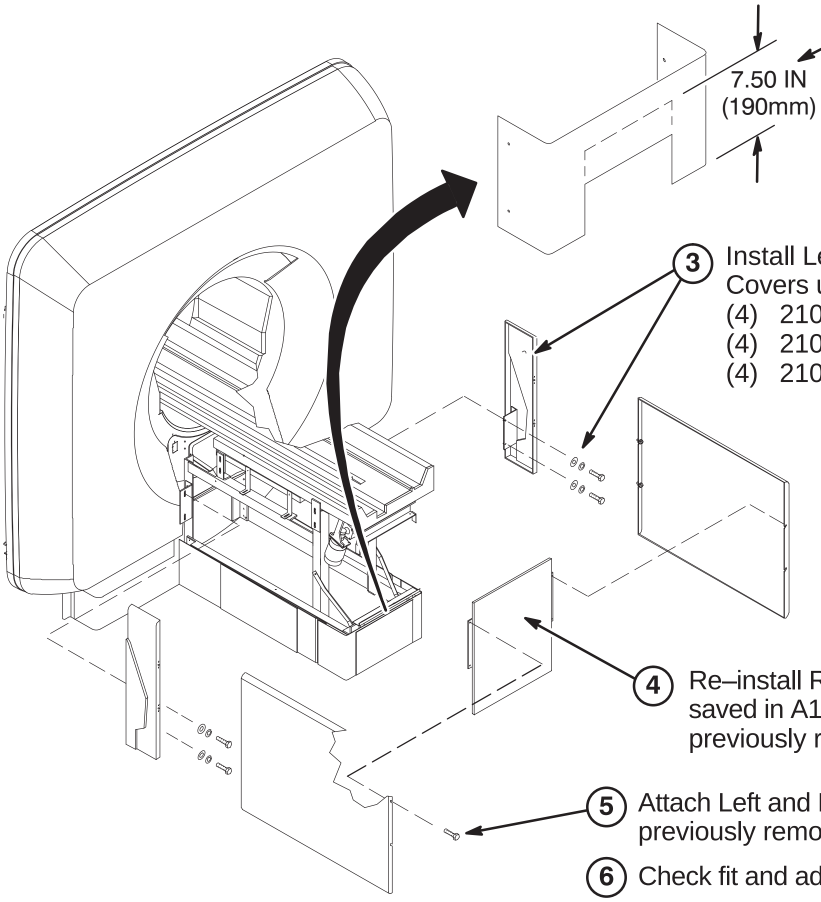
Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.

8–6–15 INSTALL ENCLOSURE SIDE COVERS

1. Re–mount existing left and right Enclosure Side covers.
2. Adjust fit of side covers by adjusting brackets in Enclosure Covers and Side Cover Brackets located on Slide Supports.
3. Close front and rear covers and fasten top and bottom rubber flex latches.
4. Check alignment of side covers. The gaps between covers should be about 0.5 in. (13mm) or less. Adjust cover positions as required.
5. Adjust bumpers and rubber flex latch keeper positions as required.
6. Fasten forks to lower side supports using previously removed hardware.



8–6–16 INSTALL REAR PEDESTAL COVERS



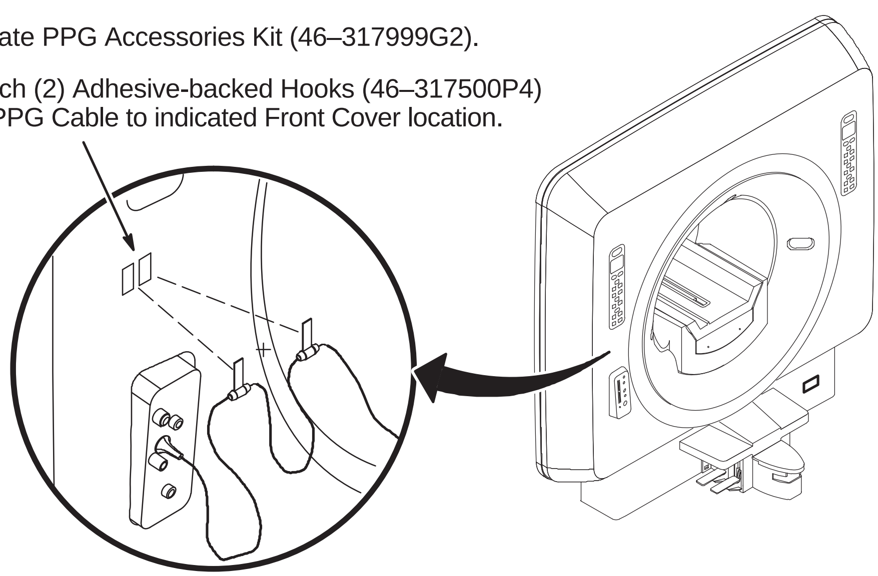
1. Remove Rear Trim Cover from room and cut opening to height indicated. Remove sharp edges.
2. Re–install Base Trim Covers with saved hardware.
3. Install Left (2131044) and Right (2131045) Side Covers using:

(4)	2109878–3	M6 SST Flat Washer
(4)	2109880–3	M6 SST Lock Washer
(4)	2109866–13	M6 x 25 SST Hex Screw

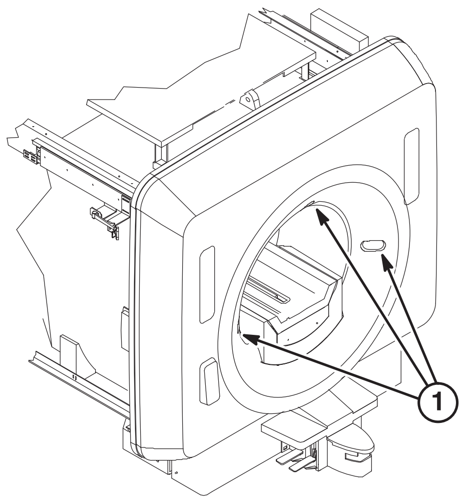
4. Re–install Rear Pedestal Back Cover Asm saved in A1–Step ① to Rear Pedestal with previously removed hardware.
5. Attach Left and Right Rear Pedestal Side Covers with previously remove hardware.
6. Check fit and adjust as required.

8–6–17 INSTALL PPG CABLE HOOKS ON FRONT COVER

1. Locate PPG Accessories Kit (46–317999G2).
2. Attach (2) Adhesive-backed Hooks (46–317500P4) for PPG Cable to indicated Front Cover location.



8–7 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

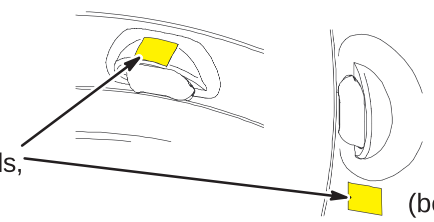


A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

English

For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



2 If enclosure does not have labels, attach new labels as indicated.

(both sides)

French

German

Portuguese

Spanish

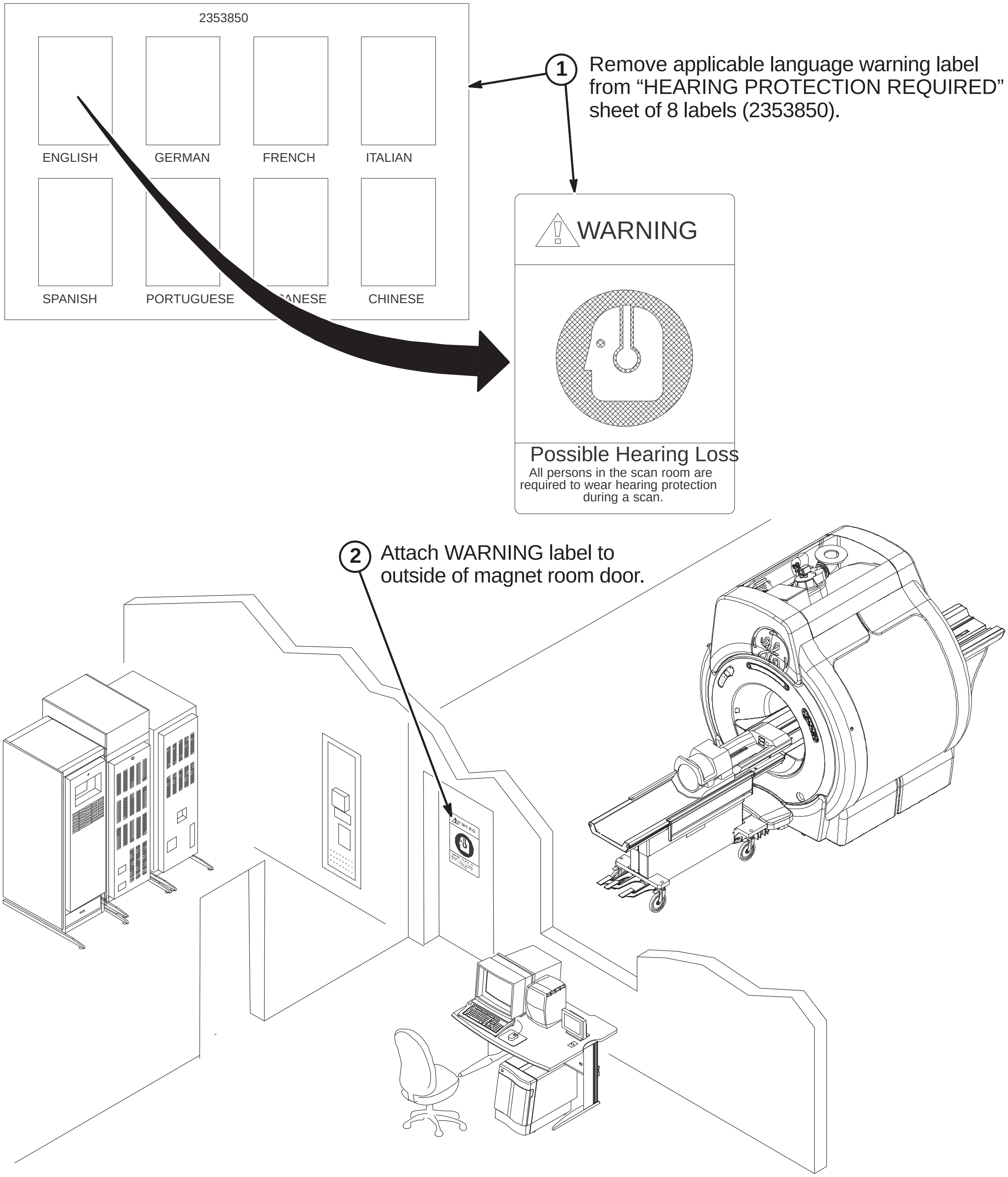
Italian

Swedish

Japanese

Chinese

8–8 INSTALL HEARING LOSS WARNING SIGN



SECTION 9 – MAGNISHIELD ADVANTAGE UPGRADE

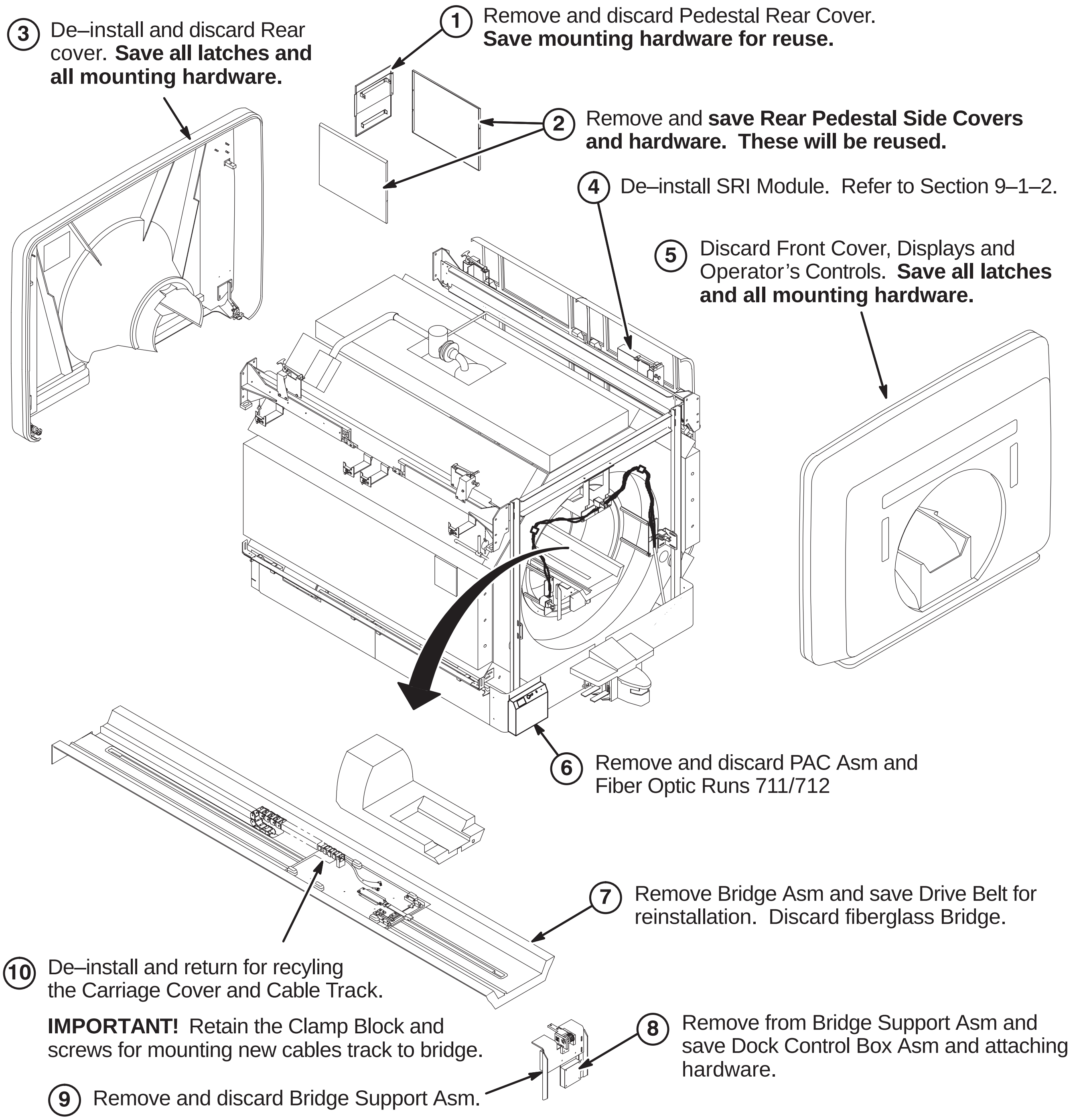
TABLE OF CONTENTS

9–1 DE–INSTALL ADVANTAGE MAGNISHIELD MAGNET ENCLOSURE

Refer to Section 4, Table 4–1 for information on cable removal. Also refer to applicable cable map for additional information on cable removal.

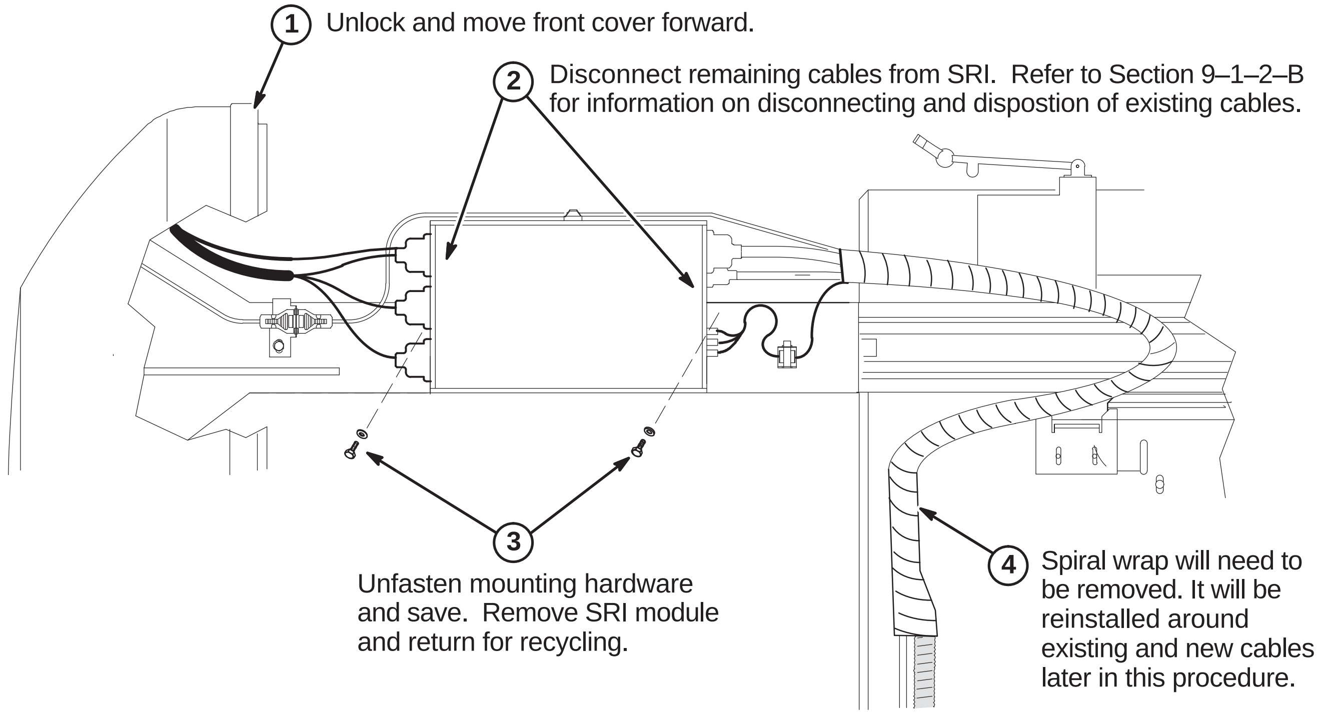
Discarded Material Return Policy – Refer to Section for procedures on how to handle all de–installed material that is designated as “discard” in the following procedures.

9–1–1 DE–INSTALL COVERS, BRIDGE, BRIDGE SUPPORT, DSIM/SRI AND PAC ASM

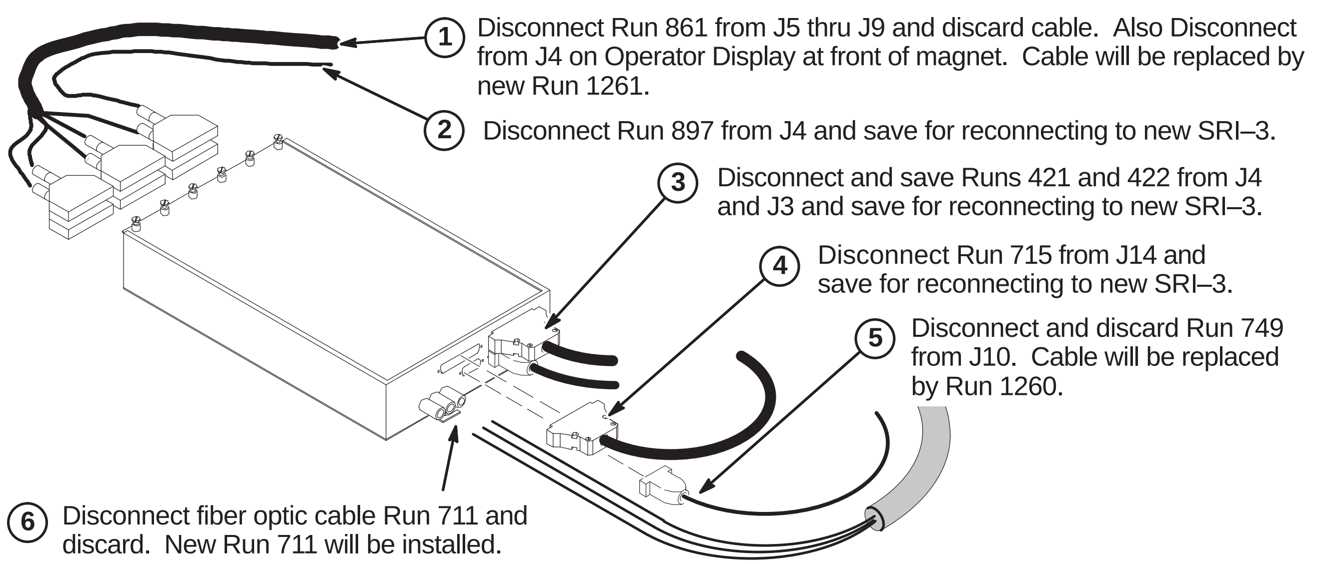


9–1–2 REMOVE EXISTING SRI MODULE

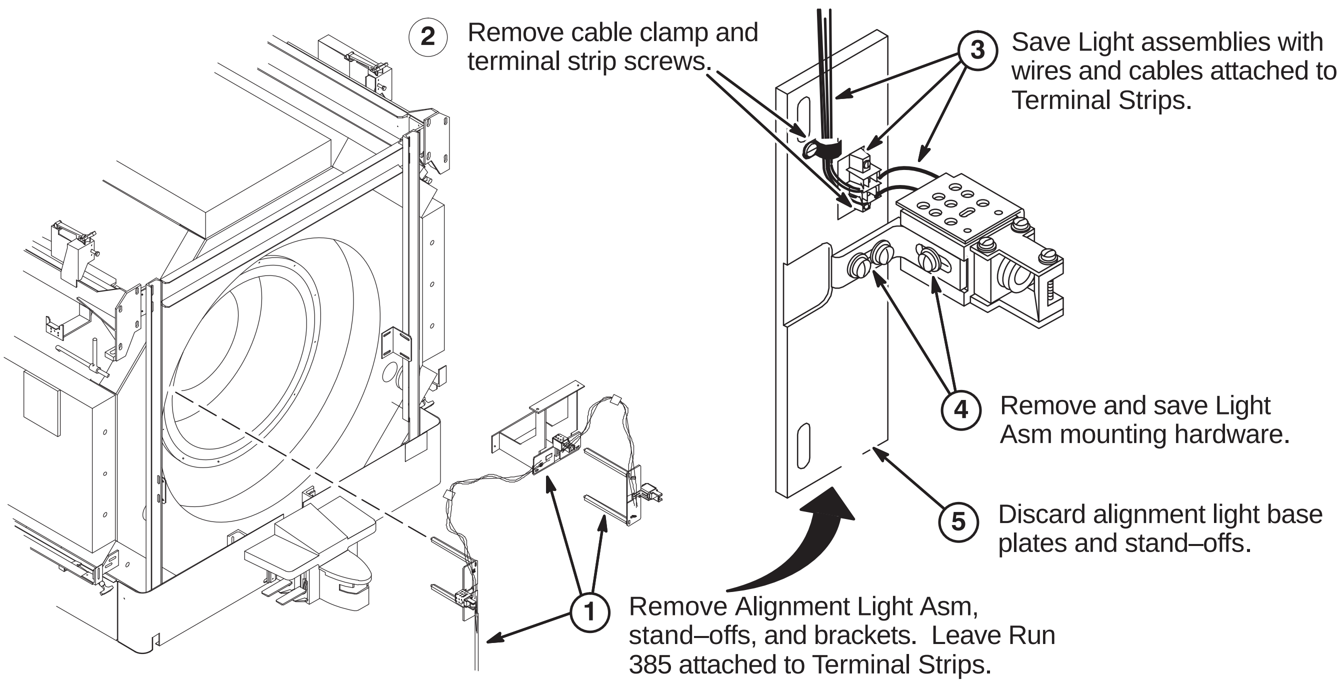
9–1–2–A De–Install SRI Module



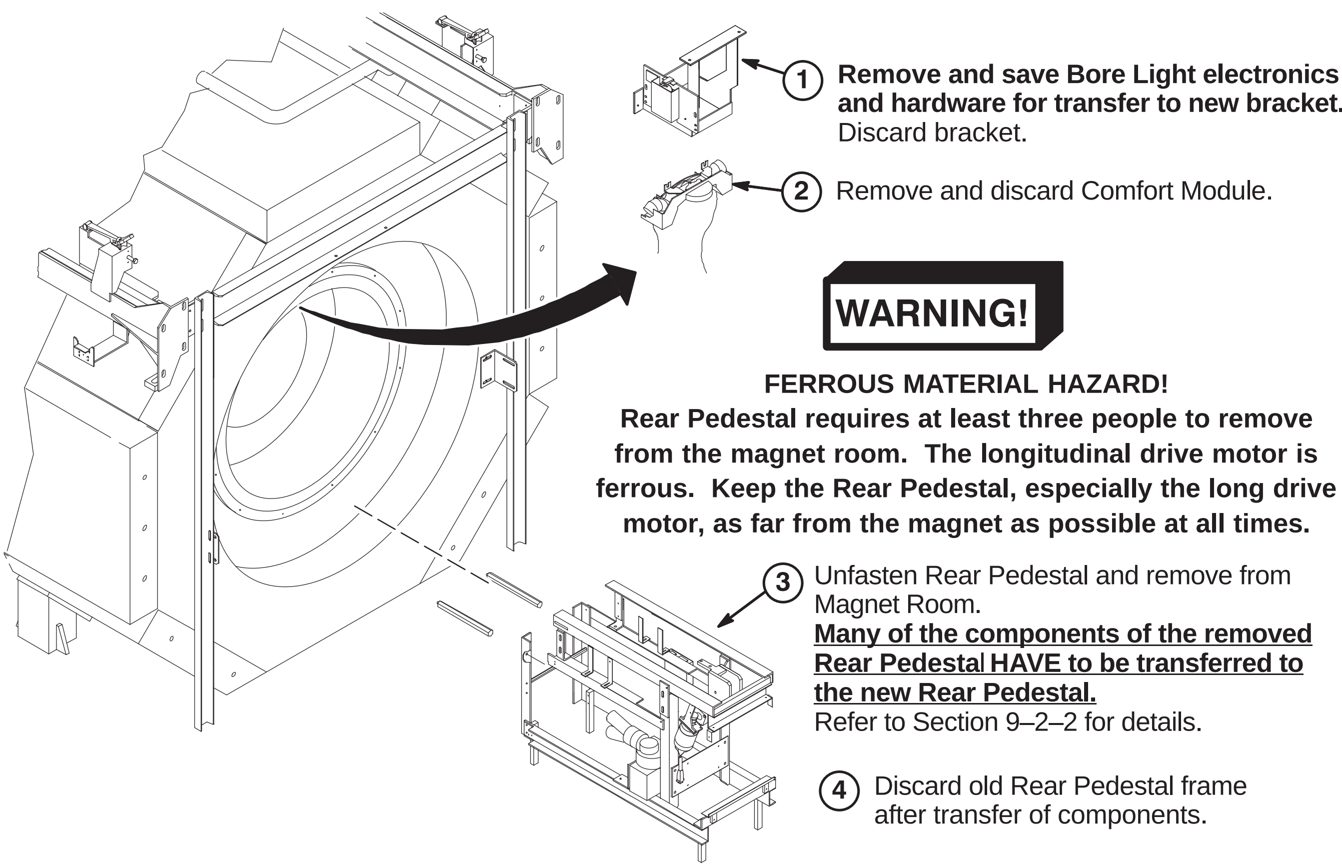
9–1–2–B SRI Cable Disposition



9–1–3 REMOVE ALIGNMENT LIGHTS



9–1–4 REMOVE REAR PEDESTAL, BORE LIGHT, & COMFORT MODULE

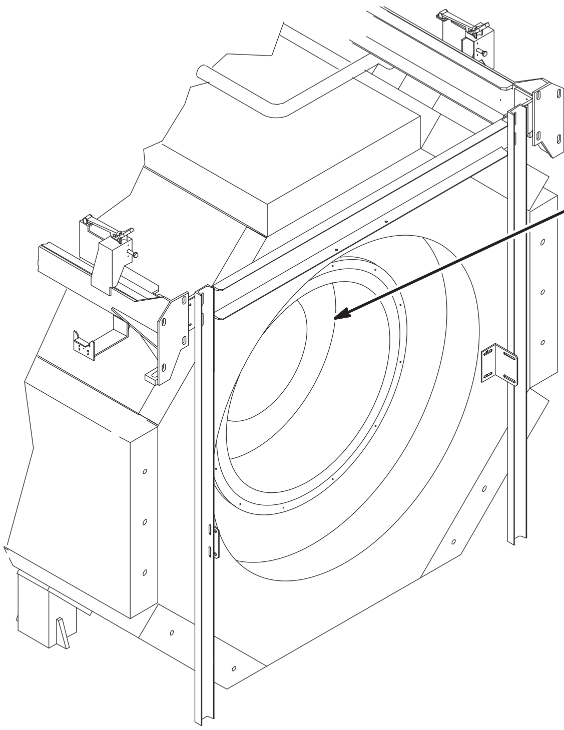


WARNING!

FERROUS MATERIAL HAZARD!
Rear Pedestal requires at least three people to remove from the magnet room. The longitudinal drive motor is ferrous. Keep the Rear Pedestal, especially the long drive motor, as far from the magnet as possible at all times.

9–1–5 BODY COIL REPLACEMENT

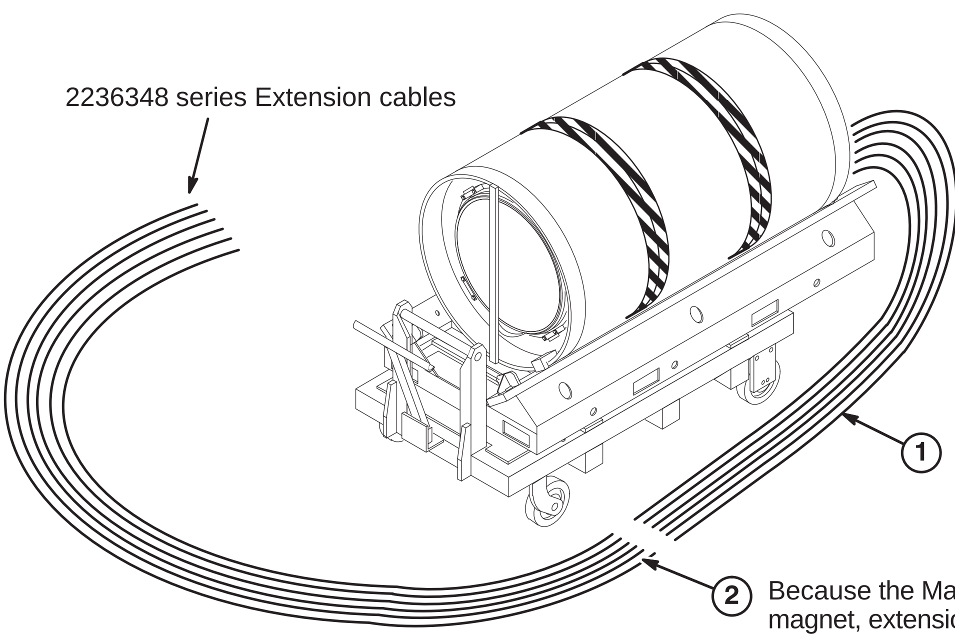
9–1–5–A Coil Removal Instructions



1. Remove Gradient Body Coil.

Note: Refer to procedures in Section **13** and Gradient Coil Replacement procedures on the applicable MR Service Methods CD–ROM

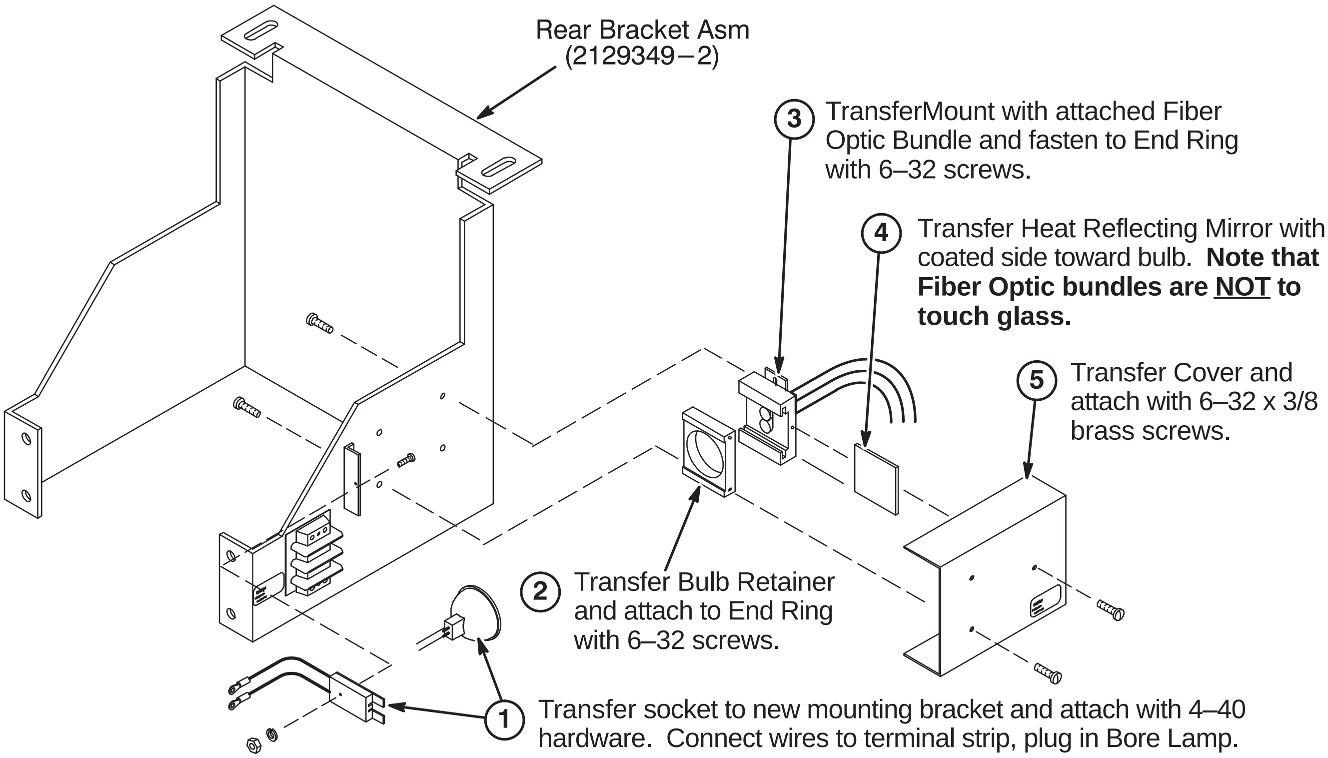
9–1–5–B BRM–D Gradient Coil Pre–Installation Instructions



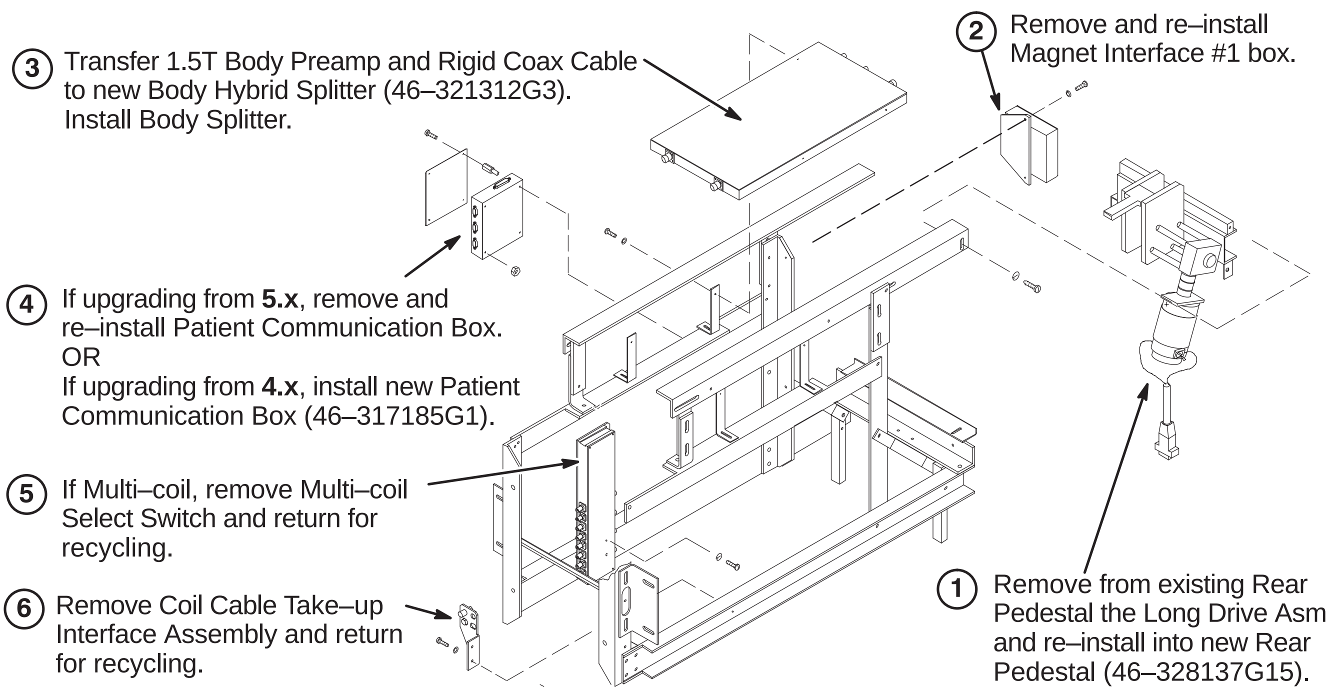
1. The new BRM–D coil will have long gradient cable leads.
2. Because the Magnishield is not an Active–Shield magnet, extension gradient leads have to be added. The procedure for attaching the extension leads is located in Section **13–2** must be completed outside of the magnet room.

9–2 TRANSFER DE–INSTALLED PARTS TO NEW COMPONENTS

9–2–1 TRANSFER & ASSEMBLE BORE LIGHT TO NEW REAR BRACKET ASM



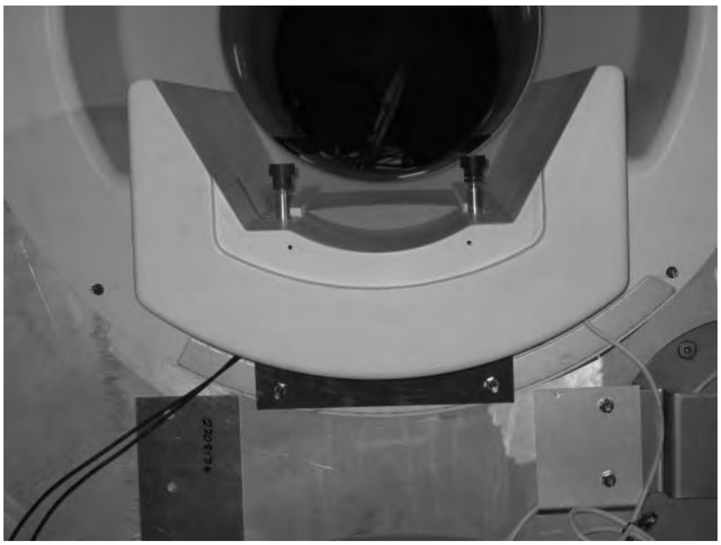
9–2–2 TRANSFER EXISTING COMPONENTS TO NEW REAR PEDESTAL FRAME



9–3 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

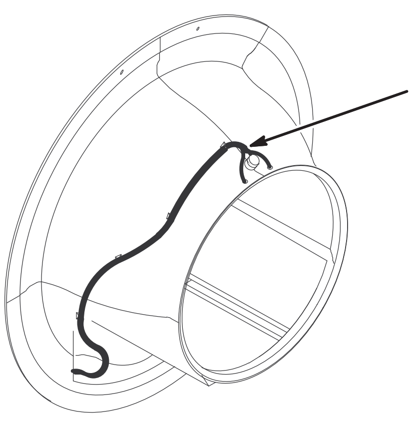
Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) has been installed. This kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It is required for allmagnets with Horizon style enclosures. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and it must be installed at this point of the EXCITE HD ugprade.

Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2401984, *S–Series Magnet Bridge Support* (shown below) and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.



Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade,* for instructions on installing this kit.

9–4 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL



The new Front End Bell is equipped with a Dual Temperature Sensor cable, Rune 749, that has to be replaced with a new cable, Run 1260 (5123812).

EXCITE HD requires Run 1260 to be installed.

Refer to Sections **10–3–7–B, 10–3–8,** and **10–3–9** for instructions on removal of existing cable and installing the new cable.

9–5 INSTALL MAGNET MONITOR

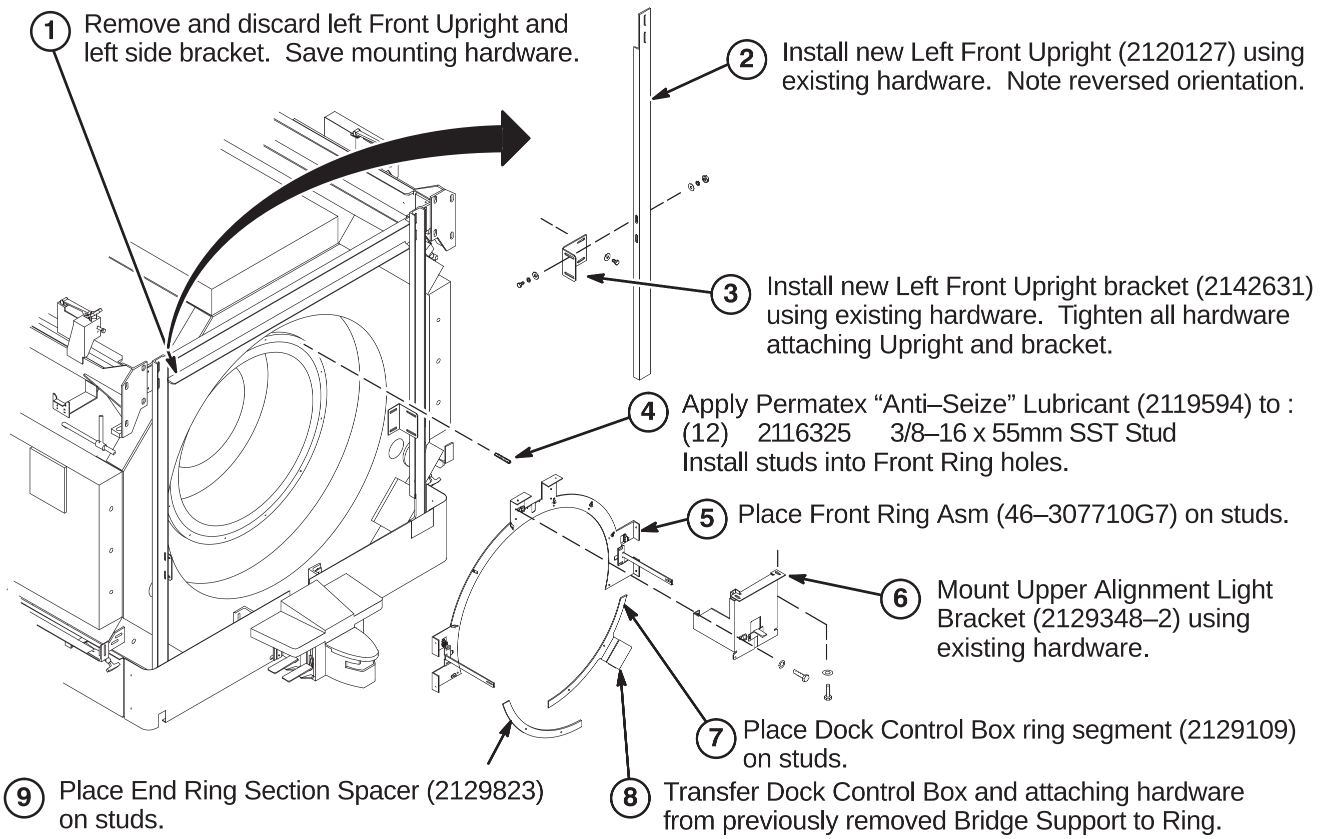
EXCITE HD requries Magnet Monitor on all systems. For this upgrade, all the parts and instructions are included in parts collector 5127838, Magnet Monitor Kit for Non–LCC Systems. Refer to Direction 2230681, included with collector, for instructions on installing magnet monitor on this magnet.

9–6 INSTALL NEW MAGNISHIELD COVERS, COMPONENTS, AND CABLES

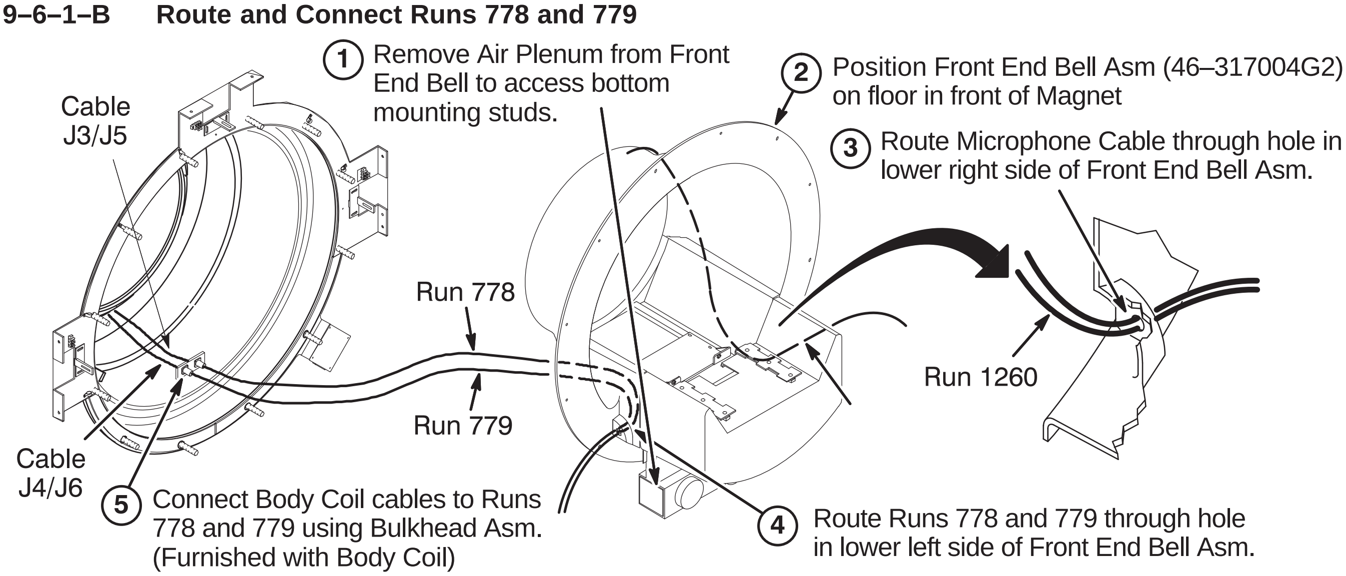
The new Gradient Coil must be installed before starting the installation of new parts referred to in the following steps. Refer to Section **13** for Gradient Coil installation instructions.

9–6–1 INSTALL COMPONENTS AT FRONT OF MAGNET

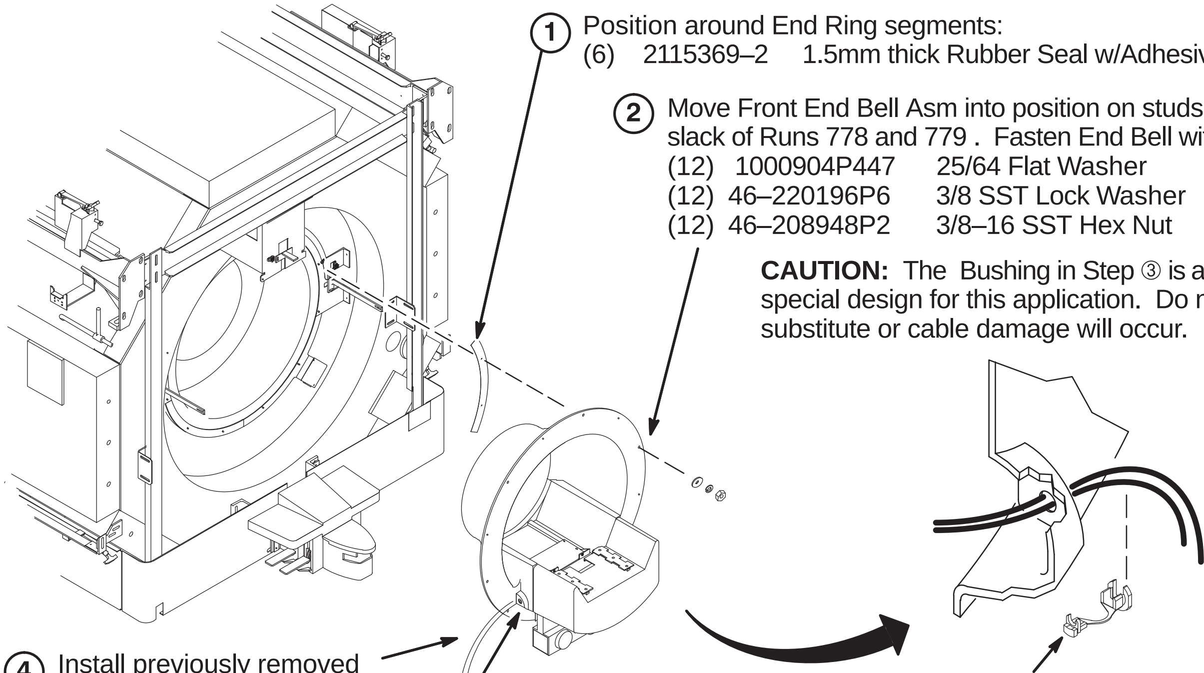
9–6–1–A Install Front Ring Asm and Vertical Upright



9–6–1–B Route and Connect Runs 778 and 779



9–6–1–C Install Front Gasket and Mount End Bell



1. Position around End Ring segments:
(6) 2115369–2 1.5mm thick Rubber Seal w/Adhesive

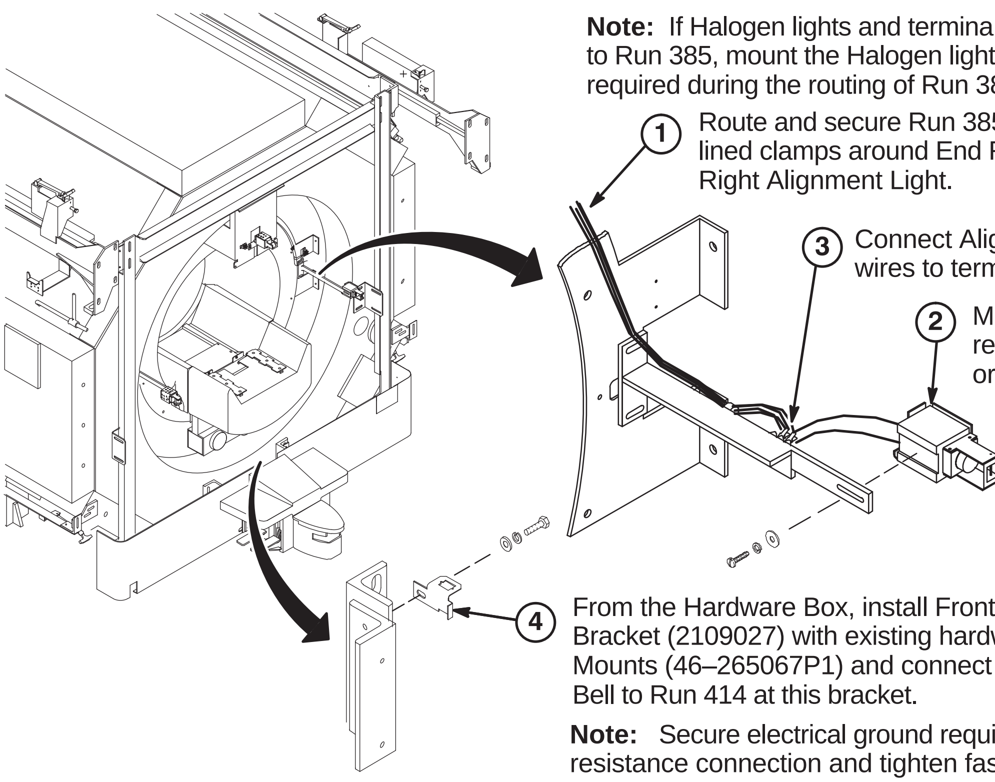
2. Move Front End Bell Asm into position on studs. Adjust slack of Runs 778 and 779 . Fasten End Bell with:
(12) 1000904P447 25/64 Flat Washer
(12) 46–220196P6 3/8 SST Lock Washer
(12) 46–208948P2 3/8–16 SST Hex Nut

CAUTION: The Bushing in Step ③ is a special design for this application. Do not substitute or cable damage will occur.

3. Place Bushings over Microphone Cable and Run 1260 on right side and Runs 778 and 779 on left side and secure into holes in End Bell Asm.

4. Install previously removed Air Plenum.

9–6–1–D Install/Connect Alignment Lights & Microphone Bracket



Note: If Halogen lights and terminal strips are still attached to Run 385, mount the Halogen lights and terminal strips as required during the routing of Run 385.

1. Route and secure Run 385 through rubber lined clamps around End Ring towards Right Alignment Light.

2. Mount the three previously removed alignment lights using original mounting hardware.

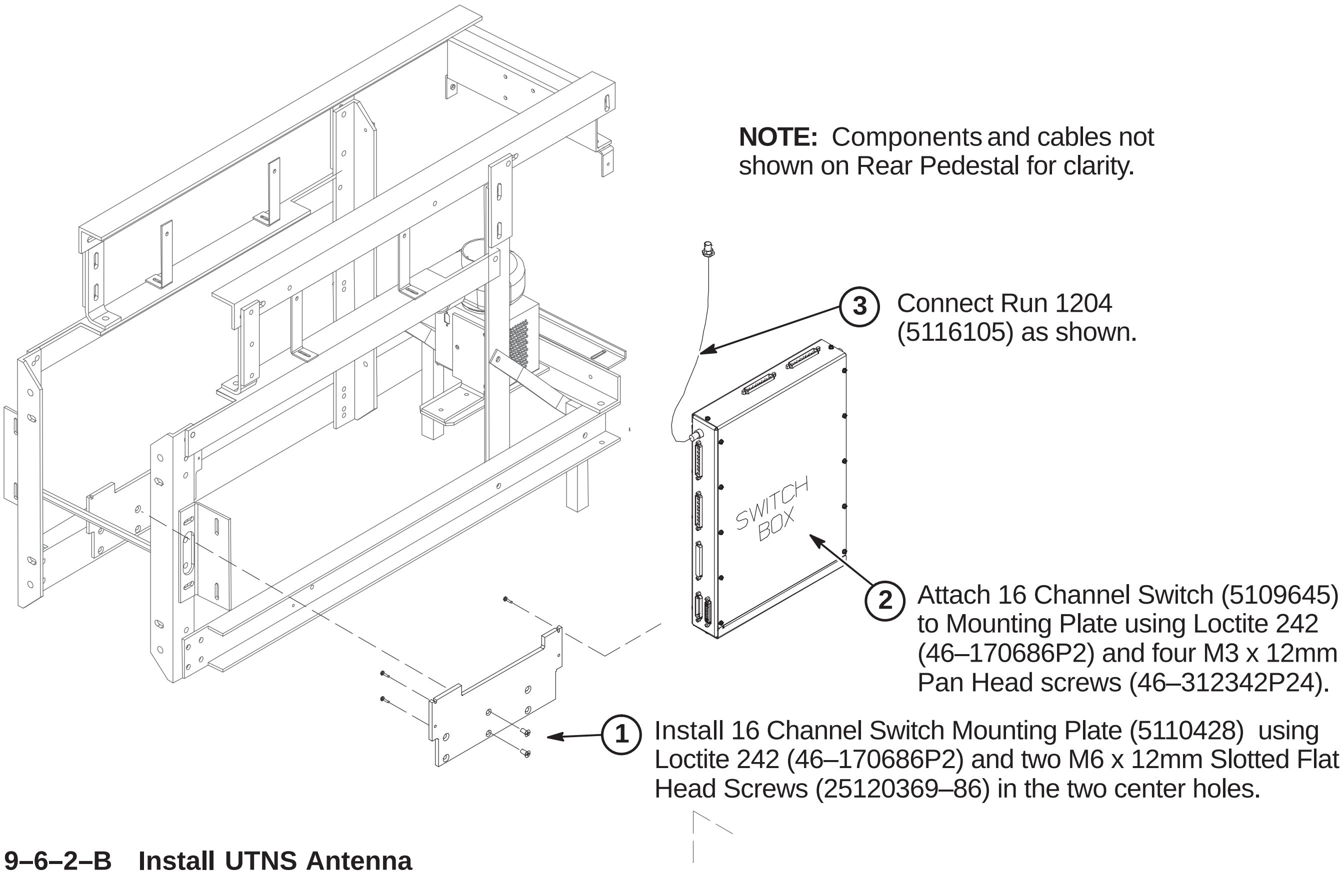
3. Connect Alignment Light wires to terminal strip.

4. From the Hardware Box, install Front Microphone Connector Bracket (2109027) with existing hardware. Insert (2) Cable Mounts (46–265067P1) and connect Microphone cable from End Bell to Run 414 at this bracket.

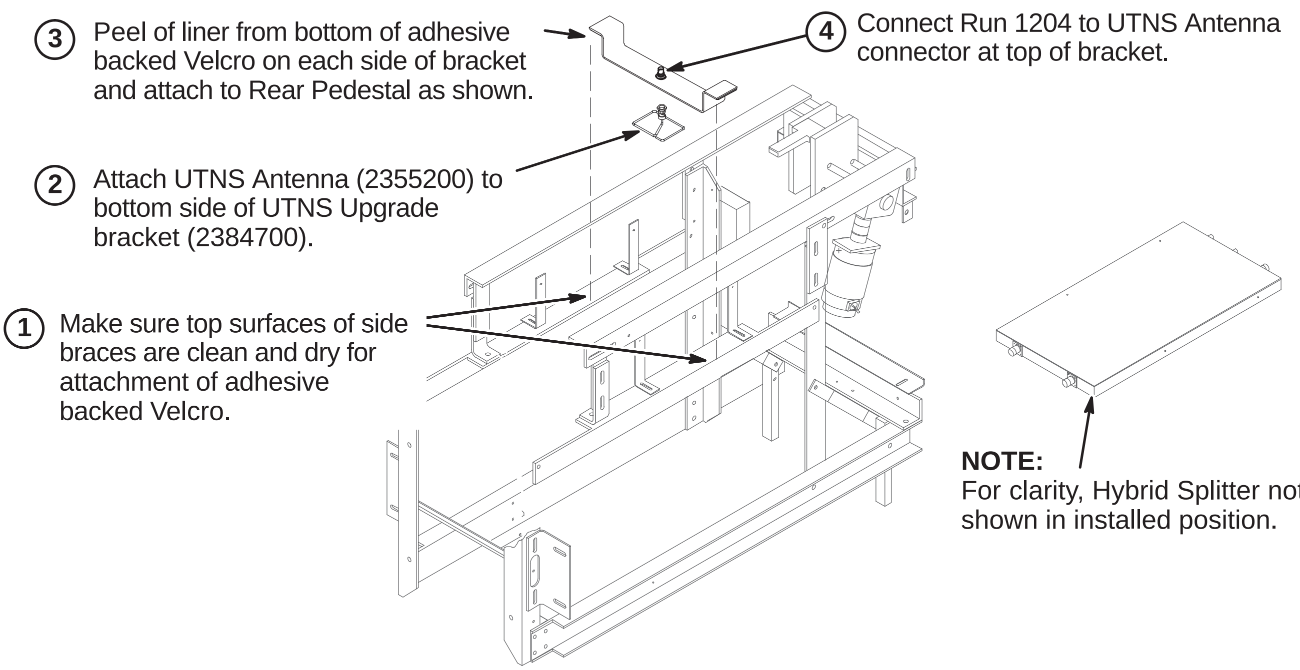
Note: Secure electrical ground required. Insure low resistance connection and tighten fasteners securely.

9–6–2 INSTALL NEW COMPONENTS ON REAR PEDESTAL

9–6–2–A Install 16 Channel Switch

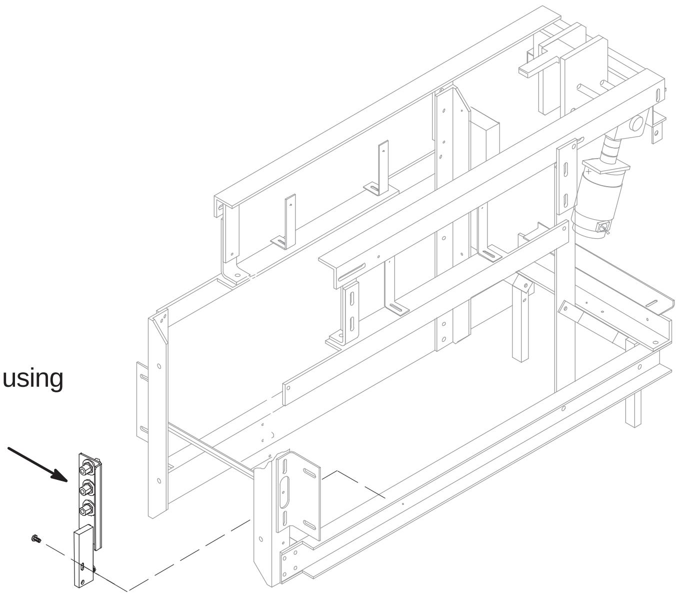


9–6–2–B Install UTNS Antenna



9–6–2–C Install Bulkhead Interface

1 Install new Bulkhead Interface (5113250) using Loctite 242 (46–170686P2) and two M6 x 12mm Pan Head Screws (2109873–24).



9–6–3 HYBRID SPLITTER UPGRADE

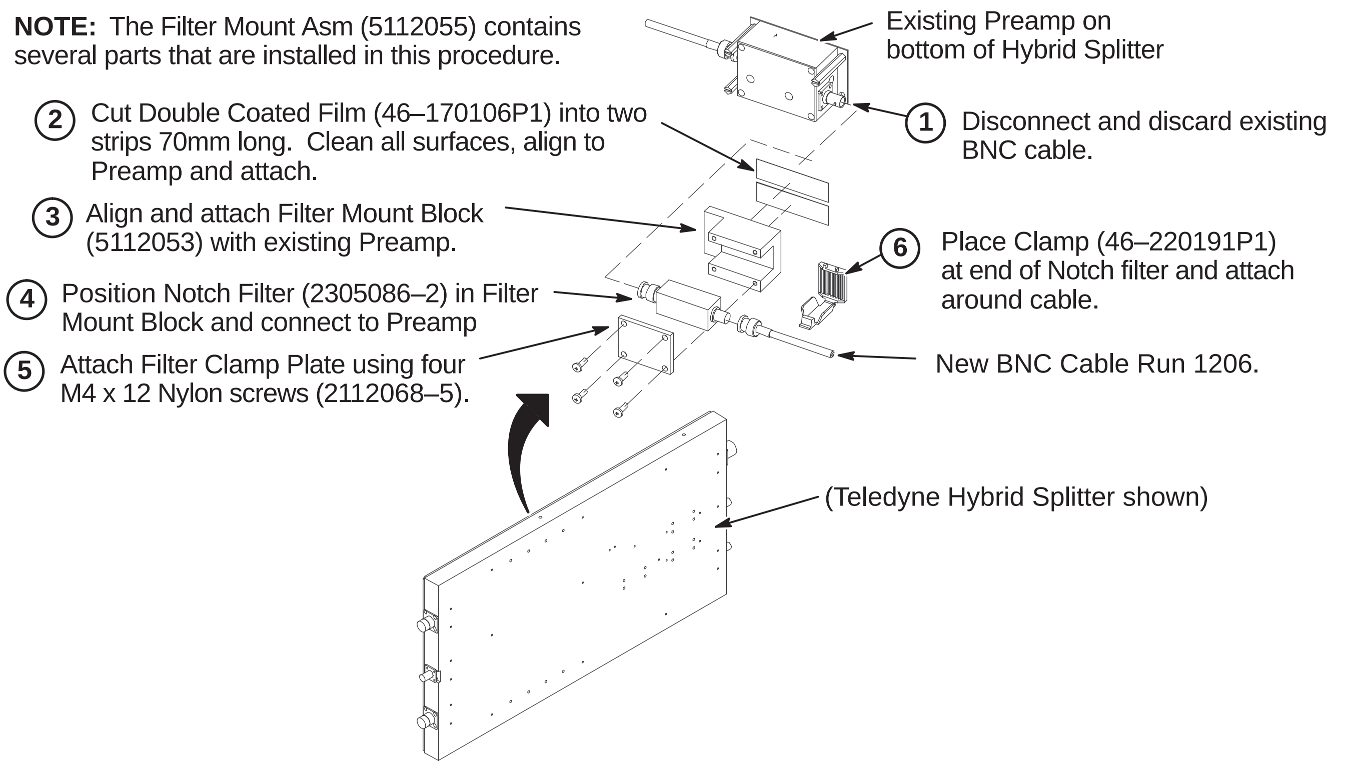
Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

- Refer to Section 9–6–3–A for instructions on installing the Notch Filter and mounting block
- Refer to Section 9–6–3–B for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 9–6–3–C for instructions on installing Isolation Filters on systems with Teledyne Splitter

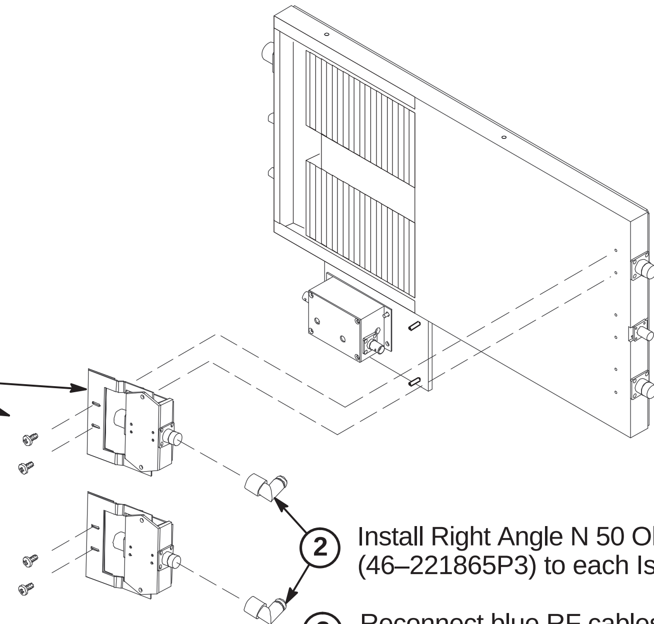
The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils.

9–6–3–A Install Notch Filter and Filter Mount Block Asm

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.



9–6–3–B Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)



FOR BRM's ONLY

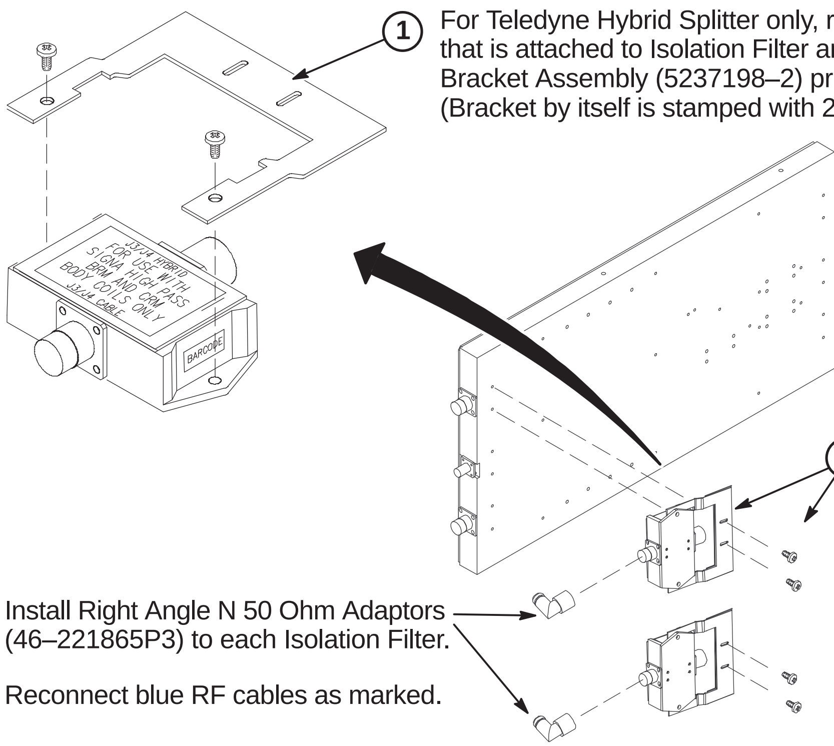
1 Attach and connect each Ground Isolation Filter Module (5118952–2) to the Hybrid Splitter using two screws (46–220184P5).

Do NOT install on TRM gradient coil systems.

2 Install Right Angle N 50 Ohm Adaptors (46–221865P3) to each Isolation Filter.

3 Reconnect blue RF cables as marked.

9–6–3–C Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)



1 For Teledyne Hybrid Splitter only, remove existing bracket that is attached to Isolation Filter and replace with Mounting Bracket Assembly (5237198–2) provided with upgrade kit. (Bracket by itself is stamped with 2254629)

FOR BRM's ONLY

2 Attach and connect each Ground Isolation Filter Module (5118952) to the Hybrid Splitter using two screws (46–220184P5) for each module.

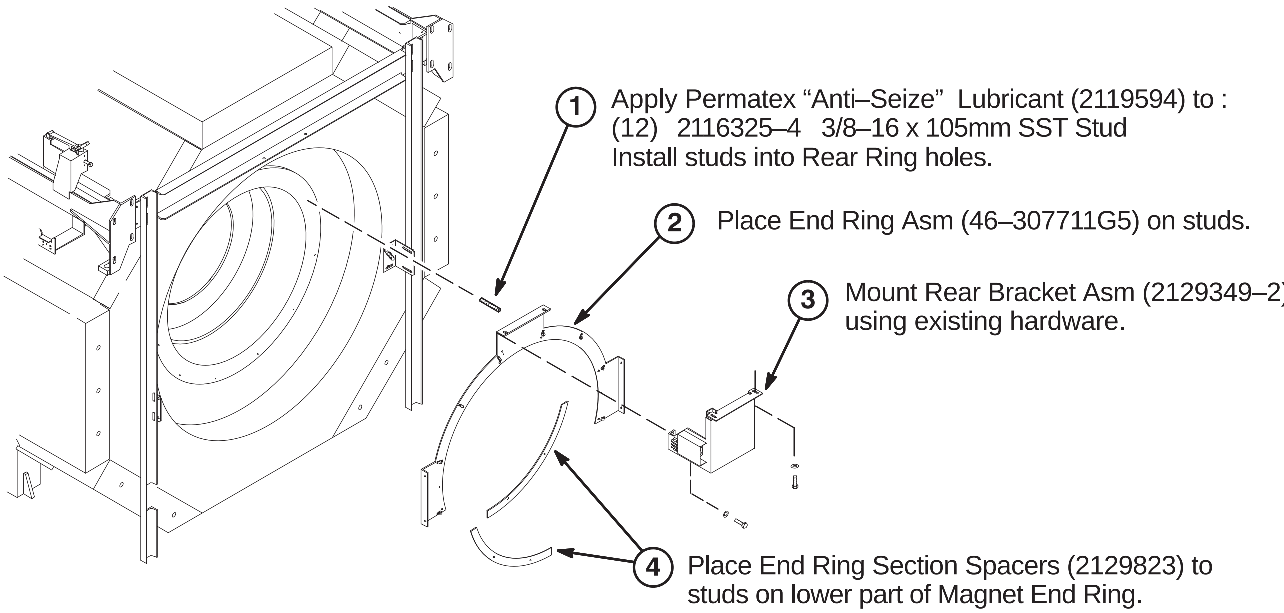
Do NOT install on TRM gradient coil systems.

3 Install Right Angle N 50 Ohm Adaptors (46–221865P3) to each Isolation Filter.

4 Reconnect blue RF cables as marked.

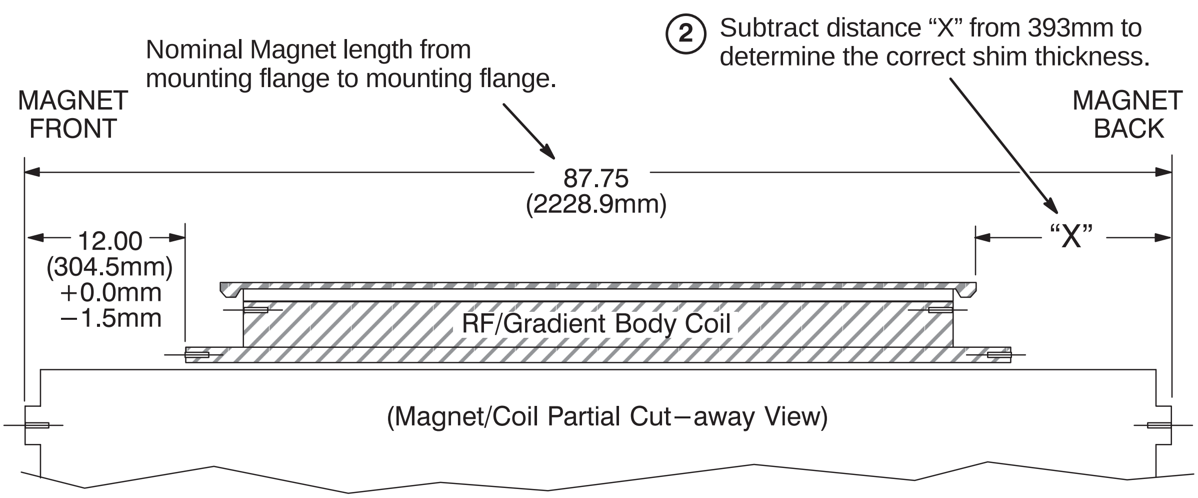
9–6–4 INSTALL COMPONENTS AT FRONT AND REAR OF MAGNET

9–6–4–A Install Rear End Rings and Bracket Asm



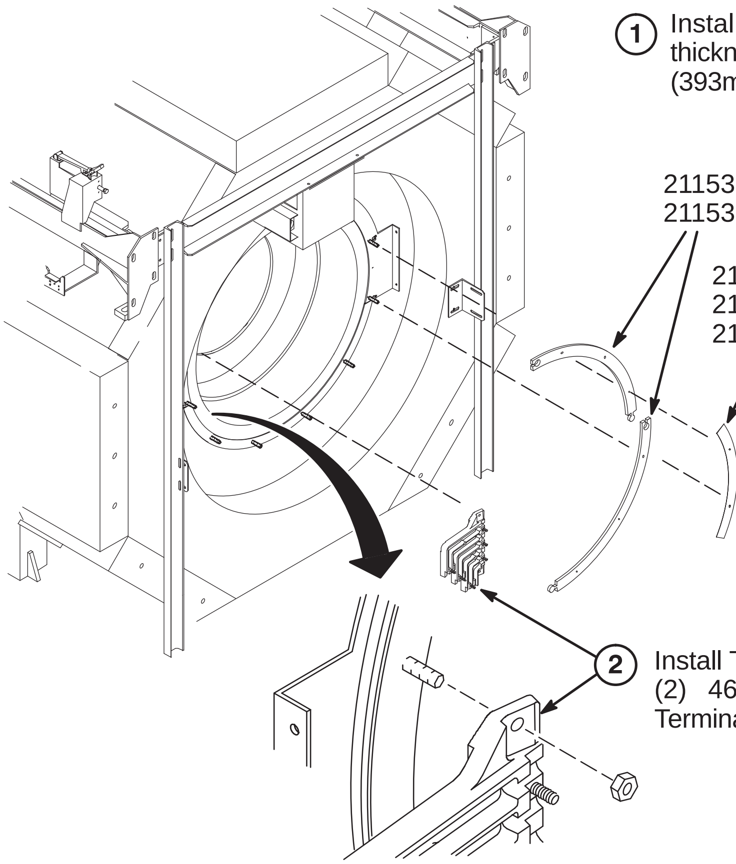
9–6–4–B Formula for Determining Rear Spacers

1. With RF/Gradient Body Coil in place, measure in millimeters the distance from RF Tube to mounting flange on Magnet. This is distance “X”. **Make sure the distance is measured to the Magnet mounting flange, not the Rear Frame Asm End Ring installed in Section 9–6–4–A.**



Note: Values shown here are for GE S–II, S–III, S–IV, S–V and Magnishield Magnet only.

9–6–4–C Install Rear Spacers and Gradient Terminal Block



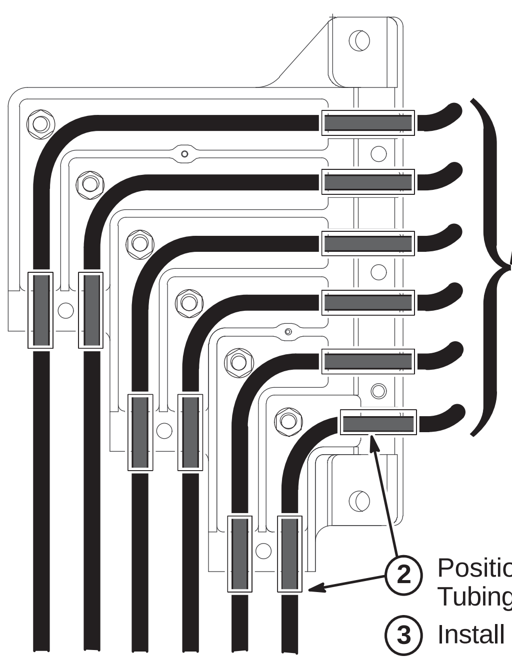
1. Install the following Rubber Seals as required to obtain a thickness derived from the formula in the previous step: (393mm – “X” = Shim Thickness)

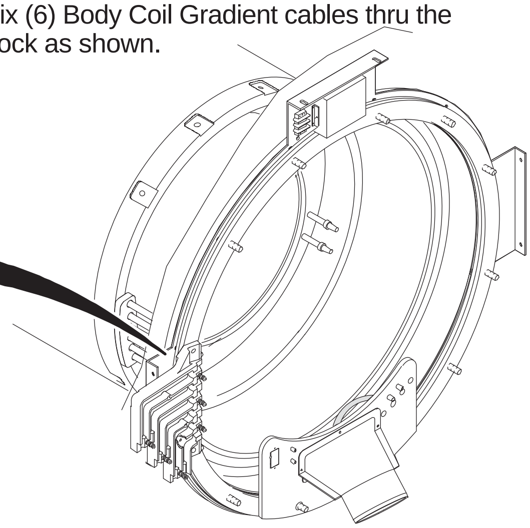
2115370	12.7mm Thick Interlocking Rubber Seal
2115370–2	7.9mm Thick Interlocking Rubber Seal

2115369	3/4mm Thick Rubber Seal w/adhesive
2115369–2	1.5mm Thick Rubber Seal w/adhesive
2115369–3	2.25mm Thick Rubber Seal w/adhesive

2. Install Terminal Block to magnet End Ring with:
(2) 46–208948P2 3/8–16 SST Hex Nut
Terminal Block Access Cover fits over top of this nut.

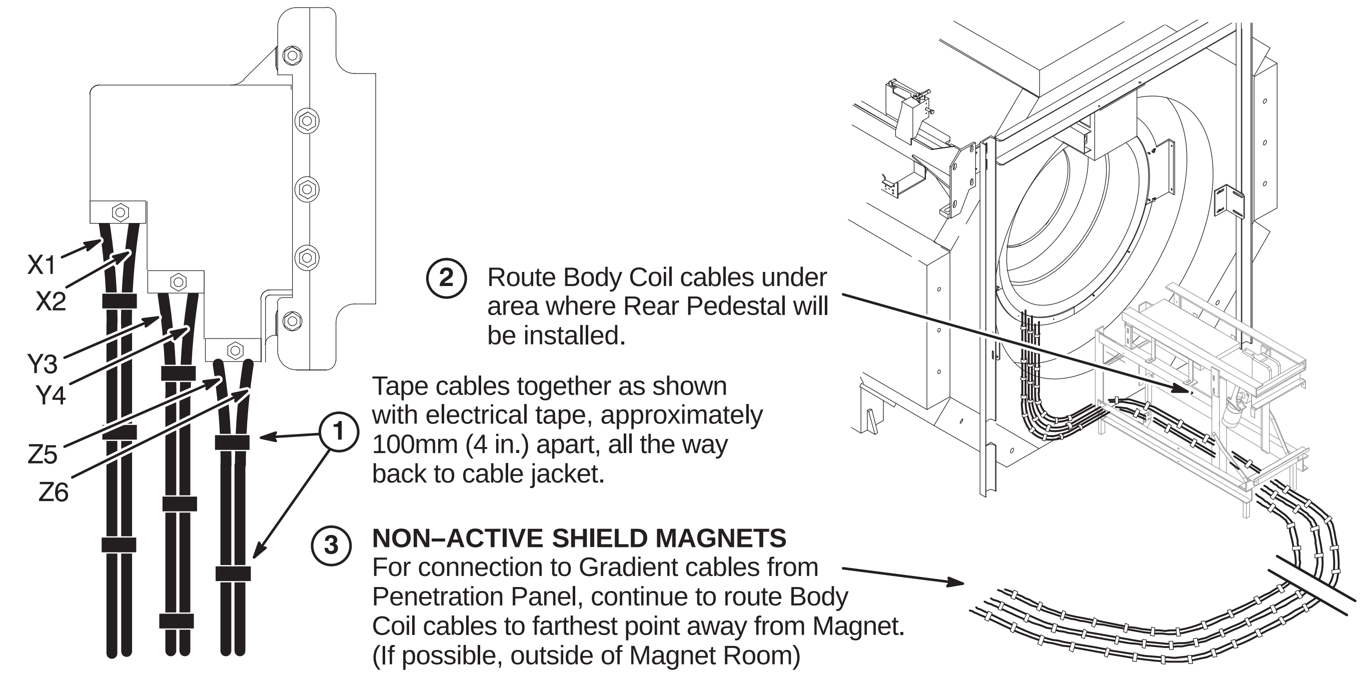
9–6–4–D Route and Install Gradient Cables in Terminal Block





1. Route the six (6) Body Coil Gradient cables thru the Terminal Block as shown.
2. Position the twelve (12) pieces of PVC Clear Tubing (46–221986P3) as shown.
3. Install Terminal Block strain reliefs over PVC tubing. Install cover.

9–6–4–E Secure X, Y, and Z Gradient Cables

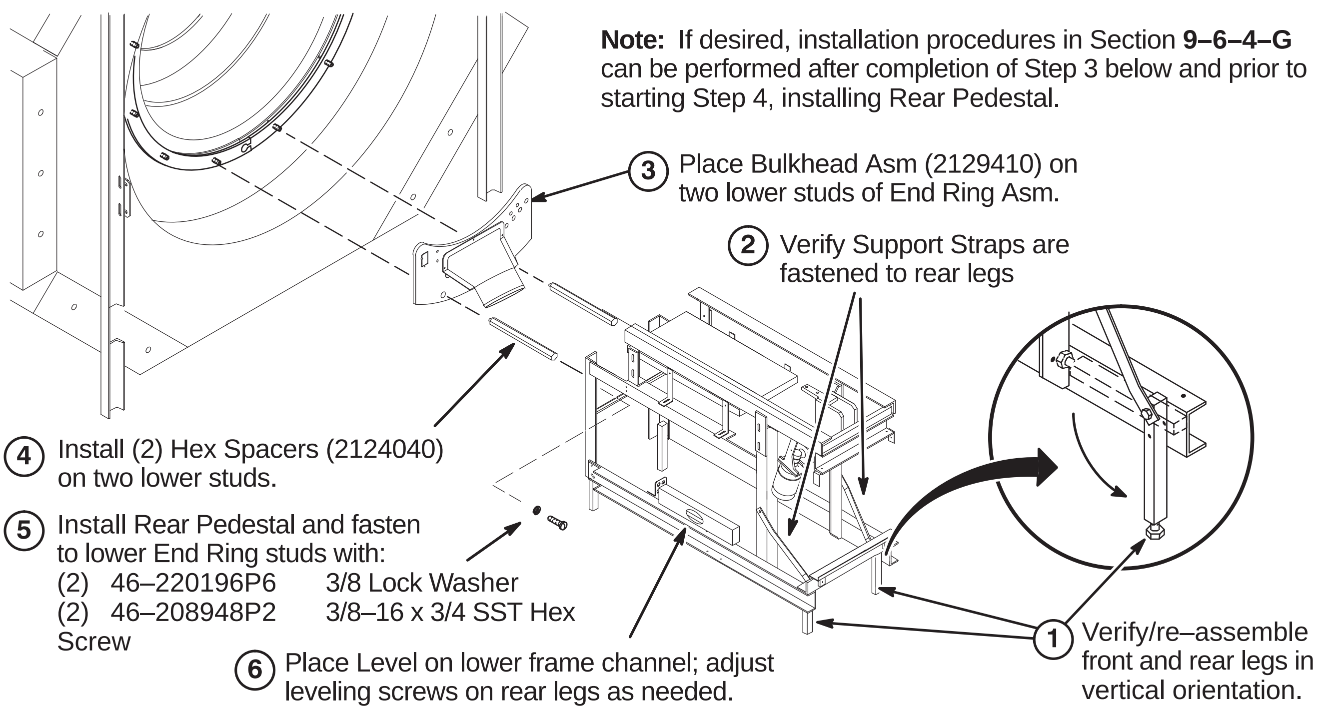


9–6–4–F Install Rear Bulkhead and Rear Pedestal

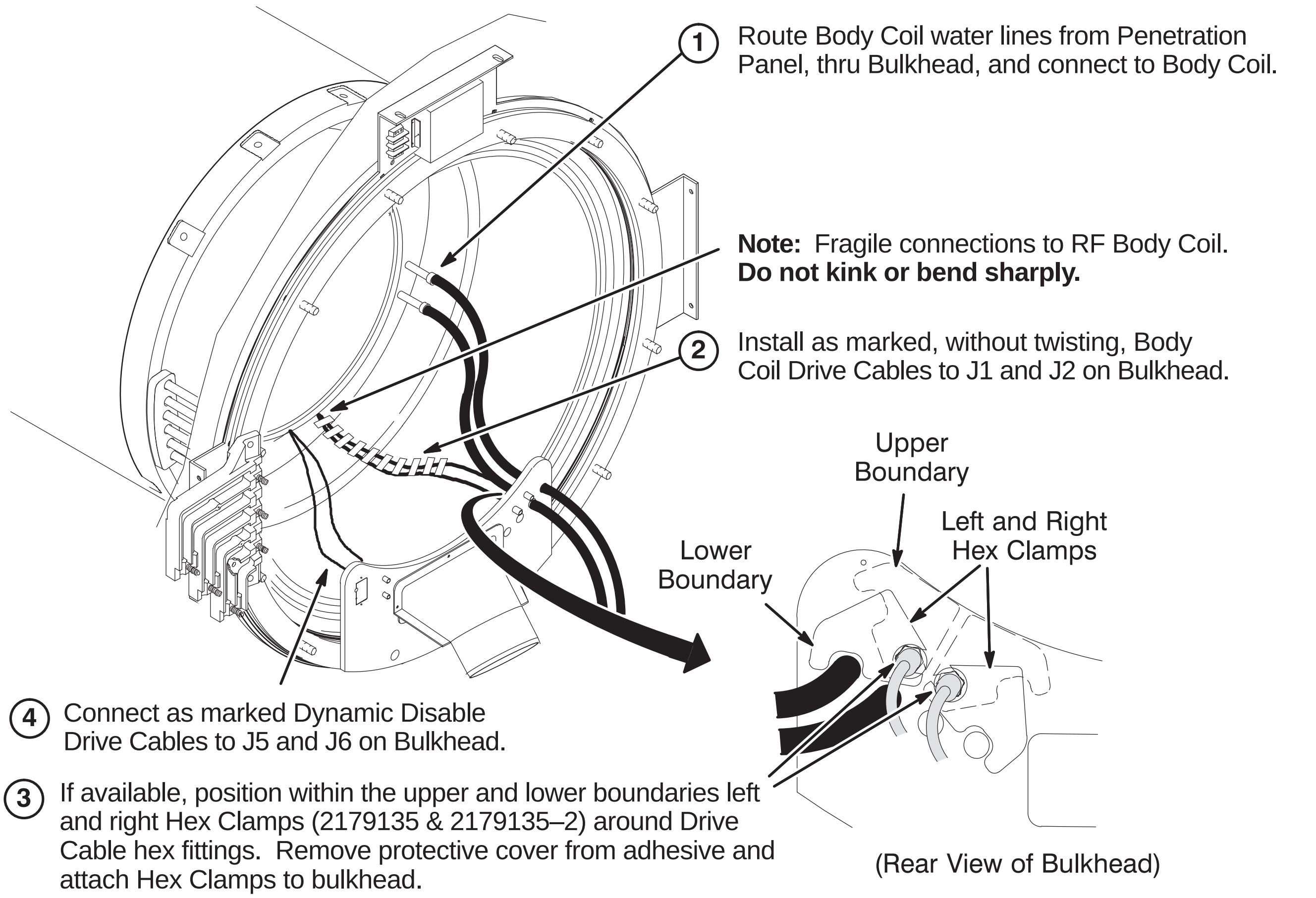
WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

Note: If desired, installation procedures in Section **9–6–4–G** can be performed after completion of Step 3 below and prior to starting Step 4, installing Rear Pedestal.

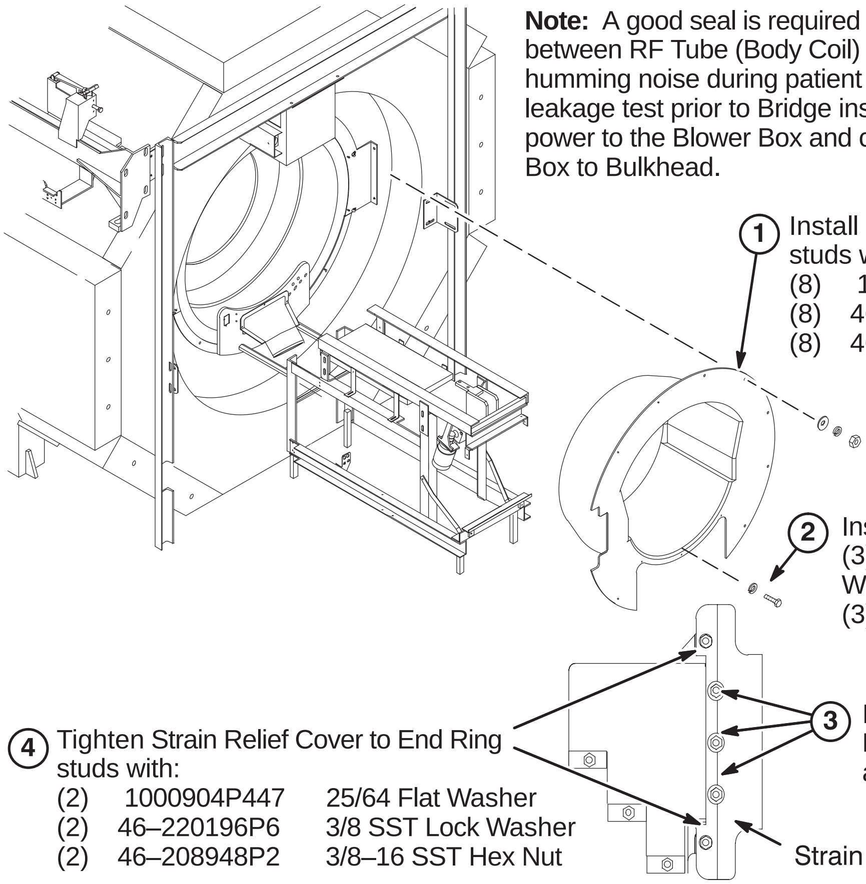


9–6–4–G Route and Connect Bulkhead Cables and Hoses



9–6–4–H Install Rear End Bell

Note: A good seal is required to prevent air leaks. Any air leaks between RF Tube (Body Coil) and End Bell joint can cause a loud humming noise during patient scanning. If possible, conduct an air leakage test prior to Bridge installation. This would require connecting power to the Blower Box and connecting 6” blower hose from Blower Box to Bulkhead.



1. Install Rear End Bell (2113076) and fasten to studs with:

(8)	1000904P447	25/64 Flat Washer
(8)	46–220196P6	3/8 SST Lock Washer
(8)	46–208948P2	3/8–16 SST Hex Nut

2. Install Rear End Bell to Bulkhead Asm with:

(3)	2109878–3	M6 SST Flat Washer
(3)	2109866–13	M6 SST Hex Screw

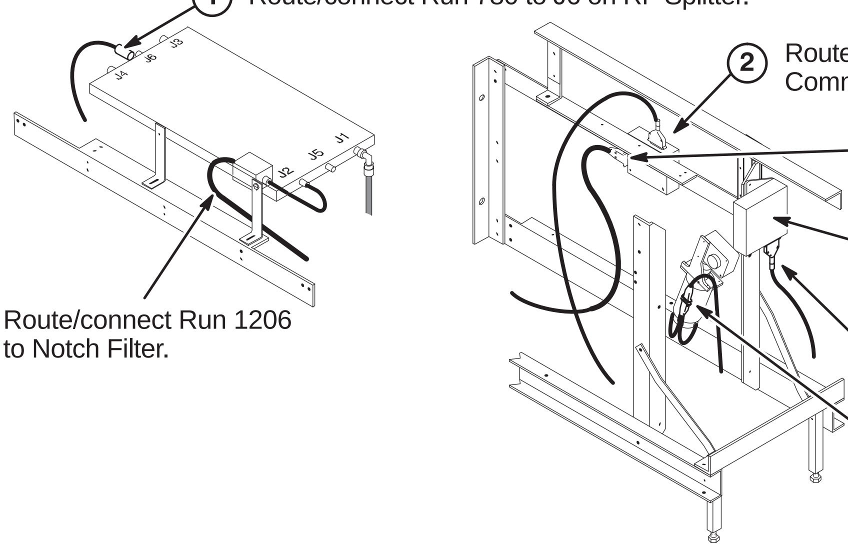
3. Install Strain Relief Cover to Terminal Block using hardware provided with assembly.

4. Tighten Strain Relief Cover to End Ring studs with:

(2)	1000904P447	25/64 Flat Washer
(2)	46–220196P6	3/8 SST Lock Washer
(2)	46–208948P2	3/8–16 SST Hex Nut

Strain Relief Cover

9–6–4–I Connect Runs 282, 318, 414, 422, 780, and 1206



1. Route/connect Run 780 to J6 on RF Splitter.
2. Route/connect Run 282 to J6 on Patient Communications Box (MG3 A2).
3. Route/connect Run 414 to J9.

Magnet Interface #1 (MG3 A1)

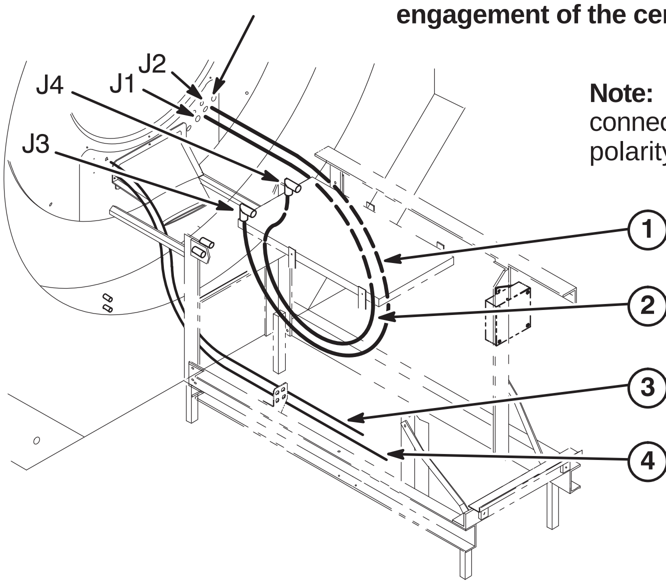
4. Route/connect Run 422 to J4 on Magnet Interface #1.
5. Route/connect Run 318 to short cable on Longitudinal Motor Drive (MG3 A7).
6. Route/connect Run 1206 to Notch Filter.

9–6–4–J Route/Connect Runs 743, 744, 781, and 782

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

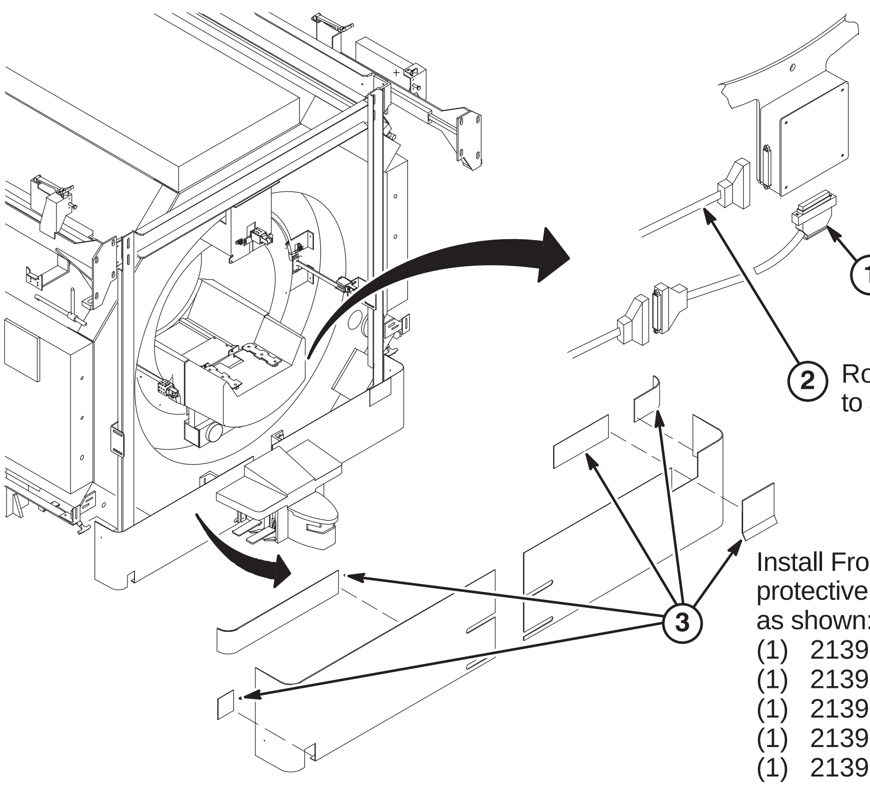
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

1. Route and connect Run 744 (2105691–5) from J1 on the Bulkhead to J4 on the RF Splitter.
2. Route and connect Run 743 (2105691–4) from J2 on the Bulkhead to J3 on the RF Splitter.
3. Route and connect Run 781 (46–328000G919) from J5 on the Bulkhead to J75 on the Penetration Panel.
4. Route and connect Run 782 (46–328000G920) from J6 on the Bulkhead to J76 on the Penetration Panel.

9–6–4–K Connect Runs 421 and Dock Cable & Install Skirt Upgrade Kit

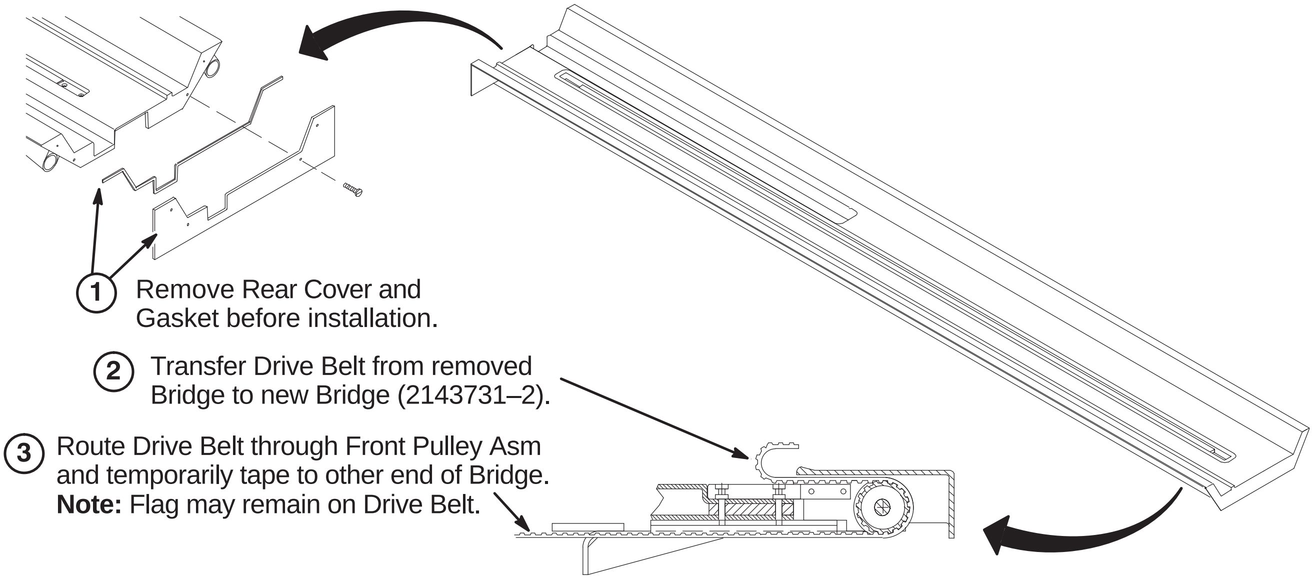


1. Attach Cable Extension (2143072) to Dock Cable and connect to J2 on Dock Limit Module (MG3 A31).
2. Route/connect Run 421 from SRI Module to J1 on Dock Limit Module (MG3 A31).
3. Install Front Skirt Upgrade Kit (2139555). Remove protective coating from following parts and install as shown:

(1)	2139555–2	Vacuum Break Port Access
(1)	2139555–3	Trim for Front Skirt
(1)	2139555–4	Front Skirt Wraparound
(1)	2139555–5	Front Skirt Top Plate
(1)	2139555–6	Front Skirt Corner Trim

9–6–5 INSTALL BRIDGE

9–6–5–A Bridge Assembly Preparation Before Installing in Magnet

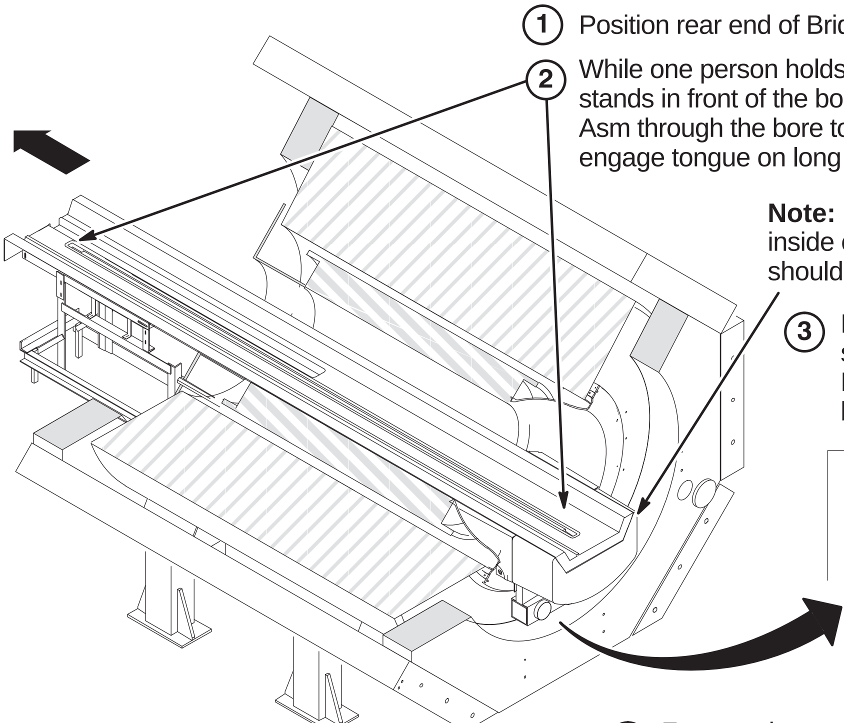


1. Remove Rear Cover and Gasket before installation.
2. Transfer Drive Belt from removed Bridge to new Bridge (2143731–2).
3. Route Drive Belt through Front Pulley Asm and temporarily tape to other end of Bridge. **Note:** Flag may remain on Drive Belt.

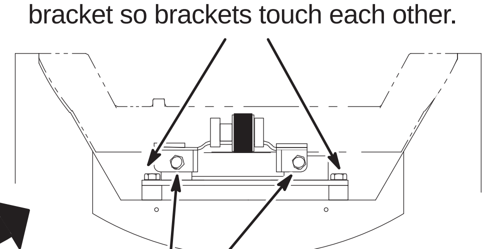
9–6–5–B Securing Front of Bridge

1. Position rear end of Bridge Asm at front of Body Coil Bore.
2. While one person holds holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.

Note: Bridge should be resting on Rubber Strips inside of Body Coil and front surface of Bridge should be flush with front surface of End Bell.

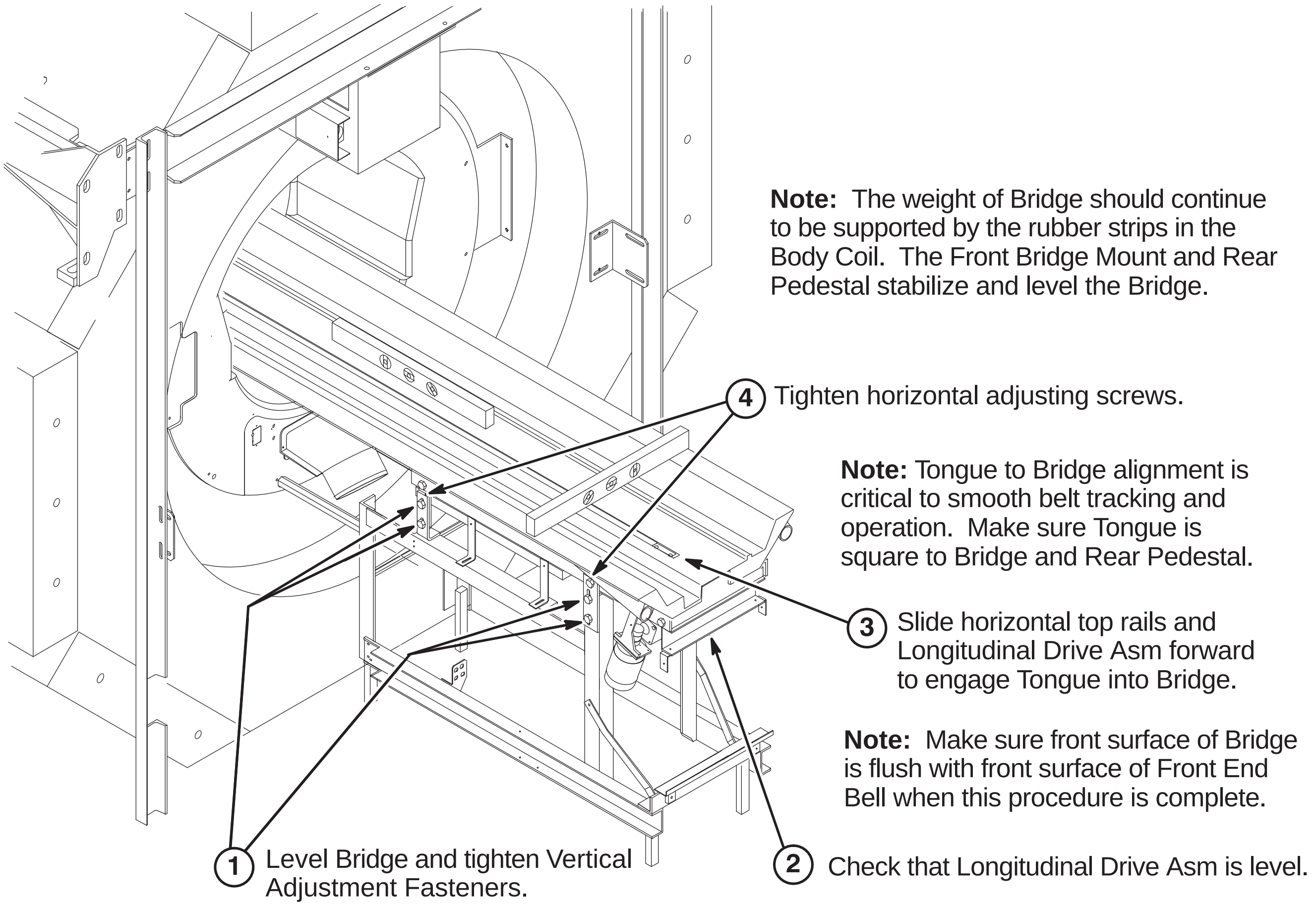


3. Bracket attached to bottom of Bridge should be touching bracket attached to End Bell. If needed, adjust End Bell bracket so brackets touch each other.

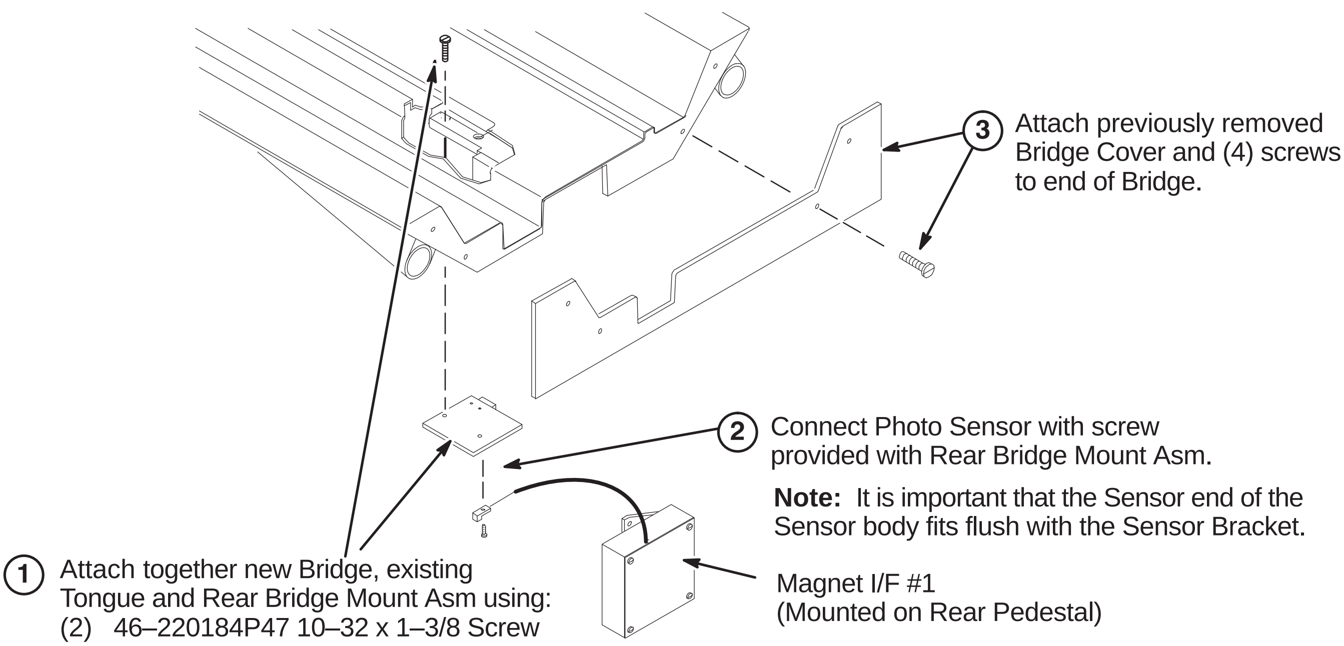


4. From underneath the End Bell extension fasten front end of Bridge Asm to Front End Bell using:
(2) 2109878–4 M10 SST Flat Washer
(2) 2109880–4 M10 SST Lock Washer
(2) 2109866–20 M10 x 25 SST Hex Screw

9–6–5–C Leveling Bridge

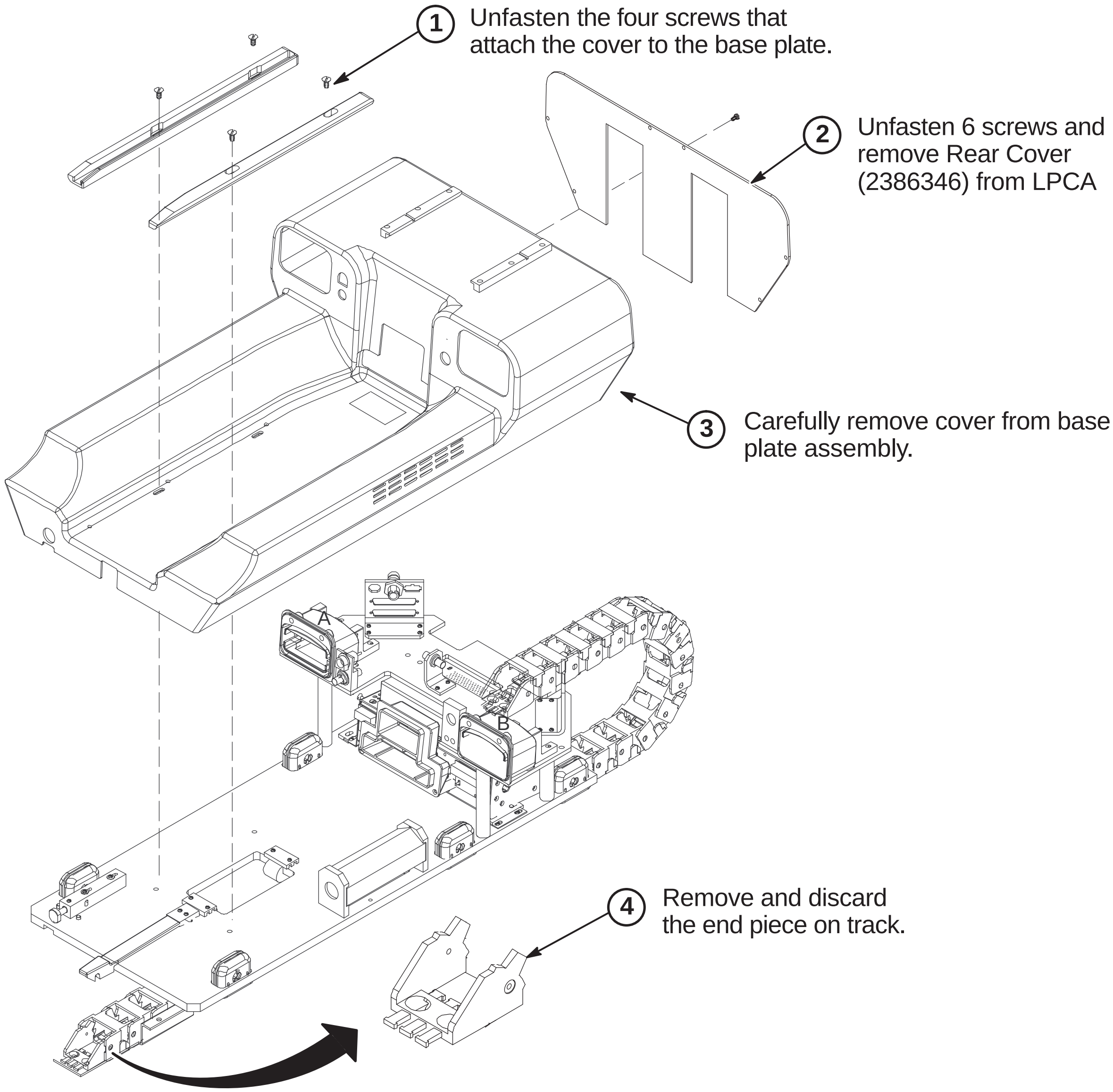


9–6–5–D Connect Photo Sensor & Bridge End Cover

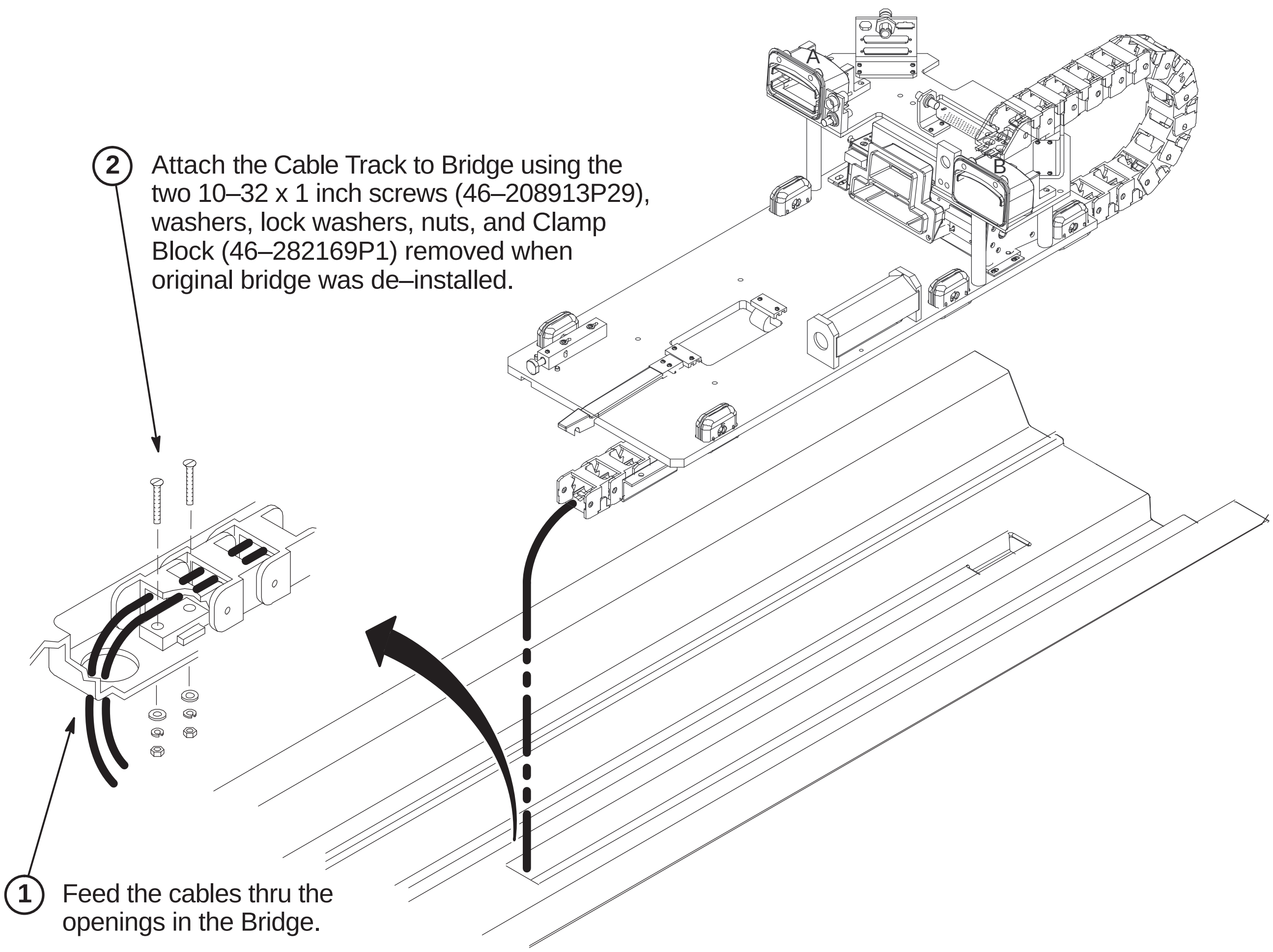


9–6–6 INSTALL LPCA AND CABLE TRACK

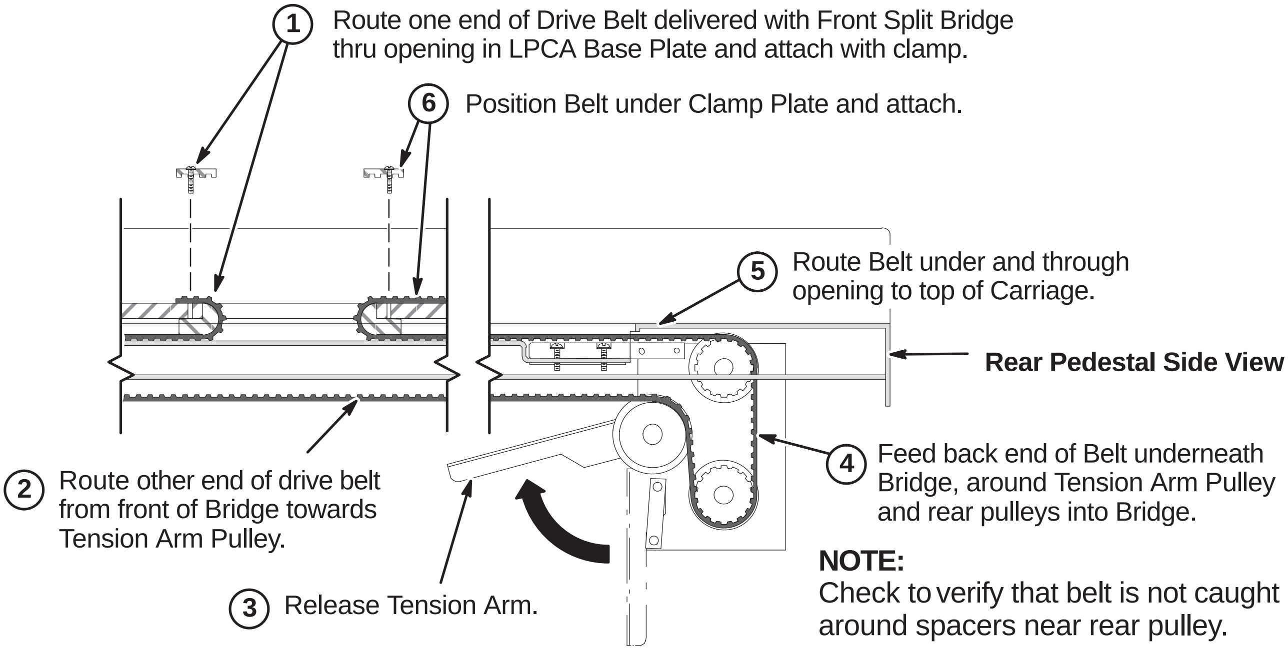
9–6–6–A Remove LPCA Cover



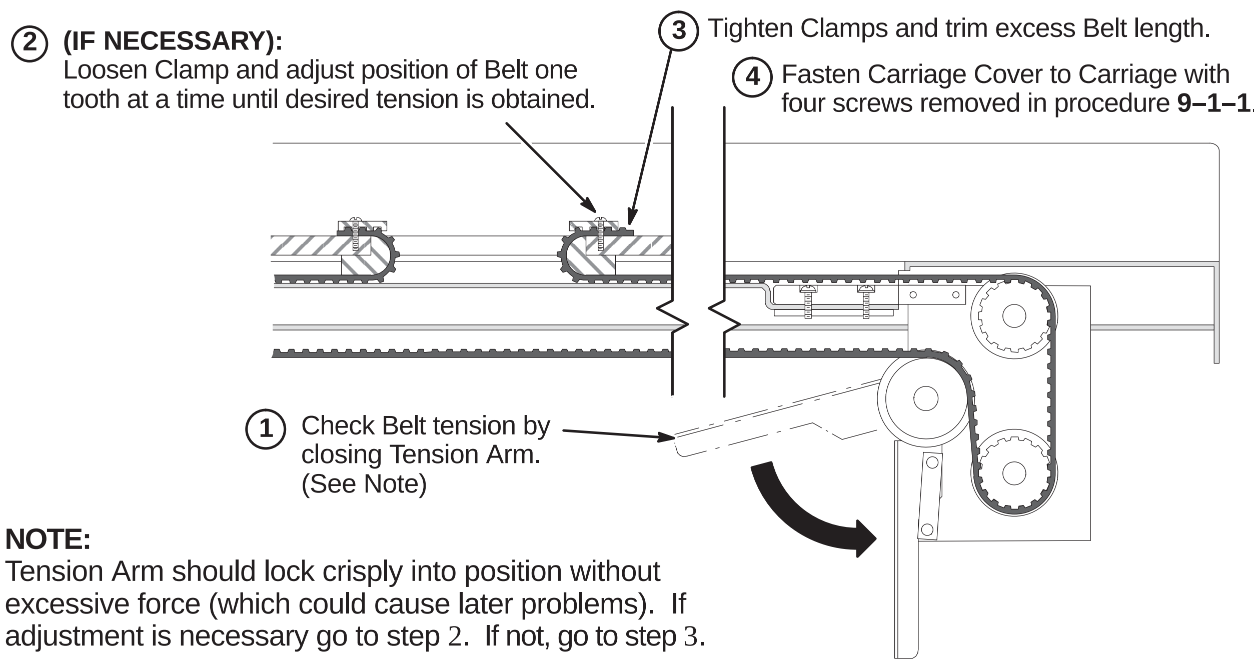
9–6–6–B Attach Cable Track to Bridge



9–6–6–C Route and Attach Drive Belt



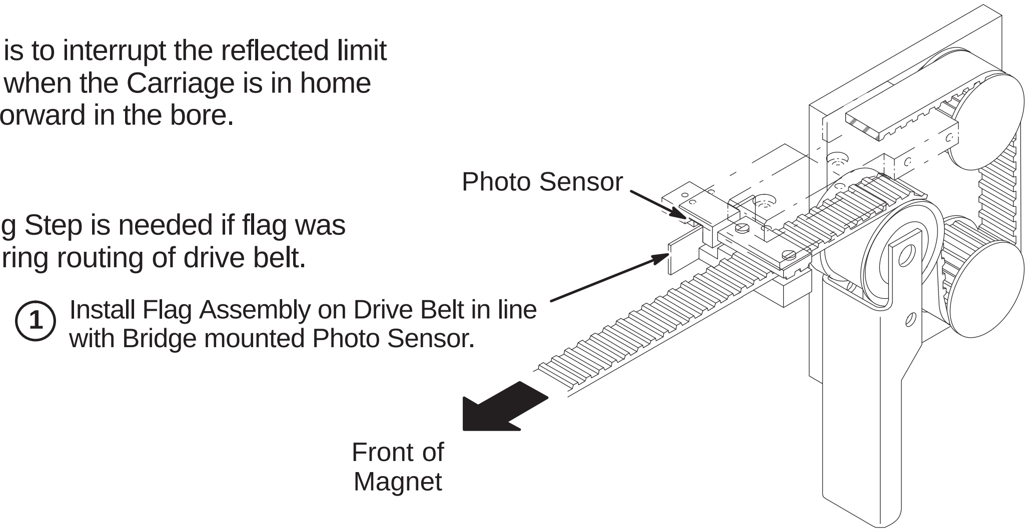
9–6–6–D Final Adjustment of Drive Belt



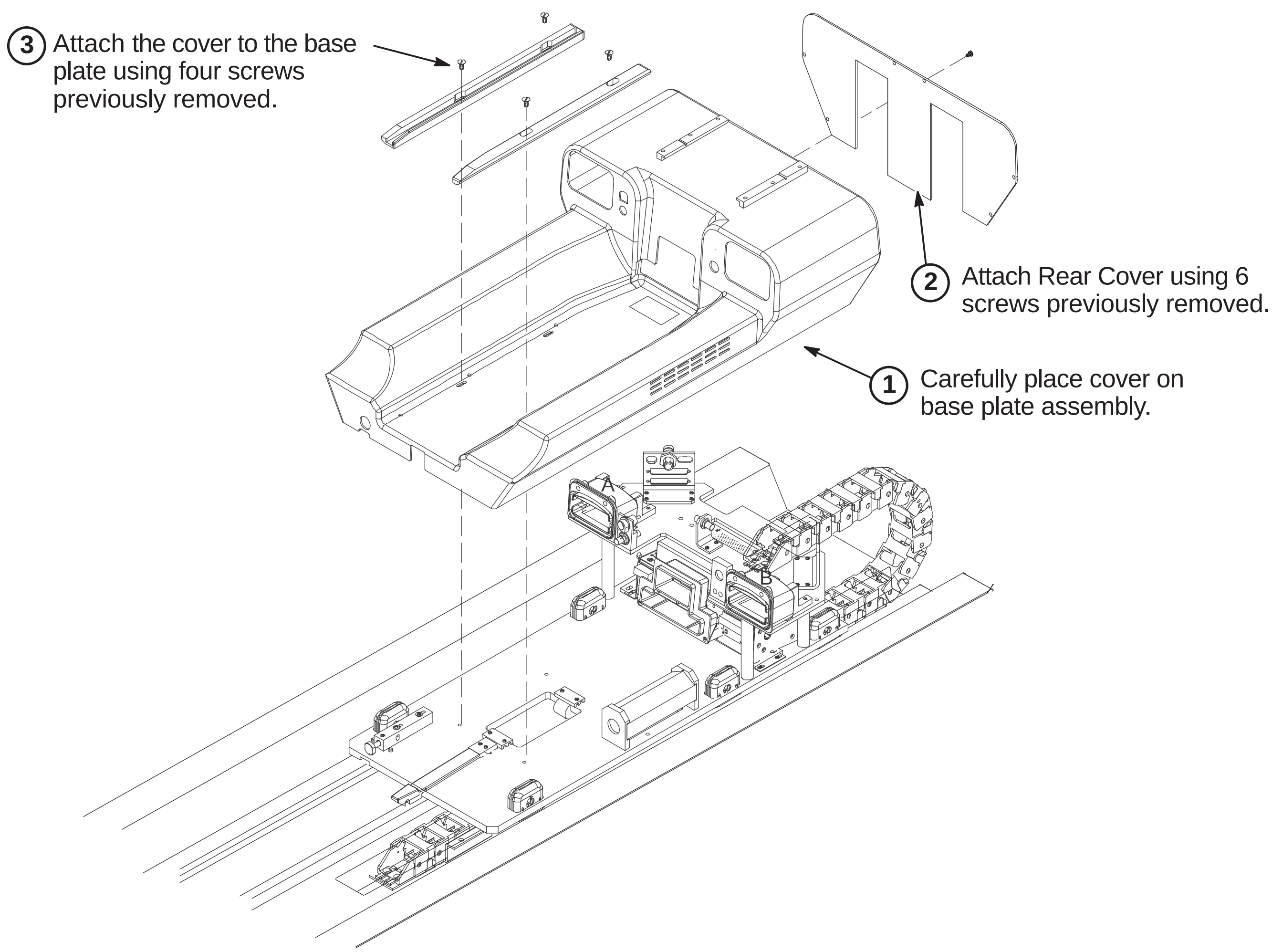
9–6–6–E Installation of Sensor Flag

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

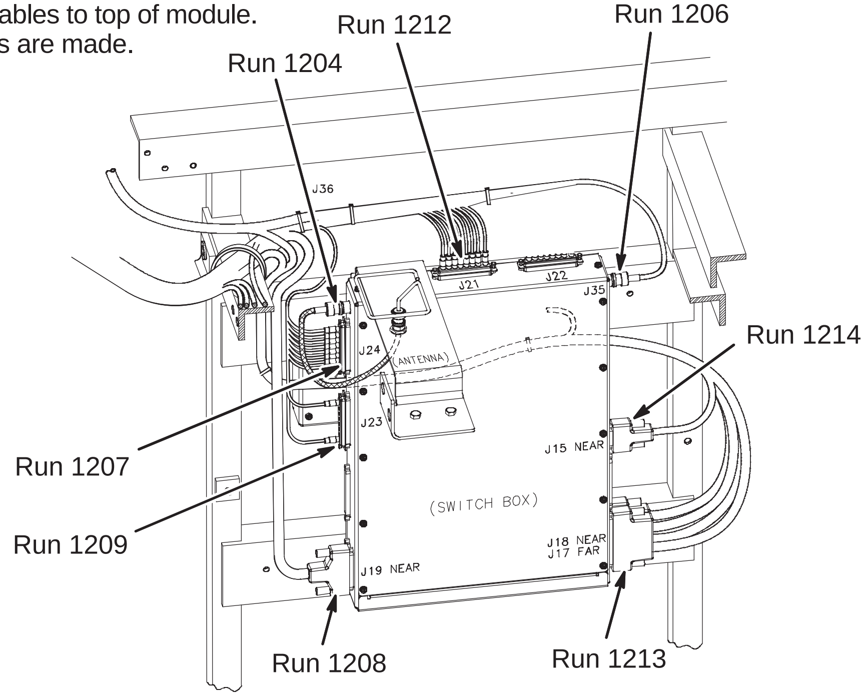


9–6–6–F Installation of LPCA Cover



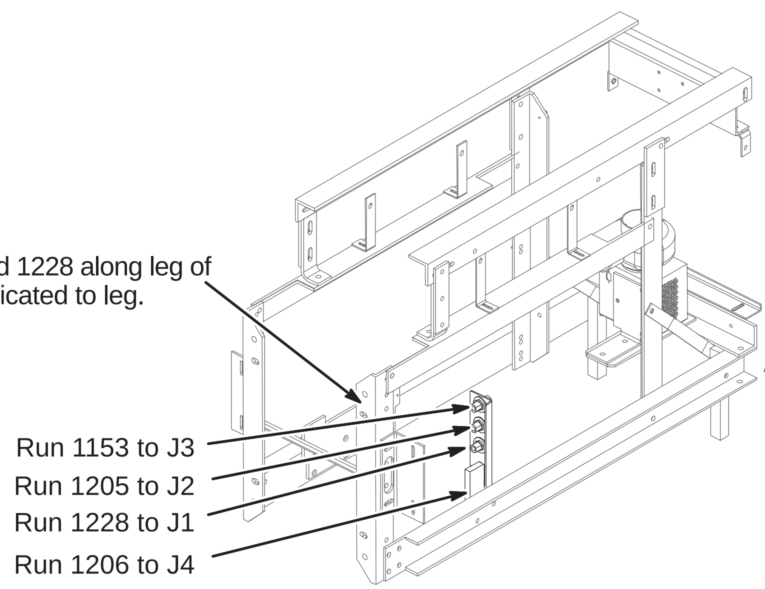
9–6–6–G Connect Cables from Cable Track to 16 Channel Switch Module

1. 16 Channnel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re–mount module after connections are made.
2. Connect cable Runs as indicated.



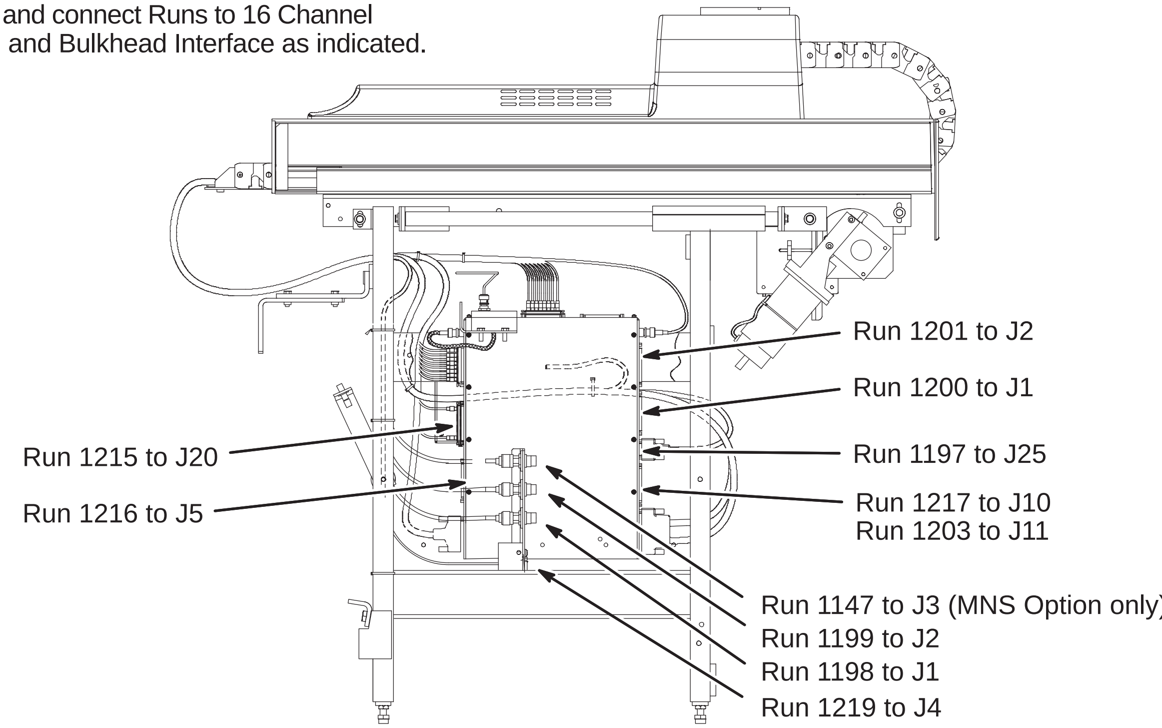
9–6–6–H Connect Cables from Cable Track to Bulkhead Interface

1. Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty–wrap as indicated to leg.
2. Connect Runs to Bulkhead Interface as indicated.



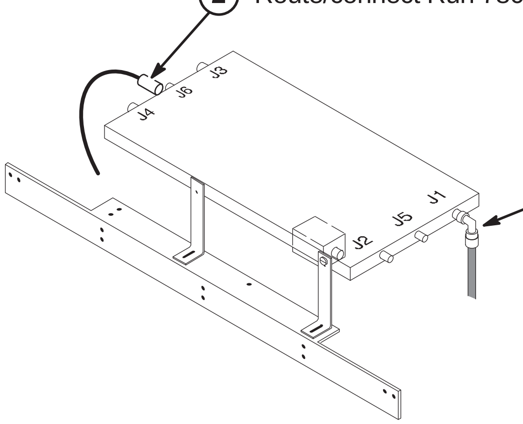
9–6–7 CONNECT REMAINING CABLES TO REAR PEDESTAL

1. Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

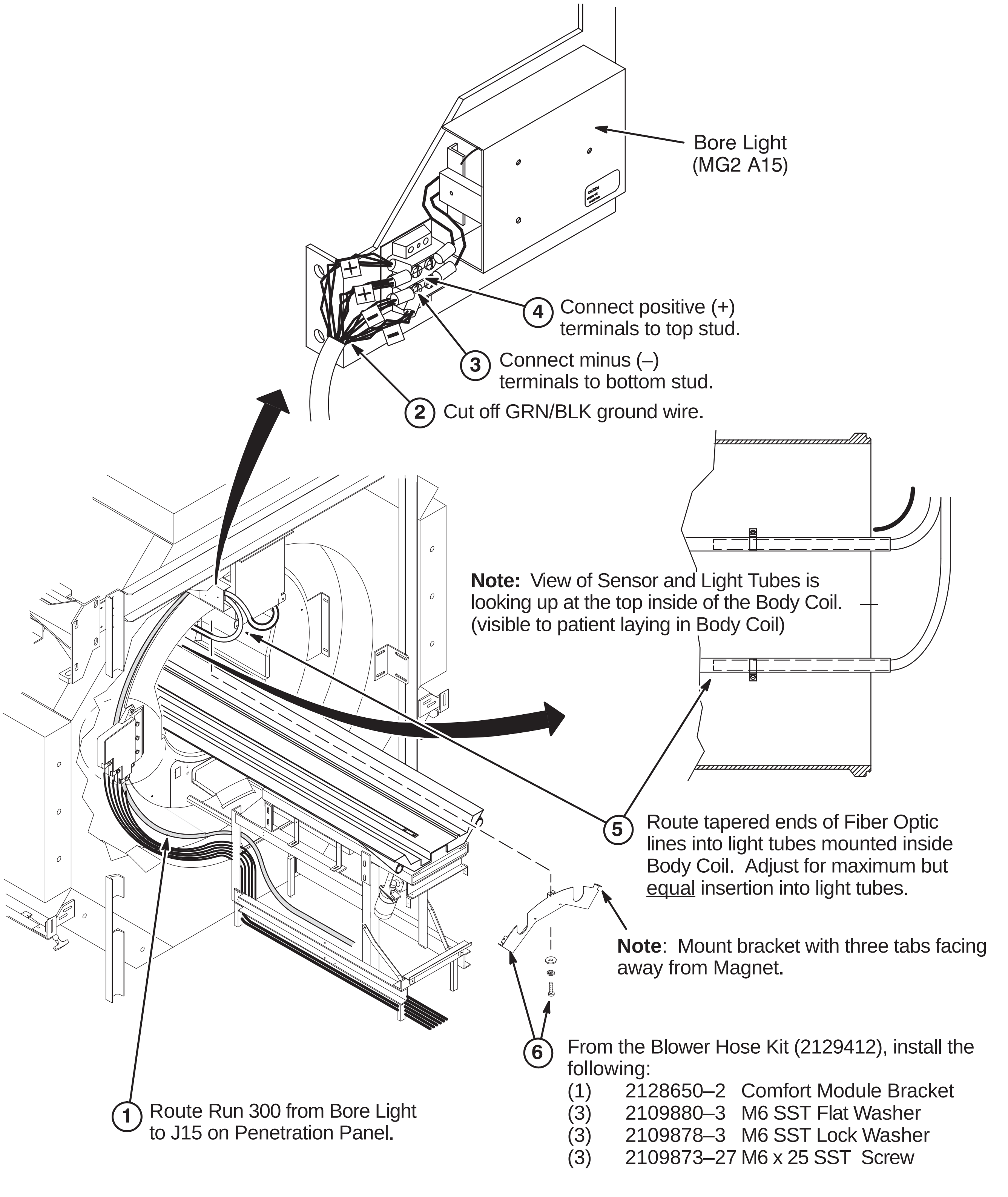


2. Route/connect Run 780 to J6 on RF Splitter.

3. Connect Run 1250 to J1 on RF Splitter:

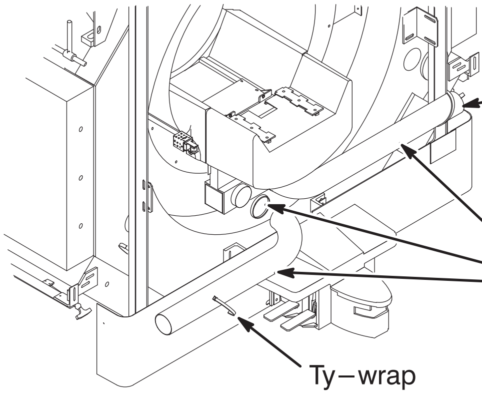


9–6–8 ROUTE AND CONNECT RUN 300 AND BORE LIGHT FIBER OPTIC LINES



9–6–9 INSTALL PATIENT COMFORT AIR SYSTEM

9–6–9–A Connecting Front Air Plenum Hoses (2129412)



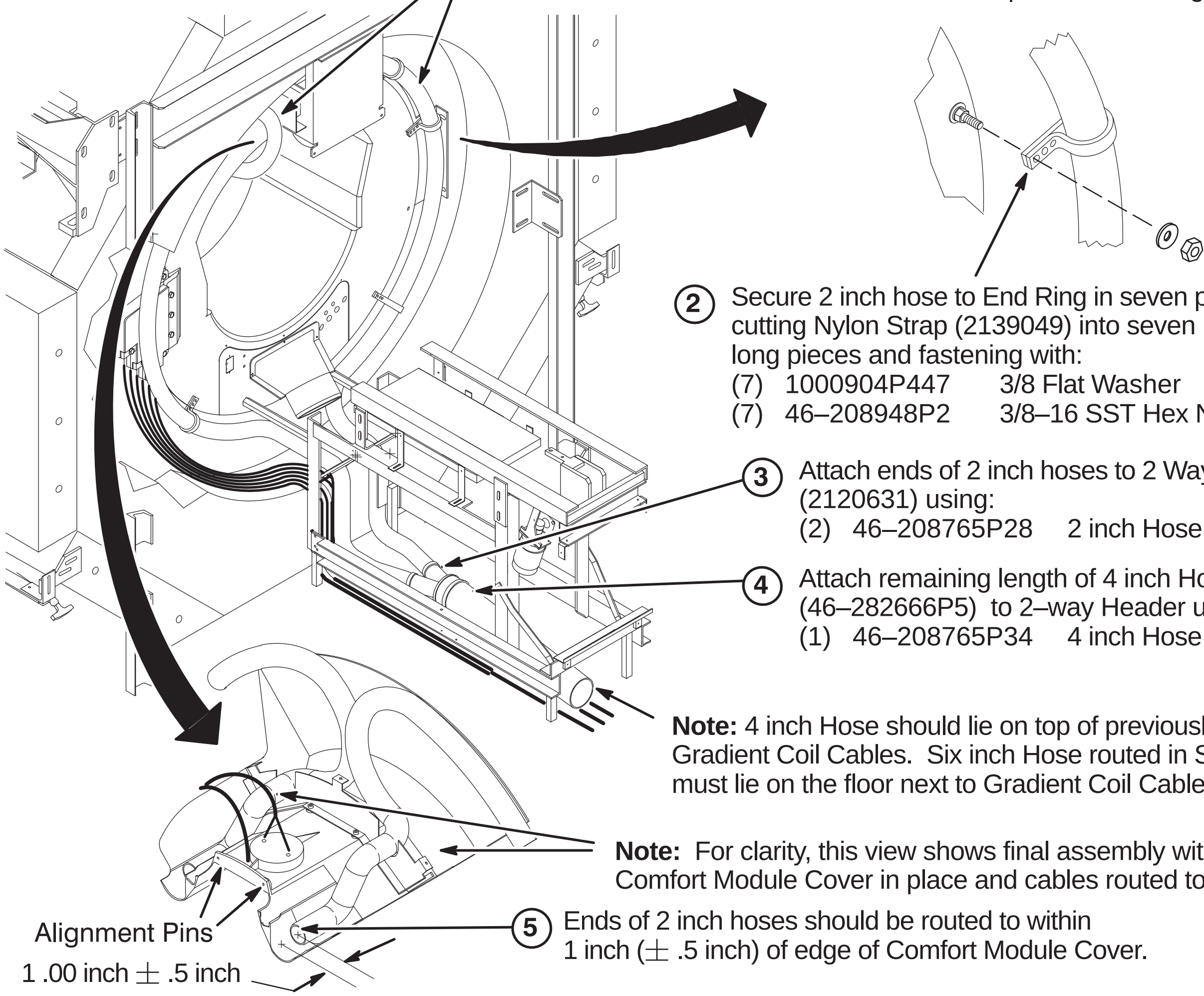
2 Ty–wrap each hose to Vertical Uprights on each side.

Note: All the air hoses and clamps in the following steps are located in the Blower Hose Kit (2129412).

1 Cut two (2) 38 inch lengths of 4 inch Hose (46–282666P5) and attach one length to each side of Front Air Plenum using:
(2) 46–208765P34 4 inch Hose Clamp

9–6–9–B Routing/Connection of Patient Comfort Air Hoses

1 Route the (2) 2 inch diameter hoses (46–282668P18) as shown from below the Rear Pedestal, around and to the top of the End Ring.



2 Secure 2 inch hose to End Ring in seven places by cutting Nylon Strap (2139049) into seven 12 inch long pieces and fastening with:
(7) 1000904P447 3/8 Flat Washer
(7) 46–208948P2 3/8–16 SST Hex Nut

3 Attach ends of 2 inch hoses to 2 Way Header (2120631) using:
(2) 46–208765P28 2 inch Hose Clamps

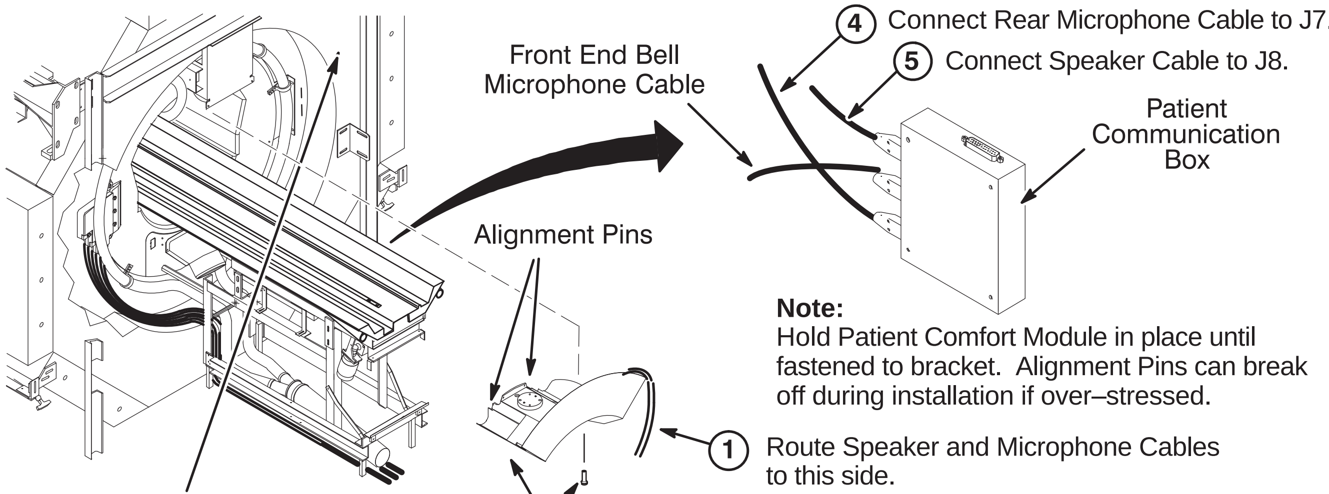
4 Attach remaining length of 4 inch Hose (46–282666P5) to 2–way Header using:
(1) 46–208765P34 4 inch Hose Clamp

Note: 4 inch Hose should lie on top of previously routed Gradient Coil Cables. Six inch Hose routed in Step K4 must lie on the floor next to Gradient Coil Cables.

Note: For clarity, this view shows final assembly with Comfort Module Cover in place and cables routed to side.

5 Ends of 2 inch hoses should be routed to within 1 inch (± .5 inch) of edge of Comfort Module Cover.

9–6–9–C Install Patient Comfort Module and Cables



3 Route Speaker and Microphone Cables along side 2 Inch Air Hose to Patient Communication Box (MG3 A2) on side of Rear Pedestal.

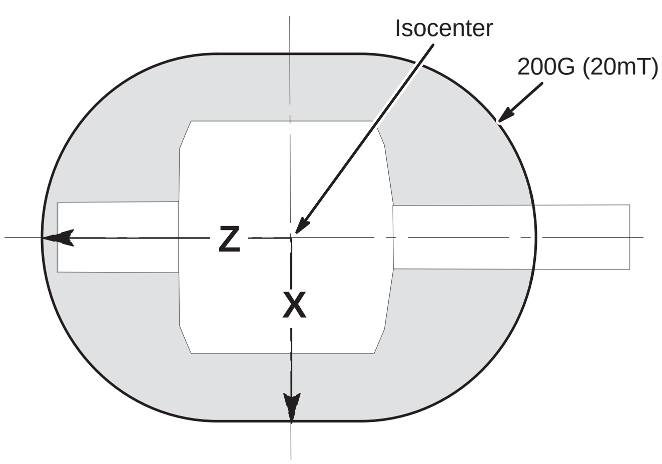
2 Install Patient Comfort Module (2128650) by aligning pins on module with two holes in Rear End Bell and attach to Comfort Module Bracket using:
(2) 2109873–27 M6 x 25 SST Pan Head Screw

9–6–9–D 200 Gauss Zone Limit for Movement of Blower Box

DANGER!!

THE BLOWER BOX IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.

Illustration is not to scale



Approximate Distance from isocenter to 200G Line		
	X	Z
1.5T	150 cm (59 in)[1]	205 cm (81 in)[1]
1.0T & Lower	145 cm (57 in)[1]	195 cm (77 in)[1]
Unshielded Magnet	Measure [2]	

Note 1:
These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed.

Note 2:
For unshielded magnets, the magnetic field must be measured using a gauss meter.

9–6–9–E Installation of Blower Box

WARNING!

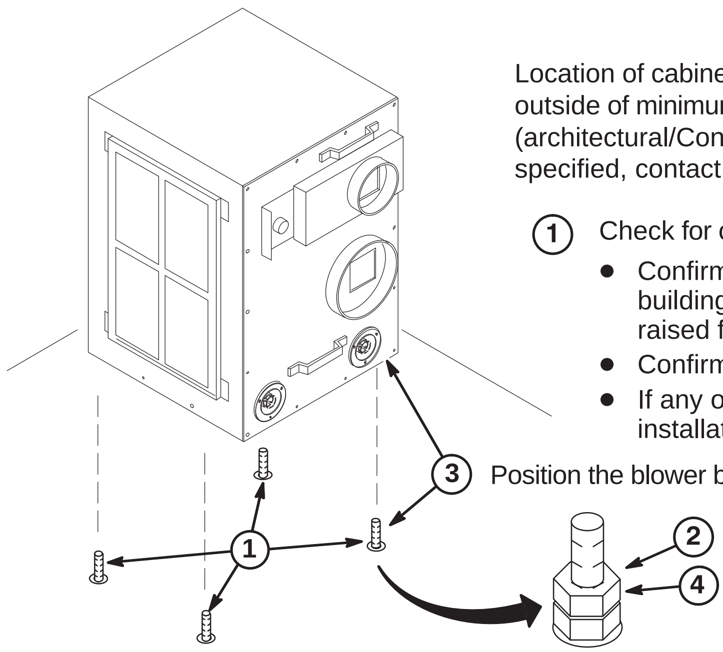
RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).



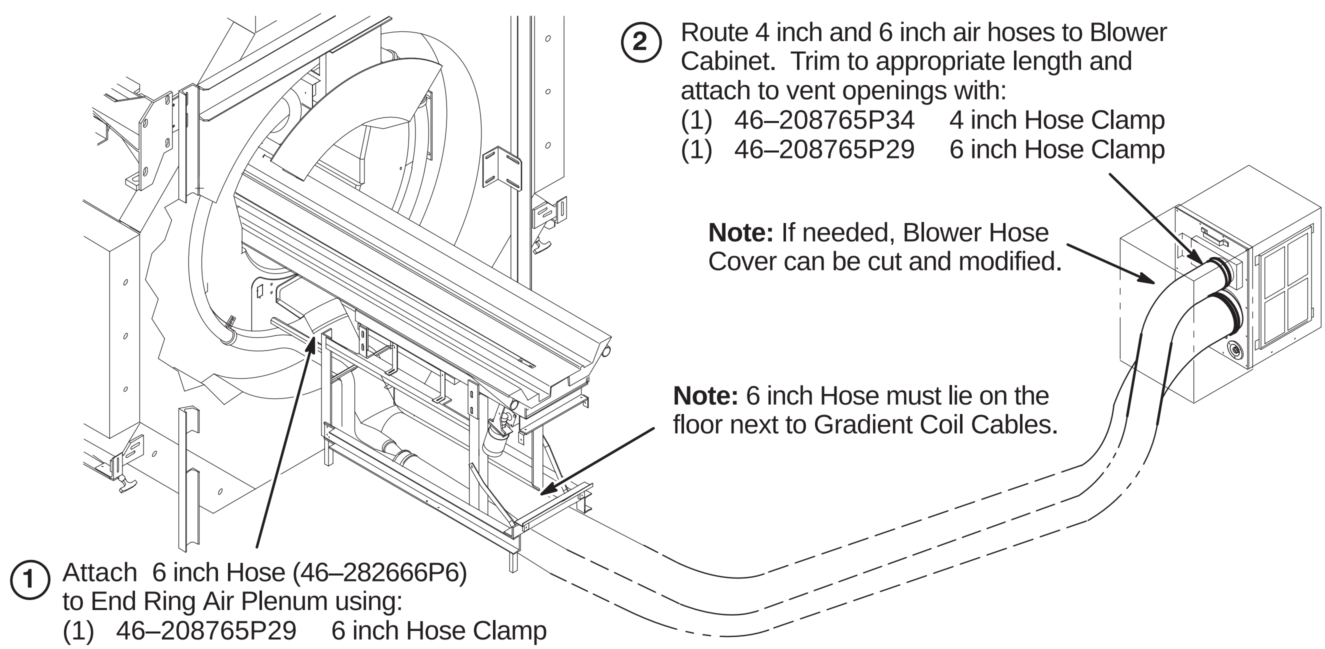
1 Check for correct installation of the anchors:
- Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
- Confirm there are at least four (4) anchors.
- If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.

3 Position the blower box on top of anchors.

2 Remove any mounting hardware from the anchors.

4 Secure the blower box using the hardware provided by the RF shield room vendor to attach.

9–6–9–F Install Blower Box Air Hoses



2 Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:
(1) 46–208765P34 4 inch Hose Clamp
(1) 46–208765P29 6 inch Hose Clamp

Note: If needed, Blower Hose Cover can be cut and modified.

Note: 6 inch Hose must lie on the floor next to Gradient Coil Cables.

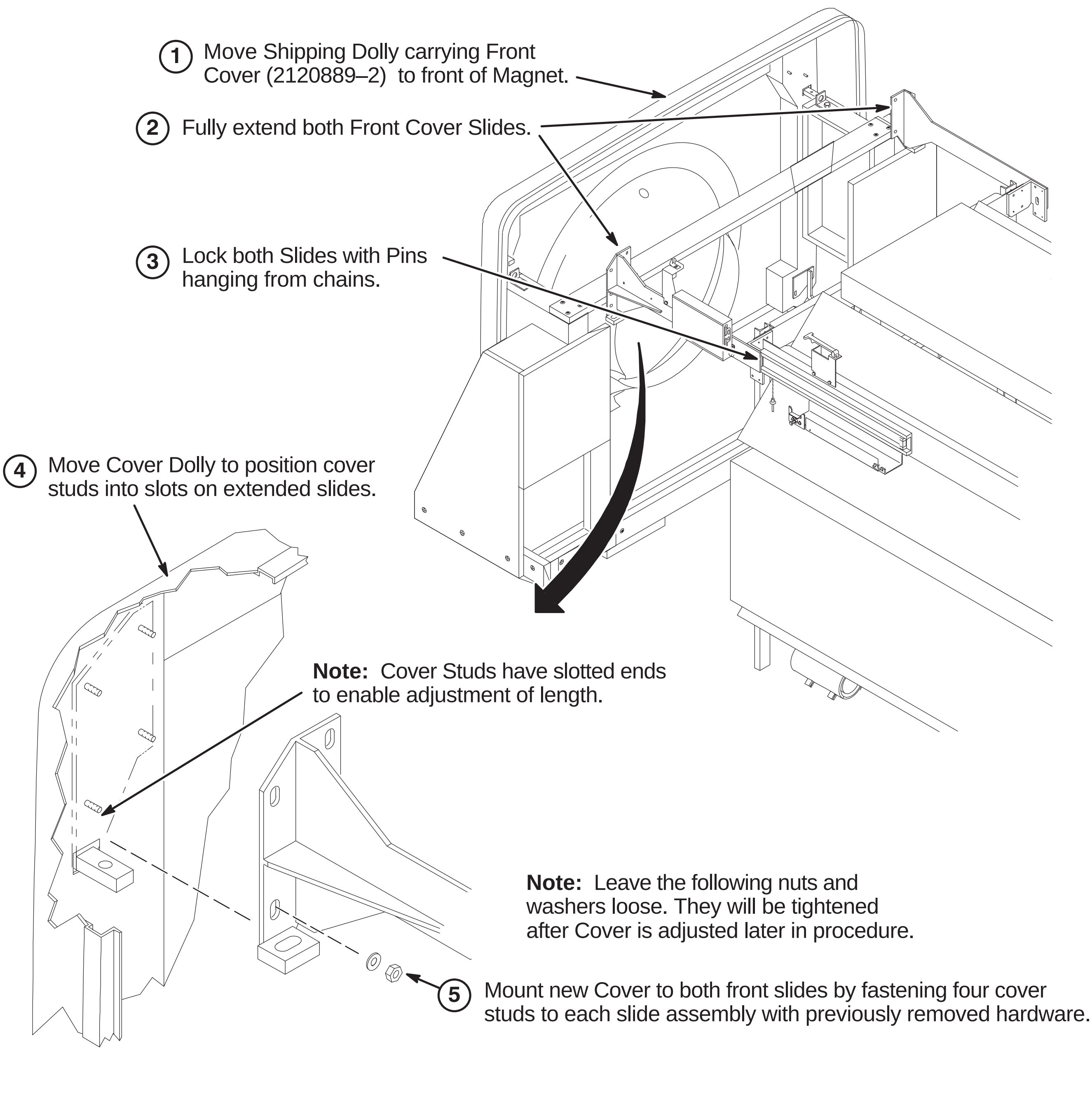
1 Attach 6 inch Hose (46–282666P6) to End Ring Air Plenum using:
(1) 46–208765P29 6 inch Hose Clamp

9–6–10 INSTALL FRONT COVER

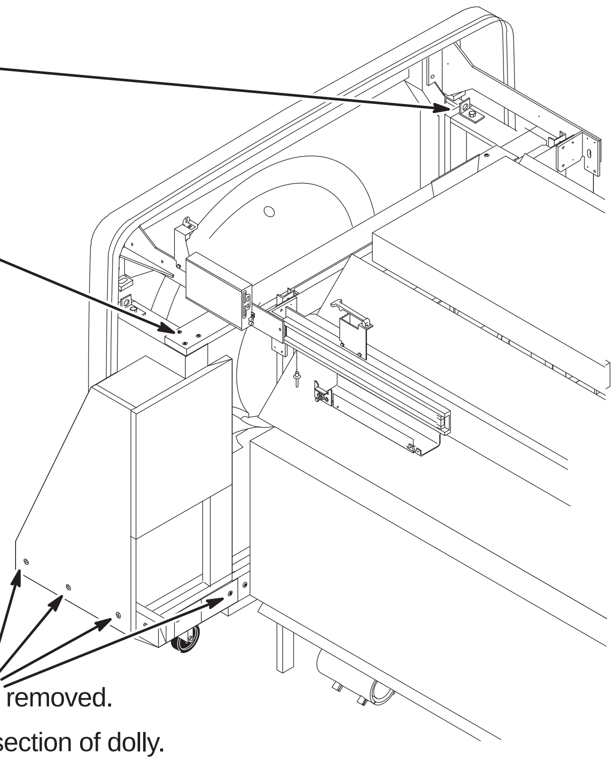
9–6–10–A Mount Front Cover

DANGER!!

FERROUS MATERIAL HAZARD! THE COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE FRONT COVER FROM DOLLY AND CARRY COVER INTO POSITION AT FRONT OF MAGNET.



9–6–10–B Disassemble Cover Dolly



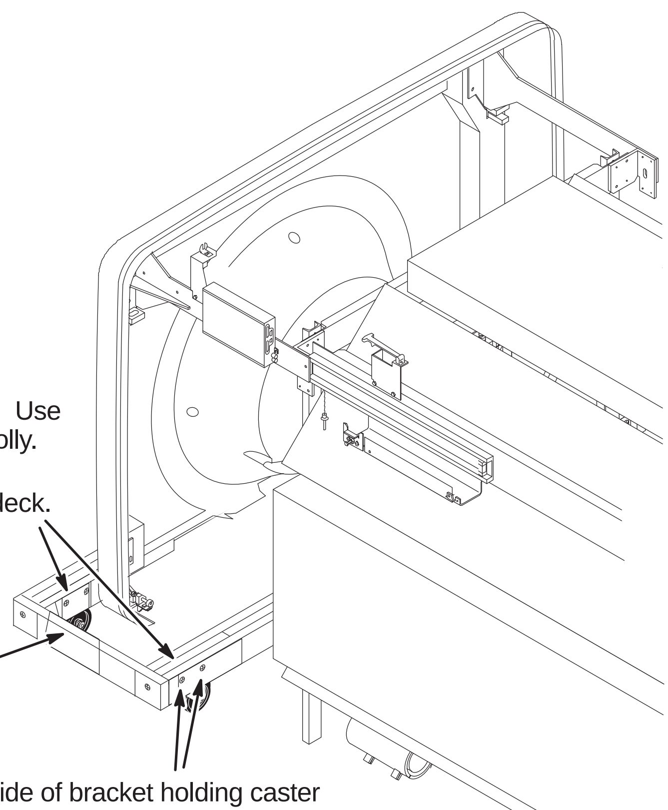
1. Remove bolts from each side of dolly that were used for attaching cover to dolly.

2. Remove three screws from each end of cross brace and remove brace.

3. Remove screws holding dolly side section. Side section will slide down when last bolt is removed.

   Repeat above for removal of opposite side section of dolly.

9–6–10–C Remove Cover Dolly



6. Place remaining dolly parts on recycle pile.

5. Place the four bracket and caster assemblies in pre–addressed box provided with dolly for shipment.

4. With casters removed, slide frame away from magnet. Use care not to damage floor when sliding and removing dolly.

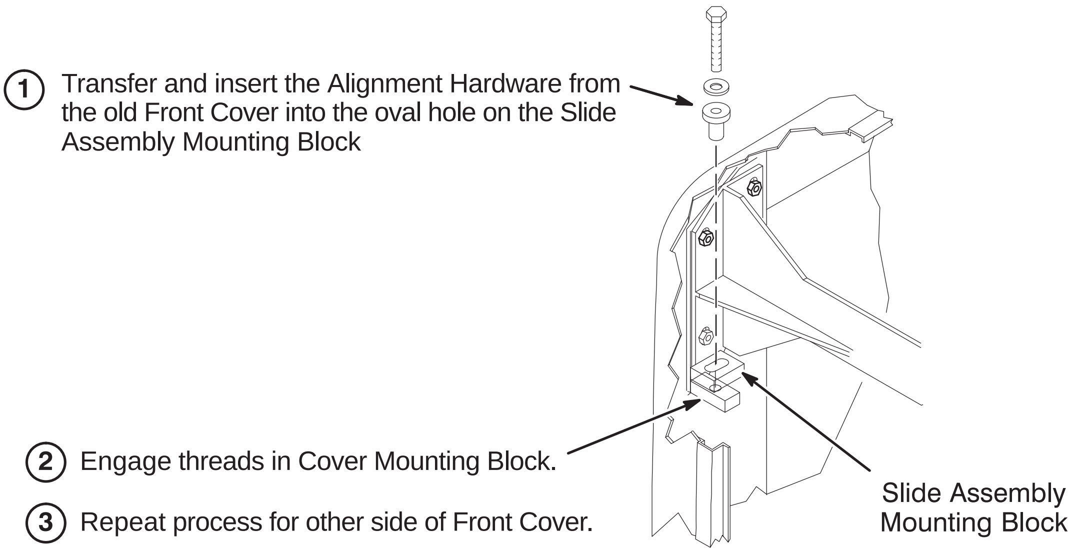
3. With end lifted, remove caster assemblies from dolly deck.

   Repeat process for other end of dolly deck.

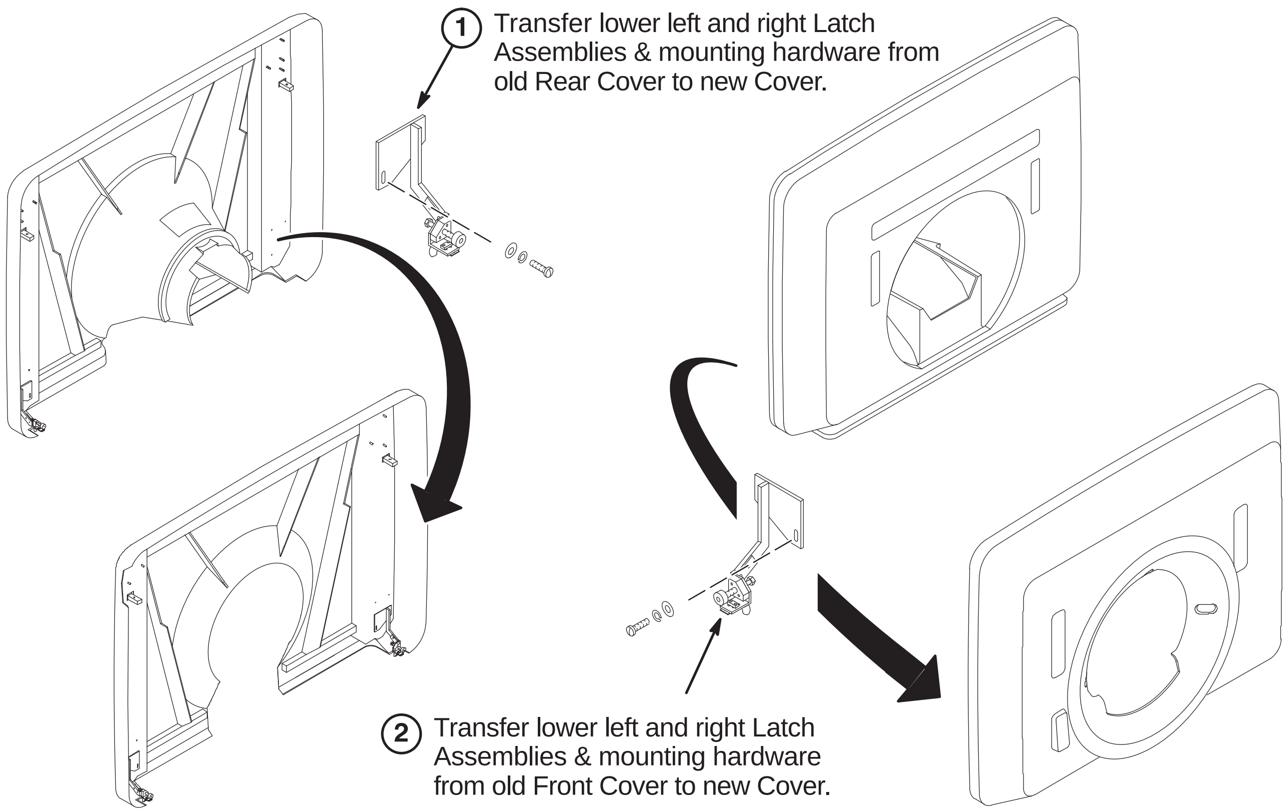
2. Lift this end of dolly deck.

1. Remove screws from each side of bracket holding caster assembly to dolly deck. Repeat process for other three corners.

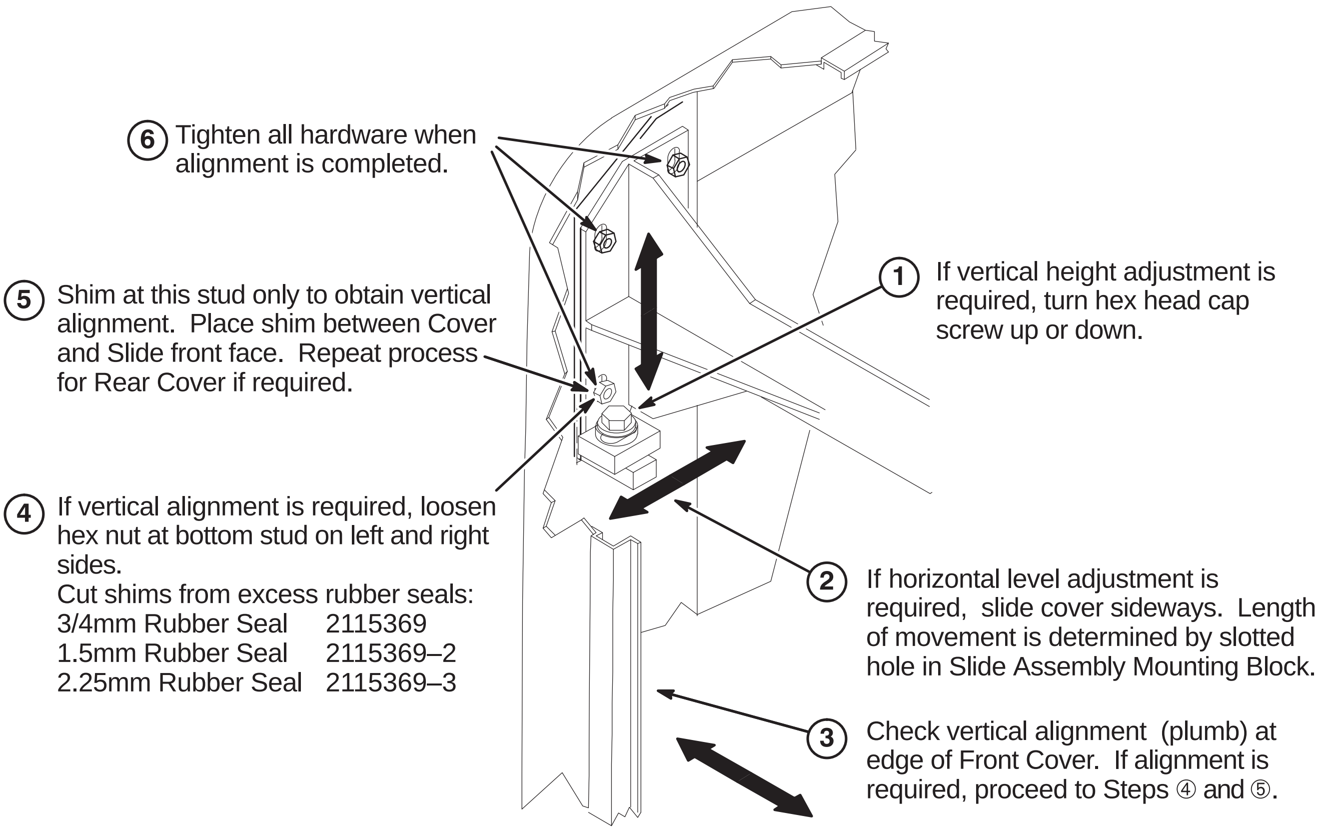
9–6–10–D Install Front Cover Alignment Hardware



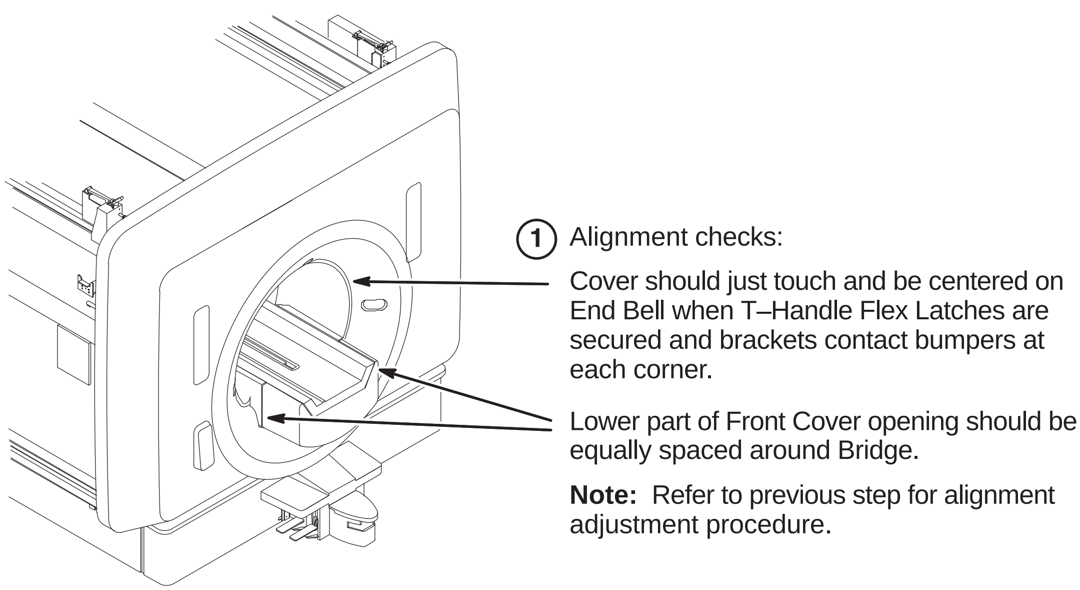
9–6–10–E Transfer Lower Latch Assemblies to New Front & Rear Covers



9–6–10–F Adjust Front Cover Alignment



9–6–10–G Align Front Cover



9–6–11 INSTALL REAR COVER

DANGER!!

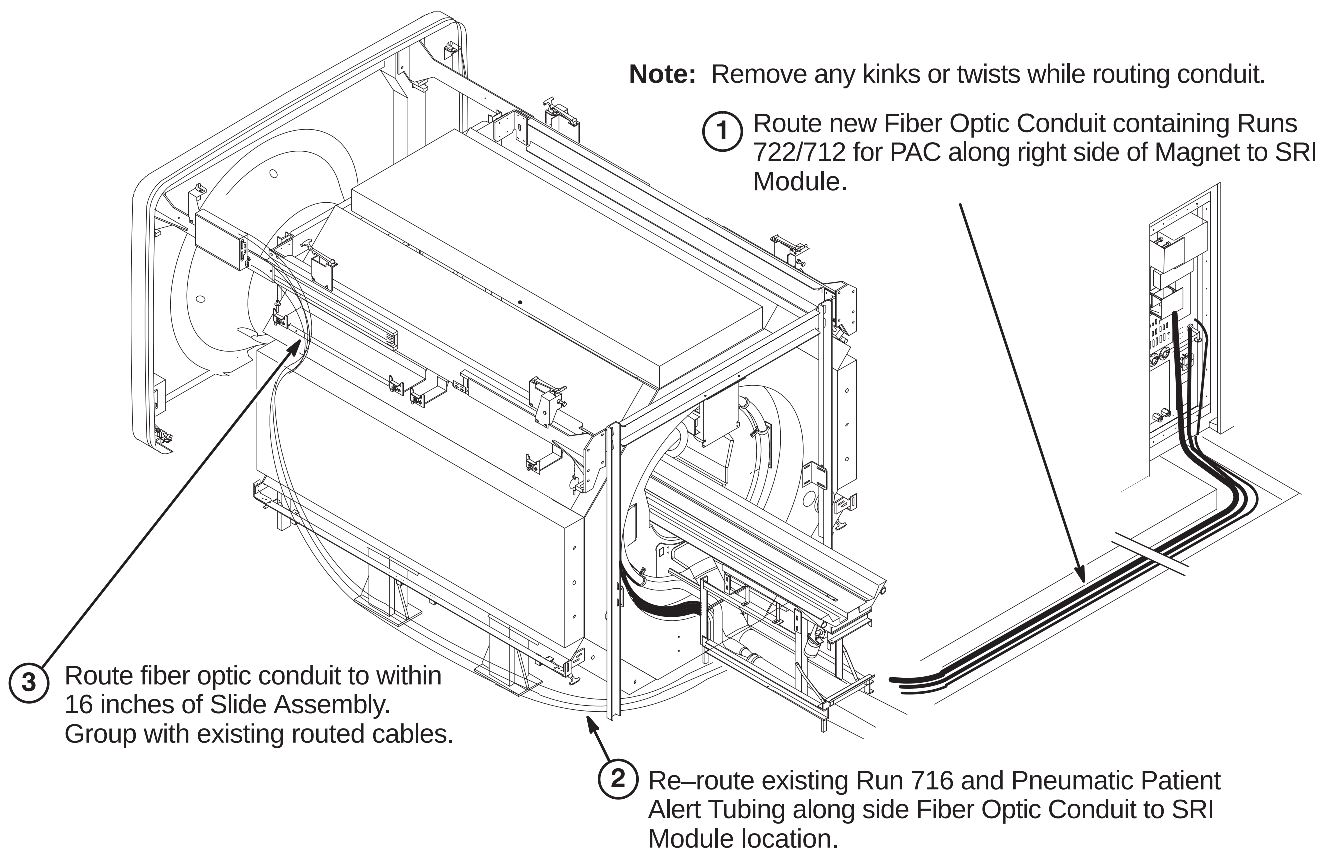
FERROUS MATERIAL HAZARD! THE REAR COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE COVER FROM DOLLY AND CARRY COVER INTO POSITION AT REAR OF MAGNET.

THE STEPS ARE THE SAME AS THE JUST COMPLETED FRONT COVER INSTALLATION PROCEDURE
Note: Experience has shown that the Rear Cover does not require shimming for vertical levelness.

9–6–12 ROUTE AND CONNECT RUNS 711/712, 716, AND PATIENT ALERT TUBING

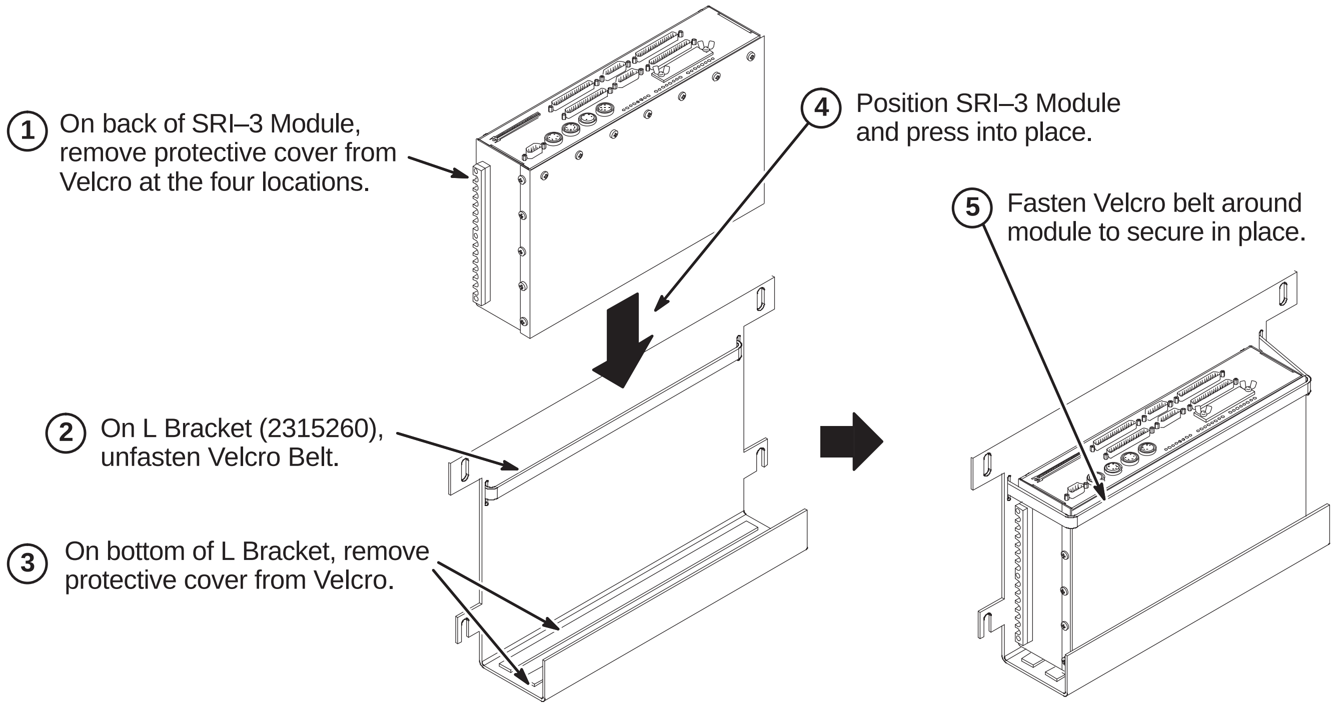
CAUTION

EQUIPMENT DAMAGE POSSIBILITY. Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend to radius smaller than two inches. Do not pull connector from cable. Avoid scratching connector ends. Keep ends protected until ready to connect.

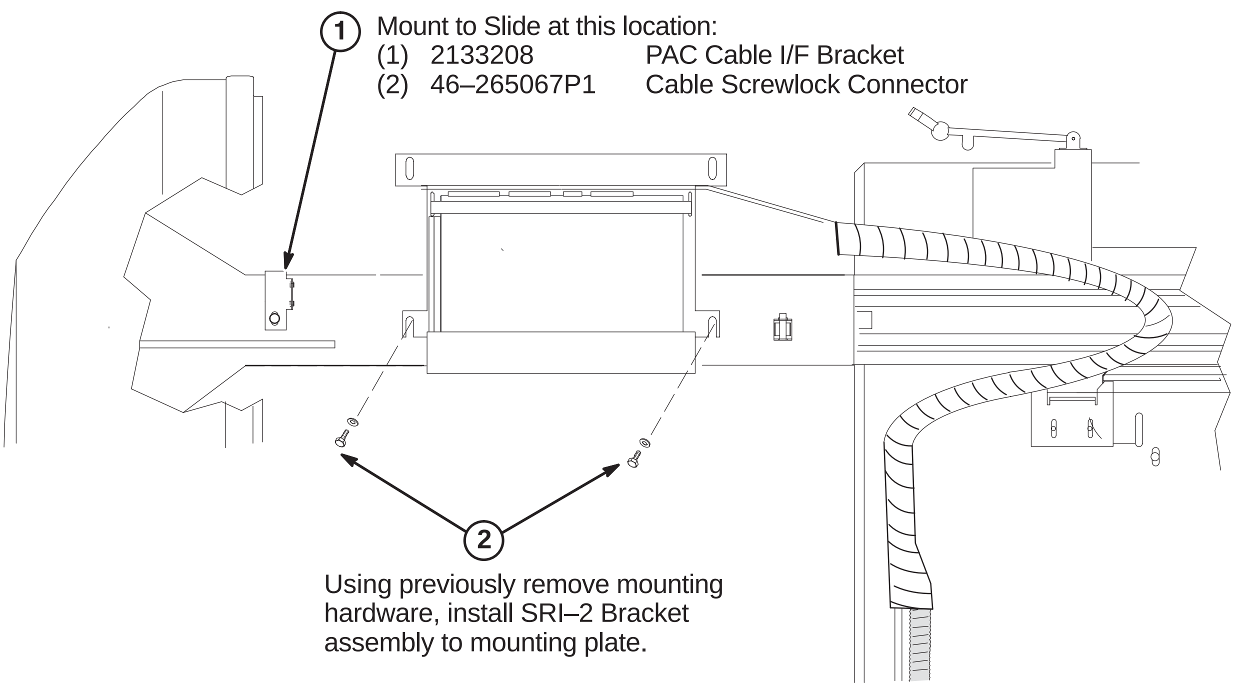


9–6–13 INSTALL SRI–3 MODULE

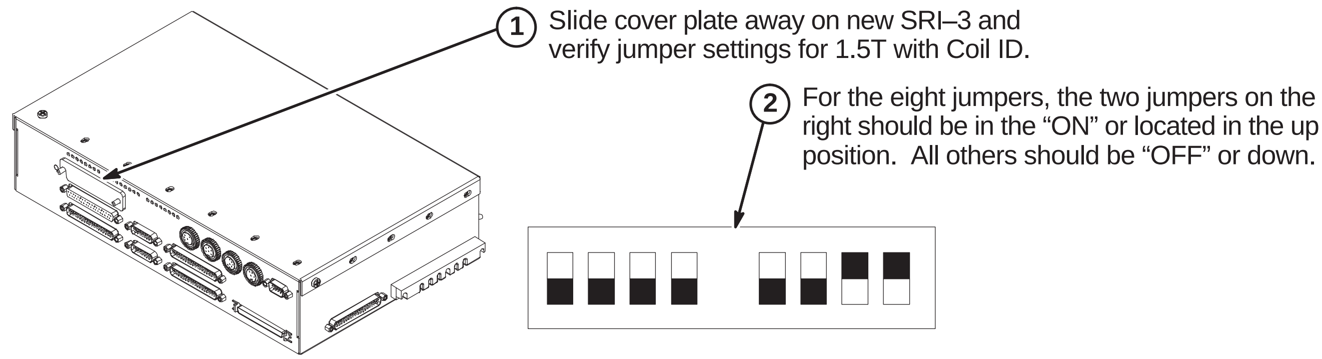
9–6–13–A Position SRI–3 in L Bracket



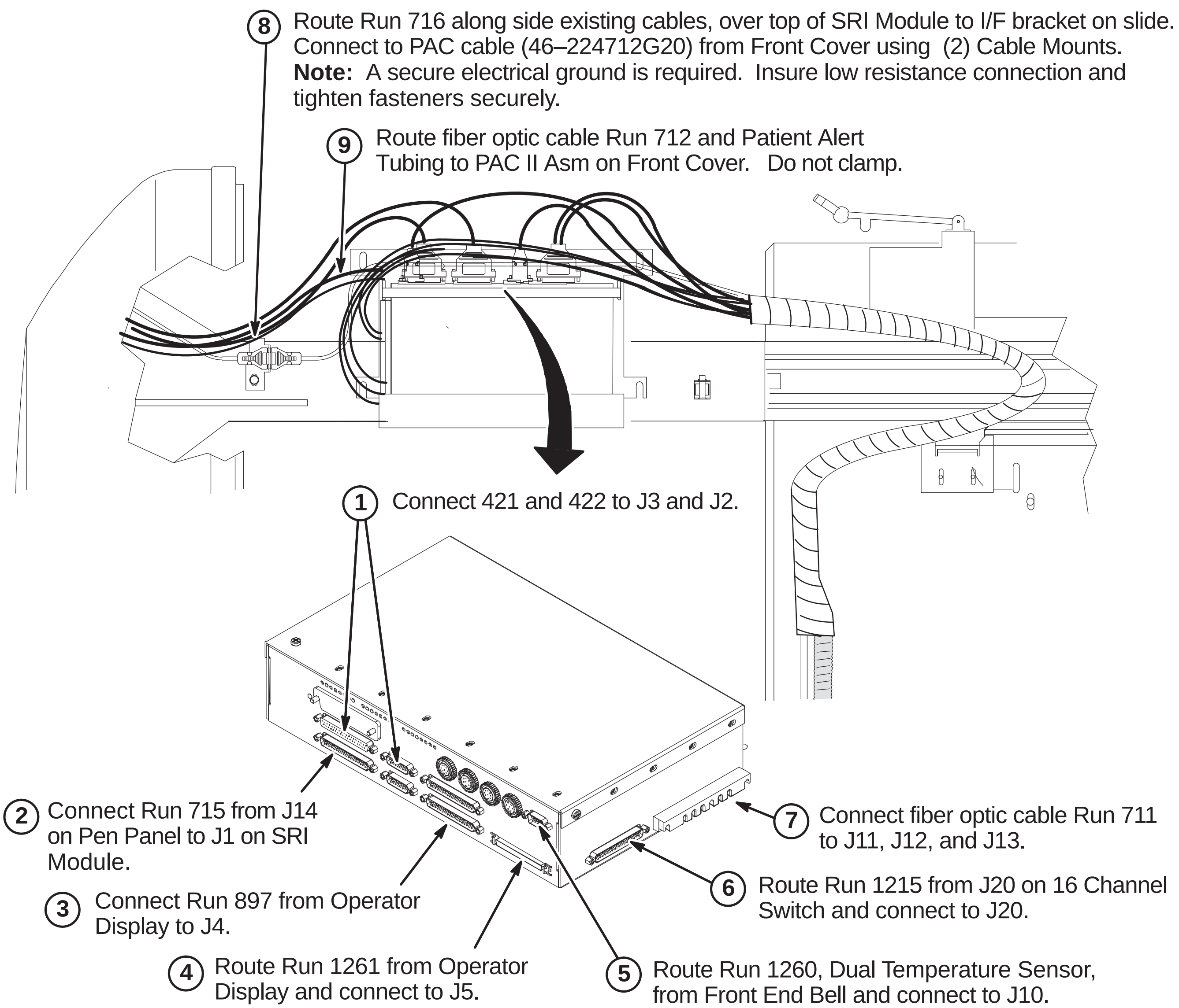
9–6–13–B Install SRI–3 on Magnet



9–6–13–C Verify SRI–3 Jumper Settings

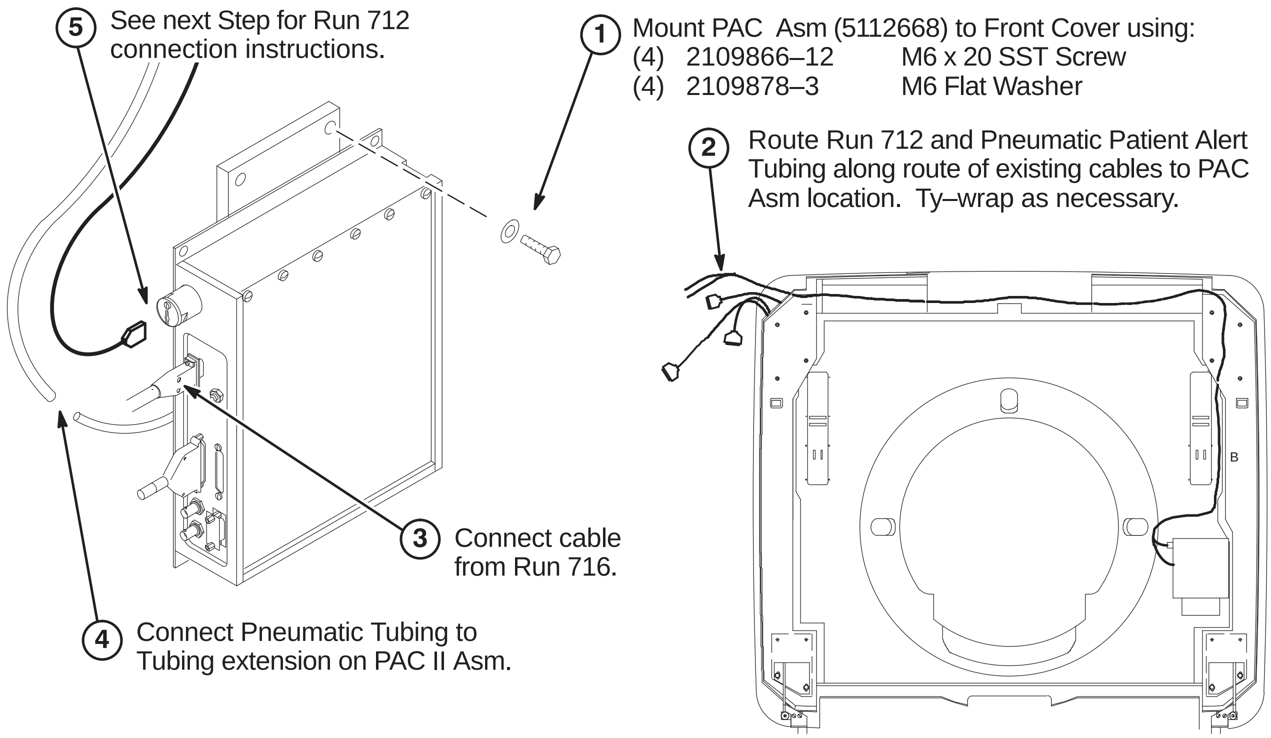


9–6–13–D Connect Cables at SRI–3 Module Location

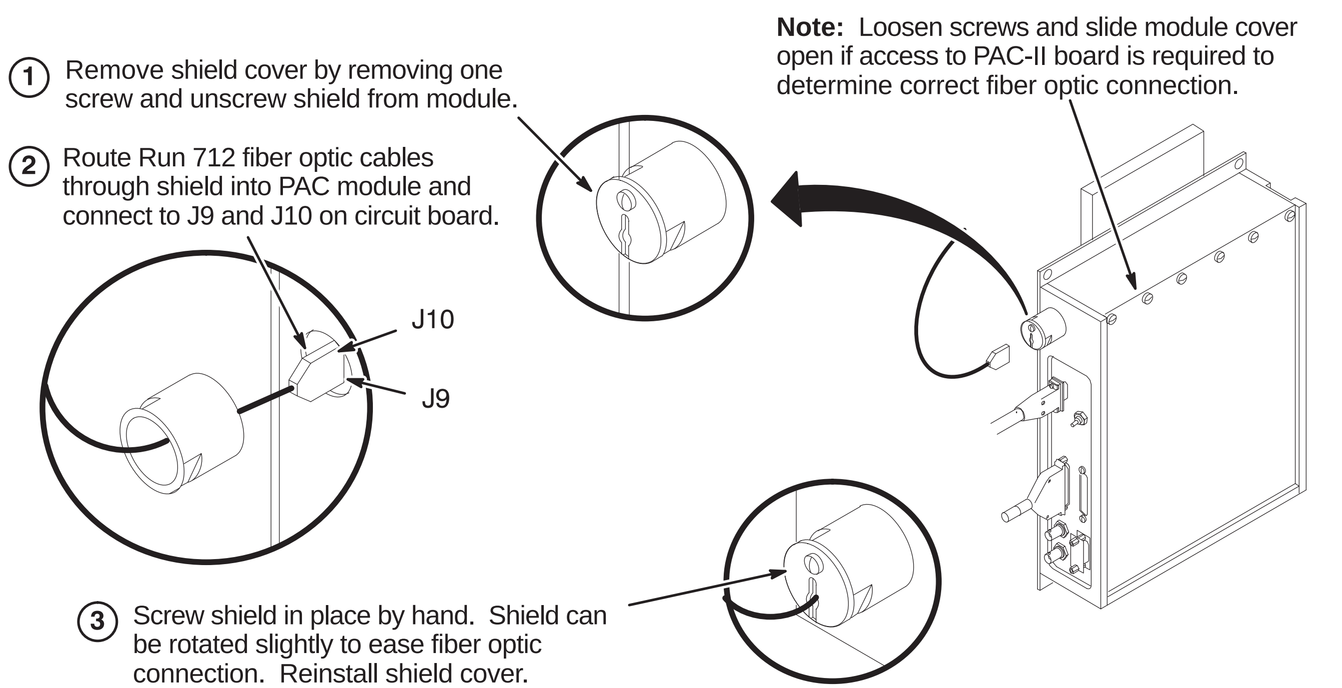


9–6–14 INSTALL NEW PAC MODULE

9–6–14–A Connect Run 712, 716, & Pneumatic Patient Alert Tubing

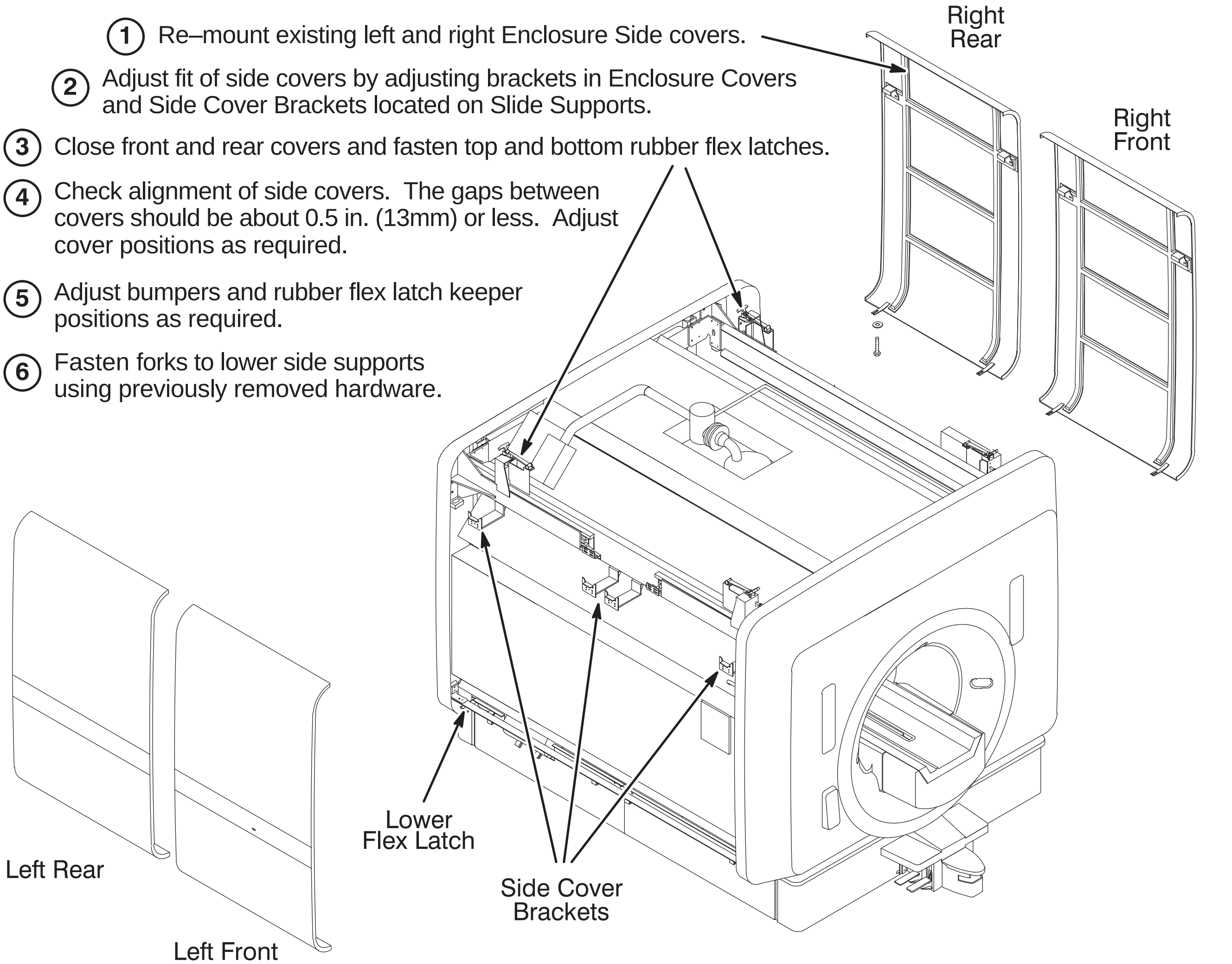


9–6–14–B Connect Run 712

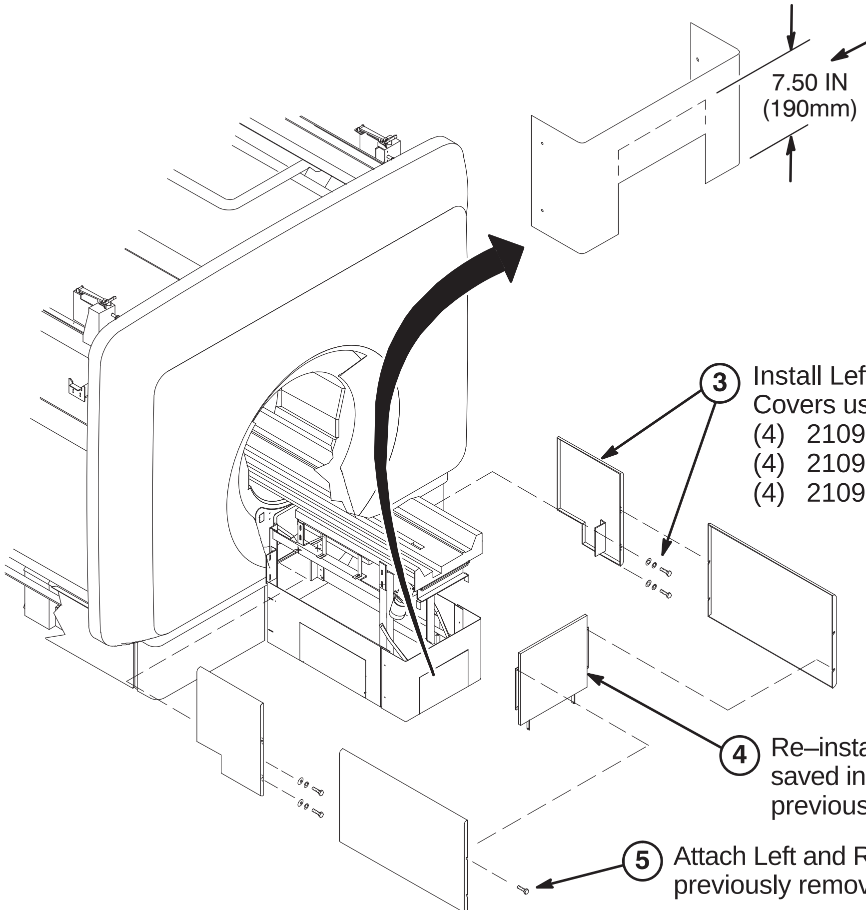


9–6–15 INSTALL ENCLOSURE SIDE COVERS

1. Re–mount existing left and right Enclosure Side covers.
2. Adjust fit of side covers by adjusting brackets in Enclosure Covers and Side Cover Brackets located on Slide Supports.
3. Close front and rear covers and fasten top and bottom rubber flex latches.
4. Check alignment of side covers. The gaps between covers should be about 0.5 in. (13mm) or less. Adjust cover positions as required.
5. Adjust bumpers and rubber flex latch keeper positions as required.
6. Fasten forks to lower side supports using previously removed hardware.



9–6–16 INSTALL REAR PEDESTAL COVERS



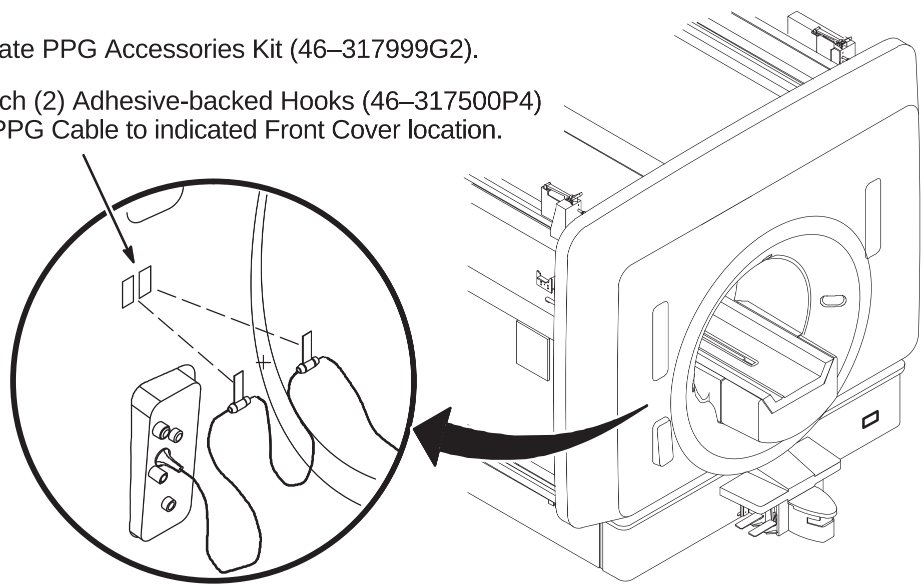
1. Remove Rear Trim Cover from room and cut opening to height indicated. Remove sharp edges.
2. Re–install Base Trim Covers with saved hardware.
3. Install Left (2131044) and Right (2131045) Side Covers using:

(4)	2109878–3	M6 SST Flat Washer
(4)	2109880–3	M6 SST Lock Washer
(4)	2109866–13	M6 x 25 SST Hex Screw

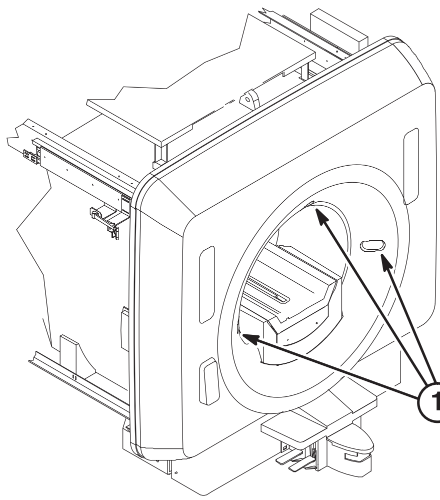
4. Re–install Rear Pedestal Back Cover Asm saved in A1–Step ① to Rear Pedestal with previously removed hardware.
5. Attach Left and Right Rear Pedestal Side Covers with previously removed hardware.
6. Check fit and adjust as required.

9–6–17 INSTALL PPG CABLE HOOKS ON FRONT COVER

1. Locate PPG Accessories Kit (46–317999G2).
2. Attach (2) Adhesive-backed Hooks (46–317500P4) for PPG Cable to indicated Front Cover location.



9–7 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

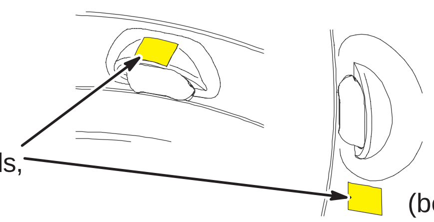


A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

English

1 For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



2 If enclosure does not have labels, attach new labels as indicated.

(both sides)

French

German

Portuguese

Spanish

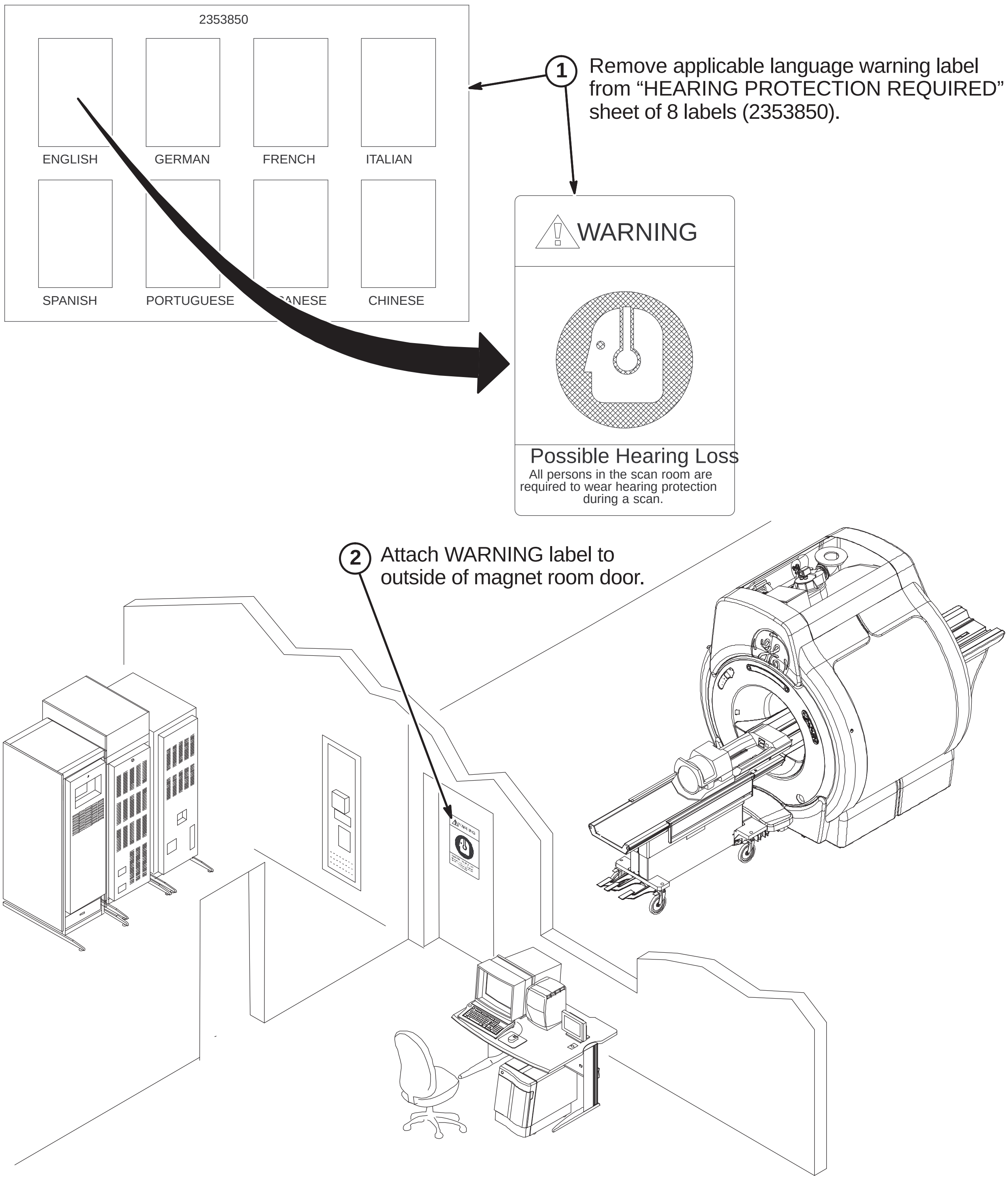
Italian

Swedish

Japanese

Chinese

9–8 INSTALL HEARING LOSS WARNING SIGN



SECTION 10 – DUAL TEMPERATURE SENSOR UPGRADE

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10–1 INTRODUCTION

The Thermal Sensor Upgrade improves bore temperature sensing reliability by adding a second temperature sensor.

The following illustrations show an S3 magnet with a Horizon style enclosure. The instructions apply to any magnet enclosure. If needed, refer to the applicable magnet enclosure upgrade procedure in this manual for additional information on removal and installation of enclosure covers, bridge, and end bell for the installation on the new dual sensor cable, Run 1260.

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!

10–2 TOOLS AND EQUIPMENT

The following tools and equipment are required to perform this procedure:

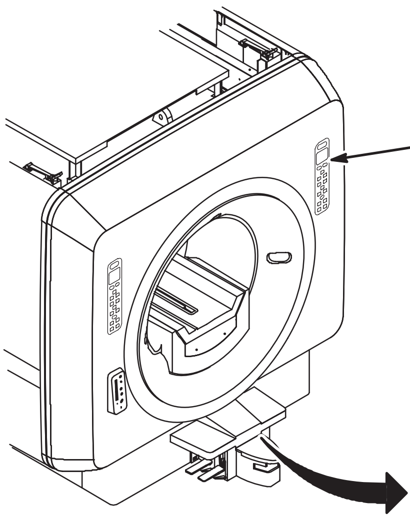
- 3mm Allen Wrench
- Drill bit, 7 mm or 5/16 inch
- Drill
- Dykes
- Digital Multimeter
- Lockout/Tagout equipment
- Tape, electrical or duct
- Permanent marker, fine–tip
- Slot–head screwdriver, 5/16” blade
- Nut driver, 5/16”
- Static–grounding wrist strap
- Copper tape (for replacement only)

The following items are provided in the Thermal Sensor Upgrade Kit, P/N 2287391

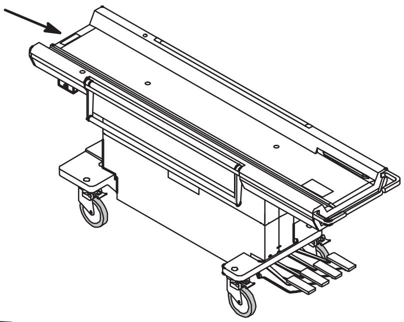
- 46–208747P1 Cable ties
- 46–282264P1 Epoxy adhesive
- 2298033 Applicator
- 46–320187P1 Nozzle

10–3 UPGRADE FOR HORIZON STYLE MAGNET ENCLOSURE

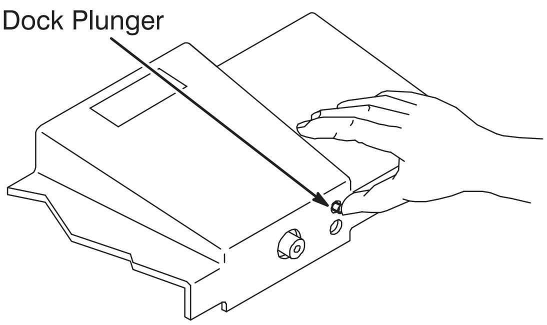
10–3–1 TABLE REMOVAL AND CARRIAGE POSITIONING



1. Undock Patient Transport. Move Patient Transport out of the way to allow enough room for removal of the magnet end bell.

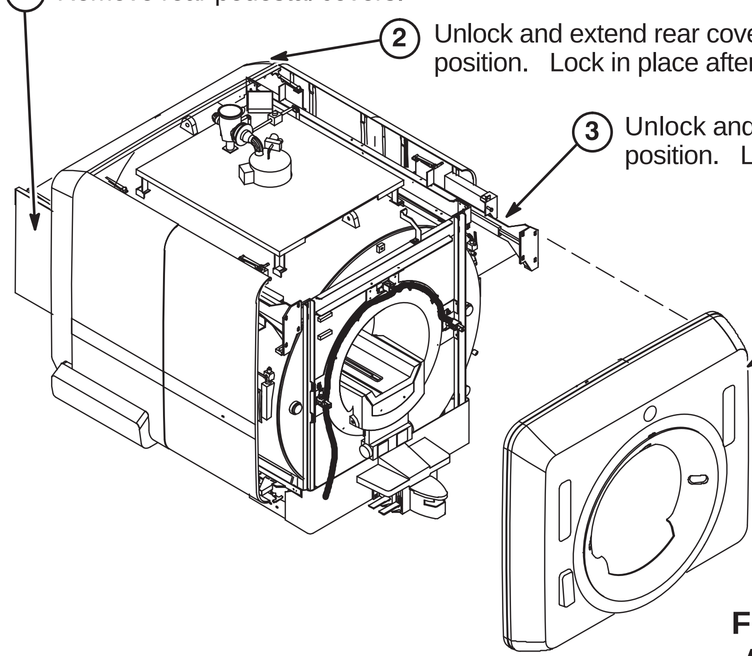


2. While holding in the patient transport dock plunger (below), move the carriage to the rear of the magnet by pressing IN FAST button on the magnet enclosure.



10–3–2 REMOVAL OF ENCLOSURE COVERS

1. Remove rear pedestal covers.
2. Unlock and extend rear cover to service position. Lock in place after extension.
3. Unlock and extend front cover to service position. Lock in place after extension.

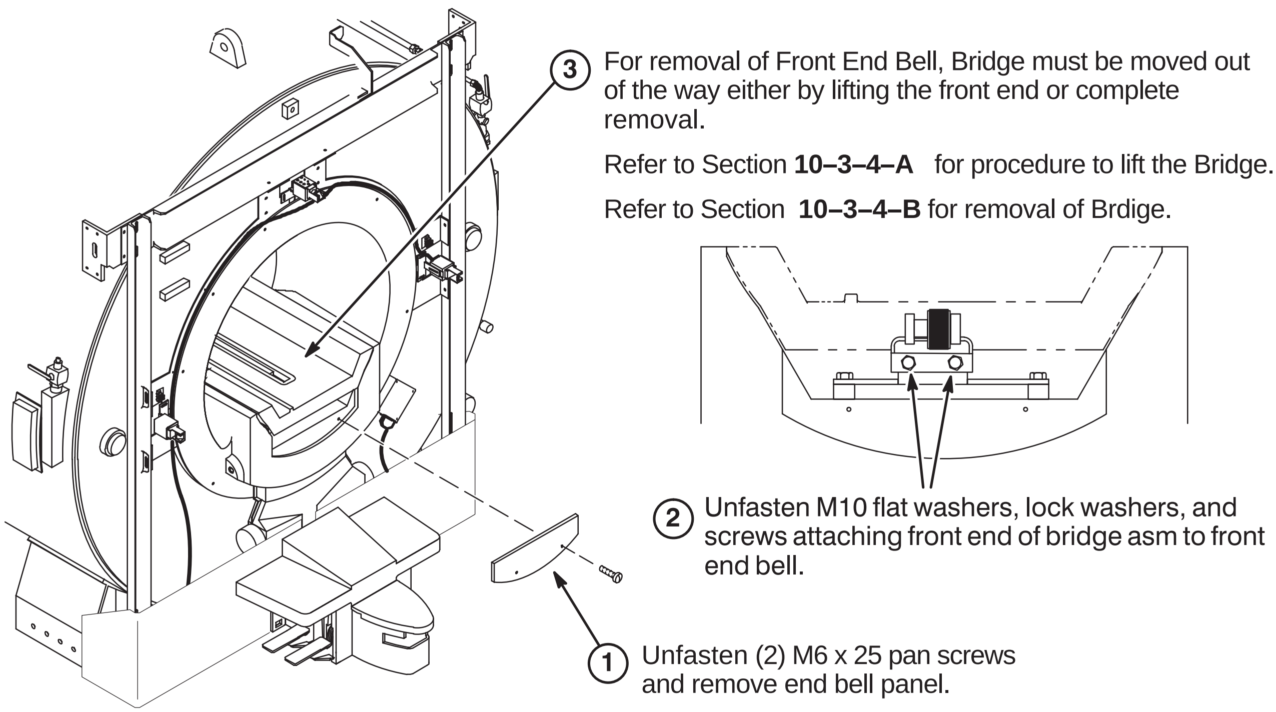


4. For ease of removal of Front End Bell, front cover may need to be removed. If so, make sure to follow WARNING below.

WARNING!

PERSONAL INJURY AND/OR EQUIPMENT DAMAGE POSSIBILITY! FRONT ENCLOSURE COVER IS HEAVY AND CUMBERSOME. AT LEAST TWO PEOPLE ARE REQUIRED TO LIFT AND MOVE IT; THREE IS PREFERABLE.

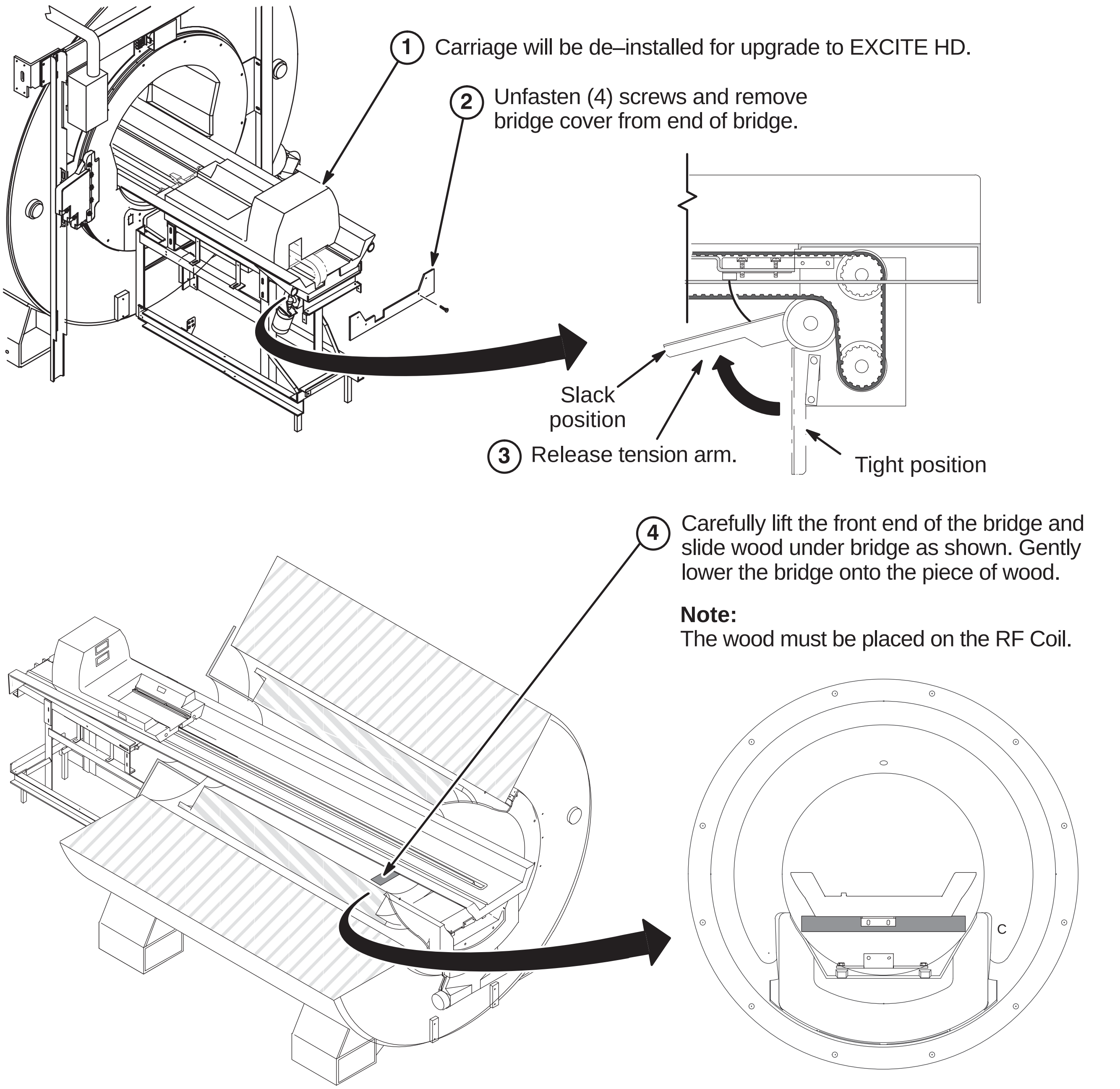
10–3–3 REMOVAL OF FRONT END BELL PANEL



10–3–4 BRIDGE HANDLING OPTIONS PRIOR TO REMOVAL OF FRONT END BELL

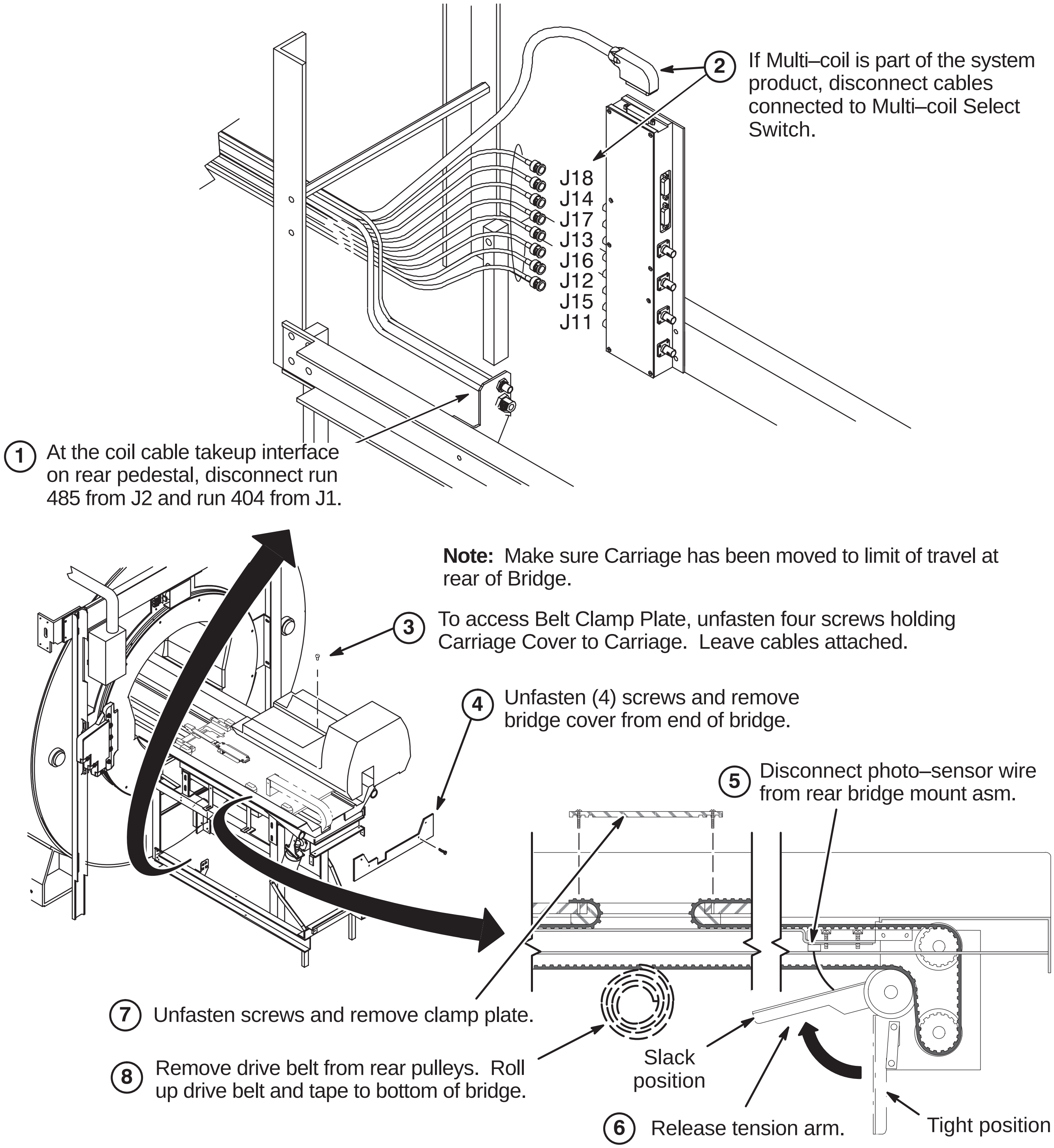
The Bridge needs to be repositioned or removed prior to removal of Front End Bell. Review sections **10–3–4–A** and **10–3–4–B**. Determine which method would best fit the needs of your site.

10–3–4–A Lifting Front End Of Bridge



10–3–4–B Removal of Bridge

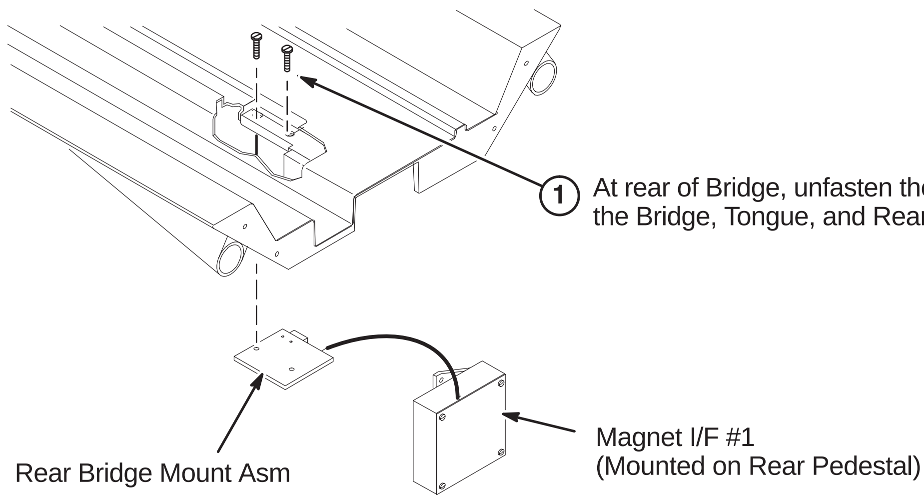
The ugprade to EXCITE HD requries a new LPCA with Cable Track and 16 Channel Switch. The de–installation instructions for existing components being replaced are located in the applicable magnet enclosure upgrade procedure in this manual. The steps below should be part of these de–installation instructions.



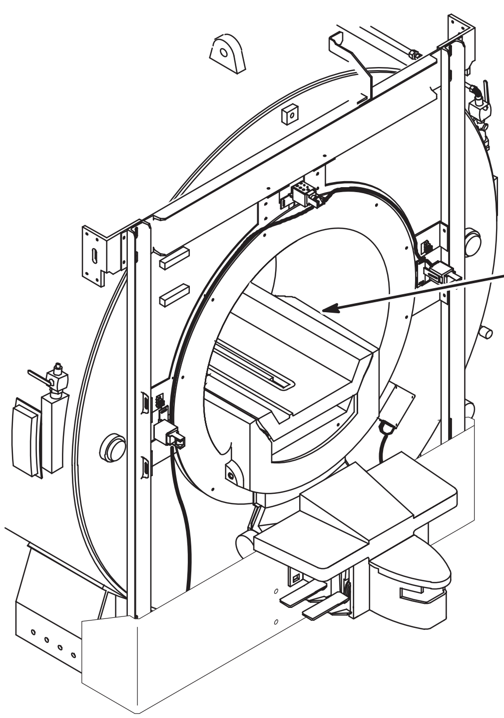
10–3–4–B Removal of Bridge (continued)

CAUTION

Possible equipment damage! Be careful not to damage the home flag on the drive belt when removing or replacing the bridge. The home flag is fragile and can easily be damaged.



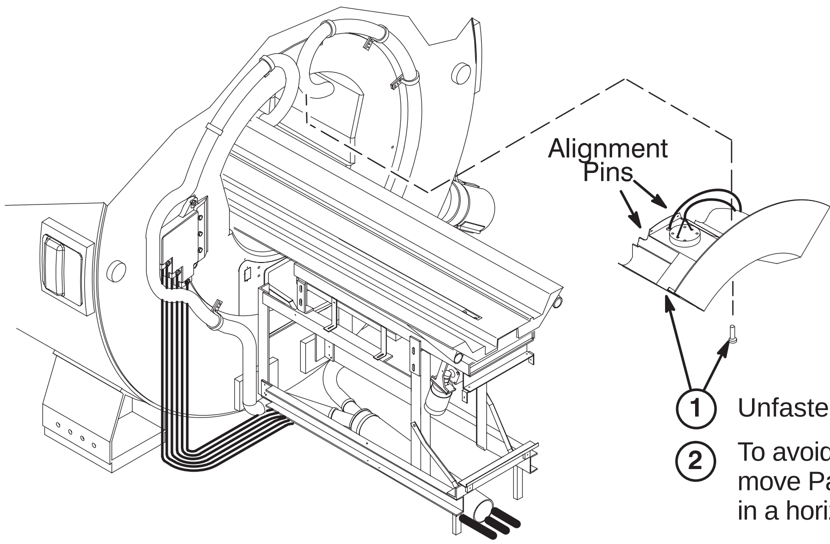
1. At rear of Bridge, unfasten the two screws attaching the Bridge, Tongue, and Rear Bridge Mount Asm.



2. To remove bridge, one person holds the front end, lifts slightly, and carefully pulls the bridge forward. The other person stands in front of the magnet next to the bridge to guide the bridge asm through the bore.
3. Place bridge aside in a safe area to prevent damage.

L

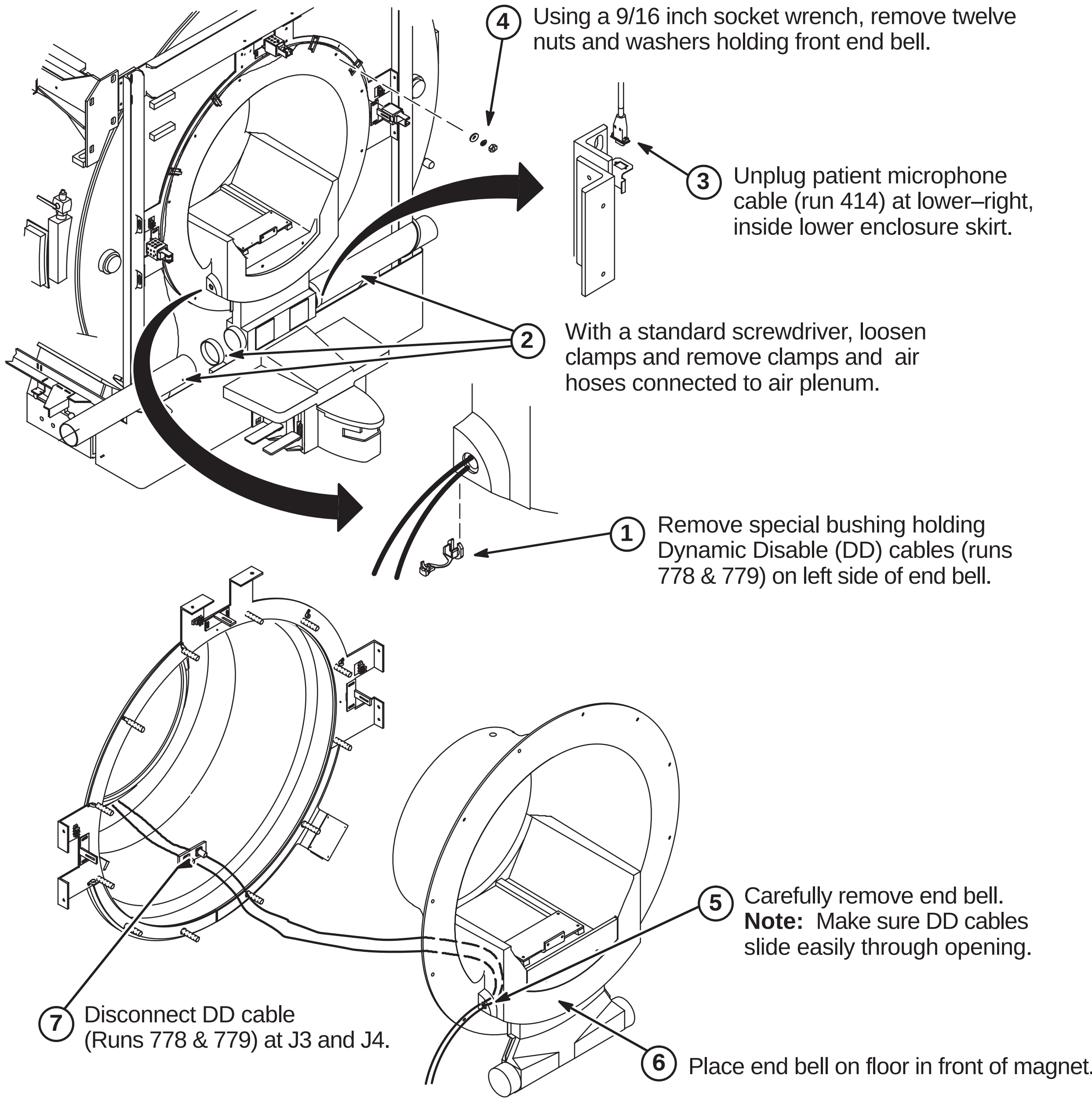
10–3–5 REMOVE PATIENT COMFORT MODULE



Note:
Hold Patient Comfort Module in place until mounting screws in Step 1 are removed. Alignment Pins can break off during removal if over–stressed.

1. Unfasten two M6 x 25 SST Pan Head Screws.
2. To avoid breaking alignment pins during removal, move Patient Comfort Module away from End Bell in a horizontal path.

10–3–6 FRONT END BELL REMOVAL

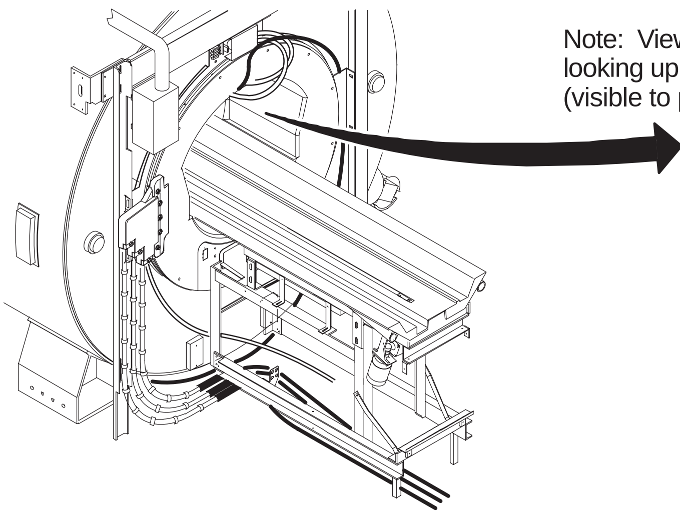


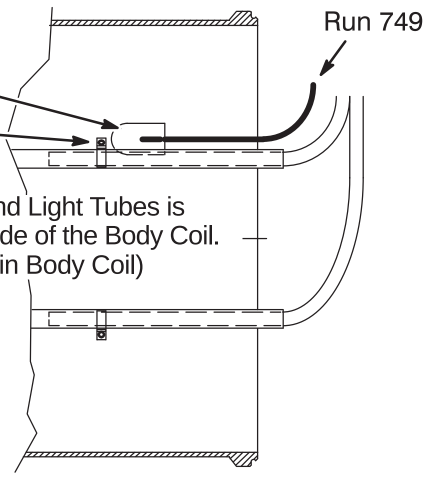
10–3–7 TEMPERATURE SENSOR CABLE REMOVAL

The site can have the existing Temperature Sensor Cable routed to the Rear end bell (earlier location) or Front end bell (later location). Either way, the cable needs to be removed and replaced with new cable, Run 1260. One of the two following procedures will be applicable to your site.

10–3–7–A Remove Existing Single Temperature Sensor Cable

1. Unfasten existing Temperature Sensor Cable (Run 749) from top inside of Body Coil. It may be attached with:
   adhesive fastener (as shown)
   OR
   with the Light and Cable clamp





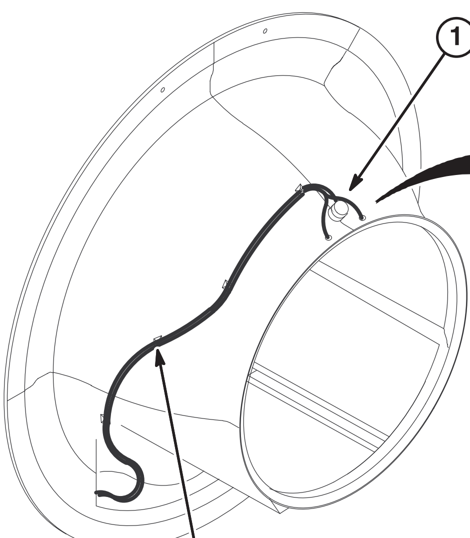
Note: View of Sensor and Light Tubes is looking up at the top inside of the Body Coil. (visible to patient laying in Body Coil)

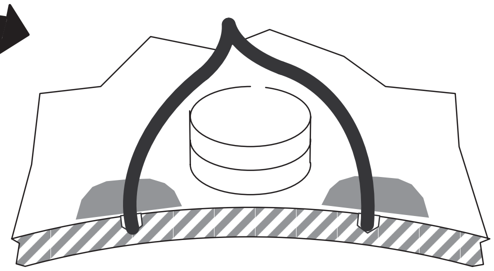
2. Disconnect other end of Run 749 from J10 on SRI module.
3. Remove and discard cable.

10–3–7–B Remove Existing Dual Temperature Sensor Cable

The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.

1. Remove existing Dual Thermal Sensor Cable from end bell. Take care not to damage end bell.



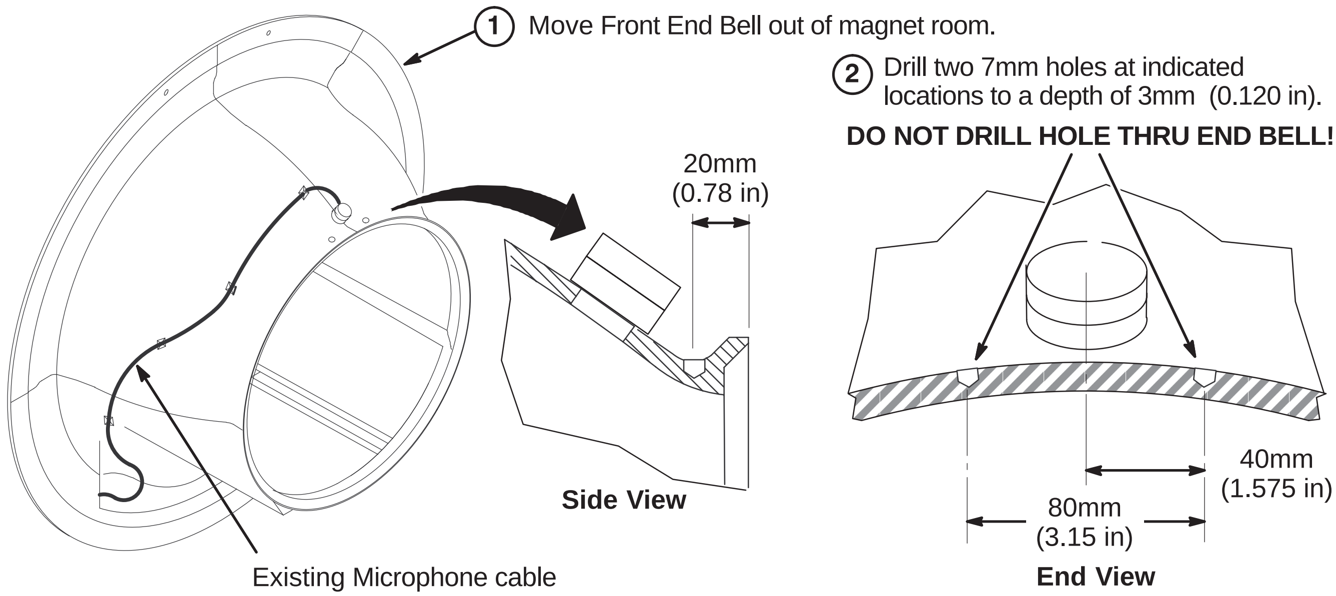


2. Cut cable ties and remove Dual Thermal Sensor Cable from Front End Bell.

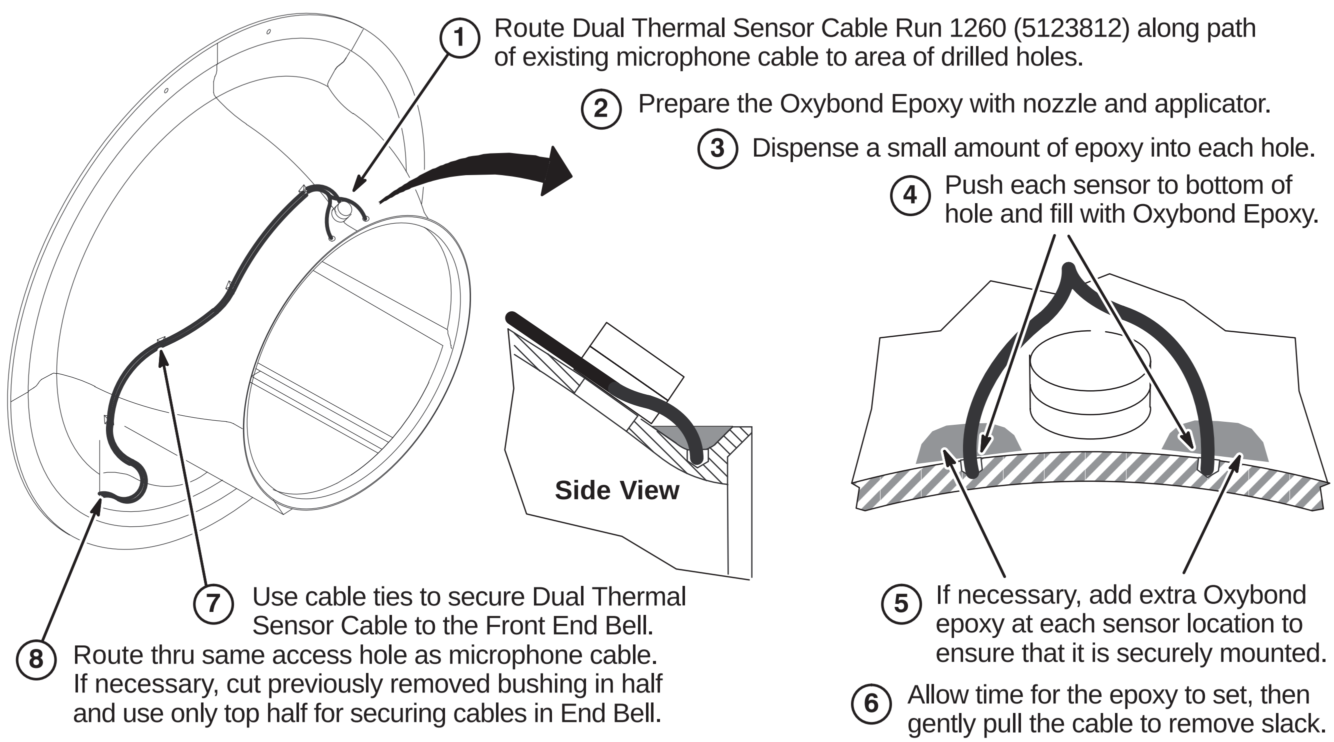
10–3–8 PREPARE END BELL FOR INSTALLATION OF DUAL THERMAL SENSOR CABLE

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!



10–3–9 ATTACH DUAL THERMAL SENSOR CABLE

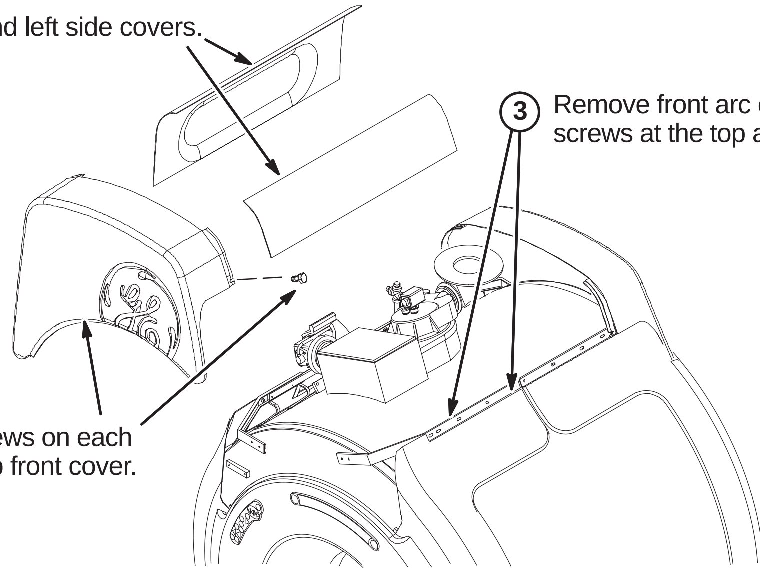


10–4 UPGRADE FOR WideOpen STYLE MAGNET ENCLOSURE

10–4–1 FRONT END BELL REMOVAL

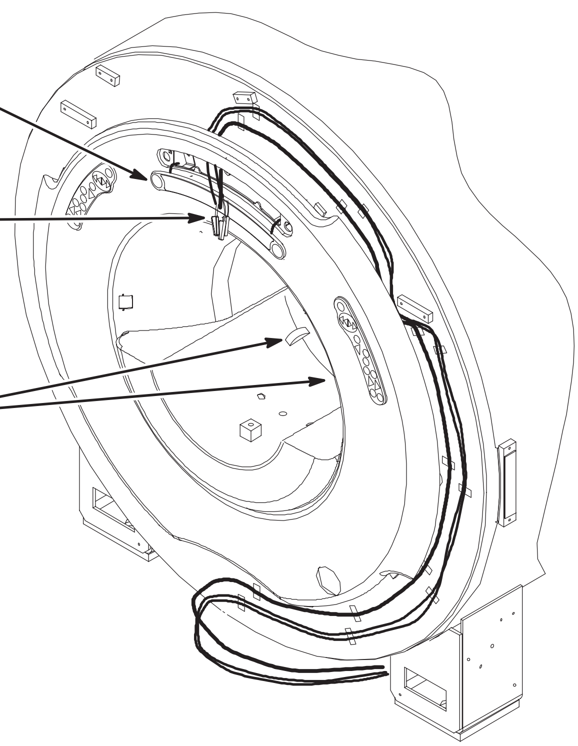
10–4–1–A Remove WideOpen Enclosure Covers

1. Remove tight and left side covers.
2. Remove screws on each side, then top front cover.
3. Remove front arc covers by removing two screws at the top and sliding cover to the side.

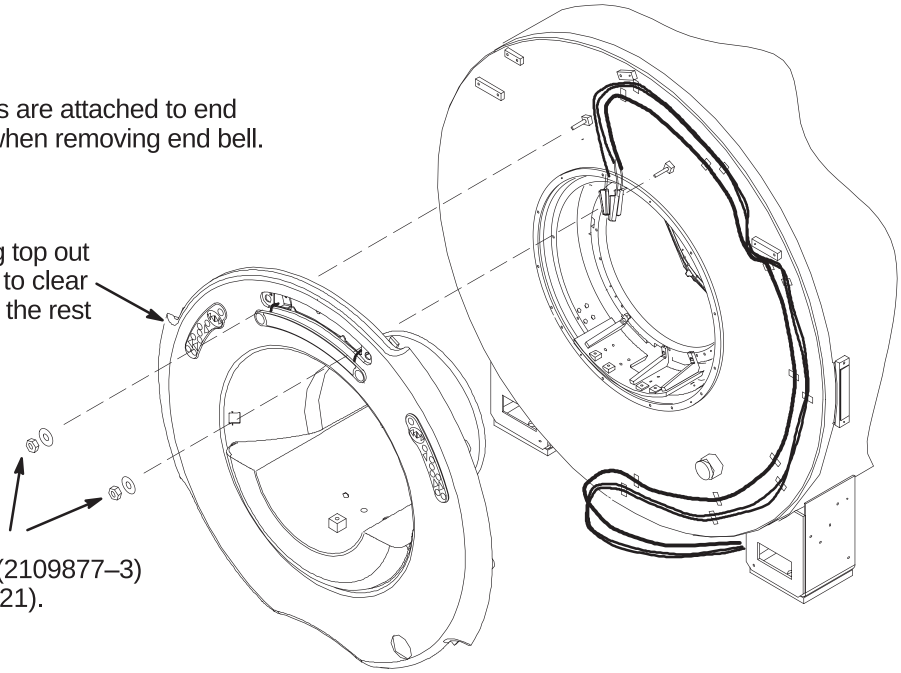


10–4–1–B Disconnect Operator Control Panel

1. Insert 3mm allen wrench into hole in Control Panel and turn counter clockwise 1/2 turn.
2. Carefully, using a small screwdriver, pry the Operator Display away from the front cover.
3. Disconnect the three cables attached to the Operator Display.
4. Lift open the two latches connecting the front end bell to the RF Coil.

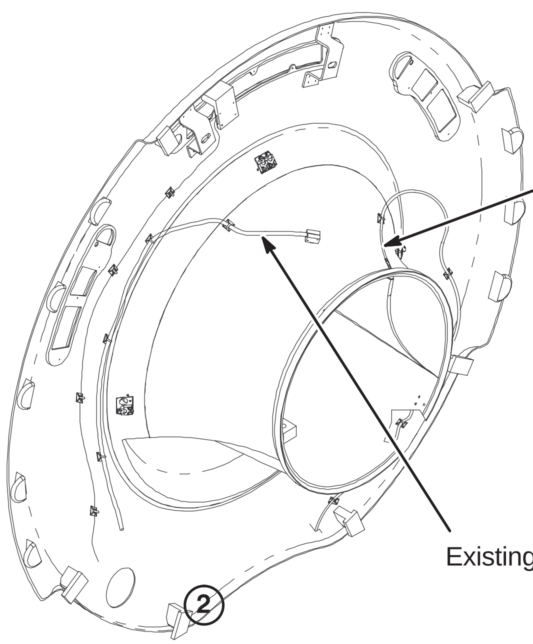


10–4–1–C Remove Front End Bell



10–4–2 TEMPERATURE SENSOR CABLE REMOVAL

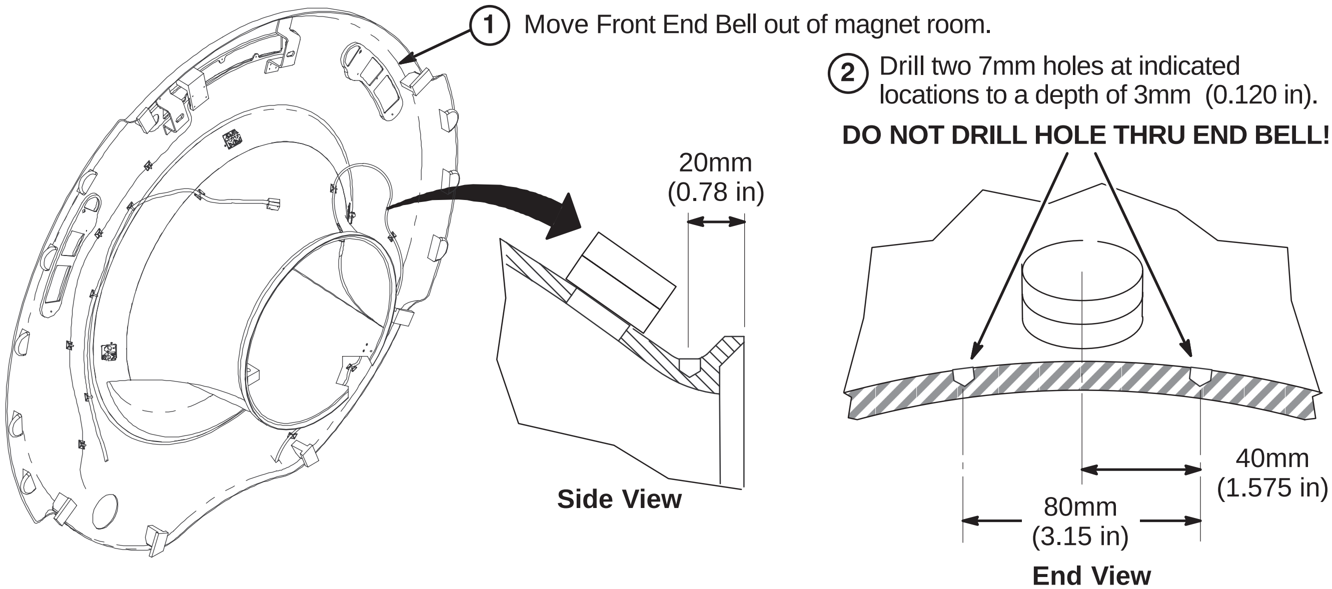
The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.



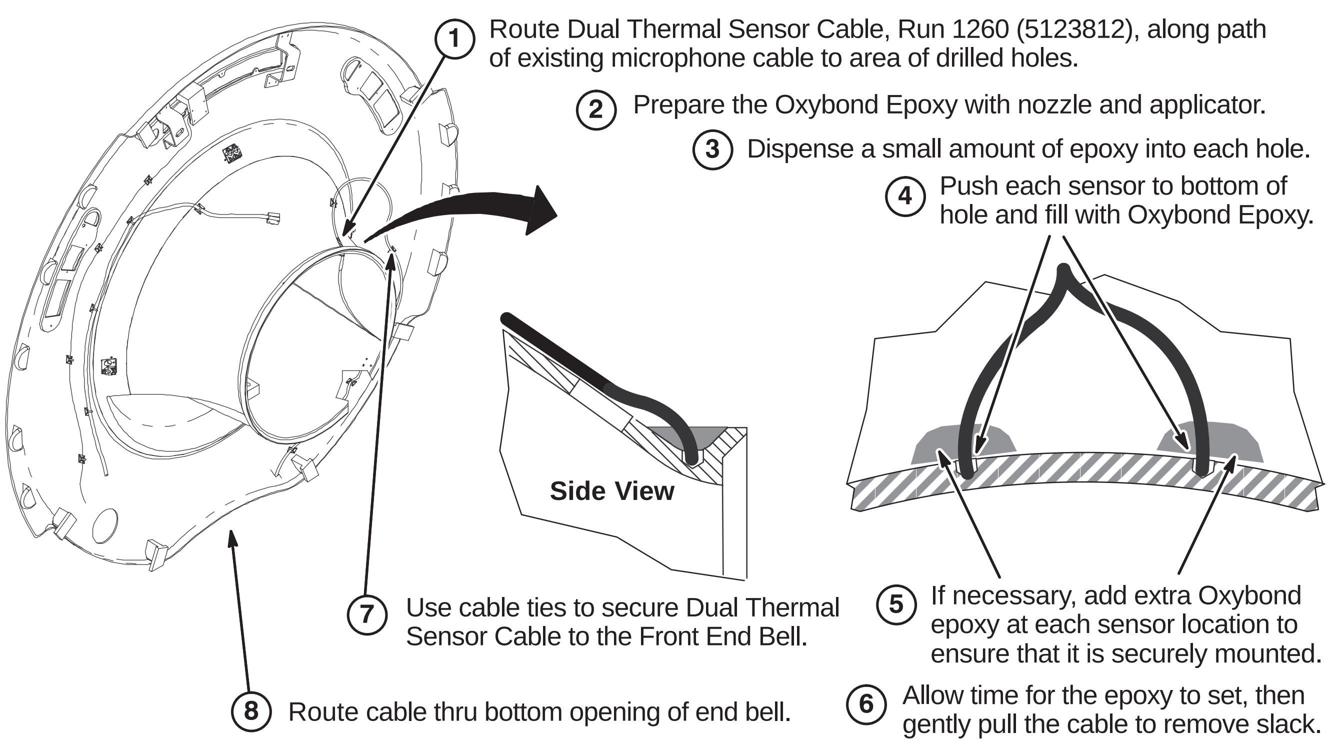
10–4–3 PREPARE END BELL FOR INSTALLATION OF DUAL THERMAL SENSOR CABLE

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!



10–4–4 ATTACH DUAL THERMAL SENSOR CABLE



10–5 REASSEMBLY OF MAGNET ENCLOSURE

For system replacing only the Temperature Sensor Cable:

10–5–1 HORIZON STYLE ENCLOSURE

Reverse the Front End Bell Removal procedure (Sections **10–3–6** to **10–3–2**) to reinstall the end bell, comfort module, and bridge.

10–5–2 WideOpen STYLE ENCLOSURE

Reverse the Front End Bell Removal procedure (Section **10–4–1**) to reinstall the end bell, comfort module, and bridge.

SECTION 11 – CRADLE RELEASE UPGRADE

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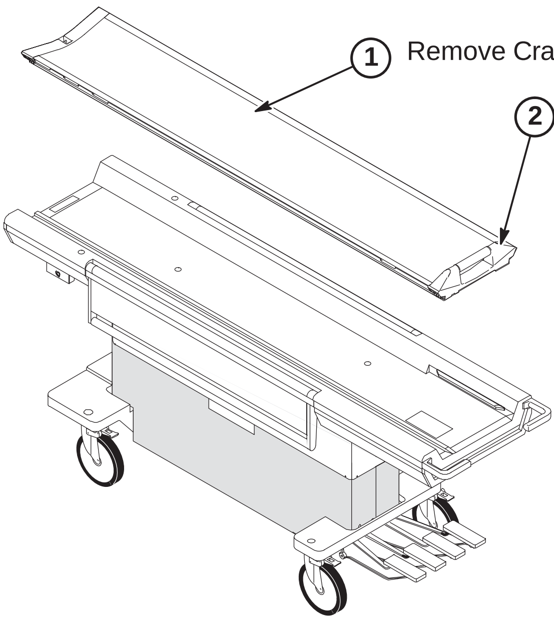
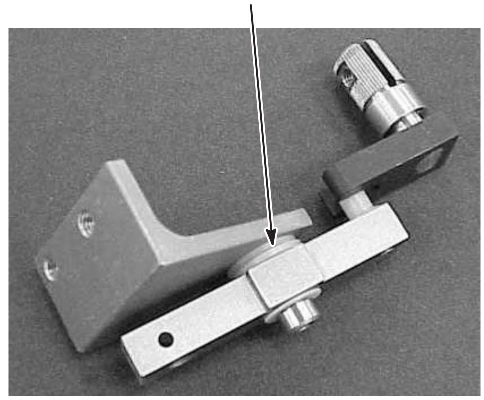
11–1 PATIENT TABLE CRADLE RELEASE UPGRADE

This upgrade is performed to improve the release mechanism and improve image quality.

11–1–1 PREPARE CRADLE FOR UPGRADE

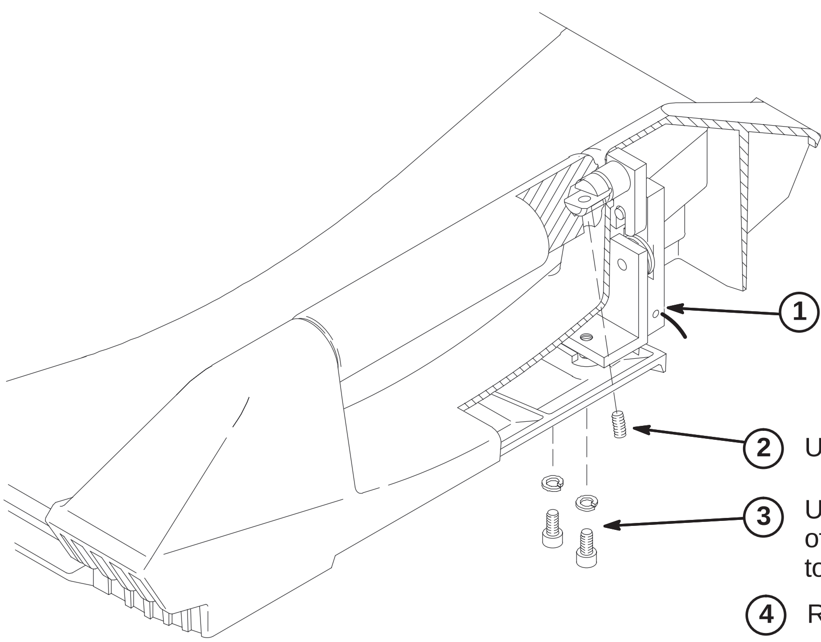


1. Remove Cradle from patient table.
2. Position Cradle to access release mechanism located under this corner of cradle.
3. The new mechnism is designed with two washers and two bushings inserted into the lever. The lever has also been widened in the center portion.



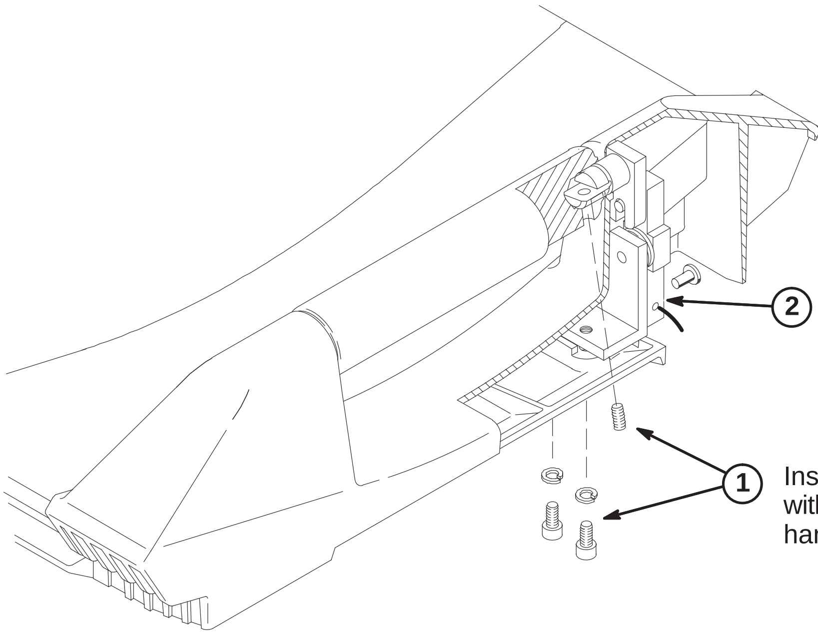
11–1–2 UPGRADE PROCEDURE

11–1–2–A Remove Existing Release Mechanism



1. Unfasten set screw at bottom of lever and remove Kevlar cord from lever. Retain set screw and flat head rivet for transfer to new mechanism if required.
2. Unfasten set screw at bottom of cradle handle.
3. Unfasten two screws and washers at bottom of cradle that attach the existing mechanism to cradle.
4. Remove old release mechanism.

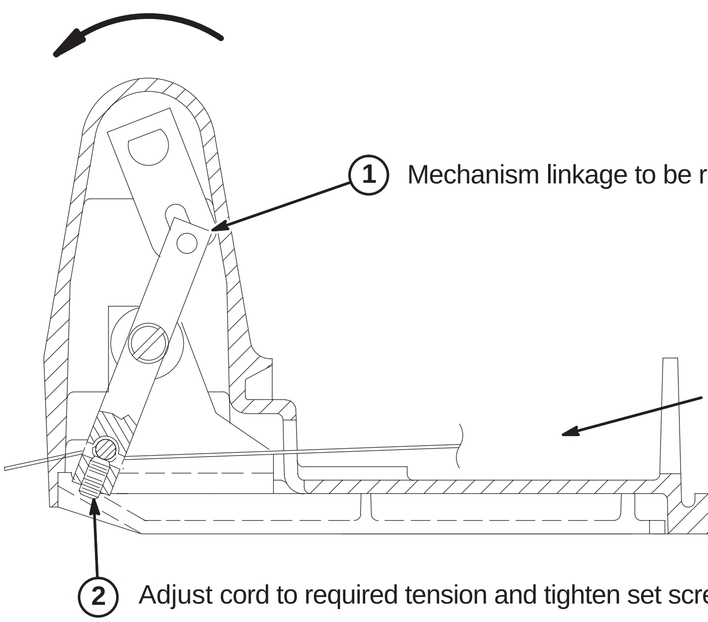
11–1–2–B Install New Release Mechanism



2 Insert Kevlar cord in lever and tighten with set screw. Refer to Procedure for cord installation and tightening.

1 Install new release mechanism and fasten with two lower screws and set screw in handle.

11–1–2–C Attach Cord



1 Mechanism linkage to be rotated as shown when tightening cord.

2 Adjust cord to required tension and tighten set screw. If needed, trim excess cord.

SECTION 12 – WATER CHILLER INSTALLATION

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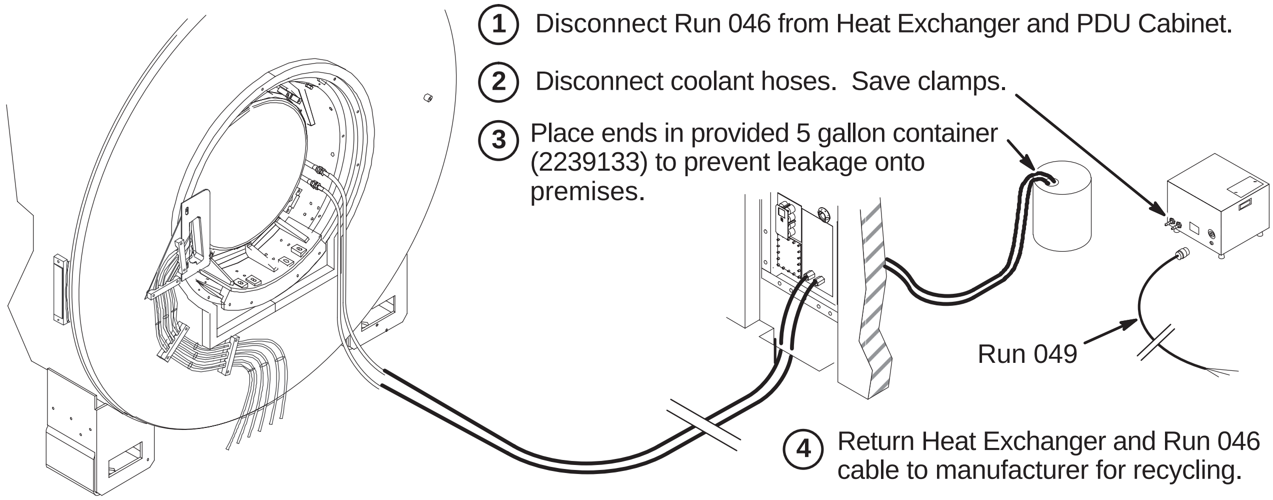
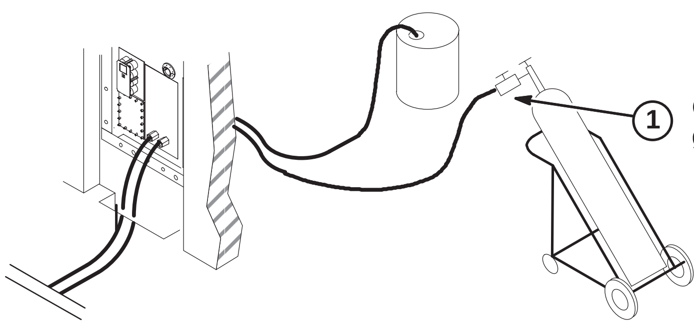
12–1 DISCONNECT HEAT EXCHANGER



1. Disconnect Run 046 from Heat Exchanger and PDU Cabinet.
2. Disconnect coolant hoses. Save clamps.
3. Place ends in provided 5 gallon container (2239133) to prevent leakage onto premises.
4. Return Heat Exchanger and Run 046 cable to manufacturer for recycling.

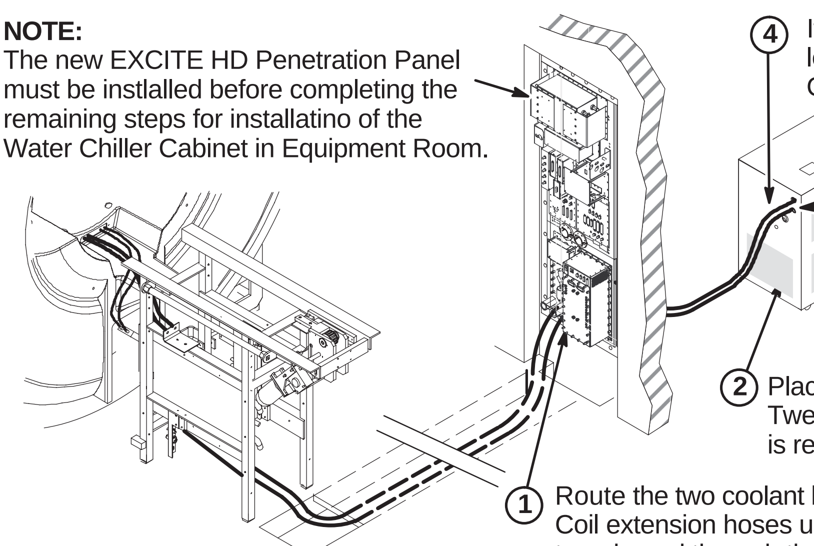
12–2 DRAIN COOLANT FROM COOLANT HOSES



1. Connect one of the hoses to locally obtained nitrogen gas cylinder.
2. Carefully open valve on cylinder and purge coolant from hoses into 5 gallon container.
3. When complete, close and seal 5 gallon container for return for recycling,

12–3 POSITION WATER CHILLER CABINET AND CONNECT HOSES

NOTE:
The new EXCITE HD Penetration Panel must be instllalled before completing the remaining steps for installatino of the Water Chiller Cabinet in Equipment Room.

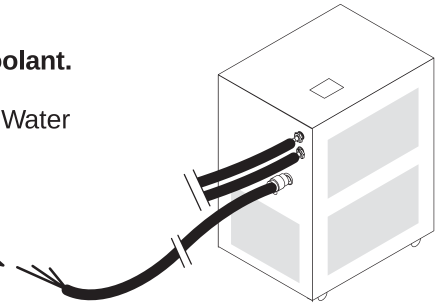


4. If required, cut coolant hoses to appropriate length and connect to cabinet with Hose Clamps (2128542–2).
3. Remove two protective plugs. Wrap threads with Teflon tape and install provided $^{1}/_{2}$ inch hose barb fittings in these connections.
2. Place Water Chiller Cabinet in Equipment Room. Twenty–four inch (610mm) clearance on all sides is required for unobstructed flow of cooling air.
1. Route the two coolant hoses that were connected to the Body Coil extension hoses under the Rear Pedestal, through the cable trough, and through the wave guides on the Penetration Panel.

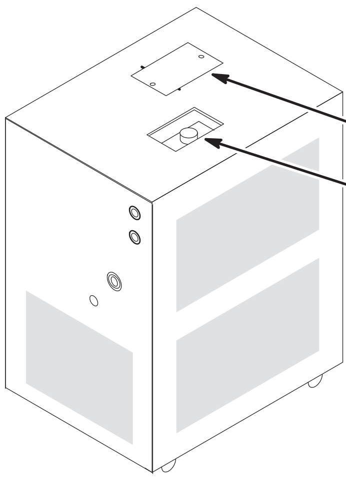
12–4 INSTALL POWER CORD

CAUTION:
Do not start Water Chiller pump before filling reservoir with coolant.
Consult vendor Water Chiller manual –
Installation, Operation, and Service Manual that is shipped with the Water Chiller.

1. Connect power cable Run 970 (2280289) to PDU as marked.

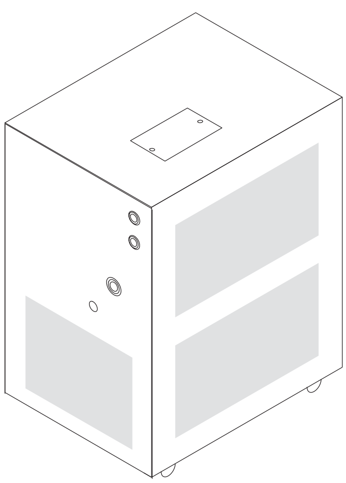


12–5 FILLING THE SYSTEM WITH COOLANT



1. Locate the twelve (12) gallons of Body Coil Coolant (2138791) that was shipped with System
2. Remove cover.
3. Consult vendor Water Chiller manual – *Installation: Filling Requirements*.
Initial Filling of Reservoir, and the following steps for the proper procedure on priming and filling the system with coolant.

12–6 STARTING WATER CHILLER



1. Consult vendor Water Chiller manual – *Operation: Start Up.* Follow instructions in manual for starting water chiller operation.
2. Verify unit is set to control at 20° C.
3. If supplied, attach plastic sleeve to side of Water Chiller and insert vendor manual. This will allow easy access to any future service/maintenance procedures. If plastic sleeve is not present, attach vendor manual to Water Chiller or store it with other service manuals on site.

SECTION 13 – BRM–D & CRM BODY COIL INSTALLATION

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13–1 BODY COIL PRE–INSTALLATION INFORMATION

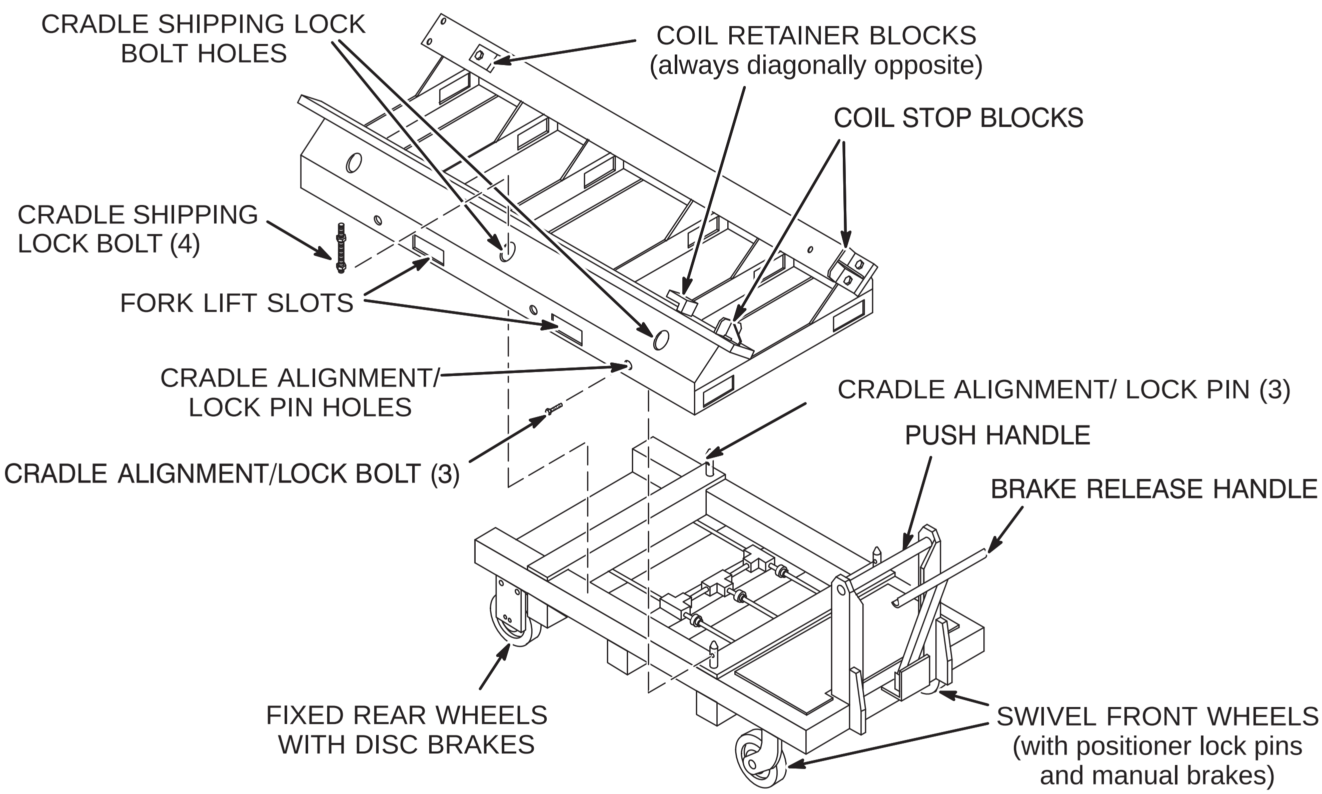
Refer to procedures in *Signa EXCITE HD 1.5T Service Methods CD–ROM,* Direction 5133315 Replacement>Coil Replacements>Gradient Coil Replacement for Non–Cx Magnets, for instructions on de–installing the existing Body Coil.

The upgrade body coils may be delivered on a cart/cradle or on a shipping pallet (cradle).

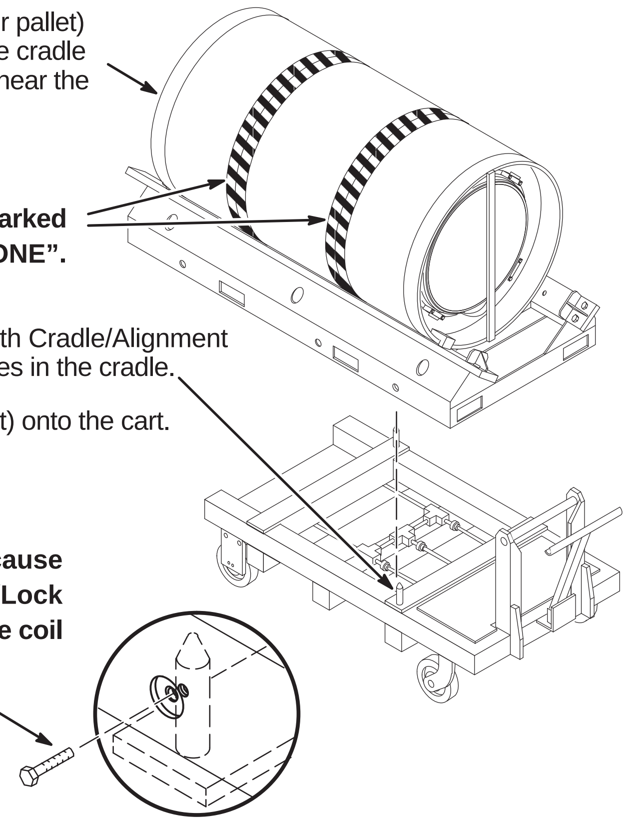
- **If body coil is delivered on a coil cart and cradle combination:**
  The body coil is transported to the magnet room. Review Section **13–1–1**, skip Section **13–1–2**, and proceed to Section **13–1–3**.
- **If body coil is delivered on a pallet (or cradle) for use with a coil cart:**
  The body coil is loaded on the regional coil cart. Review Section **13–1–1** and proceed to Section **13–1–2**.

13–1–1 BODY COIL CART AND CRADLE COMPONENTS

The Coil Cart and Cradle combination, as shown below is used to install the Body Coil in the magnet bore. The body coil may also be shipped on a disposable wood and metal pallet which performs the functions of the metal cradle shown below.



13–1–2 LOADING BODY COIL AND CRADLE OR PALLET ON COIL CART



1. Have a fork lift truck raise the body coil and cradle (or pallet) combination. The gradient leads are at the end of the cradle away from the handle, and the Coil Stop Blocks are near the handle.

CAUTION

Do not lift or apply pressure to area of Body Coil marked with black/yellow tape labelled "NO LOAD ZONE". Doing so will damage the Body Coil.

2. Position cart beneath cradle with Cradle/Alignment Lock Pins aligned with their holes in the cradle.
3. Lower body coil cradle (or pallet) onto the cart.

CAUTION

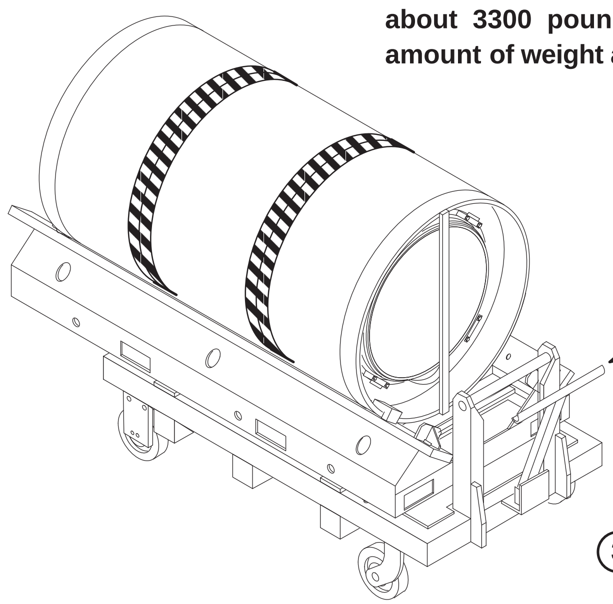
Not having the cradle anchored to the cart could cause the coil/cradle to tip. The three Cradle Alignment/Lock Bolts MUST be in place before attempting to move the coil cart.

4. Insert the three Cradle Alignment/Lock Bolts.

13–1–3 USING CART TO TRANSPORT BODY COIL

CAUTION

Avoid possible bodily injury. The combined coil/cradle/cart weigh about 3300 pounds. To avoid physical injury, and to control this amount of weight as it moves, a minimum of four people must be used.



1. Check route to magnet room and verify route clearance and sufficient manpower to overcome inclines.
2. Place plywood sheets or metal plates along route to distribute weight and protect floor surface from damage.

Note: The Brake Release Handle controls the disc brakes on the fixed rear wheels. It must be pushed against the Push Handle in order to move the cart.

3. Roll coil cart to outside of magnet room for connection of 10 foot (305cm) Gradient extension cables.

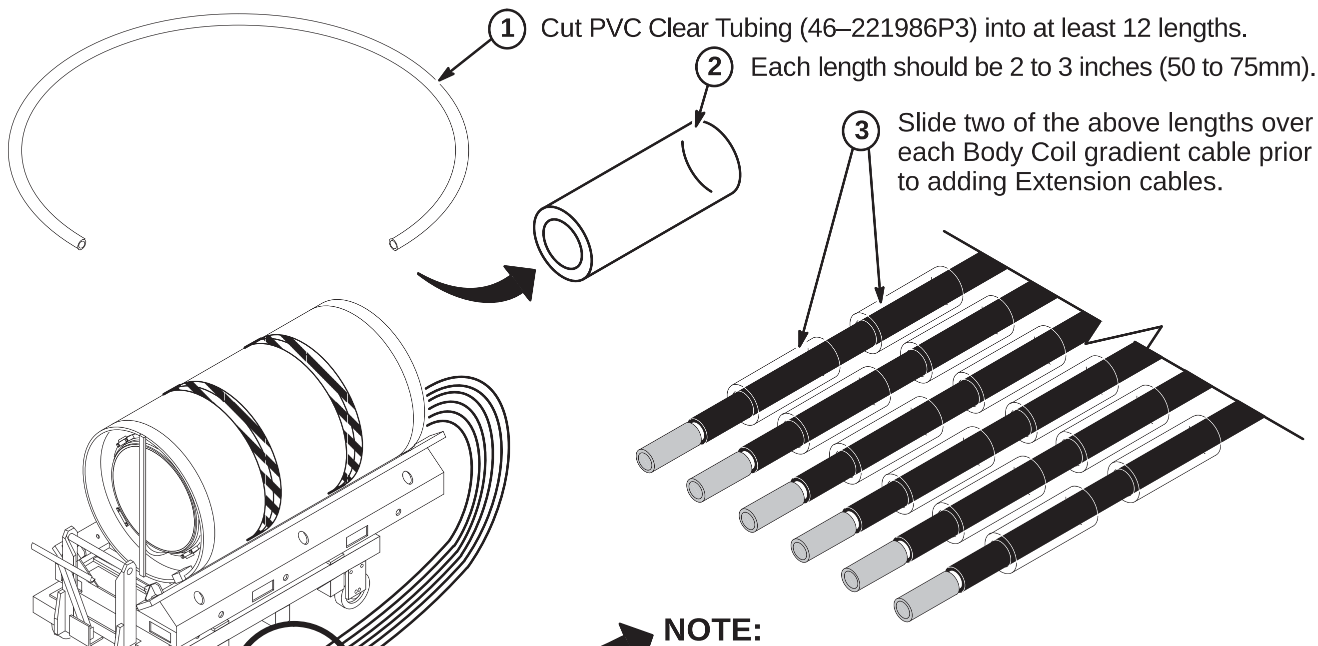
13–2 EXTENSION CABLE INSTALLATION FOR NON–ACTIVELY SHIELDED MAGNETS

WARNING!

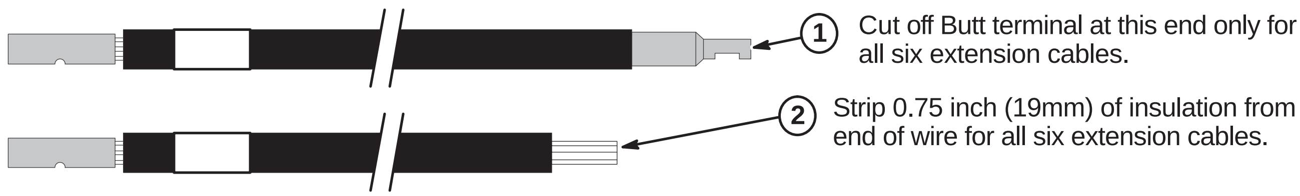
FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

<u>IF MAGNET IS AT FULL FIELD – ATTACHING THE 2119600 SERIES EXTENSION CABLES TO THE BODY COIL GRADIENT CABLES MUST BE COMPLETED OUTSIDE THE MAGNET ROOM.</u>

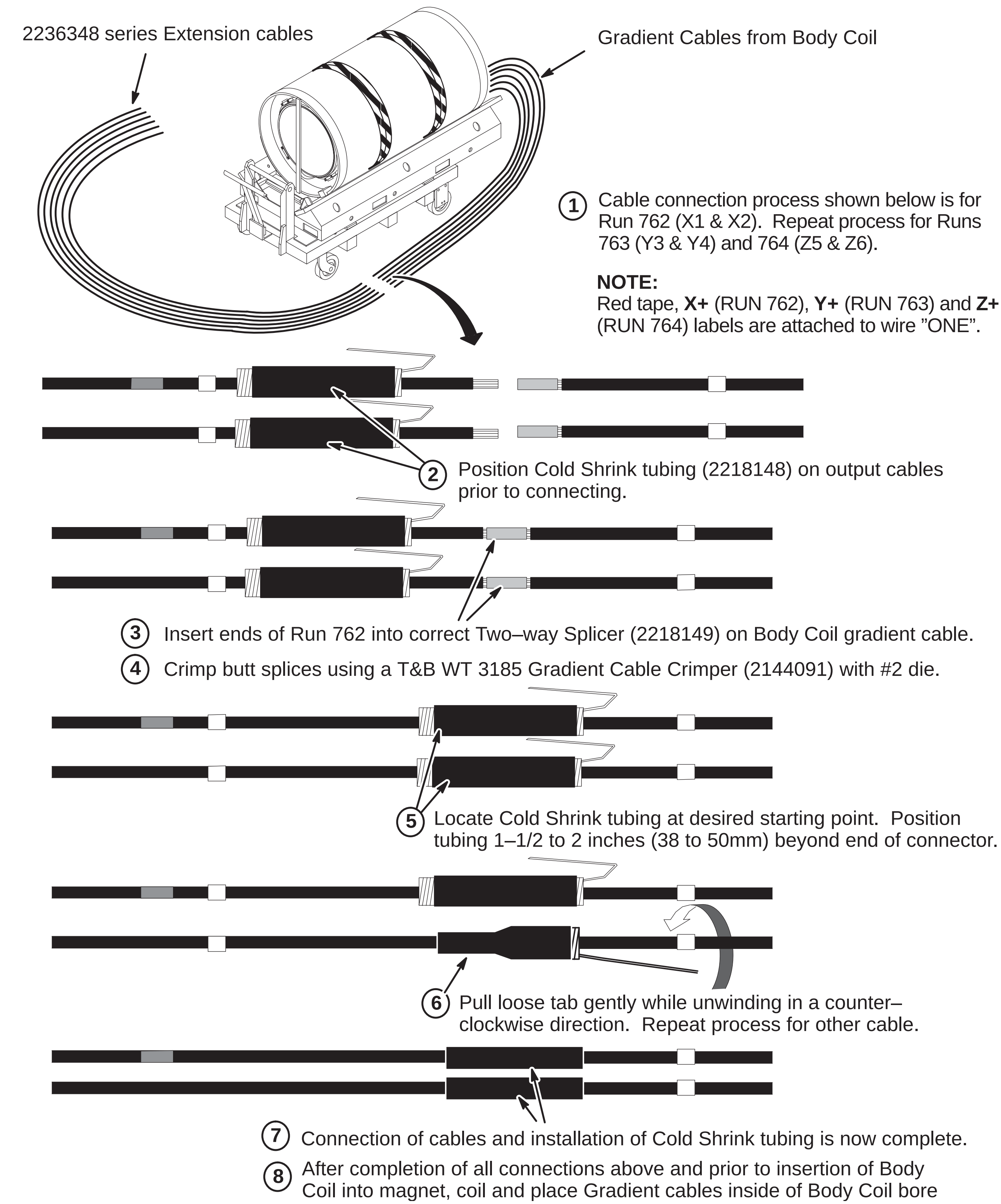
13–2–1 PRE–INSTALLATION OF PVC CLEAR TUBING



13–2–2 CABLE TERMINATIONS FOR GRADIENT EXTENSION CABLES

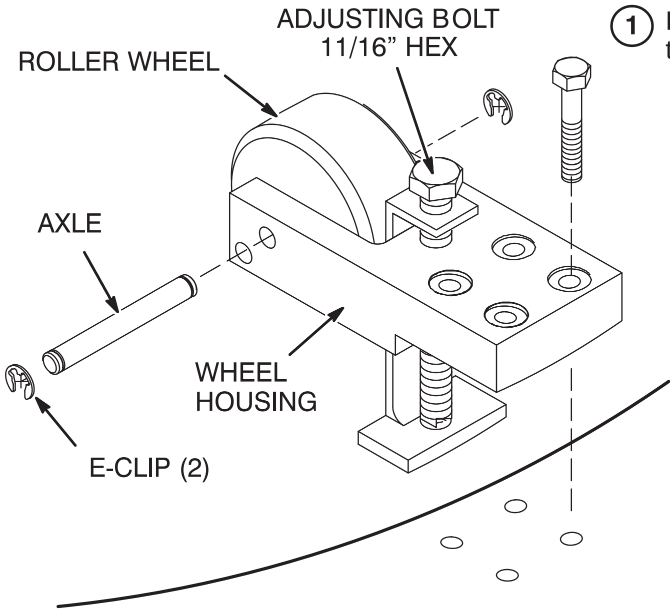


13–2–3 CONNECTION OF BODY COIL GRADIENT CABLES TO EXTENSION CABLES



13–3 INSTALL BODY COIL IN MAGNET BORE

13–3–1 INSTALLATION OF ROLLER WHEEL ASSEMBLIES



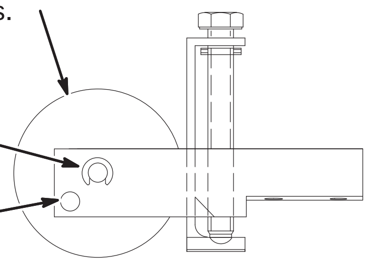
1. Remove the two Coil Retainer Blocks (used to secure the coil to the cradle or pallet during shipment).

   Note: The following steps are used to mount the four Roller Wheel Assemblies to the coil, two at each end.

2. Back the adjusting bolt all the way out by turning it counterclockwise.
3. Remove one of the e–clips and remove wheel and axle.
4. Place Wheel Housing in position over the four threaded holes on the coil and fasten with four bolts.
5. Turn the adjusting bolt clockwise on all four Roller Wheel Assemblies to raise the coil to the point that the wheels and axles can be installed.
6. Note which position to use and install wheels and axles.

Use upper hole for **GE** magnets with passive or resistive shim and **Oxford** magnet.

Use lower hole for **GE** magnets without passive or resistive shim.

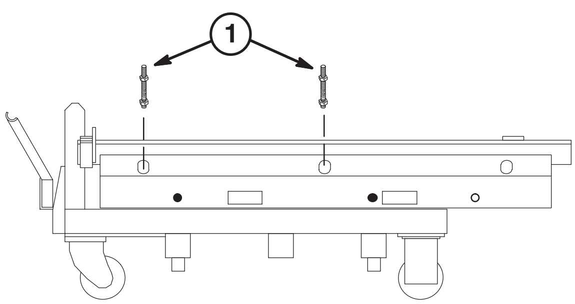


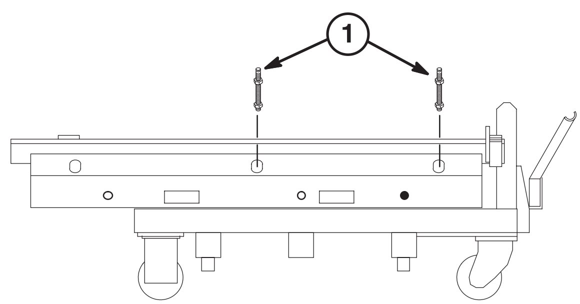
13–3–2 REMOVE CRADLE SHIPPING LOCKING BOLTS

CAUTION

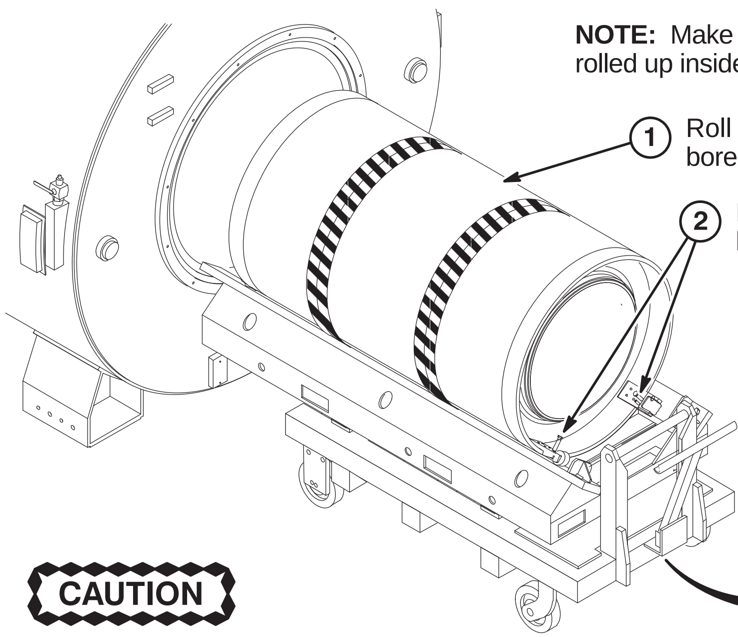
Attempting to raise the body coil with cradle shipping lock bolts installed will damage the elevating mechanism. The four Lock Bolts MUST be removed before attempting to raise the coil cart cradle.

1. Remove the four Cradle Shipping Lock Bolts from the cart (two bolts on each side). This <u>must</u> be done before the Lift Assembly can be used to raise the cradle. These bolts belong with the cart; do not misplace them.



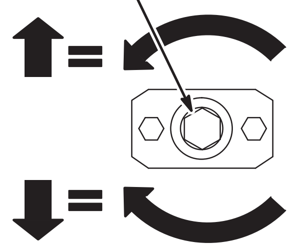


13–3–3 POSITION BODY COIL AT FRONT OF MAGNET BORE



NOTE: Make sure that all Body Coil gradient cables are rolled up inside the Body Coil prior to insertion into magnet.

1. Roll the cart into position at the front of the bore for unloading the coil into the magnet.
2. Lower the wheels on the Roller Wheel Assemblies by turning the adjusting screws counterclockwise.
3. Use a 19–mm socket on the Lift Assembly Height Adjuster, turn clockwise (see caution) to raise the cradle even with the magnet bore.

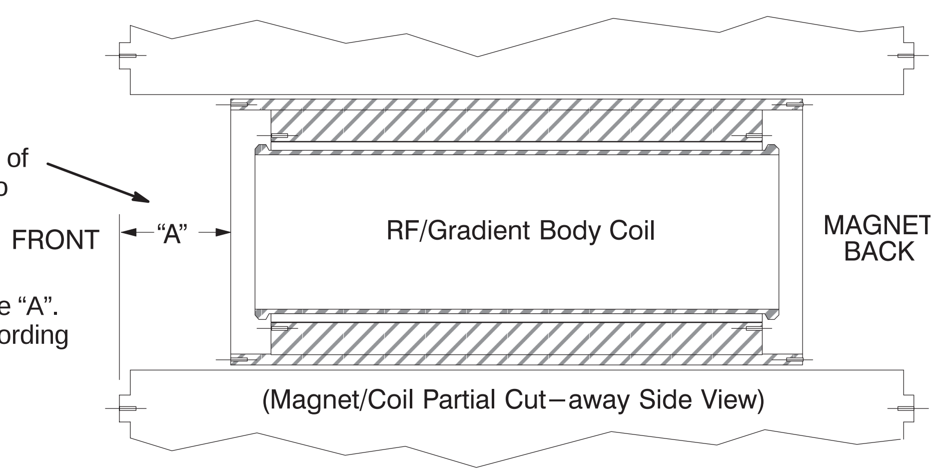


CAUTION

EQUIPMENT DAMAGE POSSIBILITY
Early carts had left–handed threads on the Lift Assembly Height Adjuster, i.e., clockwise rotation lowered the assembly, counterclockwise raised it. Gradually, these carts are being modified for right–handed threads. Be sure to check the cart that you are about to use. Damage to the mechanism cane occur if the bolt is turned the wrong way with the full weight of a coil and cradle on it.

13–3–4 LOCATE POSITION OF BODY COIL IN MAGNET BORE

1. Slowly and carefully roll the Body Coil into the magnet bore. **Do not roll Body Coil past rear end of magnet bore.**
2. Measure distance from surface of front Magnet mounting flange to edge of outer gradient coil. (see Suggestion below)
3. Refer to table below for distance "A". Select the correct distance according to Magnet type.



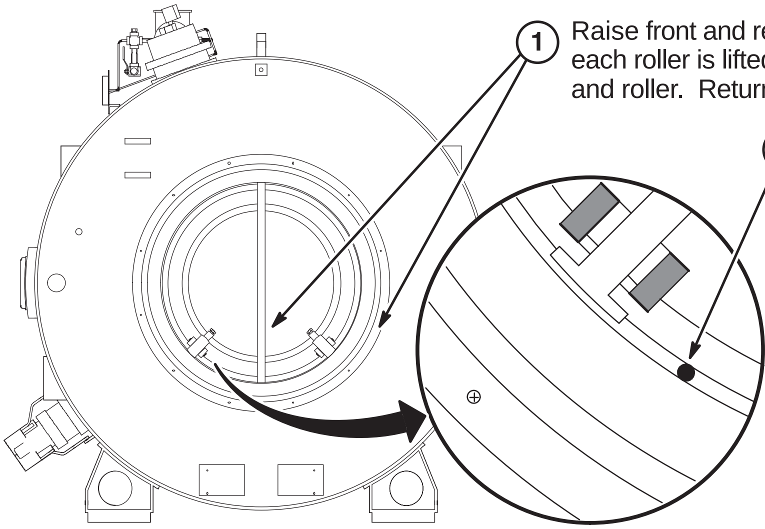
(Magnet/Coil Partial Cut–away Side View)

Body Coil	BRM–D	CRM
"A" Distance	12.00 Inch (304.5mm)	10.43 Inch (264.5mm)
TOLERANCE: +0.0mm, –1.5mm		

13–4 ALIGN BODY COIL IN MAGNET BORE WITH SHIM COIL

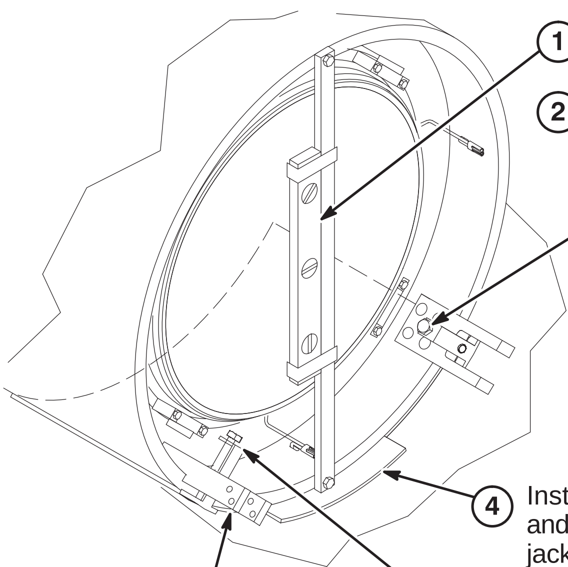
Perform this procedure if site has Shim Coil. If site does not have shim coil, go to Section **13–5**.

13–4–1 ROTATION ALIGNMENT OF BODY COIL IN GE MAGNET WITH SHIM COIL



1. Raise front and rear of Body Coil with jack screws so that each roller is lifted off of the shim coil. Remove each axle and roller. Return to coil cart accessory kit.
2. Insert (4) small rods, two in front and two in back, between Body Coil and Shim Coil, approximately as shown.
3. Holding the rods in place, lower the Body Coil so that it rests on the rods.

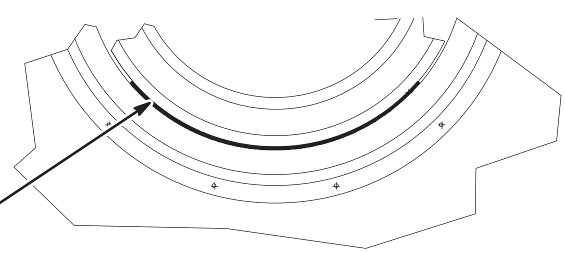
13–4–2 CENTERING BODY COIL IN GE MAGNET BORE WITH SHIM COIL



1. Tape a level to stainless steel bar (10767) that comes with new coil (to which the bias leads are attached).
2. Watching the appropriate bubble on the level, rotate the coil until the bubble is centered. Coil is now aligned along the X–axis.
3. Being careful not to further rotate coil, turn each adjusting bolt clockwise to raise the Body Coil just high enough until roller pins can be removed; remove roller pins.

   Note: Changes in Level Reading
   As each adjusting bolt is turned, the bubble in the level will shift slightly; this is only temporary, and should not affect the established levelness once the Roller Wheel Assemblies are removed.
4. Install (1) 6.5mm thick rubber pad (2126201) between Body Coil and Shim Coil at each end. Slide pad sideways to get between jackscrews. Then rotate 90° and position pad so that main edge of pad is flush with edge of coil, and the tab protudes.
5. Turn adjusting bolts counterclockwise until coil rests on the pads. Coil is now aligned along the Y–axis (vertical).
6. Remove each mounting block with jack screw asm and return them to the coil cart accessory kit.

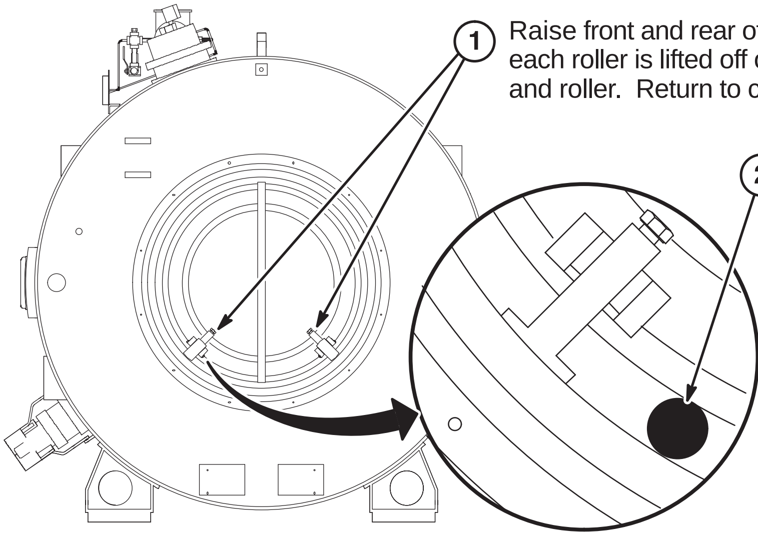
Note: Distance between top of Body Coil and top of Magnet bore should be equal to distance from bottom of Body Coil to bottom of Magnet bore.



Typical front and rear view.

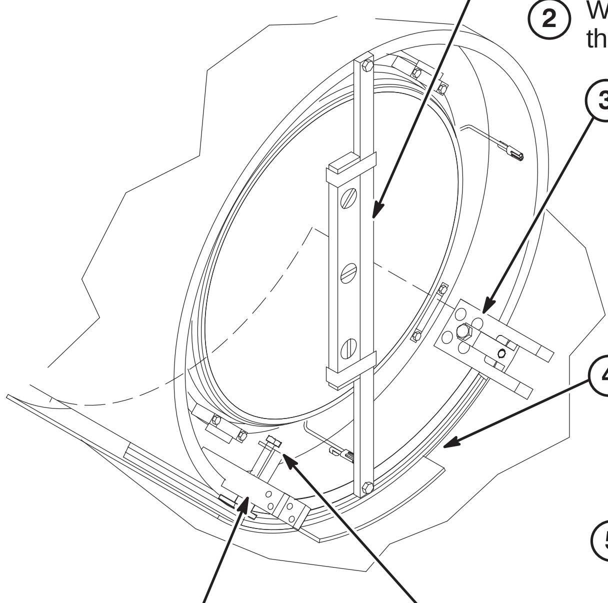
13–5 ALIGN BODY COIL IN MAGNET BORE <u>WITHOUT</u> SHIM COIL

13–5–1 ROTATION ALIGNMENT OF BODY COIL IN GE MAGNET <u>WITHOUT</u> SHIM COIL



1. Raise front and rear of Body Coil with jack screws so that each roller is lifted off of the shim coil. Remove each axle and roller. Return to coil cart accessory kit.
2. Insert (4) large rods, two in front and two in back, between Body Coil and Shim Coil, approximately as shown.
3. Holding the rods in place, lower the Body Coil so that it rests on the rods.

13–5–2 CENTERING BODY COIL IN GE MAGNET BORE <u>WITHOUT</u> SHIM COIL



1. Tape a level to stainless steel bar (10767) that comes with new coil (to which the bias leads are attached).
2. Watching the appropriate bubble on the level, rotate the coil until the bubble is centered. Coil is now aligned along the X–axis.
3. Being careful not to further rotate coil, turn each adjusting bolt clockwise to raise the Body Coil just high enough until roller pins can be removed; remove roller pins.

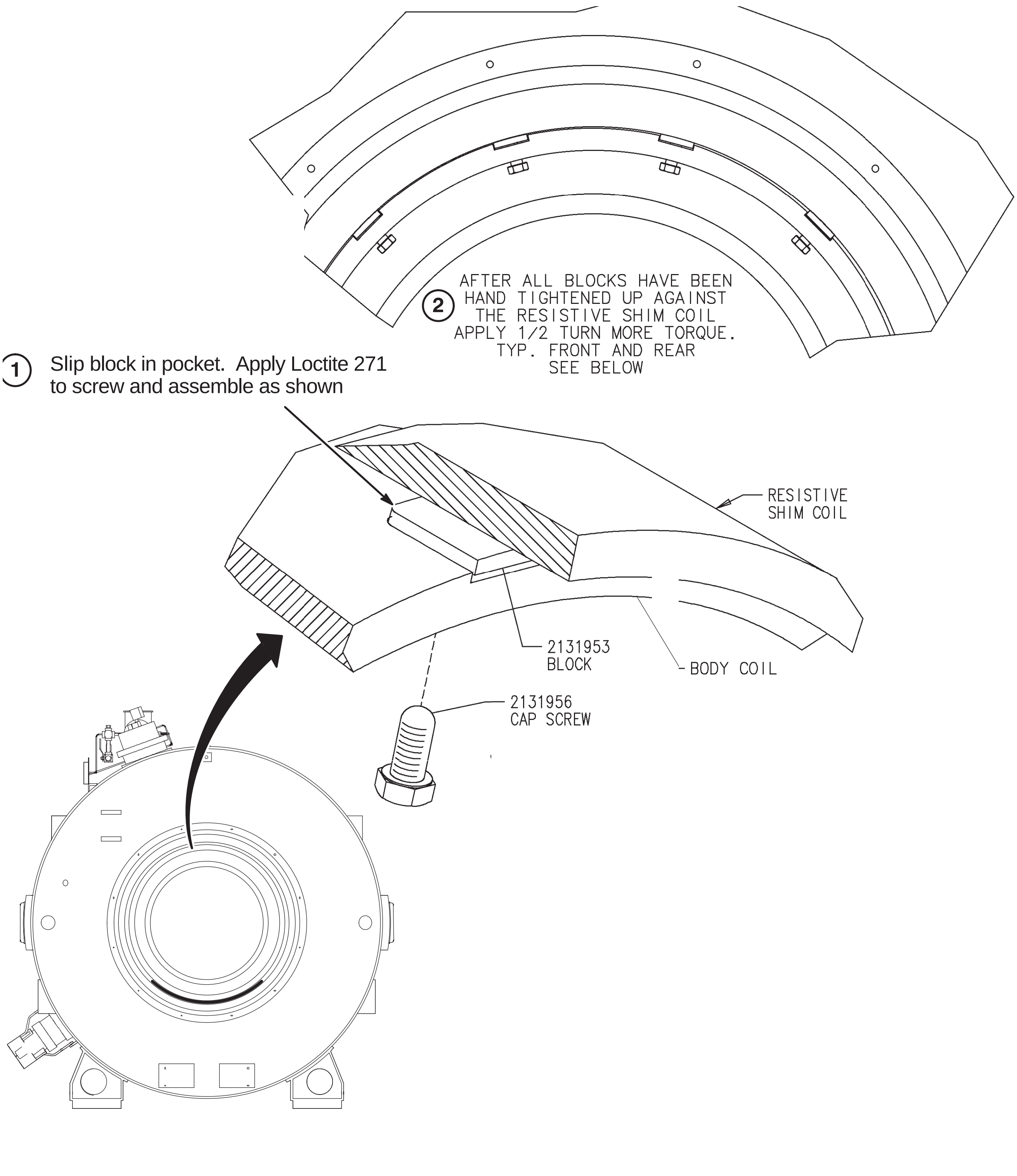
   Note: Changes in Level Reading
   As each adjusting bolt is turned, the bubble in the level will shift slightly; this is only temporary, and should not affect the established levelness once the Roller Wheel Assemblies are removed.
4. Install (3) 9.5mm rubber pads (2135234) and (1) 6.5mm thick rubber pad (2126201) between Body Coil and Shim Coil at each end. Slide pad sideways to get between jackscrews. Then rotate 90° and position pad so that main edge of pad is flush with edge of coil, and the tab protudes.
5. **Note:** Distance between top of Body Coil and top of Magnet bore should be equal to distance from bottom of Body Coil to bottom of Magnet bore.

   If distances are not equal, raise the Body Coil and insert new combination of rubber pads to achieve the correct thickness. Two extra 3.2mm rubber pads (2135234–2) are provided for this purpose.
6. Turn adjusting bolts counterclockwise until coil rests on the pads. Coil is now aligned along the Y–axis (vertical).
7. Remove each mounting block with jack screw asm and return them to the coil cart accessory kit.

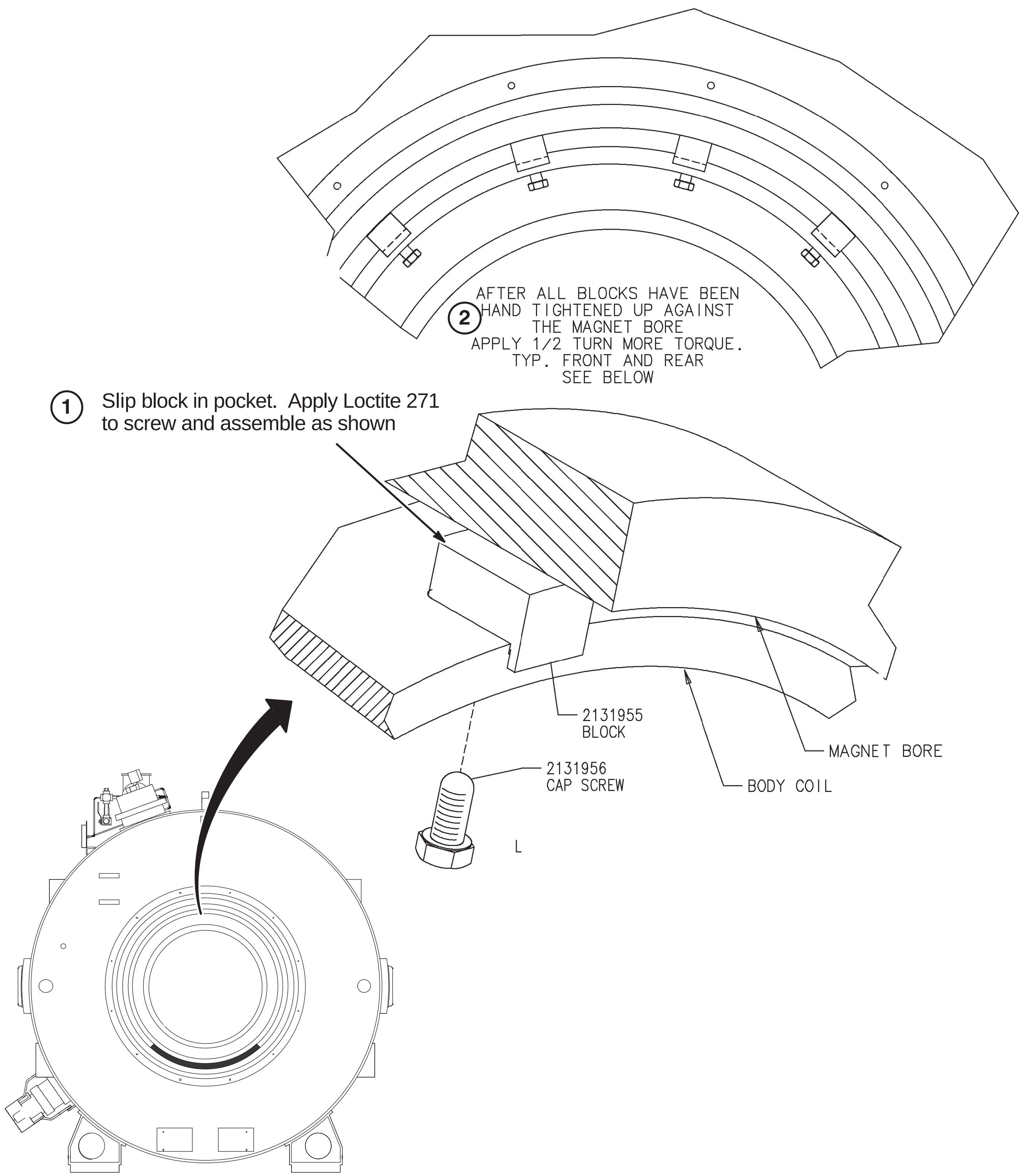
13–6 INSTALL RADIAL SUPPORT

Depending on site configuration, perform one of the following procedures.

13–6–1 INSTALL RADIAL SUPPORT IN GE MAGNET WITH 900mm DIAMETER BORE



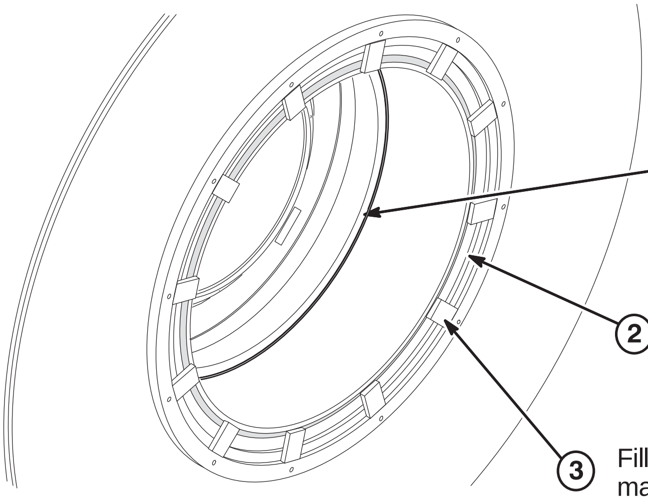
13–6–2 INSTALL RADIAL SUPPORT IN GE MAGNET WITH 950mm DIAMETER BORE



13–7 INSTALL BODY COIL INSULATION

Depending on site configuration, perform one of the following procedures.

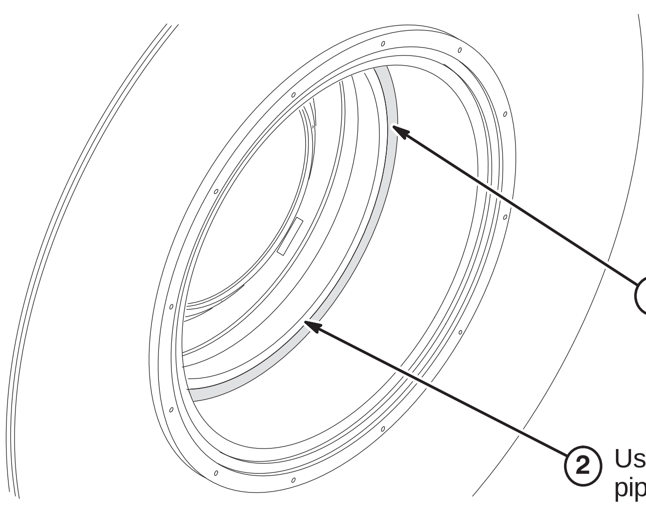
13–7–1 INSTALLING INSULATION IN MAGNET WITH PASSIVE OR RESISTIVE SHIM



Note:
Rope caulk, pipe insulation and acoustical collar are installed after body coil centering.

1. Fill the gap between the Passive Shim Coil and outer Gradient Coil at each end. Use one, two, or three strips (rolled together) of rope caulk (46–287118P1) around entire perimeter.
2. Fill the gap between the Magnet bore and the Passive Shim Coil at each end. Use 8 in. (205 mm) sections of Pipe Insulation (46–320868P1) around entire perimeter.
3. Fill the gap between the Passive Shim drum stops and magnet bore. Use rope caulk (46–287118P1) to provide an air–tight seal around entire perimeter at the back of the magnet. This is to assure maximum air flow between the RF coil and the Gradient Coil.

13–7–2 INSTALLING INSULATION IN MAGNET WITHOUT PASSIVE OR RESISTIVE SHIM

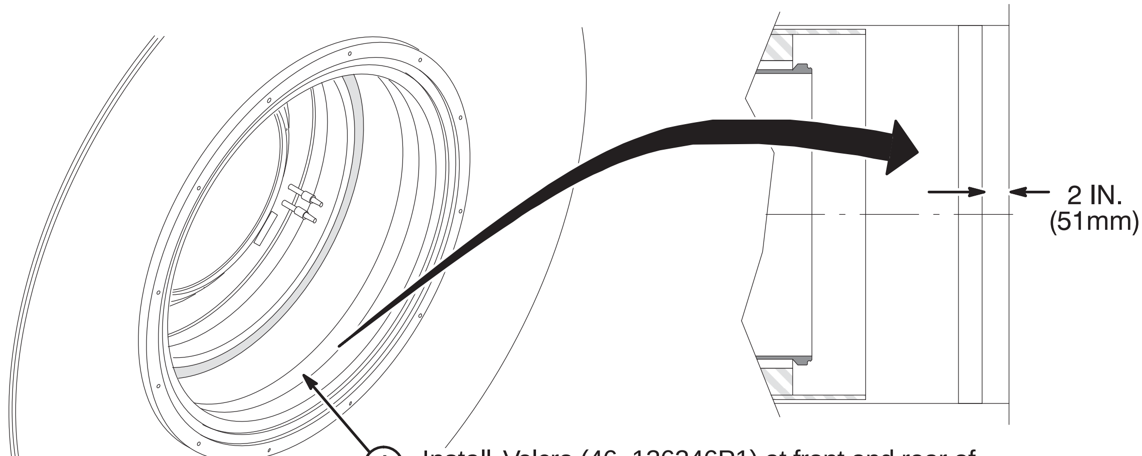


Note:
Rope caulk, pipe insulation, and acoustical collars are installed after body coil centering.

1. Fill the gap between the Magnet bore and outside of Gradient Coil at each end. Use Pipe Insulation (46–320868P1) and rope caulk (46–287118P1) around entire perimeter.
2. Use rope caulk (46–287118P1) as needed to supplement pipe insulation required for an air– tight seal between gradient coil and magnet bore at back of magnet. This is to assure maximum cooling air flow between the RF coil and the Gradient Coil.

13–8 INSTALL ACOUSTIC COLLAR

13–8–1 INSTALLATION OF VELCRO IN MAGNET BORE

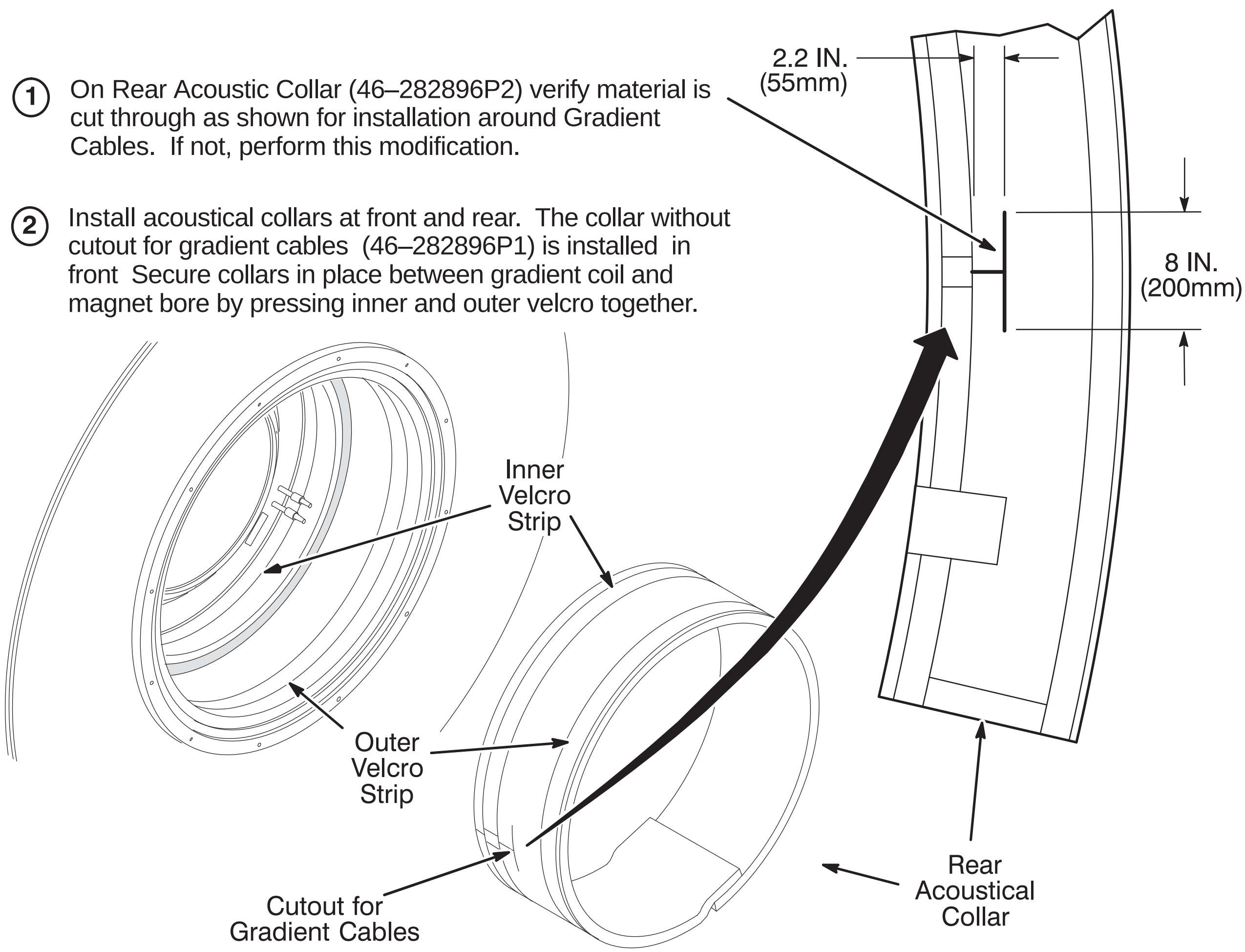


1 Install Velcro (46–136246P1) at front and rear of Magnet bore as shown, 2 inches from the oute edge.

13–8–2 ATTACH ACOUSTIC COLLAR

1 On Rear Acoustic Collar (46–282896P2) verify material is cut through as shown for installation around Gradient Cables. If not, perform this modification.

2 Install acoustical collars at front and rear. The collar without cutout for gradient cables (46–282896P1) is installed in front Secure collars in place between gradient coil and magnet bore by pressing inner and outer velcro together.



SECTION 14 – OPERATOR WORKSPACE

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14–1 INTRODUCTION

The equipment de–installed for the EXCITE HD upgrade depends on the existing site configuration. The general guideline for removal of equipment is:

- Advantage 4.x – De–install Operator Console and return for recycling
- Advantage 5.x or Horizon 5.x – De–install Operator Console and Computer. Determine if DAT and/or MOD drives are to be transferred to new GOC Operator Workspace.
- LX with Indigo – De–install modules on Operator Workspace table and Computer cabinet. Determine if DAT and/or MOD drives are to be transferred to new GOC Operator Workspace.
- LX with Octane – De–install indicated modules on Operator Workspace table and Computer cabinet. Determine if DAT and/or MOD drives are to be transferred to new GOC Operator Workspace.

Refer to applicable de–installation instructions in next section for removal of obsolete equipment.

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

14–2 DE–INSTALLATION OF OPERATOR WORKSPACE

14–2–1 LX WITH OCTANE OPERATOR WORKSPACE

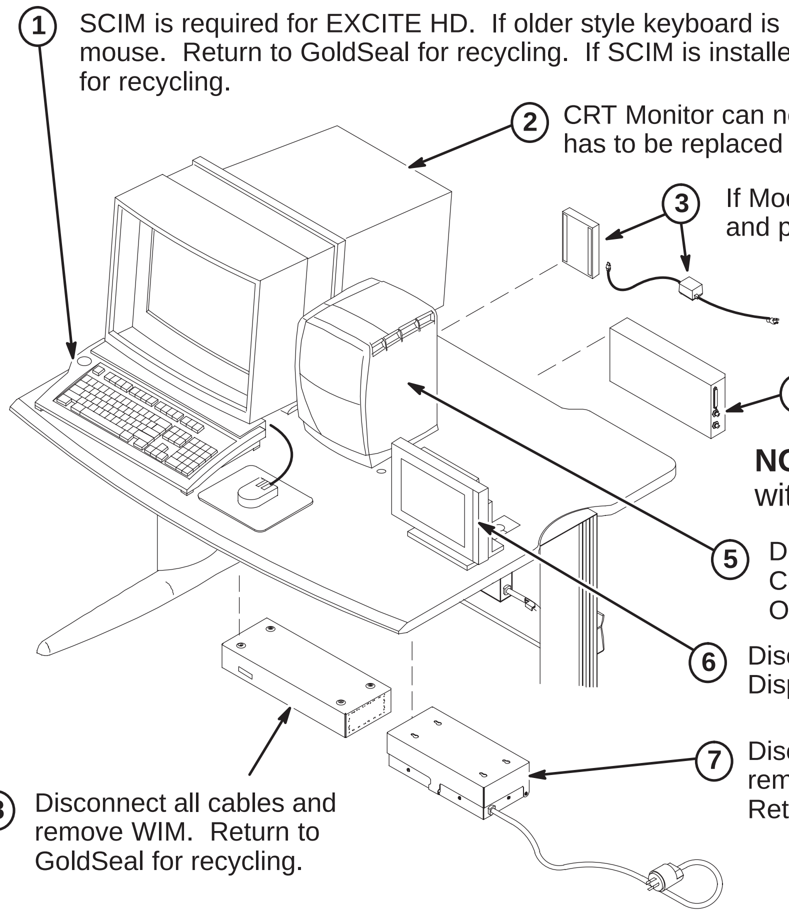
Confirm that all cables are disconnected from OperatorWorkspace and Computer. Perform the following procedure for return of the OW modules and Computer. Table will remain as part of upgrade.

14–2–1–1 De–Install Octane OW Table Modules

1. SCIM is required for EXCITE HD. If older style keyboard is part of existing site, remove keyboard and mouse. Return to GoldSeal for recycling. If SCIM is installed, retain for EXCITE HD but return cable for recycling.
2. CRT Monitor can not be used with EXCITE HD. Monitor has to be replaced by new LCD Host Monitor if it is a CRT.
3. If Modem is part of existing site, remove Modem and power cord and transfer to GOC Cabinet.
4. Remove DASM and cable to filming device and transfer to GOC Cabinet.

NOTE: DASM–1 is not compatible with the Linux PC, must use DASM–2.

5. Disconnect all cables from back of Octane Computer as indicated below and return Octane and cables to GoldSeal for recycling.
6. Disconnect all cables and remove LCD Display. Return to GoldSeal for recycling.
7. Disconnect all power cables and remove Power Distribution Box. Return to GoldSeal for recycling.
8. Disconnect all cables and remove WIM. Return to GoldSeal for recycling.



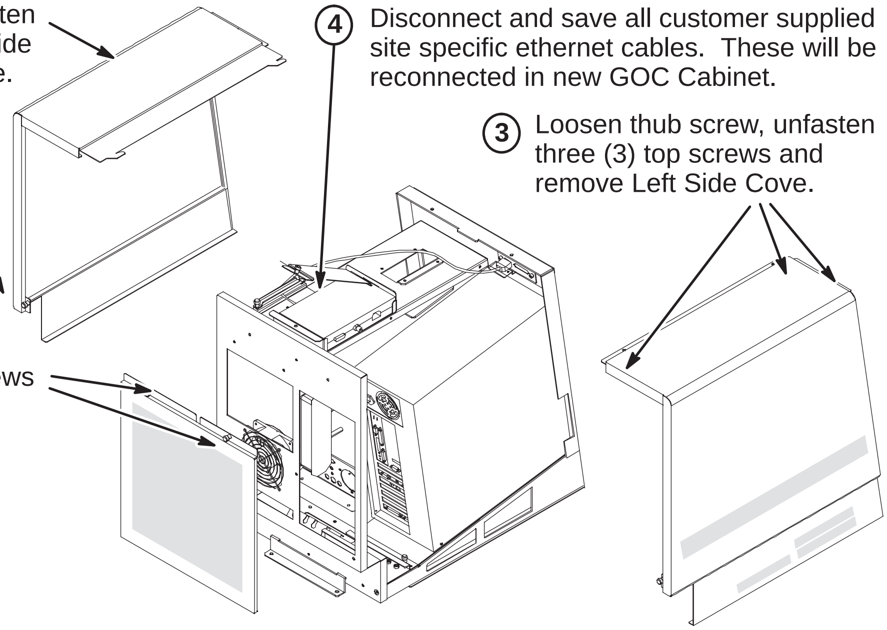
14–2–1–2 De–Install Octane Duplex Computer Cabinet

(2) Loosen thumb screw, unfasten top screw and slide Right Side Cover backwards to remove.

(4) Disconnect and save all customer supplied site specific ethernet cables. These will be reconnected in new GOC Cabinet.

(3) Loosen thub screw, unfasten three (3) top screws and remove Left Side Cove.

(1) Loosen (2) two thumb screws and remove Rear Cover.



(5) Refer to Section 14–3–1 for instructions on transferring Legacy Archive modules to EXCITE HD.

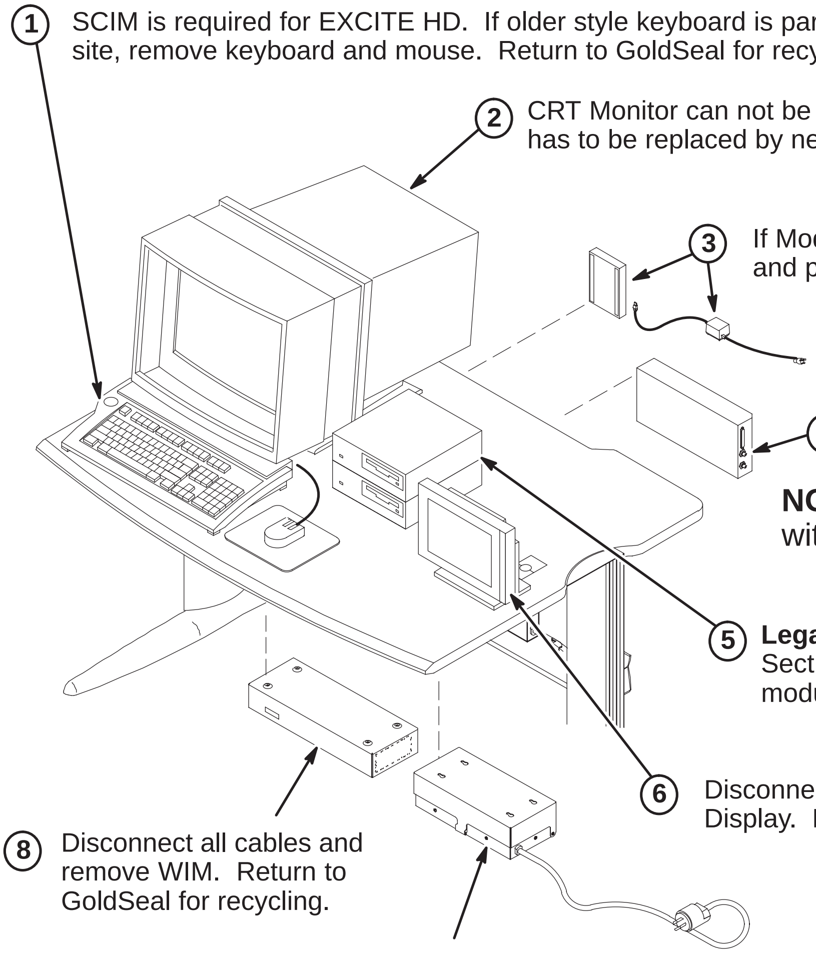
(6) The remaining Duplex cables will not be used on the new GOC Cabinet and must be returned for recyling to Gold Seal.

(7) The Duplex must be returned for recyling to Gold Seal. Duplex can be inserted onto skid that was used to ship the GOC by removing the white foam on the bottom that supported the GOC.

14–2–2 LX WITH INDIGO OPERATOR WORKSPACE

Confirm that all cables are disconnected from OperatorWorkspace and Computer. Perform the following procedure for return of the OW modules and Computer. Table will remain as part of upgrade.

14–2–2–1 De–Install indigo OW Table Modules



1. SCIM is required for EXCITE HD. If older style keyboard is part of existing site, remove keyboard and mouse. Return to GoldSeal for recycling.
2. CRT Monitor can not be used with EXCITE HD. Monitor has to be replaced by new LCD Host Monitor if it is a CRT.
3. If Modem is part of existing site, remove Modem and power cord and transfer to GOC Cabinet.
4. Remove DASM and cable to filming device and transfer to GOC Cabinet.

NOTE: DASM–1 is not compatible with the Linux PC, must use DASM–2.

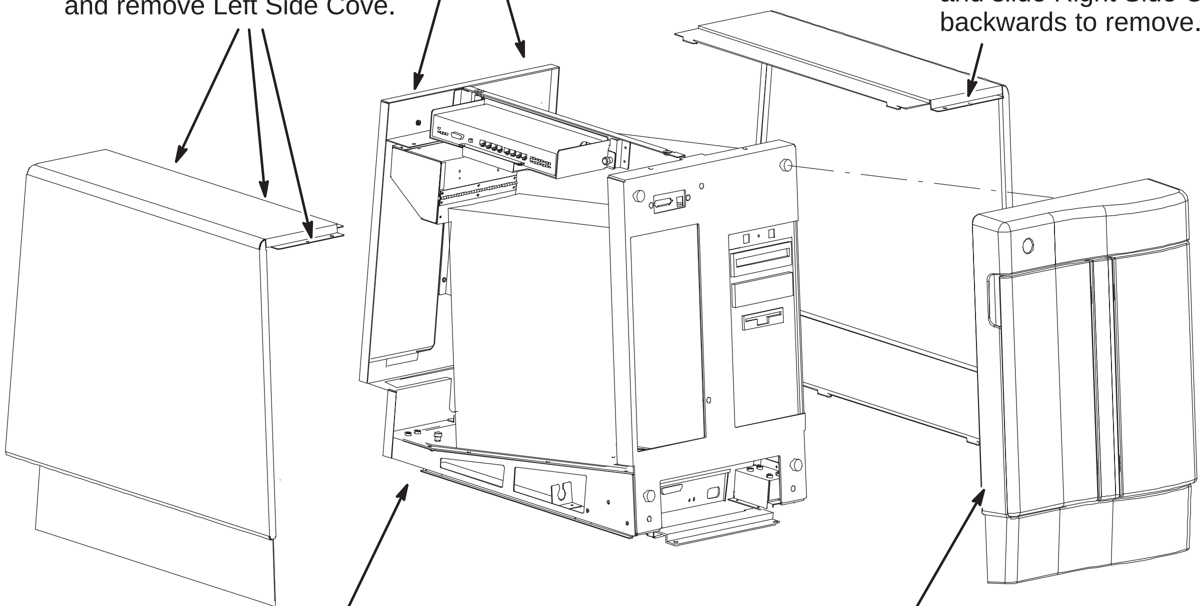
5. **Legacy Modules:** Refer to instructions in Section 14–3–2 for tranfer of DAT and MOD modules to EXCITE HD system.
6. Disconnect all cables and remove LCD Display. Return to GoldSeal for recycling.
7. Disconnect all power cables and remove Power Distribution Box. Return to GoldSeal for recycling.
8. Disconnect all cables and remove WIM. Return to GoldSeal for recycling.

14–2–2–2 De–Install Indigo Duplex Computer Cabinet

1 Loosen (2) two thumb screws and remove Rear Cover.

4 Loosen thumb screw, unfasten three (3) screws and remove Left Side Cove.

3 Loosen thumb screw at lower rear side, unfasten top screw and slide Right Side Cover backwards to remove.



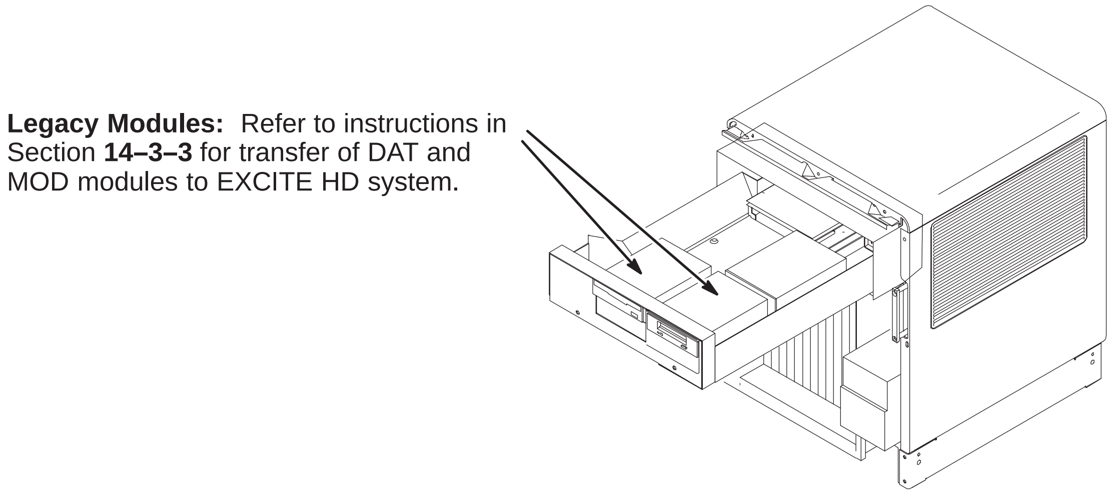
5 Disconnect all cables and return to GoldSeal for recycling.

2 Remove Front Cover by grasping each side of cover and pulling forward.

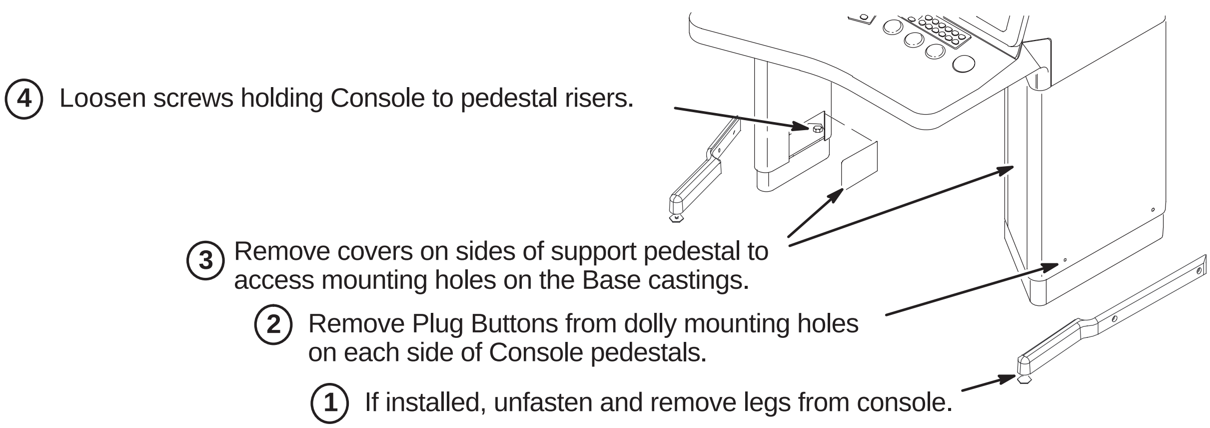
6 The Duplex cabinet must be returned to GoldSeal for recycling. Duplex can be inserted onto skid that was used to ship the GOC by removing the white foam on the bottom that supported the GOC.

14–2–3 ADVANTAGE 5.x OR HORIZON 5.x OPERATOR CONSOLE AND COMPUTER

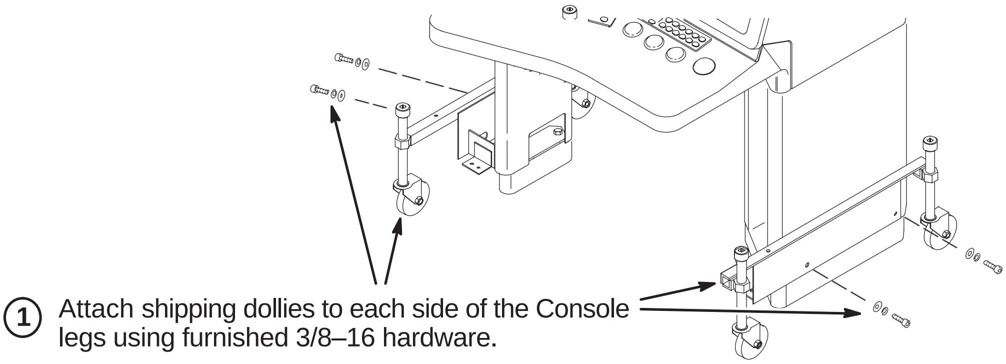
Confirm that all cables are disconnected from Operator Console and Computer. Complete the following procedure for return of the 5.x Operator Console and Computer.



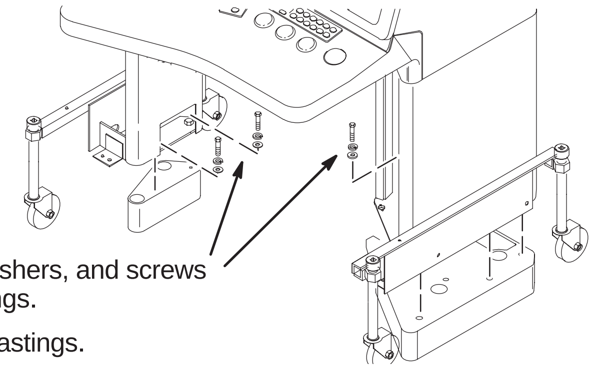
14–2–3–1 Prepare Operator Console Removal



14–2–3–2 Attaching Dolly To Operator Console

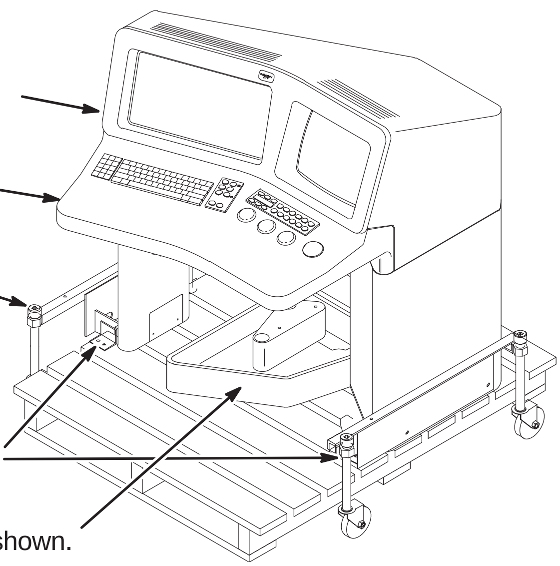


14–2–3–3 Removing Operator Console



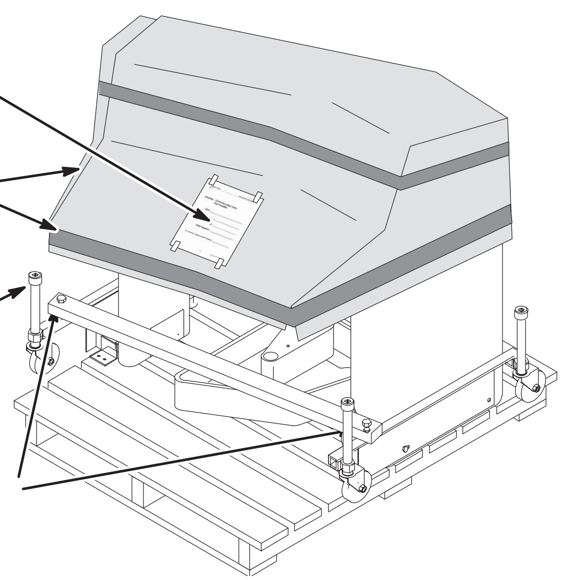
1. Remove lock washers, flat washers, and screws connecting legs to base castings.
2. Raise console up over base castings.

14–2–3–4 Fastening Operator Console To Shipping Pallet



1. Carefully roll Operator Console over an available shipping pallet.
2. Position Console so that Keyboard Housing does not extend beyond shipping pallet.
3. Use a 1/2 inch socket wrench to turn the jack screws and lower the Operator Console onto pallet.
4. Locate (4) four provided 5/16 x 1–1/2 long lag bolts.
5. Install two of the lag bolts thru each shipping dolly bracket to fasten to shipping pallet.
6. Nest together Riser castings on pallet as shown.

14–2–3–5 Final Preperation Before Shipping

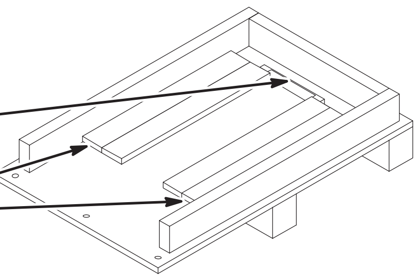


4. Fill out *GEMS Recycling Form* and attach to top of cardboard covering console.
3. Using any available cardboard (i.e. empty shipping boxes), cover Operator Console and tape in place. This is to prevent damage when shipping console back to manufacturer.
2. Make sure that all four wheel assemblies are completely raised. If not, use 1/2 inch socket wrench to turn jack screws.
1. Install provided dolly stiffener with furnished 1/4–20 x 1–1/2 long screws and nuts.

14–2–3–6 Preparing Shipping Pallet

Confirm that all cables are disconnected from Computer. Perform the following procedure for return of 5.x Computer reusing the shipping pallet that the new GOC Computer Cabinet arrived in.

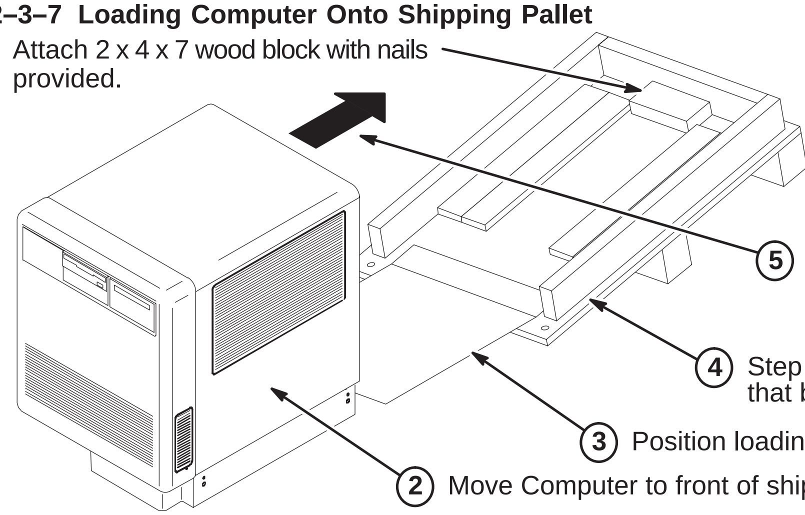
1. Locate the Operator Workspace Cabinet shipping pallet.
2. Remove existing bracket.
3. Re-position the protective foam on each side from a vertical to a horizontal position between the side of the pallet and the existing horizontal foam.



14–2–3–7 Loading Computer Onto Shipping Pallet

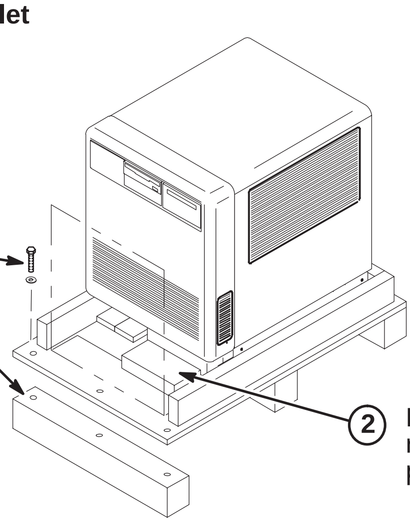
1. Attach 2 x 4 x 7 wood block with nails provided.
2. Move Computer to front of shipping pallet.
3. Position loading board at front of pallet.
4. Step on front corner of shipping pallet so that bottom board of pallet touches floor.
5. With another person, carefully roll Computer onto shipping pallet so it will rest on the horizontal foam pieces.

Note:
Pallet will return to horizontal position as Computer is rolled toward back of pallet.



14–2–3–8 Reassemble Shipping Pallet

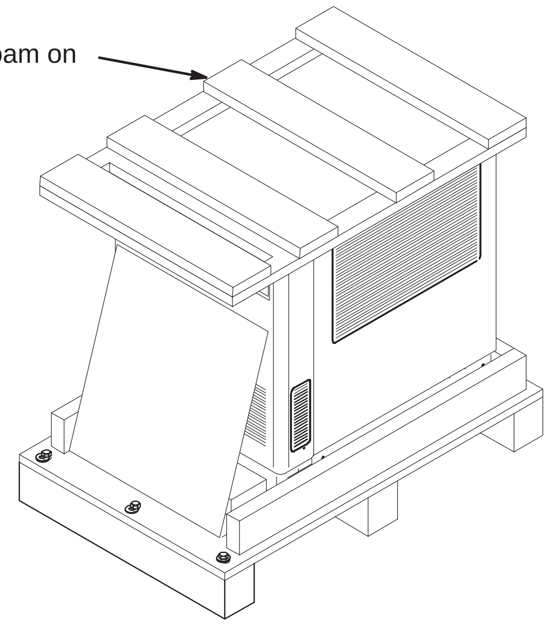
1. Re-attach wood block to pallet using the three previously removed fasteners.
2. Place 2 x 4 wood block against front right side of Computer and attach to pallet with nails provided.



14–2–3–9 Installing Computer Top Shipping Foam

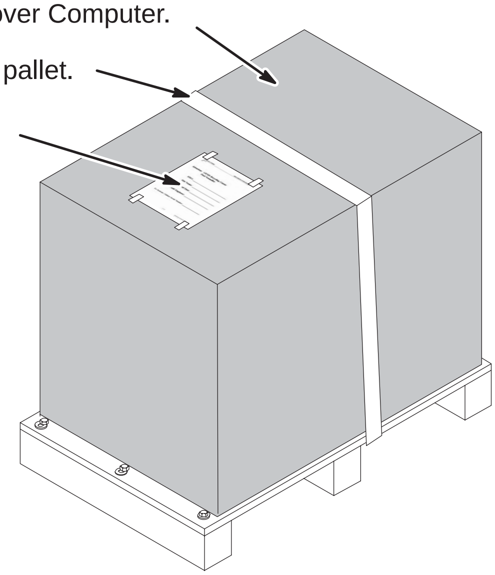
1. Install the Operator Workspace Cabinet shipping foam on top of Computer.

   Note: Shipping foam insert will not fit as well as it did with the new Operator Workspace Cabinet.



14–2–3–10 Final Preperation Before Shipping

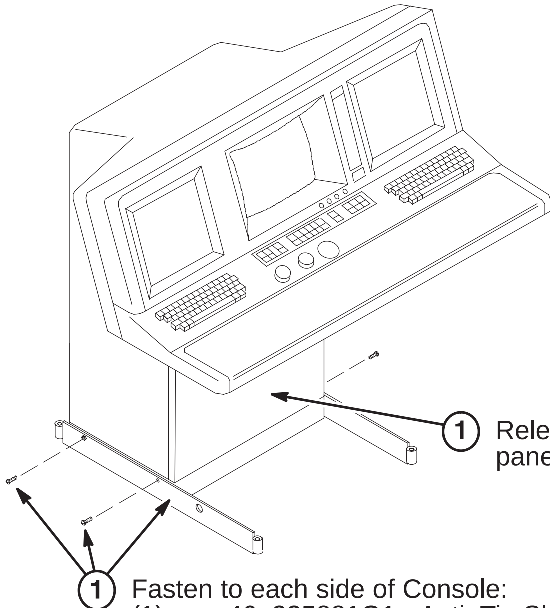
1. Place Operator Workspace shipping box over Computer.
2. Secure to shipping pallet.
3. Fill out *GEMS Recycling Form* and attach to top of shipping box.



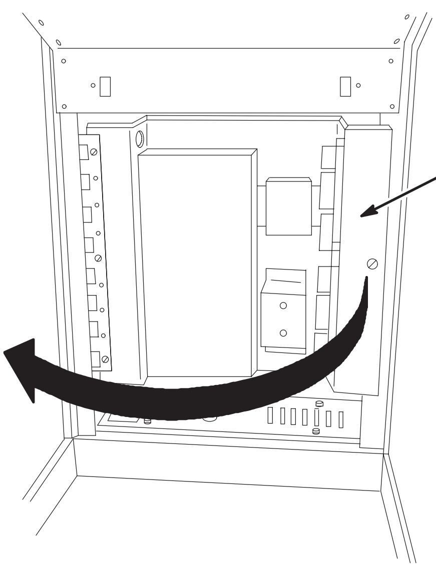
14–2–4 ADVANTAGE 4.x OPERATOR CONSOLE

Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of the 4.x Operator Console.

14–2–4–1 Install Operator Console Shipping Legs



14–2–4–2 Operator Console Removal



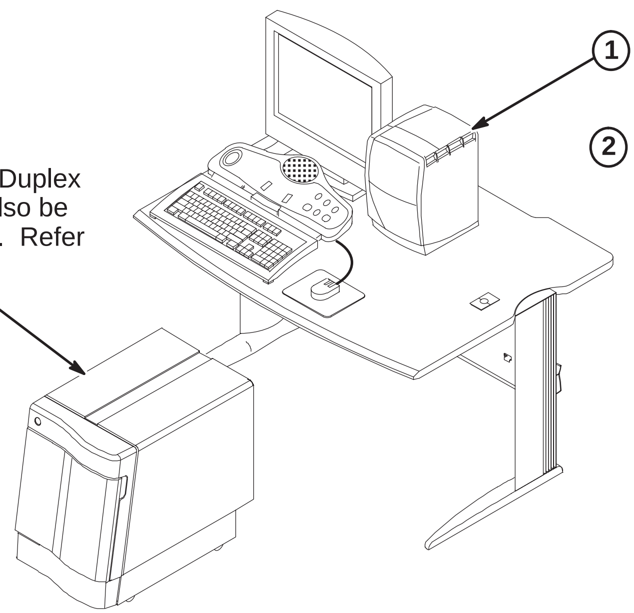
14–3 LEGACY ARCHIVE MEDIA SUPPORT

EXCITE HD does NOT support Legacy Archive Media, such as DAT and Pioneer MOD Archive Devices. Complete the applicable procedure for transfer of Legacy Archive Media for use with EXCITE HD.

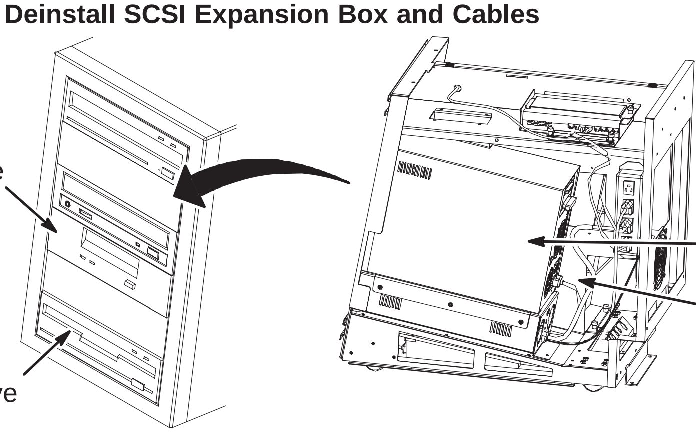
14–3–1 LEGACY MODULE UPGRADE FROM OCTANE

LX sites with Octane computers will retain their Octane computer and legacy Media Devices. This upgrade would require a Monitor (M1000NT) and Keyboard (M1000MN).

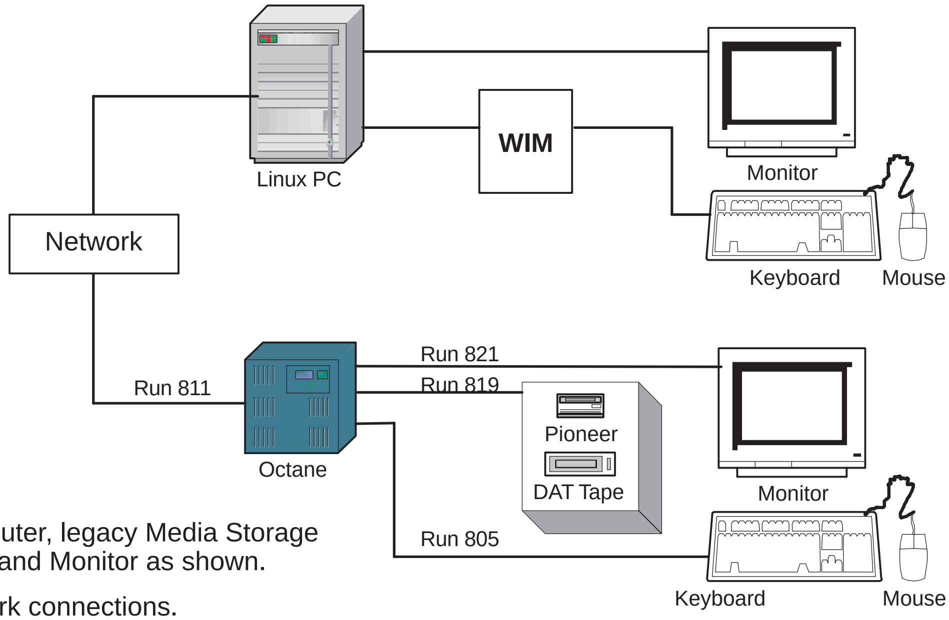
14–3–1–1 Deinstall Octane Computer



14–3–1–2 Deinstall SCSI Expansion Box and Cables



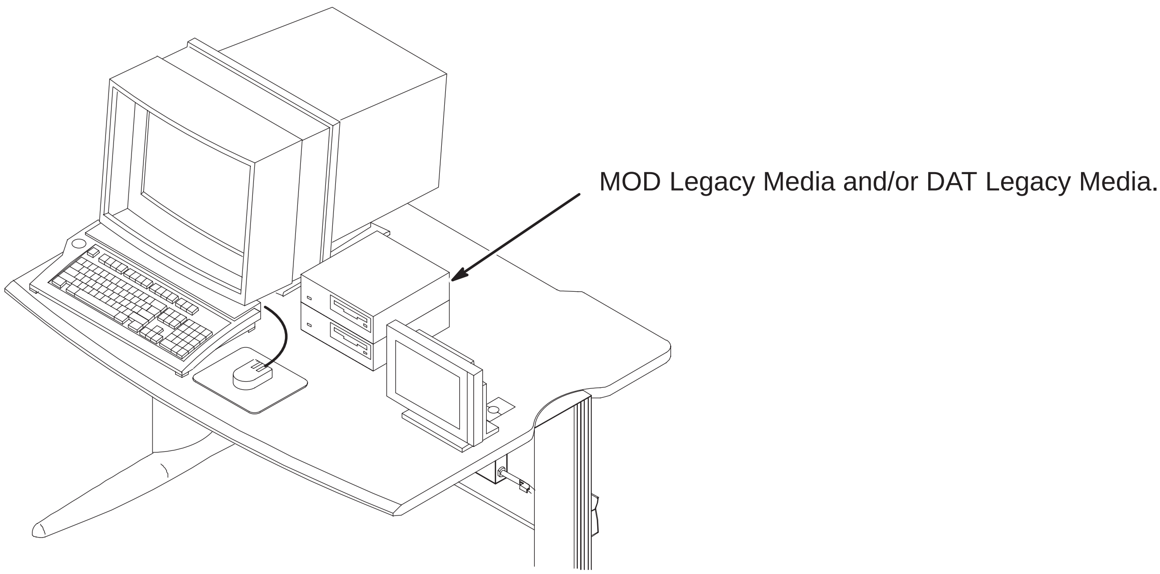
14–3–1–3 Hardware Configuration Interconnect



1. Connect the Octane Computer, legacy Media Storage Device, Keyboard, Mouse and Monitor as shown.
2. Connect power and Network connections.
3. After Octane computer is powered up and switched ON:
   1. Login as ’root’
   2. cd /export/home
   3. touch **.SHOW_MODE**
   4. Logout
   5. Login as ’signa’
4. The ’Octane’ system will now boot up in **SHOW_MODE** which will enable to further setup the DICOM Networknig to the different hosts to transfer images.

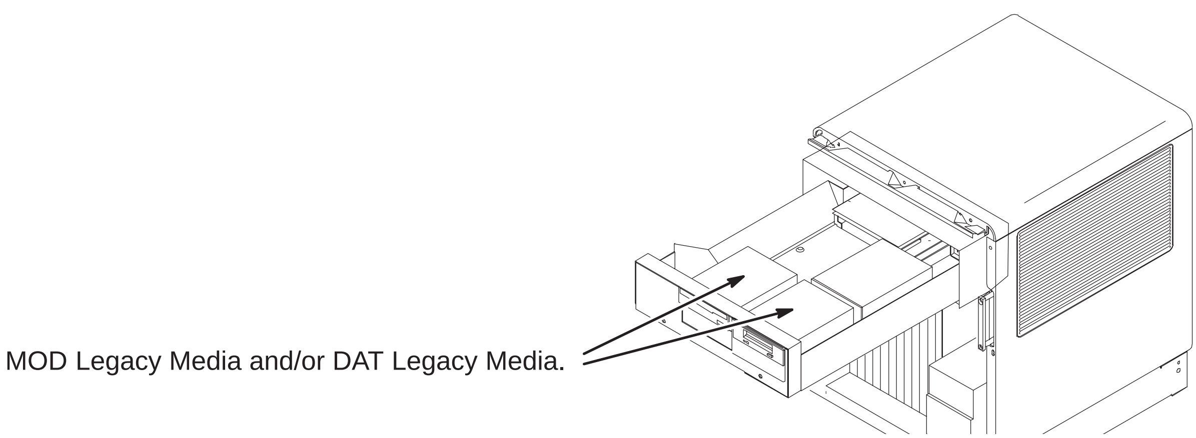
14–3–2 LEGACY ARCHIVING FOR INDIGO

At this time there is no upgrade path for customers requesting to retain Legacy Archiving capability that own systems with host computers prior to the Octane platform. Instruct the Customer to contact sales to discuss possible options.



14–3–3 LEGACY ARCHIVING FOR ADVANTAGE 5.x OR HORIZON 5.x

At this time there is no upgrade path for customers requesting to retain Legacy Archiving capability that own systems with host computers prior to the Octane platform. Instruct the Customer to contact sales to discuss possible options.

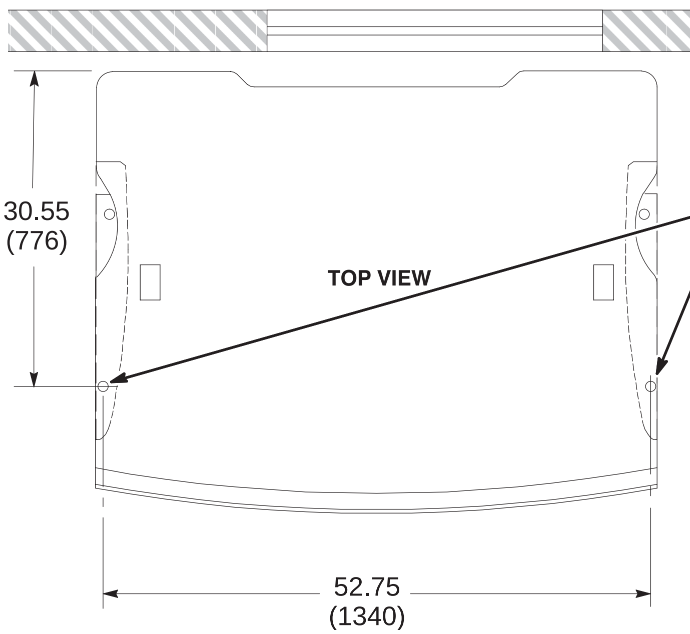


14–4 OPERATOR WORKSPACE UPGRADE FOR EXCITE HD

14–4–1 OPERATOR WORKSPACE TABLE INSTALLATION (If Needed)

A pre–LX site will not have an installed Operator Workspace Table. Refer to room layout plan and consult with Site customer for exact location of new Operator Workspace Table. If the site is an LX system, the table will be retained for this upgrade.

14–4–1–1 Operator Workspace Table Anchor Installation

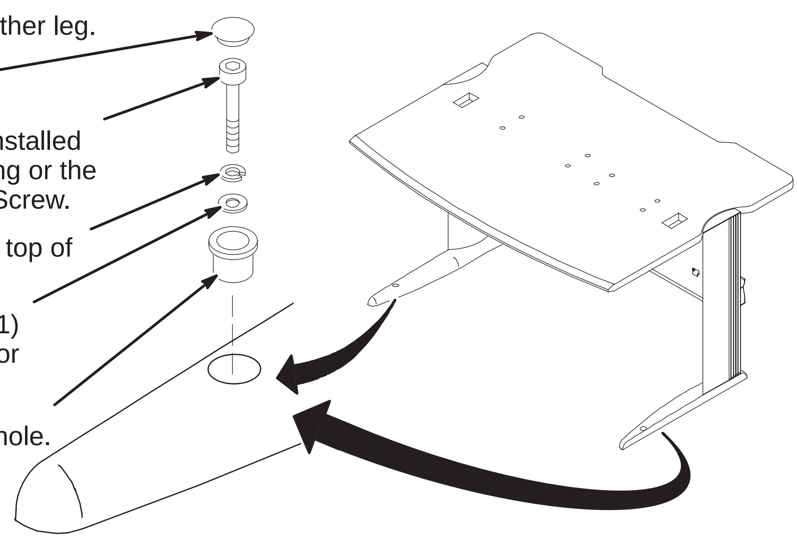


1. Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table.
2. Remove and discard the two (2) cover plugs. Locate and mark on floor the two hole locations at the front of the table legs.
3. Move table to prepare for installing the floor anchors in Steps 4 and 5.
4. At each hole marked in Step 2, drill a .38 inch hole and insert 1/4–20 x .375 diameter anchor (2168253).
5. Secure anchor in place with supplied Setting Tool (2168781).

14–4–1–2 Securing Operator Workspace Table to Floor

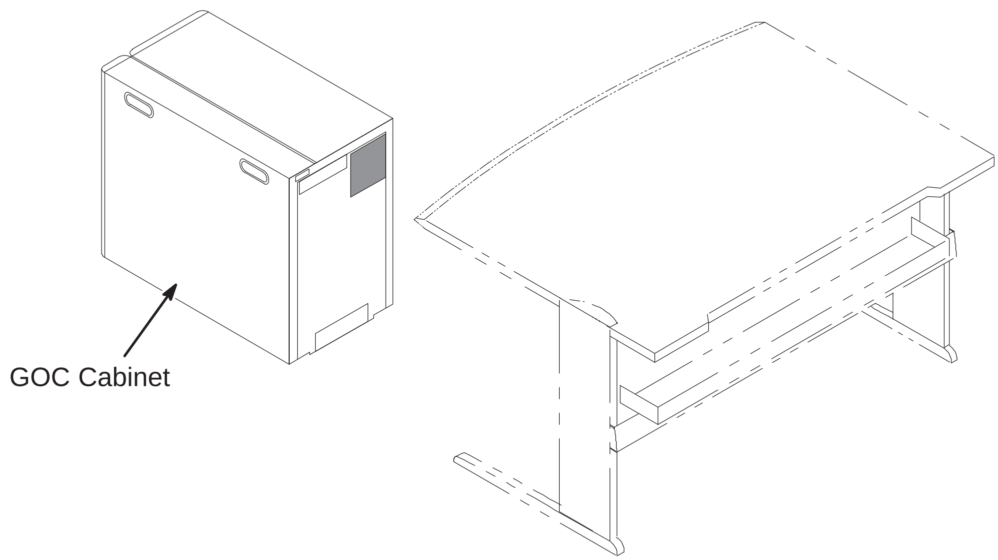
Items installed are part of Color Monitor Table Tie Down Kit, 2154928.

6. Repeat anchoring process below for other leg.
5. Install black nylon Plug (2168254).
4. Depending upon depth of previously installed anchor, install either the 1–3/4 inch long or the 2 inch long 1/4 x 20 Hex Socket Cap Screw.
3. Insert one Lock Washer (2168252) on top of Flat Washer(s).
2. Insert as many Flat Washers (2168251) as required (3 provided for each leg) for securing table to floor.
1. Insert Bushing (2151454) in table leg hole.



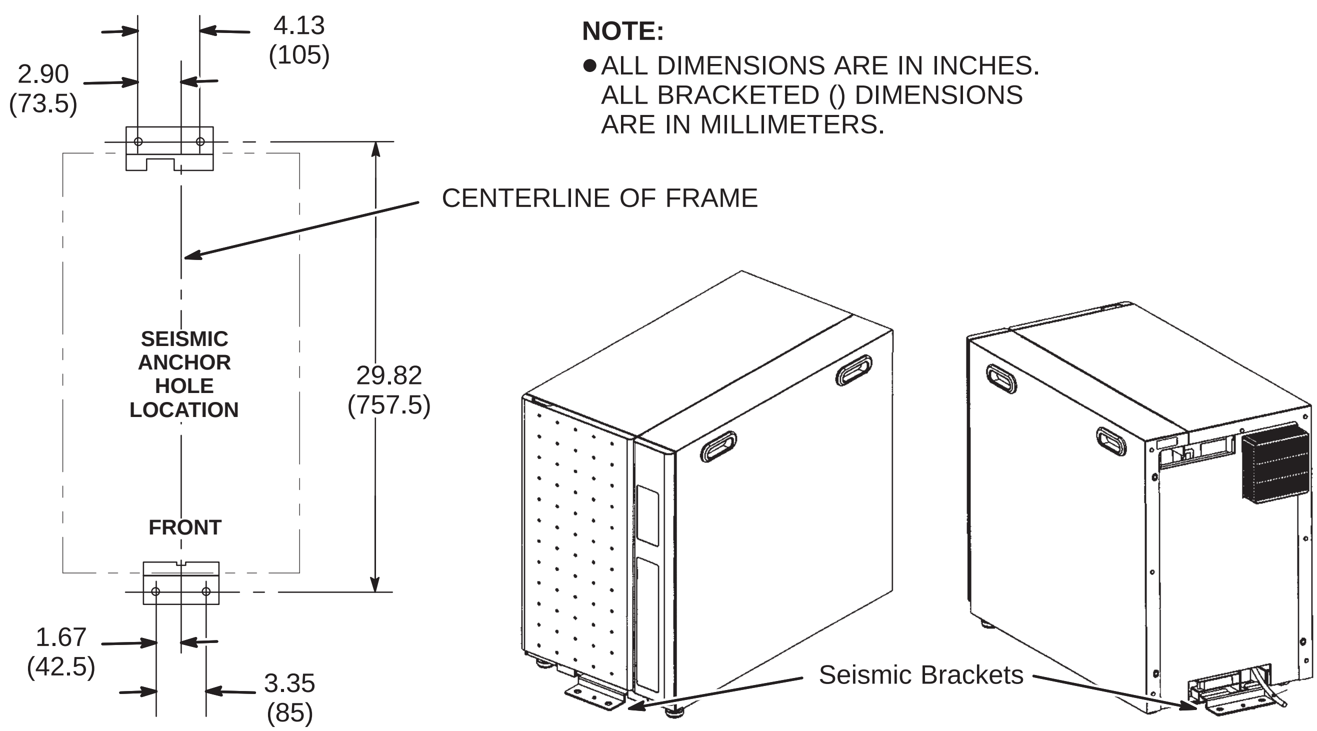
14–4–2 POSITION NEW GOC COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



14–4–3 SEISMIC ANCHORING (IF REQUIRED)

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre–installation manual. Illustration below indicates hole location for installing brackets.

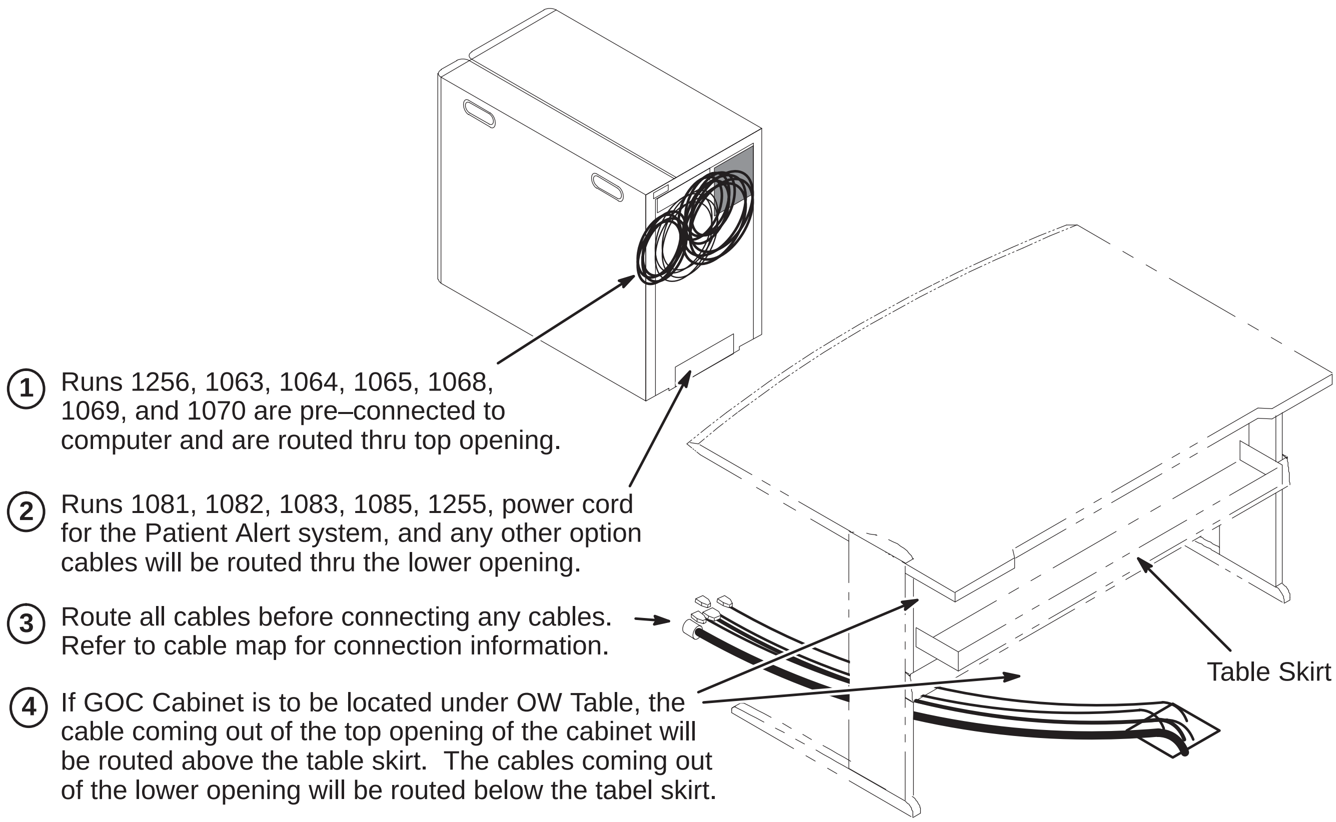


14–4–4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with seven cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

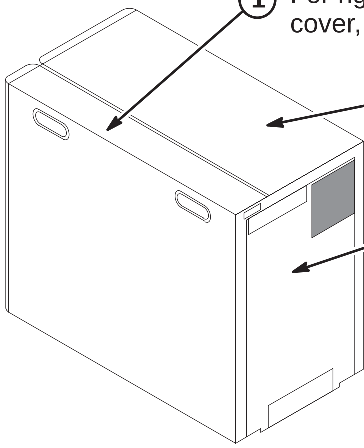
- Three power cords
- Two video ouput cables
- Keyboard output cable
- SCSI output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

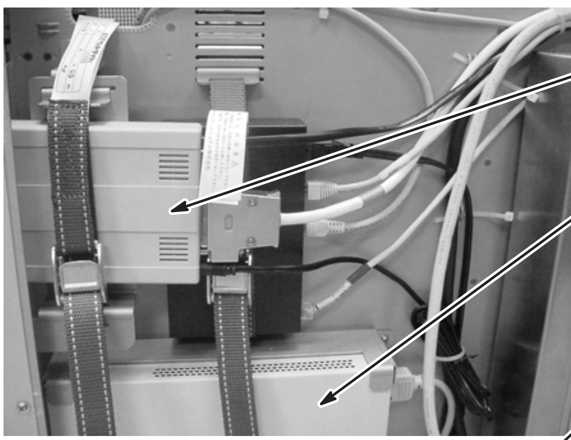


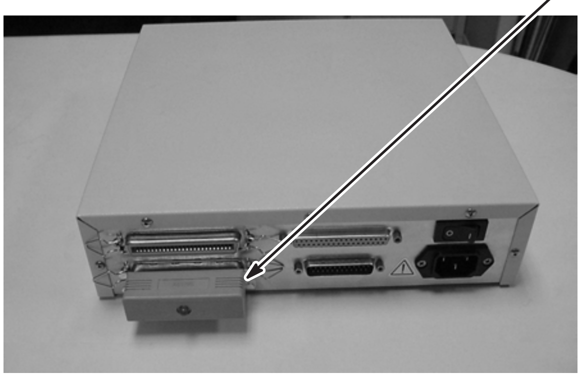
14–4–5 GOC CABINET COVER REMOVAL

1. For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
2. For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
3. Loosen four screws holding the back cover. Lift and remove cover.

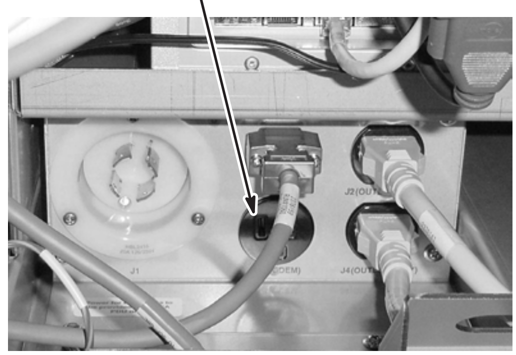


14–4–6 INSTALL MODEM AND DASM IN GOC CABINET (If Applicable)



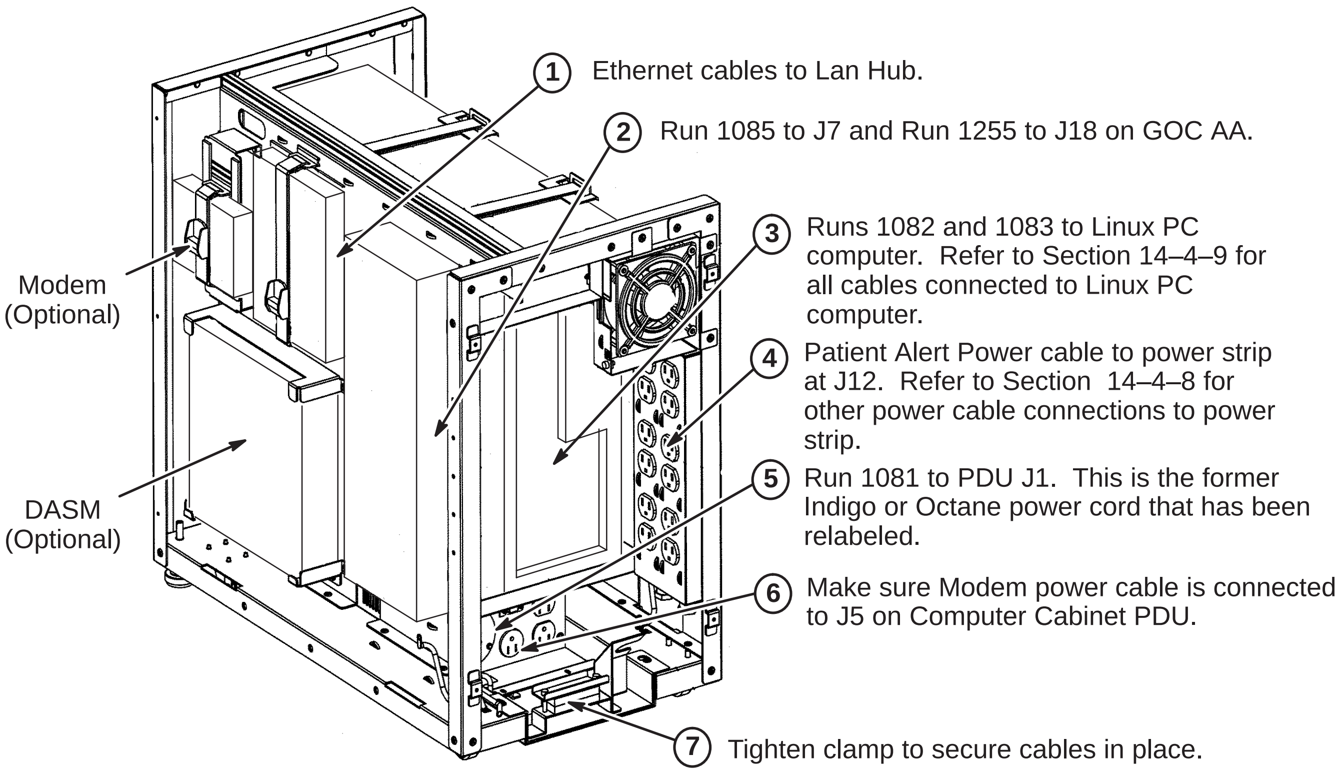


1. Mount Modem to GOC so that the underside is facing out to cover. Route customer provided phone lineand connect to Modem.
2. Mount DASM at this location. Route and connect cable from filming device.
3. Cable routed between DASM or Modem to the Linux PC are provided. Connect Run 1066 to Modem and Run 1067 to DASM.
4. If applicable, install the SCSI Terminator onto the bottom port of the DASM.
5. Verify that the **Active SCSI Terminator** located on the bottom of the DASM is in the **OFF** position.
6. Modem power cord must be connected to J5 on GOC PDU Module



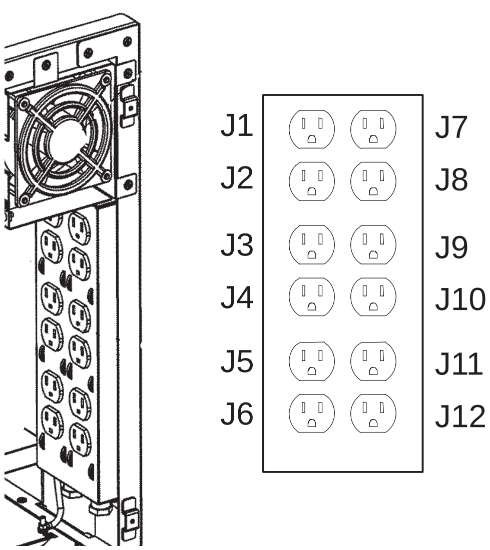
14–4–7 ROUTE CABLES FOR LOWER OPENING

Refer to cable map for information on cables to be connected thru lower opening.



14–4–8 POWER STRIP CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

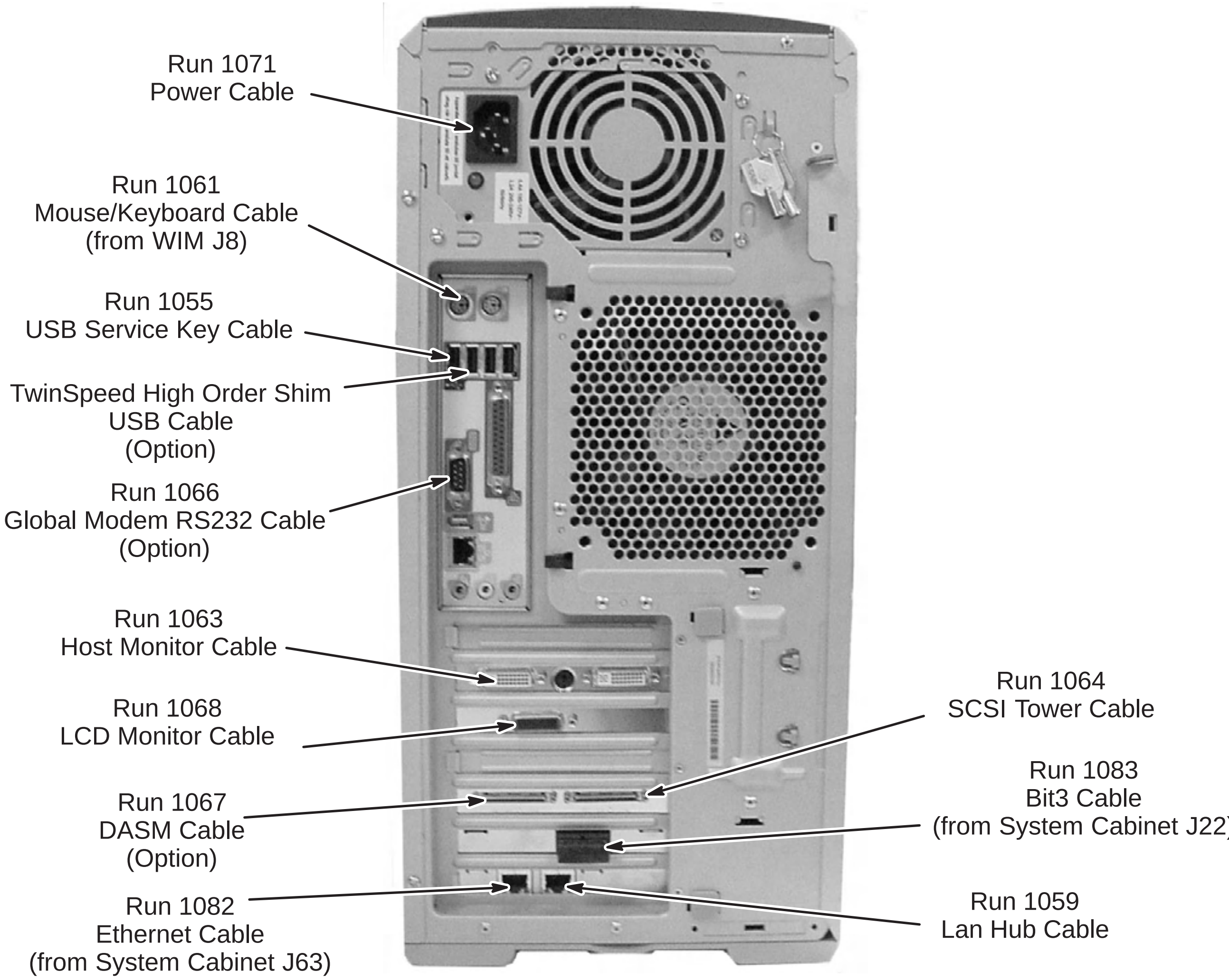


POSITION	NAME	MAX AMP CIRCUIT 1	MAX AMP CIRCUIT 2
J1	Host Comp	6.4	
J2	Host LCD	0.8	
J3	Fan	0.2	
J4			
J5			
J6			
J7	LAN Hub		0.2
J8	2nd LCD		0.8
J9			
J10	DASM		0.6
J11	SCSI Tower		7.0
J12	Patient Alert		0.2
TOTAL		7.4	8.8

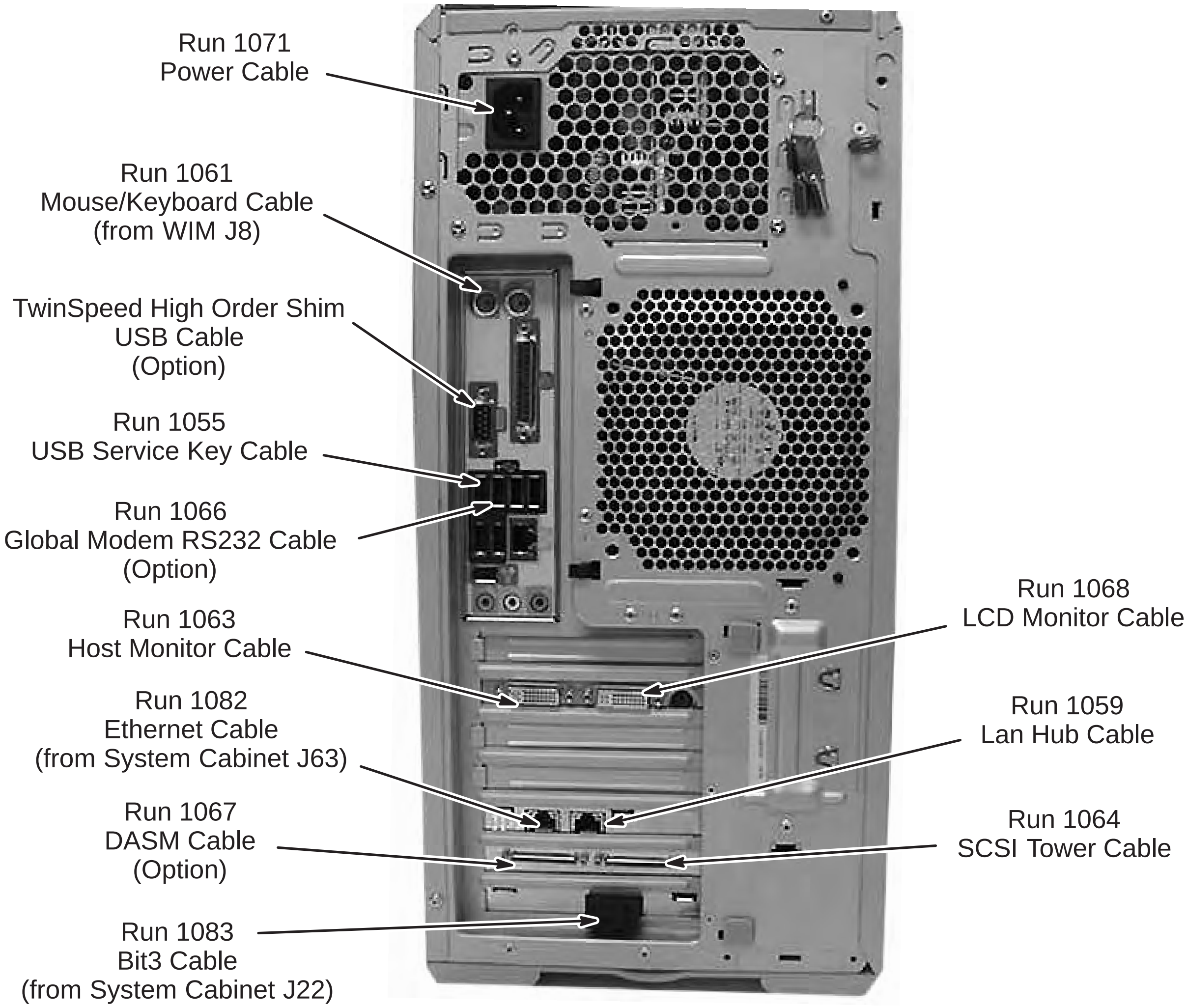
14–4–9 LINUX PC CABLE CONNECTIONS

The connection points of cables routed to the HP 8000 and the HP 8200 PC's are described in the following two sections. All cables shown are preconnected except for Runs 1082 and 1083, which have to be routed and connected at site. Refer to the applicable PC for your site.

14–4–9–1 HP 8000 Connection Points



14–4–9–2 HP 8200 Connection Points



14–4–10 INSTALL SCIM/KEYBOARD

The Scan–Control Intercom Module (SCIM) is the current keyboard on all Signa LX and OpenSpeed systems. Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.

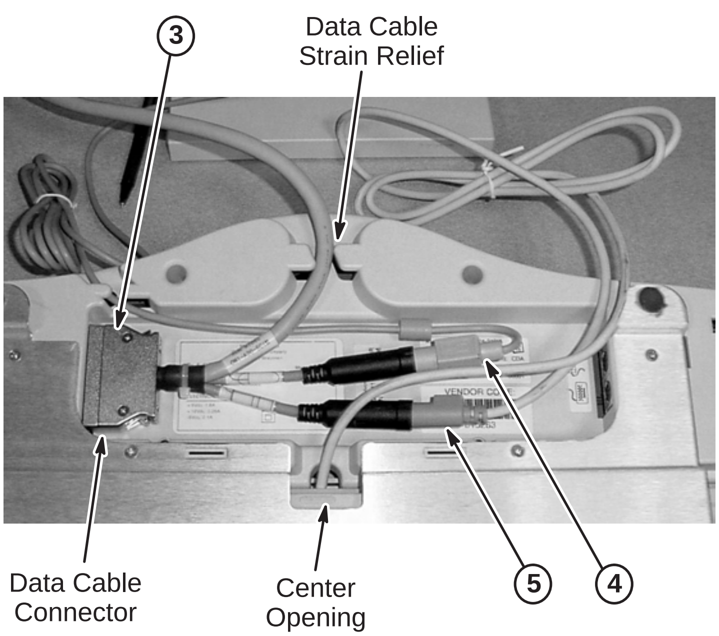
1. Remove the SCIM, Data Cable, and Keyboard From the box.

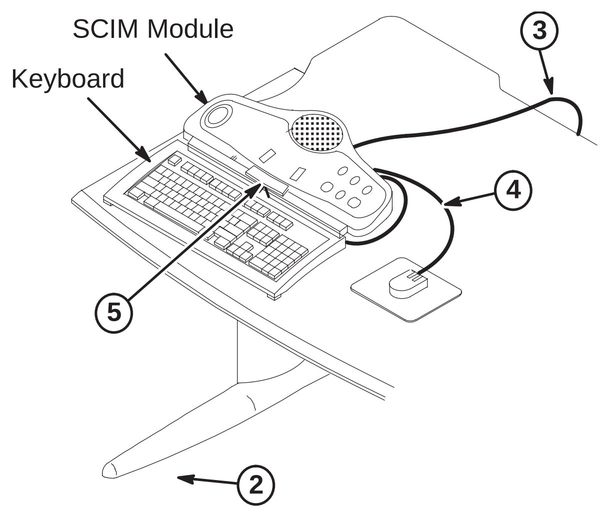
SCIM Module

Data Cable

Keyboard

2. Turn the SCIM unit up side down exposing the cable connections.
3. Route data cable Run 1262, preconnected to the GOC Cabinet and is located on back side of cabinet, from Computer Cabinet to bottom side of SCIM Module. Route thru data cable strain relief and connect as marked. (Run 805, supplied with SCIM Kit (2307175) is not used for EXCITE)
4. Attach Run 1262 cable marked with mouse icon to the Mouse. Route the cable thru the strain relief notches at the rear of the SCIM.
5. Guide the keyboard cable thru the center opening and attach to the jack on Run 1262 with the keyboard icon. (Store excess cable in the base of the SCM by tucking it in the depressed area).

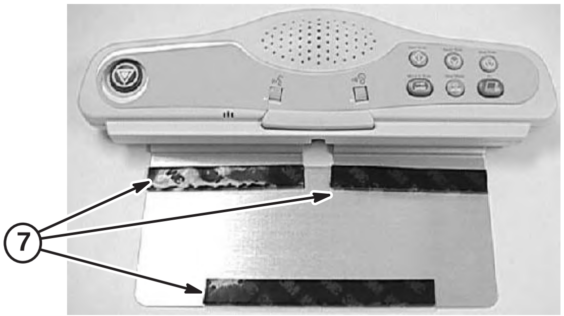


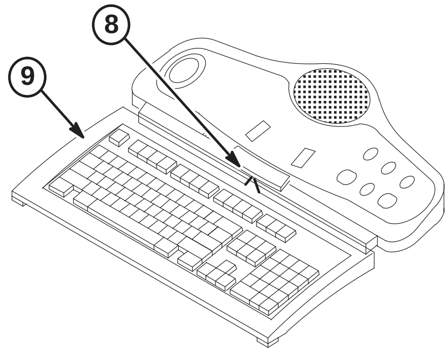


6. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
7. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface (see illustration on next page).

14–4–10 INSTALL SCIM/KEYBOARD (Continued)

8. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
9. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.

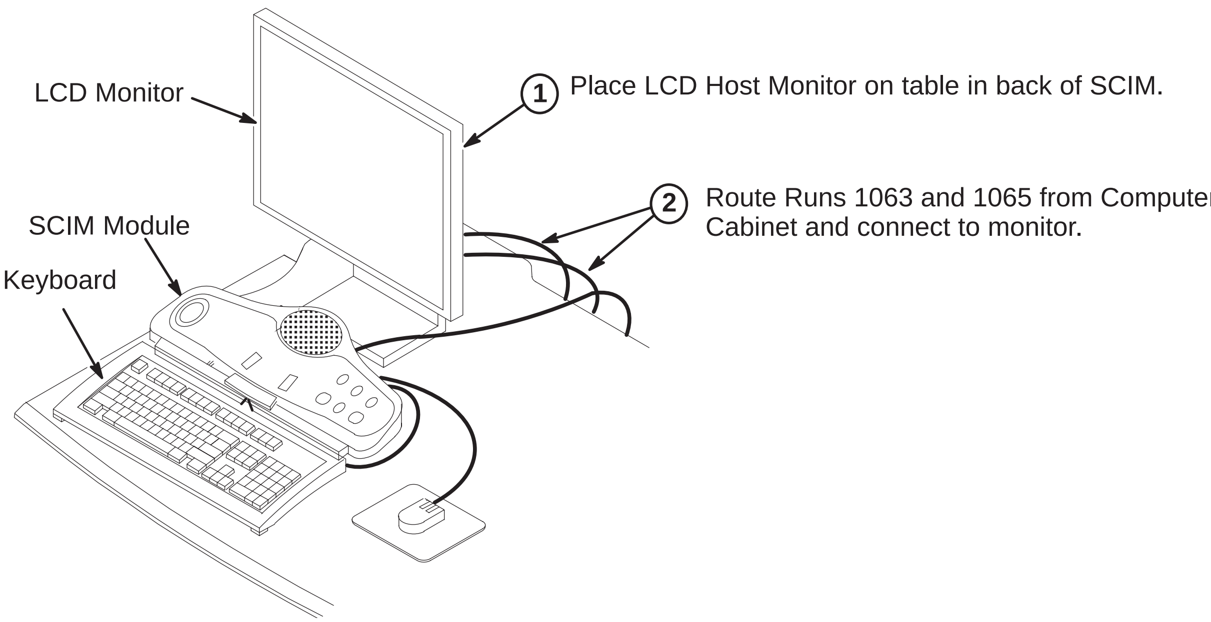




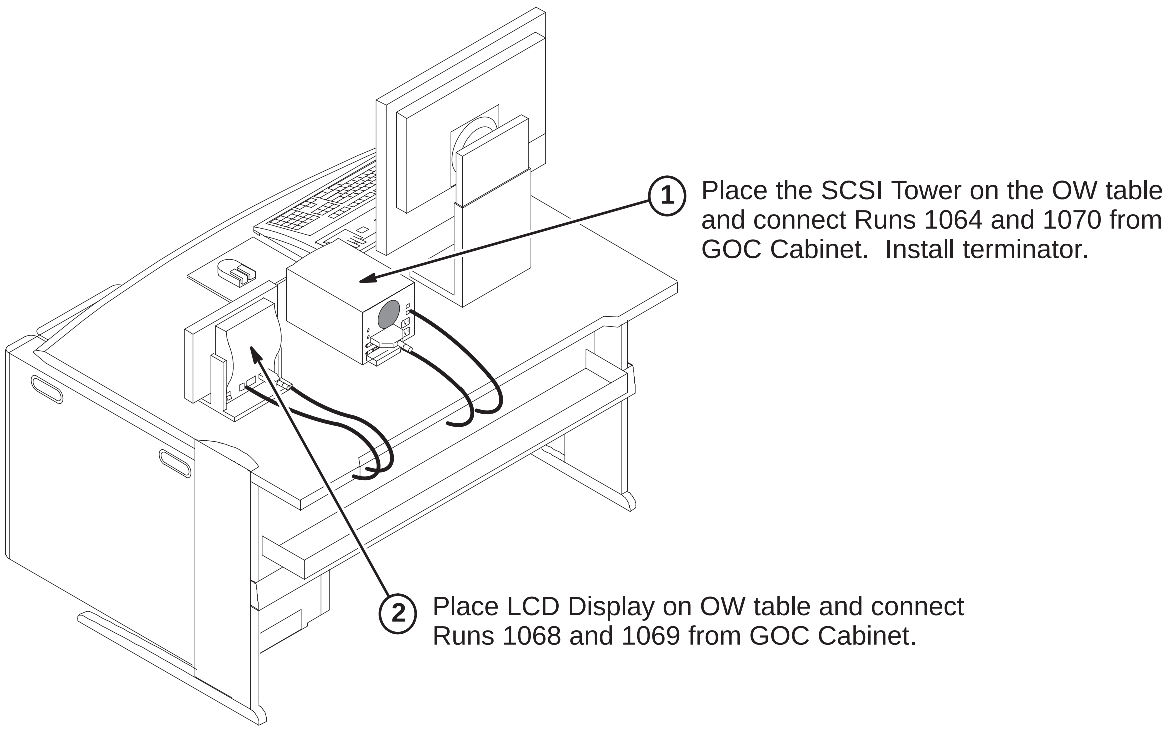
Note

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

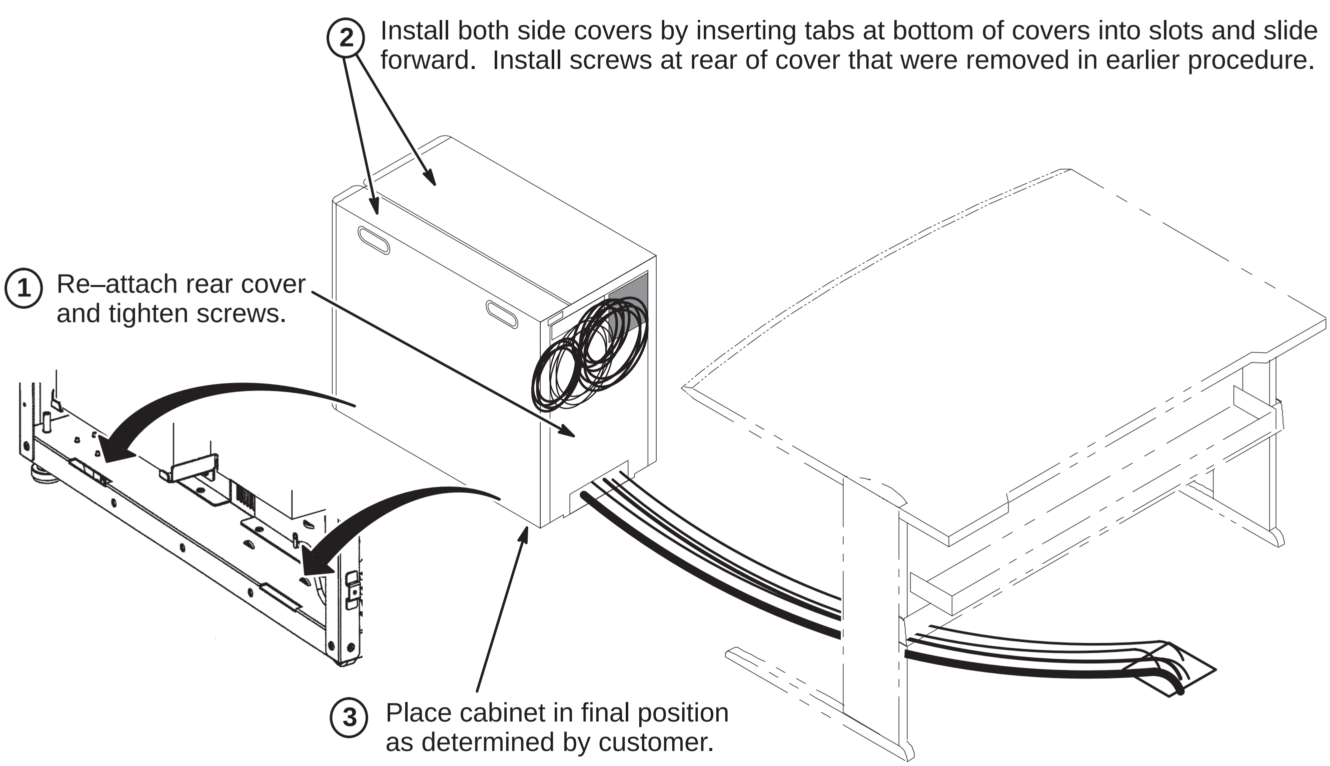
14–4–11 INSTALL MONITOR



14–4–12 INSTALL SCSI TOWER AND LCD DISPLAY



14–4–13 FINAL POSITIONING OF GOC CABINET



14–5 RETURN OF REPLACED PARTS

Refer to Section 1–7 for instructions on returning discarded parts.

1. Remove the white Styrofoam from the bottom of the GOC skid. The Duplex will now fit onto the bottom of the GOC skid. Place the removed Duplex onto the GOC Skid. Properly secure the top cardboard box onto the skid using straps. Return to GoldSeal.
2. Return the Octane, Indigo, and older computer systems to GoldSeal.
3. Return all cables that have been replaced by this upgrade to GoldSeal.

14–6 POWER ON INFORMATION

IMPORTANT
When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note
When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the WIM and Modem power.

SECTION 15 – HFD/PDU CABINET

TABLE OF CONTENTS

15–1 OVERVIEW

EXCITE HD systems require a new HFD/PDU cabinet or an HFD–S/PDU upgraded cabinet.

- Sites that have a pre–ACGD/PDU system will remove existing cabinets and install a new HFD/PDU cabinet. Refer to Sections **15–2** and **15–3**. Refer to Section **4–7–1** for connection of cables to HFD/PDU.
- Sites that have an ACGD/PDU installed will upgrade the cabinet to an HFD–S/PDU. To upgrade to HFD–S, M3000PT (1.5T non–TwinSpeed) or M3000PW (1.5T TwinSpeed), must be installed. Refer to Direction 5131778, *ACGD to HFD–S Upgrade Installation,* supplied with the catalogs for ugprade instructions. Also refer to Section **15–4** for connection of EXCITE HD cables to PDU.

All de–installed cabinets are to be returned for recycling. Refer to applicable cable removal map and applicable column in cable removal Table 4–1 for disconnection and removal of designated cables for system being de–installed.

DANGER!!

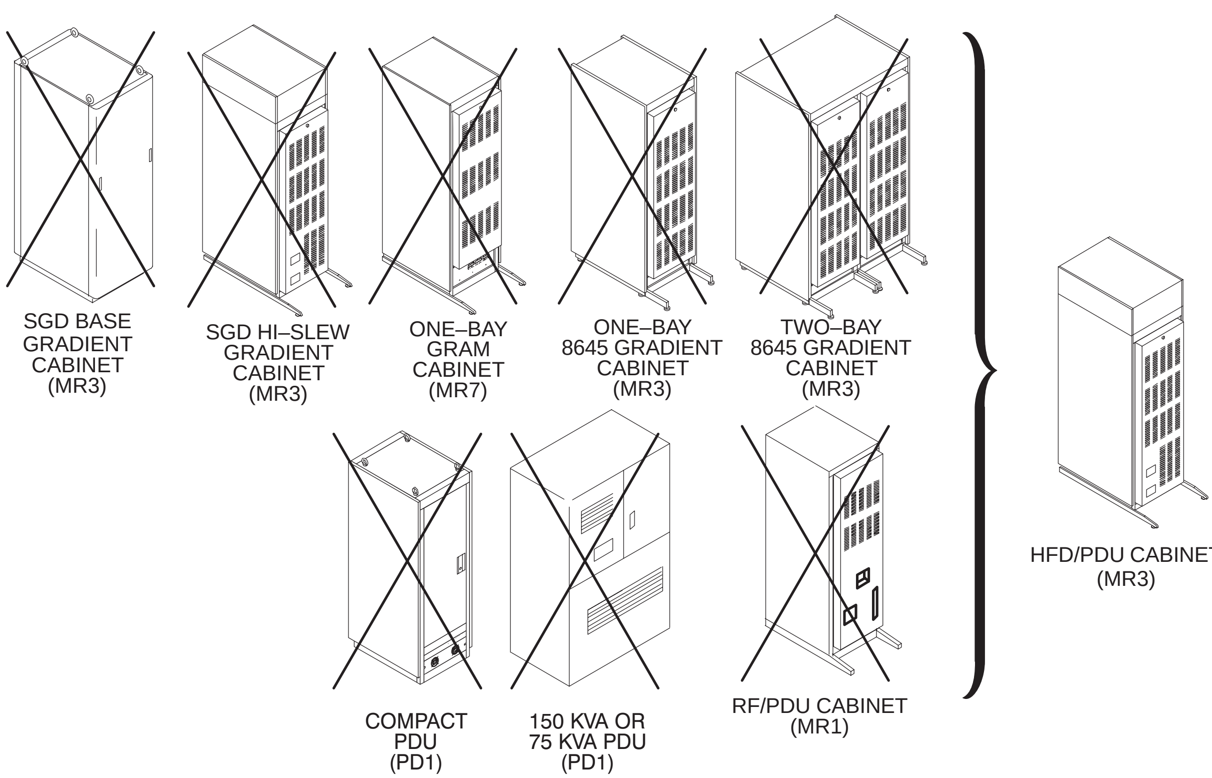
LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

15–1–1 HFD/PDU CABINET REPLACEMENT

The HFD/PDU Cabinet will replace one or more of the cabinets shown below.

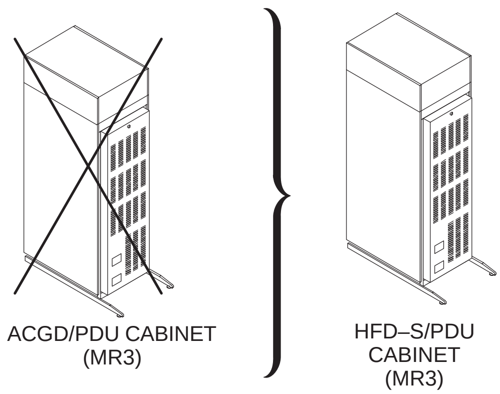
Positioning of cabinets is critical for installing cables and post–installation system operation. Refer to architectural site layouts for:

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)



15–1–2 HFD–S/PDU CABINET REPLACEMENT

The ACGD/PDU Cabinet will be retained and upgraded to an HFD–S/PDU Cabinet.



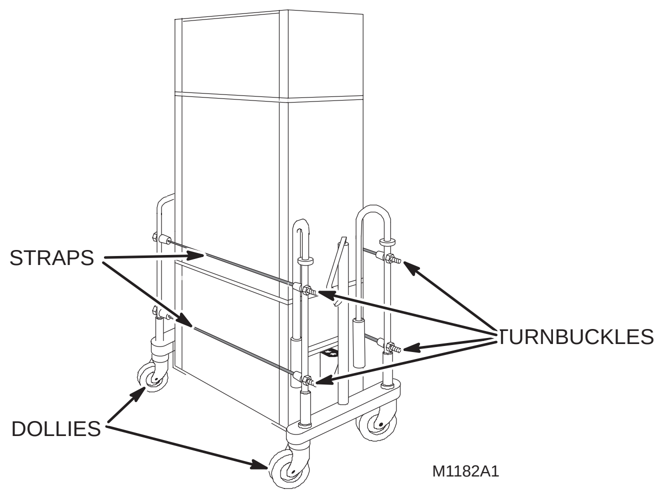
15–2 DE–INSTALL FULL–SIZE PDU (If installed on site)

15–2–1 SPECIAL HANDLING INSTRUCTIONS

All cabinets to be removed, except the Full–size PDU, have built–in casters for ease of movement. The full–size PDU does not have casters and will have to be moved with two shipping dollies placed on each side of the PDU and clamped in place using straps with turnbuckles as shown in illustration below. If dollies are not provided, the Full–size PDU will have to removed by riggers or other locally supplied equipment moving personnel.

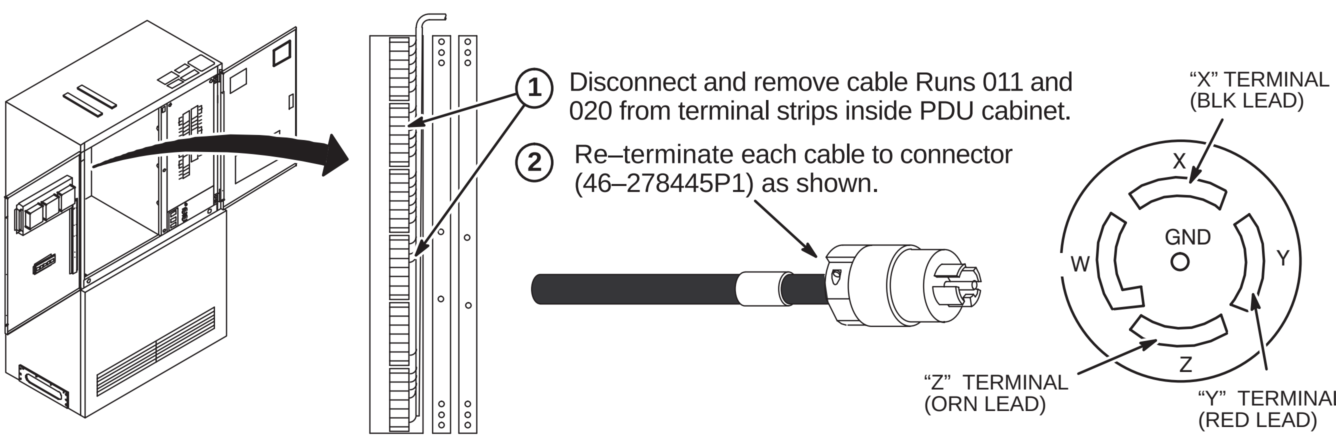
CAUTION

PDU weight is about 1000 pounds (455 kilograms). Plywood or steel plates may be used to distribute weight and protect floor surface from overload damage.



15–2–2 RETAIN RUNS 011 AND 020

If part of existing system, the shim and magnet power supply cables (Runs 011 and 020) need to be re–used for the new HFD/PDU cabinet. **DO NOT DISCARD RUNS 011 AND 020.** Refer to Illustration below for instructions on removing and reterminating Runs 011 and 020 for use with new HFD/PDU cabinet.



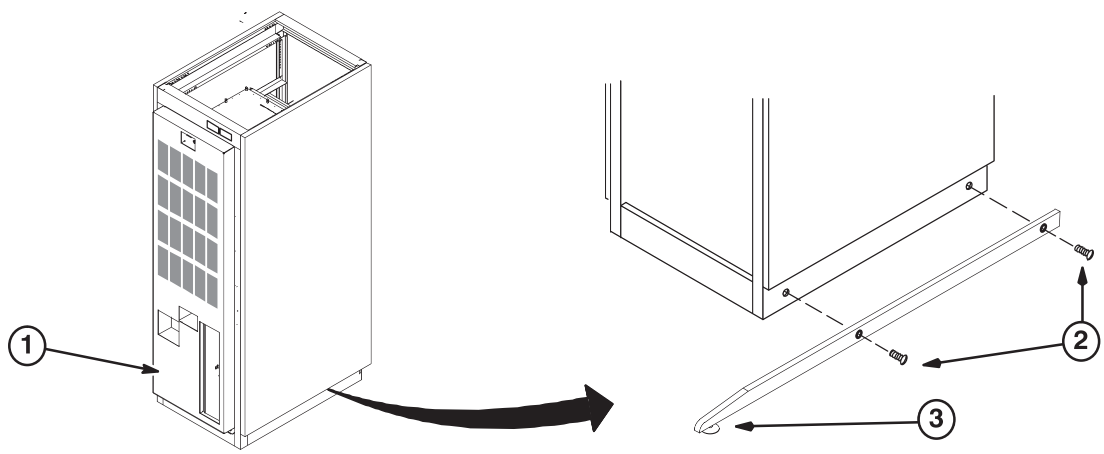
15–3 INSTALL HFD/PDU CABINET

15–3–1 POSITION HFD/PDU CABINET

WARNING!

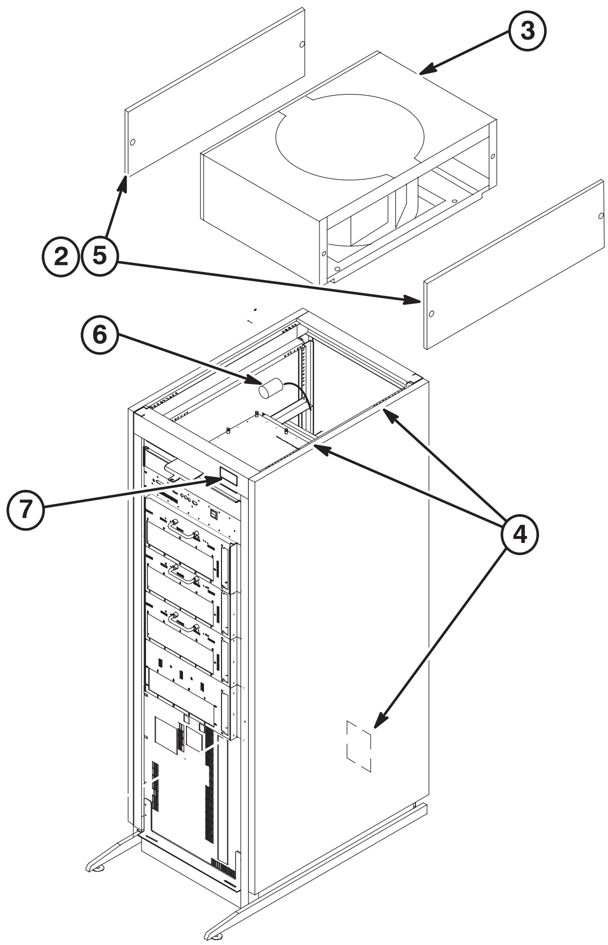
DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

1. Move cabinet to final position as indicated by site layout drawings.
2. Attach the anti–tip bars to cabinet. (Anti–tip bars are shipped separately)
3. Adjust leveling pads to level cabinet.



15–3–2 INSTALL FAN ASSEMBLY AND ATTACH CABINET IDENTIFICATION LABEL

1. Remove Fan Assembly from shipping container.
2. Remove two side trim panels (two screws from each).
3. Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.
4. Secure Fan Assembly in place with (4) four screws and washers, two on each side of cabinet. The fasteners are provided in a plastic bag located in PDU module. Rear PDU covers will need to be removed for access to plastic bags with fasteners.
5. Install two side trim panels with previously removed screws.
6. Connect cable from SGA Power Supply to the fan connector.
7. Make sure the "GRADIENT/PDU" cabinet magnet, part of the Cabinet ID Label Sheet (5126718), is attached to the front of cabinet as shown.

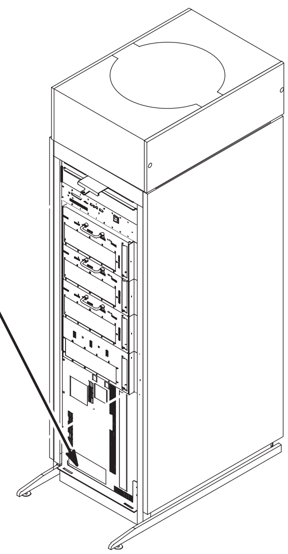


DANGER!!

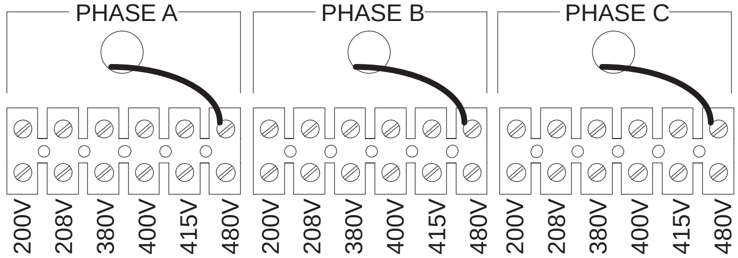
LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

15–3–3 INPUT VOLTAGE SELECTION

1. Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
2. Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
3. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.

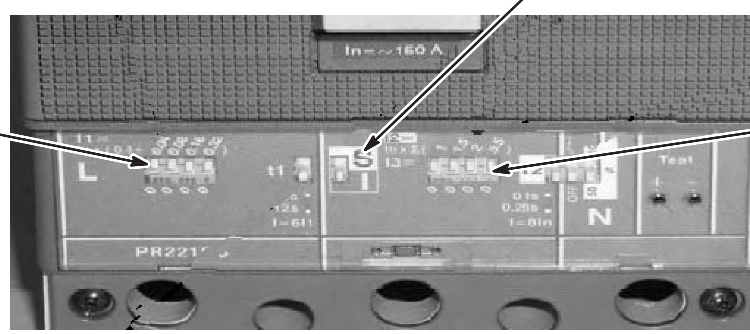


(SHOWN WIRED FOR 480V)



15–3–4 CIRCUIT BREAKER DIP SWITCH SETTINGS

1. Table below shows the position of the DIP switches for each input voltage selection. Set DIP switches according to input voltage. The white portion indicates the switch.
2. The "S/I" switch must be in the "I", down position



INPUT VOLTAGE	200	208	380	400	415	480
"L" SWITCH SETTINGS						
"I" SWITCH SETTINGS						

15–3–5 CABLE CONNECTIONS TO HFD/PDU CABINET

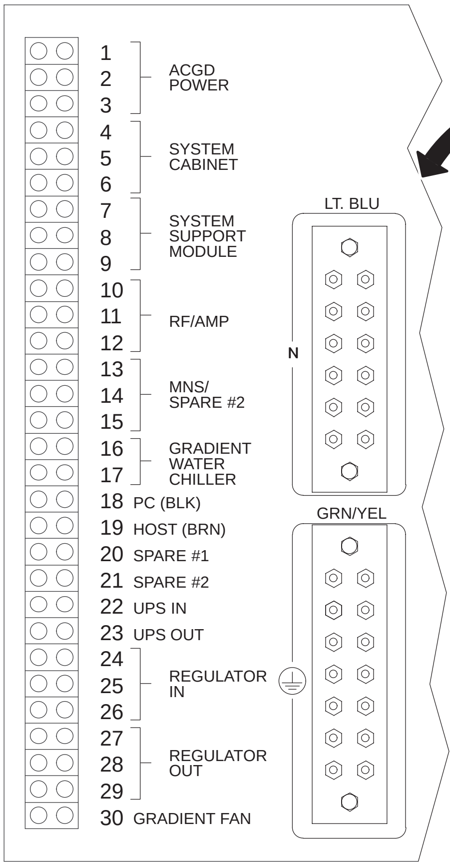
Refer to applicable cable “added” map for information on connecting new and retained cables to cabinet.

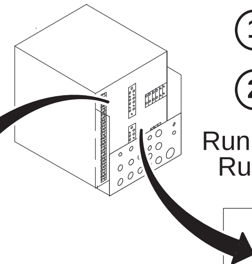
For gradient cables:

- Non–TwinSpeed sites will install new gradient cables in the equipment room.
- TwinSpeed sites will re–use the gradient cables that have not been disconnected from the retained TAC Cabinet. The other ends of the retained gradient cables will be connected to the new Penetration Panel and HFD/PDU Cabinet.

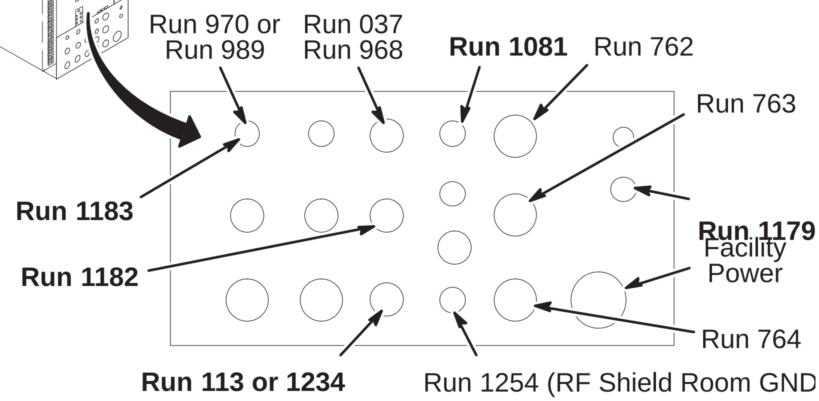
For all other cable connections, refer to applicable cable installation instructions in Section 4.

15–4 POWER CABLE CONNECTIONS TO HFD–S/PDU





1. Route new cable Runs 1081, 1179, 1182, 1183, and 1113 or1234 thru interface panel as marked.
2. Route any option Runs as marked.



WARNING!

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD–S/PDU CABINET.

3. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
4. Tighten clamps of interface panel to secure cables in place.

	CONNECT TO	RUN	PHASE	DESTINATION	
NON–TwinSpeed	4,5,6	968	ϕ A, B, C	RFS Cabinet	3ϕ, Neutral, GND
	7	1182	ϕ A	PHPS Module	Single ϕ Neutral, GND
	10,11,12	1234	ϕ A, B, C	RF Amp in RFS Cabinet	3ϕ, Neutral, GND
	16,17	970	ϕ A,B	Water Chiller	Single ϕ GND
	18,19	1081	ϕ C,A	Operator Workspace	Single ϕ, Neutral, GND
	30	1183	ϕ A	Blower Box	Single ϕ, Neutral, GND
	GRN/YEL	037		System Cabinet Ground Wire	
	GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)	
TwinSpeed	4,5,6	968	ϕ A, B, C	RFS Cabinet	3ϕ, Neutral, GND
	7	1182	ϕ A	PHPS Module	Single ϕ Neutral, GND
	10,11,12	1113	ϕ A, B, C	NB RF Amp Cabinet	3ϕ, Neutral, GND
	16,17	989	ϕ A,C	TAC Cabinet	Single ϕ, GND
	18,19	1081	ϕ C,A	Operator Workspace	Single ϕ, Neutral, GND
	(MDP)* or 30 (PDU)		ϕ A or ϕ A	AirSys Water Cooled Cabinet or AirSys Heat Exchanger	Single ϕ, Neutral, GND / Single ϕ, Neutral, GND
	GRN/YEL	037		System Cabinet Ground Wire	
	GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)	

COLOR CODE: for remainder of cables:
3ϕ = Black, Red, Orange Neutral = Light blue Single ϕ = Brown, Black Gnd = Grn/Yel

* Connection to Main Disconnect Panel to be completed by customer electrician.

SECTION 16 – RFS CABINET

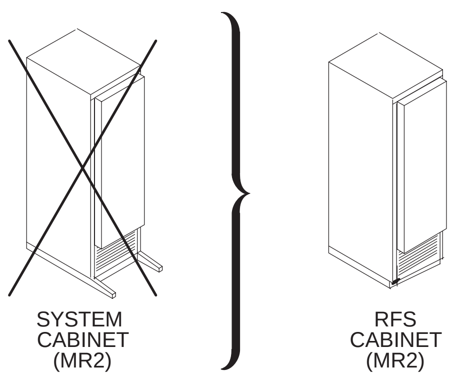
TABLE OF CONTENTS

16–1 OVERVIEW

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

The RFS Cabinet will replace the System cabinet shown below.



The de–installed System cabinet is to be returned for recycling. Refer to cable removal map in Section 4 for disconnection and removal of designated cables.

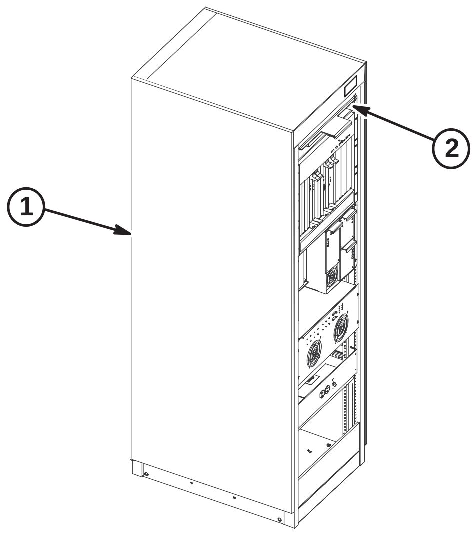
Positioning of cabinets is critical for installing cables and post–installation system operation. Refer to architectural site layouts for:

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

16–2 POSITION RFS CABINET

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

1. Position RF/System cabinet to final position as indicated by site layout drawings.
2. From the Cabinet ID Label Sheet (5126718), locate the ”RF/SYSTEMS CABINET” magnet and attach to front of cabinet as shown.



16–3 CABLE CONNECTIONS TO RFS CABINET

Refer to applicable cable “added” map in Section 4 for information on connecting new and retained cables to cabinet.

SECTION 17 – COMPLETION

17–1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non–GE personnel. Check off boxes at the front of each item below when complete.

- ❑ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ❑ Resolve shipment shortages
- ❑ Resolve omissions by mechanical contractors
- ❑ Have the on–site GE Healthcare Field Engineer, Service Engineer, or Project Mangager of Installation (PMI) to confirm that the blower box has been anchored properly and installed outside the minimum service area.
- ❑ Make sure all rating plates are installed
- ❑ Recheck all wiring connections using cable map (Section 4)
- ❑ Correct for wiring errors if necessary
- ❑ Complete Sections 17–2, 17–3, and 17–4.

17–2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible.** Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 17–3. Recycling Operation will route them to appropriate location.

17–3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1–8. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

17–4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1–9.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD–ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 11.x has been made before returning system to the users. Turn site over to applications who will instruct users.

17–5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD–ROM, Installation Calibration Flow, and perform the following:
   - PDU Voltage Checks Compact > PDU E–Stop Check
   - Startup Checks Compact
   - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.

SECTION 18 – APPENDIX

TABLE OF CONTENTS

18–1 PRE–EXCITE TO SIGNA EXCITE HD SYSTEM UPGRADE CONFIGURATIONS

18–1–1 SIGNA LX, HORIZON, 5.x, 4.x to EXCITE HD UPGRADES OVERVIEW

The EXCITE HD (Release 12.x) upgrade content for an existing 1.5T Signa LX, Signa Horizon, Signa 5.x, or Signa 4.x system is dependent on the configuration of the system being upgraded and the final product configuration. Table 18–1 lists illustrations which shows a high level summary of Release 12.x upgrade system equipment/cabinets.

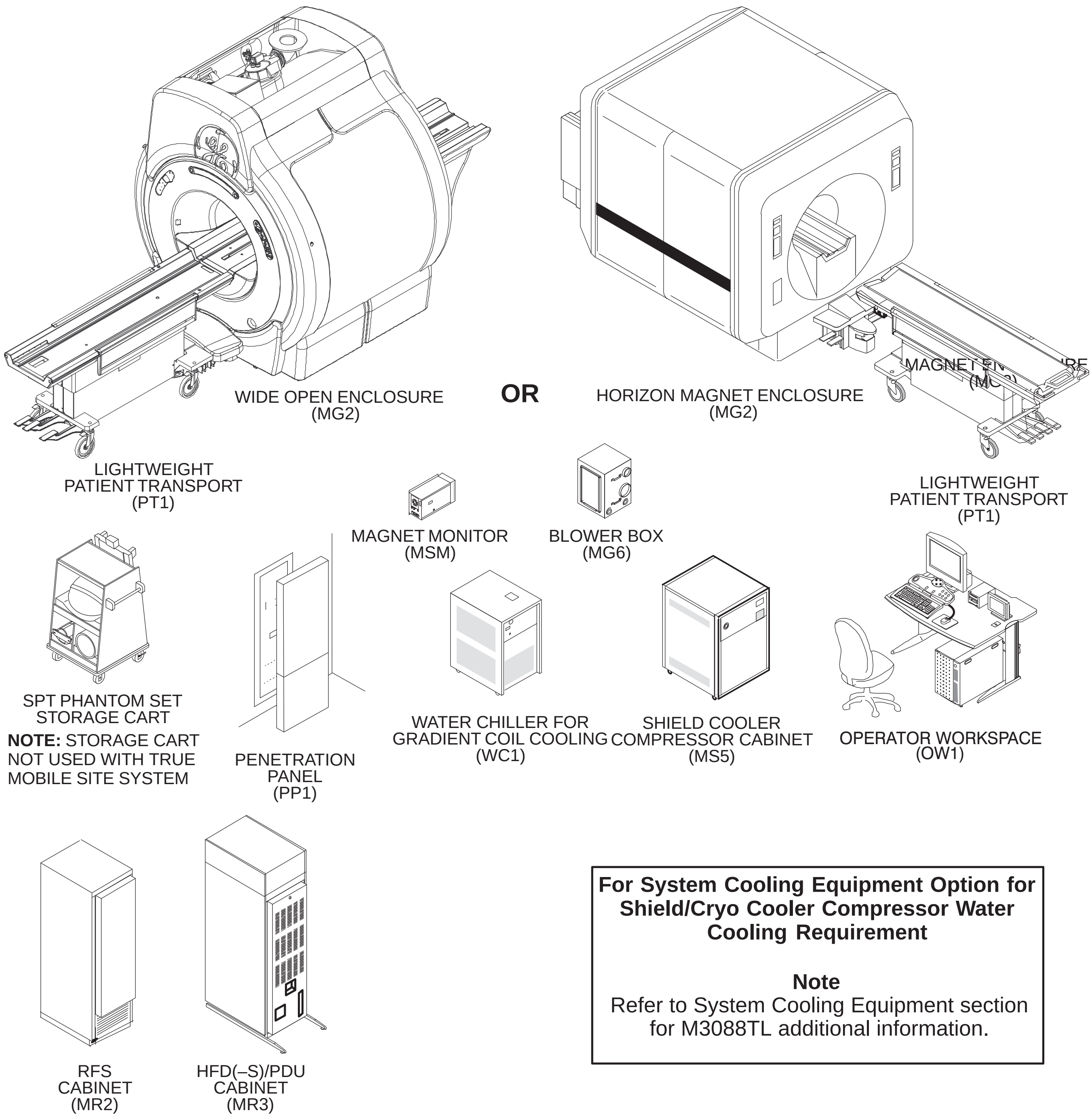
TABLE 18–1
Pre–EXCITE to Release 12.x Upgrade Impact

System Configurations		Final System Major Equipment
Existing Pre–EXCITE	**Upgraded Release 12.x**	
Signa LX, Horizon, 5.x, or 4.x	Signa EXCITE HD 1.5T	Illustration 18–4
Signa 1.5T TwinSpeed (9.x)	Signa EXCITE HD 1.5T TwinSpeed	Illustration 18–5

Refer to the sub–section apropriate to the upgrade being planned:

- Section 18–1–2: Pre–EXCITE with LCC Magnet to EXCITE HD Upgrade
- Section 18–1–3: Pre–EXCITE with Non–LCC Magnet to EXCITE HD Upgrade

18–1–1 SIGNA LX, HORIZON, 5.x, 4.x to EXCITE HD UPGRADES OVERVIEW (Continued)

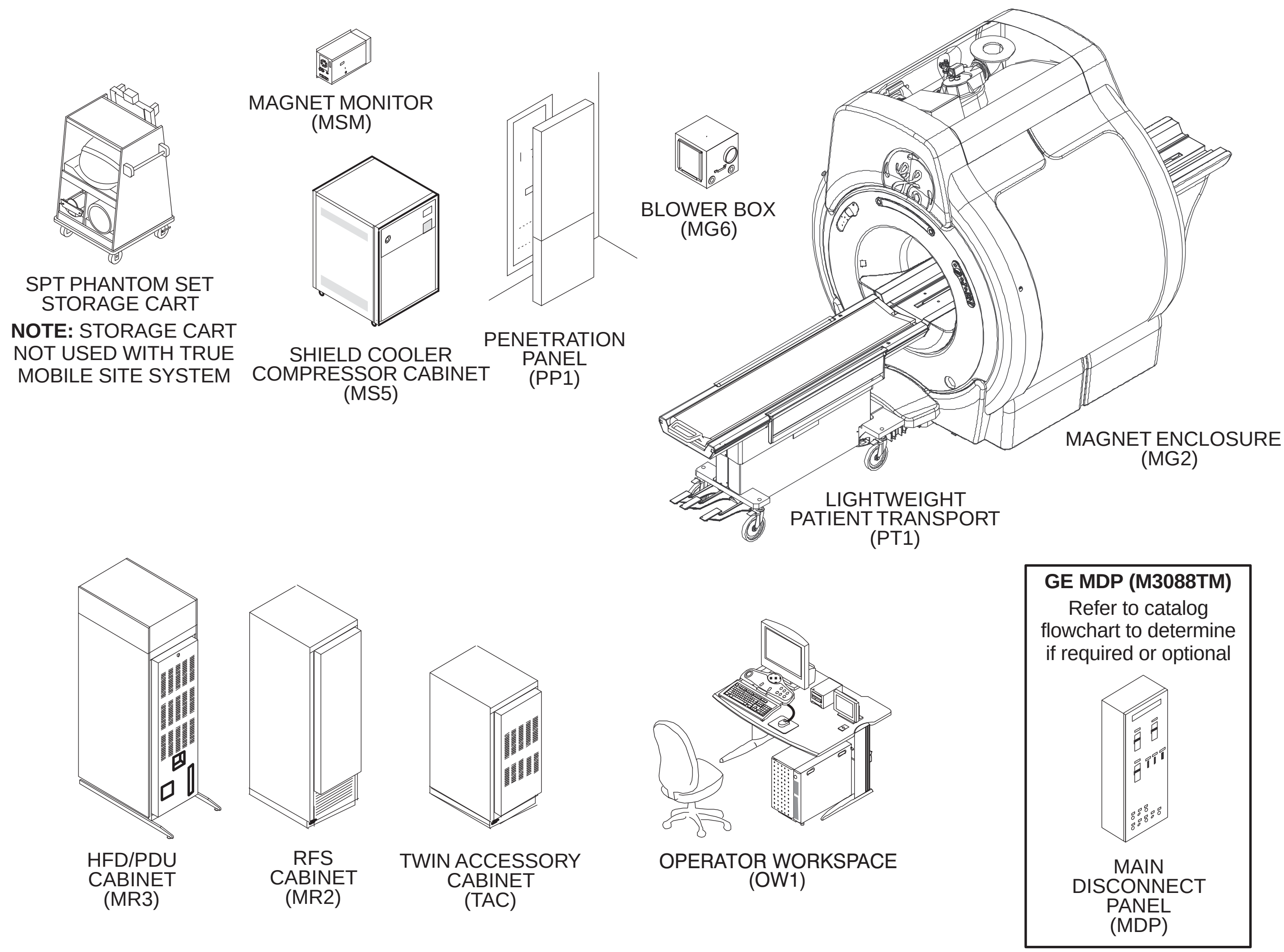


For System Cooling Equipment Option for Shield/Cryo Cooler Compressor Water Cooling Requirement

Note
Refer to System Cooling Equipment section for M3088TL additional information.

ILLUSTRATION 18–4
PRE–EXCITE UPGRADED TO Release 12.x NON–TWINSPEED SYSTEM MAJOR EQUIPMENT

18–1–1 SIGNA LX, HORIZON, 5.x, 4.x to EXCITE HD UPGRADES OVERVIEW (Continued)



For Existing TwinSpeed Systems Upgraded to EXCITE HD: Gradient Coil Cooling Equipment

Existing TwinSpeed systems retain their exisiting Gradient Coil water cooling equipment and Shield Cooler Compressor water cooling equipment.

For Systems Upgraded to TwinSpeed with EXCITE HD Upgrade: Gradient Coil Cooling Equipment

Note

Refer to System Cooling Equipment section for M3088TL and M3088TK additional information.

ILLUSTRATION 18–5

PRE–EXCITE TwinSpeed UPGRADED TO Signa EXCITE 1.5T TwinSpeed Release 12.x System

18–1–2 PRE–EXCITE WITH LCC MAGNET TO EXCITE HD UPGRADE

Signa LX with 1.5T LCC Magnet upgrade to Signa EXCITE HD catalogs are listing in Illustration 18–6. Illustrations 18–7 and 18–8 shows the equipment/cabinets modified, replaced, and added by the listed catalogs.

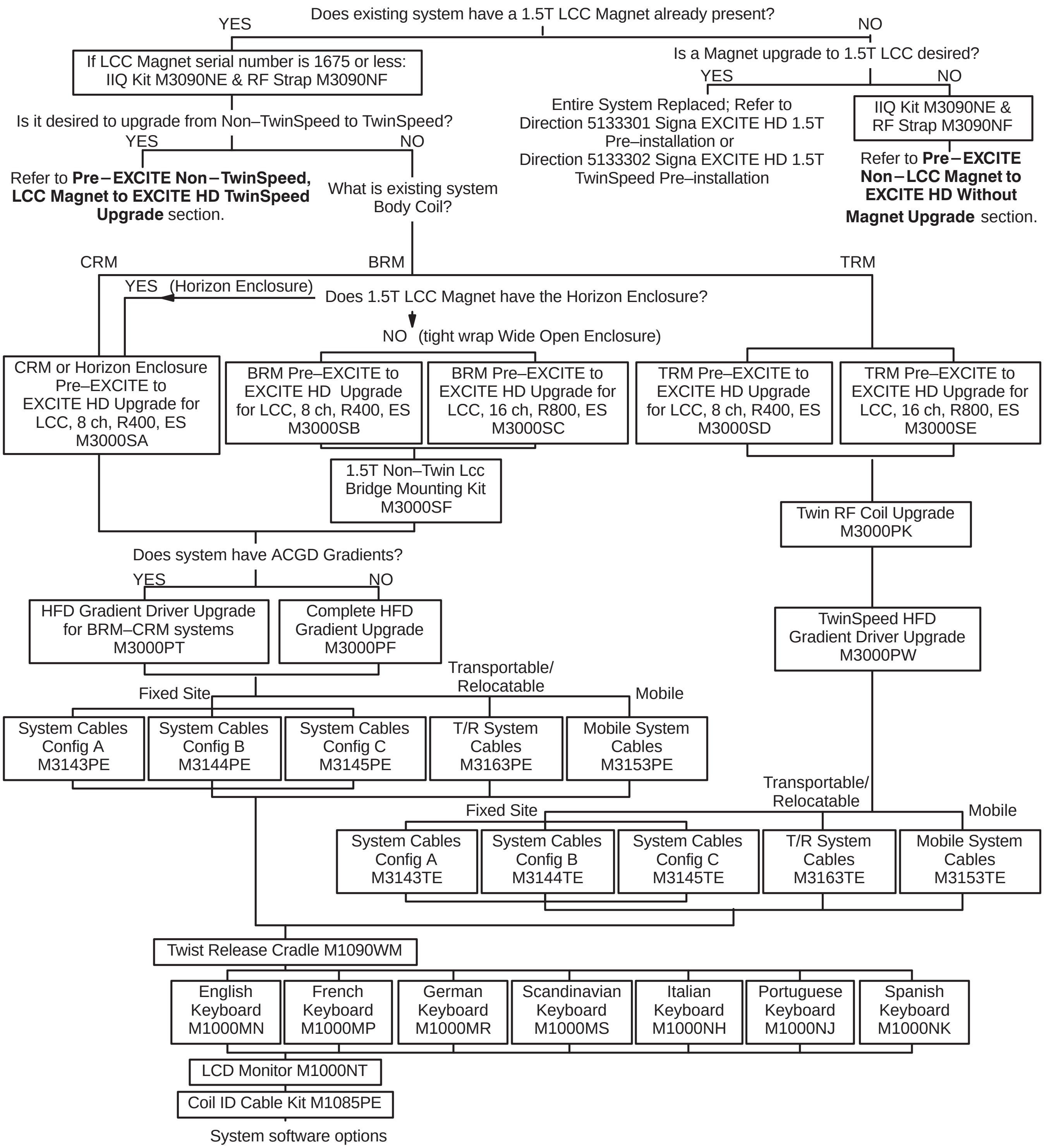


ILLUSTRATION 18–6
SIGNA LX WITH 1.5T LCC MAGNET TO SIGNA EXCITE HD UPGRADE CATALOGS

18–1–2 PRE–EXCITE WITH LCC MAGNET TO EXCITE HD UPGRADE (Continued)

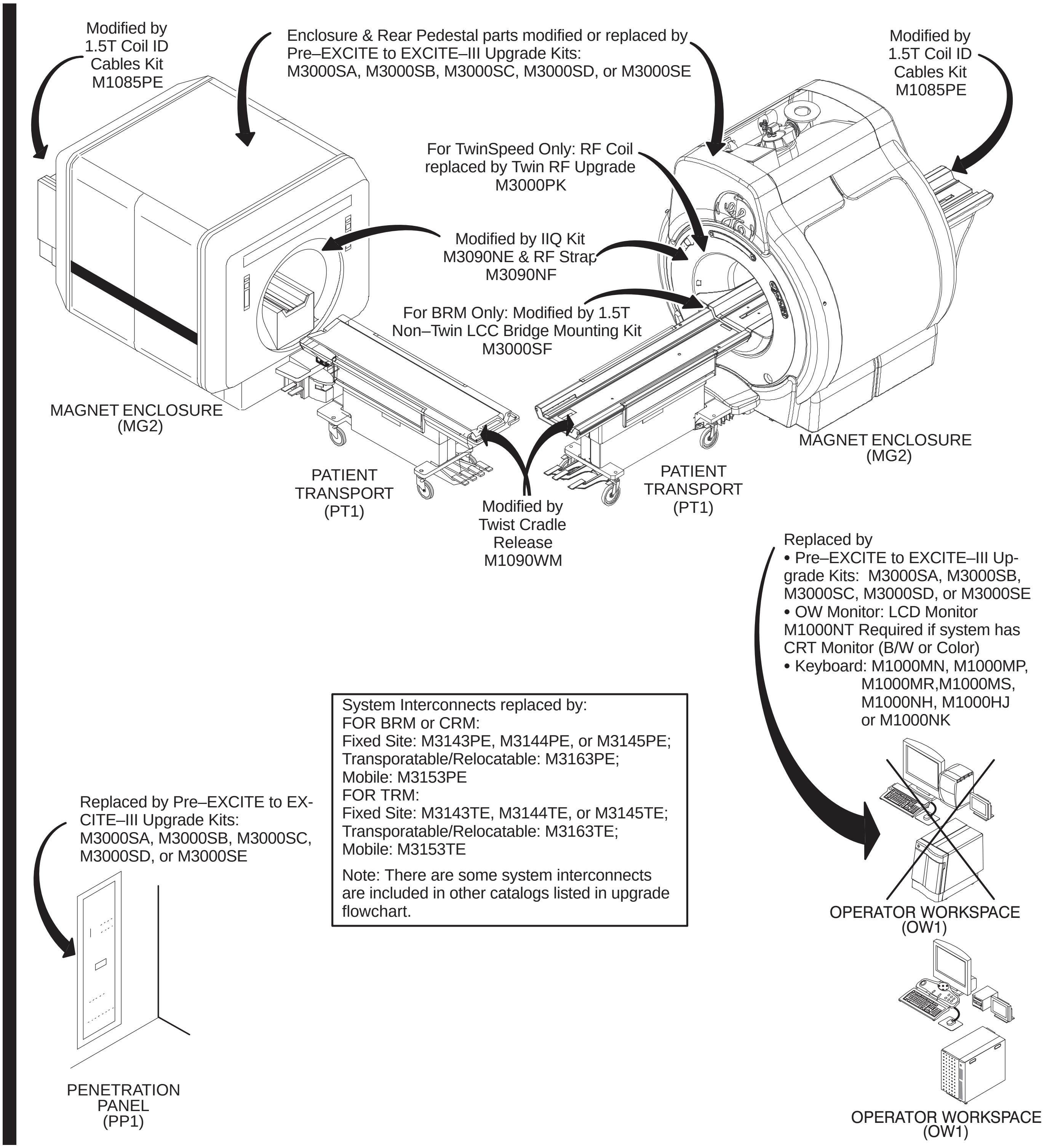


ILLUSTRATION 18–7
SIGNA MR/I & SIGNA LX WITH 1.5T LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–1–2 PRE–EXCITE WITH LCC MAGNET TO EXCITE HD UPGRADE (Continued)

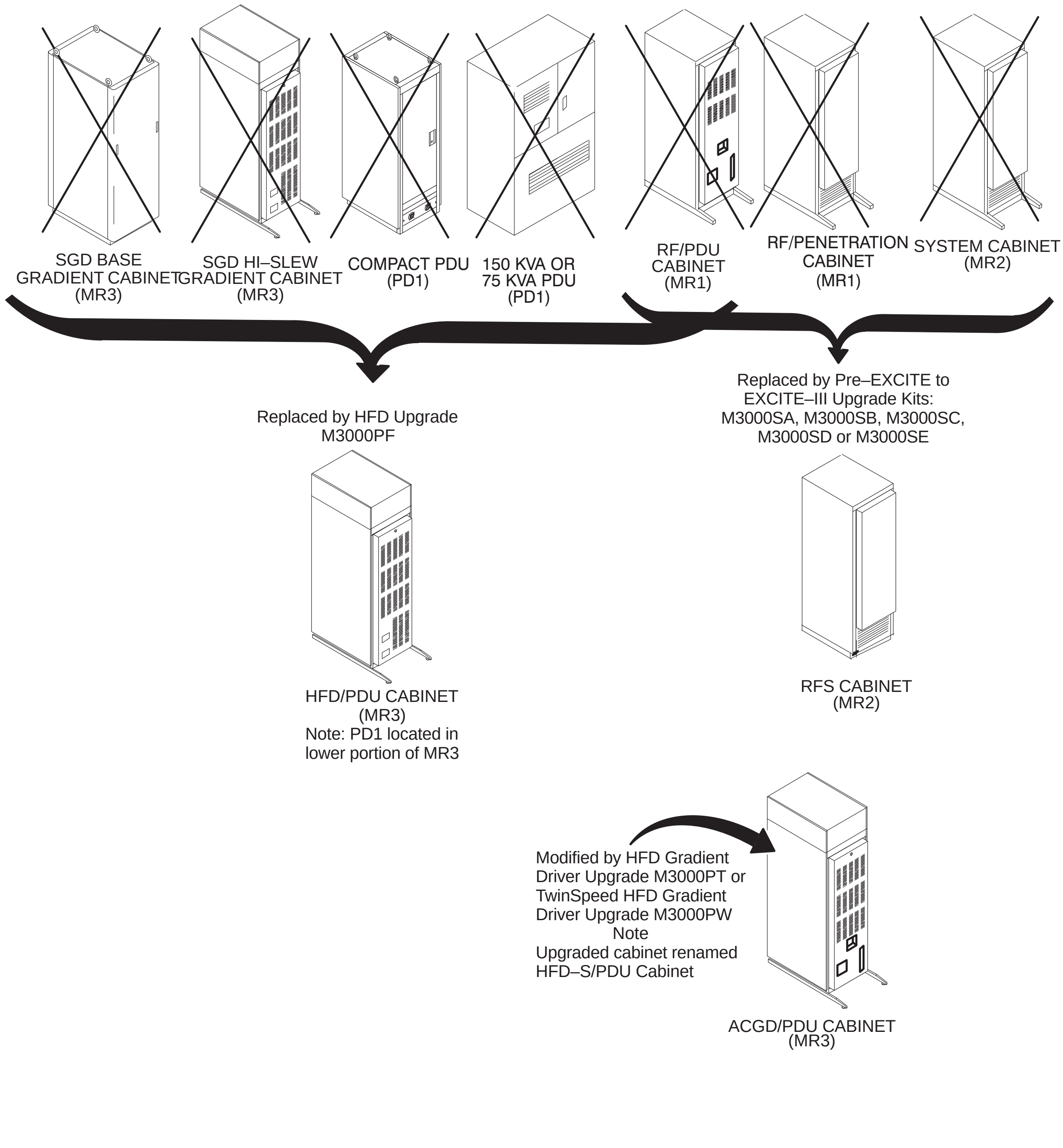


ILLUSTRATION 18–8

SIGNA MR/I & SIGNA LX WITH 1.5T LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE

Signa LX, Signa Horizon, Signa Advantage 5.x, and Signa Advantage 4.x with non–LCC Magnet upgrade to Signa EXCITE HD catalogs are listed Illustration 18–9. The equipment replaced and modified by the catalogs listed in the flowchart are shown in the illustrations as noted below:

- Signa LX see Illustrations 18–10 and 18–11
- Signa Horizon (5.x) see Illustrations 18–12 and 18–13
- Signa Advantage 5.x see Illustrations 18–14 and 18–15
- Signa Advantage 4.x see Illustrations 18–16 and 18–17.

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE (Continued)

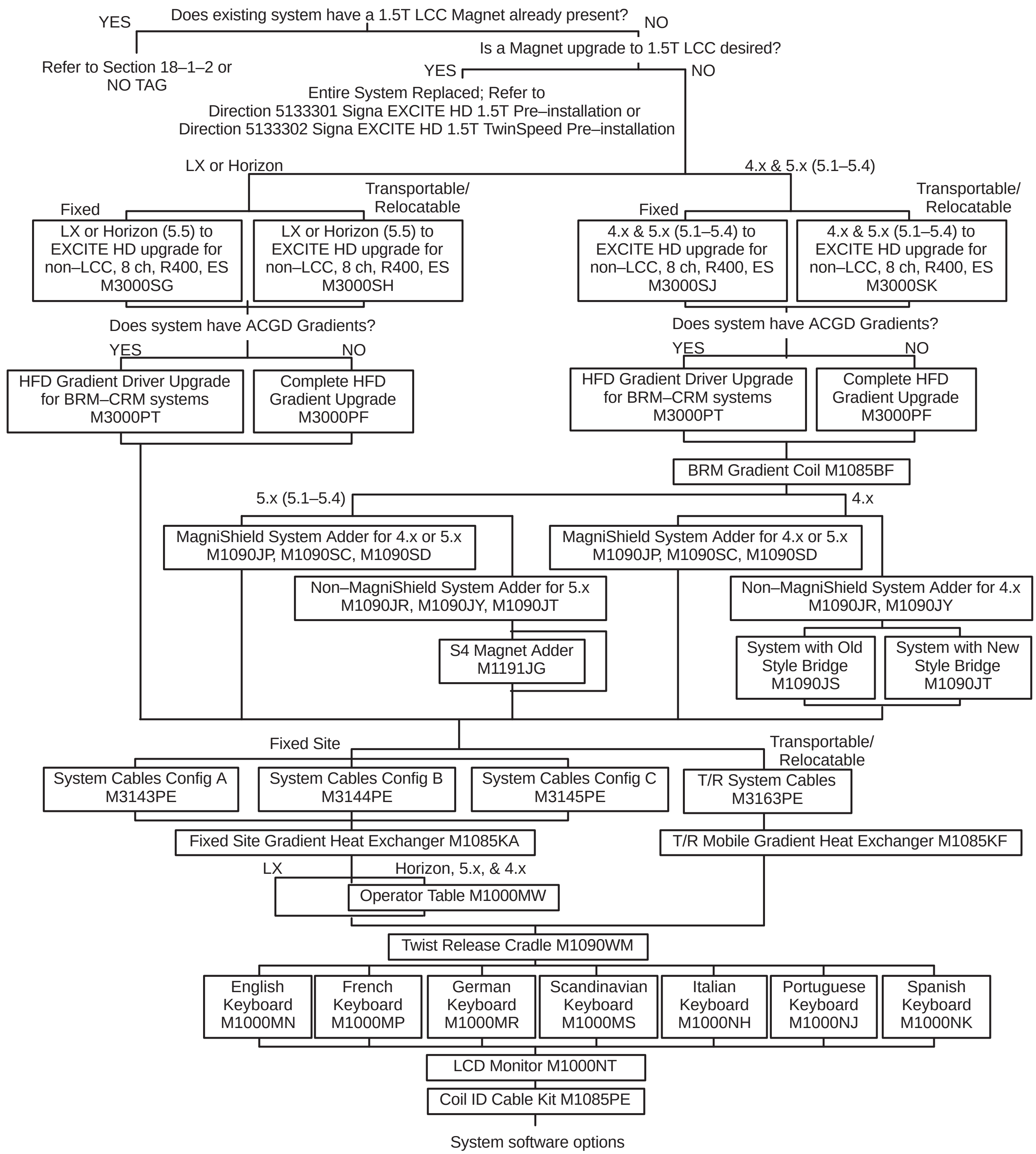


ILLUSTRATION 18–9
Pre–EXCITE WITH NON–LCC MAGNET TO SIGNA EXCITE HD UPGRADE CATALOGS

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE (Continued)

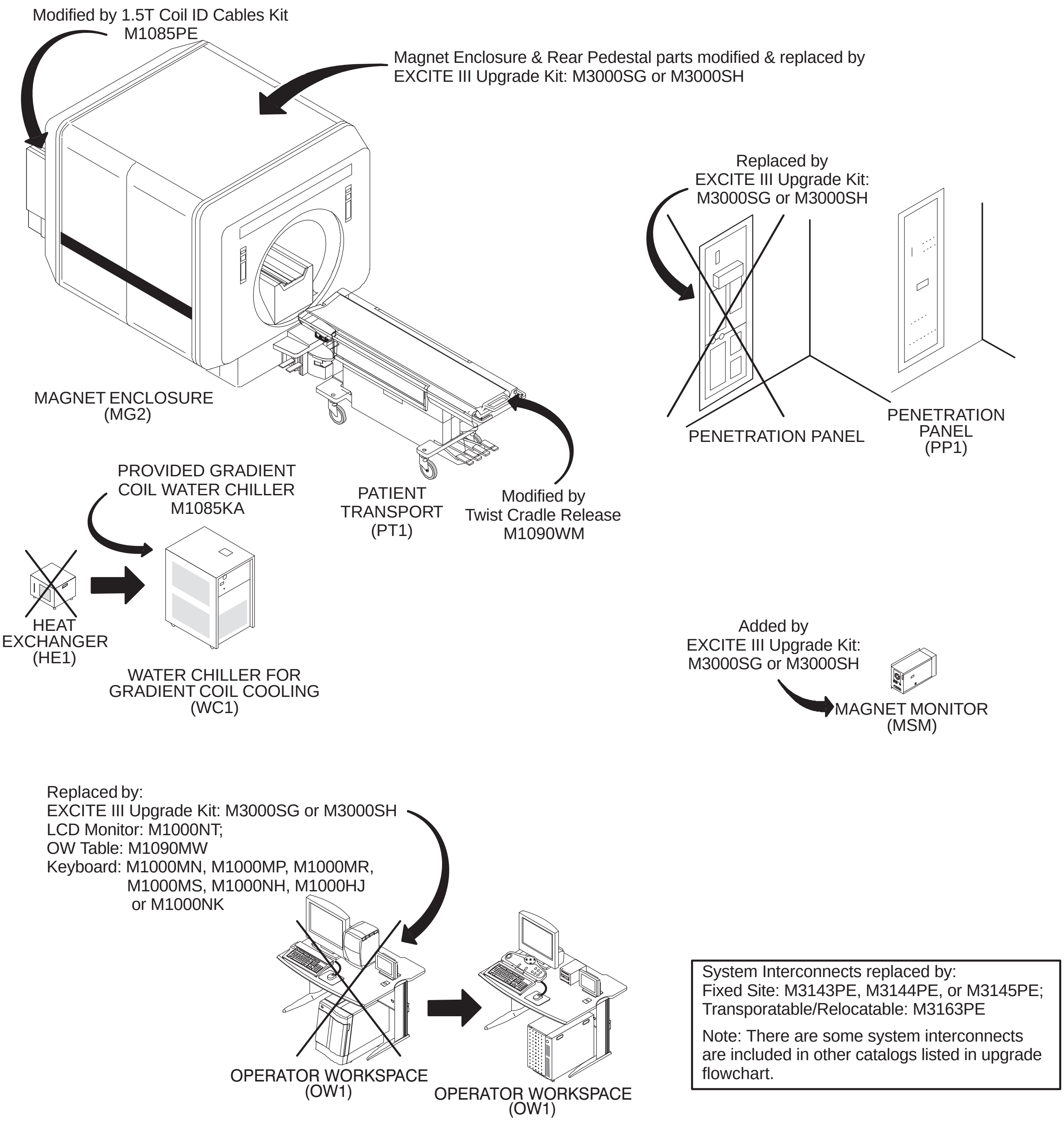


ILLUSTRATION 18–10

SIGNA LX EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE (Continued)

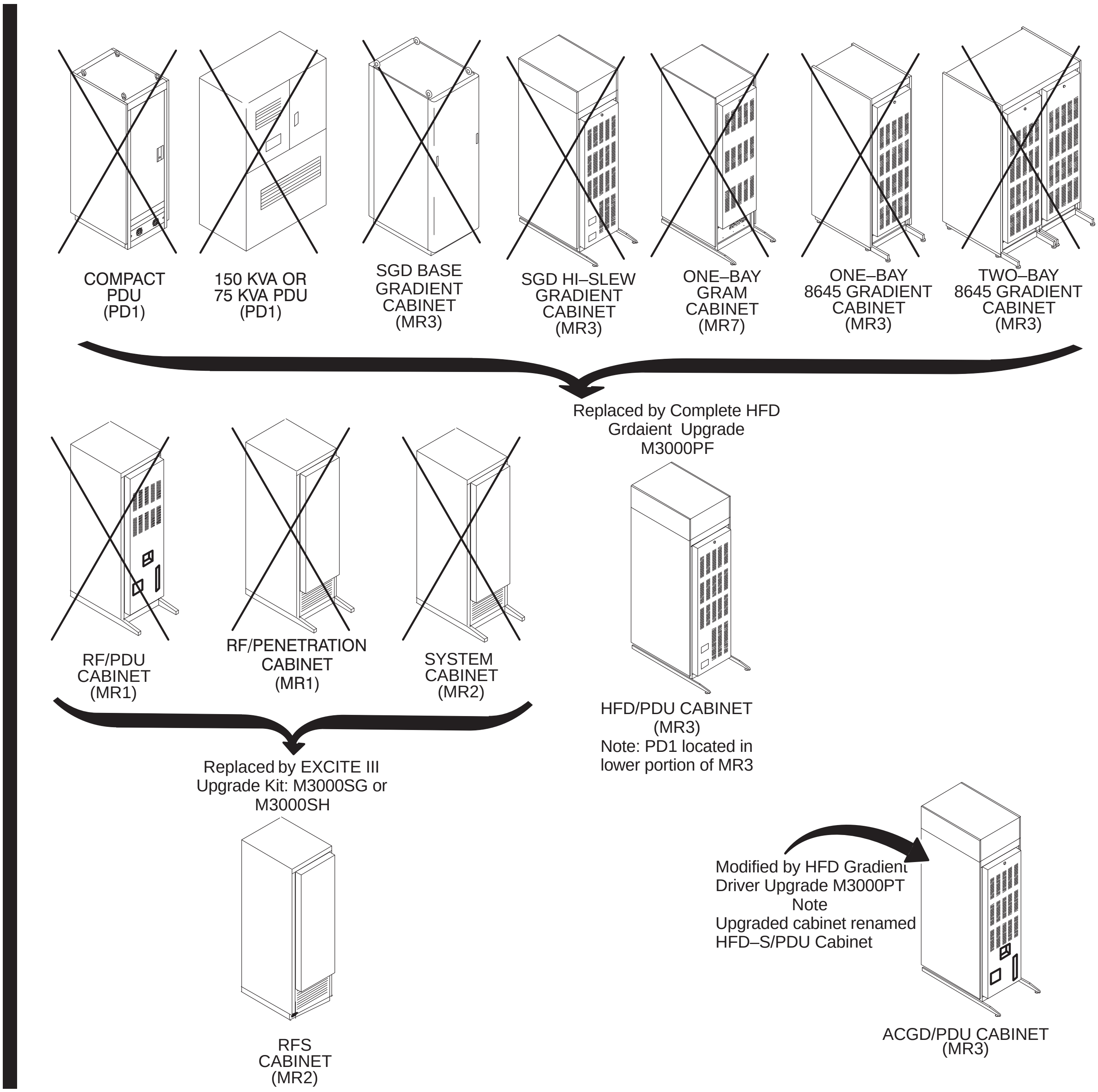


ILLUSTRATION 18–11
SIGNA LX EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE (Continued)

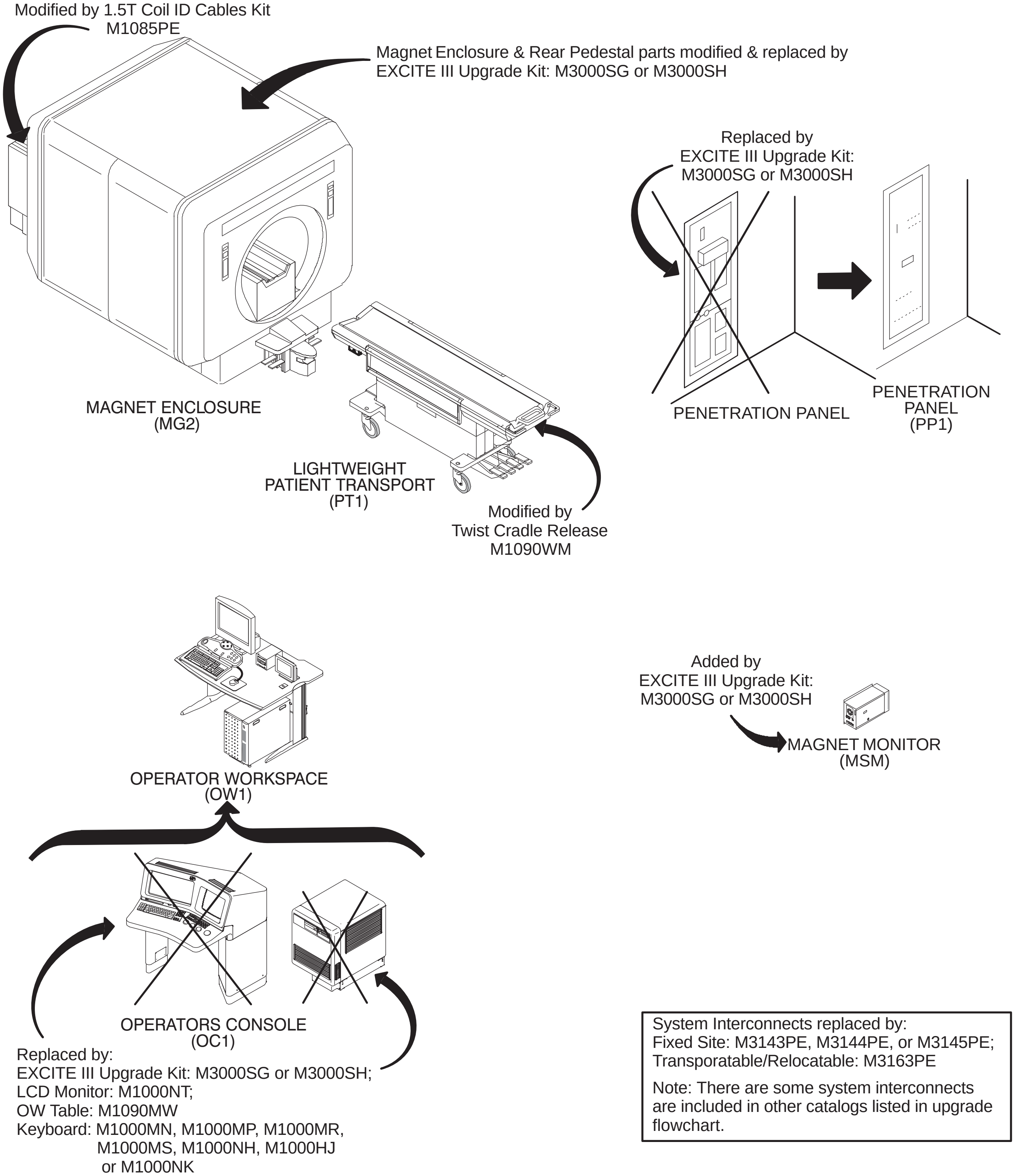


ILLUSTRATION 18–12
SIGNA HORIZON (5.X) EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE (Continued)

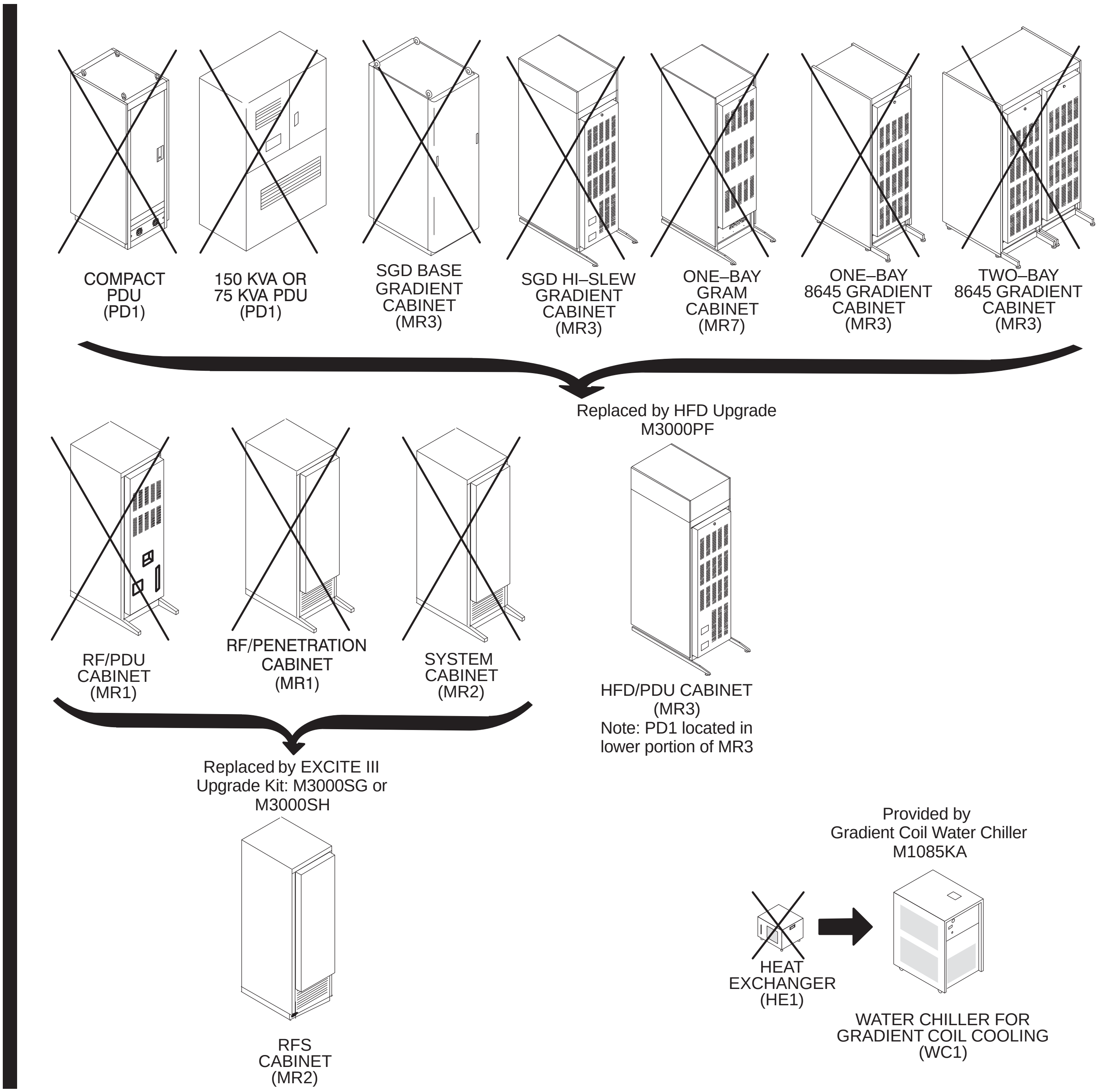


ILLUSTRATION 18–13

SIGNA HORIZON (5.X) EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE (Continued)

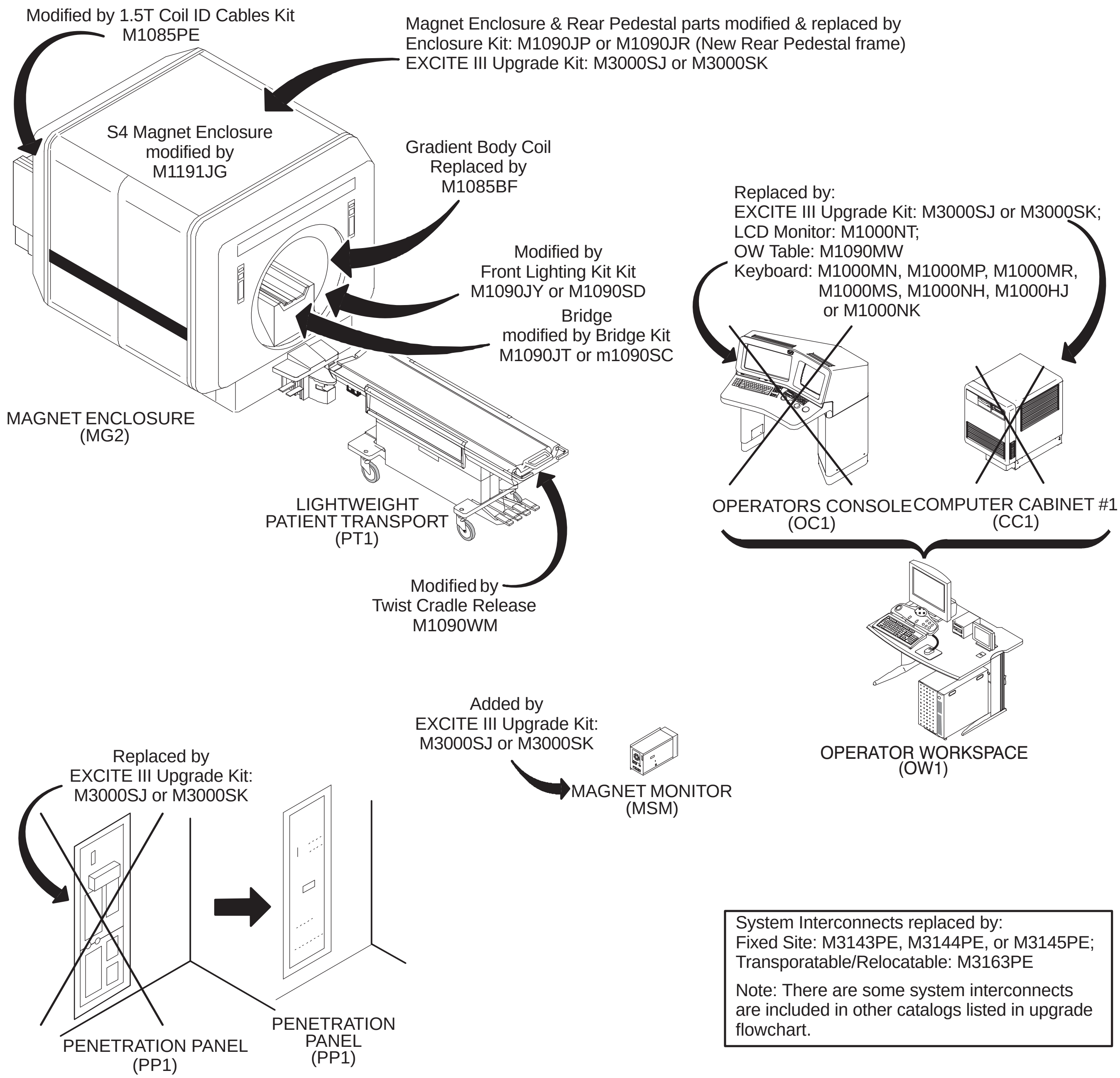


ILLUSTRATION 18–14
SIGNA ADVANTAGE 5.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE (Continued)

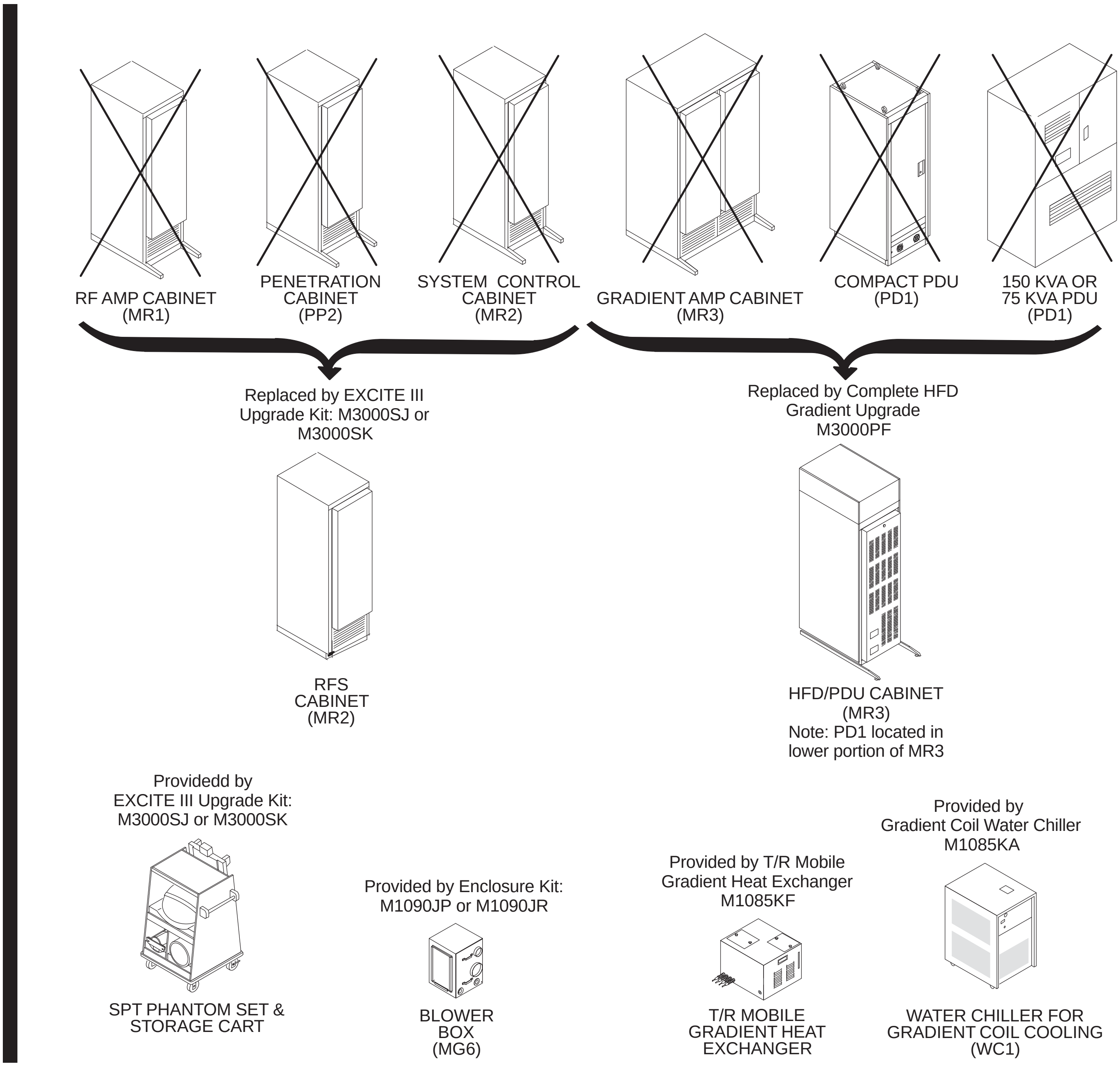


ILLUSTRATION 18–15
SIGNA ADVANTAGE 5.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE (Continued)

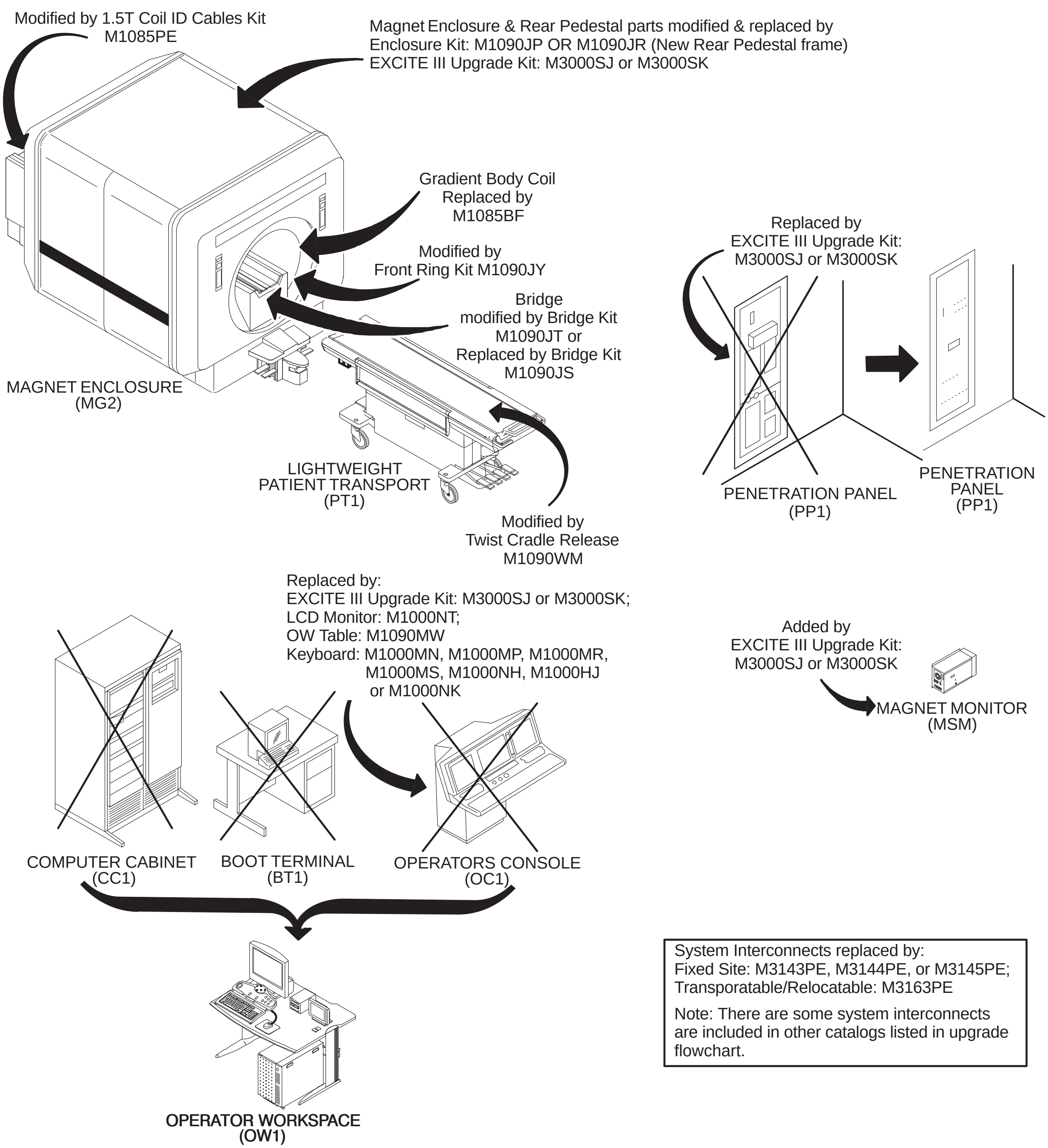


ILLUSTRATION 18–16
SIGNA ADVANTAGE 4.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–1–3 PRE–EXCITE NON–LCC MAGNET TO EXCITE HD WITHOUT MAGNET UPGRADE (Continued)

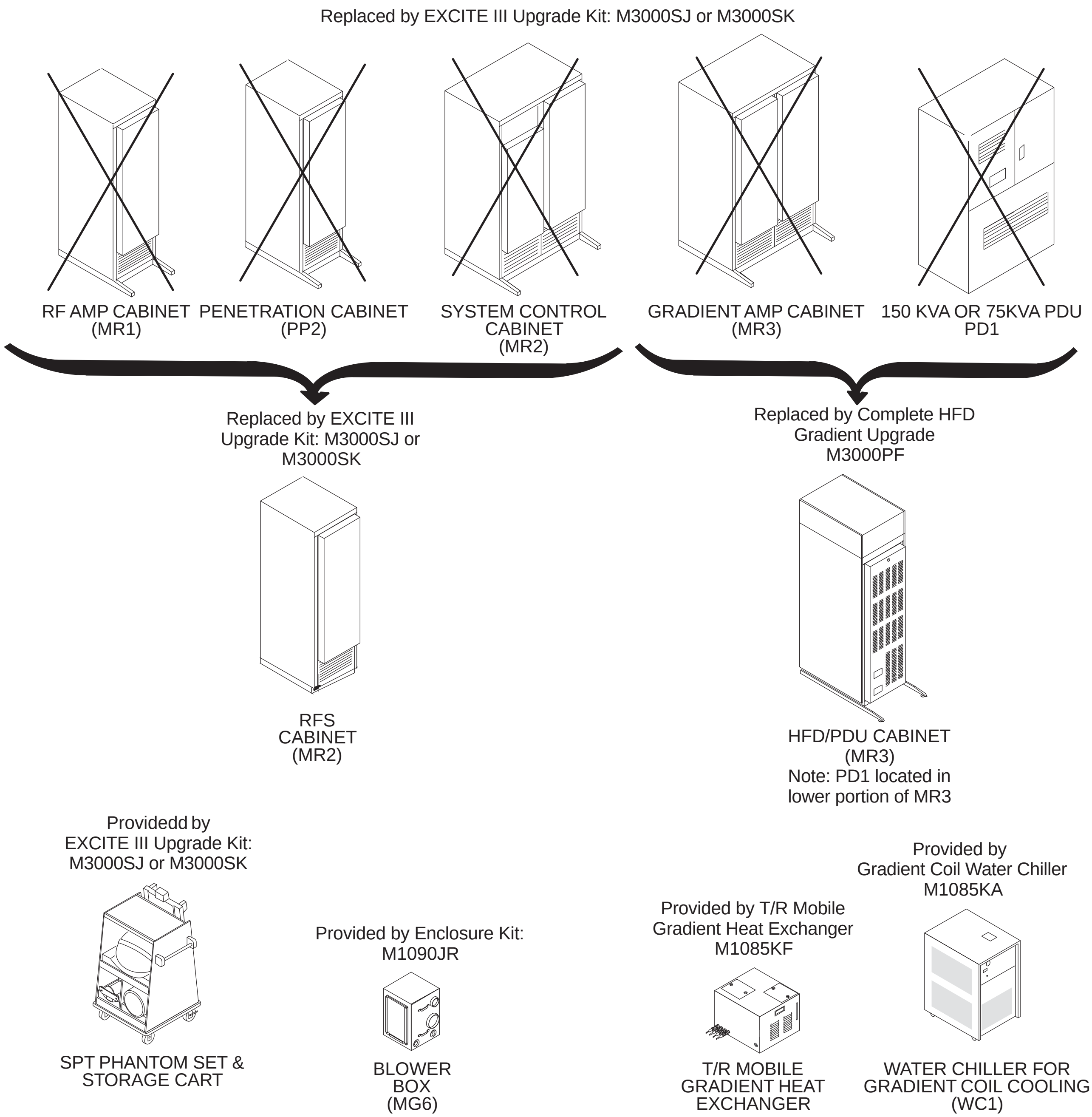


ILLUSTRATION 18–17

SIGNA ADVANTAGE 4.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

18–2 SYSTEM COOLING EQUIPMENT

The following system cooling equipment is available with the MR system as indicated in the upgrade flowcharts:

- 50 and 60 Hz Water Chiller (WC1): The WC1 is a 4KW Water–to–Air Chiller for Gradient Coil cooling. Fixed site must order Water Chiller.
- Heat Exchanger (HE1): The HE1 is a Heat Exchanger for Gradient Coil cooling on T/R Mobiile units
- MR Common Chiller (MRCC): The MRCC is a single–loop 10KW 50/60 Hz water chiller that can be used to cool either Shield/Cryo Cooler Compressor or the TwinSpeed Gradient Coil. The MRCC can be installed either indoors or outdoors. There is a Remote Panel to be installed in the MR system Equipment Room that can stop or restart the chiller and can display supply water temperature, chiller setpoints and alarms. The MRCC power is supplied by the MR system Main Disconnect Panel (MDP). This chiller will be serviced by a third party vendor.
- Gradient Water Heat Exchanger (GWHX): The GWHX is a 10KW 50/60 Hz water–to–water heat exchanger that is used for cooling the TwinSpeed Gradient Coil. The GWHX is for indoor use only. The GWHX requires facility provided chilled water. The GWHX power is supplied by the MR system Power Distribution Unit (PDU). This heat exchanger will be serviced by GE Service. When the GWHX is used then the customer must supply water cooling for the Shield/Cryo Cooler Compressor.

Note

TRM Gradient Coil water cooling must be supplied by cooling equipment (MRCC or GWHX) supplied with the MR system to prevent contamination/damage to the coil and for proper image quality.

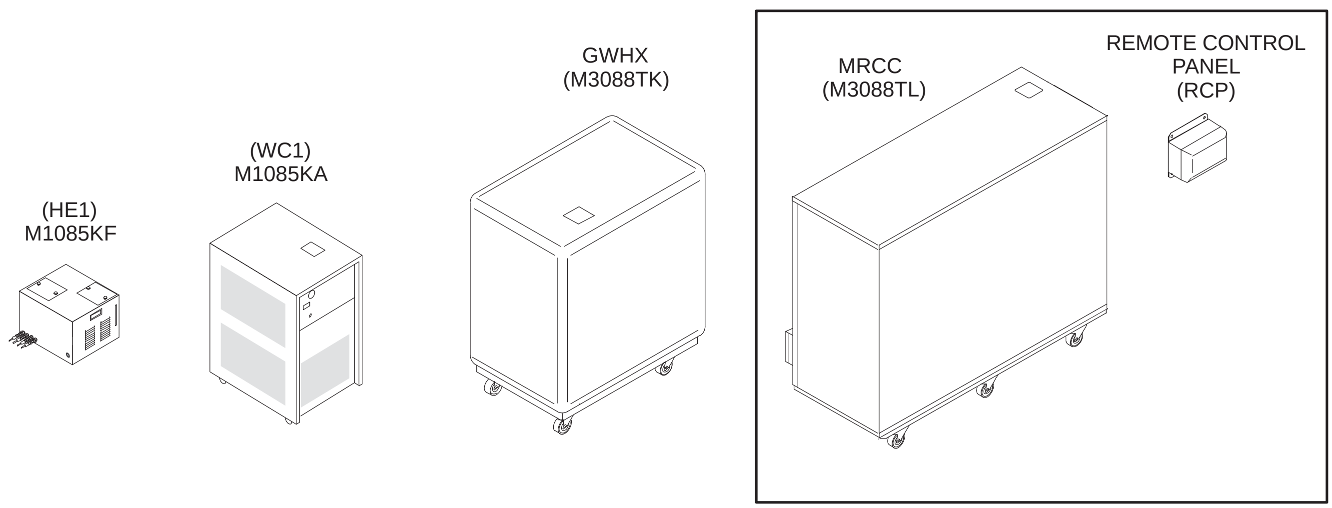


ILLUSTRATION 18–18
SYSTEM COOLING EQUIPMENT SELECTIONS

18–3 CABLE IDENTIFICATION

Cables delivered to site for installation have specifications dependent on its usage. The following tables contain information on cable runs for this system as it was originally installed.

The table below provides definitions for the following cable tables.

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
RUN	The run number as seen on the delivered cable	The vendor cable may not have run numbers.
“FROM” & “TO”	Labels the destination of the cable	
DESCRIPTION	Could include information on length, number of conductors, type of cable, and end plug type	Total length of cable delivered to site, type of cable specifies power from data.
CABLE TYPE	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continously applied to a wire	Specific to the core cable manufacturer given.
ACTUAL VOLTAGE	Nominal for continuous; maximum for variable voltage	If voltage varies during use, the maximum is given.
TEMP RATING	Maximum temperature insulating material my be used without loss of basic properties	Specific to the core cable manufacturer given.
FLAME RATING	Measure of material’s ability to support combustion	Specific to the core cable manufacturer given.
REMARKS	Describe the cable’s use in the MR system	

18–3–1 PDU POWER AND GROUND CABLES

The cables listed in Table 18–2 are connected at one end to the PDU module in the HFD/PDU Cabinet.

TABLE 18–2
PDU POWER AND GROUND CABLES

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
037	PD1 GND	MR2 A12 GND	36 ft Ground Wire (#10)	SO	600	0	105	VW–1	System Cabinet ground
040	MSI	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	208Y	105	VW–1	Magnet Ground Cable
045	PP1 GND	MG6 GND	41 ft Ground Wire (#10)	SO	600	0	105	VW–1	Blower Cabinet Separate ground
968	MR1 A12 J1	MR3 PD1	35 ft Power (5–#8)	SO	600	208Y	75	FT–4	RFS Cabinet Power
970	WC1	MR3 PD1	82 ft Power (3–#10)	STO	600	208	90	FT–4	HFD Water Chiller Power
988	PP1 A7 GND	TAC GND	60 ft. Ground (#8)	SO	600	0	80	VW–1	TAC to PP1 ground (3)
989	MR3 PD1	TAC J8	25 ft. Power (3–#10)	SO	600	120	90	FT–4	TAC Power
1081	OW1 A19 J1	MR3 PD1	65.5 ft Power (4–#10)	SO	600	120	75	FT–4	Operator Workspace Power
1182	PP1 PHPS PWR	MR3 PD1	61 ft. Power (3–#10)	AWM	600	120	90	FT–4	PHPS Module Power
1183	PP1 F7, F8	MR3 PD1	41 ft Power (3–#10)	SO	600	0	105	VW–1	Blower Cabinet Power
1234	MR2 RF AMP	MR3 PD1	35.5 ft Power (4–#8)	SO	600	208	90	FT–4	RF Amplifier Power
1254	MR3 PD1	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	208Y	105	VW–1	Magnet Ground Cable
N/A	PD1 MAG/SHIM OUTLET	MAG/SHIM POWER SUPPLY	50 ft Power (5–#8) With 30A Plug and Receptacle						Cable for PDU to GE Magnet Power Supply or SC Shim Supply Power

18–3–2 MAGNET ROOM CABLES NOT CUT TO LENGTH

The cables listed in Table 18–3 are routed and connected in the Magnet Room and are not cut to length.

TABLE 18–3
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
282	PP1 J12	MG3 A2 J6	45 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Patient Intercom
297	PP1 J18	EO1	90 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Emergency off switch
300	PP1 J15	MG2 A15	60 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Bore Light power
318	PP1 J11	MG3 A7	45 ft. 9 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Long Drive Motor power
385	PP1 J20	MG2 A30	67 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	Patient Alignment Light power
387	PP1 J47	MG2 A29 J1	60 ft. 25 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Table Elevation motor power
414	MG3 A35	MG3 A2 J9	30 ft. 9–pin Sub–D	CI3	300	<30	80	FT–4	Front Cover Microphone
421	MG2 A33 J1	MG3 A31 J1	15 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	Dock Limit
422	MG2 A33 J2	MG3 A1 J4	25 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	Long Drive Encoder/Limit Sw
458	PP1 J17	OM3	9 pin Sub–D	CI3	300	<30	80	FT–4	Optional Remote O2 Sensor
715	PP1 J14	MG2 A33 J1	65 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	SRI Power
716	PP1 J89	MG2 A11 A1 J2	65 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	PAC Power
743	MG2 A12 A1 J2	MG3 A3 A1 J3	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW–1	Body Coil Drive
744	MG2 A12 A1 J1	MG3 A3 A1 J4	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW–1	Body Coil Drive
778	PP1 A11 J72	MG2 A12 A2 J3	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil frnt)
779	PP1 A11 J73	MG2 A12 A2 J4	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil frnt)
780	PP1 A11 J74	MG3 A3 A3 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Inductive Drive
781	PP1 A11 J75	MG2 A12 A2 J5	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil rear)
782	PP1 A11 J76	MG2 A12 A2 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil rear)
784	PP1 F5, F6	MG6 J1	40 ft. 2 cond power cable	SO	600	120	80	FT–4	Patient Cooling Blower power
786	PP1 F7, F8	MG6 J2	40 ft. 2 cond power cable	SO	600	120	80	FT–4	Gradient cooling blower power
848	MG2 A11 J4	MG2 A4 J5	16 ft. 25 pin Sub–D/rt. angl	CI3	300	<30	80	FT–4	Operator Display to PAC II
862	MG2 A4 J1	MG2 A5 J1	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW–1	Operator Display to Rt. Op Cntrl
863	MG2 A4 J2	MG2 A5 J2	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW–1	Operator Display to Lt. Op Cntrl
893	MG3 A3 J2	MG3 A3 A5 INPUT	6.6 Inch BNC		1900 VRMS	<30	80	VW–1	Spliiter to Preamp Cable
897	MG2 A4 J3	MG2 A33 J4	20 ft. 37 pin Sub–D/26 pin	CI3	300	<30	80	FT–4	Operator Display to SRI

18–3–2 MAGNET ROOM CABLES NOT CUT TO LENGTH (Continued)

TABLE 18–3 (CONT’D)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1152	PP1 A5 J1	MG2 A12 A3 P1	29.5 ft. 37 pin CPC to 16 pin	PLTC	300		105	FT–4	Shim Filter Cable
1197	PP1 RFTXSW J150	MG3 16CHSW J25	49 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	
1200	PP1 J200	MG3 16CHSW J1	8–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	
1201	PP1 J201	MG3 16CHSW J2	8–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	
1202	PP1 J202	MG3 A17 J3	2–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	
1203	PP1 MC SW FLTR J203	MG3 16CHSW J11	49 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	
1215	MG3 16CHSW J20	MG2 SRI J20	37 pin Sub–D	CI3	300	<30	80	FT–4	
1216	PP1 A16 J77	MG3 A36 J5	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	Monitor to Compressor I/F
1217	PP1 A15 J78	MG3 A36 J10	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	SRI Power
1250	MG3 A3 J1	CONNECT TO RUN 257							
1260	MG2 A33 J10	MG2 A36	35 ft 3 cond w/ 9–Sub D	CI3	300	<30	105	FT–4	Bore Temperature Sensor
1261	MG2 A4 J4	MG2 A33 J5	20 ft. Data Cable (80–#28)	CI2	300	<30	80	VW–1	SRI Splitter to Oper Display

18–3–3 MAGNET MONITORING CABLES

The Magnet Monitoring cables listed in Table 18–4 are delivered with the Magnet.

TABLE 18–4
MAGNET MONITORING CABLES DELIVERED WITH MAGNET

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
823	MSM1 A1 J10	MR2 A11 J24	Data Cable (15–#22)	CI3	300	<30	80	FT–4	Monitor to System Cabinet I/F
824	PP1 J10	FJ1	Data Cable (9–#22)	CI3	300	<30	80	FT–4	Monitor Sensor I/F to Penetration Panel
825	PP1 J48	MSM1 A1 J8	Data Cable (25–#22)	CI3	300	<30	80	FT–4	Magnet Monitor connector to Penetration Panel
826	MSM1 A1 J9	FJ3	Data Cable (15–#22)	CI3	300	<30	80	FT–4	Monitor to Compressor I/F
827	FJ3	MS5 A5 A1 J1 / FJ4	Data Cable (8–#22) & Data Cable (4–#22)	CI3	300	<30	80	FT–4	Compressor I/F
828	PP1 J10	MA1 A3 A1 P403	Data Cable (9–#22)	CI3	300	<30	80	FT–4	Penetration Panel to Sensor
829	PP1 J48	MS1 FJ2	Data Cable (25–#22)	CI3	300	<30	80	FT–4	Penetration Panel to Magnet I/F
830	MS1 FJ2	MS1 A2 J2/ MS1 A5 A2 J1/ MG3 TB1	3cond/3cond/25cond	CI3	300	<30	80	FT–4	Magnet I/F
831	FJ1	MSM1 A1 J7	Data Cable (8–#24)	CI3	300	<30	80	FT–4	Monitor to Sensor I/F
832	MS1 A2 J1	MS1 SLEEVE/ MS1 COLDHEAD	Data Cable (8–#22)	CM	300	<30	80	FT–4	Buffer Amp to Cooling Sleeves
833	FJ4	MS5 A1 A6 JR	Data Cable (6–#24)	CM	300	<30	80	FT–4	Compressor to Compressor I/F
939	MDP UPS	MSM4	6 ft. Power (5–#15)	SO	600	120	85	VW–1	MDP to UPS
940	MSM3 RS232	MSM1 J1	6 ft. Data Cable (9–#22)	CI3	300	<30	80	FT–4	Monitor to Modem
1077	MSM1 J5	OW1 A2 A4 PRT7	80 ft. Cat 5 Ethernet	CI3	300	<30	80	FT–4	Monitor to OW Ethernet
	MR2 A11 J24	MR2 IPG J9/ MR2 PDU J5							System Cabinet I/F

18–3–4 GE MAGNET CABLES

The Magnet cables listed in Table 18–5 arrive on site with the magnet and are installed at the time of Magnet delivery.

TABLE 18–5
GE MAGNET CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
601	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp cables, Power Assembly Positive 4/0
602	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp cables, Power Assembly Negative 4/0
603	MS7	MS1 A3 A1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp and Shim Cable Assembly
604	MS7 or MS8	MS1 A3 A1	Data Cable (12–#22)	CI3	150	60 VDC	80	FT–4	Switch Heaters
606	MS4	MS1 A3 A1	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Emergency Run Down Unit
609	MS7 OR MS8	PDU	Power Cable (5–#16)	SO	300	208Y	105	FT–4	Feeder Cables
621	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW–1	Supply Flexible Gas Line (PP1 J60 or J61)
622	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW–1	Return Flexible Gas Line (PP1 J60 or J61)
623	MS5 A1	PP1 F1, F2, F3, F4	Power Cable (4–#18)	SO	600	200	105	VW–1	Compressor to Pen Panel Cable (equipment room)
624	MS1 A2	PP1 F1, F2, F3, F4	Power Cable (4–#18)	SO	600	200	105	VW–1	Cold Head Control Cable

18–3–5 FIBER OPTIC CABLES

The Fiber Optic cables listed in Table 18–6 are routed and connected in the Magnet Room and the Equipment Room.

TABLE 18–6
FIBER OPTIC CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
711/ 712	PP1–J91	MG2 A33 and MG2 A11 A1	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	One duplex fiber optic cables and one simplex fiber optic cable and One duplex plastic fiber optic cable for PAC–II
1045	MR2 A11 J13	PP1 A17 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
1083	MR2 A11 J22	OW1 A15 XMT1	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1179	MR2 A11 J17	MR3 A11 J2	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic cables

18–3–6 EQUIPMENT ROOM AND MAGNET ROOM CABLES AND HOSES CUT TO LENGTH

The Gradient cables and the RF Head and Body cables listed in Table 10–12 will be routed, cut, and terminated at designated cable ends according to instructions found in the CABLE INSTALLATION section of this manual.

TABLE 10–12

EQUIPMENT ROOM AND MAGNET ROOM CABLES & HOSES CUT TO LENGTH

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
235	CONNECT TO RUN 1259	PP1 J44	Times Microwave LMR–900FR	CATVR	5000 VDC		85	FT–4	1.5T Body Coil Run created from kit
257	PP1 J44	CONNECT TO RUN 1250	Times Microwave LMR–900FR	CATVR	5000 VDC		85	FT–4	1.5T Body Coil Transmit
536	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
537	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
762	MR3 CABI/F J3	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT–4	X Gradient output
762	MG3 A32 1/2	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT–4	X Gradient output
763	MR3 CABI/F J4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT–4	Y Gradient output
763	MG3 A32 3/4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT–4	Y Gradient output
764	MR3 CABI/F J5	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT–4	Z Gradient output
764	MG3 A32 5/6	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT–4	Z Gradient output
976	MR3 CABI/F J3	TAC ±X	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin X Gradient Power
977	MR3 CABI/F J4	TAC ±Y	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin Y Gradient Power
978	MR3 CABI/F J5	TAC ±Z	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin Z Gradient Power
979	TAC X+,WB–,ZM–	PP1X+,WB–,ZM–	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin X Gradient Filter
980	TAC Y+,WB–,ZM–	PP1Y+,WB–,ZM–	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin Y Gradient Filter
981	TAC Z+,WB–,ZM–	PP1Z+,WB–,ZM–	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin Z Gradient Filter
982	PP1 ZZM+,ZM–	MG2 A12 A1 7/8	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin Zzm Gradient Output
983	PP1 YZM+,ZM–	MG2 A12 A1 9/10	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin Yzm Gradient Output
984	PP1 XZM+,ZM–	MG2 A12 A1 11/12	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin Xzm Gradient Output
985	PP1 ZWB+,WB–	MG2 A12 A1 1/2	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin ZWB Gradient Output
986	PP1 YWB+,WB–	MG2 A12 A1 3/4	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin YWB Gradient Output
987	PP1 XWB+,WB–	MG2 A12 A1 5/6	Conductor (3–#2)	STO	1000	600	90	FT–4	Twin XWB Gradient Output
1198	PP1 RFTXSW J151	MG3 PEDI/F J1	Times Microwave LMR–600FR	CATVR	4000 VDC		85	FT–4	RF Head Transmit
1199	PP1 RX TX SW J153	MG3 PEDI/F J2	Times Microwave LMR–600FR	CATVR	4000 VDC		85	FT–4	RF Head Transmit
1238	PP1 J4	CONNECT TO RUN 1249	Times Microwave LMR–600FR	CATVR	6000 VDC		85	FT–4	RF Head Transmit

18–3–7 EQUIPMENT AND OPERATOR ROOM CABLES

The cables listed in Table 10–13 are routed and connected between Penetration Pnael, cabinets in equipment room and Operator Workspace.

TABLE 10–13
EQUIPMENT & OPERATOR ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
296	FACILITY DISCNCT	PP1 J18	45 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Emergency off switch
457	OM1	PP1 J17	9 pin Sub–D	CI3	300	<30	80	FT–4	Optional O2 Monitor cable
611	TAC J22	PP1 SHIM							Resiitive Shim P.S. Option
701	MR2 A11 J15	RF Door switch	70 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	RF Door switch
971	MR2 A11 J23	MR3 PD1 J2	40 ft 15 pin Sub–D	CI3	300	<30	80	FT–4	PDU Status (ACGD)
974	MR3 PD1 J5	TAC J12	49 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Twin GIP/GP
1056	OW1 A17 PRT6	OW1–A12 ETHRNET	3.3 ft. Ethernet Cat 5	CI3	30	<30	80	FT–4	Laptop Access cable
1059	OW1 A17 PRT2	OW1–A15 ETHRNET	5.9 ft. Ethernet Cat 5	CI3	30	<30	80	FT–4	Lan Hub cable
1060	OW1 A19 J3	OW1 A21 J17	3.9 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	GOCAA Power cord
1061	OW1 A15 KYBD/ MS	OW1 A21 J8	2.9 ft. 25 pin Sub–D 2 PS/2s	CI3	300	<30	90	CL–2	Keyboard/Mouse cable
1063	OW1 A15 Video 1	OW1 A6 HOST LCD	9.9 ft. 29 pin DVI, 15 pin VGA	CI3	300	<30	80	VW–1	Host LCD Display Video cable
1064	OW1 A15 SCSI 1	OW1 A16 SCSI TOWER	9.9 ft. SCSI cable 50	CI3	300	<30	80	VW–1	SCSI Tower Data cable
1065	OW1 A11 J2	OW1 A6 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	Host LCD Display Power cable
1066	OW1 A15	OW1 A13	3.9 ft. 25/9 pn Sub–D	CM	300	<30	80	FT–4	Modem I/O cable
1067	OW1 A15 SCSI 2	OW1 A5 DASM	5.9 ft. SCSI cable 50	CI3	30	<30	80	VW–1	DASM I/O cable
1068	OW1 A15 Video2	OW1 A3 LCD	9.9 ft. Data (15–#28)	CI3	300	<30	80	VW–1	LCD Display Video cable
1069	OW1 A11 J8	OW1 A3 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	LCD Display Power cable
1070	OW1 A11 J11	OW1 A16 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	SCSI Tower Power cable
1071	OW1 A11 J1	OW1 A15 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	Host Computer Power cable
1074	OW1 A19 J2	OW1 A11	3.3 ft. Power (3–#14)	SVT	600	120	105	VW–1	Power Strip cord
1075	OW1 A19 J4	OW1 A11	3.3 ft. Power (3–#14)	SVT	600	120	105	VW–1	Power Strip cord
1082	OW1 A15 ETH SLOT 2	MR2 A11 J63	75 ft. Ethernet Cat 5	CI3	300	<30	80	FT–4	EXCITE Ethernet C'cable
1085	OW1 A21 J7	PP1 J12	65 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	Intercom
1180	MR3 J7	MR2 A11 J56	26 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Can
1181	MR3 J8	PP1 PHPS J154	65 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Can
1184	MR2 A11 J125	PP1 PHPS J32	65 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	
1185	MR2 A11 J200	PP1 J200	8–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	
1186	MR2 A11 J201	PP1 J201	8–Coax 8W8 Combo D	CATV E	1500	<30	85	FT–4	
1187	MR2 A11 J202	PP1 J202	2–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	

18–3–7 EQUIPMENT AND OPERATOR ROOM CABLES (Continued)

TABLE 10–13 (CONT'D)
EQUIPMENT & OPERATOR ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1188	MR2 DRV J3	PP1 A1 J77	65 ft 15 pin Sub–D	CI3	300	<30	80	FT–4	16 Channel SW Cntrl Power
1189	MR2 DRV J5	PP1 A17 J78	65 ft 37 pin Sub–D	CI3	300	<30	80	FT–4	MC Drive
1190	MR2 DRV J10	PP1 J203	65 ft 37 pin Sub–D	CI3	300	<30	80	FT–4	MC Drive
1191	MR2 DRV J21	PP1 A11 J92	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1192	MR2 DRV J22	PP1 A11 J93	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1193	MR2 DRV J23	PP1 A11 J94	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1194	MR2 DRV J24	PP1 A11 J95	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1195	MR2 DRV J25	PP1 A11 J96	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1196	PP1 A17 J1	PP1 PHPS J35	2 ft 15 pin Sub–D	CI3	300	<30	80	FT–4	Repeater Board Power
1233	TAC J116 (J5)	PP1 PHPS J155	30 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Can
1249	MR8 J3	CONNECT TO RUN 1238	5 ft. Times Microwave LMR	CATV R	6000 VDC		85	FT–4	RF Extension cable
1255	OW1 A21 J18	MR2 A11 J26	60 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	E–stop
1256	OW1 A20, A7, A8	OW1 A21 J6	13.2 ft. 50 pin SCSI, 9 pin Sub–D	CI3	300	<30	80	VW–1	SCIM, Mouse, Keyboard cable
1259	MR2 RF AMP J4	CONNECT TO RUN 235							

18–3–8 CABLE SPARES (FRU'S)

TABLE 10–14
EQUIPMENT & OPERATOR ROOM RUNS

CABLE PART NUMBER	CABLE RUN
46–244710G30	Run 1227 Run 1236
46–244712G30	Run 1188
46–244713G30	Run 1181
46–244714G30	Run 1180 Run 1181 Run 1233
46–258877G30	Run 1189 Run 1190

18–3–9 EXCITE 4KW AND 8KW MNS CABLES (Optional)

TABLE 10–15
EXCITE 4KW AND 8KW MNS RUNS (optional)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1129	CPC J8	MR3 PD1	35 ft. Power (3–#10)	STO	600	208	90	FT–4	4KW CPC Amp Power cable
1130	CPC J3	MR2 A11 J52	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW–1	4KW MNS cable
1136	CPC J9	MR2 A11 J4	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW–1	4KW MNS cable
1137	RUN 1158	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT–4	4KW MNS cable
1138	MR6 J12	MR3 PD1	65 ft. Power (5–#8)	SO	600	208Y	75	FT–4	BB RF Amp Power Cable
1139	AMT J3	Mr2 A11 J52	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW–1	8KW MNS cable
1140	MR1 J123	MR6 J1	30 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	8KW MNS cable
1141	MR1 J102	AMT J5	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW–1	8KW MNS cable
1142	MR1 J103	AMT J6	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW–1	8KW MNS cable
1143	MR1 J104	AMT J7	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW–1	8KW MNS cable
1144	MR1 J105	AMT J8	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW–1	8KW MNS cable
1145	MR2 A11 J4	AMT J11	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW–1	8KW MNS cable
1146	RUN 1159	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT–4	8KW MNS cable
1147	MG3 A17 J3	PP! J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT–4	4KW & 8KW MNS cable
1158	RUN 1137	MR1 CPC J2	5 ft. Times Microwave LMR 400FR	CATV R	2500V DC		85	FT–4	4KW MNS Extension cable
1248	RUN 1146	MR9 J4	5 ft. Times Microwave LMR 400FR	CATV R	2500V DC		85	FT–4	8KW MNS Extension cable

18–4 FIBER OPTIC TERMINATION INSTRUCTIONS

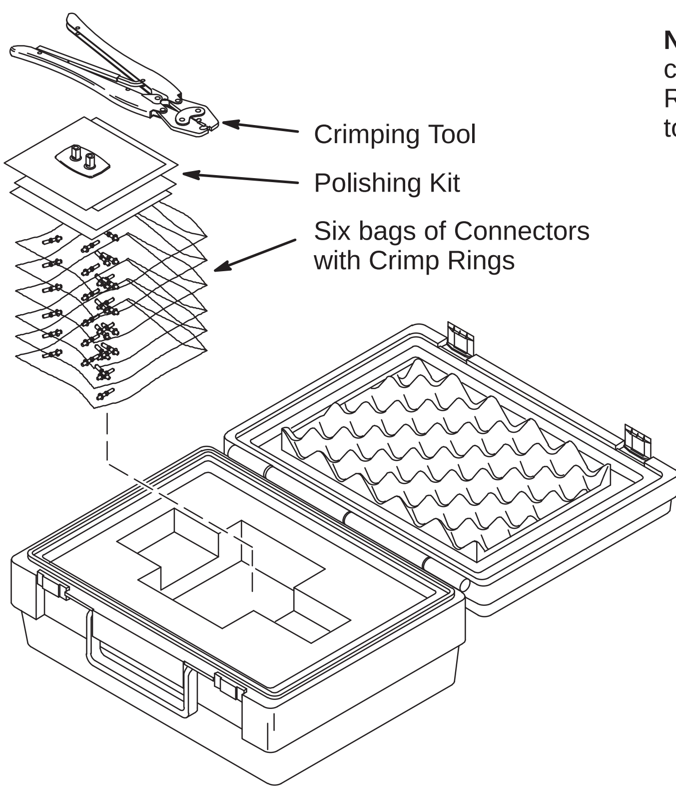
WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

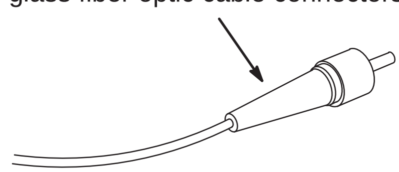
18–4–1 FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.

1. The Fiber Optic Repair Kit, 46–301450G1 shown below, contains all the necessary materials to replace a damaged connector.
2. The “Spares Kit”, delivered with the Shipping Collector, contains a 46–320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.
3. Remove the end of fiber optic cable to be re–terminated from magnet room if possible.
4. f it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.



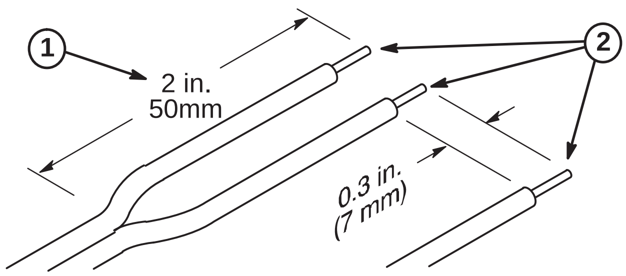
Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



18–4–2 STRIPPING FIBER OPTIC OUTER JACKET

The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

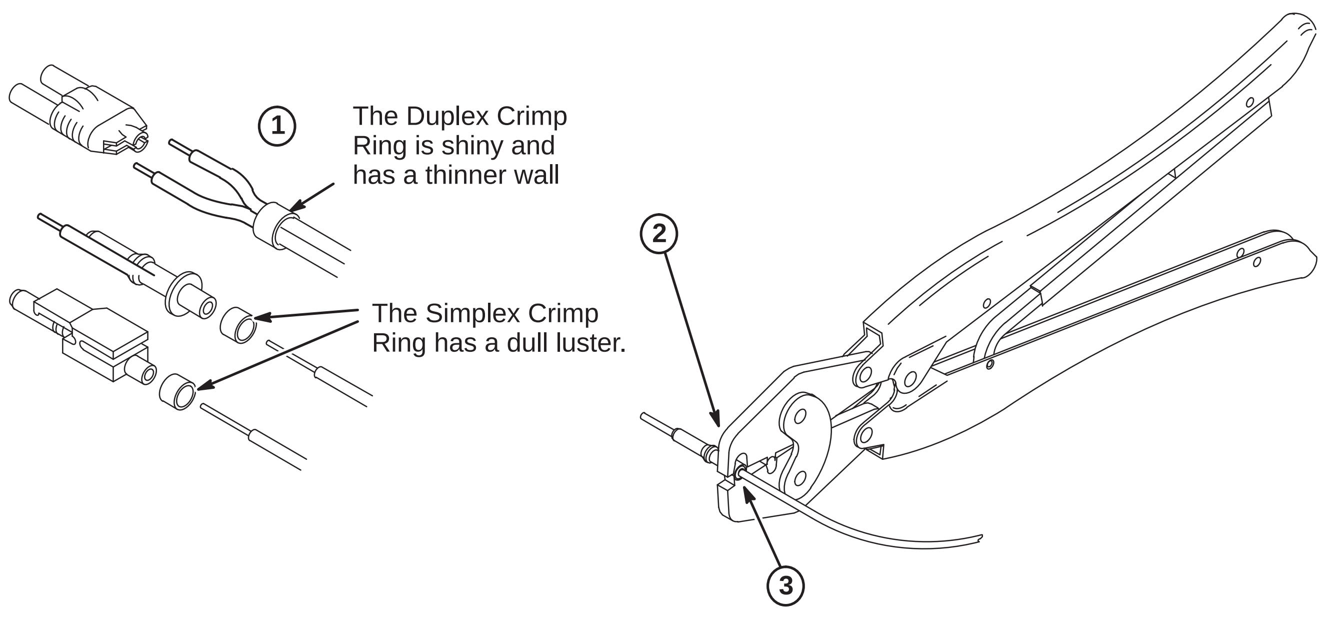
1. For Duplex Cables, seperate the two fibers approximately 2 in. (50mm) back from the ends.
2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.



18–4–3 CRIMPING CONNECTOR TO FIBER OPTIC CABLE

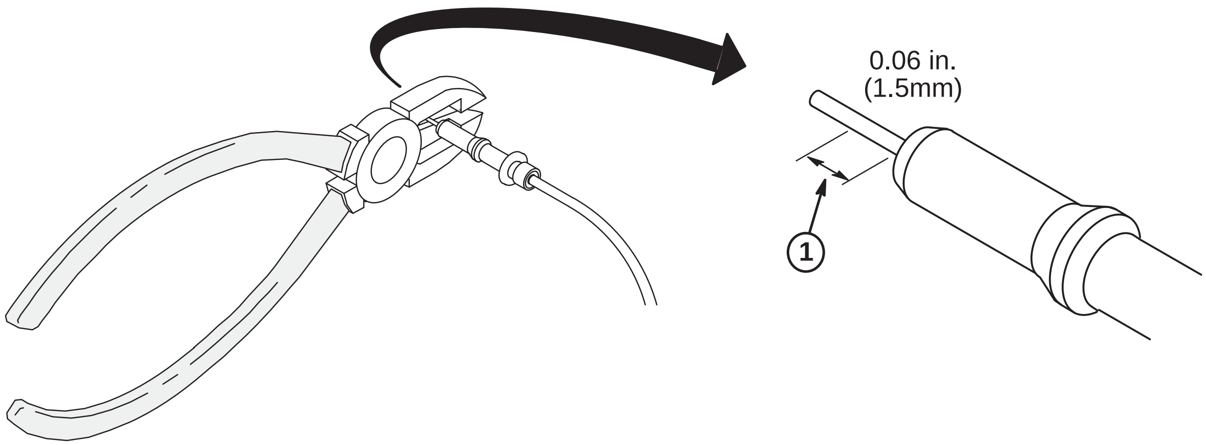
The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

1. Place the Crimp Ring and Connector over the end of the cable.
2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.
3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.



18–4–4 TRIMMING EXCESS FIBER FROM CONNECTOR

1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 inch (1–1/2mm).



18–4–5 POLISHING FIBER OPTIC END

The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.